

Chemical Reaction Engineering Module

User's Guide

Chemical Reaction Engineering Module User's Guide

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User's Guide Introduction

This guide describes the Chemical Reaction Engineering Module, an optional package that extends the COMSOL Multiphysics® modeling environment with customized physics interfaces and functionality for the analysis of mass transport, chemical reactions, thermodynamic properties, and other features that are important for chemical engineering simulation.

This chapter introduces you to the capabilities of the module. A summary of the physics interfaces and where you can find documentation and model examples is also included. The last section is a brief overview with links to each chapter in this guide.

- [About the Chemical Reaction Engineering Module](#)
- [Overview of the User's Guide](#)

About the Chemical Reaction Engineering Module

In this section:

- [The Scope of the Chemical Reaction Engineering Module](#)
- [The Chemical Reaction Engineering Module Physics Interface Guide](#)
- [Common Physics Interface and Feature Settings and Nodes](#)
- [The Material Database](#)
- [Where Do I Access the Documentation and Application Libraries?](#)

The Scope of the Chemical Reaction Engineering Module

The Chemical Reaction Engineering Module is tailor-made for the modeling of chemical systems primarily affected by chemical composition, reaction kinetics, fluid flow, and temperature as functions of space, time, and each other. It has a number of physics interfaces to model chemical reaction kinetics, mass transport in dilute, concentrated, and electric potential-affected solutions, laminar and porous media flows, and energy transport.

Included in these physics interfaces are the kinetic expressions for the reacting systems and models for the definition of mass transport. A variety of ready-made expressions are also accessible in order to calculate a system's thermodynamic and transport properties.

Like all COMSOL modules, the physics interfaces described in this guide include all the steps available for the modeling process, which are described in detail in the *COMSOL Multiphysics Reference Manual* (see [Where Do I Access the Documentation and Application Libraries?](#)), for example:

- *Definitions* of parameters and model variables.
- Creating, importing, and manipulating a *geometry*.
- Specifying the chemical and transport *material properties*.
- Defining the reaction formulas and *physics* in the system and on boundaries, and coupling them to other features.

- *Meshing* a modeling domain with appropriate consideration given to the reaction system's behavior.
- Solving the equations that describe a system for stationary or dynamic behavior, or as a parametric or optimization *study*.
- Analyzing *results* to present for further use.

Once a model is defined, you can go back and make changes to all the branches listed above, while maintaining consistency in the other definitions throughout. You can restart the solver, for example, using the existing solution as an initial guess or even alter the geometry, while the equations and boundary conditions are kept consistent through the associative geometry feature. It is also useful to review the *Introduction to the Chemical Reaction Engineering Module* included with the module's documentation.

While a major focus of this module is on chemical reactors and reacting systems, it is also extensively used for systems where mass transport is the major component. This includes unit operations equipment, separation and mixing processes, corrosion, chromatography, and electrophoresis. The module is also widely used for educational purposes including courses about chemical engineering, chemical reaction engineering, electrochemical engineering, biotechnology, and transport phenomena.

In addition to its application in traditional chemical industries, it is a popular tool for investigating clean technology processes (for example, catalytic monoliths and reactive filters), applications such as microlaboratories in biotechnology, and in the development of sensors and equipment in analytical chemistry.



The notations and structure in this module were inspired by the book *Transport Phenomena* by Bird, Stewart, and Lightfoot. The work by H. Scott Fogler, *Elements of Chemical Reaction Engineering*, was also used as an inspiration.

The Chemical Reaction Engineering Module Physics Interface Guide

The appropriate physics interface for modeling a system is first chosen from the *Model Wizard*. The wizard includes physics branches, AC/DC, Acoustics, Chemical Species Transport, Fluid Flow, Heat Transfer, Structural Mechanics, and Mathematics, as well as the core physics interfaces included with COMSOL Multiphysics (Electrostatics, for example). The Heat Transfer in Fluids, Laminar Flow, and Transport of Diluted Species interfaces are also included in the base package but have increased functionality

with this module. See the *COMSOL Multiphysics Reference Manual* for details pertaining to the base package.

	• Modeling Fluid Flow • Modeling Heat Transfer • Creating a New Model in the <i>COMSOL Multiphysics Reference Manual</i>
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When one or several physics interfaces are chosen from the **Model Wizard** (or if you open the **Add Study** window), you select an analysis type (stationary, dynamic, or parametric) and then the modeling interfaces are available as a nodes in the **Model Builder** along with all the other nodes required for modeling (Definitions, Geometry, and so forth).

By adding another physics interface, you can account for a phenomenon not previously described in a model. To do this, right-click a **Component** node in the **Model Builder** to open the **Add Physics** window. You can do this at any stage during the modeling process. This action still retains the existing geometry, equations, boundary conditions, and current solution, which you can build upon for further development of the model.

The table below lists all the physics interfaces specifically available or enhanced with this module (in addition to the basic COMSOL Multiphysics license). These physics interfaces form the backbone of the module and are based on the equations for chemical reaction kinetics, chemical species transport, as well as for fluid flow and heat transfer in porous media fundamentals.

	In the <i>COMSOL Multiphysics Reference Manual</i>: • Studies and Solvers • The Physics Interfaces • For a list of all the core physics interfaces included with a COMSOL Multiphysics license, see Physics Interface Guide.
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PHYSICS INTERFACE	ICON	TAG	SPACE DIMENSION	AVAILABLE STUDY TYPE
 Chemical Species Transport				
Transport of Diluted Species ¹		tds	all dimensions	stationary; time dependent
Transport of Concentrated Species		tcs	all dimensions	stationary; time dependent
Chemistry		chem	all dimensions	stationary; time dependent
Reaction Engineering		re	0D	time dependent; stationary plug flow
Nernst–Planck Equations		npe	all dimensions	stationary; time dependent
Nernst–Planck–Poisson Equations		tds+es	all dimensions	stationary; time dependent; stationary source sweep; small-signal analysis, frequency domain
Electrophoretic Transport		el	all dimensions	stationary; stationary with initialization; time dependent; time dependent with initialization
Transport of Diluted Species in Porous Media		tds	all dimensions	stationary; time dependent
Transport of Concentrated Species in Porous Media		tcs	all dimensions	stationary; time dependent
Surface Reactions		sr	all dimensions	stationary (3D, 2D, and 2D axisymmetric models only); time dependent
Transport of Diluted Species in Fractures		dsf	3D, 2D, 2D axisymmetric	stationary; time dependent
 Reacting Flow				
Laminar Flow, Diluted Species ¹		—	3D, 2D, 2D axisymmetric	stationary; time dependent

PHYSICS INTERFACE	ICON	TAG	SPACE DIMENSION	AVAILABLE STUDY TYPE
Laminar Flow, Concentrated Species		—	3D, 2D, 2D axisymmetric	stationary; time dependent
 Reacting Flow in Porous Media				
Transport of Diluted Species		—	3D, 2D, 2D axisymmetric	stationary; time dependent
Transport of Concentrated Species		—	3D, 2D, 2D axisymmetric	stationary; time dependent
Transport of Diluted Species, Porous Catalyst		—	3D, 2D, 2D axisymmetric	stationary; time dependent
Transport of Concentrated Species, Porous Catalyst		—	3D, 2D, 2D axisymmetric	stationary; time dependent
Transport of Diluted Species, Packed Bed		—	3D, 2D, 2D axisymmetric	stationary; time dependent
Transport of Concentrated Species, Packed Bed		—	3D, 2D, 2D axisymmetric	stationary; time dependent
Transport of Diluted Species, Packed Bed, Shrinking Core Model		—	3D, 2D, 2D axisymmetric	stationary; time dependent
 Nonisothermal Reacting Flow				
Brinkman Equations		—	3D, 2D, 2D axisymmetric	stationary; time dependent
Laminar Flow		—	3D, 2D, 2D axisymmetric	stationary; time dependent
 Dispersed Two-Phase Flow with Species Transport				
Laminar Flow		—	3D, 2D,	stationary; time dependent
 Vapor–Liquid Equilibrium				

PHYSICS INTERFACE	ICON	TAG	SPACE DIMENSION	AVAILABLE STUDY TYPE
Laminar Vapor Flow		—	3D, 2D, 2D axisymmetric	stationary; time dependent
Laminar Two-Phase Flow		—	3D, 2D, 2D axisymmetric	stationary; time dependent
Precipitation and Crystallization				
Size-Based Population Balance		pbsb	all dimensions	stationary; time dependent
Laminar Two-Phase Flow		—	3D, 2D, 2D axisymmetric, 0D	stationary; time dependent
Fluid Flow				
Single-Phase Flow				
Creeping Flow		spf	3D, 2D, 2D axisymmetric	stationary; time dependent
Laminar Flow ¹		spf	3D, 2D, 2D axisymmetric	stationary; time dependent
Porous Media and Subsurface Flow				
Brinkman Equations		br	3D, 2D, 2D axisymmetric	stationary; time dependent
Darcy's Law		dl	all dimensions	stationary; time dependent
Free and Porous Media Flow, Brinkman		fp	3D, 2D, 2D axisymmetric	stationary; time dependent
Free and Porous Media Flow, Darcy		—	3D, 2D, 2D axisymmetric	stationary; time dependent

PHYSICS INTERFACE	ICON	TAG	SPACE DIMENSION	AVAILABLE STUDY TYPE
 Nonisothermal Flow				
Brinkman Equations		—	3D, 2D, 2D axisymmetric	stationary; time dependent; stationary, one-way NITF; time dependent, one-way NITF
 Heat Transfer				
Heat Transfer in Fluids [†]		ht	all dimensions	stationary; time dependent
Heat Transfer in Solids and Fluids [†]		ht	all dimensions	stationary; time dependent
Heat Transfer in Porous Media		ht	all dimensions	stationary; time dependent
[†] This physics interface is included with the core COMSOL Multiphysics software but has added functionality for this module.				

The Material Database

The Chemical Reaction Engineering Module includes an additional **Liquids and Gases** material database with temperature-dependent fluid dynamic and thermal properties.

	• Liquids and Gases Material Library • Materials
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Common Physics Interface and Feature Settings and Nodes

There are several common settings and sections available for the physics interfaces and feature nodes. Some of these sections also have similar settings or are implemented in the same way no matter the physics interface or feature being used. There are also some physics feature nodes that display in COMSOL Multiphysics.

In each module's documentation, only unique or extra information is included; standard information and procedures are centralized in the *COMSOL Multiphysics Reference Manual*.



In the *COMSOL Multiphysics Reference Manual* see [Table 2-4](#) for links to common sections and [Table 2-5](#) to common feature nodes. You can also search for information: press F1 to open the **Help** window or Ctrl+F1 to open the **Documentation** window.

Where Do I Access the Documentation and Application Libraries?

A number of online resources have more information about COMSOL, including licensing and technical information. The electronic documentation, topic-based (or context-based) help, and the Application Libraries are all accessed through the COMSOL Desktop.



If you are reading the documentation as a PDF file on your computer, the [blue links](#) do not work to open an application or content referenced in a different guide. However, if you are using the Help system in COMSOL Multiphysics, these links work to open other modules, application examples, and documentation sets.

THE DOCUMENTATION AND ONLINE HELP

The *COMSOL Multiphysics Reference Manual* describes the core physics interfaces and functionality included with the COMSOL Multiphysics license. This book also has instructions on how to use COMSOL Multiphysics and how to access the electronic Documentation and Help content.

Opening Topic-Based Help

The Help window is useful as it is connected to the features in the COMSOL Desktop. To learn more about a node in the Model Builder, or a window on the Desktop, click to highlight a node or window, then press F1 to open the Help window, which then

displays information about that feature (or click a node in the Model Builder followed by the **Help** button (). This is called *topic-based* (or *context*) *help*.

	To open the Help window: • In the Model Builder, Application Builder, or Physics Builder, click a node or window and then press F1. • On any toolbar (for example, Home, Definitions, or Geometry), hover the mouse over a button (for example, Add Physics or Build All) and then press F1. • From the File menu, click Help (). • In the upper-right corner of the COMSOL Desktop, click the Help () button.
 	To open the Help window: • In the Model Builder or Physics Builder, click a node or window and then press F1. • In the main toolbar, click the Help () button. • From the main menu, select Help > Help.

Opening the Documentation Window

 	To open the Documentation window: • Press Ctrl+F1. • In the main toolbar, click the Documentation () button. • From the main menu, select Help > Documentation.
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THE APPLICATION LIBRARIES WINDOW

Each model or application includes documentation with the theoretical background and step-by-step instructions to create a model or application. The models and applications are available in COMSOL Multiphysics as MPH files that you can open for further investigation. You can use the step-by-step instructions and the actual models as templates for your own modeling. In most models, SI units are used to describe the relevant properties, parameters, and dimensions, but other unit systems are available.

Once the Application Libraries window is opened, you can search by name or browse under a module folder name. Click to view a summary of the model or application and its properties, including options to open it or its associated PDF document.

Opening the Application Libraries Window

To open the **Application Libraries** window ():

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Overview of the User's Guide

The *Chemical Reaction Engineering Module User's Guide* is a consummate guide to using COMSOL Multiphysics and the add-on module Chemical Reaction Engineering Module. The information in this guide is specific for this module. Instructions how to use COMSOL in general are included with the *COMSOL Multiphysics Reference Manual*. The *Introduction to the Chemical Reaction Engineering Module* contains a quick-start guide to help you get started using this module.



As detailed in the section [Where Do I Access the Documentation and Application Libraries?](#) this information can also be searched from the COMSOL Multiphysics software **Help** menu.

THE CHEMISTRY AND REACTION ENGINEERING INTERFACES

The [Reaction Engineering Interface](#), which is found under the Chemical Species Transport branch, has all of the tools required to simulate chemical reaction kinetics, is described in detail for the [Theory for the Reaction Engineering and Chemistry Interfaces](#).

THE CHEMICAL SPECIES TRANSPORT INTERFACES

The transport and conversion of chemical compounds is denoted as chemical species transport. The [Chemical Species Transport Interfaces](#) chapter describes the Transport of Diluted Species, Transport of Diluted Species in Porous Media, and Transport of Concentrated Species interfaces, which are used for the simulation of chemical reactions, and mass chemical compounds transport through diffusion, convection, and electromigration. It also includes the Surface Reactions interface, which models reactions involving surface adsorbed species and species in the bulk of a reacting surface. The Nernst-Planck, The Electrophoretic Transport, Reacting Flow, Laminar Flow, and Transport of Diluted Species in Porous Media interfaces are also described in this chapter.

The chapter [Chemical Species Transport Interfaces](#) also describes the following multiphysics interfaces included in the Chemical Reaction Engineering Module: Reacting Flow, Reacting Flow in Porous Media, Nonisothermal Reacting Flow, Dispersed Two-Phase Flow with Species Transport, Vapor-Liquid Equilibrium, and Precipitation and Crystallization.

[Overview of Chemical Species Transport Interfaces](#) helps you select the best physics interface to use. The rest of the chapter describes the physics interfaces in detail as well as the underlying theory.

THE FLUID FLOW INTERFACES

The [Fluid Flow Interfaces](#) chapter describes the physics interfaces under the Porous Media and Subsurface Flow branch, which are the Darcy's Law, Brinkman Equations, Free and Porous Media Flow, Brinkman and Free and Porous Media Flow, Darcy interfaces. To help you select the physics interface to use, see [Modeling Fluid Flow](#).

THE HEAT TRANSFER INTERFACES

The chapter describes [The Heat Transfer in Porous Media Interface](#) and includes a section to help choose the right physics interface ([Modeling Heat Transfer](#)).

THERMODYNAMICS INPUT

The chapter [Thermodynamics](#) describes how to use the thermodynamics functionality to define thermo-physical and transport properties. The properties in turn can be used when simulating chemical reaction systems, or any type of transport model involving mass transfer, fluid flow, or heat transfer.

The Chemistry and Reaction Engineering Interfaces

This chapter describes the **Chemistry (chem)** and **Reaction Engineering (re)** interfaces, available in the **Chemical Species Transport** branch ().

In this chapter:

- [Overview of the Reaction Engineering and Chemistry Interfaces](#)
- [Theory for the Reaction Engineering and Chemistry Interfaces](#)
- [The Reaction Engineering Interface](#)
- [The Chemistry Interface](#)
- [Study Steps for the Reaction Engineering Interface](#)

See also [Chemical Species Transport Interfaces](#) for the additional physics interfaces in the **Chemical Species Transport** branch ().

Overview of the Reaction Engineering and Chemistry Interfaces

The **Reaction Engineering** and **Chemistry** interfaces share many common features and functionality for defining species and reactions. The main difference is that the **Reaction Engineering** interface always is defined within the context of a (zero dimensional) *reactor model*, solving for a number of species dependent variables and an optional heat balance, whereas the **Chemistry** interfaces solely defines a number of *variable expressions*, based on species and reactions, that can be used when coupling the **Chemistry** interface to, for instance, a space-dependent mass transport model.

THE REACTION ENGINEERING INTERFACE

The **Reaction Engineering Interface** () includes all of the tools required to simulate chemical reaction kinetics in well-defined environments. This includes the automatic set up and simulation of reversible, equilibrium, and irreversible reactions in volumes or on surfaces. A library of predefined reactors is also available, including batch, continuous stirred-tank, semibatch, and plug flow reactors (see [Reactor Types in the Reaction Engineering Interface](#)). In addition, the Parameter Estimation feature makes it possible to evaluate kinetic and model parameters from experimental data.



Use the [Generate Space-Dependent Model](#) node to create a space-dependent model based on your model created in the Reaction Engineering interface. This automatically sets up the mass balances and source terms for systems where reactions occur, with a copy of all and your original species and reaction definitions recreated in separate a Chemistry node.

The Reaction Engineering interface can also be used for defining and simulating thermodynamic data.

Interfaces to internal or external thermodynamic packages (such as CAPE-OPEN compliant packages), and an import feature for CHEMKIN[®] files, mean that the thermodynamic properties within reaction kinetics, as well as mass and heat transfer applications, can be described by empirical relations and algebraic expressions. Also see the [Theory for the Reaction Engineering and Chemistry Interfaces](#).

THE CHEMISTRY INTERFACE

The [Chemistry Interface](#) () is a tool for generating a set of variables to be used for modeling chemical species and reactions systems. The variables are generated from species and reaction properties and can be divided in two categories:

- Rate expressions and heat sources for use in mass and heat balances
- Material property variables (mixture density, diffusivities, viscosity, and so on) for use in space-dependent transport equations.

The **Chemistry** interface is also created when the [Generate Space-Dependent Model](#) feature is used in [The Reaction Engineering Interface](#), collecting all mixture variables and properties for use in a space-dependent model.

Using the Reaction Node

In each [Reaction](#) node, specify the reaction **Formula** (chemical equation) and **Reaction type** of a single chemical equation.

For each reaction formula entered, specify the reaction type with a delimiter separating the two sides of the equation:

- $\rightleftharpoons$ denotes a reversible reaction
- $\Rightarrow$ denotes an irreversible reaction
- $=$ denotes a reaction at chemical equilibrium

It is also possible to adjust the reaction type with the **Reaction type** list (Figure 2-1).

Settings
Reaction

Label: 1: roh + H+ => ren + H2O + H+

Reaction Formula Balance

Formula:
roh + H+ => ren + H2O + H+ Apply

Reaction type:
Irreversible

$$R_i = \sum_j \nu_{ij} r_j$$

Reaction Rate

Mass action law

$$r_j = k_j^f \prod_{i \in \text{react}} c_i^{-\nu_{ij}}$$

Overall forward reaction order: 2

Rate Constants

☐ Use Arrhenius expressions

Forward rate constant:
 k^f 7.45e-3[m³/(s*mol)] m³/(mol*s)

Figure 2-1: After adding a Reaction node, enter the chemical reaction formula (chemical equation) and specify the reaction type.

Right-click any **Reaction** node to disable and enable the corresponding node in the **Model Builder**. Creating reaction subsets in this way is a straightforward approach to investigate the influence of individual reactions on the overall reaction system. Species that take part only in deactivated reactions are automatically deactivated, as indicated by the unavailable **Species** feature nodes. If a reaction is deleted, the interface automatically deletes those species that take part only in the deleted reactions.

ABOUT REACTION IDENTIFICATION NUMBERS, VARIABLE SCOPE, AND LABELING

When reaction formulas are entered, each reaction is associated with a unique ID (identification) number (Figure 2-2). The ID number is not a row number but rather an absolute reference number to a given reaction. This means that a reaction keeps its ID number, even if reactions are deleted with a lower number. Furthermore, constants

and expressions within fields associated with a given reaction are also indexed with the reaction number.

The **Name** prepended to the variable name is called the variable scope. Defaults are **re** and **chem** for the Reaction Engineering and Chemistry interface, respectively. All variables set up by the interface are given the same scope. The variable scope can be useful to couple the equations of different physics interfaces.

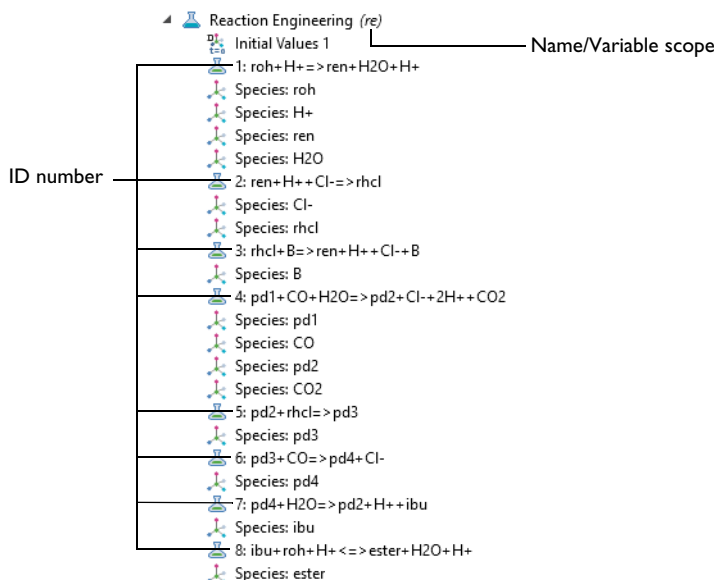


Figure 2-2: The ID number of each Reaction node is displayed in front of the reaction formula.

As a general labeling rule, the variable name that refers to the contents of a field associated with a **Reaction** node is given by the physics interface **Name**, followed by the field name, and ends with the reaction ID number. For example, the contents of the reaction rate field **r** for Reaction 1 is assigned the variable name **re.r_1**.



For an example of how to use variables with scope, see *Tank Series with Feedback Control*: Application Library path
**Chemical_Reaction_Engineering_Module/Ideal_Tank_Reactors/
 tankinseries_control**

Using the Species Node

The [Reaction Engineering Interface](#) and [The Chemistry Interface](#) generate **Species** nodes in the **Model Builder** based on the chemical equations entered in the **Settings** window for **Reaction**.

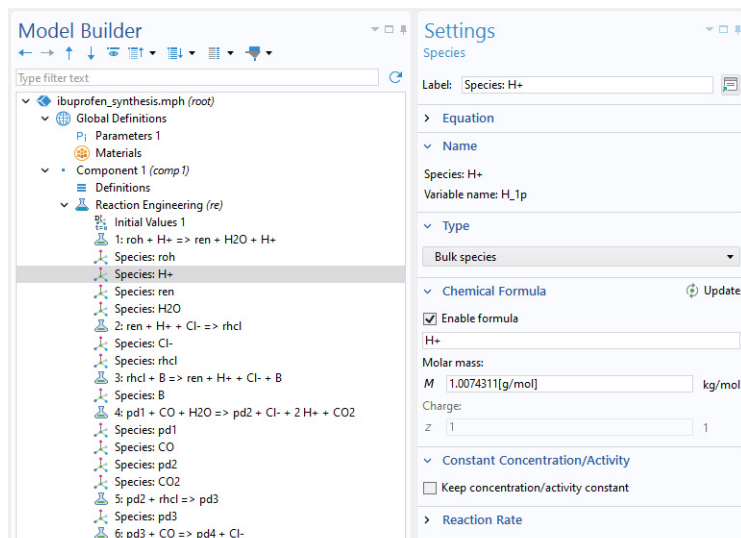


Figure 2-3: Species nodes are generated automatically as chemical equations are entered in the Reaction nodes.

As with [Reaction](#) features you can add, remove, or deactivate [Species](#) features by right-clicking a node in the Model Builder. Deactivation of a species automatically deactivates any reactions in which the species is participating.

In the **Reaction Rate** section it is possible to alter the definition of the reaction rate of the species. This will override the settings in the [Reaction](#) feature, where the rate is defined by the stoichiometry of the reaction(s) in which the species participates.

It is also possible to add individual **Species** nodes with either reactive or nonreactive species.

CHOOSING THE SPECIES TYPE

For each species entered, it is possible to change its characteristics by selecting its species type — **Bulk species**, **Surface species**, or **Solvent**.

The **Type** selection has implications in the calculation of thermodynamic and transport properties.

Bulk species and **Surface species**, defined per reactor volume and area, respectively, set the mixture's physical properties dependent upon its composition. However, configuring a species as a **Solvent** sets the physical properties of the reacting fluid mixture equal to those of the solvent species; specifically, its density, heat capacity, viscosity, and thermal conductivity. The interface also implements a solute-solvent approximation for the interaction of species in the fluid and describes the transport properties accordingly. In material balances this means that the diffusion coefficient is independent of any of the solute's concentrations, because every solute only interacts with solvent molecules, regardless of the concentration. In addition, the convective term in the flux of species is directly given by the velocity field of the solvent multiplied by the solute concentration.

The **Type** has implications on the [Generate Space-Dependent Model](#) procedure since it determines whether interfaces using a solvent-solute approximation of the reacting fluid mixture (as in [The Transport of Diluted Species Interface](#)) or a full multicomponent description of the reacting fluid mixture (as in [The Transport of Concentrated Species Interface](#)) should be generated.

Furthermore, the **Type** affects the reaction kinetics. **Solvent** sets the species' concentration to a constant value (the initial species concentration). The Reacting Engineering interface does not formulate a mass balance for the solvent species. This setting corresponds to situations where the solvent does not take part in chemical reactions at all, or where it reacts but is present in large excess.

VALID SPECIES NAMES

Valid species names consist of a case-sensitive combination of letters, numbers, charge indication and phase indication. With the exception of the “+” and “-” signs, the caret character “^”, underscore “_”, and the enclosing characters “()[]{}”, special characters are not allowed. In addition, the first character in a species name must be a letter, or enclosing character, to avoid confusion with the stoichiometric coefficients in the reaction formula.

Charge is indicated in one of the following ways:

- Fe++, Fe(++), Fe(2+)
- Fe^++, Fe^(++), Fe^(2+), Fe^2+

The charge indication, for example (++), is placed last in the species name, but before any phase indication. The charge is automatically read from the species **Name** field, or **Chemical Formula** field, and entered in the **Charge** field in the associated Species feature.

A valid phase indication consists of a valid species name followed by a valid phase. The phase should be placed last. For example, water in gaseous form is written “H₂O(g)”, and a divalent cation of iron in water can be written “Fe²⁺(aq)”. Valid phases are:

- (g), for gas or vapor
- (l), for liquid
- (s), for solid
- (aq), for aqueous solution
- (ads), for adsorbed/surface species
- (cd), for condensed phase (solid or liquid)
- (fl), for fluid (gas or liquid)
- (lc), for crystalline liquid
- (cr), for crystalline solid
- (am), amorphous solid
- (vit), vitreous substance (glass)
- (mon), for monomeric form
- (pol), for polymeric form
- (sln), for solution

Using the phases **(ads)**, **(s)**, **(l)**, **(aq)**, or **(g)** creates adsorbed, solid, liquid, aqueous, or gaseous species with variable names; `speciesname_surf`, `speciesname_solid`, `speciesname_liquid`, `speciesname_aq`, or `speciesname_gas`, respectively. The **Type** field in the Species feature, is filled in automatically depending on which phase that was used. The phases **(s)**, **(l)**, and **(g)** give rise to **Bulk species**, while a species with phase indication **(ads)** results in a **Surface species**. **(aq)** gives rise to aqueous species. Note that the [Equilibrium Species](#) remains the same despite the presence of any gaseous or liquid species in the system.

Similar to the labeling rule applying to Reaction nodes, the variable name referring to the contents of a field associated with a Species node is given by the interface **Name**, followed by the field name, and ending with the species name. For example, the contents of the **Rate expression** field **R** for the species roh is assigned the variable name `re.R_roh` (for the Reaction Engineering interface). Access the definition of all the variables used by a specific node by displaying the **Equation View** node. To display the

node, click the **Show More Options** button () and select **Equation View** in the **Show More Options** dialog.



- Theory for the Reaction Engineering and Chemistry Interfaces
- Variable Naming Convention and Namespace

Using the Equation View Node — Reactions and Species

To display the **Equation View** node, click the **Show More Options** button () and select **Equation View** in the **Show More Options** dialog. Expand any node, for example the **Reaction** or **Species** nodes, and click the **Equation View** to see the page where the definition of all the variables used by a specific node are listed (Figure 2-4).

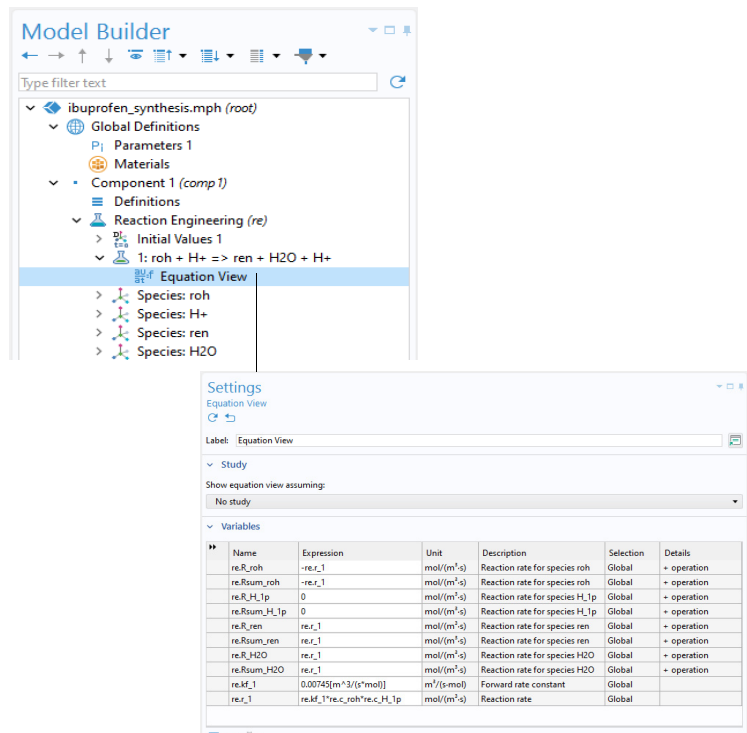


Figure 2-4: The definition of variables used by a node are accessible through the Equation View.

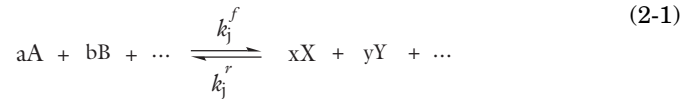
Theory for the Reaction Engineering and Chemistry Interfaces

The [Reaction Engineering Interface](#) and [The Chemistry Interface](#) theory is described in this section.

- [Reactor Types in the Reaction Engineering Interface](#)
- [Transport Properties](#)
- [CHEMKIN Data and NASA Polynomials](#)
- [Working with Predefined Expressions](#)
- [References for the Reaction Engineering Interface](#)

Reaction Kinetics and Rate Expressions

The modeling of reactions in COMSOL uses the mass action law by default. Consider a general reaction belonging to a set of j reactions and involving i species:



For such a reaction set, the reaction rates r_j (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$) can be described by the mass action law:

$$r_j = k_j^f \prod_{i \in \text{react}} c_i^{-\nu_{ij}} - k_j^r \prod_{i \in \text{prod}} c_i^{\nu_{ij}} \quad (2-2)$$

Here, k_j^f and k_j^r denote the forward and reverse rate constants, respectively. The concentration of species i is denoted as c_i (SI unit: mol/m^3). The stoichiometric coefficients are denoted ν_{ij} , and are defined as being negative for reactants and positive for products. In practice, a reaction seldom involves more than two species colliding in a reacting step, which means that a kinetic expression is usually of order 2 or less (with respect to the involved concentrations).

In addition to the concentration dependence, the temperature dependence of reaction rates can be included by using the predefined Arrhenius expression for the rate constants:

$$k = A \left(\frac{T}{T_{\text{ref}}} \right)^n \exp \left(-\frac{E}{R_g T} \right) \quad (2-3)$$

Here, A denotes the frequency factor, n the temperature exponent, E the activation energy (SI unit: J/mol) and R_g the gas constant, 8.314 J/(mol·K). The pre-exponential factor, including the frequency factor A and the temperature factor T^n , is given the units (m³/mol) ^{$\alpha-1$} /s, where α is the order of the reaction (with respect to volumetric concentrations).

The default settings for the reaction given by [Equation 2-1](#) and assuming equilibrium, yields the equilibrium expression in [Equation 2-4](#):

$$K_c = \frac{c_X^x c_Y^y \dots}{c_A^a c_B^b \dots} \quad (2-4)$$

In the case of reversible reactions, it is also possible to make use of equilibrium relationships in order to indirectly calculate reaction constants. For example, calculate the reverse rate constant from the equilibrium condition in this way:

$$k^r = \frac{k^f}{K_c} \quad (2-5)$$

The Equilibrium Constant

In general, chemical equilibrium is reached when the Gibbs free energy of the system is minimized. The Gibbs free energy of is defined as

$$G(T, P) = U + PV - TS = H - TS \quad (2-6)$$

where U is the internal energy, P is pressure, V is volume, T is temperature, S is entropy and H is enthalpy of the system. For a closed system

$$dU = \delta Q + \delta W = TdS - PdV \quad (2-7)$$

where δQ is (reversible) heat transfer to the fluid and δW is (pressure) work in the system. The change in Gibbs free energy can be written as

$$dG = TdS - PdV + PdV + VdP - TdS - SdT = VdP - SdT \quad (2-8)$$

Chemical potential μ_i is defined as partial molar Gibbs energy for species

$$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T, P, n_i \neq n_j} \quad (2-9)$$

where n is the number of mole of species i in the system. At constant temperature, this expression can be integrated as a function of pressure

$$\int_{\mu_i^o}^{\mu_i} d\mu = \int_{P^o}^{P_i} \bar{v}_i dP \quad (2-10)$$

where v is molar volume and μ_i^o is chemical potential of species at standard state. For an ideal gas this can be expressed as

$$\mu_i - \mu_i^o = R_g T \ln \left(\frac{P_i}{P^o} \right) \quad (2-11)$$

The equilibrium criterion can be written as

$$\sum_i \nu_i \mu_i = 0 \quad (2-12)$$

where ν_i is the stoichiometric coefficient and μ_i is the chemical potential of species i in the reaction. We can evaluate the chemical potential as partial molar Gibbs energy of species in mixture as

$$\mu_i - \mu_i^o = R_g T \ln \left(\frac{\hat{f}_i}{f_i^o} \right) \quad (2-13)$$

where $\hat{f}_i$ is the fugacity of species i in the mixture, and f_i^o is the fugacity of pure species in the standard state. Equation 2-12 can be rewritten as

$$\sum_i \nu_i \mu_i^o + R_g T \ln \prod_i \left(\frac{\hat{f}_i}{f_i^o} \right)^{\nu_i} = 0 \quad (2-14)$$

The equilibrium constant for the reaction, K , is defined as

$$K(T) = \exp \left(\frac{-\Delta G_{\text{rxn}}^o}{RT} \right) = \prod_i a_i^{\nu_i} = \prod_i \left(\frac{\hat{f}_i}{f_i^o} \right)^{\nu_i} \quad (2-15)$$

where a_i is the activity of species in the system.

ΔG_{rxn}^o is the Gibbs free energy of reaction (including formation terms) which is defined, in accordance with Equation 2-6, as

$$\Delta G_{rxn}^o = \Delta H_{rxn}^o - T\Delta S_{rxn}^o \quad (2-16)$$

the enthalpy of reaction, ΔH_{rxn}^o , and entropy of reaction, ΔS_{rxn}^o , both at a given temperature T is defined as

$$\Delta H_{rxn}^o(T) = \sum_i \nu_i \left(\Delta H_{f,i}^o + \int_{T^o}^T C_{p,i} dT \right) \quad (2-17)$$

$$\Delta S_{rxn}^o(T) = \sum_i \nu_i \left(S_{abs,i} + \int_{T^o}^T \frac{C_{p,i}}{T} dT \right) \quad (2-18)$$

Here T^o is the temperature at standard state. $\Delta H_{f,i}^o$ and $S_{abs,i}$ are the standard enthalpy of formation and absolute entropy for each species (these data are available in the COMSOL database).

ACTIVITY

Activity of species, a_i is defined by Equation 2-13 as

$$a_i = \exp \left(\frac{\mu_i - \mu_i^o}{R T} \right) \quad (2-19)$$

Activity depends on the choice of an arbitrary standard state. The standard state of pure species is usually at 10^5 Pa and for solute in solution is based on hypothetical molality or amount concentration also referred as infinite dilute behavior.

The activity of a species in a mixture is expressed by relationship between dimensionless activity coefficients, γ_i , and measured amount of the species in the system.

$$a_i = \gamma_{x,i} x_i = \gamma_{c,i} \frac{c_i}{c_{ref}} \quad (2-20)$$

Activity coefficients are usually estimated by [Activity Coefficient Models](#).

Gas Phase

The standard state is the pure species at ideal gas condition, 1 atm and the equilibrium temperature. Activity of species in mixture is expressed by:

$$a_i = \frac{\hat{f}_i}{f_i^o} = \frac{y_i \hat{\phi}_i P}{P^o} \quad (2-21)$$

where $\hat{f}_i$ and $\hat{\phi}_i$ are the fugacity and fugacity coefficient of species i in the mixture.

Liquid Phase

The standard state is pure liquid species at 1 atm and equilibrium temperature. The fugacity of a species in a mixture is given by

$$\hat{f}_i = y_i \gamma_i f_i \quad (2-22)$$

where γ_i is the activity coefficient of species in the mixture and f_i is the fugacity of pure species at the equilibrium temperature and pressure. The activity is expressed by

$$a_i = \frac{y_i \gamma_i f_i}{f_i^o} \quad (2-23)$$

where f_i^o is the fugacity of pure species at the equilibrium temperature and 1 atm. The ratio f_i/f_i^o is given by

$$\frac{f_i}{f_i^o} = \exp\left(\frac{v_i}{R_g T}(P - P_i^{\text{sat}})\right) \quad (2-24)$$

where v_i is the partial molar volume of species and P^{sat} is species saturated vapor pressure. For liquids it is a weak function of pressure, and can be assumed to be 1 unless at high pressure.

Dilute Solutions

The concentration can in the case of nonideal mixtures be replaced with the activity. In these interfaces, the dimensionless activity (a_i) depend on species concentration (c_i), activity coefficient (γ_i), and the standard state concentration ($c_{0s} = 1 \text{ mol/m}^3$).

$$a_i = \frac{c_i}{c_{0s}} \gamma_i \quad (2-25)$$

Additionally, an effective species concentration ($c_{e,i}$) (SI unit: mol/m^3) is used in the reaction rates ([Equation 2-2](#)) when activities are utilized.

$$c_{e,i} = c_i \gamma_i = a_i c_{0s} \quad (2-26)$$

AUTOMATICALLY DEFINED EQUILIBRIUM CONSTANTS

Below, the automatically defined equilibrium constants are defined in terms of the equilibrium expression K_c

$$K_c = \prod_i c_i^{v_i} = \frac{\prod_{i \in \text{prod}} c_i^{v_i}}{\prod_{i \in \text{react}} c_i^{-v_i}} \quad (2-27)$$

Gas Phase

Inserting Equation 2-21 in Equation 2-15 gives:

$$K(T) = K_{\text{eq}} \prod_i \left(\frac{\hat{\phi}_i P}{c_{\text{sum}} P^0} \right)^{v_i} \quad (2-28)$$

where for ideal mixtures, the fugacity coefficients $\hat{\phi}_i$ are equal to 1.

Liquid Phase

For low and moderate pressure, the equilibrium constant can be reformulated by substitution of Equation 2-23 to Equation 2-15 as:

$$K(T) = K_c \prod_i \left(\frac{\gamma_i}{c_{\text{sum}}} \right)^{v_i} \quad (2-29)$$

Handling of Equilibrium Reactions

EXAMPLE I

The following short example illustrates how the Reaction Engineering interface and the Chemistry interface handle equilibrium reactions in the formulation of the material balance equations.

Consider the reaction:



According to Equation 2-2 the reaction rate (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$) is formulated as:

$$r = k^f c_A - k^r c_B$$

where c_A and c_B (SI unit: mol/m³) are the concentrations of A and B , and k^f and k^r (SI unit: 1/s) are the forward and reverse rate constants, respectively. The net rate expressions with respect to species A and B are then:

$$R_A = -k^f c_A + k^r c_B = -r$$

$$R_B = k^f c_A - k^r c_B = r$$

Assuming the reaction in [Equation 2-30](#) is at equilibrium, the reaction rate r is 0:

$$r = k^f c_A - k^r c_B = 0$$

The relationship between the forward and reverse reaction rates in [Equation 2-30](#) is given by the following ratio:

$$K^{\text{eq}} = \frac{k^f}{k^r} = \frac{c_B}{c_A} \quad (2-31)$$

The Reaction Engineering interface also sets up mass balances that are solved. The general material balances for species A and B , respectively, are:

$$\frac{\partial c_A}{\partial t} = R_A = -r \quad (2-32)$$

$$\frac{\partial c_B}{\partial t} = R_B = r \quad (2-33)$$

The rate of consumption of species A equals the production rate of species B , as shown in [Equation 2-32](#) and [Equation 2-33](#).

With the combined information in [Equation 2-32](#), [Equation 2-33](#), and [Equation 2-31](#), the Reaction Engineering interface is able to define the mass balances for the equilibrium system without the reaction rate expressions. The equation system solved for becomes:

$$\frac{\partial}{\partial t}(c_A + c_B) = 0 \quad (2-34)$$

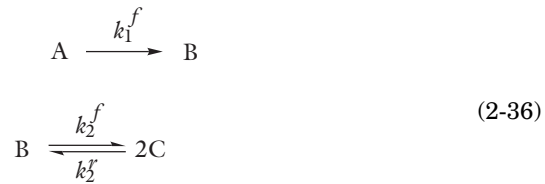
$$K^{\text{eq}} = \frac{c_B}{c_A} \quad (2-35)$$

In general, for a system of reactions contributing to k mass balances and with j reactions being at equilibrium, the reduced system of equations to be solved is composed of $k - j$ mass balances and j equilibrium expressions. The elimination process producing the above system of equations is automated, allowing simple modeling of chemical equilibrium reactions together with irreversible and/or reversible reactions.

EXAMPLE II

This example shows how equilibrium reactions are considered in the Reaction Engineering interface using the [Equilibrium Species](#) section.

If two nonequilibrium reactions are taking place in a perfectly mixed isothermal reactor of constant volume:



the corresponding mass balances are:

$$\frac{d}{dt}c_A = -r_1 \quad (2-37)$$

$$\frac{d}{dt}c_B = r_1 - r_2 \quad (2-38)$$

$$\frac{d}{dt}c_C = 2r_2 \quad (2-39)$$

Now compare [Equation 2-37](#), [Equation 2-38](#), and [Equation 2-39](#) with the balance equations that the physics interface sets up for the related chemistry, where the second reaction is instead an equilibrium reaction:



In contrast to the reversible reaction given by [Equation 2-36](#), to make use of the information contained in the equilibrium relation, the mass balances must be reformulated. Mass balances set up for the reactions given by [Equation 2-40](#) and [Equation 2-41](#) are then:

$$\frac{d}{dt}c_A = -r_1$$

$$\frac{d}{dt}(2c_B + c_C) = 2r_1 \quad (2-42)$$

$$K_2^{\text{eq}} = \frac{c_C^2}{c_B} \quad (2-43)$$

The equilibrium expression ([Equation 2-43](#)) introduces an algebraic relationship between the species' concentrations.

Two species, B or C, can be set as Predefined Dependent Species in the [Equilibrium Species](#) section. Selecting B as the dependent species solves [Equation 2-42](#) for the concentration of C, while B is computed from [Equation 2-43](#).

Reactor Types in the Reaction Engineering Interface

MASS BALANCE

The mass balances that are set up in the Reaction Engineering interface are simplified versions of the general mass transport equation. The main assumption is that the reactor is perfectly mixed, meaning that any variations in compositions within the reactor are neglected.



- [Mass Balance Equation](#)
-

ENERGY BALANCE

The energy balances solved in the Reaction Engineering interface are derived from the general energy balance of a system. The utilized equation follows the perfect mixing assumption and is dependent on the selected reactor as described below.



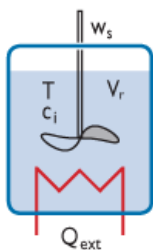
Theory for Heat Transfer

The following reactor types are available in the Reaction Engineering interface (the Chemistry interfaces does not contain reactor models):

- Batch
- Batch, Constant Volume
- CSTR, Constant Mass/Generic
- CSTR, Constant Volume
- Semibatch
- Plug Flow

BATCH

In the batch reactor no mass enters or leaves the system. Common for all reactor models is that reacting fluids in the gas phase are assumed to behave as ideal gases. Liquid mixtures are assumed to be ideal and incompressible.



The species mass balances are given by:

$$\frac{d(c_i V_r)}{dt} = V_r R_i \quad (2-44)$$

which takes into account the effect of changing volume. In Equation 2-44, c_i (SI unit: mol/m³) is the species molar concentration, V_r (SI unit: m³) denotes the reactor volume, and R_i (SI unit: mol/(m³·s)) is the species rate expression.

For an ideal reacting gas, the energy balance is:

$$V_r \sum_i c_i C_{p,i} \frac{dT}{dt} = Q + Q_{\text{ext}} + V_r \frac{dp}{dt} \quad (2-45)$$

In Equation 2-45, $C_{p,i}$ (SI unit: J/(mol·K)) is the species molar heat capacity, T (SI unit: K) is the temperature, and p (SI unit: Pa) the pressure. On the right-hand side, Q (SI unit: J/s) is the heat due to chemical reaction, and Q_{ext} (SI unit: J/s) denotes heat added to the system. The heat of reaction is:

$$Q = -V_r \sum_j H_j r_j$$

where H_j (SI unit: J/mol) is the enthalpy of reaction, and r_j (SI unit: mol/(m³·s)) the reaction rate.

For an incompressible and ideally mixed reacting liquid, the energy balance is:

$$V_r \sum_i c_i C_{p,i} \frac{dT}{dt} = Q + Q_{\text{ext}} \quad (2-46)$$



Ibuprofen Synthesis: Application Library path
**Chemical_Reaction_Engineering_Module/Ideal_Tank_Reactors/
 ibuprofen_synthesis**

BATCH, CONSTANT VOLUME

The constant volume batch reactor is the default reactor. It assumes a closed, perfectly mixed system of constant volume. The species mass balances are:

$$\frac{dc_i}{dt} = R_i$$

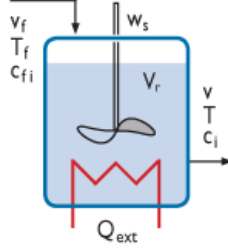
For an ideal reacting gas, the energy balance is given by Equation 2-45. For an incompressible and ideally mixed reacting liquid, the energy balance is given by Equation 2-46.



Neutralization of Chlorine in a Scrubber: Application Library path
**Chemical_Reaction_Engineering_Module/Ideal_Tank_Reactors/
 chlorine_scrubber**

CSTR, CONSTANT MASS/GENERIC

The continuous stirred tank reactor (CSTR) has reacting species entering and leaving the reactor by means of inlet and outlet streams. The reactor is again assumed to be perfectly mixed, so that the species concentrations of the exit stream are the same as the concentrations in the reactor volume.



The species mass balances for the CSTR are given by:

$$\frac{d(c_i V_r)}{dt} = \sum_m v_{f,m} c_{f,m} - v c_i + R_i V_r \quad (2-47)$$

In Equation 2-47, $c_{f,m}$ (SI unit: mol/m³) is the species molar concentration of the associated feed inlet stream $v_{f,m}$ (SI unit: m³/s). V_r (SI unit: m³) denotes the reactor volume and is a function of time.

The reactor volume as a function of time is given by:

$$\frac{dV_r}{dt} = \sum_m v_{f,m} - v + v_p \quad (2-48)$$

In Equation 2-48, v_p (SI unit: m³/s) denotes the volumetric production rate. It is given by Equation 2-49 for ideally mixed liquids and by Equation 2-50 for ideal gases.

$$v_p = V_r \sum_i \frac{R_i M_i}{\rho_i}, \quad R_i = \sum_j v_{ij} r_j \quad (2-49)$$

$$v_p = \frac{R_g T}{p} V_r \sum_i R_i \quad (2-50)$$

where v_{ij} is the stoichiometric coefficient of species i in reaction j , M_i (SI unit: kg/mol) denotes the species molecular weight, ρ_i (SI unit: kg/m³) the species density, and R_i (SI unit: mol/(m³·s)) is the reaction rate of species i .

When this reactor model is solved for constant mass, the reactor model assumes that the volumetric flow rate of the outlet stream, v (SI unit: m^3/s), is adjusted in such a way that the total mass of the reactor is held constant:

$$v = \frac{\sum v_{f,m} \rho_{f,m}}{\rho} \quad (2-51)$$

In contrast, when the model is set to be solved for generic conditions a specific outlet flow stream can be set.

The energy balance for an ideal reacting gas is:

$$V_r \sum_i c_i C_{p,i} \frac{dT}{dt} = Q + Q_{\text{ext}} + V_r \frac{dp}{dt} + \sum_m \sum_i v_{f,mi} c_{f,mi} (h_{f,mi} - h_i) \quad (2-52)$$

For an incompressible and ideally mixed reacting liquid, the energy balance is:

$$V_r \sum_i c_i C_{p,i} \frac{dT}{dt} = Q + Q_{\text{ext}} + \sum_m \sum_i v_{f,mi} c_{f,mi} (h_{f,mi} - h_i) \quad (2-53)$$



Ideal Stirred Tank Reactor System: Application Library path
**Chemical_Reaction_Engineering_Module/Ideal_Tank_Reactors/
 tank_flow_system**

CSTR, CONSTANT VOLUME

This reactor type is a CSTR reactor where the reactor volume is constant.

The species mass balances are given by:

$$\frac{d(c_i V_r)}{dt} = \sum_m v_{f,m} c_{f,m} - v c_i + R_i V_r \quad (2-54)$$

Assuming constant reactor volume:

$$\frac{dV_r}{dt} = 0$$

and setting the volumetric outlet rate to:

$$v = \sum_m v_{f,m} + v_p \quad (2-55)$$

Equation 2-54 can be rewritten as:

$$V_r \frac{dc_i}{dt} = v_{f,i} c_{f,i} - \left(\sum v_{f,i} + v_p \right) c_i + R_i V_r \quad (2-56)$$

The volumetric production rate, v_p , is defined as in Equation 2-49 and Equation 2-50.

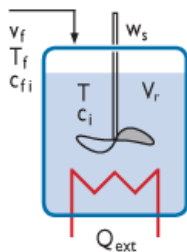
The energy balance is the same as for the CSTR with Constant Mass/Generic reactor type (Equation 2-53).



Startup of a Continuous Stirred Tank Reactor: Application Library
path **Chemical_Reaction_Engineering_Module/Tutorials/cstr_startup**

SEMI-BATCH

In the semibatch reactor, reactants enter the reactor by means of one or several feed inlet streams.



The species mass balances are given by:

$$\frac{d(c_i V_r)}{dt} = v_{f,i} c_{f,i} + R_i V_r \quad (2-57)$$

The reactor volume is a function of time given by:

$$\frac{dV_r}{dt} = \sum_m v_{f,m} + v_p \quad (2-58)$$

The volumetric production rate, v_p , is defined as in [Equation 2-49](#) and [Equation 2-50](#).

An energy balance over the Semibatch reactor results in the same energy balance expression as for the CSTR reactor types ([Equation 2-53](#)).

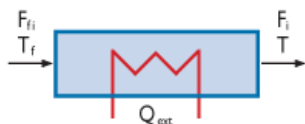


Semibatch Polymerization: Application Library path

**Chemical_Reaction_Engineering_Module/Ideal_Tank_Reactors/
semibatch_polymerization**

PLUG FLOW

In the plug flow reactor the species concentrations and the temperature vary with position. For a tubular reactor configuration, plug flow assumes concentration and temperature gradients to only develop in the axial direction but not in the radial direction of the reactor.



The species mass balances are given by:

$$\frac{dF_i}{dV} = R_i \quad (2-59)$$

where F_i (SI unit: mol/s) is the species molar flow, V (SI unit: m^3) is the reactor volume, and R_i (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$) denotes the species rate expression.

In order to evaluate the rate expressions R_i , which are functions of the species concentrations, the physics interface calculates:

$$c_i = \frac{F_i}{v}$$

where v (SI unit: m^3/s) is the volumetric flow rate.

For liquids, the volumetric flow rate is given by:

$$v = \sum_i \frac{F_i M_i}{\rho_i}$$

where M_i (SI unit: kg/mol) denotes the species molecular weight and ρ_i (SI unit: kg/m³) the species density

For ideal gases:

$$v = \frac{R_g T}{p} \sum_i F_i$$

so that

$$c_i = \frac{p}{R_g T} \frac{F_i}{\sum_i F_i}$$

where p (SI unit: Pa) is the constant pressure.

Neglecting pressure drop, the energy balance for an ideal reacting gas, as well as an incompressible and ideally mixed reacting liquid is given by:

$$\sum_i F_i C_{p,i} \frac{dT}{dV} = Q + Q_{\text{ext}} \quad (2-60)$$

Equation 2-60 is similar to the energy balance for the batch reactors (Equation 2-46), but with a reactor volume dependence instead of a time dependence.



Nonisothermal Plug-Flow Reactor: Application Library path
Chemical_Reaction_Engineering_Module/Tutorials/nonisothermal_plug_flow

Transport Properties

The Reaction Engineering Interface and The Chemistry Interface can calculate several transport properties that can be accessed in interfaces in space dependent models.

DIFFUSIVITY

The diffusivity is calculated in terms of binary diffusion coefficients. These are available for the following fluid mixtures:

Gases with or Without Solvent

The diffusion coefficients are calculated from the following expression based on kinetic gas theory (Ref. 7):

$$D_{AB} = 2.662821 \cdot 10^{-22} \cdot \frac{\sqrt{T^3(M_A + M_B)/(2 \cdot 10^3 M_A M_B)}}{p \sigma_A \sigma_B} \cdot \Omega_D^{-1} \quad (2-61)$$

Here, D_{AB} (SI unit: m^2/s) is the binary diffusion coefficient, M (SI unit: kg/mol) equals the molecular weight, T (SI unit: K) represents the temperature, p (SI unit: Pa) is the pressure, and σ (SI unit: m) equals the characteristic length of the Lennard–Jones/ Stockmayer potential. In addition, Ω_D is the collision integral, given by the relation of Neufeld and others (Ref. 6):

$$\Omega_D = \frac{c_1}{(T^*)^{c_2}} + \frac{c_3}{\exp(c_4 T^*)} + \frac{c_5}{\exp(c_6 T^*)} + \frac{c_7}{\exp(c_8 T^*)} \quad (2-62)$$

where the reduced temperature is:

$$T^* = T \frac{k_b}{\sqrt{\epsilon_A \epsilon_B}} \quad (2-63)$$

In Equation 2-62, c_x are empirical constants, μ_D is the species dipole moment value (SI unit: Cm) and ϵ/k_b (SI unit: K) the potential energy minimum value divided by Boltzmann’s constant. Tabulated data in the literature frequently lists values of ϵ/k_b rather than ϵ . It should be noted that predefined expressions for binary diffusivities only treat ideal gas mixtures. Thus, these are applicable as input only for gases at moderate pressure in multicomponent diffusive transport models.

The binary diffusivity according to Equation 2-61 is also suited for gaseous species in solvent, simply by setting either the component A or B to the solvent. The binary diffusion coefficient is in this case equal to the diffusion coefficient of the bulk species.

Liquid with Solvent

The diffusivity of a species A in a solvent B is calculated with the Wilke–Chang equation (Ref. 8):

$$D_{AB} = 3.7 \cdot 10^{-15} \frac{(\phi_B M_B)^{1/2} T}{\mu_B V_{b,A}^{0.6}} \quad (2-64)$$

where μ_B (SI unit: Ns/m^2) denotes the solvent's dynamic viscosity (see Equation 2-69), and $V_{b,A}$ (SI unit: m^3/mol) equals the molar volume at the normal boiling point of the solute species. ϕ_B is the dimensionless association factor of the solvent, which by default is set to 1.

DYNAMIC VISCOSITY

The dynamic viscosities are computed for the following fluid mixtures:

Gases with Solvent

The dynamic viscosity of the mixture is assumed to be the same as that of the solvent. This is given by the kinetic gas theory (Ref. 6 and Ref. 7), with species i being the solvent:

$$\mu = 2.669 \cdot 10^{-6} \frac{\sqrt{T(M_i \cdot 10^3)}}{\sigma_i^2} \Omega_D^{-1} \quad (2-65)$$

In Equation 2-65, μ (SI unit: Ns/m^2) represents the dynamic viscosity, and Ω_D is the dimensionless collision integral given by:

$$\begin{aligned} \Omega_D = & b_1(T/(\epsilon_i/k_b))^{-b_2} + b_3[\exp(-b_4 T/(\epsilon_i/k_b))] \\ & + b_5[\exp(-b_6 T/(\epsilon_i/k_b))] + \frac{(4.998 \cdot 10^{-40})\mu_{D,i}^4}{k_b^2 T(\epsilon_i/k_b)\sigma_i^6} \end{aligned} \quad (2-66)$$

In Equation 2-65 and Equation 2-66, b_x are empirical constants, μ_D (SI unit: Cm) the species dipole moment value, and ϵ/k_b (SI unit: K) the potential energy minimum value divided by Boltzmann's constant. Tabulated data in the literature frequently lists values of ϵ/k_b rather than ϵ , and σ (SI unit: $\text{\AA}$) is the characteristic length value.

Gases without Solvent

The dynamic viscosity of gas mixtures without solvent are calculated according to Ref. 9 with the following expression:

$$\mu = \sum_{i=1}^n \left(\frac{\mu_i}{1 + \frac{1}{x_i} \sum_{j=1, j \neq i}^n x_j \phi_{ij}} \right) \quad (2-67)$$

where ϕ_{ij} is defined as:

$$\phi_{ij} = \frac{[1 + (\mu_i/\mu_j)^{0.5}(M_j/M_i)^{0.25}]^2}{(2\sqrt{2})[1 + M_i/M_j]^{0.5}} \quad (2-68)$$

In Equation 2-67 and Equation 2-68, x_i is the molar composition and μ_i is computed with Equation 2-65 for each of the species in the mixture.

Liquid with Solvent

The dynamic viscosity of the mixture is assumed to be the same as that of the solvent. An approximate expression for the temperature dependence of the dynamic viscosity is given by (Ref. 3):

$$\mu = 10^{-3} \left((\mu_{\text{ref}} 10^3)^{-0.2661} + \frac{T - T_{\text{ref}}}{223} \right)^{-3.758} \quad (2-69)$$

where μ (SI unit: Ns/m²) is the dynamic viscosity. As inputs for Equation 2-69, the physics interface takes the reference viscosity, μ_{ref} (SI unit: Pa·s) at the reference temperature T_{ref} (SI unit: K).

THERMAL CONDUCTIVITY

The thermal conductivity is calculated as well for some types of fluid mixtures:

Gas With or Without Solvent

The predefined expression for thermal conductivity comes from the Stiel–Thodos equation (Ref. 10), which is defined as:

$$k = \frac{\mu}{M} (1.15 C_p + 0.88 R_g) \quad (2-70)$$

where k (SI unit: W/(m·K)) is the thermal conductivity and C_p (SI unit: J/(mol·K)) denotes the molar heat capacity. Equation 2-70 is a function of viscosity, μ , as given by Equation 2-65. Equation 2-70 is directly used in the case of a solvent; all parameters being those of the solvent. Without a solvent, however, the following equation is also used:

$$k = \frac{1}{2} \left[\sum_{i=1} x_i k_i + \left(\sum_{i=1} \frac{x_i}{k_i} \right)^{-1} \right]$$

where k_i is the thermal conductivity of each species i and x_i the molar composition for each of the species in the mixture.

CHEMKIN[®] files can be used to import data for simulation of complex chemical reactions in the gas phase. These input files may contain three basic sets of information — chemical kinetics, species thermodynamic properties, and species transport properties for either volumetric or surface species. Use imported data to set up and do the following:

- Solve mass and energy balances for ideal reactor systems (reactors, semi-reactors, CSTRs, and plug flow reactors).
- Evaluate species transport properties as a function of reactor conditions.
- Evaluate mass, energy, and momentum balances for space-dependent models, and transfer them to COMSOL Multiphysics.

It is possible to read the input files for kinetics, thermodynamic, and transport properties independently and use these as separate data resources. For example, if a set of reactions is entered into the Reaction Engineering interface, species thermodynamic and transport data can be supplied by reading the appropriate input files. The full functionality of the physics interface is retained even after the import procedure. This means that all expressions and all data imported into the software are available for reference and for editing.

The following CHEMKIN files are compatible:

- Transport CHEMKIN files that supply data used to compute [Transport Properties](#).
- Thermodynamic CHEMKIN files containing data for Gordon and McBride or NASA polynomials ([Ref. 4](#)). These polynomials are denoted *NASA format* and compute the species' heat capacity, molar enthalpy, and molar entropy, in the manner of:

$$C_{p,i} = R_g(a_1 + a_2T + a_3T^2 + a_4T^3 + a_5T^4) \quad (2-71)$$

$$h_i = R_g\left(a_1T + \frac{a_2}{2}T^2 + \frac{a_3}{3}T^3 + \frac{a_4}{4}T^4 + \frac{a_5}{5}T^5 + a_6\right) \quad (2-72)$$

$$s_i = R_g\left(a_1 \ln T + a_2T + \frac{a_3}{2}T^2 + \frac{a_4}{3}T^3 + \frac{a_5}{4}T^4 + a_7\right) \quad (2-73)$$

Here:

- $C_{p,i}$ (SI unit: J/(mol·K)) denotes the species' molar heat capacity,

- T (SI unit: K) is the temperature,
- R_g the ideal gas constant, 8.314 J/(mol·K),
- h_i (SI unit: J/mol) is the species' molar enthalpy, and
- s_i (SI unit: J/(mol·K)) represents its molar entropy at standard state.

From the CHEMKIN files the coefficients a_1 to a_7 are directly imported into the corresponding NASA format fields. Coefficients for NASA polynomials are available as public resources ([Ref. 11](#)).

Kinetics CHEMKIN files that can be imported in the [Reversible Reaction Group](#) feature. These consist of reaction kinetics data, such as activation energy (SI unit: J/mol).



Homogeneous Charge Compression Ignition of Methane: Application Library path **Chemical_Reaction_Engineering_Module/Ideal_Tank_Reactors/compression_ignition**

Chemical Vapor Deposition of GaAs: Application Library path **Chemical_Reaction_Engineering_Module/Reactors_with_Mass_and_Heat_Transfer/gaas_cvd**

Working with Predefined Expressions

You can choose whether to use the predefined expressions as given, use parts of a predefined set of expressions, or replace expressions entirely. A few short examples highlight the use of the predefined expressions used in the [Transport Expressions](#) and [Thermodynamic Expressions](#) sections to enter the thermodynamic properties of carbon monoxide. The sections also illustrate the possibility to adapt to different modeling situations.

CASE I

Carbon monoxide is part of a reacting mixture. You want to do several things: make use of the predefined expressions of species C_p as an input to the heat capacity of the reacting mixture ([Equation 2-71](#)); use the predefined expression h for each species to calculate the heat of reaction ([Equation 2-72](#)); and use the predefined expression s for each species to calculate the entropy of reaction ([Equation 2-73](#)) and subsequently the equilibrium constant of reaction (See [Handling of Equilibrium Reactions](#)).

To accomplish this, enter the seven coefficients of the NASA format, a_1 to a_7 , into the appropriate fields, or import a [CHEMKIN Import for Species Properties](#) thermo input file.

CASE 2

Carbon monoxide is part of a reacting mixture. You want to make use of the predefined expressions of species C_p to calculate the heat capacity of the reacting mixture ([Equation 2-98](#)).

Polynomials for C_p are available in the literature ([Ref. 2](#), [Ref. 3](#), and [Ref. 5](#)) in the frequently used form

$$C_p = R_g \sum_n a_n T^{n-1} \quad n = 1, \dots, 5$$

You can directly use the predefined expression for the species' heat capacity, [Equation 2-71](#), also given in the C_p field, by supplying coefficients in the a_1 to a_5 fields.

You also want to use the predefined expressions h for each species to calculate the heat of reaction ([Equation 2-72](#)). An option is to make use of the heat of formation at standard state (298.15 K) to calculate the coefficient a_6 . Identifying the coefficient a_6 of [Equation 2-72](#) is straightforward. In the NASA polynomial format, the species molar enthalpy is related to its heat capacity according to

$$h = \int_0^T C_p dT + h(0) \quad (2-74)$$

Inserting [Equation 2-71](#) into [Equation 2-74](#), and comparing the result with [Equation 2-72](#), shows that the term $a_6 R_g$ is identified as the species enthalpy of formation at 0 K, that is, $h(0)$. Evaluate $h(0)$ from the species enthalpy of formation at standard state temperature, $T_{std} = 298.15$ K, which is given by

$$a_6 = \frac{h(T_{std})}{R_g} - \left(a_1 T_{std} + \frac{a_2}{2} T_{std}^2 + \frac{a_3}{3} T_{std}^3 + \frac{a_4}{4} T_{std}^4 + \frac{a_5}{5} T_{std}^5 \right) \quad (2-75)$$

CASE 3

Carbon monoxide is again part of a reacting mixture. It is satisfactory to approximate C_p as being constant. Specify the species enthalpy by making use of the heat of formation at 298 K, $h(298)$, according to the well-known expression

$$h = C_p(T - 298 \text{ K}) + h(298 \text{ K})$$

References for the Reaction Engineering Interface

1. H.G. Rackett, "Equation of State for Saturated Liquids," *J. Chem. Eng. Data*, vol. 15, no. 4, pp. 514–517, 1970.
2. R.H. Perry and D.W. Green, *Perry's Chemical Engineering Handbook*, 7th ed., McGraw Hill, 1997.
3. B.E. Poling, J.M. Prausnitz, and J.P. O'Connell, *The Properties of Gases and Liquids*, 5th ed., McGraw Hill, 2000.
4. S. Gordon and B.J. McBride, *Computer Program for Calculation of Complex Chemical Equilibrium Compositions, Rocket Performance, Incident and Reflected Shocks, and Chapman-Jouquet Detonations*, NASA-SP-273, 1971.
5. M. Zabransky, V. Ruzicka Jr., V. Majer, and E.S. Dolmaski, "Heat Capacity of Liquids: Critical Review and Recommended Values," *J. Phys. Chem. Ref. Data Volume I and II, Monograph, No. 6*, 1996.
6. P.D. Neufeld, A.R. Janzen, and R.A. Aziz, "Empirical Equations to Calculate 16 of the Transport Collision Integrals $\Omega^{(l,s)*}$ for the Lennard-Jones (12-6) Potential," *J. Chem. Phys.*, vol. 57, pp. 1100–1102, 1972.
7. R.S. Brokaw, "Predicting Transport Properties of Dilute Gases," *Ind. Eng. Process Design Develop.*, vol. 8, no. 2, pp. 240–253, 1969.
8. C.R. Wilke and P. Chang, "Correlation of Diffusion Coefficients in Dilute Solutions," *AIChE J.*, vol. 1, no. 2, pp. 264–270, 1955.
9. C. R. Wilke, "A Viscosity Equation for Gas Mixtures," *J. Chem. Phys.*, vol. 18, no. 4, pp. 517–519, 1950.
10. L.I. Stiel and G. Thodos, "The Viscosity of Polar Substances in the Dense Gaseous and Liquid Regions," *AIChE J.*, vol. 10, p. 26–29, 1964.
11. See, for example, www.comsol.com/chemical-reaction-engineering-module

The Reaction Engineering Interface

The **Reaction Engineering (re)** interface (), found under the **Chemical Species Transport** branch () when adding a physics interface, is used to model several chemical reactor types and the evolution of chemical reactions over time. The mass balance and energy balance equations describing these systems assume perfect or well-defined mixing in the reactor. Essentially, the physics interface simulates, tank, and plug flow chemical reactors to investigate the behavior over time of a chemical reaction.

The reaction kinetics expressions of a reactor can be exported to a space-dependent model, using the [Generate Space-Dependent Model](#) feature. This gives you the power to simulate reacting systems as they depend on fluid flow, mass transfer, and heat transfer — in other words, including space dependencies.

Add physics features from the toolbar, or right-click **Reaction Engineering** to select features from the context menu. Many of the fields and nodes described in this section are made available when either a [Reaction](#) or a [Species](#) (or both) subnode is added to the Model Builder. Because nodes and subnodes are accessible at any time, and any change is updated throughout the model, reactions and species are often defined before the settings described in this section.

All predefined constants and expressions can be overwritten by user-defined expressions. This makes it possible to go beyond the modeling assumptions that are the defaults in this physics interface.



- [Theory for the Reaction Engineering and Chemistry Interfaces](#)
- [Features Nodes Available for the Reaction Engineering Interface](#)

The following is a description of the features and fields available in the **Settings** window for Reaction Engineering.

SETTINGS

The **Label** is the default physics interface name.

The **Name** is used primarily as a scope prefix for variables defined by the physics interface. Refer to such physics interface variables in expressions using the pattern `<name>.<variable_name>`. In order to distinguish between variables belonging to

different physics interfaces, the **name** string must be unique. Only letters, numbers, and underscores (**_**) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first physics interface in the model setup) is **re**.

EQUATION

This section displays the governing equations according to the selection of reactor types in the **Reactor** section.

REACTOR

Select a **Reactor type** to define the reaction system. The available reactor types are: **Batch**, **Batch, constant volume**, **CSTR**, **constant volume**, **CSTR, constant mass/generic**, **Semibatch**, and **Plug flow**. Each reactor type solves a mass balance based on properties typical to the type, such as reactor volume, volumetric production, or flow rate:

- **Batch**: In reactors no mass enters or leaves the system. This type can account for a variable reactor volume.
- **Batch, Constant Volume**: Same as Batch but where the reactor volume is assumed to be constant. As this is the situation for most reacting systems, this is the default condition.
- **CSTR, Constant Mass/Generic**: Continuous stirred tank reactors (CSTR) differ from batch reactors, since these allow species to enter and leave the reactor by means of feed inlet streams and outlet streams. The system's volume is allowed to change, such as in a car engine cylinder or a balloon. This reactor model can be solved either for a constant mass condition or by selecting a specific outlet flow.
- **CSTR, Constant Volume**: Same as the CSTR with constant mass/generic reactor but assumes that the volume is constant during operation.
- **Semibatch**: Semibatch reactors differ from batch reactors in that they allow reactants to enter the reactor by means of one or several feed inlet streams.
- **Plug Flow**: In the plug flow reactor, the species concentrations and the temperature vary with position. Plug flow in a tubular configuration means that concentration and temperature gradients develop only in the axial direction, but not in the radial direction.

For each **Reactor type**, additional settings are shown in [Table 2-1](#). If a surface reaction or species is included, except for the **Plug flow** reactor type, the surface reaction area is

also a parameter. The parameters and expressions are used in the mass balance equations.

TABLE 2-1: REACTOR TYPE PARAMETERS.

REACTOR TYPE	PARAMETERS TO DEFINE
Batch	Reactor volume V_r
Batch, constant volume	Reactor volume V_r
CSTR, constant mass/ generic	Volumetric production rate v_p and Volumetric outlet rate v
CSTR, constant volume	Volumetric production rate v_p and Reactor volume V_r
Semibatch	Volumetric production rate v_p
Plug flow	Volumetric flow rate v

Volumetric Rate

For **CSTR**, **constant volume** and **Semibatch** reactor types, the **Volumetric production rate** (v_p) is available. For **Automatic**, the in-built **Volumetric production rate** expression is shown. If **User defined** is selected, the expression can be changed (SI unit: m^3/s). For instance, this enables the setting of zero (0) volumetric production rate, which ignores volume changes due to reactions.

- For liquid phase reactions, the **Automatic** expression for the **Volumetric production rate** varies with the reaction rate of each species as defined by:

$$v_p = V_r \sum_i \frac{R_i M_i}{\rho_i}, \quad R_i = \sum_j v_{ij} r_j \quad (2-76)$$

The physics interface automatically inserts the stoichiometric coefficients (v_{ij}) and reaction rate expressions for each species (R_i) that depend on j number of reactions of a rate (r_j), as defined in the **Reaction** feature node. Furthermore, the values of the molar mass (M_i) and the species density (ρ_i) are automatically taken from the **Species** features.

- For gas phase reactions, the **Automatic** expression for the **Volumetric production rate** is similarly given by:

$$v_p = \frac{R_g T}{p} V_r \sum_i R_i, \quad R_i = \sum_j v_{ij} r_j \quad (2-77)$$

For **CSTR**, **constant mass/generic**, select **Volumetric rate** to either **Constant mass (Automatic)** or **Generic**. The **Constant mass (Automatic)** selection shows both the expression for **Volumetric production rate** (v_p) ([Equation 2-77](#) for gas, or

Equation 2-76 for liquid phase reactions) and **Volumetric outlet rate** (v) (Equation 2-78). The latter is derived from constant mass flow condition through the reactor:

$$v = \frac{\sum v_{f,m} \cdot \rho_{f,m}}{\rho} \quad (2-78)$$

The mixture density (ρ_f) of m number of feed inlet streams is determined in the same way as in the **Mixture Properties** section. For **Generic** both the **Volumetric rate** properties can be edited (SI unit: m^3/s). This means that it is possible to completely control the volumetric outlet rate from the CSTR.

For the **Plug flow** reactor, the **Volumetric flow rate** along the reactor is set. The **Automatic** definition computes a variable volumetric flow rate that depends on the molar flow rate of each species (F_i).

- For liquid phase reactions, the expression is:

$$v = \sum_i \frac{F_i M_i}{\rho_i} \quad (2-79)$$

- and for gas phase reactions:

$$v = \frac{R_g T}{p} \sum_i F_i. \quad (2-80)$$

The default value for p (**Reactor pressure**) is 1 atm in Equation 2-81 and it is set in the **Mixture Properties** section.

Reactor Volume

This input field sets the **Reactor volume** V_r — that is, the fluid volume in which chemical reaction takes place. The **Batch** reactor type can account for a changing volume, thus a time-dependent volume expression can be entered here.

Surface Reaction Area

Once a surface reaction, or a surface species, has been added, for all reactor types except **Plug flow**, the **Surface reaction area** settings become visible. Here, the area of the

surface on which the surface reactions take place can be defined. The surface area can either be defined directly as a parameter, or by defining the **Surface area to volume ratio**.



ENERGY BALANCE

From the list select to **Exclude** or **Include** the energy balance, which in essence determines whether the system is solved for either isothermal or nonisothermal conditions, respectively. The latter introduces the temperature as a variable in the interface.

If **Exclude** is chosen, enter a **Temperature** T for the system.

It is possible to incorporate cooling or heating of the reactor. This is done in **External heating or cooling** (Q_{ext}). Enter a negative value to account for cooling or a positive one for heating (SI unit: W). Note that for the Plug Flow reactor type, the **External heating or cooling** (Q) is defined per reactor volume (SI unit: W/m³).

MIXTURE PROPERTIES

Thermodynamics

All property parameters and property functions required by the interface can be automatically created by coupling to a system added to the **Thermodynamics** node. To do so click the **Thermodynamics** checkbox and select an existing [Thermodynamic System](#).

The **Thermodynamics** checkbox is enabled except if:

- the [Thermodynamics](#) node, including one or more systems, is not added under **Global Definitions**, or
- no species are added in the **Reaction Engineering** interface, or
- a CHEMKIN file is imported in the **Reaction Engineering** interface.

Phase

Use the **Phase** list to specify the state of aggregation of the mixture.

Density

This setting is available when the **Calculate mixture properties** checkbox in the **Calculate Transport Properties** section has been selected.

Two settings are always available for the mixture **Density: Automatic** or **User defined**. The **Thermodynamics** options is available when the interface is coupled to a [Thermodynamic System](#), and all interface species has been matched to species in the system. In this case the density is defined by a function automatically added under the thermodynamic system coupled to.

The mixture density is transferred to physics interfaces set up by the [Generate Space-Dependent Model](#) feature. The density is compiled for both multicomponent and solute-solvent solutions.

The **Automatic** setting uses the following logic:

- **Automatic** selected for **Liquid**, considers the liquid as ideal and incompressible. The liquid mixture density depends on the density of i number of pure species (ρ_i) and the species mass fraction (w_i).

$$\rho = \frac{1}{\sum_i \frac{w_i}{\rho_i}}$$

The mass fraction is given by the species concentration (c_i) and the molar mass (M_i).

$$w_i = \frac{c_i M_i}{\sum_i c_i M_i}$$

- **Automatic** set for **Gas** calculates the gas mixture density (ρ) from the concentrations (c_i) and molar masses (M_i) of the mixture species, which are automatically taken from [Species](#) features.

$$\rho = \sum_i c_i M_i \quad (2-81)$$

- If a [Type](#) is set to **Solvent** and the **Mixture** is **Liquid**, the mixture density is the same as the solvent density as defined in **Density** in **General parameters** in the Species node. When **Mixture** is **Gas**, the mixture density is calculated from [Equation 2-81](#) only for the species set as **Solvent**.

Reactor Pressure

The **Phase - Gas** setting displays the **Reactor pressure**. For all reactor types, except the **Plug flow** reactor, select either a pressure computed from the **Ideal gas law** or from any

other expression using the **User defined** option. The **Batch** and **Semibatch** reactor types also have the option to keep the reactor pressure **Constant** during reaction.

For the **Plug Flow** reactor, the **Reactor pressure** can be entered in the case of the **Reactor** section having the **Volumetric rate** set to **Automatic**, in which case the **User defined** alternative is available and a constant pressure fits the conditions.

When the **Thermodynamics** checkbox is selected in the **Mixture Properties** section and the species are fully coupled (see the section below), the reactor pressure is set to **Thermodynamics** indicating that it is automatically computed.

SPECIES MATCHING

The **Species Matching** section is activated when the **Thermodynamics** checkbox is selected in the **Mixture Properties** section. The species in the Reaction Engineering interface can be matched to species in the **Thermodynamic System**. This ensures that the arguments in the thermodynamic system functions are correctly defined.

Use the lists in the **From Thermodynamics** column to match each species in the interface to a species in the coupled thermodynamic system.

For each species matched, the required property parameters and functions are added under to the corresponding thermodynamic system.

When all species are matched, the interface is considered fully coupled and functions representing mixture properties, such as density, are also added automatically under the corresponding thermodynamic system.

CALCULATE TRANSPORT PROPERTIES

Transport properties are not utilized in the reactor equations available in the Reaction Engineering interface, where perfect mixing is assumed. However, transport properties such as diffusivity, thermal conductivity, and viscosity are of central importance when solving time- and space-dependent models. The physics interface helps to set up detailed expressions of species transport properties and transfers them automatically to the multiphysics model through the **Generate Space-Dependent Model** feature.

The most general description of a mixture is the one that treats the mixture as a multicomponent solution, where all species interact with each other. A simplified description, but still a common one, assumes that the solution has a solvent that dominates the properties of the solution. The solutes in such a solution interact only with solvent molecules.

Select the **Calculate mixture properties** checkbox to enable calculation of mixture transport properties exported from the Reaction Engineering interface.

From the list for each property, select the in-built **Automatic** expression or set a **User defined** entry. The mixture properties you can transfer for space-dependent models are:

- **Heat capacity** (c_p) (SI unit: J/(kg·K)) (this is available when the [Energy Balance](#) is set to **Include**)
- **Thermal conductivity** (k) (SI unit: W/(m·K))
- **Dynamic viscosity** (μ) (SI unit: Pa·s) (this is available when a [Type](#) is set to **Solvent**)
- **Mixture density** (ρ) (SI unit: kg/m³) (this is available in the [Mixture Properties](#) section)

All species properties needed to compute the mixture properties are entered in the **Species Transport Expression** or **Species Thermodynamic Expression** in the [Species](#) node.



Transport Expressions

EQUILIBRIUM SPECIES

This section is available when at least one equilibrium reaction has been defined, that is, when a [Reaction](#) node has **Reaction type** set to **Equilibrium**, or when an equilibrium reaction has been defined in an [Equilibrium Reaction Group](#).

For each equilibrium reaction, the set of species solved for is reduced by one. This is done by instead defining one of the participating species from the **Equilibrium expression** (in the corresponding [Reaction](#) node). Species defined in this manner are referred to as equilibrium species. By default the leftmost species in each **Reaction formula**, which is not already an equilibrium species, is selected.

The **User defined equilibrium species (comma separated)** text field can be used to manually specify which equilibrium species to use. To minimize the impact of numerical errors, it is recommended to select the species with the highest concentration as equilibrium species. The equilibrium species currently in use are shown below, but grayed out.

The **Suppress negative concentrations** checkbox exists to aid the computation of equilibrium reaction systems. A selected checkbox ensures that no negative values of concentrations are accepted as solution to the equilibrium condition.



[Example II in Handling of Equilibrium Reactions.](#)

ACTIVITY

Select the **Use activity** checkbox to solve for species activities instead of species concentrations, which is a common approach when nonideal fluids are modeled.

An activity coefficient other than 1 can be set for each species for the [Species](#) node in the **Species Concentration/Activity** section.



[Activity](#)

CHEMKIN IMPORT FOR SPECIES PROPERTIES

This section enables CHEMKIN[®] import to simulate complex chemical reactions in gas phase.

Two types of CHEMKIN input files can be imported here — **Thermo** and **Transport**, for thermodynamic properties and transport properties respectively. Properties for either volumetric or surface species are supported. Click **Browse** to locate the CHEMKIN file to be imported, then click **Import**. For Thermo the imported data is directly entered in the **NASA format** fields in the [Species Thermodynamic Expressions](#) section. For Transport the imported data is entered in the [Species Transport Expressions](#) section.



- [CHEMKIN Data and NASA Polynomials](#)
 - [Reversible Reaction Group](#)
-

ADVANCED SETTINGS

To display this section, click the **Show More Options** button () and select **Advanced Physics Options**.

The **Uniform scaling of concentration variables** checkbox is not selected by default. When selected, all concentration variables are scaled using the same scale factor in the **Study** node. Enabling uniform scaling can decrease solver time for problems involving many concentration variables.

Features Nodes Available for the Reaction Engineering Interface

The following feature nodes are available from the **Reaction Engineering** ribbon toolbar (Windows users), **Reaction Engineering** context menu (Mac or Linux users), or right-click to access the context menu (all users).

- [Initial Values](#)
- [Reaction](#)
- [Species](#)
- [Reversible Reaction Group](#)
- [Equilibrium Reaction Group](#)
- [Species Group](#)¹
- [Additional Source](#)
- [Species Activity](#)¹
- [Species Thermodynamics](#)¹
- [Feed Inlet](#)
- [Generate Space-Dependent Model](#)

¹ This is a subnode and it is only available together with a parent node.

Initial Values

The **Initial Values** node sets the initial values of the dependent variables solved for in the Reaction Engineering interface.

GENERAL PARAMETERS

For **CSTR**, **constant mass** and **Semi** reactor types, in which the reactor volume changes with reaction, an **Initial system volume** V_{r0} can be set.

For nonisothermal conditions (that is, the [Energy Balance](#) is set to **Include**), enter an **Initial temperature** T_0 for the system. For the **Plug flow** reactor a corresponding **Inlet temperature** $T_{0,in}$ is entered.

VOLUMETRIC SPECIES INITIAL VALUES

In the **Volumetric Species Initial Values** table, the **Species** column automatically includes all the species in the model. Enter the corresponding values or expression for each species initially in the **Concentration (mol/m³)** column.

For the **Plug flow** reactor an **Inlet molar flow rate** table is instead shown. Enter the values or expressions representing the inlet conditions of each species in the **Molar flow rate (mol/s)** column.

SURFACE SPECIES INITIAL VALUES

The **Surface Species Initial Values** table is displayed when surface species are present in the reactor, that is, when the **Type** is set to **Surface species** for at least one species in the reactor. In the **Surface concentration (mol/m²)** column, enter the corresponding values or expression for each species initially on the reactive surface area (A_r) within the reactor. The values in the **Site occupancy number** column are by default set to 1, edit the value if a surface species takes up more than one active site on the reactive surface.

The total surface concentration is restricted by the molar amount of available sites on the reactive surface area. This is defined by the **Density of sites** Γ_s (mol/m²) entry. Make sure that the sum of the entered initial surface concentrations do not surpass the value selected here. Note that the total amount of sites occupied of a surface species $c_{ads,i}$ is the **Surface concentration (mol/m²)** multiplied with the **Site occupancy number** $\sigma_{s,i}$. The restriction of the total surface concentration is defined as:

$$\Gamma_s \geq \sum_i c_{ads,i} \sigma_{s,i} \quad (2-82)$$

EQUILIBRIA

When one or more equilibrium reactions are present, a section which allows the user to enable **Mass-preserving initialization** is shown. When enabled, the solver will first solve for the values of the species participating in any equilibrium reaction, while disregarding any kinetic reactions. The solution to the equilibrium subproblem is then used as a consistent starting point for the time stepping. The functionality solves a log-transformed version of the problem, which in turn puts a requirement that a finite concentration is present for each building block (atom kinds, charge, and/or a combination thereof). If **Settings** is changed from **Automatic** to **Manual**, the user may change the maximum number of iterations and the absolute tolerance.

Reaction

Use the **Reaction** node () to implement a chemical reaction involving a number of chemical species. The participating species may represent any kind of molecular entities like atoms, molecules, ions, or radicals.

To add a **Reaction** node either right-click the **Reaction Engineering** node or on the **Reaction Engineering** toolbar click **Reaction**.

REACTION FORMULA

Formula

Enter a chemical equation in the **Formula** field. The chemical equation should be of the format “A + B $\rightarrow$ C + D”. Here, A and B are reactants, C and D are product species, and $\rightarrow$ denotes a reaction arrow. The participating species should be written with [Valid Species Names](#). Valid reaction arrows are “ $\rightleftharpoons$ ” for reversible reaction, “ $\Rightarrow$ ” for irreversible reaction, and “ $=$ ” for equilibrium reaction. Species can be given trivial names, or their chemical formulas can be used as names. Examples of valid formulas are; “carbon+oxygen $\Rightarrow$ carbondioxide”, and “C(s)+O₂(g) $\Rightarrow$ CO₂(g)”.

Click **Apply** to let the interface parse the reaction. This process identifies the participating species as well its stoichiometric coefficients and any charges of ionic species. A [Species](#) feature is automatically added for all participating species.

Balance ()

Use the **Balance** button to automatically calculate the stoichiometric coefficients such that the number of atoms of each kind are the same on both sides of the reaction. This turns, for example, the formula “H₂+O₂ $\Rightarrow$ H₂O” into “2H₂+O₂ $\Rightarrow$ 2H₂O”. The balancing involves parsing all participating species for elements in the periodic table. It therefore requires that all species in the **Formula** field are written either using their chemical formula, say “H₂O”, or that each species in the reaction have an enabled **Chemical Formula** field in their **Species** node. For example, it is possible to balance the formula “H₂+O₂ $\Rightarrow$ water” as long as there is a Species node with the name “water” already present, and an enabled **Chemical Formula**.

For automatic reaction balancing to be successful, requires that the problem is well posed. One example of a problem that is not well posed is “C+H₂ $\Rightarrow$ CH₄+C₂H₆”, in which case any ratio of CH₄/C₂H₆ could be obtained. Another example is “H₂ $\Rightarrow$ O₂” where not all elements are present on both sides.

Reaction Type

Select the **Reaction type** — **Reversible**, **Irreversible**, or **Equilibrium** — or edit the expression directly in the **Formula** field. In the latter case, specify the reaction type with a delimiter separating the two sides of the equation:

- $\rightleftharpoons$ denotes a **Reversible** reaction

- => denotes an **Irreversible** reaction
- = denotes a reaction at chemical **Equilibrium**

Each **Reaction type** has its own set of reaction kinetics:

- When the reaction is **Reversible** or **Irreversible**, the rate of reaction r_j , for reaction j , contributes to the change in species i in the manner of:

$$R_i = \sum_j v_{ij} r_j \quad (2-83)$$

where v_{ij} is the stoichiometric coefficient.

- In the case of an **Equilibrium** reaction, the equilibrium expression is equal to the equilibrium constant:

$$K_{eq} = K_{eq0} \quad (2-84)$$

REACTION RATE

This section is available when the **Reaction type** is either **Reversible** or **Irreversible**.

When **Mass action law** is selected (default), the rate expression is automatically derived from the stoichiometric coefficients in the reaction formula:

- For an **Irreversible** reaction type, the reaction rate becomes:

$$r_j = k_j^f \prod_{i \in \text{react}} c_i^{-v_{ij}} \quad (2-85)$$

- For a **Reversible** reaction type, the expression instead becomes:

$$r_j = k_j^f \prod_{i \in \text{react}} c_i^{-v_{ij}} - k_j^r \prod_{i \in \text{prod}} c_i^{v_{ij}} \quad (2-86)$$

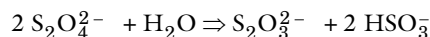
The deduced overall reaction order is shown in text below the respective equation in the **Reaction Rate** section.

Arbitrary rate expression

If the reaction order differs from the stoichiometric coefficients, or if an arbitrary rate expressions is applicable, change **Reaction Rate** to **User defined**. An expression field r appears with the default expression being that from the mass action law. Below this there are fields to set the reaction order. For a reversible reaction the reverse reaction order may be specified in addition to the forward one. The unit of the rate constant k (or frequency factor A in the case of Arrhenius behavior), is derived from the reaction

order, in SI units: $(\text{m}^3/\text{mol})^{\alpha-1}/\text{s}$, where α equals the order with respect to volumetric species. When surface species are present — identified by their “(ads)” suffix — the unit is instead given by $\text{m}^{3\alpha+2\beta-2}/\text{mol}^{\alpha+\beta-1}/\text{s}$, where β is the order with respect to surface species.

Consider for example the reaction:



The automatically deduced reaction order is three, however, in the case that water is the solvent, the order should probably be two. This is specified as follows:

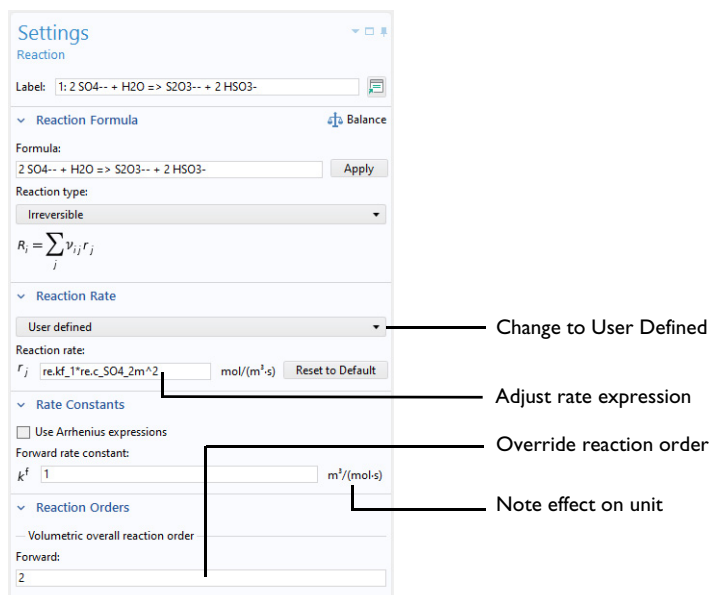


Figure 2-5: The reaction order can be manually specified for a user defined reaction rate.

RATE CONSTANTS

This section applies for **Reversible** or **Irreversible** reactions and defines the reaction rate constants used in the reaction rates.

Forward Rate Constant and Reverse Rate Constant

The **Forward rate constant** k^f is used for both **Reversible** and **Irreversible** reactions. The **Reverse rate constant** k^r is only used for **Reversible** reactions (Equation 2-86).

The SI units of the rate constants are automatically based on the order of the reaction with respect to the concentrations, as defined in the **Reaction formula**.

Specify Equilibrium Constant

The **Specify equilibrium constant** checkbox is available for **Reversible** reactions. If the checkbox is selected the rate constants are defined in a different manner with the reverse rate constant being computed from the following expression:

$$k^r = k^f / K_{eq0} \quad (2-87)$$

Thus, in this case, the forward rate constant and equilibrium constant for the reaction are needed. The **Equilibrium constant** is edited in the [Equilibrium Settings](#) section.

Use Arrhenius Expressions

When the **Use Arrhenius expressions** checkbox is selected the Arrhenius parameters are automatically used in predefined expressions for the forward and reverse rate constants k^f and k^r , respectively.

$$k^f = A^f (T/T_{ref})^{n^f} \exp(-E^f/(R_g T)) \quad (2-88)$$

$$k^r = A^r (T/T_{ref})^{n^r} \exp(-E^r/(R_g T)) \quad (2-89)$$

Specify the activation energy and the frequency factor in the Arrhenius expressions to account for temperature variations. The reference temperature, T_{ref} equals 1 K. The available fields are based on the **Reaction type** chosen in the [Reaction](#) node. Enter values or expressions for each of the following (reverse expressions are only available for reversible reactions):

- **Forward frequency factor** A^f and **Reverse frequency factor** A^r (unit depends on reaction order)
- **Forward temperature exponent** n^f and **Reverse temperature exponent** n^r
- **Forward activation energy** E^f and **Reverse activation energy** E^r (SI unit: J/mol)

EQUILIBRIUM SETTINGS

This section is available for equilibrium reactions, and for reversible reactions when the **Specify equilibrium constant** checkbox has been selected.

Equilibrium Expression

For an equilibrium reaction, specify the **Equilibrium expression**. When the **Equilibrium expression** is set to **Automatic** the following expression is used:

$$K_{eqj} = \frac{\prod_{i \in \text{prod}} c_i^{v_{ij}}}{\prod_{i \in \text{react}} c_i^{-v_{ij}}} \quad (2-90)$$

Select **User defined** from the **Equilibrium expression** list to instead enter a manually defined equilibrium expression.

Equilibrium Constant

Specify the **Equilibrium constant** K_{eq0} for an equilibrium reaction, or for a reversible reaction when the **Specify equilibrium constant** checkbox has been selected (in the **Rate Constants** section).

The **Equilibrium constant** can either be **User defined**, or automatically defined when set to **Automatic** or **Thermodynamics**.

Use the **Automatic** option to compute the equilibrium constant for an ideal system. This setting requires that the temperature is also solved for by setting **Energy Balance** to **Included**.

The **Thermodynamics** option is available when all reactions in the interface are equilibrium reactions, and the interface is fully coupled to a **Thermodynamic System** (see **Species Matching**). Use this setting to automatically compute the equilibrium constant for an ideal or nonideal system, dependent on the thermodynamic model applied for the coupled system.

Using **Automatic** or **Thermodynamics**, K_{eq0} is calculated from the Gibbs free energy of the reaction. For more details see **The Equilibrium Constant** and the **Automatically Defined Equilibrium Constants** section therein.

REACTION THERMODYNAMIC PROPERTIES

This section contains information about thermodynamic properties that relate to a selected reaction and is available when the **Energy Balance** is set to **Include**. Several **Automatic** definitions are available here.

Enthalpy of Reaction

The **Enthalpy of reaction** H (SI unit: J/mol) is calculated by the interface from species properties and the related stoichiometric coefficients:

$$H_j = \sum_{i \in \text{prod}} v_{ij} h_i - \sum_{i \in \text{react}} (-v_{ij}) h_i \quad (2-91)$$

Entropy of Reaction

The **Entropy of reaction** S (SI unit: J/(mol·K)) comes from a similar expression:

$$S_j = \sum_{i \in \text{prod}} \nu_{ij} s_i - \sum_{i \in \text{react}} (-\nu_{ij}) s_i \quad (2-92)$$

In [Equation 2-91](#) and [Equation 2-92](#), h_i and s_i are the species' molar enthalpy and molar entropy, respectively.

Enter these quantities in the **Thermodynamic Expressions** section for the [Species](#) node either by using the predefined polynomial or by providing a custom expression or constants.

The stoichiometric coefficients ν_{ij} , are defined as being negative for reactants and positive for products. Using [Equation 2-91](#) and [Equation 2-92](#) to equate the Gibbs free energy of reaction enables the equilibrium constant to be expressed according to [Handling of Equilibrium Reactions](#).

Heat Source of Reaction

The **Heat source of reaction** (SI unit: W/m³) is automatically computed from the heat of each reaction j , given by:

$$Q_j = -r_j H_j$$



Using the Reaction Node

Species

When a [Reaction](#) is defined, a **Species** node () is automatically generated for the participating reactants and products. This feature enables you to review and enter species specific information regarding chemical kinetics, thermodynamics, and transport properties.

It is also possible to add and define an individual **Species** node: on the **Reaction Engineering** toolbar click **Species** or right-click the **Reaction Engineering** node and select it from the context menu.

NAME

When a **Species** node is automatically generated using the **Formula** text field in the **Reaction** node, the species name is also automatically generated.

For a **Species** node added individually, enter a **Name** in the field and click **Apply**. By entering a name consisting of only elements from the periodic table, the chemical formula is added, and molar mass is calculated and added in the **Chemical Formula** section. The species charge, and type, are also deduced from the species name.



Valid Species Names

TYPE

Select a species type — **Bulk species**, **Surface species**, or **Solvent**.

Bulk species are solved for volumetric concentrations (SI unit: mol/m^3), while **Surface species** are solved for surface concentration (SI unit: mol/m^2). The compositions for **Bulk species** and **Solvent** (not solved for) use the syntax `c_speciesname`, while **Surface species** uses `csurf_speciesname_surf`.

Surface species is referring to species adsorbed on a reactive surface. When **Surface species** is selected, the corresponding reaction formula introduces **(ads)** after the species notation and changes the species' name to `speciesname_surf`. Additionally, the Species node name is updated in a similar fashion.

For the **Plug flow** reactor only **Bulk species** and **Solvent** are allowed.



Choosing the Species Type

CHEMICAL FORMULA

The **Chemical Formula** section contains the species chemical formula, **Molar mass**, M , and **Charge**, z .

If the name entered for the species consists of only elements from the periodic table, and optionally charge and phase indication, then the **Chemical Formula** field will be populated with the name entered in the **Name** section, and the molar mass and charge will be added to this section. The species molar mass is computed from the mass of

occurring individual elements¹. The rules for writing chemical formulas are the same as those for **Species** names.



Valid Species Names

When the species name contains parts not in the periodic table, for example when a descriptive name such as **water** is used, the molar mass is set to 0.0 kg/mol. If the species name consists of a valid charge or phase indication, then charge and type will be set automatically. In this case, the molar mass needs to be defined in order to achieve mass balanced reactions and correct definitions of mass basis properties. By entering a chemical formula consisting only of elements from the periodic table (and optionally charge and phase indication), the molar mass, and charge are derived automatically.

When needed, the **Molar mass** can be edited in the corresponding text field. Editing the **Chemical formula** and pressing apply will override the content in the molar mass and charge fields. The **Charge** field cannot be edited as long as a chemical formula is enabled.

With an enabled **Chemical Formula** it is possible to use the trivial name of a species and balance reactions. Having added the Species **water**, **oxygen**, and **hydrogen**, and filled in H₂O, O, and H in their respective chemical formula sections, the **Formula** oxygen+hydrogen=water is balanced into oxygen+2hydrogen=water by clicking **Balance** in the **Reaction** node.

It is possible to specify the species density ρ when the fluid **Mixture** is specified as **Liquid**. The default value is that of water at 293 K.

REACTION RATE

Change the **Automatic** default setting to **User defined** to use a species reaction rate other than the one set up in the associated **Reaction** node. For individual species, use the **User defined** option to set a reaction rate other than zero (that is, nonreactive).

Edit either the **Rate expression** (SI unit: mol/(m³·s)), the **Surface rate expression** (SI unit: mol/(m²·s)), or both. For a bulk species, both expressions appear if surface reactions are present in the reactor since the reaction of the species can depend both on bulk reaction R and surface reaction R_{ads} rates. For a surface species, only the surface reaction rate R_{ads} appears.

1. Values are averages of upper and lower bounds for each atom as published by CIAAW. Source: CIAAW. Atomic weights of the elements 2019. Available online at www.ciaaw.org.

The reaction rate is not editable if the species in question participates in an equilibrium reaction and has been selected as a **Predefined dependent species** in [Equilibrium Species](#).

CONSTANT CONCENTRATION/ACTIVITY

To account for nonideality in the fluid mixture adjust the activity coefficient in the **Activity coefficient** text field here. The section is only shown if activity instead of concentration has been chosen in the interface; that is, the **Use activities** checkbox is selected on the Reaction Engineering interface **Settings** window

Click to select the **Keep concentration/activity constant** checkbox if the species concentration or activity should be treated as constant.

TRANSPORT EXPRESSIONS

Here the required parameters to compute various transport properties can be edited. To show this section the **Calculate mixture properties** checkbox needs to be selected under the **Calculate Transport Properties** section in the **Settings** window of the physics interface.

- For a gas mixture, there are maximum five properties to consider:
 - σ , the characteristic length (unit: Å) of the Lennard–Jones/Stockmayer potential
 - ϵ/k_b (SI unit: K) the potential energy minimum value of the Lennard–Jones/Stockmayer potential, divided by Boltzmann’s constant. Tabulated data in the literature frequently lists values of ϵ/k_b rather than ϵ
 - μ_D , the dipole moment (SI unit: Debye)
 - k_i , the thermal conductivity of the gas (SI unit: W/(m·K)); and Diffusivity of the species (SI unit: m²/s). The latter two can be set to **Automatic**, when σ and ϵ/k_b have been given.
- For mixtures of gas with solvent, all properties above except the thermal conductivity is shown. This is instead calculated under the **Calculate Transport Properties** section in the Settings window for the Reaction Engineering interface. This requires that [Energy Balance](#) is set to **Included**.
- For a liquid mixture, there is only the parameter k_i , the thermal conductivity (SI unit: W/(m·K)) to edit.
- For mixtures of liquid with solvent, the thermal conductivity is shown only for the species set as **Solvent**.

THERMODYNAMIC EXPRESSIONS

The parameters utilized for calculation of thermodynamic mixture and reaction properties are set in this section. It is accessed when the [Energy Balance](#) is set to **Include** for the Reaction Engineering interface, in other words, for nonisothermal conditions.

The **Thermodynamic Expressions** are by default computed using data in the **NASA format**. In this case, enter the following to compute the species' heat capacity, C_p (SI unit: J/(mol·K)), the molar enthalpy, h (SI unit: J/mol), and the molar entropy, s (SI unit: J/(mol·K)):

- **Lower temperature limit** T_{lo}
- **Middle temperature limit** T_{mid}
- **Upper temperature limit** T_{hi}
- **Polynomial coefficients** $a_{low,k}$ and $a_{hi,k}$

Any coefficients for the thermodynamic polynomials entered into the $a_{low,k}$ fields apply to the temperatures in the range T_{lo} to T_{mid} ; coefficients entered into the $a_{hi,k}$ fields apply to temperatures in the range T_{mid} to T_{hi} range. The coefficients can also be imported in the **CHEMKIN Import for Species Properties** section in the Reaction Engineering node.

Choose the **User defined** alternative to specify C_p , h , and s directly.



[Using the Species Node](#)

[Transport Properties](#)

Reversible Reaction Group

The **Reversible Reaction Group** node () allows input of a large number of reversible reactions in the same table. All reactions are treated in the same way. More than one Reversible Reaction Group can be added to a component. The feature is typically used when importing CHEMKIN kinetics files.

Add the node from the **Reaction Engineering** toolbar or right-click **Reaction Engineering** and add it from the context menu.

For the case when reaction kinetics data are entered manually into the **Reaction table** and nonisothermal conditions apply ([Energy Balance](#) is set to **Include**). The Enthalpy of Reaction (J/mol) can be calculated from the thermodynamics properties of the species.

REACTION TABLE

The reversible reactions in the **Reaction table** are numbered and contain reactants, products, and kinetic parameters describing the reaction. Use the buttons under the **Reaction table** to add and sort the reaction details.

- In general, use the **Move Up** (), **Move Down** (), and **Delete** () buttons and the fields under tables to edit the table contents. Or right-click a table cell and select **Move Up**, **Move Down**, or **Delete**.
- The **Add** button () adds default reactant, A, and product, B, with a default. Click the corresponding field to edit the reactant, product, or parameters. After editing the **Species Group** node is also updated. It is created together with the reaction group.
- You can save the parameters to a text file to reuse in other models. Click the **Save to File** button () and enter a **File name** in the **Save to File** dialog, including the extension .txt. Click **Save** to store the parameters in a text file or in a Microsoft Excel Workbook spreadsheet if the license includes LiveLink™ for Excel®. The information is saved in space-separated columns in the same order as displayed on screen. When saving to Excel, an **Excel Save** dialog appears where you can specify the sheet and range and whether to overwrite existing data, include a header, or use a separate column for units.
- You can import or load data in files from a spreadsheet program, for example, with the **Load from File** button () and the **Load from File** dialog that appears. Data must be separated by spaces or tabs. If there is already data in the table, imported parameters are added after the last row. Move or edit rows as needed. If the license includes LiveLink™ for Excel® you can also load parameters from a Microsoft Excel Workbook spreadsheet. Then an **Excel Load** dialog appears where you can specify the sheet and range and whether to overwrite existing data. It is also possible to import from a spreadsheet containing a separate column for units.

CHEMKIN IMPORT FOR KINETICS

Click to select the **Import CHEMKIN data** checkbox to import a CHEMKIN **Kinetics input file**. To import the data directly into the table columns, click **Browse** to locate the CHEMKIN file to be imported, then click **Import**.



CHEMKIN Data and NASA Polynomials

MOVE REACTION AND SPECIES

Enter a reaction number in the **Moving reaction (with the number) from table** text field and click **Create Reaction** to remove the reaction from the **Reaction table** to the model tree. This introduces a reaction node named after the reaction number and species names and simultaneously creates corresponding species nodes. This action is powerful, since it enables all the editing possibilities available for [Reaction](#) and [Species](#) nodes.

Equilibrium Reaction Group

The **Equilibrium Reaction Group** node () allows for input of a large number of equilibrium reactions in the same table. This node functions in the same way as the [Reversible Reaction Group](#) for reversible reactions.

Add the node from the **Reaction Engineering** toolbar or right-click **Reaction Engineering** and add it from the context menu.

EQUILIBRIUM REACTION TABLE

The equilibrium reactions in the **Reaction table** are numbered and contain reactants, products, and equilibrium constants. Use the buttons under the **Reaction table** to add and sort the reaction details.

- In general, use the **Move Up** (), **Move Down** (), and **Delete** () buttons and the fields under tables to edit the table contents. Or right-click a table cell and select **Move Up**, **Move Down**, or **Delete**.
- The **Add** button () adds default reactant, A, and product, B, with a default. Click the corresponding field to edit the reactant, product, or parameters. After editing the [Species Group](#) node is also updated. It is created together with the reaction group.
- You can save the parameters to a text file to reuse in other models. Click the **Save to File** button () and enter a **File name** in the **Save to File** dialog, including the extension .txt. Click **Save** to store the parameters in a text file or in a Microsoft Excel Workbook spreadsheet if the license includes LiveLink™ for Excel®. The information is saved in space-separated columns in the same order as displayed on screen. When saving to Excel, an **Excel Save** dialog appears where you can specify the sheet and range and whether to overwrite existing data, include a header, or use a separate column for units.
- You can import or load data in files from a spreadsheet program, for example, with the **Load from File** button () and the **Load from File** dialog that appears. Data must

be separated by spaces or tabs. If there is already data in the table, imported parameters are added after the last row. Move or edit rows as needed. If the license includes LiveLink™ for Excel® you can also load parameters from a Microsoft Excel Workbook spreadsheet. Then an **Excel Load** dialog appears where you can specify the sheet and range and whether to overwrite existing data. It is also possible to import from a spreadsheet containing a separate column for units.

MOVE REACTION AND SPECIES

Enter a reaction number in the **Moving reaction (with the number) from table** text field and click **Create Reaction** to remove the reaction from the **Reaction table** to the model tree. This introduces a reaction node named after the reaction number and species names and simultaneously creates corresponding species nodes. This action is powerful, since it enables all the editing possibilities available for [Reaction](#) and [Species](#) nodes.

	<i>Homogeneous Charge Compression Ignition of Methane:</i> Application Library path Chemical_Reaction_Engineering_Module/Ideal_Tank_Reactors/compression_ignition <i>Chemical Vapor Deposition of GaAs:</i> Application Library path Chemical_Reaction_Engineering_Module/Reactors_with_Mass_and_Heat_Transfer/gaas_cvd
--	---

Species Group

The **Species Group** node () contains information on a molecular level about the volumetric species and the surface species present in the model. The **Property for Volumetric Species** or **Property for Surface Species** tables typically collect parameters from when importing CHEMKIN transport files.

This node is automatically created when either the [Reversible Reaction Group](#) or the [Equilibrium Reaction Group](#) is added. The created species group node is coupled to the corresponding reaction group feature. If a reaction group feature is removed, disabled or enabled, the coupled species group feature will follow the same actions. The species in species group feature are only from its corresponding reaction group feature.

For the case of nonisothermal reactor conditions (**Energy Balance** is set to **Include**), a **Species Thermodynamics** subnode is automatically created in which the thermodynamic properties of the species can be specified.

The **Show species name from CHEMKIN file** checkbox is available when the coupled **Reversible Reaction Group** feature is used to import a CHEMKIN reaction system. With this checkbox selected, there will be an extra column, **Name in CHEMKIN**, in both the **Property for Volumetric Species** and **Property for Surface Species** tables. The **Name in CHEMKIN** column shows the original species names from the CHEMKIN file because the species names defined there can be modified to follow the naming rules in COMSOL.

CHEMKIN

This section allows import of CHEMKIN transport files. To import the data directly into the table columns, click **Browse** to locate the CHEMKIN file to be imported, then click **Import**. Note that this section is only shown when transport properties are computed (**Calculate mixture properties** checkbox under the **Calculate Transport Properties** is selected).



CHEMKIN Data and NASA Polynomials

Additional Source

Use the **Additional Source** node () to add an additional rate expression (SI unit: mol/m^3) and/or an additional volumetric production rate (SI unit: m^3/s) to the mass balance of certain species.

Add the node from the **Reaction Engineering** toolbar or right-click **Reaction Engineering** and add it from the context menu.

ADDITIONAL VOLUMETRIC PRODUCTION RATE

An additional **Additional volumetric production rate** can be added for some of the available reactor types, namely **CSTR, constant mass/generic, CSTR, constant volume**, or **Semi**.

ADDITIONAL RATE EXPRESSION

Here, values or expressions describing additional rates can be entered for selected **Species** in the column denoted **Additional rate expression (mol/m³s)** within the **Volumetric species** table.

ADDITIONAL SURFACE SPECIES RATE

Here, values or expressions describing additional surface species rates can be entered for selected **Surface species** in the column denoted **Additional rate expression (mol/m²s)** within the **Surface species** table.

Species Activity

The **Species Activity** subnode () creates variables for the activities for all the species/surface species present in the Species Group parent feature. Edit the **Activity coefficient** field in the **Species Activity** or **Surface Species Activity** tables by clicking in these.

Species Activity is a subnode to the **Species Group** node. It is automatically generated when nonideality in the fluid mixture is accounted for; when the **Use activity** checkbox is selected in the **Activity** section.

Species Thermodynamics

The **Species Thermodynamics** node () creates variables for the enthalpies, entropies, and heat capacities for all the species/surface species present in the **Species Group** parent feature. The purpose is to compute thermodynamic mixture properties and the heat of reactions.

This node is a subnode to the **Species Group** node when the **Energy Balance** is set to **Include**.

Similar to its parent **Species Group** node, the checkbox **Show species name from CHEMKIN file** is available when its coupled **Reversible Reaction Group** feature is set to CHEMKIN import. Select this checkbox to show the original species names (under the column **Name in CHEMKIN**) from the CHEMKIN file.

THERMODYNAMICS PROPERTIES OF SPECIES

Select a **Type selection** for the thermodynamic properties. The **NASA format** alternative requires the setting of several polynomial constants and is typically used together with CHEMKIN import. The **User defined** alternative only requires input of the **Species molar enthalpy** (SI unit: J/mol) and the **Heat capacity** (SI unit: J/(mol·K)).

CHEMKIN IMPORT FOR THERMODYNAMIC DATA

This section allows import of thermodynamic CHEMKIN transport files. To import the data directly into the table columns, click **Browse** to locate the CHEMKIN file to be imported, then click **Import**. Note that this can only be done for the **NASA format** option.



- [CHEMKIN Data and NASA Polynomials](#)

Feed Inlet

The **Feed Inlet** () feature is used for adding inlet streams to the reactor.

After adding a [Reaction](#) node and setting its [Reactor](#) to **CSTR**, **constant volume**, **CSTR**, **constant mass/generic**, or **Semi**, add a **Feed Inlet** node from the **Reaction Engineering** toolbar or right-click the **Reaction Engineering** node to add it from the context menu.

FEED INLET PROPERTIES

Edit the **Volumetric feed rate** (v_f) (SI unit: m^3/s) to fit the modeling conditions.

The **Feed inlet temperature** (T_f) is required as input when nonisothermal conditions are investigated ([Energy Balance](#) is set to **Include**).

FEED INLET CONCENTRATION

This section is available when [Energy Balance](#) is set to **Exclude**.

For each of the Species, enter a value or expression in the table for the **Concentration** (mol/m^3) to set the composition of the inlet stream.

FEED INLET CONCENTRATION AND ENTHALPY

This section is available when [Energy Balance](#) is set to **Include**.

For each of the species, enter values or expressions in the table for the **Concentration** (mol/m^3) and **Species enthalpy** (J/mol) to set both the composition and enthalpy of the feed inlet stream. The column **Enthalpy** provides a combo box to select the type of enthalpy sources, that is, taken from **Species enthalpy** (a [Species](#) node) or by **User**

defined. The default values for **Species enthalpy (J/mol)** are enthalpy variable (of a [Species](#) node) expressions.



- [CSTR, Constant Mass/Generic](#)
- [CSTR, Constant Volume](#)
- [Semibatch](#)

Generate Space-Dependent Model

It is often relevant to perform an extended analysis of the reaction model. For example, to study how a reacting system's detailed geometry impacts the concentration and temperature distributions. Use the **Generate Space-Dependent Model** feature to export the properties within the Reaction Engineering interface to physics interfaces solving for fluid flow, mass transport and heat transfer in a space-dependent geometry. Properties exported from Reaction Engineering are reaction kinetics, thermodynamics, and transport parameters. The reaction kinetics are replicated in a Chemistry interface added to the space-dependent component. The physicochemical properties in Chemistry can be coupled to mass, fluid and heat transfer interfaces.

To add the **Generate Space-Dependent Model** node, find the **Reaction Engineering** toolbar and click **Generate Space-Dependent Model** () or right-click the **Reaction Engineering** node to add it from the context menu. Note that only a single **Generate Space-Dependent Model** node can be added.

SELECTING THE GEOMETRY AND PHYSICS INTERFACES

When using the **Generate Space-Dependent Model** feature, select which geometry and physics interfaces to create in the following manner:

- 1 Select the component to use: 1D, 1D Axisymmetric, 2D, 2D Axisymmetric, 3D, or an existing component in the Model Builder.
- 2 Select the physics interfaces to use in each category (chemical species transport, fluid flow and heat transfer).
- 3 Select a study type.

Then click the **Create/Refresh** button at the top of the settings window (see [Figure 2-6](#)). A new Chemistry interface is generated by default when a new geometry or a new chemical species transport interface is selected. It is also possible to add a new Chemistry interface to an existing component, for example after changing the reaction mechanism, by using the **Chemistry** list and setting the other lists to **None**.

When possible, one or several multiphysics coupling features are automatically added under the Multiphysics node in the Model Builder window. The coupling node simplifies multiphysics models by synchronizing settings between interfaces. When for example a **Transport of Diluted Species** interface is created together with a **Laminar Flow** interface, a **Reacting Flow, Diluted Species** multiphysics coupling node is also added.

To add physics interfaces to an existing component, simply select the component from the **Components to use** list, use the **Physics Interfaces** section to select additional interfaces, and click the **Create/Refresh** button.

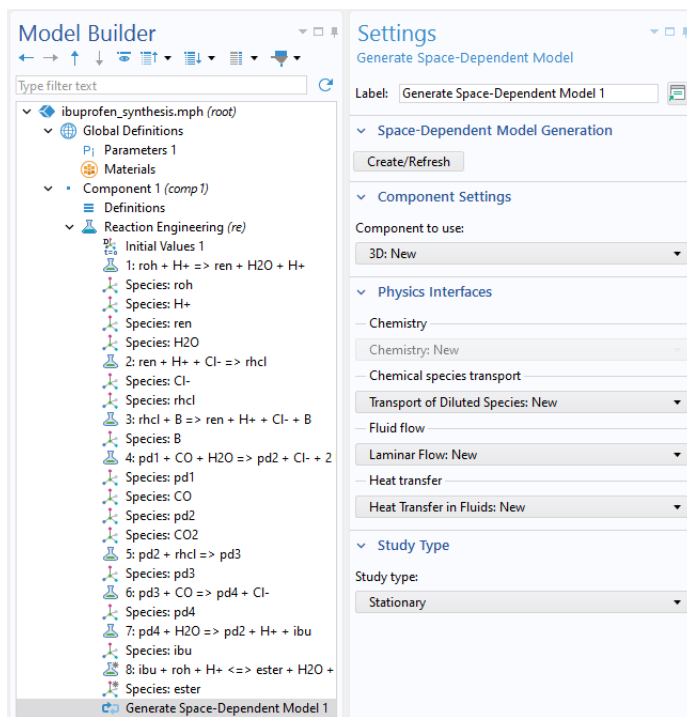


Figure 2-6: Using the Generate Space-Dependent Model node the reaction mechanism in a Reaction Engineering interface is transferred to physics interfaces in a space-dependent component (1D, 1D Axisymmetric, 2D, 2D Axisymmetric, or 3D).

COMPONENT SETTINGS

Select a **Component to use**. Either specify the space dimension of a new component — **1D**, **1D Axisymmetric**, **2D**, **2D Axisymmetric**, or **3D** — or select a component already present in the Model

Builder. The geometry selection affects the content of other interface lists. For example, the reacting flow interfaces are only available in 2D and 3D.

PHYSICS INTERFACES

Select the applicable physics interfaces to create from the **Chemical species transport**, **Fluid flow**, and **Heat transfer** lists. The interfaces available depends on the reaction system. When a solvent species is defined, a diluted solution is assumed and available interfaces for mass transport are based on the **Transport of Diluted Species** interface. When no solvent is defined, interfaces based on **Transport of Concentrated Species** are also available.

The **Chemical species transport** list contains two categories of mass transfer interfaces, as seen in [Figure 2-7](#). The top of the list contains single physics interfaces, such as **Transport of Diluted Species**. These can be used together with other single physics interfaces, from the **Fluid Flow** or **Heat Transfer** lists, to create a manual multiphysics setup.

The **Chemical species transport** list also contains predefined multiphysics interfaces, like **Reacting Flow** or **Reacting Flow in Porous Media**. Selecting one of these automatically includes multiple interfaces, as well as dedicated coupling features. These multiphysics interfaces correspond to the ones available in the **Chemical Species Transport** branch of the Model Wizard or the Add Physics window. When different versions of the interface exist, additional lists corresponding to subbranches (in the Model Wizard) are enabled. An example of a multiphysics interface version is shown in [Figure 2-8](#), where the **Porous Catalyst, Transport of Concentrated Species** version of the **Reacting Flow in Porous Media** has been selected.

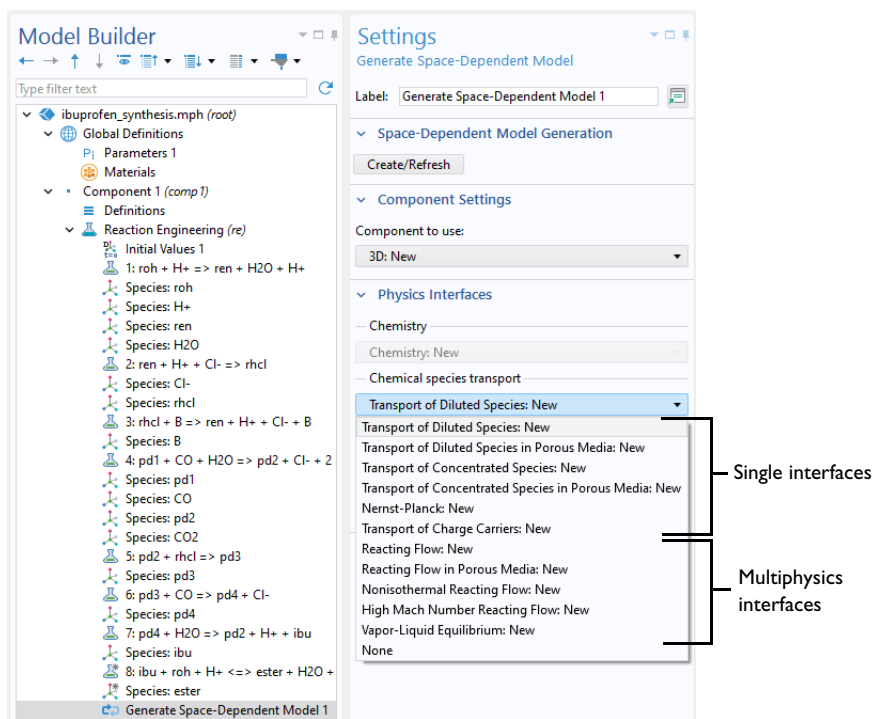


Figure 2-7: The Chemical species transport list contains both single interface entries as well as multiphysics interfaces.

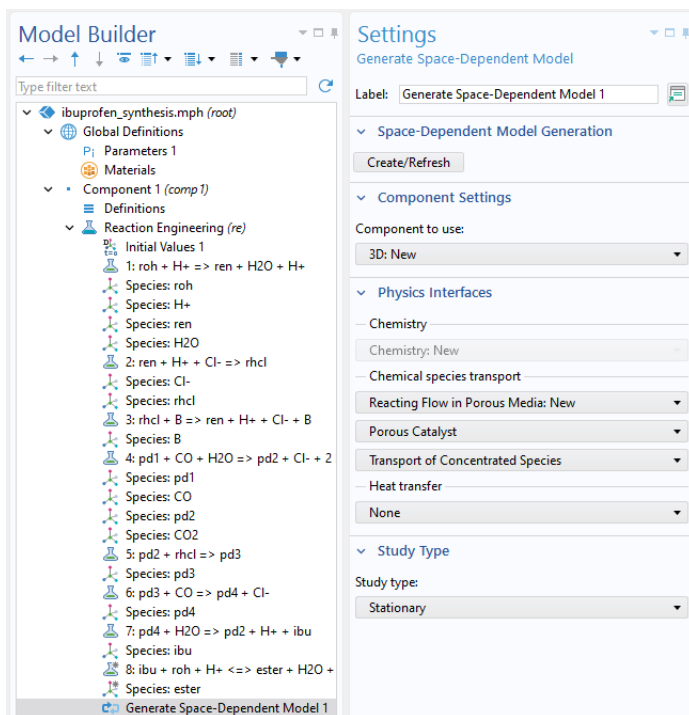


Figure 2-8: The Porous Catalyst, Transport of Concentrated Species version of the Reacting Flow in Porous Media interface is one example of a multiphysics interface that can be created from the Generate Space-Dependent Mode node.

The **Chemistry Interface** is always created and added when generating a new component or a new chemical species transfer interface. It generates global variables for the reaction kinetics, thermodynamics, and transport properties. The variables generated are available for all space-dependent interfaces. [Figure 2-9](#) displays an application where a Reactions feature uses reaction rates defined by a Chemistry interface. The syntax **chem** points to the default **Name** of the Chemistry node.

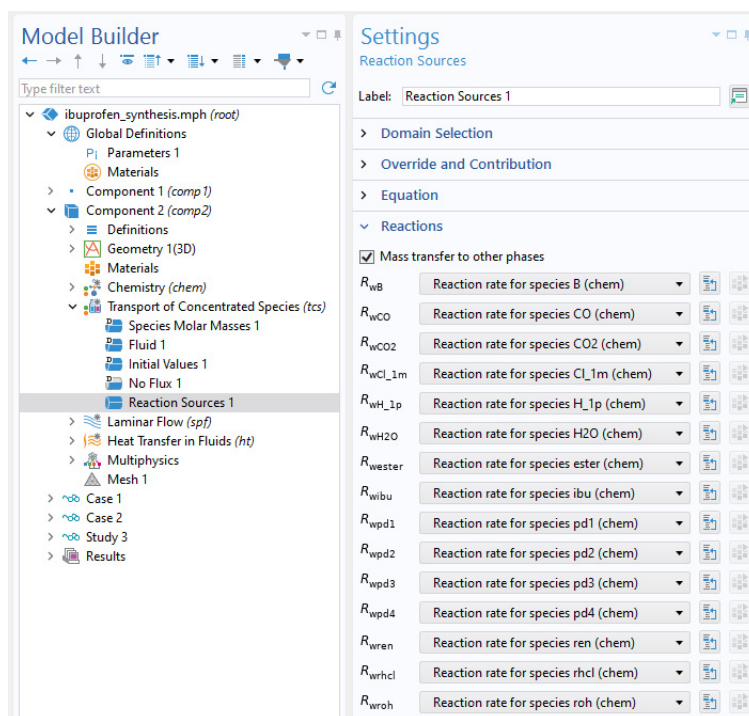


Figure 2-9: The Chemistry node generates global variables for rate variables and properties defined in the Reaction Engineering interfaces, and announces these to be used in relevant features of other space-dependent interfaces. The reaction rates within a Reaction Sources feature in a Transport of Concentrated Species interface illustrate this.

SURFACE REACTIONS

When surface species are present (that is, when the **Type** is set to **Surface species** for at least one species in the reactor), the surface reactions can be implemented in the space-dependent model in three different ways.

- When **Transport of Diluted Species in Porous Catalysts** or **Transport of Concentrated Species in Porous Catalysts** is selected, the surface reactions will be modeled using a **Surface Reaction** node under the **Porous Catalyst** feature.
- When **Transport of Diluted Species in Packed Beds** or **Transport of Concentrated Species in Packed Beds** is selected, the surface reactions will be modeled using a **Reactions** node under the subnode **Pellets** of the **Packed Bed** feature.
- When any other chemical species transport interface (except these mentioned above) is selected, the surface reactions will be modeled using a separate

Surface Reaction interface defined on the boundaries of the geometry. The surface reaction rates of species will be represented by a **Flux** feature in the created chemical species transport interface.

A feature for volumetric reactions is also added and set up, in accordance with the reaction kinetics defined in the **Reaction Engineering** interface, when clicking the **Create/Refresh** button in the **Space-Dependent Model Generation** section.

In [Figure 2-10](#), the surface reaction kinetics in a **Reacting Engineering** interface has been implemented in a **Packed Bed** feature using a **Reactions** subfeature. Note that the surface

reaction rates are defined by the **Chemistry** interface with the **Define variables for porous pellets** being checked under **Pellet Chemistry** section.

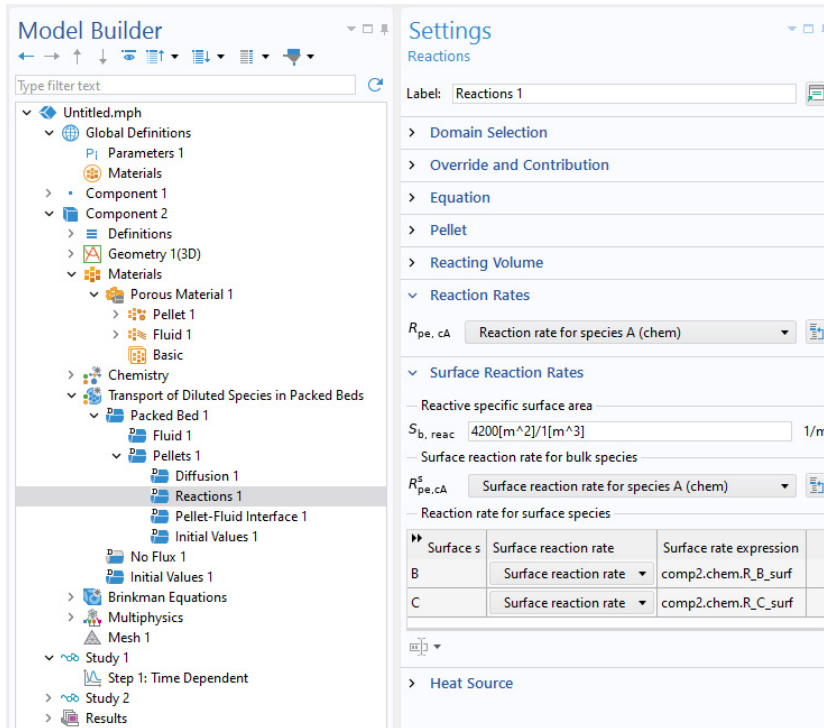


Figure 2-10: A Packed Bed feature including surface species can be automatically created, from a Reaction Engineering interface, using the Generate Space-Dependent Model feature.



- The Packed Bed Feature
- The Transport of Diluted Species Interface

Chemical Species Transport

The following single physics interfaces are available with the Chemical Reaction Engineering Module:

- The Transport of Diluted Species Interface
- The Transport of Diluted Species in Porous Media Interface
- The Transport of Concentrated Species Interface

- [The Transport of Concentrated Species in Porous Media Interface](#)
- [The Nernst–Planck Equations Interface](#)

The model generation automatically defines the dependent variables for all species. The interfaces based on transport of diluted species uses default variable names according to the syntax *c_{speciesname}*, referring to species concentrations in mole per volume. The interfaces based on transport of concentrated species use the syntax *w_{speciesname}* for default variable names, referring to the species weight fraction. The initial values for the dependent species variables in the space-dependent model are based on the initial species values in the Reaction Engineering interface.

Binary diffusion coefficients can be computed and transferred to a space-dependent model. The available diffusion models for the mass balance physics interfaces sometimes require the binary diffusion coefficients of the mixture, describing all pairwise diffusion interactions between species. Under the assumption of an ideal gas the mass transfer interface for Transport of Concentrated Species can utilize the binary diffusion coefficient. The Transport of Diluted Species and Nernst-Planck Equations interfaces assumes that the chemical species being transported are present in small amounts and are diluted in a solvent. The binary diffusion coefficient in these cases (in the presence of a solvent species) become the diffusion coefficient of the bulk species.

Fluid Flow

The model generation process can be selected to generate a Fluid Flow interface.

There are these separate momentum balance physics interfaces:

- [The Single-Phase Flow, Laminar Flow Interface](#)
- [The Darcy’s Law Interface](#) (in the *CFD Module User’s Guide*)
- [The Free and Porous Media Flow, Brinkman Interface](#) (in the *CFD Module User’s Guide*)

Species density (see [Equation 2-93](#), [Equation 2-94](#), and [Equation 2-95](#)) and dynamic viscosity can for some mixture options be transferred from the Reaction Engineering interface to the Fluid Flow interfaces.



Transport Properties



NOx Reduction in a Monolithic Reactor: Application Library path
Chemical_Reaction_Engineering_Module/Tutorials/monolith_reactor

Hydrocarbon Dehalogenation in a Tortuous Microreactor: Application
Library path **Chemical_Reaction_Engineering_Module/
Reactors_with_Mass_Transfer/tortuous_reactor**

Protein Adsorption: Application Library path
**Chemical_Reaction_Engineering_Module/Mixing_and_Separation/
protein_adsorption**

Heat Transfer

In the Reaction Engineering interface, time- and space-dependent energy balance equations can also be set up. In the model generation process the physics interface generates expressions used by the Heat Transfer interface, such as the heat generated by a chemical reaction. It also generates expressions for physical transport properties.

There are two heat transfer interfaces under **Heat transfer**:

- **Heat Transfer in Fluids**
- **Heat Transfer in Porous Media**

Several expressions for the species density, heat capacity, and thermal conductivity are available and can be transferred from the Reaction Engineering interface.

The densities are available from the [Calculate Transport Properties](#) section in the interface, where the fluid mixture properties are selected. The density depends on the [Species](#) settings and is computed as follows for:

- Liquid, if assuming the liquid to be ideal and incompressible. The liquid mixture density depends on the density of i number of pure species, ρ_i , and the species weight fraction, w_i .

$$\rho = \frac{1}{\sum_i \frac{w_i}{\rho_i}} \quad (2-93)$$

The volume fraction is given by the species concentration, c_i , and the molar mass, M_i .

$$w_i = \frac{c_i M_i}{\sum_i c_i M_i} \quad (2-94)$$

- Gas mixture density depends on the concentrations and molar masses of the mixture species.

$$\rho = \sum_i c_i M_i \quad (2-95)$$

For mixtures with solvent all values are taken from the species set as solvent.

The heat capacity, c_p (SI unit: J/(mol·kg)), of the mixture is calculated by the species' molar heat capacity, C_p (SI unit: J/(mol·kg)) according to

$$c_p = \sum_i w_i \frac{C_{p,i}}{M_i} \quad (2-96)$$

where M is the molar mass (SI unit: kg/mol) and w_i the weight fraction.

Porous media type

The **Porous media type** is available when **Transport of Diluted Species in Packed Beds** or **Transport of Concentrated Species in Packed Beds** is selected for **Chemical species transport** and the **Heat Transfer in Porous Media** is selected for **Heat transfer**. There are three options the **Porous media type**:

- **Local thermal equilibrium**
- **Local thermal nonequilibrium**
- **Packed bed**

These options are the same as that for **Porous Medium** feature under **Heat Transfer in Porous Media** interface. The heat balance equation is set up inside pellets when the **Packed bed** is selected.

STUDY TYPE

Select a **Study Type**, either **Stationary** or **Time Dependent**.

The Chemistry Interface

This physics interface can be used to create reaction kinetics and optionally compute mass transport and thermodynamic properties for direct use in 1D, 2D, or 3D models. This is similar to [The Reaction Engineering Interface](#), except that it does not solve for an ideal reactor model.

The **Chemistry (chem)** interface () is found under the **Chemical Species Transport** branch () when adding a physics interface. The Chemistry interface is also created when the [Generate Space-Dependent Model](#) feature is used in the **Reaction Engineering** interface, collecting all mixture variables and properties for use in a space-dependent model.

This physics interface is a tool for generating a set of variables to be used for modeling chemical species and reactions systems. The variables are generated from species and reaction properties and can be divided in two categories:

- Rate expressions and heat sources for use in mass and heat balances.
- Material property variables (mixture density, diffusivities, viscosity, and so on) for use in space-dependent transport equations.

Many of the fields and nodes described in this section are only made available when either a [Reaction](#) or a [Species](#) (or both) subnode is added to the Model Builder. All predefined constants and expressions can be overwritten by user-defined expressions. This makes it possible to go beyond the modeling assumptions set as defaults in this physics interface.



- [Theory for the Reaction Engineering and Chemistry Interfaces](#)
 - [Transport Properties](#)
 - [Feature Nodes Available for the Chemistry Interface](#)
-

The following is a description of the features and fields available in the **Settings** window for the Chemistry interface.

SETTINGS

The **Label** is the default physics interface name.

The **Name** is used primarily as a scope prefix for variables defined by the physics interface. Refer to such physics interface variables in expressions using the pattern

<name>.<variable_name>. In order to distinguish between variables belonging to different physics interfaces, the name string must be unique. Only letters, numbers, and underscores (_) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first physics interface in the model) is **chem**.

MODEL INPUT

This section sets the **Temperature**, **Pressure**, and **Electrode potential** (only available with a Battery Design Module, Corrosion Module, Electrochemistry Module, Electrodeposition Module, or Fuel Cell & Electrolyzer Module license) to be used by the current interface. Use the lists to select a temperature, pressure, or electrode potential defined and announced by another interface in the model. For example, when a heat transfer interface is also included, the temperature solved for is available in the **Temperature** list.

For **Temperature** or **Pressure**, you can also select **Common model input** to use a globally available common model input. In all three cases, select **User defined** to manually define the variable in question.

Mixture Properties

Type

Use this setting to define specify what kind of mixture assumption is used. For **Diluted Species** the mixture is composed of low concentration solutes present in a solvent. For **Concentrated Species** a mixture where no single species is considered to be in excess is assumed.

Select **Diluted species** from the **Type** list to use the concentration variables from a Transport of Diluted Species interface in the [Species Matching](#) section. The same setting should be used for any other interface solving for molar concentrations using a diluted species assumption. Select **Concentrated species** from the **Type** list to use the mass fractions from a Transport of Concentrated Species interface.

Thermodynamics

All property parameters and property functions required by the interface can be automatically created by coupling to a system added to the **Thermodynamics** node. To do so, click the **Thermodynamics** checkbox and select an existing [Thermodynamic System](#).

The **Thermodynamics** checkbox is enabled when the [Thermodynamics](#) node, including one or more systems, is available under **Global Definitions**. Also, the Chemistry interface needs to include at least one species.

Phase

Use the **Phase** list to specify the state of aggregation of the mixture.

Density



The mixture density settings are only available when you have selected the **Calculate mixture properties** checkbox in the **Calculate Transport Properties** section.

Density always has two settings available: **Automatic** or **User defined**. The **Thermodynamics** options is available when the interface is coupled to a [Thermodynamic System](#), and all interface species has been matched to species in the system. In this case the density is defined by a function automatically added under the thermodynamic system coupled to.

The **Automatic** setting uses the following logic:

- When **Phase** is set to **Liquid**, **Automatic** assumes that the liquid is ideal and incompressible. The liquid mixture density depends on the density of i number of pure species (ρ_i) and the species weight fraction (w_i).

$$\rho = \frac{1}{\sum_i \frac{w_i}{\rho_i}}$$

The weight fraction is given by the species concentration (c_i) and the molar mass (M_i).

$$w_i = \frac{c_i M_i}{\sum_i c_i M_i}$$

- When **Phase** is set to **Gas**, **Automatic** calculates the gas mixture density (ρ) from the concentrations (c_i) and molar masses (M_i) of the mixture species, which are automatically taken from [Species](#) features.

$$\rho = \sum_i c_i M_i \quad (2-97)$$

- If the **Type**, for one of the present species, is set to **Solvent**, and the **Mixture** is **Liquid**, the mixture density is the same as the solvent density. The species density is defined in the **General parameters** section of the corresponding Species node. When **Mixture** is **Gas**, the mixture density is calculated from Equation 2-97 only for the species set as **Solvent**.

SPECIES MATCHING

This section is used to match species solved for in space dependent physics interfaces to the species defined in Chemistry. By doing so the variables for the reaction kinetics and the mixture properties defined by Chemistry can be applied to study chemically reacting or composition dependent systems.

Species Solved For

Use this list to populate the inputs in the **Bulk species** table with dependent variables from a certain mass transfer interface. The **Species solved for** list contains all present interfaces of the **Type** selected in the **Mixture Properties** section. For example, when the **Type** is **Diluted species**, all Transport of Diluted Species interfaces or variants thereof, such as Transport of Diluted Species in Porous Media interface, are available in the list.

Bulk Species

Use the **Bulk species** table to specify the concentrations to be used as arguments in reaction kinetics variables, for example the reaction rate. Reaction kinetics variables are generated by the species features (**Species** and **Species Group**) and by the reaction features (**Reaction**, **Reversible Reaction Group**, and **Equilibrium Reaction Group**).

The bulk species concentrations are also used in the mixture properties, both for transport properties like the density and viscosity, and thermodynamic properties like the enthalpy or heat capacity.

When a mass transfer interface has been selected in the **Species solved for** list, use the **Molar concentration** column to select one of the dependent variables in the selected interface to the corresponding **Species** in Chemistry. When using a concentrated species interface (defined by the **Type** list in the **Mixture Properties** section), instead use the **Mass fraction** column to select a dependent variable for each species in Chemistry. For a consistent set up, the species molar mass in the Transport of Concentrated Species interface is updated to the one defined in the Chemistry interface.

The **Molar concentration** and **Mass fraction** columns also includes a **User defined** option, in which case a constant, parameter, or variable expression can be entered in the **Value** column.

Thermodynamics Coupling

When the **Thermodynamics** is enabled, the species in the Chemistry interface can be coupled to the species in the **Thermodynamic System**. This is needed to ensure that arguments for the thermodynamic functions are correctly defined. Use the lists in the **From Thermodynamics** column to match each species in the Chemistry to a species in the coupled thermodynamic system. For each thermodynamics-coupled species, the required property parameters and functions are added under the corresponding thermodynamic system. When all species are matched, the Chemistry is considered fully coupled and functions representing mixture properties, such as the density, are also added automatically under the same thermodynamic system.

Surface Species

The **Surface Species** table is available when there is at least one surface species (with “(ads)” as suffix).

Similar to the table for **Bulk Species**, the surface species can be matched (or coupled) directly to the surface species dependent variables if the Chemistry is coupled to a mass transfer interface where the surface concentration is solved. There are three cases where mass transfer interfaces solve surface species concentration (s).

- **Surface Reactions** physics interface;
- **Packed Bed** feature (with surface species transport in pellet) available in the **Transport of Diluted Species or Transport of Concentrated Species**;
- **Porous Catalyst** feature (with surface species transport in catalyst) available in the **Transport of Diluted Species or Transport of Concentrated Species**.

The last column **Value(mol/m²)** will be automatically set to **Solved for** when a surface species dependent variable is selected under the column **Surface concentration**.

Otherwise for a User defined being for **Surface concentration** column, specify the concentration of surface species **Value(mol/m²)** column.

The species present in this section will not be used to define mixture properties.

Solid Species

The **Solid Species** table is available when there is at least one solid species (with “(s)” as suffix). When present, use the table to specify the concentration of solid species.

Solid species may affect reaction kinetics, but does not contribute to mixture properties.

Aqueous Species

The **Aqueous Species** table is available when there is at least one aqueous species (with “(aq)” as suffix). When present, use the table to specify the concentration of aqueous species.

Aqueous species may affect reaction kinetics, but does not contribute to mixture properties.

Equilibrium Reaction Rate Table

This table is only available when equilibrium reactions are present in the interface. Use the **Reaction rate** column to specify the rate for each equilibrium reaction. By doing so the postprocessing variable for the total reaction rate for each species, of form `chem.Rsum_species`, will be updated correctly. For a mass transport interface, say Transport of Diluted Species, the reaction rate needed for an equilibrium reaction is typically a dependent variable. In that case, the name of the variable can be found in the **Shape Function** section of the **Equation View** of the node.

When the Chemistry interface is created using from the **Generate Space-Dependent Model** the table is automatically set up in accordance with the added equilibrium reaction nodes.

CALCULATE TRANSPORT PROPERTIES

Select the **Calculate mixture properties** checkbox (selected as default) to calculate mixture properties that can be picked up in the space-dependent model interfaces. The properties that can be calculated are shown beneath the checkbox. There are four properties, heat capacity, ratio of specific heats, thermal conductivity and dynamic viscosity. For these properties, the **Automatic** and **User defined** are always available, the **Thermodynamics** is available when all bulk species are coupled to a **Thermodynamic System** (fully coupled). Consider also if the built-in **Automatic** expressions fit the model or if **User defined** expressions are more suitable.

The density and diffusion coefficient are always calculated.

It could be costly to calculate mixture properties by **Automatic** or **Thermodynamics** if there are a lot of species. The calculations of mixture properties are set to the **User defined** when the number of species from an imported file (a CHEMKIN file or a reaction file) exceeds the maximum number (default 200). The maximum number can be changed by entering a new value in the **Maximum number of species for mixture**

property models field. This field is visible when a **Reversible Reaction Group** has the checkbox **Import CHEMKIN data** selected.



Transport Properties

ACTIVITY

This section is available when the **Thermodynamics** checkbox is cleared.

Select the **Use activity** checkbox to solve for species activities instead of species concentrations, which is a common approach when nonideal fluids are modeled.

An activity coefficient other than 1 can be set for each species for the [Species](#) node in the **Species Concentration/Activity** section.



Activity

CHEMKIN IMPORT FOR SPECIES PROPERTIES

It is available when the **Thermodynamics** checkbox is cleared.

This section enables CHEMKIN[®] import to simulate complex chemical reactions in gas phase.

Two types of CHEMKIN input files can be imported here: **Thermo** and **Transport**, for thermodynamic properties and transport properties, respectively. Properties for either volumetric or surface species are supported. Click **Browse** to locate the CHEMKIN file to be imported, then click **Import**. For **Thermo**, the imported data is directly entered in the **NASA format** fields in the [Species](#) node's **Thermodynamic Expressions** section; for **Transport**, the imported data is entered in its **Transport Expressions** section.



- [CHEMKIN Data and NASA Polynomials](#)
- [Reversible Reaction Group](#)

PELLET CHEMISTRY

Select the checkbox **Define variables for porous pellets** when the **Chemistry** interface is used to model transport and reactions in porous pellets using the **Packed Bed** feature. In this case an extra dimension, used to represent solid pellets, or particles, is present

in each point of the domain. In this extra dimension both bulk and surface species concentrations are solved for. For correct pellet chemistry make sure to input the intraparticle concentrations in the **Species Matching** section. When **Define variables for porous pellets** is selected, reaction rates as well as the mixture properties defined by **Chemistry** become available in the **Packed Bed** node and its subnodes.

For more information see the [Packed Bed \(Moving Packed Bed\)](#) feature in the Transport of Diluted Species interface.

Feature Nodes Available for the Chemistry Interface

The following feature nodes are available from the **Chemistry** ribbon toolbar (Windows users), **Chemistry** context menu (Mac or Linux users), or right-click to access the context menu (all users).

- [Equilibrium Reaction Group](#)
- [Reaction](#)
- [Reversible Reaction Group](#)
- [Species](#)
- [Species Activity](#)
- [Species Group](#)
- [Species Thermodynamics](#)¹

Reaction

Use the **Reaction** node () to define a chemical reaction where one or several species form other chemical species.

To add a **Reaction** node either right-click the **Chemistry** node, or click **Reaction** in the **Chemistry** toolbar.



Using the Reaction Node

REACTION FORMULA

Formula

Enter a chemical equation in the **Formula** field. The chemical equation should be of the format “A + B $\rightarrow$ C + D”. Here, A and B are reactants, C and D are product species, and $\rightarrow$ denotes a reaction arrow. The participating species should be written with [Valid Species Names](#). Valid reaction arrows are “ $\rightleftharpoons$ ” for reversible reaction, “ $\Rightarrow$ ” for

irreversible reaction, and “=” for equilibrium reaction. Species can be given trivial names, or their chemical formulas can be used as names. Examples of valid formulas are; “carbon+oxygen=>carbondioxide”, and “C(s)+O2(g)=>CO2(g)”.

Click **Apply** to make the interface examine the species taking part in the chemical equation, and automatically add the associated **Species** features to the Model Builder.

Balance ()

Use the **Balance** button to automatically calculate the stoichiometric coefficients such that the number of atoms of each kind are the same on both sides of the reaction. This turns, for example, the formula “H2+O2=>H2O” into “2H2+O2=>2H2O”. The balancing involves parsing all participating species for elements in the periodic table. It therefore requires that all species in the **Formula** field are written either using their chemical formula, say “H2O”, or that each species in the reaction have an enabled **Chemical Formula** field in their **Species** node. For example, it is possible to balance the formula “H2+O2=>water” as long as there is a Species node with the name “water” already present, and an enabled **Chemical Formula**.

For automatic reaction balancing to be successful, requires that the problem is well posed. One example of a problem that is not well posed is “C+H2=>CH4+C2H6”, in which case any ratio of CH₄/C₂H₆ could be obtained. Another example is “H2=>O2” where not all elements are present on both sides.

Reaction Type

Select the **Reaction type** — **Reversible**, **Irreversible**, or **Equilibrium** — or edit the expression directly in the **Formula** field. In the latter case, specify the reaction type with a delimiter separating the two sides of the equation:

- <=> denotes a **Reversible** reaction
- => denotes an **Irreversible** reaction
- = denotes a reaction at chemical **Equilibrium**

Each **Reaction type** has its own set of reaction kinetics:

- If the reaction is **Reversible** or **Irreversible**, the reaction rate for reaction i contributes to the change in species i as follows:

$$R_i = \sum_j v_{ij} r_j \quad (2-98)$$

where v_{ij} is the stoichiometric coefficient.

- If it is an **Equilibrium** reaction, the equilibrium expression is equal to the equilibrium constant:

$$K_{\text{eq}} = K_{\text{eq}0}$$

REACTION RATE

This section is available when the **Reaction type** is either **Reversible** or **Irreversible**.

When **Mass action law** is selected (default), the rate expression is automatically derived from the stoichiometric coefficients in the reaction formula:

- For an **Irreversible** reaction type, the reaction rate becomes:

$$r_j = k_j^f \prod_{i \in \text{react}} c_i^{-v_{ij}} \quad (2-99)$$

- For a **Reversible** reaction type, the expression instead becomes:

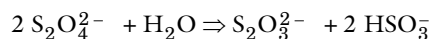
$$r_j = k_j^f \prod_{i \in \text{react}} c_i^{-v_{ij}} - k_j^r \prod_{i \in \text{prod}} c_i^{v_{ij}} \quad (2-100)$$

The deduced overall reaction order is shown in text below the respective equation in the **Reaction Rate** section.

Arbitrary Rate Expression

If the reaction order differs from the stoichiometric coefficients, or if an arbitrary rate expressions is applicable, change **Reaction Rate** to **User defined**. An expression field r appears with the default expression being that from the mass action law. Below this there are fields to set the reaction order. For a reversible reaction the reverse reaction order may be specified in addition to the forward one. The unit of the rate constant k (or frequency factor A in the case of Arrhenius behavior), is derived from the reaction order, in SI units: $(\text{m}^3/\text{mol})^{\alpha-1}/\text{s}$, where α equals the order with respect to volumetric species. When surface species are present — identified by their “(ads)” suffix — the unit is instead given by $\text{m}^{3\alpha+2\beta-2}/\text{mol}^{\alpha+\beta-1}/\text{s}$, where β is the order with respect to surface species.

Consider for example the reaction:



The automatically deduced reaction order is three, however, in the case that water is the solvent, the order should probably be two. This is specified as follows:

The screenshot shows the 'Settings' dialog for a reaction. The reaction formula is $2 \text{SO}_4^{2-} + \text{H}_2\text{O} \Rightarrow \text{S}_2\text{O}_3^{2-} + 2 \text{HSO}_3^-$. The reaction type is 'Irreversible'. The reaction rate is set to 'User defined' with the expression $r_j = k_f \cdot [\text{re.c.SO}_4\text{2m}]^2$. The units are $\text{mol}/(\text{m}^3\cdot\text{s})$. The forward rate constant k^f is set to 1 with units $\text{m}^3/(\text{mol}\cdot\text{s})$. The volumetric overall reaction order is set to 2.

Annotations in the image:

- Change to User Defined (points to the 'User defined' dropdown)
- Adjust rate expression (points to the rate expression input field)
- Override reaction order (points to the 'Forward rate constant' input field)
- Note effect on unit (points to the units input field)

Figure 2-11: The reaction order can be manually specified for a user defined reaction rate.

RATE CONSTANTS

This section applies for **Reversible** or **Irreversible** reactions and defines the reaction rate constants used in the reaction rates.

Forward Rate Constant and Reverse Rate Constant

The **Forward rate constant** k^f is used for both **Reversible** and **Irreversible** reactions. The **Reverse rate constant** k^r is only used for **Reversible** reactions (Equation 2-99).

The SI units of the rate constants are automatically based on the order of the reaction with respect to the concentrations, as defined in the **Reaction formula**.

Specify Equilibrium Constant

The **Specify equilibrium constant** checkbox is available for **Reversible** reactions. If the checkbox is selected the rate constants are defined in a different manner with the reverse rate constant being computed from the following expression:

$$k^r = k^f / K_{\text{eq}0}$$

Thus, in this case, the forward rate constant and equilibrium constant for the reaction are needed. The **Equilibrium constant** is edited in the [Equilibrium Settings](#) section.

Use Arrhenius Expressions

When the **Use Arrhenius expressions** checkbox is selected the Arrhenius parameters are automatically used in predefined expressions for the forward and reverse rate constants k^f and k^r , respectively.

$$k^f = A^f (T/T_{\text{ref}})^{n^f} \exp(-E^f/(R_g T))$$

$$k^r = A^r (T/T_{\text{ref}})^{n^r} \exp(-E^r/(R_g T))$$

Specify the activation energy and the frequency factor in the Arrhenius expressions to account for temperature variations. The reference temperature, T_{ref} equals 1 K. The available fields are based on the **Reaction type** chosen in the [Reaction](#) node. Enter values or expressions for each of the following (reverse expressions are only available for reversible reactions):

- **Forward frequency factor** A^f and **Reverse frequency factor** A^r (unit depends on reaction order)
- **Forward temperature exponent** n^f and **Reverse temperature exponent** n^r
- **Forward activation energy** E^f and **Reverse activation energy** E^r (SI unit: J/mol)

EQUILIBRIUM SETTINGS

This section is available for equilibrium reactions, and for reversible reactions when the **Specify equilibrium constant** checkbox has been selected.

Equilibrium Expression

For an equilibrium reaction, specify the **Equilibrium expression**. When the **Equilibrium expression** is set to **Automatic** the following expression is used:

$$K_{\text{eq},j} = \frac{\prod_{i \in \text{prod}} c_i^{v_{ij}}}{\prod_{i \in \text{react}} c_i^{-v_{ij}}}$$

Select **User defined** from the **Equilibrium expression** list to instead enter a manually defined equilibrium expression.

Equilibrium Constant

Specify the **Equilibrium constant** K_{eq0} for an equilibrium reaction, or for a reversible reaction when the **Specify equilibrium constant** checkbox has been selected (in the **Rate Constants** section).

The **Equilibrium constant** can either be **User defined**, or automatically defined when set to **Automatic** or **Thermodynamics**.

Use the **Automatic** option to compute the equilibrium constant for an ideal system.

The **Thermodynamics** option is available when all reactions in the interface are equilibrium reactions, and the interface is fully coupled to a [Thermodynamic System](#) (see [Species Matching](#)). Use this setting to automatically compute the equilibrium constant for an ideal or nonideal system, dependent on the thermodynamic model applied for the coupled system.

Using **Automatic** or **Thermodynamics**, K_{eq0} is calculated from the Gibbs free energy of the reaction. For more details see [The Equilibrium Constant](#) and the [Automatically Defined Equilibrium Constants](#) section therein.

REACTION THERMODYNAMIC PROPERTIES

This section contains information about thermodynamic properties that relate to a selected reaction. Several **Automatic** definitions are available here.

Enthalpy of Reaction

The **Enthalpy of reaction** H (SI unit: J/mol) is calculated by the interface from species properties and the related stoichiometric coefficients:

$$H_j = \sum_{i \in \text{prod}} \nu_{ij} h_i - \sum_{i \in \text{react}} (-\nu_{ij}) h_i \quad (2-101)$$

Entropy of Reaction

The **Entropy of reaction** S (SI unit: J/(mol·K)) comes from a similar expression:

$$S_j = \sum_{i \in \text{prod}} \nu_{ij} s_i - \sum_{i \in \text{react}} (-\nu_{ij}) s_i \quad (2-102)$$

In [Equation 2-101](#) and [Equation 2-102](#), h_i and s_i are the species' molar enthalpy and molar entropy, respectively.

Enter these quantities in the **Species Thermodynamic Expressions** section for the [Species](#) node either by using the predefined polynomial or by providing a custom expression or constants.

The stoichiometric coefficients, ν_{ij} , are defined as being negative for reactants and positive for products. Using [Equation 2-101](#) and [Equation 2-102](#) to equate the Gibbs free energy of reaction enables the equilibrium constant to be expressed according to [Equation 2-101](#).

Heat Source of Reaction

The **Heat source of reaction** (SI unit: W/m^3) is automatically computed from the heat of each reaction j , given by

$$Q_j = -H_j r_j$$

TURBULENT FLOW

When a reaction occur in a fluid within the turbulent flow regime, the reaction rate may be affected by the turbulent flow. Use the settings in this section to model reactions affected by turbulent flow.

The **Turbulent Flow** section is available when the below conditions are met:



- 1** The mixture **Type** is **Concentrated species** in the **Mixture Properties** section
- 2** Reversible or irreversible bulk reaction is considered
- 3** **Use activity** not selected in the **Activity** section
- 4** **Define variables for porous pellets** is not selected in the **Pellet Chemistry** section

Turbulent Flow

When the **Turbulent-reaction model** is set to **None**, laminar flow is assumed and the reaction source terms are defined from the prescribed reaction rates.

When the **Turbulent-reaction model** is set to **Eddy-dissipation**, turbulent flow will be accounted for in the reaction mass sources in the manner described in the [Turbulent Reactions](#) section, and in [The Reaction Source Term for Turbulent Flow](#), in the *CFD Module User's Guide*. In this case, enter values for the **Turbulent reaction model parameters** α_{ED} and β_{ED} (dimensionless).

The **Eddy-dissipation** model also requires an estimation of the turbulent mixing time of the fluid flow turbulence. When a Fluid Flow interface defining it is present in the

model, it can be selected from the **Turbulence time scale** list. For example, select **Turbulence time scale (spf/fp1)** to use the time scale defined by the **Fluid Properties** node fp1 in a Turbulent Flow, k - ϵ interface with the **Name** set to spf.

Assume Infinitely Fast Reaction This checkbox is available when a valid turbulence model is selected. With the checkbox selected, the reaction rate is limited by the turbulent mixing, and the settings for kinetic reaction rates in the **Rate Constants** and **Reaction Orders** is not available.

Regularization

Select **Reaction rate regularization** to regularize the individual rate expressions added to each species. If the mass fraction for a reactant species ω_i becomes smaller than its **Damping limit**, ω_i^{dl} , the rate expression, for species ω_i is reduced linearly. If $\omega_i \leq 0$ for a reactant species, the reaction rate contribution to that species is completely removed. Similarly, if the mass fraction for a product species ω_j becomes larger than $1 - \omega_j^{\text{dl}}$, the rate expression added is damped linearly. If $\omega_j \geq 1$ for a product species, the reaction rate contribution to that species is completely removed.

The default value for the **Damping limit**, ω_i^{dl} , is 10^{-6} , which is appropriate for most applications, but can require adjustment when working with for example catalytic trace species.

Regularization of the rate expressions adds considerable stability to the reaction expressions, but it should ideally only be used as a means to reach convergence. When reaction zones are not adequately resolved on the mesh, regularization can affect the mass balance.



This section is available under the same conditions as those of the **Turbulent Flow** section.

Species

When a **Reaction** is defined, a **Species** node () is automatically generated for the participating reactants and products. This feature enables you to review and enter species specific information regarding chemical kinetics, thermodynamics, and transport properties.

It is also possible to add and define an individual **Species** node: in the **Chemistry** toolbar click **Species** or right-click the **Chemistry** node and select it from the context menu.

NAME

When a **Species** node is automatically generated using the **Formula** text field for the Reaction node, the **Name** is also automatically generated.

For a **Species** node added individually, enter a **Name** in the field and click **Apply**. By entering a name consisting of elements from the periodic table, the molar mass is calculated and added in the **Chemical Formula** section. The species charge, and type, are also deduced from the species name.



Valid Species Names

TYPE

Select a species type — **Bulk species**, **Surface species**, or **Solvent**. The latter is only available when a diluted solution is assumed. That is when **Type** is set to **Diluted Species** in the **Mixture Properties** section (in the interface level).

Bulk species and **Solvent** are solved for volumetric concentrations (SI unit: mol/m^3), while **Surface species** are solved for surface concentration (SI unit: mol/m^2). The compositions for **Bulk species** and **Solvent** use the syntax `c_speciesname`, while **Surface species** uses `csurf_speciesname_surf`.

When **Surface species** is selected, the corresponding reaction formula introduces **(ads)** after the species notation and changes the species' name to `speciesname_surf`. Additionally, the Species node name is updated in a similar fashion.

CHEMICAL FORMULA

The **Chemical Formula** section contains the species chemical formula, **Molar mass**, M , and **Charge**, z .

If the name entered for the species consists of only elements from the periodic table, and optionally charge and phase indication, then the **Chemical Formula** field will be populated with the name entered in the **Name** section, and the molar mass and charge will be added to this section. The species molar mass is computed from the mass of occurring individual elements². The rules for writing chemical formulas are the same as those for **Species** names.



Valid Species Names

When the species name contains parts not in the periodic table, for example when a descriptive name such as **water** is used, the molar mass is set to 0.0 kg/mol and the charge to zero. In this case, the molar mass needs to be defined in order to achieve mass balanced reactions and correct definitions of mass basis properties. By entering a chemical formula consisting only of elements from the periodic table (and optionally charge and phase indication), the molar mass, and charge are derived automatically.

When needed, the **Molar mass** can be edited in the corresponding text field. Editing the **Chemical formula** and pressing apply will override the molar mass and charge fields. The **Charge** field cannot be edited as long as a chemical formula is enabled.

With an enabled **Chemical Formula** it is possible to use the trivial name of a species and balance reactions. Having added the Species **water**, **oxygen**, and **hydrogen**, and filled in H₂O, O, and H in their respective chemical formula sections, the **Formula** oxygen+hydrogen=water is balanced into oxygen+2hydrogen=water by clicking **Balance** in the **Reaction** node.

It is possible to specify the species density ρ when the fluid **Mixture** is specified as **Liquid**. The default value is that of water at 293 K.

REACTION RATE

Change the **Automatic** default setting to **User defined** to use a species reaction rate other than the one set up in the associated **Reaction** node. For individual species, use the **User defined** option to set a reaction rate other than zero (that is, nonreactive).

Edit either the **Rate expression** (SI unit: mol/(m³·s)), the **Surface rate expression** (SI unit: mol/(m²·s)), or both. For a bulk species, both expressions appear if surface reactions are present since the reaction of the species can depend both on bulk reaction R and surface reaction R_{ads} rates. For a surface species, only the surface reaction rate R_{ads} appears.

ADDITIONAL SOURCE

The **Additional Source** section is available in order to include additional rate contribution for the species to the reaction kinetics. When the **Additional source** checkbox is selected, add an **Additional rate expression** in the text field (SI unit: mol/m³).

2. Values are averages of upper and lower bounds for each atom as published by CIAAW. Source: CIAAW. Atomic weights of the elements 2019. Available online at www.ciaaw.org.

CONSTANT CONCENTRATION/ACTIVITY

To account for nonideality in the fluid mixture, adjust the activity coefficient in the **Activity coefficient** input field. This input field is only shown if activity instead of concentration has been chosen in the interface, that is, the **Use activity** checkbox is selected on the Chemistry interface **Settings** window

Click to select the **Keep concentration/activity constant** checkbox if the species concentration or activity should be treated as constant.

TRANSPORT EXPRESSIONS

Here the required parameters to compute various transport properties can be edited. To show this section the **Calculate mixture properties** checkbox needs to be selected under the **Calculate Transport Properties** section.

- For a gas mixture, there are maximum five properties to consider: σ , the characteristic length (unit: Å) of the Lennard–Jones/Stockmayer potential; ϵ/k_b , the energy minimum (SI unit: K) of the Lennard–Jones/Stockmayer potential; μ_D , the dipole moment (SI unit: Debye); k_i , the thermal conductivity of the gas (SI unit: W/(m·K)); and Diffusivity of the species (SI unit: m²/s). The latter two can be set as **Automatic**, when theoretical Transport Parameters expressions are available.
- For mixtures of gas with solvent, all properties above except the thermal conductivity is shown. This is instead calculated under the **Calculate mixture properties** section in the Settings window for the Chemistry interface.
- For a liquid mixture, there is only the parameter k_i , the thermal conductivity (SI unit: W/(m·K)) to edit.
- For mixtures of liquid with solvent, the thermal conductivity is shown only for the species set as **Solvent**.

THERMODYNAMIC EXPRESSIONS

The parameters utilized for calculation of thermodynamic mixture and reaction properties are set in this section.

The **Thermodynamic Expressions** are by default computed using data input in the **NASA format**. In this case, enter the following to compute the species' heat capacity, C_p (SI unit: J/(mol·K)), the molar enthalpy, h (SI unit: J/mol), and the molar entropy, s (SI unit: J/(mol·K)):

- **Lower temperature limit** T_{lo}
- **Middle temperature limit** T_{mid}

- **Upper temperature limit** T_{hi}
- **Polynomial coefficients** $a_{low,k}$ and $a_{hi,k}$

Any coefficients for the thermodynamic polynomials entered into the $a_{low,k}$ fields apply to the temperatures in the range T_{lo} to T_{mid} ; coefficients entered into the $a_{hi,k}$ fields apply to temperatures in the range T_{mid} to T_{hi} range. The coefficients can also be imported in the **CHEMKIN Import for Species Properties** section in the Chemistry node.

Choose the **User defined** alternative to specify C_p , h , and s directly.



Using the Species Node

Reversible Reaction Group

The **Reversible Reaction Group** node () allows input of a large number of reversible reactions in the same table. All reactions are treated in the same way. More than one Reversible Reaction Group can be added to a component. The feature is typically used when importing CHEMKIN kinetics files.

Add the node from the **Chemistry** toolbar or right-click **Chemistry** and add it from the context menu.

For the case when reaction kinetics data are entered manually into the **Reaction table** and temperature-dependent reaction kinetics apply. The Enthalpy of Reaction (J/mol) can be calculated from the thermodynamics properties of the species.

REACTION TABLE

The reversible reactions in the **Reaction table** are numbered and contain reactants, products, and kinetic parameters describing the reaction. Use the buttons under the **Reaction table** to add and sort the reaction details.

- In general, use the **Move Up** (), **Move Down** (), and **Delete** () buttons and the fields under tables to edit the table contents. Or right-click a table cell and select **Move Up**, **Move Down**, or **Delete**.
- The **Add** button () adds default reactant, A, and product, B, with a default. Click the corresponding field to edit the reactant, product, or parameters. After editing the **Species Group** node is also updated. It is created together with the reaction group.

- You can save the parameters to a text file to reuse in other models. Click the **Save to File** button () and enter a **File name** in the **Save to File** dialog, including the extension .txt. Click **Save** to store the parameters in a text file or in a Microsoft Excel Workbook spreadsheet if the license includes LiveLink™ for Excel®. The information is saved in space-separated columns in the same order as displayed on screen. When saving to Excel, an **Excel Save** dialog appears where you can specify the sheet and range and whether to overwrite existing data or include a header.
- You can import or load data in files from a spreadsheet program, for example, with the **Load from File** button () and the **Load from File** dialog that appears. Data must be separated by spaces or tabs. If there is already data in the table, imported parameters are added after the last row. Move or edit rows as needed. If the license includes LiveLink™ for Excel® you can also load parameters from a Microsoft Excel Workbook spreadsheet. Then an **Excel Load** dialog appears where you can specify the sheet and range and whether to overwrite existing data. It is also possible to import from a spreadsheet containing a separate column for units.

CREATE REACTION AND SPECIES

Enter a reaction number in the **Moving reaction (with the number) from table** text field and click **Create Reaction** to remove the reaction from the **Reaction table** to the model tree. This introduces a reaction node named after the reaction number and species names and simultaneously creates corresponding species nodes. This action is powerful, since it enables all the editing possibilities available for [Reaction](#) and [Species](#) nodes.

CHEMKIN IMPORT FOR KINETICS

Click to select the **Import CHEMKIN data** checkbox to import a CHEMKIN **Kinetics input file**. To import the data directly into the table columns, click **Browse** to locate the CHEMKIN file to be imported, then click **Import**.



[CHEMKIN Data and NASA Polynomials](#)

Equilibrium Reaction Group

The **Equilibrium Reaction Group** node () allows for input of a large number of equilibrium reactions in the same table. This node functions in the same way as the [Reversible Reaction Group](#) for reversible reactions.

Add the node from the **Chemistry** toolbar or right-click **Chemistry** and add it from the context menu.

REACTION TABLE

The equilibrium reactions in the table are numbered and contain reactants, products, and kinetic parameters describing the reaction. Use the buttons under the **Reaction table** to add and sort the reaction details.

- In general, use the **Move Up** (), **Move Down** (), and **Delete** () buttons and the fields under tables to edit the table contents. Or right-click a table cell and select **Move Up**, **Move Down**, or **Delete**.
- The **Add** button () adds default reactant, A, and product, B, with a default. Click the corresponding field to edit the reactant, product, or parameters. After editing the **Species Group** node is also updated. It is created together with the reaction group.
- You can save the parameters to a text file to reuse in other models. Click the **Save to File** button () and enter a **File name** in the **Save to File** dialog, including the extension .txt. Click **Save** to store the parameters in a text file or in a Microsoft Excel Workbook spreadsheet if the license includes LiveLink™ for Excel®. The information is saved in space-separated columns in the same order as displayed on screen. When saving to Excel, an **Excel Save** dialog appears where you can specify the sheet and range and whether to overwrite existing data or include a header.
- You can import or load data in files from a spreadsheet program, for example, with the **Load from File** button () and the **Load from File** dialog that appears. Data must be separated by spaces or tabs. If there is already data in the table, imported parameters are added after the last row. Move or edit rows as needed. If the license includes LiveLink™ for Excel® you can also load parameters from a Microsoft Excel Workbook spreadsheet. Then an **Excel Load** dialog appears where you can specify the sheet and range and whether to overwrite existing data. It is also possible to import from a spreadsheet containing a separate column for units.



Chemical Vapor Deposition of GaAs: Application Library path
**Chemical_Reaction_Engineering_Module/
Reactors_with_Mass_and_Heat_Transfer/gaas_cvd**

Species Group

The **Species Group** node () contains information on a molecular level about the volumetric species and the surface species present in the coupled reaction group features. The **Property for Volumetric Species** or **Property for Surface Species** tables typically collect parameters from when importing CHEMKIN transport files.

This node is automatically added when either the [Reversible Reaction Group](#) or the [Equilibrium Reaction Group](#) is used.

The **Show species name from CHEMKIN file** checkbox is available when the coupled [Reversible Reaction Group](#) feature is used to import a CHEMKIN reaction system. With this checkbox selected, there will be an extra column, **Name in CHEMKIN**, in both the **Property for Volumetric Species** and **Property for Surface Species** tables. The **Name in CHEMKIN** column shows the original species names from the CHEMKIN file because the species names defined there can be modified to follow the naming rules in COMSOL.

For the case of temperature-dependent reaction kinetics, a [Species Thermodynamics](#) subnode is automatically created in which the thermodynamic properties of the species can be specified.

CHEMKIN

This section allows import of CHEMKIN transport files. To import the data directly into the table columns, click **Browse** to locate the CHEMKIN file to be imported, then click **Import**. Note that this section is only shown when transport properties are computed (**Calculate mixture properties** checkbox under the **Calculate Transport Properties** is selected).



- [CHEMKIN Data and NASA Polynomials](#)
-

Species Activity

The **Species Activity** node () creates variables for the activities of all the species/surface species present in the Species Group parent feature. Edit the **Activity coefficient** field in the **Species Activity** or **Surface Species Activity** tables by clicking in these.

Species Activity is a subnode to the [Species Group](#) node. It is automatically generated when nonideality in the fluid mixture is accounted for; when the **Use activity** checkbox is selected in the **Activity** section.

Species Thermodynamics

The **Species Thermodynamics** node () creates variables for the enthalpies, entropies, and heat capacities for all the species/surface species present in the [Species Group](#) parent feature. The purpose is to compute thermodynamic mixture properties and the heat of reactions.

This node is a subnode to the [Species Group](#) node.

THERMODYNAMIC PROPERTIES OF SPECIES

Select a **Type selection** for the thermodynamic properties. The **NASA format** alternative requires specifying several polynomial constants and is typically used together with CHEMKIN import. The **User defined** alternative only requires input of the **Species molar enthalpy** (J/mol) and the **Heat capacity** (J/(mol·K)).

Similar to its parent [Species Group](#) node, the checkbox **Show species name from CHEMKIN file** is available when its coupled [Reversible Reaction Group](#) feature is set to CHEMKIN import. Select this checkbox to show the original species names (under the column **Name in CHEMKIN**) from the CHEMKIN file.

CHEMKIN IMPORT FOR THERMODYNAMIC DATA

This section allows import of thermodynamic CHEMKIN transport files. To import the data directly into the table columns, click **Browse** to locate the CHEMKIN file to be imported, then click **Import**. Note that this can only be done for the **NASA format** option.



- [CHEMKIN Data and NASA Polynomials](#)
-

Study Steps for the Reaction Engineering Interface

Reactor Types and Solver Study Steps

The predefined **CSTR**, **Batch**, and **Semibatch** reactor types available in the Reaction Engineering interface set up equations describing how species concentrations in a perfectly mixed system change with time. **Time Dependent** is the correct study type to use for these reactor types. The simulation time is specified in the settings of the **Time Dependent** study step.

The predefined **Plug flow** reactor type sets up equations for stationary conditions describing how species molar flow rates (SI unit: mol/s) vary with the reactor volume. **Stationary Plug Flow** is the correct study type to use for this reactor type. The reactor volume is specified in the **Stationary Plug Flow** study step feature

Note that selecting the **Reaction Engineering** interface and the **Time Dependent** study sets up the **Batch, constant volume** reactor type by default. The **Time Dependent** study is also suitable for all **CSTR** reactors and the **Semibatch** reactor. However, if the reactor type is changed to **Plug flow**, then the Time Dependent study step needs to be replaced with a **Stationary Plug Flow** step. Combining a Reaction Engineering interface with the **Stationary Plug Flow** study does not set up a **Plug flow** reactor type by default. In that situation, the **Plug Flow** reactor type needs to be set manually in the Reaction Engineering interface.

Chemical Species Transport Interfaces

The **Chemical Species Transport** branch () includes a number of subbranches to describe mass transport and chemical reactions. The physics interfaces are summarized in the first section, and then described in detail throughout this chapter. All physics interfaces include features for defining arbitrary reaction kinetics, to be used as mass sources and sinks, as well as arbitrary mass transport parameters.

- [Overview of Chemical Species Transport Interfaces](#)
- [The Transport of Diluted Species Interface](#)
- [The Transport of Diluted Species in Porous Media Interface](#)
- [The Transport of Diluted Species in Fractures Interface](#)
- [The Transport of Concentrated Species Interface](#)
- [The Transport of Concentrated Species in Porous Media Interface](#)
- [The Nernst–Planck Equations Interface](#)
- [The Nernst–Planck–Poisson Equations Interface](#)
- [The Electrophoretic Transport Interface](#)
- [The Size-Based Population Balance Interface](#)
- [The Vapor–Liquid Equilibrium Multiphysics Interfaces](#)

- The Reacting Flow, Concentrated Species Multiphysics Interface
- The Reacting Flow, Diluted Species Multiphysics Interface
- The Reacting Flow in Porous Media Multiphysics Interfaces
- The Nonisothermal Reacting Flow Multiphysics Interfaces
- The Dispersed Two-Phase Flow with Species Transport Multiphysics Interfaces
- The Precipitation and Crystallization Multiphysics Interfaces
- The Surface Reactions Interface
- The Transport of Diluted Species in Fractures Interface
- Theory for the Transport of Diluted Species Interface
- Theory for the Transport of Concentrated Species Interface
- Theory for the Electrophoretic Transport Interface
- Theory for the Surface Reactions Interface
- Theory for the Nernst–Planck Equations Interface
- Theory for the Size-Based Population Balance Interface
- Theory for the Vapor–Liquid Equilibrium Interfaces
- Theory for the Reacting Laminar Flow Interface

For a detailed description and background of [The Chemistry Interface](#) and [The Reaction Engineering Interface](#) see [The Chemistry and Reaction Engineering Interfaces](#) chapter.

Overview of Chemical Species Transport Interfaces

In this section:

- [Available Physics Interfaces](#)
- [Coupling to Other Physics Interfaces](#)
- [Adding a Chemical Species Transport Interface and Specifying the Number of Species](#)
- [Available Result Templates](#)

Available Physics Interfaces

THE TRANSPORT OF DILUTED SPECIES, AND THE TRANSPORT OF DILUTED SPECIES IN POROUS MEDIA INTERFACES

The mass transfer equations account for chemical species transport by diffusion, convection, and migration due to an electric field (electrokinetic flow)—either alone or in combination. [The Transport of Diluted Species Interface](#) () is an enhanced version of the physics interface included with the basic COMSOL Multiphysics license and is applicable for solutions (either fluid or solid) where the transported species have concentrations at least one order of magnitude less than the solvent. The settings for this physics interface can be chosen so as to simulate chemical species transport through diffusion (Fick's law), convection (when coupled to fluid flow), and migration (when coupled to an electric field — electrokinetic flow). Also see the [The Transport of Diluted Species Interface](#).

[The Transport of Diluted Species in Porous Media Interface](#) () is tailored for the modeling of solute transport in porous media. The physics interface supports cases where either the solid phase substrate is exclusively immobile, or when a gas-filling medium is also assumed to be immobile. Also see the [Mass Balance Equation for Transport of Diluted Species in Porous Media](#).

THE TRANSPORT OF CONCENTRATED SPECIES, AND THE TRANSPORT OF CONCENTRATED IN POROUS MEDIA SPECIES INTERFACES

[The Transport of Concentrated Species Interface](#) () is used for modeling transport in gas or liquid mixtures where no component is clearly dominant. Often the

concentrations of the participating species are of the same order of magnitude, and the molecular effects of respective species on each other needs to be considered. This physics interface supports transport through Fickian diffusion, a mixture-average diffusion model, and as described by the Maxwell-Stefan equations. These take into account the diffusion properties of all species with respect to each other. Convective transport and transport by migration in electric fields are also accounted for by this physics interface. Also see the [Theory for the Transport of Concentrated Species Interface](#).

The [Transport of Concentrated Species in Porous Media Interface](#) () is tailored for the modeling of gas or liquids mixtures in porous media.

THE NERNST-PLANCK EQUATIONS INTERFACE

The [Nernst-Planck Equations Interface](#) () is used to compute concentrations in electrolytes subjected to electric fields. It defines the material balances and transport of ions and neutral species in combination with the electroneutrality condition. The physics interface can be used to model generic electrochemical cells with significant concentration gradients of the current-carrying species (ions). Also see the [Theory for the Nernst-Planck Equations Interface](#).

THE NERNST-PLANCK-POISSON EQUATIONS INTERFACE

The [Nernst-Planck-Poisson Equations Interface](#) () describes transport of electrolyte species without assuming local electroneutrality. This allows for simulating charge separation that typically arises close to an electrode surface, where ions in the electrolyte are attracted and repelled by unscreened excess charge on the electrode.

THE ELECTROPHORETIC TRANSPORT INTERFACE

The [Electrophoretic Transport Interface](#) () is used to compute the potential and concentration fields of a dilute solute in an aqueous solvent. The interface combines the Nernst-Planck equations, with electroneutrality and dissociation equilibria for weak acids, bases, and ampholytes, as well as the water auto-ionization reaction. It can be used to model various electrophoresis modes, such as zone electrophoresis, isotachophoresis, isoelectric focusing, and moving boundary electrophoresis. Also see the [Theory for the Electrophoretic Transport Interface](#)

THE REACTING FLOW INTERFACES

The **Chemical Species Transport > Reacting Flow** branch contains multiphysics interface that combines the equations for fluid flow with equations governing mass transport and reactions in fluid mixtures. Multiphysics interfaces for diluted solutions, as well as

for concentrated solutions are available. The physics interfaces support laminar flow and, depending on the licensed products, turbulent flow.

THE REACTING FLOW IN POROUS MEDIA INTERFACES

The **Chemical Species Transport > Reacting Flow in Porous Media** branch contains multiphysics interfaces that combine equations for fluid flow and chemical composition of a gas or liquid moving through the interstices of a porous medium. The branch also included dedicated interfaces for modeling porous catalysts and packed bed reactors. Apart from porous media, the system may also include regions with free flow.

THE REACTING NONISOTHERMAL FLOW INTERFACES

The **Chemical Species Transport > Nonisothermal Reacting Flow** branch contains multiphysics interface that combines the equations for fluid flow, heat transfer, and mass transport and reactions in fluid mixtures. These physics interfaces support laminar flow and, depending on the licensed products, turbulent flow, as well as flow through porous media.

THE DISPERSED TWO-PHASE LAMINAR FLOW WITH SPECIES TRANSPORT INTERFACE

The **Dispersed Two-Phase Laminar Flow with Species Transport Interface** () can be used to simulate species transport in a two-phase flow involving a liquid and a dispersed phase consisting of bubbles or droplets. Transport and reactions in both phases as well as transfer between them can be studied.

THE VAPOR-LIQUID EQUILIBRIUM INTERFACES

The Vapor-Liquid Equilibrium multiphysics interfaces can be used to model vapor transport in multicomponent mixtures in contact with liquids. The **Laminar Vapor Flow** () version includes features for evaporation and condensation at vapor-liquid interfaces. The **Laminar Two-Phase Flow** () version can be used to also track the motion of the interface resulting from the mass transfer.

THE PRECIPITATION AND CRYSTALLIZATION IN FLUID FLOW INTERFACE

The **Precipitation and Crystallization in Fluid Flow Interface** () can be used to model the size distribution of particles or that precipitate out of a fluid solution. It can be for example be used to study crystals in a liquid, gas bubbles in a liquid, or droplets in a gas. The **Size-Based Population Balance Interface** included solves for the discretized size distribution of particles, with support for growth and nucleation terms.

THE SURFACE REACTIONS INTERFACE

The [Surface Reactions Interface](#) () models reactions involving surface adsorbed species and species in the bulk of a reacting surface. The physics interface is typically active on a model boundary and is coupled to a mass transport physics interface active on a model domain. The Surface Reactions interface can be used together with other Chemical Species Transport interfaces. Predefined expressions for the growth velocity of the reacting surface makes it easy to set up models with moving boundaries. Also see the [Theory for the Surface Reactions Interface](#).

THE TRANSPORT OF DILUTED SPECIES IN FRACTURES INTERFACE

The [Transport of Diluted Species in Fractures Interface](#) () is used to model transport of a solute species along thin porous fractures, by accounting for convection, diffusion, dispersion, and chemical reactions. The fractures are defined by boundaries in 2D and 3D. The mass transport equation solved along the fractures is the tangential differential form of the convection–diffusion–reaction equation.

The next sections, [Coupling to Other Physics Interfaces](#) and [Adding a Chemical Species Transport Interface and Specifying the Number of Species](#) provide more information to help you start modeling.

Coupling to Other Physics Interfaces

When using this module you are often simulating applications that can be described by the mass transport interfaces in the Chemical Species Transport branch. Yet, as a lot of mass transport is dependent on convection, couplings to Fluid Flow interfaces are required. COMSOL Multiphysics and the Chemical Reaction Engineering Module include physics interfaces for laminar flow as well as porous media flow. More advanced descriptions of fluid flow, such as turbulent flow and multiphase flow, require the CFD Module.

Most chemical reactions or other type of material processing, such as casting, either require or produce heat, which in turn affects both the reaction and other physical processes connected to the system. Heat transfer through conduction and convection, when flow is laminar, as well as through porous media are supported in this module. More extensive description of heat transfer, such as in turbulent flow or involving radiation, require the Heat Transfer Module. The Heat Transfer Module also includes turbulent flow.

COMSOL Multiphysics easily supports simulations of electrostatics or DC-based physical phenomena, even if conductivity is nonlinear. If the electric field is AC/DC in

nature, or if the system is affected by electromagnetic waves, then the AC/DC Module and RF Module have appropriate physics interfaces for these phenomena. Some applications of electrochemical reactions, particularly in electrochemical power source applications, are better handled by the Battery Design Module.

Adding a Chemical Species Transport Interface and Specifying the Number of Species

A chemical species transport interface can be added when first creating a new model, or at any time during the modeling process.



Creating a New Model in the *COMSOL Multiphysics Reference Manual*

- 1 To add a physics interface:
 - Select **New** to open the **Model Wizard**, after selecting a space dimension, go to the **Select Physics** page.
 - From the **Home** toolbar click **Add Physics** (). Or right-click the **Component** node in the **Model Builder** and select **Add Physics**. Go to the **Add Physics** window.
- 2 Under **Chemical Species Transport**, navigate to the physics interface to add and double-click it.

There are other ways to add a physics interface depending on whether you are in the Model Builder or Add Physics window:

 - In the **Model Wizard**, click **Add** or right-click and select **Add Physics** (). The physics interface displays under **Added physics interfaces**.
 - In the **Add Physics** window, click **Add to Component** () or right-click and select **Add to Component**.
- 3 Specify the number of species (concentrations or mass fractions) and the names:
 - In the **Model Wizard**, on the **Review Physics Interface** page under **Dependent Variables**.
 - In the **Add Physics** window, click to expand the **Dependent Variables** section.
 - After adding the physics interface, you can also edit this information—click the node in the **Model Builder**, then, in the **Settings** window under **Dependent Variables**, specify the information.

- 4 Continue by adding more physics interfaces and specifying the number of species (concentrations or mass fractions) that are to be simulated in a mass transport interface when adding that interface.
- 5 In the **Dependent Variables** section, enter the **Number of species**. To add a single species, click the **Add Concentration** button () underneath the table or enter a value into the **Number of species** field. Click the **Remove Concentration** button () underneath the table if required.
- 6 The **Transport of Concentrated Species** interface and the **Nernst–Planck Equations** interface both need to contain at least two species (the default). Also edit the strings or names directly in the table. The names must be unique for all species (and all other dependent variables) in the model.
- 7 For the **Nernst–Planck Equations** interface, an electric potential variable is also required.



The **Reaction Engineering** interface can automatically set up the reacting species and their dependent variables. If you plan on using chemical reactions from a Reaction Engineering interface in a species transport interface, the *Generate Space Dependent Model* feature is recommended to set up the transport interface.

Available Result Templates

A number of results templates with predefined plots or evaluation groups are available when using the Chemical Species Transport interfaces. The results templates are available in the **Result Templates** window after computing the solution.

SPECIES PLOT ARRAYS

When solving a model with multiple species, one or more plot arrays are added to the **Result Templates** window. The plots show the concentrations of up to four species in each array plot, with surface plots for 2D or 3D.

MASS AND MOLE BALANCE EVALUATION GROUPS

When the **Mass transfer in porous media** checkbox is not selected, evaluation groups showing the mass conservation are added to the **Result Templates** window. The evaluation groups are not available when solving for multiple inner solutions, for example, when adding inner solutions with the **Auxiliary sweep** study extension. When the results are time-dependent, the accuracy of the mass or mole balance depends on

the output times. The simplest way to get results representative of a time-dependent simulation is to select **Steps taken by solver** from the **Times to store** list in the settings of the solver.

The Transport of Diluted Species Interface

The **Transport of Diluted Species (tds)** interface () , found under the **Chemical Species Transport** branch () , is used to calculate the concentration field of a dilute solute in a solvent. Transport and reactions of the species dissolved in a gas, liquid, or solid can be handled with this interface. The driving forces for transport can be diffusion by Fick's law, convection when coupled to a flow field, and migration, when coupled to an electric field.

The interface supports simulation of transport by convection and diffusion in 1D, 2D, and 3D as well as for axisymmetric components in 1D and 2D. The dependent variable is the molar concentration, c . Modeling multiple species transport is possible, whereby the physics interface solves for the molar concentration, c_i , of each species i .

SETTINGS

The **Label** is the default physics interface name.

The **Name** is used primarily as a scope prefix for variables defined by the physics interface. Refer to such physics interface variables in expressions using the pattern `<name>.<variable_name>`. In order to distinguish between variables belonging to different physics interfaces, the `name` string must be unique. Only letters, numbers, and underscores (`_`) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first physics interface in the model) is `tds`.

DOMAIN SELECTION

If any part of the model geometry should not partake in the mass transfer model, remove that part from the selection list.

OUT-OF-PLANE THICKNESS

For 2D components, the **Thickness** field (default value: 1 m) defines a parameter for the thickness of the geometry perpendicular to the two-dimensional cross-section. Both constant and varying thicknesses are supported. The value of this parameter is used, among other things, to automatically calculate molar flow rates from the total molar flux.

CROSS-SECTIONAL AREA

For 1D components, enter a **Cross-sectional area** A_c (SI unit: m^2) to define a parameter for the area of the geometry perpendicular to the 1D component. Both constant and varying areas are supported. The value of this parameter is used, among other things, to automatically calculate molar flow rates from the total molar flux. The default is 1 m^2 .

TRANSPORT MECHANISMS

Mass transport due to diffusion is always included. Use the checkboxes available under **Additional transport mechanisms** to control other transport mechanisms.

Note: Some of the additional transport mechanisms listed below are only available in certain products. For details see www.comsol.com/products/specifications/.

- By default, the **Convection** checkbox is selected. Clear the checkbox to disable convective transport.
- Select the **Migration in electric field** checkbox to activate transport of ionic species in an electric field. See further the theory section [Adding Transport Through Migration](#).



Transport of ionic species in an electric field is available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

Mass Transport in Porous Media

The **Mass transport in porous media** checkbox activates functionality specific to species transport in porous media:

- [Porous Medium](#)
- [Unsaturated Porous Medium](#)
- [Porous Electrode Coupling](#)
- [Packed Bed \(Moving Packed Bed\)](#)
- [Porous Catalyst](#)

- Volatilization
- Species Source



Note: Mass transport in porous media is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

CONSISTENT STABILIZATION

To display this section, click the **Show** button () and select **Stabilization**.

- When the **Crosswind diffusion** checkbox is selected, a weak term that reduces spurious oscillations is added to the transport equation. The resulting equation system is always nonlinear. There are two options for the **Crosswind diffusion type**:
 - **Do Carmo and Galeão** — the default option. This type of crosswind diffusion reduces undershoots and overshoots to a minimum but can in rare cases give equation systems that are difficult to fully converge.
 - **Codina** — this option is less diffusive compared to the Do Carmo and Galeão option but can result in more undershoots and overshoots. It is also less effective for anisotropic meshes. The Codina option activates a text field for the **Lower gradient limit** g_{lim} . Its defaults to $0.1 [\text{mol}/\text{m}^3] / \text{tds.helem}$, where tds.helem is the local element size.
- For both consistent stabilization methods, select an **Equation residual**. **Approximate residual** is the default and means that derivatives of the diffusion tensor components are neglected. This setting is usually accurate enough and is computationally faster. If required, select **Full residual** instead.

INCONSISTENT STABILIZATION

To display this section, click the **Show** button () and select **Stabilization**. By default, the **Isotropic diffusion** checkbox is not selected, because this type of stabilization adds artificial diffusion and affects the accuracy of the original problem. However, this option can be used to get a good initial guess for under resolved problems.

ADVANCED SETTINGS

To display this section, click the **Show** button () and select **Advanced Physics Options**. Normally these settings do not need to be changed. Select a **Material Balance Form** — **Nonconservative** (the default) or **Conservative**. The conservative formulation should be used for compressible flow. See [Material Balance Formulation](#) for more information.

DISCRETIZATION

To display all settings available in this section, click the **Show** button () and select **Advanced Physics Options**.

The **Compute boundary fluxes** checkbox is activated by default so that COMSOL Multiphysics computes predefined accurate boundary flux variables. When this option is selected, the solver computes variables storing accurate boundary fluxes from each boundary into the adjacent domain.

If the checkbox is cleared, the COMSOL Multiphysics software instead computes the flux variables from the dependent variables using extrapolation, which is less accurate in postprocessing results but does not create extra dependent variables on the boundaries for the fluxes.

The flux variables affected in the interface are:

- ndflux_c (where c is the dependent variable for the concentration). This is the normal diffusive flux and corresponds to the boundary flux when diffusion is the only contribution to the flux term.
- ntflux_c (where c is the dependent variable for the concentration). This is the normal total flux and corresponds to the boundary flux plus additional transport terms, for example, the convective flux when you use the nonconservative form.

Also the **Apply smoothing to boundary fluxes** checkbox is available if the previous checkbox is selected. The smoothing can provide a more well-behaved flux value close to singularities.

For details about the boundary fluxes settings, see [Computing Accurate Fluxes](#) in the *COMSOL Multiphysics Reference Manual*.

The **Value type when using splitting of complex variables** setting should in most pure mass transfer problems be set to **Real**, which is the default. It makes sure that the dependent variable does not get affected by small imaginary contributions, which can occur, for example, when combining a Time Dependent or Stationary study with a frequency-domain study. For more information, see [Splitting Complex-Valued Variables](#) in the *COMSOL Multiphysics Reference Manual*.

DEPENDENT VARIABLES

The dependent variable name is the **Concentration** c by default. The names must be unique with respect to all other dependent variables in the component.

Add or remove species variables in the model and also change the names of the dependent variables that represent the species concentrations.

Enter the **Number of species**. Use the **Add concentration** () and **Remove concentration** () buttons as needed.

FURTHER READING

	• The Transport of Diluted Species Interface • Numerical Stabilization in the <i>COMSOL Multiphysics Reference Manual</i>. • In the <i>COMSOL Multiphysics Reference Manual</i>, see Table 2-4 for links to common sections and Table 2-5 for common feature nodes. You can also search for information: press F1 to open the Help window or Ctrl+F1 to open the Documentation window.
	• <i>Effective Diffusivity in Porous Materials</i>: Application Library path COMSOL_Multiphysics/Diffusion/effective_diffusivity • <i>Micromixer</i>: Application Library path COMSOL_Multiphysics/Fluid_Dynamics/micromixer

The Transport of Diluted Species in Porous Media Interface

This interface (), found under the **Chemical Species Transport** branch (), is used to calculate the species concentration and transport in free and porous media. The interface is the same as the *Transport of Diluted Species* interface but it uses other defaults: The [Mass Transport in Porous Media](#) property is selected, and a [Porous Medium](#) node is added by default. The interface includes reaction rate expressions and solute sources for modeling of species transport and reaction in porous media.

This interface is dedicated to modeling transport in porous media, including immobile and mobile phases, where the chemical species may be subjected to diffusion, convection, migration, dispersion, adsorption, and volatilization in porous media. It supports cases where either the solid phase substrate is exclusively immobile, or when a gas-filling medium is also assumed to be immobile.

It applies to one or more diluted species or solutes that move primarily within a fluid that fills (saturated) or partially fills (unsaturated) the voids in a solid porous medium.

The pore space not filled with fluid contains an immobile gas phase. Models including a combination of porous media types can be studied.

The main feature nodes are the **Porous Medium** and **Unsaturated Porous Medium** nodes, which add the equations for the species concentrations and provide an interface for defining the properties of the porous media, as well as additional properties governing adsorption, volatilization, dispersion and diffusion, migration, and the velocity field to model convection.

The physics interface can be used for stationary and time-dependent analysis.

When this physics interface is added, these default nodes are also added to the **Model Builder** — **Porous Medium, No Flux** (the default boundary condition), and **Initial Values**. Then, from the **Physics** toolbar, add other nodes that implement, for example, boundary conditions, reaction rate expressions, and species sources. You can also right-click **Transport of Diluted Species in Porous Media** to select physics features from the context menu.

SETTINGS

The rest of the settings are the same as [The Transport of Diluted Species Interface](#).

FURTHER READING

	• Mass Balance Equation for Transport of Diluted Species in Porous Media • Domain, Boundary, and Pair Nodes for the Transport of Diluted Species Interface
	• <i>Variably Saturated Flow and Transport — Sorbing Solute:</i> Application Library path Subsurface_Flow_Module/Solute_Transport/sorbing_solute Web link: www.comsol.com/model/variably-saturated-flow-and-transport-sorbing-solute-490

The Transport of Diluted Species in Porous Catalysts Interface

This interface () , found under the **Reacting Flow in Porous Media** multiphysics node () , on the **Chemical Species Transport** branch () , is used to study the impact of adsorption, and heterogeneous reactions, as a gas or liquid containing solute species is

moving through the interstices of a porous medium. Adsorption and desorption of bulk species onto, or from, the fluid–solid interface can be studied, as well as heterogeneous reactions involving both bulk species and surface species bound to the fluid–solid interface. Volumetric reactions occurring in the pores can also be included. The interface is the same as the *Transport of Diluted Species* interface, but it has a [Porous Catalyst](#) node added by default. All other features for transport and reactions in porous media are also always available.

The main feature of the interface is the **Porous Catalyst** node which adds the governing equations for the concentrations of all solute species in the bulk, and the equations for surface species. It has two default subnodes, **Fluid** and **Porous Matrix**, that in turn are used to define the physical properties of each phase, and to prescribe the transport properties. The latter are typically the flow field, and when applicable, the electric field. The **Adsorption** and **Surface Reaction (Porous Catalyst)** subnodes can be added to implement adsorption/desorption and heterogeneous reactions occurring on the fluid–solid interface.

When this physics interface is added, the following default nodes are also added to the **Model Builder — Porous Catalyst, No Flux** (the default boundary condition), and **Initial Values**. Then, from the **Physics toolbar**, add other nodes that implement, for example, boundary conditions, reaction rate expressions, and species sources. You can also right-click **Transport of Diluted Species in Porous Catalysts** to select physics features from the context menu.

SETTINGS

The rest of the settings are the same as in [The Transport of Diluted Species Interface](#).

The Transport of Diluted Species in Packed Beds Interface

This interface () , found under the **Reacting Flow in Porous Media** multiphysics node () , on the **Chemical Species Transport** branch () , is used to model reactors filled with catalyst pellets. The interparticle macropores (void between pellets), and the intraparticle micropores (pores inside the pellets) result in a system with bimodal porosity. The **Transport of Diluted Species in Packed Beds** interface can be used to study interparticle and intraparticle mass transport, as well as reactions, in detail. Heterogeneous reactions and as volumetric reactions, both in the pore fluid inside the pellets and in the bulk fluid in the bed, can also be studied.

The pellets are available in each point of the bed domain, and are defined using an extra dimension describing the pellet radius. The interface is the same as the *Transport of*

Diluted Species interface, but it has a [Packed Bed \(Moving Packed Bed\)](#) node added by default. All other features for transport and reactions in porous media are also always available.

The main feature of the interface is the **Packed Bed** node which adds the governing equations for the concentrations of all bulk species, and the equations for surface species. It has two subnodes, **Fluid** and **Pellets**, that in turn are used to define the physical properties corresponding to each phase. The latter node is also used to define volumetric and heterogeneous intraparticle reactions. The **Packed Bed** node requires the use of a [Porous Material](#) (described in the *COMSOL Multiphysics Reference Manual*) for defining phase-specific properties, and for sharing them among different physics interfaces. A packed bed containing several pellet types can be modeled by adding multiple **Pellets** nodes to the **Porous Material**.

When this physics interface is added, the following default nodes are also added to the **Model Builder** — **Packed Bed**, **No Flux** (the default boundary condition), and **Initial Values**. From the **Physics** toolbar, add other nodes that implement, for example, boundary conditions, reaction rate expressions, and species sources. Optionally, right-click **Transport of Diluted Species in Packed Beds** to select physics features from the context menu.

SETTINGS

The rest of the settings are the same as [The Transport of Diluted Species Interface](#).

Domain, Boundary, and Pair Nodes for the Transport of Diluted Species Interface

[The Transport of Diluted Species Interface](#) has the following domain, boundary, point, and pair nodes, listed in alphabetical order, available from the **Physics** ribbon toolbar (Windows users), **Physics** context menu (Mac or Linux users), or by right-clicking to access the context menu (all users).



- To add a node, go to the **Physics** toolbar, no matter what operating system you are using.
- Contextual subnodes (attributes) are available by clicking the parent node in the Model Builder, and then selecting the subnode from the **Attributes** menu.

- Adsorption
- Adsorption (Porous Catalyst)
- Bed Inlet
- Bed Outlet
- Bed Wall
- Concentration
- Continuity on Interior Boundaries
- Diffusion (Transport of Diluted Species)
- Dispersion
- Electrode Surface Coupling
- Equilibrium Reaction
- Fast Irreversible Surface Reaction
- Fluid
- Fluid (Porous Medium)
- Fluid (Porous Catalyst)
- Flux
- Flux Discontinuity
- Fracture
- Gas
- Heterogeneous Reactions
- Inflow
- Initial Values
- Initial Values (Pellets)
- Line Mass Source
- Liquid
- Mass-Based Concentrations
- No Flux
- Open Boundary
- Out-of-Plane Flux
- Outflow
- Packed Bed (Moving Packed Bed)
- Partition Condition
- Pellet–Fluid Interface
- Pellets
- Periodic Condition
- Point Mass Source
- Porous Catalyst
- Porous Layer Reactions
- Porous Matrix
- Porous Matrix (Porous Catalyst)
- Porous Medium
- Porous Electrode Coupling
- Reaction Coefficients
- Reactions
- Reactions (Pellets)
- Shrinking Core Reactions
- Solid
- Species Properties
- Species Source
- Surface Reactions
- Surface Reaction (Porous Catalyst)
- Surface Equilibrium Reaction
- Symmetry
- Thin Diffusion Barrier
- Thin Impermeable Barrier
- Unsaturated Porous Medium
- Volatilization

Some features require certain add-on modules. For details see www.comsol.com/products/specifications/



For axisymmetric components, COMSOL Multiphysics takes the axial symmetry boundaries (at $r = 0$) into account and automatically adds an **Axial Symmetry** node that is valid on boundaries representing the symmetry axis.



In the *COMSOL Multiphysics Reference Manual*, see [Table 2-4](#) for links to common sections and [Table 2-5](#) for common feature nodes. You can also search for information: press F1 to open the **Help** window or Ctrl+F1 to open the **Documentation** window.

Prescribing Conditions on Fluid–Solid Interfaces

Defining conditions on interfaces between different phases, typically a wall in contact with a fluid, it is worthwhile to be careful with how variables in the conditions are evaluated. COMSOL Multiphysics supports that variables are defined differently on adjacent domains (or any other geometry dimension). Evaluating a variable with differing definitions, the mean value is returned on a boundary separating the different domains. The density, pressure, or temperature are examples of variables that may differ in a fluid and a solid. In order to specify on which side a variable is evaluated, the [up and down](#) operators (described in the *COMSOL Multiphysics Reference Manual*) can be used.

Species Properties

Use this node to define parameters specific to each of the species. It is available when the **Migration in electric field** checkbox is selected in the [Transport Mechanisms](#) section of the interface.

The node will change its label depending on what inputs are available in the node. If only **Charge** is available, it will be labeled “Species charges”.

CHARGE

Enter the charge number z_c (dimensionless, but requires a plus or minus sign) for each species.

ACTIVITY

This section is available when the **Species Activity** combo box is set to **Debye–Hückel** or **User defined**.

If **Debye–Hückel** is selected, enter the ion size number, a_0 , for each species. If **User defined** is selected, enter the activity coefficient, f , for each species.

Fluid

The settings in this node are dependent on the checkboxes selected under **Transport Mechanisms** in the **Settings** window for the Transport of Diluted Species interface. It includes only the sections required by the activated transport mechanisms. It has all the equations defining transport of diluted species as well as inputs for the material properties.

MODEL INPUTS

The temperature model input is always available. Select the source of the **Temperature**. For **User defined**, enter a value or expression for the temperature in the input field. This input option is always available.

You can also select the temperature solved for by a Heat Transfer interface added to the model component. These physics interfaces are available for selection in the **Temperature** list.

CONVECTION

If transport by convection is active, the velocity field of the solvent needs to be specified. Select the source of the **Velocity field**. For **User defined**, enter values or expressions for the velocity components in the input fields. This input option is always available.

You can also select the velocity field solved for by a Fluid Flow interface added to the model component. These physics interfaces are available for selection in the **Velocity field** list.

DIFFUSION

Use this section to specify diffusion coefficients describing the diffusion of each species in the solvent fluid.

Use the **Source** list to select diffusion coefficients defined in a material or a Chemistry interface.

Select **Material** to use a diffusion coefficient in a material available in the model.

User-defined property groups including a diffusion coefficient output property are available for selection in the **Fluid diffusion coefficient** list.

Select **Chemistry** to use a diffusion coefficient from a Chemistry interface added to model component. Diffusion coefficients are automatically defined when **Calculate transport properties** is selected on the interface level of a Chemistry interface. All defined diffusion coefficients are available for selection in the **Fluid diffusion coefficient** list.

For **User defined**, enter a value or expression for the **Fluid diffusion coefficient** D_c of each species in the corresponding input field. This can be a scalar value for isotropic diffusion or a tensor describing anisotropic diffusion. Select the appropriate tensor type — **Isotropic**, **Diagonal**, **Symmetric**, or **Full** that describes the diffusion transport, and then enter the values in the corresponding element (one value for each species).

Note that multiple species, as well as Migration in Electric fields (described below) is only available for certain COMSOL Multiphysics add-on products. See details: www.comsol.com/products/specifications/.

MIGRATION IN ELECTRIC FIELD

This section is available when the **Migration in electric field** checkbox is selected. From the **Electric potential** list, select the source of the electric field.

- Enter a value or expression for the **Electric potential** V , which is **User defined**; this input option is always available.
- Select the electric potential solved by an AC/DC-based interface that has also been added to the model.
- Select the electric potential defined or solved by Electrochemistry interface that has been added to the component.

By default the **Mobility** is set to be calculated based on the species diffusivity and the temperature using the **Nernst–Einstein relation**. For **User defined**, and under **Mobility**, select the appropriate scalar or tensor type — **Isotropic**, **Diagonal**, **Symmetric**, or **Full** — and type in the value of expression of the mobility $u_{m,c}$.

Enter the **Charge number** z_c (dimensionless, but requires a plus or minus sign) for each species.

The temperature (if you are using mobilities based on the Nernst–Einstein relation) is taken from **Model Inputs** section.

Note that the migration in electric fields feature is only available in some COMSOL products. See details: www.comsol.com/products/specifications/.

EXAMPLE MODELS

	• <i>Separation Through Dialysis</i>: Application Library path Chemical_Reaction_Engineering_Module/Mixing_and_Separation/dialysis Web link: www.comsol.com/model/separation-through-dialysis-258
	• <i>Transport in an Electrokinetic Valve</i>: Application Library path Microfluidics_Module/Fluid_Flow/electrokinetic_valve Web link: www.comsol.com/model/transport-in-an-electrokinetic-valve-603

Solid

Use this node to describe diffusion in a solid phase domain. In this feature, the activities of the species are assumed to be ideal.

DIFFUSION

Specify the diffusion coefficients describing the diffusion of each species in the solid.

Enter a value or expression for the **Diffusion coefficient** D_c of each species in the corresponding input field. This can be a scalar value for isotropic diffusion or a tensor describing anisotropic diffusion. Select the appropriate tensor type — **Isotropic**, **Diagonal**, or **Symmetric** that describes the diffusion transport, and then enter the values in the corresponding element (one value for each species).

ADVANCED

This section is only available with the **Advanced Physics Options** selected and when the model contains a split between the material frame and the spatial frame. A split of these frames occurs when using a **Moving Mesh** node or a Structural Mechanics interface like **Solid Mechanics**. In such a model, the reference geometry (the material frame) differs from the deformed geometry (the spatial frame). The diffusive transport of species is solved for on the material frame, but the equation is adjusted for geometry deformations so that species conservation is achieved on the spatial frame. Use the **Diffusion model** to define the frame of reference for the diffusion tensor.

Select **Reference geometry based (Lagrangian)** to specify the diffusion tensor in the undeformed geometry, implying that the diffusion is defined in the material frame (indicated by coordinates X, Y, Z). This is the preferred setting for an anisotropic

diffusion in combination with solid body translation or rotation. Also in general when the diffusion tensor components have been measured in an undeformed geometry.

Select **Deformed geometry based (Eulerian)** to specify the diffusion tensor in the deformed geometry. This means that the diffusion is defined in the spatial frame (indicated by coordinates x, y, z). Use this setting when the diffusion tensor components have been measured in a deformed geometry.

The **Advanced Physics Options** is found in the **Show More Options** dialog, accessed by clicking the **Show More Options** button ().

Note that multiple species are only available for certain COMSOL Multiphysics add-on products. See details: www.comsol.com/products/specifications/.

Initial Values

The **Initial Values** node specifies the initial values for the concentration of each species. These serve as an initial guess for a stationary solver or as initial conditions for a transient simulation.

DOMAIN SELECTION

If there are several types of domains with different initial values defined, it might be necessary to remove some domains from the selection. These are then defined in an additional **Initial Values** node.

INITIAL VALUES

Enter a value or expression for the initial value of the **Concentration** or concentrations, c_i . This also serves as a starting guess for stationary problems.

Mass-Based Concentrations

Use the **Mass-Based Concentrations** node to add postprocessing variables for mass-based concentrations (SI unit: kg/m^3) and mass fractions (dimensionless) for all species.

MIXTURE PROPERTIES

The default **Solvent density** ρ_{solvent} is taken **From material**. For **User defined**, enter a value or expression manually. Define the **Molar mass** of each species, which is needed to calculate the mass-based concentration.

Use the **Reactions** node to account for the consumption or production of species through chemical reactions. Define the rate expressions as required.

DOMAIN SELECTION

From the **Selection** list, choose the domains on which to define rate expression or expressions that govern the source term in the transport equations.

Several reaction nodes can be used to account for different reactions in different parts for the modeling geometry.

REACTION RATES

The **Chemistry** list shows all the Chemistry interfaces that are coupled to the current mass transport interface. The coupling conditions for the Chemistry interfaces are:

- Active
- **Define variables for porous pellets** (under the **Pellet Chemistry** section) is cleared
- **Species solved for** (under the **Species Matching** section) set to the current mass transport interface

The reaction rates can be taken from a coupled Chemistry interface automatically or by specifying a rate expression R_i (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$) for species i :

- With a **Chemistry** interface selected, the species mass sources will be set to the corresponding reaction rates in the **Chemistry** according to the **Species Matching** table. In the case of partial coupling, the mass sources for the uncoupled species will be set to 0.
- **User defined.** Enter a value or expression in the field. Note that if you have the *Chemistry* interface available, provided with the *Chemical Reaction Engineering Module*, the reaction rate expressions can be generated automatically and picked up from the lists. For an example, see the application *Fine Chemical Production in a Plate Reactor* as linked below.

REACTING VOLUME

This section is only available when the [Mass Transport in Porous Media](#) property is available and selected. See www.comsol.com/products/specifications/ for more details on availability.

When specifying reaction rates for a species in porous media, the specified reaction rate may have the basis of the total volume, the pore volume, or in the case of using the

volume of a particular phase (applicable when modeling an unsaturated porous medium).

- For **Total volume**, the reaction expressions, in $\text{mol}/(\text{m}^3 \cdot \text{s})$, are specified per unit volume of the model domain (multiplied by unity).
- For **Pore volume**, the reaction expressions, in $\text{mol}/(\text{m}^3 \cdot \text{s})$, are specified per unit volume of total pore space. The reaction expressions will be multiplied by the domain porosity, ϵ_p . (ϵ_p equals unity for nonporous domains).
- For **Liquid phase**, the reaction expressions, in $\text{mol}/(\text{m}^3 \cdot \text{s})$, are specified per unit volume of liquid in the pore space. The expressions will be multiplied by the liquid volume fraction θ_l . (θ_l equals the porosity (ϵ_p) on Porous Medium domains).
- For **Gas phase**, the reaction expressions, in $\text{mol}/(\text{m}^3 \cdot \text{s})$, are multiplied by the gas volume fraction $\theta_g = \epsilon_p - \theta_l$. θ_g equals 0 for Porous Medium domains.

HEAT SOURCE

The heat source will be defined as reaction heat from whole Chemistry if the **Chemistry** list is set to a coupled chemistry interface, otherwise it is defined as the specified value (SI unit: W/m^3).

FURTHER READING

See the theory chapter on chemical species transport, starting with the section *Mass Balance Equation*.



- *Fine Chemical Production in a Plate Reactor*: Application Library path **Chemical_Reaction_Engineering_Module/Reactors_with_Mass_and_Heat_Transfer/plate_reactor**
Web link: www.comsol.com/model/fine-chemical-production-in-a-plate-reactor-8589

No Flux

This node is the default boundary condition on exterior boundaries. It should be used on boundaries across which there is no mass flux, typically solid walls where no surface reactions occur. The condition applied for each species corresponds to

$$-\mathbf{n} \cdot (-D \nabla c) = 0$$

where $\mathbf{n}$ denotes the outward pointing normal of the boundary. When the mass transport includes migration of ionic species, the no flux condition is:

$$-\mathbf{n} \cdot (-D\nabla c - zu_{\text{m}}Fc\nabla\phi) = 0$$

CONVECTION

By default, the feature prescribes a vanishing flux due to diffusion and migration in an electric field. This is the appropriate no flux condition when the relative convective velocity at the boundary is zero. When the fluid velocity at the boundary is not equal to that of the boundary, it is often convenient to prescribe the total flux including the convection. To do this select **Include** in the **Convection** section.

When including the convection, the no flux condition prescribed is

$$-\mathbf{n} \cdot (-D\nabla c + \mathbf{u}c) = 0$$

or

$$-\mathbf{n} \cdot (-D\nabla c - zu_{\text{m}}Fc\nabla\phi + \mathbf{u}c) = 0$$

when migration of ionic species is included.

Inflow

Use this node to specify all species concentrations at an inlet boundary.

If you want to specify the concentration of a subset of the partaking species, this can be done by using the [Concentration](#) node instead.

For the **Electroanalysis** interface, this node is available when you select the **Convection** checkbox on the physics interface **Settings** window.

CONCENTRATION

For the concentration of each species $c_{0,e}$ (SI unit: mol/m^3), enter a value or expression.

BOUNDARY CONDITION TYPE

The option **Concentration constraint** constrains the concentration values on the boundary by the use of pointwise constraints.

The other option, **Flux (Danckwerts)** can be used when the concentration at the boundary is not known, or when it varies in a nontrivial manner. This may, for example, be useful when reactions with high reaction rates occur in the vicinity of the inlet. In this case the concentration far upstream of the boundary is instead prescribed. The **Flux (Danckwerts)** condition prescribes the total flux defined by the upstream concentration and the fluid velocity at the boundary.

CONSTRAINT SETTINGS

To display this section, click the **Show More Options** button () and select **Advanced Physics Options**. You can find details about the different constraint settings in the section [Constraint Reaction Terms](#) in the *COMSOL Multiphysics Reference Manual*.

FURTHER READING

See the theory chapter in the section [Danckwerts Inflow Boundary Condition](#).

Outflow

Apply this condition at outlets boundaries where species should be transported out of the model domain by fluid motion or by an electric field (in the case of ions). It is assumed that convection and migration (in an electric field) are the dominating transport mechanisms across the boundary, and therefore that the diffusive transport can be ignored, that is:

$$\mathbf{n} \cdot (-D\nabla c) = 0$$

Note that the **Convection** or the **Migration in electric field** transport mechanisms needs to be included for this node to be available.

Concentration

This condition node adds a boundary condition for the species concentration. For example, a $c = c_0$ condition specifies the concentration of species c .

CONCENTRATION

Individually specify the concentration for each species. Select the checkbox for the **Species** to specify the concentration, and then enter a value or expression in the

corresponding field. To use another boundary condition for a specific species, click to clear the checkbox for the concentration of that species.

When defining conditions on boundaries between different phases, see the section [Prescribing Conditions on Fluid–Solid Interfaces](#).

CONSTRAINT SETTINGS

To display this section, click the **Show** button () and select **Advanced Physics Options**. You can find details about the different constraint settings in the section [Constraint Reaction Terms](#) in the *COMSOL Multiphysics Reference Manual*.

Flux

This node can be used to specify the species molar flux across a boundary. The flux can for example occur due to chemical reactions or a phase change at the boundary. The flux can also represent the transport to or from a surrounding environment currently not included model.

The prescribed flux of a species c is by default defined as

$$-\mathbf{n} \cdot (-D\nabla c) = J_0$$

where $\mathbf{n}$ denotes the outward pointing normal of the boundary. When the mass transport includes migration of ionic species, the flux is defined as

$$-\mathbf{n} \cdot (-D\nabla c - zu_m Fc\nabla\phi) = J_0$$

The flux prescribed, J_0 , can include any arbitrary user-specified expression. It can be constant or a function of a dependent variable or independent variable. Common examples are a flux dependent of the concentration, temperature, pressure, or the electric potential ϕ .

When defining conditions on boundaries between different phases, see the section [Prescribing Conditions on Fluid–Solid Interfaces](#).

CONVECTION

By default, the flux due to diffusion and migration in an electric field is prescribed. This is the appropriate flux condition when the relative velocity at the boundary is zero. When the fluid velocity is not equal to that of the boundary, it is often convenient to prescribe the total flux, including the convection. To do this select **Include** in the **Convection** section.

When including the convection, the prescribed flux is defined as

$$-\mathbf{n} \cdot (-D\nabla c + \mathbf{u}c) = J_0$$

or

$$-\mathbf{n} \cdot (-D\nabla c - zu_m Fc \nabla \phi + \mathbf{u}c) = J_0$$

when migration of ionic species is included.

INWARD FLUX

Select the Species checkbox for the species for which to specify the flux, and enter a value or expression for the inward flux in the corresponding field. Use a minus sign when specifying a flux directed out of the system. To use another boundary condition for a specific species, click to clear the checkbox for that species.

External convection

Set **Flux type** to **External convection** to prescribe a flux to or from an exterior domain (not modeled) assumed to include convection. The exterior can for example include a forced convection to control the temperature or to increase the mass transport. In this case the prescribed mass flux corresponds to

$$J_0 = k_c(c_b - c)$$

where k_c is a mass transfer coefficient and c_b is the bulk concentration, the typical concentration far into the surrounding exterior domain.

Symmetry

The **Symmetry** node can be used to represent boundaries where the species concentration is symmetric, that is, where there is no mass flux across the boundary.

This boundary condition is identical to that of the [No Flux](#) node.

Flux Discontinuity

This node represents a discontinuity in the mass flux across an interior boundary:

$$-\mathbf{n} \cdot [(\mathbf{J} + \mathbf{u}c)_u - (\mathbf{J} + \mathbf{u}c)_d] = N_0 \quad \mathbf{J} = -D\nabla c - zu_m Fc \nabla \phi$$

where the value N_0 (SI unit: $\text{mol}/(\text{m}^2\cdot\text{s})$) specifies the jump in total flux at the boundary. This can be used to model a boundary source, for example a surface reaction, adsorption, or desorption.

FLUX DISCONTINUITY

In this section the jump in species flux (or surface source) is specified.

Select the Species checkbox for the species to specify and enter a value or expression for the material flux jump in the corresponding field. To use a different boundary condition for a specific species, click to clear the checkbox for the flux discontinuity of that species.

Partition Condition

The **Partition Condition** node can be used to prescribe the relation between the concentration of a solute species in two adjoining immiscible phases. It can for example be used on interior boundaries separating two liquid phases, a gas-liquid interface, or on a boundary separating a liquid phase and a solid or porous media.

Select **Partition coefficient** from the **Relation** list to prescribe a linear relationship between the concentration on either side of a boundary. For a species concentration c_i , the ratio between the concentration on the upside and on the downside of the boundary ($c_{i,u}$ and $c_{i,d}$ respectively) is then defined in terms of a partition coefficient K_i in the manner of

$$K_i = \frac{c_{i,u}}{c_{i,d}}$$

Use the associated input field to prescribe the partition coefficient for each species.

Select **User defined** from the **Relation** list to set up a nonlinear relation between the up and downside concentrations. For such relations $f(c_i)_u = f(c_i)_d$, where $f(c_i)_u$ is the expression enforced on the upside. Use the associated input fields to enter the upside and downside expressions for each species.

The up and downside of the selected boundary is indicated with a red arrow in the Graphics window. The arrow points from the downside into the upside. Select the **Reverse direction** checkbox to reverse the direction of the arrow on the selected

boundary, and update the definition of the up and downside concentrations accordingly.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

FURTHER READING

For an example of using a partition condition, see this application example:



Separation Through Dialysis: Application Library path
Chemical_Reaction_Engineering_Module/Mixing_and_Separation/dialysis

Periodic Condition

The **Periodic Condition** node can be used to define periodicity for the mass transport between two sets of boundaries. The node prescribes continuity in the concentration and the mass flux between the “source” and the “destination” side, respectively. Note that these names are arbitrary and does not influence the direction in which mass is transported. It is dictated by mass transfer equations in the adjacent domains.

BOUNDARY SELECTION

The node can be activated on more than two boundaries, in which case the feature tries to identify two separate surfaces that each consist of one or several connected boundaries.

DESTINATION SELECTION

For more complex geometries, it might be necessary to specify the destination selection manually. To do so, right-click the **Periodic Condition** node and choose **Manual Destination Selection**. You can then specify the boundaries that constitute the destination surfaces in the **Destination Selection** section.

CONSTRAINT SETTINGS

To display this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog. This section contains settings for specifying the type of constraint and whether to use a pointwise or weak constraint. See

[Constraint Settings](#) in the *COMSOL Multiphysics Reference Manual* for more information.

ORIENTATION OF SOURCE

To display this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog. For information about the **Orientation of Source** section, see [Orientation of Source and Destination](#) in the *COMSOL Multiphysics Reference Manual*.

ORIENTATION OF DESTINATION

This section appears if the setting for **Transform to intermediate map** in the **Orientation of Source** section is changed from the default value, **Automatic**, and **Advanced Physics Options** is selected in the **Show More Options** dialog. For information about the **Orientation of Destination** section, see [Orientation of Source and Destination](#) in the *COMSOL Multiphysics Reference Manual*.

MAPPING BETWEEN SOURCE AND DESTINATION

To display this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog. For information about the **Mapping Between Source and Destination** section, see [Mapping Between Source and Destination](#) in the *COMSOL Multiphysics Reference Manual*.

FURTHER READING

For an example of using a periodic condition, see this application example:



The KdV Equation and Solitons: Application Library path
COMSOL_Multiphysics/Equation_Based/kdv_equation

Line Mass Source

The **Line Mass Source** feature models mass flow originating from a tube or line region with an infinitely small radius.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

SELECTION

The **Line Mass Source** feature is available for all dimensions, but the applicable selection differs between the dimensions.

MODEL DIMENSION	APPLICABLE GEOMETRICAL ENTITY
2D	Points
2D Axisymmetry	Points not on the symmetry axis and the symmetry axis
3D	Edges

SPECIES SOURCE

Enter the source strength, $\dot{q}_{l,c}$, for each species (SI unit: mol/(m·s)). A positive value results in species injection from the line into the computational domain, and a negative value means that the species is removed from the computational domain.

Line sources located on a boundary affect the adjacent computational domains. This effect makes the physical strength of a line source located in a symmetry plane twice the given strength.

FURTHER READING

See the section [Mass Sources for Species Transport](#).

Point Mass Source

The **Point Mass Source** feature models mass flow originating from an infinitely small domain around a point.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

SPECIES SOURCE

Enter the source strength, $\dot{q}_{p,c}$, for each species (SI unit: mol/s). A positive value results in species injection from the point into the computational domain, and a negative value means that the species is removed from the computational domain.

Point sources located on a boundary or on an edge affect the adjacent computational domains. This has the effect, for example, that the physical strength of a point source located in a symmetry plane is twice the given strength.

FURTHER READING

See the section [Mass Sources for Species Transport](#).

Open Boundary

Use this node to set up mass transport across boundaries where both convective inflow and outflow can occur. On the parts of the boundary where fluid flows into the domain, an exterior species concentration is prescribed. On the remaining parts, where fluid flows out of the domain, a condition equivalent to the **Outflow** node is instead prescribed.

The direction of the flow across the boundary is typically calculated by a fluid flow interface and is provided as a model input to the Transport of Diluted Species interface.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

EXTERIOR CONCENTRATION

Enter a value or expression for the **Exterior concentration**.

Thin Diffusion Barrier

Use this boundary condition to model a thin layer through which mass is transported by diffusion only. The node is applicable on interior boundaries and can be used to avoid meshing thin structures.

THIN DIFFUSION BARRIER

Specify the **Layer thickness**, d_s , and input a **Diffusion coefficient**, $D_{s,c}$, for each of the species included.

Thin Impermeable Barrier

This feature models a thin mass transfer barrier. It is available on interior boundaries and introduces a discontinuity in the concentration across the boundary. On each side, a no-flux condition is prescribed for the mass transport implying that it acts as a barrier. The feature can be used to avoid meshing thin structures.

Solving a model involving coupled fluid flow and mass transfer, the Thin Impermeable Barrier feature can be combined with an Interior Wall feature in order to model a thin solid wall.

Equilibrium Reaction

Use this node to model a reaction where the kinetics is so fast that the equilibrium condition is fulfilled at all times. The node solves for an additional degree of freedom (the reaction rate R_{eq}) to fulfill the equilibrium condition at all times in all space coordinates.

If the **Apply equilibrium condition on inflow boundaries** checkbox is selected, the specified inflow concentration values in all active **Inflow** boundary nodes for the physics interface are modified to comply with the equilibrium condition.



- A necessary requirement for this feature to be available is that two or more species are solved for by the interface.
- This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

EQUILIBRIUM CONDITION

The **Chemistry** list shows two ways to define an equilibrium reaction.

- A coupled chemistry — The **Equilibrium reaction formula** is available when the **Chemistry** list is set to a coupled chemistry interface. All the equilibrium reaction formulas in the coupled chemistry will appear in the formula list. Select one of them, then all coefficients (stoichiometric coefficients, equilibrium constant) will be taken from the coupled reaction automatically. There would be no any source term contributions if the **Equilibrium reaction formula** list is set to **None**.
- **None** — If **None** is selected from the **Chemistry** list, an equilibrium reaction will be defined by settings from fields of **Equilibrium definition**, **Equilibrium constant**, and **Unit activity concentration**.

The default value for **Equilibrium definition** list is **Equilibrium constant** or select **User defined**. For either option, the **Apply equilibrium condition on inflow boundaries** checkbox is selected by default.

For **Equilibrium constant**, enter an **Equilibrium constant** K_{eq} (dimensionless). Also enter a value or expression for the **Unit activity concentration** C_{a0} (SI unit: mol/m³).

Selecting **Equilibrium constant** defines an equilibrium condition based on the stoichiometric coefficients, the species activities, and the law of mass action.

For **User defined**, enter an **Equilibrium expression** E_{eq} (dimensionless).

STOICHIOMETRIC COEFFICIENTS

This section is available when the **Chemistry** list is set to **None**.

Enter a value for the stoichiometric coefficient ν_c (dimensionless). The default is 0. Use negative values for reactants and positive values for products in the modeled reaction.

Species with a stoichiometric coefficient value of 0 are not affected by the **Equilibrium Reaction** node.

HEAT SOURCE

The heat source will be defined from reaction heat from this equilibrium reaction.

- An equilibrium reaction defined from a **Chemistry** (from the **Equilibrium reaction formula** list). The heat source is defined as the same as that of equilibrium reaction in Chemistry.
- **None** selected in the **Chemistry** list. Enter a value or expression for the heat source (SI unit: W/m³).

FURTHER READING

See

- [Equilibrium Reaction Theory](#)

Surface Reactions

The **Surface Reactions** node can be used to account for the species boundary flux due to chemical reactions occurring on a surface (heterogeneous reactions). For a domain species participating in a surface reaction, the boundary flux corresponds to the reaction rate at the surface.

SURFACE REACTION RATE

Specify the surface reaction rate J_0 of each species resulting from the reactions. Note that if you have the Chemistry interface available, provided with the Chemical

Reaction Engineering Module, the reaction rate expressions can be automatically generated and picked up from the lists.

When defining conditions on boundaries between different phases, see the section [Prescribing Conditions on Fluid–Solid Interfaces](#).

FURTHER READING

For an example of using the Surface Reactions node, see this application example:



Chemical Vapor Deposition of GaAs: Application Library path
**Chemical_Reaction_Engineering_Module/
Reactors_with_Mass_and_Heat_Transfer/gaas_cvd**

Surface Equilibrium Reaction

Use this node to model an equilibrium reaction on a boundary (surface). The settings for this node are similar to [Equilibrium Reaction](#). Note that a necessary requirement for this feature to be available is that two or more species are solved for by the interface.

When defining conditions on boundaries between different phases, see the section [Prescribing Conditions on Fluid–Solid Interfaces](#).



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

Fast Irreversible Surface Reaction

This boundary node defines an irreversible reaction where the kinetics is so fast that the only factor limiting the reaction rate is the transport of a species to the reacting surface.

The node will set the **Rate limiting species concentration to zero** at the boundary, and balance the fluxes of the species participating in the reaction and the current densities according to the Stoichiometric Coefficients settings.

When defining conditions on boundaries between different phases, see the section [Prescribing Conditions on Fluid–Solid Interfaces](#).



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

Porous Electrode Coupling

Use this node to add a molar source in a domain that is coupled to one or multiple [Porous Electrode Reaction](#) nodes of an Electrochemistry Interface.

The molar source is calculated from the number of electrons, stoichiometric coefficients, and volumetric current densities of the coupled porous electrode reactions specified in the **Reaction Coefficients** subnodes.

In the Transport of Concentrated Species interface, the molar sources (or sinks) are multiplied by the species molar masses to obtain the corresponding mass sources.

Additional [Reaction Coefficients](#) subnodes are available from the context menu (right-click the parent node) as well as from the **Physics** toolbar, **Attributes** menu.

Note that if you are also modeling the momentum transport and expect a nonnegligible total mass source or sink, which is often the case in gas diffusion electrodes, you need to also add a corresponding mass source/sink in the Fluid Flow interface. (A **Reacting Flow** multiphysics node adds the associated source/sink term in a Fluid Flow interface automatically.)



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

Reaction Coefficients

Add this node to the [Electrode Surface Coupling](#) and [Porous Electrode Coupling](#) features to define molar fluxes and sources based on electrode current densities in an **Electrochemistry** interface.

The molar flux or source is proportional to the stoichiometric coefficients and the current density according to Faraday's law.

Current densities from **Electrode Reaction** (i_{loc} , SI unit: A/m²) or **Porous Electrode Reaction** nodes (i_v , SI unit: A/m³) of any **Electrochemistry** interface in the model are available for selection as the **Coupled reaction**, and user-defined expressions are also supported.

Enter the **Number of participating electrons** n_m (dimensionless) and the **Stoichiometric coefficient** v_e (dimensionless) as explained in the theory section linked below.

Use multiple subnodes to couple to multiple reactions.

Electrode Surface Coupling

Use this node to define a flux boundary condition based on current densities of one or multiple **Electrode Reaction** nodes in an **Electrochemistry** interface.

The flux is proportional to the current densities and the stoichiometric coefficients according to Faraday's law as defined by summation over the **Reaction Coefficients** subnodes.

Note that if you are also modeling the momentum transport and expect a nonnegligible total mass flux over the boundary, which is often the case for gas diffusion electrodes, you need to also add a corresponding **Electrode Surface Coupling** node in the **Fluid Flow** interface.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

Porous Medium

Use this node to model the concentration of diluted species transported by a solvent (mobile fluid phase) through interstices in a solid porous medium. In addition to transport due to convection and diffusion, the node contains functionality to include species evolution through adsorption and dispersion.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

Species transport through a porous medium is affected both by properties of the fluid phase, and as well as properties of the solid matrix. These properties are defined in the [Fluid \(Porous Medium\)](#) and [Porous Matrix](#) subnodes respectively. The Porous Medium node supports material properties using a [Porous Material](#) node (described in the *COMSOL Multiphysics Reference Manual*), where properties are defined per phase in a similar manner.

Add an [Adsorption](#) or a [Dispersion](#) subfeature to the **Porous Medium** to account for the corresponding transport mechanism.

Fluid (Porous Medium)

Use this node to specify the mass transfer in the mobile fluid solvent filling the pores of a porous medium. It is used as a subnode to [Porous Medium](#), [Porous Catalyst](#), and [Packed Bed \(Moving Packed Bed\)](#).

MODEL INPUTS

The temperature model input is always available. By default, the **Temperature** model input is set to **Common model input**, and the temperature is controlled from [Default Model Inputs](#) under **Global Definitions** or by a locally defined [Model Input](#). If a Heat Transfer interface is included in the component, it controls the temperature **Common model input**. Alternatively, the temperature field can be selected from another physics interface. All physics interfaces have their own tags (**Name**). For example, if a Heat Transfer in Fluids interface is included in the component, the **Temperature (ht)** option is available for T .

You can also select **User defined** from the **Temperature** model input in order to manually prescribe T .

CONVECTION

If transport by convection is active, the velocity field of the solvent needs to be specified. For **User defined**, enter values or expressions for the velocity components in the input fields. This input option is always available. You can also select the velocity field solved for by a Fluid Flow interface added to the model component. These physics interfaces are available for selection in the **Velocity field** list.

DIFFUSION

Use this section to specify diffusion coefficients describing the diffusion of each species in the solvent fluid.

Use the **Source** list to select to pick up diffusion coefficients defined in a material or a Chemistry interface.

Select **Material**, and a **Fluid material** to use a diffusion coefficient in a material available in the model. User-defined property groups including a diffusion coefficient output property are available for selection in the **Fluid diffusion coefficient** list.

Select **Chemistry** to use a diffusion coefficient from a Chemistry interface added to model component. Diffusion coefficients are automatically defined when **Calculate transport properties** is selected on the interface level of a Chemistry interface. All defined diffusion coefficients are available for selection in the **Fluid diffusion coefficient** list.

For **User defined**, enter a value or expression for the **Fluid diffusion coefficient** $D_{F,i}$ of each species in the corresponding input field.

In a porous medium the diffusivity is reduced due to the fact that the solid grains impede Brownian motion. Select an **Effective diffusivity model** to account for the reduced diffusivity. The available models are **Millington and Quirk model** (the default), **Bruggeman model**, **Tortuosity model**, or **No correction**. For **Tortuosity model**, enter a value for the tortuosity $\tau_{F,i}$ (dimensionless).

Note that multiple species, as well as Migration in Electric fields (described below) is only available for certain COMSOL Multiphysics add-on products. For details, see www.comsol.com/products/specifications/.

MIGRATION IN ELECTRIC FIELD

This section is available when the **Migration in electric field** checkbox is selected. From the **Electric potential** list, select the source of the electric field.

- For **User defined**, enter a value or expression for the **Electric potential** V . This input option is always available.
- Select the electric potential solved by an AC/DC-based interface that has added to the component.
- Select the electric potential defined or solved for by an Electrochemistry interface added to the component.

By default the **Mobility** is set to be calculated based on the species effective diffusivity and the temperature using the **Nernst–Einstein relation**. For **User defined**, select the appropriate scalar or tensor type — **Isotropic**, **Diagonal**, **Symmetric**, or **Full** — and type in the value or expression of the effective mobility $u_{me,i}$.

Enter the **Charge number** z_c for each species.

Porous Matrix

This node sets the porosity when modeling transport of diluted species in a [Porous Medium](#), an [Unsaturated Porous Medium](#), or a [Porous Catalyst](#). The porosity corresponds to the volume fraction occupied by the species containing fluid.

- By default the porosity of the solid matrix, ϵ_p , is taken **From material**. The **Porous Matrix** node supports the use of a **Porous Material** added to the Materials branch. In this case the porosity is defined in the manner of

$$\epsilon_p = 1 - \sum_i \theta_{s,i} - \sum_i \theta_{\text{imf},i}$$

where $\theta_{s,i}$ and $\theta_{\text{imf},i}$ are the volume fractions given in **Solid** and **Immobile Fluids** subnodes under the **Porous Material** node.

- Select **From pellet bed densities** to compute the porosity using the (dry bulk) **Bed density** ρ_b and the (single phase) **Pellet density** ρ_{pe} . The porosity is then defined from

$$\epsilon_p = 1 - \frac{\rho_b}{\rho_{pe}}$$

- For **User defined**, enter a value or expression for the porosity.

Dispersion

Local variations in the velocity as the fluid flows around solid particles lead to mechanical mixing, referred to as dispersion. Use this feature to account for dispersion in a [Porous Medium](#) or a [Unsaturated Porous Medium](#).

This subfeature is available when both the **Mass transfer in porous media mass transfer** checkbox and the **Convection** checkbox are selected in the **Settings** window for the physics interface.

Select the **Specify dispersion for each species individually** checkbox to specify the dispersion tensor D_D (SI unit: m^2/s) for each species separately. When not selected the same dispersion tensor D_D is used for all species.

Select an option from the **Dispersion tensor** list — **Dispersivity** or **User defined**.

Select **Dispersivity** to specify the dispersion in terms of dispersivities (SI unit: m). Select an option from the **Dispersivity model** list: **Isotropic** (the default) or **Transverse isotropic**

based on the properties of the porous media. For isotropic porous media, specify the longitudinal and transverse dispersivities. For transverse isotropic porous media, specify the longitudinal, horizontal transverse, and vertical transverse dispersivities.

For **User defined**, specify the dispersion components in terms of constants or expressions. Select **Isotropic**, **Diagonal**, **Symmetric**, or **Full** to enable the appropriate tensor components.

Unsaturated Porous Medium

Use this node to model the concentration of diluted species transported by a solvent (mobile fluid phase) through interstices in a solid porous medium. In addition to transport due to convection and diffusion, the node contains functionality to include species evolution through adsorption and dispersion.

Use this node to model the concentration of diluted species transported by a liquid (mobile fluid phase) in a partially filled solid porous medium. The interstices of the porous medium contains the liquid carrier phase and gas pockets. Apart from convection and diffusion, the node contains functionality to include species evolution through adsorption, dispersion, and volatilization.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

The properties of each phase present are defined using the [Liquid](#), the [Gas](#), and the [Porous Matrix](#) subnodes respectively. The Unsaturated Porous Medium node supports material properties using a [Porous Material](#) node (described in the *COMSOL Multiphysics Reference Manual*), where properties are defined per phase in a similar manner.

Add an [Adsorption](#) or a [Dispersion](#) subfeature to the **Unsaturated Porous Medium** to account for the corresponding transport mechanism.

Liquid

Use this node to specify the mass transfer in the mobile liquid solvent present in the pores of the [Unsaturated Porous Medium](#).

MODEL INPUTS

The temperature model input is always available. By default, the **Temperature** model input is set to **Common model input**, and the temperature is controlled from [Default Model Inputs](#) under **Global Definitions** or by a locally defined [Model Input](#). If a Heat Transfer interface is included in the component, it controls the temperature **Common model input**. Alternatively, the temperature field can be selected from another physics interface. All physics interfaces have their own tags (**Name**). For example, if a Heat Transfer in Fluids interface is included in the component, the **Temperature (ht)** option is available for T .

You can also select **User defined** from the **Temperature** model input in order to manually prescribe T .

SATURATION

Select **Saturation** or **Liquid volume fraction** from the list.

For **Saturation**, enter a value for s (dimensionless) between 0 and 1. The liquid volume fraction is then computed from the saturation and porosity as $\theta_l = \varepsilon_p s$.

For **Liquid volume fraction**, enter a value for θ_l (dimensionless) between 0 and the value of the porosity.

Select a **Liquid fraction time change**: **Liquid fraction constant in time** (the default), **Time change in liquid fraction**, or **Time change in pressure head**.

- For **Time change in fluid fraction**, enter $d\theta/dt$ (SI unit: 1/s).
- For **Time change in pressure head**, enter dH_p/dt (SI unit: m/s) and a **Specific moisture capacity** C_m (SI unit: 1/m). If a Darcy's Law interface is included in the component, the time change in pressure head solved for can be selected.

CONVECTION

If transport by convection is active, the velocity field of the solvent needs to be specified. For **User defined**, enter values or expressions for the velocity components in the input fields. This input option is always available. You can also select the velocity field solved for by a Fluid Flow interface added to the model component. These physics interfaces are available for selection in the **Velocity field** list.

DIFFUSION

Use this section to specify diffusion coefficients describing the diffusion of each species in the solvent liquid.

Use the **Source** list to select to pick up diffusion coefficients defined in a material or a Chemistry interface.

Select **Material**, and a **Liquid material** to use a diffusion coefficient in a material available in the model. User-defined property groups including a diffusion coefficient output property are available for selection in the **Liquid diffusion coefficient** list.

Select **Chemistry** to use a diffusion coefficient from a Chemistry interface added to model component. Diffusion coefficients are automatically defined when **Calculate transport properties** is selected on the interface level of a Chemistry interface. All defined diffusion coefficients are available for selection in the **Liquid diffusion coefficient** list.

For **User defined**, enter a value or expression for the **Liquid diffusion coefficient** $D_{L,i}$ of each species in the corresponding input field.

In a porous medium the diffusivity is reduced due to the fact that the solid grains impede Brownian motion. Select an **Effective diffusivity model, liquid** to account for the reduced diffusivity in the liquid. The available models are **Millington and Quirk model** (the default), **Bruggeman model**, **Tortuosity model**, or **No correction**. For **Tortuosity model**, enter a value for the tortuosity $\tau_{L,i}$ (dimensionless).

Note that multiple species, as well as Migration in Electric fields (described below) is only available for certain COMSOL Multiphysics add-on products. For details see: www.comsol.com/products/specifications/.

MIGRATION IN ELECTRIC FIELD

This section is available when the **Migration in electric field** checkbox is selected. From the **Electric potential** list, select the source of the electric field.

- For **User defined**, enter a value or expression for the **Electric potential** V . This input option is always available.
- Select the electric potential solved by an AC/DC-based interface that has added to the component.
- Select the electric potential defined or solved for by an Electrochemistry interface added to the component.

By default the **Mobility** is set to be calculated based on the species effective diffusivity and the temperature using the **Nernst–Einstein relation**. For **User defined**, select the appropriate scalar or tensor type — **Isotropic**, **Diagonal**, **Symmetric**, or **Full** — and type in the value or expression of the effective mobility $u_{me,i}$.

Enter the **Charge number** z_c for each species.

Gas

Use this node to specify the mass transfer in the gas phase present in the pores of the [Unsaturated Porous Medium](#).

DIFFUSION

The settings for the diffusion of each species in the gas phase are identical to those in the [Liquid](#) subnode.

VOLATILIZATION

Enter a value for the volatilization isotherm $k_{G,c}$ (dimensionless) for each species.

Adsorption

Use this node to model adsorption of the (fluid phase) solute species onto the surface of the porous matrix. It is available as a subnode to the [Porous Medium](#) and the [Unsaturated Porous Medium](#) nodes.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

MATRIX PROPERTIES

The density of the porous media is needed when modeling adsorption to the surface of the porous matrix. Choose to input either the **Dry bulk density** ρ , or the **Solid phase density** ρ_s . The former is the density of the porous matrix including empty pores, while the latter corresponds to the density of the pure solid phase. The density can be defined from the domain material by selecting **From material**, or from a user defined expression. When a **Porous Material** is used on the selection, the density will be requested from a **Solid** subfeature. Several **Solid** features can be used to model a homogeneous mixture of several solid components.

ADSORPTION

Select an **Adsorption isotherm** — **Langmuir**, **Freundlich**, **Toth**, **BET**, or **User defined** to specify how to compute c_P , the amount of species sorbed to the solid phase (moles per unit dry weight of the solid):

- For **Langmuir**:

$$c_P = c_{P\max} \frac{K_L c}{1 + K_L c} \quad , \quad K_P = \frac{\partial c_P}{\partial c} = \frac{K_L c_{P\max}}{(1 + K_L c)^2}$$

Enter a **Langmuir constant** $k_{L,c}$ (SI unit: m^3/mol) and an **Adsorption maximum** $c_{p,\max,c}$ (SI unit: mol/kg).

- For **Freundlich**:

$$c_P = K_F \left(\frac{c}{c_{\text{ref}}} \right)^N \quad , \quad K_P = \frac{\partial c_P}{\partial c} = N \frac{c_P}{c}$$

Enter a **Freundlich constant** $k_{F,c}$ (SI unit: mol/kg), a **Freundlich exponent** $N_{F,c}$ (dimensionless), and a **Reference concentration** $c_{\text{ref},c}$ (SI unit: mol/m^3).

- For **Toth**:

$$c_P = c_{P\max} \frac{b_T c}{(1 + (b_T c)^{N_T})^{1/N_T}}$$

and

$$K_P = \frac{\partial c_P}{\partial c} = c_{P\max} b_T (1 + (b_T c)^{N_T})^{-\left(1 + \frac{1}{N_T}\right)}$$

Enter a **Toth constant** $b_{T,c}$ (SI unit: m^3/mol), a **Toth exponent** $N_{T,c}$ (dimensionless), and an **Adsorption maximum** $c_{p,\max,c}$ (SI unit: mol/kg).

- For **BET (Brunauer–Emmett–Teller)**:

$$c_P = \frac{K_B c_0 c}{(c_S - c) \left(1 + (K_B - 1) \frac{c}{c_S} \right)}$$

and

$$K_P = \frac{\partial c_P}{\partial c} = \frac{K_B c_0 c_S ((K_B - 1)c^2 + c_S^2)}{(c - c_S)^2 (c_S + (K_B - 1)c)^2}$$

Enter a **BET constant** $K_{B,c}$ (dimensionless), a **Monolayer adsorption capacity** $c_{0,c}$ (SI unit: mol/kg), and an **Saturation concentration** $c_{S,c}$ (SI unit: mol/m³).

- For **User defined** enter an **Adsorption isotherm** $c_{P,c}$ (SI unit: mol/kg):

$$c_P = f(c)$$

For more information, see [Adsorption](#) in the theory section.

FURTHER READING

See the theory chapter in the section [Mass Balance Equation for Transport of Diluted Species in Porous Media](#).

Volatilization

This feature is available when the **Mass transfer in porous media** checkbox is selected in the **Settings** window for the physics interface.

Use this feature to model mass transfer at the boundary due to volatilization. The species dissolved in the liquid are assumed to be vaporized at the boundary, and transported into the surrounding bulk region due to convection and diffusion. The mass transfer at the boundary is defined as

$$-\mathbf{n} \cdot \mathbf{J}_c = -h_c (k_{G,c} c - c_{\text{Gatm},c})$$

where h_c is the mass transfer coefficient, $k_{G,c}$ the volatilization coefficient, and $c_{\text{Gatm},c}$ the concentration in the surrounding atmosphere.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

VOLATILIZATION

Enter a **Mass transfer coefficient** h_c defining the transfer into the surrounding media. This can be given by boundary layer theory. When assuming that no convective flow

is present in the surrounding, the mass transfer coefficient can be defined from the gas diffusion coefficient D_{Gc} and the thickness of the diffusion layer d_s in the manner of

$$h_c = \frac{D_{Gc}}{d_s}$$

Also give the atmospheric concentration for each species, $c_{\text{Gatm},c}$. The **Volatilization** coefficient $k_{G,c}$ for each species are taken from the adjacent [Unsaturated Porous Medium](#) domain.

Packed Bed (Moving Packed Bed)

Use this feature to model a packed bed reactor filled with solid catalytic or reactant pellets. The bed pellets are porous, forming a bimodal porosity in the reactor. In addition to mass transfer and reactions in the fluid between the pellets (interparticle processes), the same processes can also be studied inside the pellets (intraparticle). The pellets are available in each point of the bed domain, and are defined using an extra dimension describing the pellet radius.

The **Packed Bed** includes two subnodes, **Fluid** and **Pellets**, which in turn are used to define the physical properties of the corresponding phase. The latter node is also used to define volumetric reactions, as well as heterogeneous reactions, occurring within the porous pellets.

The **Packed Bed** node requires a [Porous Material](#) (described in the *COMSOL Multiphysics Reference Manual*) on the domains occupied by the bed. Using the porous material it is possible to model a bed comprised of any number of pellet types by adding multiple pellet nodes.

For information on the equations solved in the pore space and inside the pellets see the section [The Packed Bed Feature](#), in the [The Transport of Diluted Species Interface](#) chapter. When using the corresponding feature for concentrated mixtures, see [The Packed Bed Feature](#) in the [Theory for the Transport of Concentrated Species Interface](#) chapter.

For an application using the Packed Bed feature, see



- *A Multiscale 3D Packed Bed Reactor*: Application Library path **Chemical_Reaction_Engineering_Module/Reactors_with_Porous_Catalysts/packed_bed_reactor_3d**
Web link: www.comsol.com/model/a-multiscale-3d-packed-bed-reactor-17019

DOMAIN SELECTION

From the **Selection** list, choose the domains with porous material. Note that all domains chosen needs to be occupied by a **Porous Material**.

BED PROPERTIES

Use the **Porosity** list to specify the bed porosity, that is the volume fraction not occupied by the pellets. Select **From material** to use the bed porosity defined by the **Porous Material**. Select **User defined** to enter a constant or an expression.

BED TYPE

Use the **Bed type** list to specify the bed type: **Packed bed** or **Moving packed bed**. If you choose **Moving packed bed**, enter the bed velocity in the **Bed velocity field**.

There are two subnodes for the **Packed Bed** reactor: **Fluid** and **Pellets**. For the **Moving packed bed** reactor, in addition to the two subnodes, there are other three subnodes: **Bed Inlet**, **Bed Outlet**, and **Bed Wall**.

MOVING PACKED BED

The solid bed, comprised of solid catalytic or reactant pellets, moves relative to the reactor. For the mass transport inside the solid pellets, there is an extra convective contribution. For species i , this term can be expressed as

$$\mathbf{u}_{mb} \cdot \nabla(\epsilon_{pe}(1 - \epsilon_p)\epsilon_{pe,i})$$

where $\mathbf{u}_{mb}$ denotes the bed moving velocity, ϵ_{pe} is the pellet porosity, and $(1 - \epsilon_p)$ is the pellet volume fraction of the bed.

The solid pellet bed is rigid, the moving bed velocity is the same in the reactor.

Pellets

Use the **Pellets** node to define intraparticle transport and reactions in a packed bed.

For the Transport of Diluted Species, there are two types of pellets, acting as porous catalyst and reactant (Shrinking Core model) respectively. For the porous catalyst pellets, the node focuses on the diffusion and reactions inside pellet; while for the Shrinking Core pellets, the pellet is shrinking with the proceeding of the solid species consuming reactions. The node focuses on the core consuming reactions on the core surface besides the diffusion and reactions inside the porous reacted layer.

DOMAIN SELECTION

The pellets in use are listed in the domain selection. This corresponds to all **Pellet** nodes added to the **Porous Material** on the domains defining the packed bed.

PELLET TEMPERATURE (TRANSPORT OF CONCENTRATED SPECIES)

This section is only available in a **Transport of Concentrated Species** interface, where the intraparticle flow velocity and pressure is solved for.

Use this section to specify the temperature in the pellets. Select **From fluid** to use the temperature in the fluid surrounding the pellets. When the pellet temperature is solved for, for example using a **Heat Transfer in Pellets** interface, the corresponding temperature is also available.

SURFACE SPECIES

This section is only available in a **Transport of Diluted Species** interface with **Pellet Type** selected as **Porous Catalyst** and a **Transport of Concentrated Species** interface.

Add species to this table to account for surface reactions occurring inside the pellets. The surface species added are dependent variables and their concentrations (amount per area) will be solved for together with that of the other (bulk) species inside the pellets. The surface reaction rate for each surface species is defined in the **Reactions** subnode.

When surface reactions are defined in a **Chemistry** interface, these surface species should be coupled to those defined in the **Pellets** node. This is achieved in the **Settings** window of said **Chemistry** interface. First, select the **Define variables for porous pellets** checkbox in the **Pellet Chemistry** section. Then choose the relevant packed bed interface in the **Species solved for** list in the **Species Matching** section. Finally, type in the dependent variable name in the **Surface species** table. The dependent variable is found in the **Shape Functions** section in the **Equation View** for **Pellets**.

When the checkbox in the **Pellet Chemistry** section is used, the **Chemistry** interface describes the chemistry in the interior of the pellets. To also study the chemistry in the

bulk fluid surrounding the pellets in the **Packed Bed**, add an additional **Chemistry** interface and keep said checkbox cleared.

When using the **Transport of Concentrated Species** interface, the molar mass of each surface species is required to convert between molar and mass basis. For a surface species already defined in a **Chemistry** interface, say $\text{O}_2(\text{ads})$, select **Molar mass for $\text{O}_2(\text{ads})$** for the corresponding surface species (using the column **Molar mass from**). Select **User defined** to instead enter a constant value or an expression in the column **Molar mass (kg/mol)**.

SOLID CORE

This section is only available in a **Transport of Diluted Species** interface with **Pellet Type** selected as **Shrinking Core model**.

The **Pellet density** specifies the density of unreacted core. The default selection is **User defined**. It can also be **From material**.

There can be some solid substances (supporting material, for example) inside the pellet core that do not participate in the core consuming reactions. These substances are classified as matrix. Enter the mass fraction for **Mass fraction of inert matrix**. Its default value is 0.

Select or enter the molar mass and mass fraction for solid species in **Solid core species** table. The molar mass can be picked up from Chemistry (with **Define variables for porous pellets** being selected under **Pellet Chemistry** section) from column **Molar mass from**. If **User defined** is selected from **Molar mass from** column, then the corresponding cell under column **Molar mass (kg/mol)** is enabled, where you can enter the expression for the molar mass. Finally, enter the mass fraction for each solid species under the column **Solid mass fraction (1)**. The mass fractions in the table will be normalized to 1, then multiplied by $(1 - w_{\text{matrix}})$.

REACTED LAYER

This section is only available in a **Transport of Diluted Species** interface with **Pellet Type** selected as **Shrinking Core model**.

Enter a porosity for the porous reacted layer. The default selection is **User defined**. It can also be **From material**.

Diffusion (Transport of Diluted Species)

Use this node to define species diffusion properties inside porous pellets of a packed bed.

DOMAIN SELECTION

The node inherits all valid domains from its parent feature. For the invalid domains, **not applicable** is shown.

DIFFUSION

Select **Diffusion model** — **Millington and Quirk model** (the default), **Bruggeman model**, **Tortuosity model**, or **User defined** to describe the effective correction of the diffusion coefficient in the pellet. In the case of the **Tortuosity model**, a tortuosity factor for the tortuosity τ_{pe} within the pellet is required. Use this section to specify diffusion models and diffusion coefficients inside the reactive pellets.

From the **Diffusion coefficient** list, select diffusion coefficient for different species. If a **User defined** diffusion model is selected, an **Effective diffusion coefficient** $D_{peff,c}$ (SI unit: m^2/s) is entered. The default value is $1 \cdot 10^{-9} m^2/s$.

Diffusion coefficients calculated in Chemistry are available in the list when Chemistry (with the **Define variables for porous pellets** selected under Pellet Chemistry section) is coupled to the Packed Bed feature.

Shrinking Core Reactions

Use this node to define reactions occurring on the surface of the solid pellet core.

The pellet core can consist of one or more solid phase species. Core consumption occurs when the solid species are consumed by surface reactions on the core outer surface (facing the porous layer). Core consuming reactions can also involve bulk species. In this case the bulk species are removed or produced at the core surface.

DOMAIN SELECTION

The node inherits all valid domains from its parent feature. For invalid domains, **not applicable** is shown.

PELLET

Select **All** to apply the core consumption reactions for all available pellets types. Or, select one of the pellet types in the list.

CORE CONSUMING REACTIONS

If there is one solid species only, the core shrinking rate depends on the consumed rate of this species. If there are more than one solid species, the core shrinking rate depends on the weighted mass consumed rate of all species, assuming that the same consumed rate is held for all species along the core axial direction.

In the **Core species reaction rate** table, select the reaction rate (SI unit: $\text{mol}/(\text{m}^2 \cdot \text{s})$) for solid species under the column **Surface reaction rate**. Select the **Define variables for porous pellets** checkbox in the **Pellet Chemistry** section to couple the reaction rate to the chemistry. With **User defined** selected from the column **Surface reaction rate**, enter a reaction-rate expression in the column **Surface rate expression ($\text{mol}/(\text{m}^2 \cdot \text{s})$)**.

For bulk species that take part in the core consuming reactions, or other surface reactions on the core surface, the reaction rate is some kind of molar flux between core surface and reacted layer.

Select **Bulk species reaction rate** (or molar flux, SI unit: $\text{mol}/(\text{m}^2 \cdot \text{s})$) for bulk species i from the $J_{\text{pe},i}$ list. The reaction rate can be picked up from the chemistry by selecting the **Define variables for porous pellets** checkbox. Enter a reaction-rate expression when **User defined** is selected from the list.

HEAT SOURCE

Specify the **Heat source** (Q_{RC}) evolved from core surface reactions. When multiple pellets types are present, all respective heat source are included.

The heat sources are most conveniently picked up from a surface reaction in a Chemistry interface (with the **Define variables for porous pellets** selected in the **Pellet Chemistry** section). In that case, the heat source expression can be selected from the list. Otherwise, you can enter an expression in the text field.

The heat source can be applied in a Heat Transfer in Packed Beds interface by using a Heat Source node added to the Pellets feature.

The total heat source (including heat from Porous Layer Reactions nodes) can be applied in a Heat Transfer in Porous Media interface by using a Heat Source feature.

Porous Layer Reactions

The porous layer (reacted) extends to the core center with the proceeding of the core consuming reactions. During the process, bulk species (gas) diffuse from base geometry into the reacted layer (in pellet), and at the same time the products from the consuming reactions (on core surface) diffuse through the reacted layer out into base

geometry. This node describes the reactions for these species during the diffusion process.

Several Porous Layer Reactions nodes can be used to account for different reaction mechanisms in porous layer and different pellets in the porous material.

DOMAIN SELECTION

The domain selection inherits all valid domains from its parent feature. For the invalid domains, it shows **not applicable**.

PELLET

Select **All** or one of existing pellets for the node.

REACTING VOLUME

A reaction takes usually place in the porous part (fluid) of the pellet, but can also in the whole volume of the pellet. When specifying reaction rate for a bulk species, the specified reaction rate has the basis of the pore volume, or the total volume. With the **Pore volume** selected, the reaction expression in $\text{mol}/(\text{m}^3 \cdot \text{s})$ will be multiplied by porous layer's porosity ε_{pr} .

REACTION RATES

Select a reaction rate (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$) for bulk species i from $R_{\text{pe},i}$ list. Enter a reaction rate expression when **User defined** is selected from the list. Alternatively, if reaction rate expressions have been defined in a Chemistry interface (with the **Define variables for porous pellets** selected under Pellet Chemistry section), they can be chosen from the list.

HEAT SOURCE

Specify the **Heat source** originating from the heat (Q_{pe}) of the bulk reactions inside the porous layer. With several pellets being added, their heat sources will be summed up.

The heat sources are most conveniently picked up from a Chemistry interface (with the **Define variables for porous pellets** selected under Pellet Chemistry section) that defines the reaction rate and the heat of reactions. In that case, the heat source expression can be selected from the list. Otherwise you can enter an expression in the text field.

The heat source (on extra dimension) in a Porous Layer Reactions node can be picked by Heat Source feature under the Pellets feature of heat transfer interface. The total heat source (including heat from Shrinking Core Reactions nodes, on base domain) can be used by a Heat Source feature in any of the heat transfer interfaces.

DOMAIN SELECTION

The domain selection inherits all valid domains from its parent feature. For the invalid domains, it shows **not applicable**.

Several Reactions nodes can be used to account for different reaction mechanisms in different domains and pellets in the porous material.

PELLET

This section is invisible for default Reactions node. It is visible for later added Reactions nodes.

Select **All** or one of existing pellets for the nondefault Reactions nodes.

REACTING VOLUME

A reaction takes usually place in the porous part (fluid) of the pellet, but can also in the whole volume of the pellet. When specifying reaction rate for a bulk species, the specified reaction rate has the basis of the pore volume, or the total volume. With the **Pore volume** selected, the reaction expression in $\text{mol}/(\text{m}^3 \cdot \text{s})$ will be multiplied by pellet's porosity ε_{pe} .

REACTION RATES

Select a reaction rate (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$) for bulk species i from $R_{\text{pe},i}$ list. Enter a reaction rate expression when **User defined** is selected from the list. Alternatively, if reaction rate expressions have been defined in a Chemistry interface (with the **Define variables for porous pellets** selected under Pellet Chemistry section), they can be chosen from the list.

SURFACE REACTION RATES

The section is available when one or more surface species have been added in the **Surface Species** section of its parent feature (Pellets).

Specify the **Reactive specific surface area** (total surface reaction area per cubic meter) $S_{\text{b, reac}}$ for all surface reactions.

Under **Surface reaction rate for bulk species**, specify the rate expression $R_{\text{pe},i}^s$ (SI unit: $\text{mol}/(\text{m}^2 \cdot \text{s})$) for the surface reaction rate of each bulk species i participating in the surface reaction. Furthermore, specify the surface reaction rates for the surface species in the **Reaction rate for surface species** table.

If several pellet types have been defined, one set of surface reaction rates per pellet type can be set by selecting specific pellet under **Pellet** section. If surface reaction rate expressions have been defined in a Chemistry interface (with the **Define variables for porous pellets** selected under Pellet Chemistry section), they can be chosen from the list.

HEAT SOURCE

Specify the **Heat source** originating from the heat of chemical reactions inside the pellet. Both heat source Q_{pe} from bulk reactions in the fluid, and heat source Q_{spe} (available when there is one or more surface species) resulting from surface reactions can be defined. With several pellets being added, their heat sources will be summed up.

The heat sources are most conveniently picked up from a Chemistry interface (with the **Define variables for porous pellets** selected under Pellet Chemistry section) that defines the reaction rate and the heat of reactions. In that case, the heat source expression can be selected from the list. Otherwise you can enter an expression in the text field.

The heat source (inside pellet, on extra dimension) in a Reactions node can be picked by Heat Source feature under the Pellets feature of heat transfer interface. The total heat source (from all pellets, in the base domain) of all Reactions nodes can be used by a Heat Source feature in any of the heat transfer interfaces.

Initial Values (Pellets)

DOMAIN SELECTION

The domain selection inherits all valid domains from its parent feature. For the invalid domains, it shows **not applicable**.

Several Initial Values nodes can be used to set different initial values for different domains and pellets.

PELLET

This section is invisible for default Initial Values node. It is visible for later added Initial Values nodes.

Select **All** or one of existing pellets for the nondefault Initial Values nodes.

INITIAL VALUES

Specify the initial value c_{pe0_i} (SI unit: mol/m^3) for species i in the initial value list.

SURFACE CONCENTRATION INITIAL VALUES

Specify the initial values for surface species under the column **Surface concentration (mol/m²)** in the table.

Pellet–Fluid Interface

DOMAIN SELECTION

The domain selection inherits all valid domains from its parent feature. For the invalid domains, it shows **not applicable**.

Several Pellet–Fluid Interface nodes can be used to specify different boundary conditions between fluid and pellet out-surface for different domains and pellets.

PELLET

This section is invisible for the default Pellet–Fluid Interface node. It is visible for Pellet–Fluid Interface nodes if several Pellet–Fluid Interface nodes are present.

Select **All** or one of existing pellets for the nondefault Pellet–Fluid Interface nodes.

PELLET–FLUID INTERFACE

The **Coupling type** list is selectable for the default Pellet–Fluid Interface node, but it is disabled for nondefault Pellet–Fluid Interface nodes because it is supposed that there should be only one type of boundary condition in a packed bed.

There are two **Coupling type** options for the coupling of concentration between the pellet internals and the surrounding fluid:

- **Continuous concentration**, assuming that all resistance to mass transfer to/from the pellet is within the pellet and no resistance to pellet–fluid mass transfer is on the bulk fluid side. The concentration in the fluid will thus be equal to that in the pellet pore just at the pellet surface: $c_{\text{pe},i} = c_i$. This constraint also automatically ensures flux continuity between the internal pellet domain and the free fluid domain through so-called reaction forces in the finite element formulation.
- **Film resistance (mass flux)**: The flux of mass across the pellet–fluid interface into the pellet is assumed to be rate determined on the bulk fluid side, by film resistance. The resistance is expressed in terms of a film mass transfer coefficient, $h_{D,i}$, such that the inward flux is

$$N_{i,\text{inward}} = h_{D,i}(c_i - c_{\text{pe},i})$$

For the **Active specific surface area** (SI unit: m^{-1}) select either the **Automatic** setting that calculates the specific surface area from the pellet shape, or **User defined**.

Specify the **Mass transfer parameters** — **Automatic** (the default), or **User defined** to define the film mass transfer coefficient.

- **Mass transfer parameters** with **Automatic** selected. Specify one of the models from the **Sherwood number expression** list to compute the Sherwood number. There are three models available: **Frössling**, **Rosner**, and **Garner and Keey**. The Frössling equation is the default and probably the most commonly used for packed spheres. All of these are based on the dimensionless Reynolds (**Re**) and Schmidt (**Sc**) numbers, which are computed from **Density** and **Dynamic viscosity**. Both of these two properties are specified in the Fluid node under the Packed Bed feature.
- **Mass transfer parameters** with **User defined** selected. Specify the type of **Parameter** — **Mass transfer coefficient** (the default), or **Sherwood number**, and enter the selected quantity in the corresponding text field.

Fluid (Porous Catalyst)

Use this feature to specify mass transfer properties in the fluid filling the pores of a porous catalyst medium.

The settings of this node are similar to those of the [Fluid \(Porous Medium\)](#) node of a porous medium.

Heterogeneous Reactions

Inside a multiphase system, mass can be transported from one phase to another one, and chemical reaction can take place among different phases. This kind of mass source term can be described by the **Heterogeneous Reaction** node.

DOMAIN SELECTION

From the **Selection** list, choose the domains on which to define rate expression or expressions that govern the source term in the transport equations.

REACTION RATES

The **Chemistry** list shows all the **Chemistry** interfaces that are coupled to the current mass transport interface. The coupling conditions and the reaction rate variables taken from the **Chemistry** interface are the same as that in the [Reactions](#) node.

With **User defined** selected in the **Chemistry** list, add a mas source term R_i (SI unit: mol/(m³·s)) for each species, or pick up a reaction rate variable from a **Chemistry** interface.

REACTING VOLUME

See the [Reactions](#) node.

HEAT SOURCE

See the [Reactions](#) node.

Bed Inlet

This node specifies an input for the pore concentrations in the pellets in a moving packed bed. Together with it, a [Bed Outlet](#) node should be also added. The solid pellet bed is rigid, the **Bed Inlet** and **Bed Outlet** are usually set to opposite boundaries.

BOUNDARY SELECTION

Select boundary where the solid pellet bed moves into the reactor. These boundaries belong to the pellet domains which are defined in **Porous Material**.

INLET PELLET CONCENTRATIONS

This section is available for the catalytic pellet bed. There are three ways to specify the pellet inlet conditions.

- **From bed fluid conditions.** The species concentrations inside the pellet are the same as that in the fluid (between pellets).
- **From pellet initial values.** The species concentrations inside the pellets are the same as the initial values under the **Pellets** node.
- **User defined.** Enter the inlet concentrations $c_{pe0,i}$ (SI unit: mol/m³) in the **Concentration** text fields.

The surface species concentration table is available if there is any surface reaction in the pellet and the inlet conditions is set to **From bed fluid conditions** or **User defined**. Enter the inlet surface concentration (SI unit: mol/m²) in the surface species table.

The surface species concentration table is not available when the inlet conditions is set to **From pellet initial values**. In this case, the inlet surface concentrations are taken from the **Initial Values** node.

INITIAL VALUES

The section is available for the Shrinking Core Model. Enter the factor rc_{0f} in the **Initial value factor for core radius** field.

Bed Outlet

BOUNDARY SELECTION

Select boundary where the solid pellet bed leaves the reactor. These boundaries belong to the pellet domains which are defined in **Porous Material**.

The bed moving velocity is always equal to the velocity at the [Bed Inlet](#).

Bed Wall

It is assumed that there are no penetration and no viscous effects at the walls.

Porous Catalyst

Use this feature to model adsorption, desorption, and heterogeneous reactions, as a gas or liquid is moving through the interstices of a catalytic porous medium. The mass transfer in the fluid inside the porous matrix is treated using the Linear Driving Force model and a lumped mass transfer coefficient.

A catalyst is a substance that increases the rate of a reaction without itself being consumed. The catalytic process usually consists of the following steps:

- The adsorbate transfers across the fluid–adsorbent boundary layer to the adsorbent surface
- The adsorbate is adsorbed on the adsorbent surface
- Chemical reactions on the surface involving the adsorbate
- Desorption of reaction products (adsorbate) from the adsorbent surface into the fluid–adsorbent boundary layer
- Adsorbate transfers across the boundary layer to the bulk fluid.

Species transport through a porous medium is affected both by properties of the fluid phase, as well as by properties of the solid matrix. These properties are defined in the **Fluid** and **Porous Matrix** subnodes respectively.

Add an **Adsorption** or a **Surface Reaction** subfeature to the **Porous Catalyst** to model adsorption or heterogeneous reactions. To model volumetric reactions in the fluid phase (bulk), add a **Reactions** subnode under **Fluid**.

MODEL INPUT

Specify the temperature to be used in the catalyst. The temperature will be used for the adsorption isotherms and the rate of surface reactions.

- By default, the **Common model input** option corresponds to the `minput.T` variable, set to 293.15 K by default. To edit the `minput.T` variable, click the **Go to Source** button (), and in the **Default Model Inputs** node under **Global Definitions**, set a value for the **Temperature** in the **Expression for remaining selection** section.
- Select **User defined** to enter a value or an expression for the temperature (SI unit: K). This input is always available.
- If available, select a temperature defined by a Heat Transfer interface present in the model. For example, select **Temperature (ht)** to use the temperature defined by the Heat Transfer in Porous Media interface using the Name `ht`.

REACTIVE SPECIFIC SURFACE AREA

Enter a specific surface area for the adsorption and surface reaction processes. The default value is $1 \cdot 10^7$ 1/m.

CONCENTRATION INPUT

Select a **Concentration input — Molar concentration (mol/(m²))** (the default) or **Amount per solid mass (mol/kg)**. This parameter determines the unit for initial values of adsorbed species and user-defined surface species.

ADSORBED SPECIES

Specify which of the bulk species that can be adsorbed to the catalyst (adsorbent). The surface concentrations (SI unit: mol/m²) of the adsorbed species, named `cads_j` corresponding to bulk species `j`, are solved for. The rate of adsorption is defined in the **Adsorption** subnode. With an adsorbed species being selected, its initial value field is activated. Enter the initial value in accordance with the **Concentration input**.

SURFACE SPECIES

There could be extra surface species besides the adsorbed bulk species and their corresponding adsorbed sites on adsorbent surface. These additional surface species could take part in catalytic reactions. The surface reaction rate is defined in the **Surface Reaction** subnode.

Add a user-defined surface species in the **Surface species** table by clicking the **Add** button under the table. The concentration (SI unit: mol/m^2) of all surface species added are solved for. To define the surface reaction kinetics add a **Surface Reaction** subnode.

When using the Transport of Concentrated Species interface, the molar mass of each surface species is required to convert between molar and mass basis. For a surface species already defined in a Chemistry interface, say $\text{O}_2(\text{ads})$, select **Molar mass for $\text{O}_2(\text{ads})$** for the corresponding surface species (using the column **Molar mass from**). Select **User defined** to instead enter a constant or expression in the column **Molar mass (kg/mol)**.

Porous Matrix (Porous Catalyst)

DOMAIN SELECTION

The domain selection inherits all valid domains from its parent feature.

MATRIX PROPERTIES

Specify the porous material porosity from **Porosity** — **From material** (the default), **User defined**. The default value for **User defined** is 0.3.

The density of the porous material can be defined either from the **Dry bulk density** — **From material** (the default), **User-defined**, or from **Solid phase density** — **From material** (the default), **User-defined**. The default values of **User-defined** for both **Dry bulk density** and **Solid phase density** are $1400 \text{ kg}/\text{m}^3$.

Adsorption (Porous Catalyst)

The adsorption process can be regarded as being under equilibrium if the adsorption rate is very fast and mass transfer resistance is negligible. In such case, the adsorption depends only on the amount of adsorbent, concentration (or pressure) of adsorbate and temperature. This kind of isotherm adsorption process is implemented in the Adsorption subfeature of the Porous Medium and Unsaturated Porous Medium features.

In general, the adsorption process deviates to some extent from its equilibrium state. With the assumption that the species adsorption processes are independent of each other, the **Adsorption** node uses the Linear Driving Force (LDF) model to define the adsorbed species-solid (catalyst) mass transfer rate with regard to an adsorption isotherm. The overall adsorption rate ($\text{mol}/(\text{m}^3 \cdot \text{s})$) is expressed as:

$$R_{\text{ads},i} = h_{\text{LDF},i}(c_{\text{ads},i}S_{\text{area}} - c_{\text{P},e,i}P_b), \quad (3-1)$$

Here the adsorption isotherm, $c_{\text{P},e,i}$ (mol/kg), represents the equilibrium value of bulk species i adsorbed per dry mass of adsorbent. $c_{\text{ads},i}$ is the local value of the adsorbed species concentration (mol/m²). The available specific surface, and the adsorbent bulk density are denoted by S_{reac} (1/m) and ρ_b (kg/m³) respectively. $h_{\text{LDF},i}$ is a lumped mass transfer coefficient, which can be estimated from empirical equations based on the mass transfer mechanism. The mass transfer rate is added as a source or sink term in the equation for the bulk species concentration. The adsorbed species concentration is solved for in the manner of:

$$\frac{\partial c_{\text{ads},i}}{\partial t} = \frac{R_{\text{ads},i}}{S_{\text{area}}} \quad (3-2)$$

DOMAIN SELECTION

The domain selection inherits all the domain selections from its parent feature. Several adsorption features can be added to the same domain.

ADSORPTION

Adsorption parameters can be set only for the species that are selected under **Adsorbed Species** section in the **Porous Catalyst** feature.

Select the **Adsorption isotherm for individual species** checkbox to set different adsorption isotherm models for each adsorbed species. The equations for the different models are shown in the **Equation** section. Select the adsorption isotherm from the **Adsorption isotherm** list. Choose among **Langmuir**, **Freundlich**, **Toth**, **BET**, or **User defined**.

With **Adsorption isotherm for individual species** cleared, all species use the same adsorption isotherm model. In this case, **Adsorption isotherm** is available directly below the **Adsorption isotherm for individual species** checkbox.

Use the **Species** checkbox to choose the species that are adsorbed. For each selected adsorbate, there are three parameters to fill in; **Adsorption isotherm** (available when the checkbox **Adsorption isotherm for individual species** is selected), adsorption model specific parameters and **Mass transfer coefficient**.

Select an adsorption isotherm for each adsorbate (available with **Adsorption isotherm for individual species** being selected).

The adsorption model specific parameters for species i are:

- **Langmuir:**

- 1 **Langmuir constant** $k_{L,i}$ (SI unit: m^3/mol)
- 2 **Adsorption maximum** $c_{p,\max,i}$ (SI unit: mol/kg).
- **Freundlich:**
 - 1 **Freundlich constant** $k_{F,i}$ (SI unit: mol/kg)
 - 2 **Freundlich exponent** $N_{F,i}$ (dimensionless)
 - 3 **Reference concentration** $c_{\text{ref},i}$ (SI unit: mol/m^3).
- **Toth:**
 - 1 **Adsorption maximum** $c_{p,\max,i}$ (SI unit: mol/kg)
 - 2 **Toth constant** $b_{T,i}$ (SI unit: m^3/mol)
 - 3 **Toth exponent** $N_{T,i}$ (dimensionless).
- **BET (Brunauer–Emmett–Teller):**
 - 1 **Monolayer adsorption capacity** $c_{0,i}$ (SI unit: mol/kg)
 - 2 **Saturation concentration** $c_{S,i}$ (SI unit: mol/m^3)
 - 3 **BET constant** $K_{B,i}$ (dimensionless).
- **User defined:**
 - 1 **Adsorption isotherm** $c_{P,i}$ (SI unit: mol/kg):

The last parameter to enter for each adsorbate is the **Mass transfer coefficient** h_i (SI unit: m/s).

Surface Reaction (Porous Catalyst)

The **Surface Reaction** subnode under **Porous Catalyst** is used to define surface reaction source terms to the mass transport within the catalyst.

DOMAIN SELECTION

From the **Selection** list, choose the domains (from its parent feature) on which the surface reaction rate is defined.

Several Surface Reaction nodes can be used to account for different surface reactions in different parts for the modeling geometry.

SURFACE REACTION RATE FOR BULK SPECIES

The bulk species (defined as dependent variables in physics interface) can also take part in a surface reaction.

Select a surface reaction rate (SI unit: $\text{mol}/(\text{m}^2\cdot\text{s})$) for species i from R_i^s list. Enter a reaction rate expression when **User defined** is selected from the list. Alternatively, if surface reaction rate expressions have been defined in a Chemistry interface, they can be chosen from the list.

SURFACE REACTION RATE FOR ADSORBED SPECIES

This section is available when at least one species is selected under the Adsorbed Species section in the Porous Catalyst feature.

Select a surface reaction rate (SI unit: $\text{mol}/(\text{m}^2\cdot\text{s})$) for adsorbed species cads_i from $R_{\text{cads}_i}^s$ list. If **User defined** is selected from the list, enter a surface reaction rate expression. Alternatively, if surface reaction rate expressions have been defined in a Chemistry interface, they can be chosen from the list.

REACTION RATE FOR SURFACE SPECIES

This section is available when there is at least one species added to the surface species table under the Surface Species section in the Porous Catalyst feature.

In the surface reaction rate table, there are three columns, **Surface species**, **Surface reaction rate**, and **Surface rate expression ($\text{mol}/(\text{m}^2\cdot\text{s})$)**.

Select a surface reaction rate (SI unit: $\text{mol}/(\text{m}^2\cdot\text{s})$) from the column **Surface reaction rate** for a surface species which is defined in the parent feature. With **User defined** being selected, the table cell at the same row under column **Surface rate expression ($\text{mol}/(\text{m}^2\cdot\text{s})$)** is editable (otherwise it is read-only), and enter a surface reaction rate expression. Alternatively, if surface reaction rate expressions have been defined in a Chemistry interface, they can be chosen from the list under column **Surface reaction rate**.

Species Source

In order to account for consumption or production of species in porous domains, the **Species Source** node adds source terms expressions S_i to the right-hand side of the species transport equations.

DOMAIN SELECTION

From the **Selection** list, choose the domains on which to define rate expression or expressions that govern the source term in the transport equations.

If there are several types of domains, with subsequent and different reactions occurring within them, it might be necessary to remove some domains from the selection. These are then defined in an additional **Species Source** node.

SPECIES SOURCE

Add a source term S_i (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$) for each of the species solved for. Enter a value or expression in the field of the corresponding species.

Hygroscopic Swelling

The **Hygroscopic Swelling** multiphysics coupling node () is used for moisture concentration coupling between the Solid Mechanics interface and either the Transport of Diluted Species or Transport of Diluted Species in Porous Media interfaces.

Hygroscopic swelling is an effect of internal strain caused by changes in moisture content. This volumetric strain can be written as

$$\epsilon_{\text{hs}} = \beta_{\text{h}} M_{\text{m}} (c_{\text{mo}} - c_{\text{mo,ref}})$$

where β_{h} is the coefficient of hygroscopic swelling, M_{m} is the molar mass, c_{mo} is the moisture concentration, and $c_{\text{mo,ref}}$ is the strain-free reference concentration.

This feature requires a license of either the MEMS Module or the Structural Mechanics Module. The multiphysics feature will appear automatically if both the Transport of Diluted Species and the Solid Mechanics interfaces are added to the same component. For the most current information about licensing, please see www.comsol.com/products/specifications/.

FURTHER READING

More information about how to use hygroscopic swelling can be found in the [Hygroscopic Swelling](#) section in the *Structural Mechanics Module User's Guide*.

More information about multiphysics coupling nodes can be found in the section [The Multiphysics Branch](#) in the *COMSOL Multiphysics Reference Manual*.

Out-of-Plane Flux

The out-of-plane flux node is used to prescribe a flux in an out-of-plane domain, and it can be used to reduce a model geometry to 2D, or even 1D, when the concentration variation is small in one or more directions. This could be the case for example when

the object to model is thin or slender. Figure 3-1 shows examples of possible situations in which this type of geometry reduction can be applied.

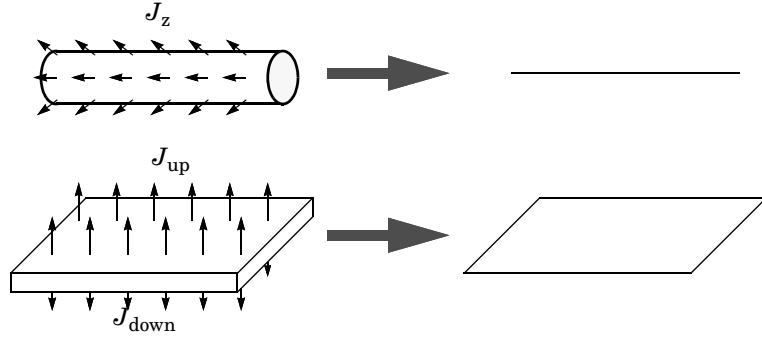


Figure 3-1: Geometry reduction from 3D to 1D (top) and from 3D to 2D (bottom).

For a 1D component this node adds a single out-of-plane molar flux $J_{0,z,i}$ for species i . For a 2D component two fluxes can be prescribed for each species; $J_{0,u,i}$ for the upside of the domain, and $J_{0,d,i}$ for the downside of the domain.

The fluxes are added as a source contribution to the right-hand side of the transport equation. For example, in 2D components with diffusion, convection, and out-of-plane flux present, the mass transfer of species i is defined by the mass conservation equation

$$\frac{\partial c_i}{\partial t} + \nabla \cdot \mathbf{J}_i + \mathbf{u} \cdot \nabla c_i = R_i + S_{\text{opf},i}, \quad (3-3)$$

where $S_{\text{opf},i}$ is the out-of-plane source for species i

$$S_{\text{opf},i} = \frac{J_{0,i}}{d_z} = \frac{J_{0,u,i} + J_{0,d,i}}{d_z}.$$

For external convection on the upside and the downside of the domain, the out-of-plane flux is

$$J_{0,i} = k_{c,u,i}(c_{b,u,i} - c_i) + k_{c,d,i}(c_{b,d,i} - c_i)$$

where k_c is a mass transfer coefficient and c_b is the bulk concentration, the typical concentration far into the surrounding exterior domain. The prescribed flux, J_0 , can

include any arbitrary user-specified expressions. It can be a constant or a function of a dependent variable or independent variables.

UPSIDE INWARD FLUX

For 1D components, enter the **cross-sectional perimeter** P_c to get the out-of-plane flux

$$J_{0,i} = P_c J_{0,z,i}$$

The default value of P_c is the circumference. Either keep the default value, for a circular cross-section shape, or edit the value to get a user-defined shape of the out-of-plane cross-section.

The available flux type options are **General inward flux** and **External convection**. Select the **Species** checkbox for the species for which to specify the flux, and enter a value or expression for the inward flux in the corresponding field. Use a minus sign when specifying a flux directed out of the system. To use another boundary condition for a specific species, click to clear the checkbox for that species.

Set **Flux type** to **External convection** to prescribe a flux to or from an exterior domain (not modeled) assumed to include convection. The exterior can for example include a forced convection to control the temperature or to increase the mass transport. In this case the prescribed mass flux corresponds to

$$J_0 = k_c(c_b - c)$$

where k_c is a mass transfer coefficient and c_b is the bulk concentration, the typical concentration far into the surrounding exterior domain.

DOWNSIDE INWARD FLUX

The available options are **General inward flux** and **External convection**. The settings are the same as for the [Upside Inward Flux](#) section.

FURTHER READING

For an example of using the Out-of-Plane Flux node, see this application example:



Cross-Flow Mass Transfer in a Thin Domain: Application Library path
Chemical_Reaction_Engineering_Module/Tutorials/thin_domain

Fracture

Use this node to model mass transport along thin fractures in porous media. The node assumes that the transport in the tangential direction along the fracture is dominant as a result of lower flow resistance.

The Fracture feature has two default subfeatures — the [Fluid \(Fracture\)](#) subfeature where convection and diffusion are specified and the [Fracture Material](#) subfeature where the porosity ϵ_p is defined.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

FRACTURE PROPERTIES

Specify a value for the **Fracture thickness** d_f .

For the **Fracture** boundary condition, there are a number of **Attributes** available which can be accessed either via the **Physics** Ribbon toolbar (for Windows) or by right-clicking on the **Fracture** node. These attributes are identical to the [Boundary, Edge, Point, and Pair Nodes for the Transport of Diluted Species in Fractures Interface](#).

Fluid (Fracture)

Use this node to specify the mass transfer in the mobile fluid solvent filling the pores of a fracture.

It uses the same input options as the [Fluid \(Porous Medium\)](#) subfeature under the [Porous Medium](#) feature. There are further details about the settings.

MODEL INPUTS

The temperature model input is always available. By default, the **Temperature** model input is set to **Common model input**, and the temperature is controlled from **Default Model Inputs** under **Global Definitions** or by a locally defined **Model Input**. If a Heat Transfer interface is included in the component, it controls the temperature **Common model input**. Alternatively, the temperature field can be selected from another physics interface. All physics interfaces have their own tags (**Name**). For example, if a Heat Transfer in Fluids interface is included in the component, the **Temperature (ht)** option is available for T.

CONVECTION

If transport by convection is active, the velocity field of the solvent needs to be specified. Select an option from the **Velocity field** list to specify the convective velocity along the fracture. For a consistent model, use a Fracture feature in a Darcy's Law interface to compute the fluid flow velocity in the fracture.

For **User defined**, enter values or expressions for the velocity components in the table shown

DIFFUSION

Use this section to specify diffusion coefficients describing the diffusion of each species in the solvent fluid.



The **Diffusion** settings are similar to those for the diffusion coefficients in the [Fluid \(Porous Medium\)](#) node under [Porous Medium](#).

Fracture Material

Specify the **Porosity** ε_p (dimensionless) of the porous matrix. This is by default taken **From material**. Select **From pellet bed densities** to compute the porosity from the (dry bulk) **Bed density** ρ_b and the (single phase) **Pellet density** ρ_{pe} through the relation

$$\varepsilon_p = 1 - \frac{\rho_b}{\rho_{pe}}$$

For **User defined**, enter a value or expression for the porosity.

The Transport of Diluted Species in Fractures Interface

The **Transport of Diluted Species in Fractures (dsf)** interface () , found under the **Chemical Species Transport** branch () , is used to model the transport of a solute species along thin porous fractures, taking into account diffusion, dispersion, convection, and chemical reactions. The fractures are defined by boundaries and the solute species is assumed to be present in a solvent.

The interface supports simulation of species transport along boundaries in 2D and 3D, and axisymmetric components in 2D. The dependent variable is the molar concentration, c . Modeling multiple species transport is possible, whereby the physics interface solves for the molar concentration, c_i , of each species i .



This interface is only available in a limited set of add-on products. For a detailed overview of which interfaces are available in each product, visit www.comsol.com/products/specifications/

SETTINGS

The **Label** is the default physics interface name.

The **Name** is used primarily as a scope prefix for variables defined by the physics interface. Refer to such physics interface variables in expressions using the pattern `<name>.<variable_name>`. In order to distinguish between variables belonging to different physics interfaces, the `name` string must be unique. Only letters, numbers, and underscores (`_`) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first physics interface in the model) is `dsf`.

BOUNDARY SELECTION

If the model geometry includes boundaries that should not be included in the mass transfer simulation, remove those from the selection list.

OUT-OF-PLANE THICKNESS

For 2D components, the **Thickness** field (default value: 1 m) defines a parameter for the thickness of the geometry perpendicular to the two-dimensional cross-section. Both constant and varying thicknesses are supported. The value of this parameter is used,

among other things, to automatically calculate molar flow rates from the total molar flux.

TRANSPORT MECHANISMS

Use the **Convection** checkbox, available under **Additional transport mechanisms**, to control whether to also include convective transport.

CONSISTENT STABILIZATION

To display this section, click the **Show** button () and select **Stabilization**. Use this section to control the application of the available consistent stabilization methods; **Streamline diffusion** and **Crosswind diffusion**.

- When the **Crosswind diffusion** checkbox is selected, a weak term that reduces spurious oscillations is added to the transport equation. The resulting equation system is always nonlinear. There are two options for the **Crosswind diffusion type**:
 - **Do Carmo and Galeão** — the default option. This type of crosswind diffusion reduces undershoots and overshoots to a minimum but can in rare cases give equation systems that are difficult to fully converge.
 - **Codina**. This option is less diffusive compared to the Do Carmo and Galeão option but can result in more undershoots and overshoots. It is also less effective for anisotropic meshes. The Codina option activates a text field for the **Lower gradient limit** g_{lim} . Its default value is $0.1 [\text{mol}/\text{m}^3) / \text{tds}.\text{helem}$, where $\text{tds}.\text{helem}$ is the local element size.
- For both consistent stabilization methods select an **Equation residual**. **Approximate residual** is the default and means that derivatives of the diffusion tensor components are neglected. This setting is usually accurate enough and is computationally faster. If required, select **Full residual** instead.

INCONSISTENT STABILIZATION

To display this section, click the **Show** button () and select **Stabilization**. By default, the **Isotropic diffusion** checkbox is not selected, because this type of stabilization adds artificial diffusion and affects the accuracy of the original problem. However, this option can be used to get a good initial guess for under resolved problems.

DISCRETIZATION

To display all settings available in this section, click the **Show** button () and select **Advanced Physics Options**.

The **Value type when using splitting of complex variables** setting should in most pure mass transfer problems be set to **Real**, which is the default. It makes sure that the dependent variable does not get affected by small imaginary contributions, which can occur, for example, when combining a Time Dependent or Stationary study with a frequency-domain study. For more information, see [Splitting Complex-Valued Variables](#) in the *COMSOL Multiphysics Reference Manual*.

DEPENDENT VARIABLES

The dependent variable name is **Concentration c** by default. A dependent variable name must be unique with respect to all other dependent variables in the component.

Add or remove species variables in the model and also change the names of the dependent variables that represent the species concentrations.

Enter the **Number of species**. Use the **Add concentration** () and **Remove concentration** () buttons as needed.

FURTHER READING



- [Mass Transport in Fractures](#) in the theory section.
- [Numerical Stabilization](#) in the *COMSOL Multiphysics Reference Manual*.
- [Domain, Boundary, and Pair Nodes for the Transport of Diluted Species Interface](#)
- In the *COMSOL Multiphysics Reference Manual*, see [Table 2-4](#) for links to common sections and [Table 2-5](#) for common feature nodes. You can also search for information: press F1 to open the **Help** window or Ctrl+F1 to open the **Documentation** window.

Boundary, Edge, Point, and Pair Nodes for the Transport of Diluted Species in Fractures Interface

The [Transport of Diluted Species Interface](#) has the following boundary, edge, point, and pair nodes, listed in alphabetical order, available from the **Physics** ribbon toolbar

(Windows users), **Physics** context menu (Mac or Linux users), or by right-clicking to access the context menu (all users).

- [Concentration](#)
- [Flux](#)
- [Fracture](#)
- [Inflow](#)
- [Initial Values](#)
- [No Flux](#)
- [Outflow](#)
- [Reactions](#)
- [Species Source](#)

Furthermore, there are the following attributes to the **Fracture** node

- [Adsorption](#)
- [Dispersion](#)

Adsorption

Use this node to model adsorption of the fluid phase species onto the porous media surface of the fracture.

MATRIX PROPERTIES

Use the **Porous material** list to define a material specifying the matrix properties on the current selection. By default the **Boundary material** is used as [The Transport of Diluted Species Interface](#) is active on boundaries only. The **Density** ρ of the porous media is needed when modeling adsorption to the surface of the porous matrix. By default the option **From material** is selected.

ADSORPTION

Select an **Adsorption isotherm** — **Langmuir** (the default), **Freundlich**, **Toth**, **BET**, or **User defined** to specify how to compute c_p , the amount of species sorbed to the solid phase (moles per unit dry weight of the solid). Also activate the checkbox for the species c_i for which adsorption takes place.

- For **Langmuir**:

$$c_P = c_{P\max} \frac{K_L c}{1 + K_L c} \quad \frac{\partial c_P}{\partial c} = \frac{K_L c_{P\max}}{(1 + K_L c)^2}$$

Enter a **Langmuir constant** $k_{L,c}$ (SI unit: m^3/mol) and an **Adsorption maximum** $c_{P,\max,c}$ (SI unit: mol/kg):

- For **Freundlich**:

$$c_P = K_F \left(\frac{c}{c_{\text{ref}}} \right)^{N_F} \quad \frac{\partial c_P}{\partial c} = N_F \frac{c_P}{c} \quad \text{Freundlich}$$

Enter a **Freundlich constant** $k_{F,c}$ (SI unit: mol/kg), a **Freundlich exponent** $N_{F,c}$ (dimensionless), and a **Reference concentration** $c_{\text{ref},c}$ (SI unit: mol/m^3).

- For **Toth**:

$$c_P = c_{P\max} \frac{b_T c}{(1 + (b_T c)^{N_T})^{1/N_T}}$$

and

$$\frac{\partial c_P}{\partial c} = c_{P\max} b_T (1 + (b_T c)^{N_T})^{-\left(1 + \frac{1}{N_T}\right)}$$

Enter a **Toth constant** $b_{T,c}$ (SI unit: m^3/mol), a **Toth exponent** $N_{T,c}$ (dimensionless), and an **Adsorption maximum** $c_{P,\max,c}$ (SI unit: mol/kg).

- For **BET (Brunauer–Emmett–Teller)**:

$$c_P = \frac{K_B c_0 c}{(c_S - c) \left(1 + (K_B - 1) \frac{c}{c_S} \right)}$$

and

$$\frac{K_B c_0 c_S ((K_B - 1) c^2 + c_S^2)}{(c - c_S)^2 (c_S + (K_B - 1) c)^2}$$

Enter a **BET constant** $K_{B,c}$ (dimensionless), a **Monolayer adsorption capacity** $c_{0,c}$ (SI unit: mol/kg), and a **Saturation concentration** $c_{S,c}$ (SI unit: mol/m^3).

- For **User defined**:

$$c_p = f(c)$$

Enter an adsorption isotherm $c_{p,c}$ (SI unit: mol/kg).

For more information, see [Adsorption](#) in the theory section.

FURTHER READING

See the theory chapter in the section [Mass Balance Equation for Transport of Diluted Species in Porous Media](#).

Concentration

Use this node to specify the species concentration on a fracture boundary (applied in points in 2D and along edges in 3D). For example, a $c = c_0$ condition specifies the concentration of species c .

CONCENTRATION

Individually specify the concentration for each species. Select the checkbox for the **Species** to specify the concentration, and then enter a value or expression in the corresponding field. To use another boundary condition for a specific species, click to clear the checkbox for the concentration of that species.

CONSTRAINT SETTINGS

To display this section, click the **Show** button () and select **Advanced Physics Options**. You can find details about the different constraint settings in the section [Constraint Reaction Terms](#) in the *COMSOL Multiphysics Reference Manual*.

Dispersion

Local variations in the velocity as the fluid flows around solid particles lead to mechanical mixing, referred to as dispersion. Use this feature to account for dispersion in a Fracture.

This subfeature is available when the **Convection** checkbox is selected in the **Settings** window for the physics interface.

Select the **Specify dispersion for each species individually** checkbox to specify the dispersion tensor D_D (SI unit: m^2/s) for each species separately. When not selected the same dispersion tensor D_D is used for all species.

Select an option from the **Dispersion tensor** list — **Dispersivity** or **User defined**.

Select **Dispersivity** to specify the dispersion in terms of dispersivities (SI unit: m). Select an option from the **Dispersivity model** list: **Isotropic** (the default) or **Transverse isotropic** based on the properties of the porous media. For isotropic porous media, specify the longitudinal and transverse dispersivities. For transverse isotropic porous media, specify the longitudinal, horizontal transverse, and vertical transverse dispersivities.

For **User defined**, specify the dispersion components in terms of constants or expressions. Select **Isotropic**, **Diagonal**, **Symmetric**, or **Full** to enable the appropriate tensor components.

Flux

This node can be used to specify the species flux across a boundary of a porous fracture (applied in points in 2D and along edges in 3D). The flux of species c is defined as

$$\mathbf{n} \cdot d_{\text{fr}}(D_c \nabla c) = d_{\text{fr}} N_0$$

where N_0 is an arbitrary user-specified flux expression. For example, N_0 can represent a flux due to chemical reactions, or a phase change. A positive N_0 implies that the concentration inside the fracture increases.

INWARD FLUX

The available options are **General inward flux** and **External convection**. If the latter is chosen, define a **Mass transfer coefficient** and a **Bulk concentration**. Specify the flux of each species individually. To use another boundary condition for a specific species, make sure that the checkbox for the mass fraction of that species is cleared.

Inflow

Use this node to specify all species concentrations at a fracture inlet. The condition is applied in points in 2D and along edges in 3D.

If you want to specify the concentration of a subset of the partaking species, this can be done by using the [Concentration](#) node instead.

CONCENTRATION

For the concentration of each species $c_{0,c}$ (SI unit: mol/m³), enter a value or expression.

BOUNDARY CONDITION TYPE

This section in the settings is only available for some products. Search for “Inflow” on the page: www.comsol.com/products/specifications/ for more details on availability.

The option **Concentration constraint** constrains the concentration values on the boundary by the use of pointwise constraints. The other option, **Flux (Danckwerts)** can be more stable and fast to solve when high reaction rates are anticipated in the vicinity of the inlet. Oscillations on the solutions can also be avoided in such cases. The latter condition uses a flux boundary condition based on the velocity across the boundary and the concentration values. See further details in the theory section.

CONSTRAINT SETTINGS

To display this section, click the **Show** button () and select **Advanced Physics Options**. You can find details about the different constraint settings in the section [Constraint Reaction Terms](#) in the *COMSOL Multiphysics Reference Manual*.

FURTHER READING

See the theory chapter in the section [Danckwerts Inflow Boundary Condition](#).

No Flux

This node can be used to specify that the species flux across a boundary of a porous fracture is zero. The condition is applied in points in 2D and along edges in 3D.

Outflow

Set this condition at fracture outlets where species are transported out of the model domain by fluid motion. The condition is applied in points in 2D and along edges in 3D. It is assumed that convection is the dominating transport mechanism across outflow boundaries, and therefore that diffusive transport can be ignored, that is:

$$\mathbf{n} \cdot (-D_c \nabla c) = 0$$

Reactions

Use the **Reactions** node to account for the consumption or production of species through chemical reactions in the fracture. Define the rate expressions as required.

BOUNDARY SELECTION

From the **Selection** list, choose the boundaries on which to define rate expression or expressions that govern the source term in the transport equations.

Several reaction nodes can be used to account for different reactions in different parts of the fracture.

REACTION RATES

Add a rate expression R_i for species i . Enter a value or expression in the field. Note that if you have the Chemistry interface available, included with the Chemical Reaction Engineering Module, the reaction rate expressions can be automatically generated and picked up from the lists.

REACTING VOLUME

When specifying reaction rates for a species in a fracture, the specified reaction rate may have the basis of the pore volume of the fracture, or the total volume.

- For **Total volume**, the reaction expressions in are specified per unit volume of the fracture. The reaction expressions will be multiplied by the fracture thickness d_{fr} .
- For **Pore volume**, the reaction expressions in $\text{mol}/(\text{m}^3 \cdot \text{s})$ are specified per unit volume of total pore space in the fracture. The reaction expressions will be multiplied by the fracture thickness d_{fr} and the fracture porosity, ϵ_{p} .

Species Source

In order to account for consumption or production of species in a fracture, the **Species Source** node adds source terms expressions S_i to the right-hand side of the species transport equations.

BOUNDARY SELECTION

From the **Selection** list, choose the boundaries on which to define expressions that govern the source term in the transport equations.

If there are several different parts of the fracture, with subsequent and different sources occurring within them, it might be necessary to remove some boundaries from the selection. The sources in these can then be defined using an additional **Species Source** node.

SPECIES SOURCE

Add a source term S_i for each of the species solved for. Enter a value or expression in the field of the corresponding species.

The Transport of Diluted Species in Moving Packed Beds Interface

This interface () , found under the **Chemical Species Transport** branch () > **Reacting Flow in Porous Media** multiphysics node () > **Moving Packed Bed** group node, is used to model reactors where the bed, comprised of packed porous catalytic pellets, move relative to the reactor. The interparticle macropores (void between pellets), and the intraparticle micropores (pores inside the pellets) result in a system with bimodal porosity. The **Transport of Diluted Species in Moving Packed Beds** interface can be used to study interparticle and intraparticle mass transport, as well as reactions, while the solid pellet bed is moving downward, counter-currently to the moving fluid between pellets.

The main feature of the interface is the **Packed Bed (Moving Packed Bed)** node. With the **Bed Type** parameter being changed from **Packed bed** to **Moving packed bed**, the **Transport of Diluted Species in Packed Beds** interface switches to the **Transport of Diluted Species in Moving Packed Beds** interface. For the **Transport of Diluted Species in Moving Packed Beds** interface, there is an extra convection term in the mass transport governing equation,

$$\epsilon_{pe} \frac{\partial c_{pe,i}}{\partial t} + \frac{1}{r^2 r_{pe}} \frac{\partial}{\partial r} \left(-r^2 D_{pe,i} \frac{\partial c_{pe,i}}{\partial r} \right) + \mathbf{u}_{mb} \cdot \nabla (\epsilon_{pe} (1 - \epsilon_p) c_{pe,i}) = R_{pe,i}$$

This extra term

$$\mathbf{u}_{mb} \cdot \nabla (\epsilon_{pe} (1 - \epsilon_p) c_{pe,i})$$

is contributed by the solid pellet bed moving.

See [The Transport of Diluted Species in Packed Beds Interface](#).

The Transport of Diluted Species in Moving Packed Beds, Shrinking Core

This interface () , found under the **Chemical Species Transport** branch () > **Reacting Flow in Porous Media** multiphysics node () > **Moving Packed Bed** group node, is used to model reactors where the bed, comprised of initially solid

reactant particles, move relative to the reactor. As the reactant particles are consumed by heterogeneous reactions from the outside inward, the solid core shrinks, leaving behind a porous layer around it. Apart from the core consuming reactions, transport and reactions inside the porous reacted layer, as well as transport in the pore space around the particles are studied.

The main feature of the interface is the [Packed Bed \(Moving Packed Bed\)](#) node. With the **Bed Type** parameter being changed from **Packed bed** to **Moving packed bed**, the **Transport of Diluted Species in Packed Beds** interface switches to the **Transport of Diluted Species in Moving Packed Beds** interface. The effect of the solid bed moving on the mass transport and reaction inside the pellets is described in [The Transport of Diluted Species in Moving Packed Beds Interface](#).

In the [Pellets](#) node which is the subnode to the **Packed Bed** node, With the **Pellet Type** parameter being changed from **Porous catalyst** to **Shrinking Core model**, the **Transport of Diluted Species in Moving Packed Beds** interface switches to the **Transport of Diluted Species in Moving Packed Beds, Shrinking Core** interface. In the **Transport of Diluted Species in Moving Packed Beds, Shrinking Core** interface, the shrinking core reaction occurs at the solid particle outer surface that moves inward into the solid interior, and the fluid (often gas) reactant diffuses through the growing reacted porous layer to the reacting surface. The mass transport and reactions in the reacted porous layer, as well as the size decrease of the unreacted solid core are described in [The Transport of Diluted Species in Packed Beds, Shrinking Core](#).

The Transport of Diluted Species in Packed Beds, Shrinking Core

This interface () , found under the **Chemical Species Transport** branch () > **Reacting Flow in Porous Media** multiphysics node () > **Packed Bed** group node, is used to model reactors filled with initially solid reactant particles. As the reactant particles are consumed by heterogeneous reactions from the outside inward, the solid core shrinks, leaving behind a porous layer around it. Apart from the core consuming reactions, transport and reactions inside the porous reacted layer, as well as transport in the pore space around the particles are studied.

The interparticle macropores (void between solid reactant particles), and the intraparticle micropores (pores inside the solid reactant particles) result in a system with bimodal porosity.

The main feature of the interface is the [Pellets](#) which is the subnode to the **Packed Bed** node. With the **Pellet Type** parameter being changed from **Porous catalyst** to **Shrinking Core model**, the **Transport of Diluted Species in Packed Beds** interface switches to the

Transport of Diluted Species in Packed Beds, Shrinking Core interface. For the **Transport of Diluted Species in Packed Beds, Shrinking Core** interface, the governing equation for core shrinking is,

$$\frac{dr_{core}}{dt} = R_{solid} \frac{M_{solid}}{\rho_{pe}}$$

where r_{core} (SI unit: m) denotes the radius of the reactant pellet, R_{solid} (SI unit: mol/(m²·s)) heterogeneous reaction rate on the reactant pellet surface, M_{solid} (SI unit: kg/mol) the molar mass of the solid reactant, and ρ_{pe} (SI unit: kg/m³) the reactant pellet density.

With the unreacted solid core decreased in size over time, the reacted porous layer grows thicker. The fluid (often gas) reactant moves from an external fluid through the reacted porous layer to the solid core surface, and the product from the heterogeneous reaction leaves the solid core surface through the reacted porous layer to the external fluid. The mass transport phenomena and possible reactions in the reacted porous layer are modeled in the same way as that in [The Transport of Diluted Species in Packed Beds Interface](#).

The Transport of Concentrated Species Interface

The **Transport of Concentrated Species (tcs)** interface () found under the **Chemical Species Transport** branch () when adding a physics interface, is used to study gaseous and liquid mixtures where the species concentrations are of the same order of magnitude and none of the species can be identified as a solvent. In this case, properties of the mixture depend on the composition, and the molecular and ionic interactions between all species need to be considered. The physics interface includes models for multicomponent diffusion, where the diffusive driving force of each species depends on the mixture composition, temperature, and pressure.

The physics interface solves for the mass fractions of all participating species. Transport through convection, diffusion, and migration in an electric field can be included.

It supports simulations of transport by convection, migration, and diffusion in 1D, 2D, and 3D as well as for axisymmetric components in 1D and 2D. The physics interface defines the equations for the species mass fractions, including a diffusion model (Mixture-averaged, Maxwell–Stefan, or Fick’s law).

The available transport mechanisms and diffusion models differs between various COMSOL products (see www.comsol.com/products/specifications/).

Some examples of what can be studied with this physics interface include:

- The evolution of a chemical species transported by convection and diffusion.
- The migration in an electric field in the case of ionic species, in mixtures and solutions that cannot be deemed as being diluted.
- Concentrated solutions or gas mixtures, where the concentration of all participating species are of the same order of magnitude, and their molecular and ionic interaction with each other therefore must be considered. This implies that the diffusive transport of a single species is dependent on the mixture composition, and possibly on the temperature, the electric potential, the pressure, or any combination.

When this physics interface is added, the following default nodes are also added in the **Model Builder** — **Fluid, No Flux**, and **Initial Values**. Then, from the **Physics** toolbar, add other nodes that implement, for example, boundary conditions and reactions. You can also right-click **Transport of Concentrated Species** to select physics features from the context menu.

SETTINGS

The **Label** is the default physics interface name.

The **Name** is used primarily as a scope prefix for variables defined by the physics interface. Refer to such physics interface variables in expressions using the pattern `<name>.<variable_name>`. In order to distinguish between variables belonging to different physics interfaces, the **name** string must be unique. Only letters, numbers, and underscores (_) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first physics interface in the model) is **tcs**.

EQUATION

The basic equation for the conservation of mass of a species i is:

$$\frac{\partial}{\partial t}(\rho\omega_i) + \nabla \cdot (\rho\omega_i\mathbf{u}) = -\nabla \cdot \mathbf{j}_i + R_i \quad (3-4)$$

Using the continuity equation, the sum of all species equations, the equation can be recast in its nonconservative form:

$$\rho \frac{\partial}{\partial t}(\omega_i) + \rho(\mathbf{u} \cdot \nabla)\omega_i = -\nabla \cdot \mathbf{j}_i + R_i \quad (3-5)$$

This form of the equation is the one used in the Transport of Concentrated Species interface. The equation displayed in the interface changes depending on the active transport mechanisms, the selected diffusion model, and the type of reactions modeled.

OUT-OF-PLANE THICKNESS

For 2D components, the **Thickness** field (default value: 1 m) defines a parameter for the thickness of the geometry perpendicular to the two-dimensional cross-section. Both constant and varying thicknesses are supported. The value of this parameter is used, among other things, to automatically calculate mass flow rates from the total mass flux.

CROSS-SECTIONAL AREA

For 1D components, enter a **Cross-sectional area** A_c (SI unit: m^2) to define a parameter for the area of the geometry perpendicular to the 1D component. Both constant and varying areas are supported. The value of this parameter is used, among other things, to automatically calculate mass flow rates from the total mass flux. The default is 1 m^2 .

TRANSPORT MECHANISMS

The Transport of Concentrated Species interface always accounts for transport due to convection and diffusion.

The available diffusion models and the additional transport mechanisms differs between various COMSOL products (see www.comsol.com/products/specifications/).

Diffusion Model

- The **Maxwell–Stefan** option employs the most detailed diffusion model, but is also the most computationally expensive. The model is intended for diffusion dominated models, and requires that the multicomponent Maxwell–Stefan diffusivities of all component pairs are known.
- The **Mixture-averaged** option is less computationally expensive than the Maxwell–Stefan model. It is a simpler model that can be used when variations in the partial pressures and temperature can be assumed to not affect the multicomponent diffusion. The model includes stabilization but requires the multicomponent Maxwell–Stefan diffusivities of all component pairs.
- The **Fick’s law** model is a general model that should be used when the diffusion is assumed Fickian, or when no multicomponent diffusivities are available. Also, when molecular diffusion is not the dominating transport mechanism and a robust but low order model is wanted, the Fick’s law options should be used. The model includes stabilization.

Additional Transport Mechanisms

Under **Additional transport mechanisms**, click to select or clear any combination of checkboxes as needed.

Migration in Electric Field

Select the **Migration in electric field** checkbox to activate migration of ionic species due to an electric field. The resulting migration term is part of the relative mass flux vector.

Mass Transport in Porous Media

The **Mass transport in porous media** checkbox activates functionality specific to species transport in porous media. When selected the following domain features are enabled:

- [Porous Medium](#)
- [Porous Catalyst](#)
- [Packed Bed \(Moving Packed Bed\)](#)
- [Porous Electrode Coupling](#)

Maxwell–Stefan Diffusion Model

When using the **Maxwell–Stefan** diffusion model the relative mass flux vector is

$$\mathbf{j}_i = -\rho\omega_i \sum_{k=1}^Q \tilde{D}_{ik} \mathbf{d}_k - \frac{D_i^T}{T} \nabla T$$

where $\tilde{D}_{ik}$ (SI unit: m²/s) are the multicomponent Fick diffusivities, $\mathbf{d}_k$ (SI unit: 1/m) is the diffusional driving force, T (SI unit: K) is the temperature, and D_i^T (SI unit: kg/(m·s)) is the thermal diffusion coefficient.

The diffusional driving force is defined as

$$\mathbf{d}_k = \nabla x_k + \frac{1}{p} \left[(x_k - \omega_k) \nabla p - \rho\omega_k \mathbf{g}_k + \omega_k \sum_{l=1}^Q \rho\omega_l \mathbf{g}_l \right] \quad (3-6)$$

where $\mathbf{g}_k$ is an external force (per unit mass) acting on species k . In the case of an ionic species, the external force due to the electric field, which is added by selecting the **Migration in electric field** checkbox, is

$$\mathbf{g}_k = -\frac{z_k F}{M_k} \nabla \phi \quad (3-7)$$

where z_k is the species charge number, F (SI unit: A·s/mol) is Faraday's constant and ϕ (SI unit: V) is the electric potential.

Mixture-Averaged Diffusion Model

When using the **Mixture-averaged** diffusion model, the diffusive flux is formulated in terms of a mixture-averaged diffusion coefficient representing the diffusion of each species into the resulting mixture. The diffusion coefficient is based on the multicomponent Maxwell–Stefan diffusivities $\tilde{D}_{ik}$. The **Mixture-averaged** diffusion model is computationally less expensive, and significantly more robust than the [Maxwell–Stefan Diffusion Model](#), but constitutes an approximation of the multicomponent flux. For information on the flux formulation in this case see [Multicomponent Diffusion: Mixture-Averaged Approximation](#).

Fick's Law Diffusion Model

When using the **Fick's law** diffusion model, the diffusive flux is formulated in terms of a Fickian diffusion coefficient. The **Fick's law** diffusion model is computationally less expensive and significantly more robust than the [Maxwell–Stefan Diffusion Model](#), but

constitutes an approximation of the multicomponent flux. For information on the flux formulation in this case see [Multispecies Diffusion: Fick's Law Approximation](#).

SPECIES

Select the species that this physics interface solves for using the mass constraint in [Equation 3-8](#) (that is, its value comes from the fact that the sum of all mass fractions must equal 1). In the **From mass constraint** list, select the preferred species. To minimize the impact of any numerical and model introduced errors, use the species with the highest concentration. By default, the first species is used.

$$\omega_1 = 1 - \sum_{i=2}^Q \omega_i \quad (3-8)$$

CONSISTENT AND INCONSISTENT STABILIZATION

To display this section, click the **Show** button () and select **Stabilization**.

- Two consistent stabilization methods are available — **Streamline diffusion** and **Crosswind diffusion**. Both are active by default.

The **Residual** setting applies to both the consistent stabilization methods.

Approximate residual is the default setting and it means that derivatives of the diffusion tensor components are neglected. This setting is usually accurate enough and computationally faster. If required, select **Full residual** instead.

- There is one inconsistent stabilization method, **Isotropic diffusion**, which is available when using the [Mixture-Averaged Diffusion Model](#) or [Fick's Law Diffusion Model](#).

ADVANCED SETTINGS

To display this section, click the **Show** button () and select **Advanced Physics Options**. Normally these settings do not need to be changed.

Regularization

From the **Regularization** list, select **On** (the default) or **Off**. When turned **On**, regularized mass fractions are calculated such that

$$0 \leq w_{i, \text{reg}} \leq 1$$

Regularized mass fractions are used for the calculation of composition-dependent material properties, such as the density.

Diffusion

The **Diffusion** settings are available for the approximate diffusion models Mixture-averaged and Fick’s law.

When the **Mixture diffusion correction** is enabled, a flux correction is added to ensure that the net diffusive flux is zero. This typically also mean that the solution becomes less sensitive to the species selected to be computed from the mass constraint in the [Species](#) section. More information on this correction is available in the theory section [Multicomponent Diffusion: Mixture-Averaged Approximation](#).

The **Diffusion flux type** list controls the whether the molecular flux is assumed proportional to the mole fraction or the mass fraction. See [Multicomponent Diffusion: Mixture-Averaged Approximation](#) or [Multispecies Diffusion: Fick’s Law Approximation](#) for information on the diffusive flux formulation.

Pseudo Time Stepping

The **Use pseudo time stepping for stationary equation form** option adds pseudo time derivatives to the equation when the **Stationary** equation form is used in order to speed up convergence. When selected, a CFL number expression should also be defined. For the default **Automatic** option, the local CFL number (from the Courant–Friedrichs–Lewy condition) is determined by a PID regulator. For more information, see [Pseudo Time Stepping for Mass Transport](#).

DISCRETIZATION

To display all settings available in this section, click the **Show** button () and select **Advanced Physics Options**.

For more information about these settings, see the *Discretization* section under [The Transport of Diluted Species Interface](#).

	In the <i>COMSOL Multiphysics Reference Manual</i> see Table 2-4 for links to common sections and Table 2-5 to common feature nodes. You can also search for information: press F1 to open the Help window or Ctrl+F1 to open the Documentation window.
	• Domain, Boundary, and Pair Nodes for the Transport of Concentrated Species Interface • Theory for the Transport of Concentrated Species Interface

	• <i>Species Transport in the Gas Diffusion Layers of a PEM</i>: Application Library path Fuel_Cell_and_Electrolyzer_Module/Fuel_Cells/pem_gdl_species_transport_2d Web link: www.comsol.com/model/species-transport-in-the-gas-diffusion-layers-of-a-pem-260
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DEPENDENT VARIABLES

Add or remove species in the model and also change the names of the dependent variables that represent the species concentrations.

Specify the **Number of species**. There must be at least two species. To add a single species, click the **Add concentration** button () under the table. To remove a species, select it in the list and click the **Remove concentration** button () under the table. Edit the names of the species directly in the table.

	The species are dependent variables, and their names must be unique with respect to all other dependent variables in the component.
---	--

	Adding a Chemical Species Transport Interface and Specifying the Number of Species
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The Transport of Concentrated Species in Porous Media Interface

This interface (), found under the **Chemical Species Transport** branch (), is used to calculate the chemical composition in a fluid mixture transported through the interstices of a porous media. The interface is the same as the *Transport of Concentrated Species* interface but it has a **Porous Medium** node added by default. All other features for porous media transport is always available.

This interface is dedicated to analyzing mass transport in porous media where the chemical species may be subjected to convection, diffusion, and migration in an electric field. The interface also includes reaction rate expressions and source terms for modeling chemical reactions in porous media. It is also possible to include regions with free flow.

The main feature of the interface is the **Porous Medium** node which adds the governing equations for the mass fractions of all present species. It has two subnodes, **Fluid** and

Porous Matrix, which in turn are used to define the physical properties corresponding to each phase, and to prescribe the transport properties. The latter are typically the flow field, and when applicable, the electric field. The **Porous Medium** node supports the use of a [Porous Material](#) (described in the *COMSOL Multiphysics Reference Manual*) for defining phase-specific properties, and for sharing them among different physics interfaces.

When this physics interface is added, the following default nodes are also added to the **Model Builder** — **Porous Medium, No Flux** (the default boundary condition), and **Initial Values**. Then, from the **Physics** toolbar, add other nodes that implement, for example, boundary conditions, reaction rate expressions, and species sources. You can also right-click **Transport of Concentrated Species in Porous Media** to select physics features from the context menu.

SETTINGS

The rest of the settings are the same as [The Transport of Concentrated Species Interface](#).

The Transport of Concentrated Species in Porous Catalysts Interface

This interface () is used to study the impact of adsorption, and heterogeneous reactions, as a gas or liquid is moving through the interstices of a porous medium. Adsorption and desorption of bulk species onto, or from, the fluid-solid interface can be studied, as well as heterogeneous reactions involving both bulk species and surface species bound to the fluid-solid interface. Volumetric reactions occurring in the pore space can also be included. The interface is the same as the *Transport of Concentrated Species* interface, but it has a [Porous Catalyst](#) node added by default. All other features for transport and reactions in porous media are also always available.

The main feature of the interface is the **Porous Catalyst** node which adds the governing equations for the mass fractions of all bulk species, and the equations for surface species. It has two subnodes, **Fluid** and **Porous Matrix**, which in turn are used to define the physical properties corresponding to each phase, and to prescribe the transport properties. The latter are typically the flow field, and when applicable, the electric field. The **Adsorption** and **Surface Reaction** subnodes can be added to implement adsorption/desorption and heterogeneous reactions occurring on the fluid-solid interface.

When this physics interface is added, the following default nodes are also added to the **Model Builder** — **Porous Catalyst, No Flux** (the default boundary condition), and **Initial Values**. Then, from the **Physics** toolbar, add other nodes that implement, for example,

boundary conditions, reaction rate expressions, and species sources. You can also right-click **Transport of Concentrated Species in Porous Catalyst** to select physics features from the context menu.

SETTINGS

The rest of the settings are the same as [The Transport of Concentrated Species Interface](#).

The Transport of Concentrated Species in Packed Beds Interface

This interface () is used to model reactors filled with catalytic pellets. The bed pellets are porous, forming a bimodal porosity in the reactor. In addition to transport in the pore space between the pellets, interparticle transport and reactions can be studied in detail. Both heterogeneous reactions and volumetric reactions can be studied within the pellets, as well as volumetric reactions in the pore space. The pellets are available in each point of the bed domain, and are defined using an extra dimension describing the pellet radius. The interface is the same as the *Transport of Concentrated Species* interface, but it has a **Packed Bed (Moving Packed Bed)** node added by default. All other features for transport and reactions in porous media are also always available.

The main feature of the interface is the **Packed Bed** node which adds the governing equations for the mass fractions of all bulk species, and the equations for surface species. It has two subnodes, **Fluid** and **Pellets**, which in turn are used to define the physical properties corresponding to each phase. The latter node is also used to define volumetric reactions, as well as heterogeneous reactions, occurring within the porous pellets. The **Packed Bed** node requires the use of a **Porous Material** (described in the *COMSOL Multiphysics Reference Manual*) for defining phase-specific properties, and for sharing them among different physics interfaces. A packed bed containing several pellet types can be modeled by adding multiple pellet nodes to the porous material.

When this physics interface is added, the following default nodes are also added to the **Model Builder** — **Packed Bed, No Flux** (the default boundary condition), and **Initial Values**. Then, from the **Physics** toolbar, add other nodes that implement, for example, boundary conditions, reaction rate expressions, and species sources. You can also right-click **Transport of Concentrated Species in Packed Beds** to select physics features from the context menu.

SETTINGS

The rest of the settings are the same as [The Transport of Concentrated Species Interface](#).

The Transport of Concentrated Species in Vapor Interface

This interface () is used to model mass transport to or from vapor mixtures in contact with liquids. The vapor transfer to or from water or any liquid mixture present on boundaries or in adjacent domains can be studied. The physics interface is the same as the *Transport of Concentrated Species* interface, but it includes three features for vapor transfer from liquids; [Vapor Inflow](#), [Vapor-Liquid Interface](#), and [Vapor-Liquid-Solution Interface](#). The first two features are added by default in the interface. All other features for transport and reactions in the Transport of Concentrated Species interface are also always available.

The Transport of Concentrated Species in Moving Packed Beds Interface

This interface () , found under the **Chemical Species Transport** branch () > **Reacting Flow in Porous Media** multiphysics node () > **Moving Packed Bed** group node, is used to model reactors where the bed, comprised of packed porous catalytic pellets, move relative to the reactor. In addition to transport in the pore space between the pellets, intraparticle transport of chemical species and reactions are also studied.

The interparticle macropores (void between pellets), and the intraparticle micropores (pores inside the pellets) result in a system with bimodal porosity. In the **Transport of Concentrated Species in Moving Packed Beds** interface, the solid pellet bed is usually moving downward, counter-currently to the moving fluid between catalytic pellets.

The main feature of the interface is the [Packed Bed \(Moving Packed Bed\)](#) node. With the **Bed Type** parameter being changed from **Packed bed** to **Moving packed bed**, the **Transport of Concentrated Species in Packed Beds** interface switches to the **Transport of Concentrated Species in Moving Packed Beds** interface. For the **Transport of concentrated Species in Moving Packed Beds** interface, in the mass transport governing equations, there is an extra convection term from the contribution of bed moving,

$$\rho \mathbf{u}_{mb} \cdot \nabla (\epsilon_{pe} (1 - \epsilon_p) \omega_{pe,i})$$

For mass fraction of species i inside the reacted porous layer,

$$\epsilon_{pe}\rho\frac{\partial\omega_{pe,i}}{\partial t} + \frac{1}{r_{pe}^2}\frac{\partial}{\partial r}(r_{pe}^2j_i) + \frac{1}{r_{pe}}\rho u_r\frac{\partial\omega_{pe,i}}{\partial r} + \rho\mathbf{u}_{mb} \bullet \nabla(\epsilon_{pe}(1-\epsilon_p)\omega_{pe,i}) = R_{pe,i}$$

For the density in the reacted porous layer,

$$\epsilon_{pe}\frac{\partial\rho_{pe}}{\partial t} + \frac{1}{r_{pe}^2}u_r\rho_{pe} + \rho\mathbf{u}_{mb} \bullet \nabla(\epsilon_{pe}(1-\epsilon_p)\omega_{pe,i}) = Q_m$$

Domain, Boundary, and Pair Nodes for the Transport of Concentrated Species Interface

The [Transport of Concentrated Species Interface](#) has these domain, boundary, and pair nodes, listed in alphabetical order, available from the **Physics** ribbon toolbar (Windows users), **Physics** context menu (Mac or Linux users), or right-click to access the context menu (all users).



In general, to add a node, go to the **Physics** toolbar, no matter what operating system you are using. Subnodes are available by clicking the parent node and selecting it from the **Attributes** menu.



In the *COMSOL Multiphysics Reference Manual* see [Table 2-4](#) for links to common sections and [Table 2-5](#) to common feature nodes. You can also search for information: press F1 to open the **Help** window or Ctrl+F1 to open the **Documentation** window.



For axisymmetric components, COMSOL Multiphysics takes the axial symmetry boundaries (at $r = 0$) into account and automatically adds an **Axial Symmetry** node to the component that is valid on the axial symmetry boundaries only.

These nodes are described in this section or as indicated:

- Adsorption
- Bed Inlet
- Bed Outlet¹
- Bed Wall¹
- Electrode Surface Coupling
- Equilibrium Reaction
- Fluid (Porous Medium)
- Flux
- Flux Discontinuity
- Inflow
- Initial Values
- Initial Values (Pellets)
- Mass Fraction
- No Flux
- Open Boundary
- Out-of-Plane Flux
- Outflow
- Packed Bed (Moving Packed Bed)¹
- Surface concentration Initial Values
- Pellets¹
- Porous Matrix
- Porous Medium
- Porous Catalyst
- Reaction
- Reactions (Pellets)
- Reaction Sources
- Species Properties
- Surface Equilibrium Reaction
- Surface Reaction
- Symmetry
- Thin Impermeable Barrier¹
- Fluid
- Transport Properties (Pellets)
- Vapor Inflow
- Vapor–Liquid Interface
- Vapor–Liquid–Solution Interface

Some features require certain add-on modules. See details www.comsol.com/products/specifications/

¹ The node is described for the Transport of Diluted Species interface.

Prescribing Conditions on Fluid–Solid Interfaces

Defining conditions on interfaces between different phases, typically a wall in contact with a fluid, it is worthwhile to be careful with how variables in the conditions are evaluated. COMSOL Multiphysics supports that variables are defined differently on adjacent domains (or any other geometry dimension). Evaluating a variable with

differing definitions, the mean value is returned on a boundary separating the different domains. The density, pressure, or temperature are examples of variables that may differ in a fluid and a solid. In order to specify on which side a variable is evaluated, the [up and down](#) operators (described in the *COMSOL Multiphysics Reference Manual*) can be used.

Species Properties

Use this node to define parameters specific to each of the species.

The node will change its label depending on what inputs are available in the node. If only **Molar Mass** is available, it will be labeled “Species Molar Masses”.

MOLAR MASS

Enter a value or expression for the **Molar mass** M_w for each species. The default value is 0.032 kg/mol, which is the molar mass of O₂ gas.

CHARGE

This section is available when the **Migration in electric field** checkbox is selected in the [Transport Mechanisms](#) section of the interface. Enter the charge number z_w (dimensionless, but requires a plus or minus sign) for each species.

Fluid

The **Fluid** node is the main node used to model mass transfer in a fluid mixture with the Transport of Concentrates species interface. The node adds the equations governing the mass fractions of all present species, and provides inputs for the transport mechanisms and for the material properties of the fluid mixture.

The settings in this node are dependent on the checkboxes selected under [Transport Mechanisms](#) in the Settings window of the Transport of Concentrated Species interface.

The options available in this feature differs between COMSOL products. (See www.comsol.com/products/specifications/).

MODEL INPUTS

Specify the temperature and pressure to be used in the physics interface. The temperature model input is used when calculating the density from the ideal gas law, but also when thermal diffusion is accounted for by supplying thermal diffusion

coefficients. The pressure model input is used in the diffusional driving force in [Equation 3-6](#) (that is, when a [Maxwell–Stefan Diffusion Model](#) is used) and when calculating the density from the ideal gas law.

Temperature

Select the source of the **Temperature** field T :

- Select **User defined** to enter a value or an expression for the temperature (SI unit: K). This input is always available.
- If required, select a temperature defined by a Heat Transfer interface present in the model (if any). For example, select **Temperature (ht)** to use the temperature defined by the Heat Transfer in Fluids interface with the ht name.

Absolute Pressure

Select the source of the **Absolute pressure** p :

- Select **User defined** to enter a value or an expression for the absolute pressure (SI unit: Pa). This input is always available.
- In addition, select a pressure defined by a Fluid Flow interface present in the model (if any). For example, select **Absolute pressure (spf)** to use the pressure defined in a Laminar Flow interface with spf as the name.

DENSITY

Define the density of the mixture and the molar masses of the participating species.

Mixture Density

Select a way to define the density from the **Mixture density** list — **Ideal gas** or **User defined**:

- For **Ideal gas**, the density is computed from the ideal gas law in the manner of:

$$\rho = \frac{pM}{R_g T}$$

Here M is the mean molar mass of the mixture and R_g is the universal gas constant. The absolute pressure, p , and temperature, T , used corresponds to the ones defined in the [Model Inputs](#) section.

- For **User defined** enter a value or expression for the **Mixture density** ρ .

CONVECTION

Select the source of the **Velocity field $\mathbf{u}$** :

- Select **User defined** to enter manually defined values or expressions for the velocity components. This input is always available.
- Select a velocity field defined by a Fluid Flow interface present in the model (if any). For example, select **Velocity field (spf)** to use the velocity field defined by the **Fluid Properties** node fp1 in a Single-Phase Flow, Laminar Flow interface with spf as the Name.

DIFFUSION

Specify the molecular and thermal diffusivities of the present species based on the selected **Diffusion model**.

When using a [Maxwell–Stefan Diffusion Model](#) or a [Mixture-Averaged Diffusion Model](#), select the **Binary diffusion input type** (Table or Matrix) and specify the **Maxwell–Stefan diffusivities** in the table or matrix, then enter the **Thermal diffusion coefficients** D_w^T .

When using a [Fick’s Law Diffusion Model](#), specify the **Diffusion coefficient** D_w^F and the **Thermal diffusion coefficients** D_w^T for each of the species.

Maxwell–Stefan Diffusivity Matrix

Using a [Maxwell–Stefan Diffusion Model](#) or a [Mixture-Averaged Diffusion Model](#), the **Maxwell–Stefan diffusivity matrix** D_{ik} (SI unit: m^2/s) can be specified by a table or matrix. For a simulation involving Q species the Maxwell–Stefan diffusivity matrix is a Q -by- Q symmetric matrix, where the diagonal components are 1. Enter values for the upper triangular components, D_{ij} , which describe the interdiffusion between species i and j . For the table input type, only upper triangular components (D_{ij}) are listed. The name of species pair consists of species in the first and second column. For the matrix input type, the numbering of the species corresponds to the order, from top to bottom, used for all the input fields for species properties (see for example the molar mass fields in the [Density](#) section). The Maxwell–Stefan diffusivity matrix is used to compute the multicomponent Fick diffusivities as described in [Multicomponent Diffusivities](#).

Diffusion Coefficient

Using a [Fick’s Law Diffusion Model](#), the diffusion is by default assumed to be isotropic and governed by one **Diffusion coefficient** D_w^F (SI unit: m^2/s) for each species. To allow for a general representation, it is also possible to use diffusion matrices (diagonal, symmetric, or anisotropic).

Thermal Diffusion Coefficient

To model thermal diffusion, prescribe the **Thermal diffusion coefficients** D_i^T (SI unit: kg/(m·s)), by entering one thermal diffusion coefficient for each species in the corresponding field. In a multicomponent mixture, the sum of the thermal diffusion coefficients is zero. The default value for all thermal diffusion coefficients is 0.

Specify the molecular and thermal diffusivities of the present species based on the selected **Diffusion model**.

MIGRATION IN ELECTRIC FIELD

This section is available when the **Migration in electric field** checkbox is selected for the Transport of Concentrated Species interface.

Electric Potential

- Select **User defined** to enter a value or expression for the electric potential. This input is always available.
- If required, select an electric potential defined by an AC/DC interface that is present in the model (if any). For example, select **Electric potential (ec)** to use the electric field defined by the Current Conservation node cucn1 in an Electric Currents interface ec.

Settings for the mobilities are used for the **Mixture-averaged** and **Fick's law** transport models. By default the mobility is set to be calculated based on the species diffusivities and the temperature using the **Nernst–Einstein relation**. To manually specify the mobilities, select **User defined** for the mobility $u_{m,c}$ (SI unit: s·mol/kg) and enter one value for each species.

The temperature (if you are using mobilities based on the Nernst–Einstein relation) is taken from the **Model Inputs** section.

Porous Medium

Use this node to model the composition in a gas or liquid mixture as it is transported through the interstices of solid porous medium. In addition to transport due to convection and diffusion, the node contains functionality to include species evolution through reactions in the fluid phase.

Species transport in a porous medium is affected both by properties of the fluid phase as well as properties of the solid matrix. These properties are defined in the [Fluid \(Porous Medium\)](#) node and the [Porous Matrix](#) node respectively. The Porous Medium

node supports material properties using a [Porous Material](#) node (described in the *COMSOL Multiphysics Reference Manual*), where properties are defined per phase in a similar manner.

Bed Inlet

This node specifies an input for the pore conditions in the solid catalytic pellets in a moving packed bed. Together with it, a [Bed Outlet](#) node should be also added. The solid pellet bed is rigid, the **Bed Inlet** and **Bed Outlet** are usually set to opposite boundaries.

BOUNDARY SELECTION

Select boundary where the solid catalytic pellet bed moves into the reactor. These boundaries belong to the pellet domains which are defined in **Porous Material**.

INLET PELLET CONCENTRATIONS

There are three ways to specify the pellet inlet conditions.

- **From bed fluid conditions.** The species mass fractions inside the catalytic pellet are the same as that in the fluid (between pellets).
- **From pellet initial values.** The species mass fractions inside the catalytic pellet are the same as the initial values under the **Pellets** node.
- **User defined.** There are three different kinds of concentration inputs from the **Mixture specification** list.
 - **Mass fractions.** Enter the inlet mass fractions in the **Mass fractions** text fields.
 - **Mole fractions.** Enter the inlet mole fractions in the **Mole fractions** text fields. The temperature and pressure will be taken from the fluid between pellets at the boundary.
 - **Molar concentrations.** Enter the inlet molar concentrations $c_{pe0,i}$ (SI unit: mol/m³) in the **Molar concentrations** text fields. The pressure will be taken from the fluid between pellets at the boundary.

The surface species concentration table is available if there is any surface reaction inside the catalytic pellet and the inlet conditions is set to **From bed fluid conditions** or **User defined**. Enter the inlet surface concentration (SI unit: mol/m²) in the surface species table.

The surface species concentration table is not available when the inlet conditions is set to **From pellet initial values**. In this case, the inlet surface concentrations are taken from the **Initial Values** node.

Transport Properties (Pellets)

Use this node to specify transport properties of the fluid mixture inside porous pellets.

DOMAIN SELECTION

The domain selection inherits all valid domains from its parent feature. For the invalid domains, **not applicable** is shown.

FLUID AND MATRIX PROPERTIES

Select **Fluid density** — **Ideal gas**, or **User defined** to enter an expression for the density ρ of the fluid inside pellets. If a density variable is defined in Chemistry with the **Define variables for porous pellets** selected (in the **Pellet Chemistry** section), the density can be selected from the list.

Select **Dynamic viscosity** — **User defined** for the fluid inside pellets. Enter a number or expression for the viscosity μ . If a viscosity variable is defined in Chemistry with the **Define variables for porous pellets** selected (in the **Pellet Chemistry** section), the viscosity can be selected from this list.

Select **Permeability** — **From material** or **User defined** for the fluid inside pellets. If **User defined** is selected, enter a permeability expression for κ . The default value is $1 \cdot 10^{-12} \text{ m}^2$.

DIFFUSION

To account for the effect of porosity in the binary diffusivities, select an **Effective diffusivity model** — **Millington and Quirk model**, **Bruggeman model**, **Tortuosity model**, or **No correction**. Using one of the first three models, the effective transport factor, f_e , is defined from the porosity and the fluid tortuosity factor in the manner of:

$$f_e = \frac{\epsilon_{pe}}{\tau_{pe}} \quad (3-9)$$

For **No correction**, the effective transport factor is set to one.

- For the **Millington and Quirk model**, the effective transport factor is $\tau_{pe} = \epsilon_{pe}^{-1/3}$.
- For the **Bruggeman model**, the effective transport factor is $\tau_{pe} = \epsilon_{pe}^{-1/2}$.
- For the **Tortuosity model**, specify the tortuosity factor τ_{pe} .

The species diffusivities are automatically adjusted for porous media transport using the effective transport factor.

Select **Binary diffusion input type** — **Table** (the default), or **Matrix** to input the binary diffusion coefficients.

- **Table.** In the **Maxwell–Stefan diffusivities** table, the first (**Species 1**) and second (**Species 2**) columns are the species names for a binary diffusion coefficient. Select a diffusivity from the **Diffusivity** column. If **User defined** is selected, enter a binary diffusion coefficient expression in the **Diffusion coefficient (m²/s)** column (this column is editable with the **User defined** selected, otherwise read only).

If there are diffusivity variables defined in the Chemistry interface (with **Define variables for porous pellets** selected in the Pellet Chemistry section), these diffusivities can be selected from the **Diffusivity** column.

- **Matrix.** In the **Maxwell–Stefan diffusivity matrix**, enter the diffusivity expressions in the upper part of the matrix. The default values are $1 \cdot 10^{-5} \text{ m}^2/\text{s}$.

Enter the thermal diffusion coefficient for different species under the **Thermal diffusion coefficient** list.

Reactions (Pellets)

DOMAIN SELECTION

The domain selection inherits all valid domains from its parent feature. For the invalid domains, **not applicable** is shown.

Several Reactions nodes can be used to account for different reaction mechanisms in different domains and pellets in the porous material.

PELLET

This section is invisible for default Reactions node. It is visible for later added Reactions nodes.

Select **All** or one of existing pellets for the nondefault Reactions nodes.

REACTING VOLUME

A reaction takes usually place in the porous part (fluid) of the pellet, but can also in the whole volume of the pellet. When specifying reaction rate for a bulk species, the specified reaction rate has the basis of the pore volume, or the total volume. With the

Pore volume selected, the reaction expression in $\text{kg}/(\text{m}^3 \cdot \text{s})$ will be multiplied by pellet's porosity ε_{pe} .

REACTION RATES

Select a reaction rate (SI unit: $\text{kg}/(\text{m}^3 \cdot \text{s})$) for bulk species i from the $R_{\text{pe},i}$ list. Enter a reaction rate expression when **User defined** is selected from the list. Alternatively, if reaction rate expressions have been defined in a Chemistry interface (with the **Define variables for porous pellets** selected under Pellet Chemistry section), they can be chosen from the list.

SURFACE REACTION RATES

The section is available when one or more surface species have been added in the **Surface Species** section of its parent feature (Pellets).

Specify the **Reactive specific surface area** (total surface reaction area per cubic meter) $S_{\text{b, reac}}$ for all surface reactions.

Under **Surface reaction rate for bulk species**, specify the rate expression $R_{\text{pe},i}^s$ (SI unit: $\text{kg}/(\text{m}^2 \cdot \text{s})$) for the surface reaction rate of each bulk species i participating in the surface reaction. Furthermore, specify the surface reaction rates for the surface species in the **Reaction rate for surface species** table.

If several pellet types have been defined, one set of surface reaction rates per pellet type can be set by selecting specific pellet under **Pellet** section. If surface reaction rate expressions have been defined in a Chemistry interface (with the **Define variables for porous pellets** selected under Pellet Chemistry section), they can be chosen from the list.

HEAT SOURCE

Specify the **Heat source** originating from the heat of chemical reactions inside the pellet. Both heat source Q_{pe} from bulk reactions in the fluid, and heat source $Q_{\text{s, pe}}$ (available when there is one or more surface species) resulting from surface reactions can be defined. With several pellets being added, their heat sources will be summed up.

The heat sources are most conveniently picked up from a Chemistry interface (with the **Define variables for porous pellets** selected under Pellet Chemistry section) that defines the reaction rate and the heat of reactions. In that case, the heat source expression can be selected from the list. Otherwise you can enter an expression in the text field.

The heat source (inside pellet, in the extra dimension) in a Reactions node can be picked by Heat Source feature under the Pellets feature of heat transfer interface. The

total heat source (from all pellets, in the base domain) of all Reactions nodes can be used by a Heat Source feature in any of the heat transfer interfaces.

Initial Values (Pellets)

DOMAIN SELECTION

The domain selection inherits all valid domains from its parent feature. For the invalid domains, it shows **not applicable**.

Several Initial Values nodes can be used to set different initial values for different domains and pellets.

PELLET

This section is invisible for default Initial Values node. It is visible for later added Initial Values nodes.

Select **All** or one of existing pellets for the nondefault Initial Values nodes.

INITIAL VALUES

Specify the **Mixture specification** — **Mass fractions** (the default), **Mole fractions**, or **Molar concentrations** for different initial value types inside the pellets.

- **Mass fractions.** Enter the **Pressure** $P_{\text{ep}0}$ (Pa) for the initial pressure value, and enter the **Mass fractions** $w_{\text{pe}0,i}$ for species i .
- **Mole fractions.** Enter the **Pressure** $P_{\text{ep}0}$ (Pa) for the initial pressure value, and enter the **Mole fractions** $x_{\text{pe}0,i}$ for species i .
- **Molar concentrations.** Enter the **Temperature** $T_{\text{pe}0}$ (SI unit: K) for the initial temperature value. Specify the **Initial pressure** — **Ideal gas** (the default), or **User defined** for different types of initial pressure. Enter the **Pressure** $P_{\text{ep}0}$ (Pa) for the initial pressure value if **User defined** is selected. Enter the **Molar concentrations** $c_{\text{pe}0,i}$ for species i .

SURFACE CONCENTRATION INITIAL VALUES

Specify the initial values for **Surface species** under the column **Surface concentration** (mol/m^2) in the table.

DOMAIN SELECTION

The domain selection inherits all valid domains from its parent feature. For the invalid domains, it shows **not applicable**.

Several Pellet–Fluid Interface nodes can be used to specify different boundary conditions between fluid and pellet out-surface for different domains and pellets.

PELLET

This section is invisible for default Pellet–Fluid Interface node. It is visible for later added Pellet–Fluid Interface nodes.

Select **All** or one of existing pellets for the nondefault Pellet–Fluid Interface nodes.

PELLET–FLUID INTERFACE

The **Coupling type** list is selectable for the default Pellet–Fluid Interface node, but it is disabled for nondefault Pellet–Fluid Interface nodes because it is supposed that there should be only one type of boundary condition in a packed bed.

There are two **Coupling type** options for the coupling of concentration between the pellet internals and the surrounding fluid:

- **Continuous concentration**, assuming that all resistance to mass transfer to/from the pellet is within the pellet and no resistance to pellet–fluid mass transfer is on the bulk fluid side. The concentration in the fluid will thus be equal to that in the pellet pore just at the pellet surface: $c_{pe,i} = c_i$. This constraint also automatically ensures flux continuity between the internal pellet domain and the free fluid domain through so-called reaction forces in the finite element formulation.
- **Film resistance (mass flux)**: The flux of mass across the Pellet–Fluid interface into the pellet is possibly rate determined on the bulk fluid side, by film resistance. The resistance is expressed in terms of a film mass transfer coefficient, h_{Di} , such that: $N_{i,inward} = h_{Di}(c_i - c_{pe,i})$. The **Film resistance (mass flux)** option computes the inward surface flux, h_{Di} is the mass transfer coefficient (SI unit: m/s) and is calculated with the default **Automatic** setting from a dimensionless **Sherwood number expression** or with **User defined** mass transfer coefficients.

The **Active specific surface area** (SI unit: m^{-1}) is required to couple the mass transfer between the pellets and the bed fluid. Select either the **Automatic** setting that calculates the specific surface area from the shape information given above, or **User defined** that is available for explicit surface area specification.

Specify the **Mass transfer parameters** — **Automatic** (the default), or **User defined** to set the mass transfer parameters.

- **Mass transfer parameters** with **Automatic** selected. Specify one of the models from the **Sherwood number expression** list to compute the Sherwood number. There are three models available: **Frössling**, **Rosner**, and **Garner and Keey**. The Frössling equation is the default and probably the most commonly used for packed spheres. All of these are based on the dimensionless Reynolds (**Re**) and Schmidt (**Sc**) numbers, which are computed from **Density** and **Dynamic viscosity**. Both of these two parameters are specified in the Fluid node under the Packed Bed feature. They can be taken **From material** or choose the **User defined** alternative. They could also be picked up from Chemistry if the species in the Chemistry are coupled to that of the current Transport of Concentrated Species interface.
- **Mass transfer parameters** with **User defined** selected. Specify the type of **Parameter** — **Mass transfer coefficient** (the default), or **Sherwood number** to be user defined. Enter the mass transfer coefficients and Sherwood numbers for **Mass transfer coefficient** and **Sherwood number**, respectively.

Porous Catalyst

The catalyst is a kind of substance that increases the rate of a reaction without itself being consumed. The catalytic process usually consists of following steps:

- Adsorbate transfers across the fluid-adsorbent boundary layer to adsorbent surface;
- Adsorbate is absorbed on the adsorbent surface;
- Chemical reaction of adsorbate on the surface;
- Desorption of reaction products (adsorbate) from the adsorbent surface into the fluid-adsorbent boundary layer;
- Adsorbate transfers from the boundary layer into bulk fluid.

if the resistance in boundary layer is neglected, the five steps above may be reduced to three steps:

- Adsorption on to the adsorbent surface;
- Surface reaction at adsorbent surface;
- Desorption from adsorbent surface.

Use this feature to model a catalytic process where it concerns surface reaction and adsorption. It has four subnodes. Two of them are default: Fluid and Porous Matrix, and other two are selective: Surface Reaction and Adsorption.

REACTIVE SPECIFIC SURFACE AREA

Enter a specific surface area for the adsorption and surface reaction processes. The default value is $1 \cdot 10^7$ 1/m.

CONCENTRATION INPUT

Select a **Concentration input** — **Molar concentration (mol/(m²))** (the default), **Amount per solid mass (mol/kg)**. This parameter determines the unit for initial values of adsorbed species and user-defined surface species.

ADSORBED SPECIES

Specify these species which are adsorbate to the catalyst (adsorbent). The concentrations (for example, *cads_i* for species *i*) of adsorbed species are defined as dependent variables and solved. The rate of adsorption is defined in Adsorptions subnode. With an adsorbed species being selected, its initial value field is activated. Enter the initial value according to the concentration type which is decided by the selected value from **Concentration input**.

SURFACE SPECIES

There could be extra surface species besides the adsorbed species (coupled to physics interface dependent variables) and their corresponding adsorbed sites on adsorbent surface. These extra surface species (also called user-defined species) could take part in catalytic reactions. The concentrations (for example, *csurf_j* for species *j*) of them are solved. The surface reaction rate is defined in the Surface Reaction subnode.

Add a user-defined surface species in the **Surface species** table by clicking the **Add** button under the table. The default name is *cs<n>* (n is a series number). The name can be changed into any valid name.

Vapor Inflow

Use this node to model a vapor inflow originating from an upstream gas mixture in contact with a liquid. The upstream liquid can either be water or a liquid solution involving any number of components.

VAPOR INFLOW

Select an entry from the **Upstream liquid system** list. The available options are **Water**, **Thermochemistry coupling**, and **User defined**.

- Use **Water** to model incoming water vapor. Saturated vapor, or vapor at a specific relative humidity can be applied.

- **Thermochemistry coupling** can be selected in models that contains a **Thermodynamic System** coupled to a **Chemistry** interface. In this case all thermodynamic properties needed (vapor pressure or fugacity, heat of evaporation, liquid phase density) are automatically added to the **Thermodynamic system**.
- Selecting **User defined** in the **Upstream liquid system** list, the vapor pressure, P_i , and for each of the species can be manually defined.

For **Water** and **User defined** input the **Upstream temperature** T_{ustr} . For **Thermochemistry coupling** also input the **Upstream absolute pressure** P_{ustr} .

Below the settings are detailed for the respective entries of the **Upstream liquid system**.

- When **Water** is selected, use the **Water vapor species** list to identify the species solved for that corresponds to water vapor. When needed select to define a **Relative humidity**.
- For **Thermochemistry coupling** also select an interface from the **Chemistry** list. In order for a **Chemistry** interface to be available, it needs fulfill the following criteria:
 - The interface should include at least one species.
 - The interface should be set to consider a concentrated mixture. This is applied by selecting **Concentrated species** in the **Type** list at the top of the **Mixture Properties** section (in the settings window of the interface).
 - The interface should be coupled to a **Thermodynamics System**. In the **Mixture Properties** section, select **Thermodynamics** and chose an appropriate entry from the **Thermodynamic system** list.
 - The selected system needs to include both a vapor and a liquid phase.
 - Gas phase mixture properties should be produced by selecting **Gas** in the **Phase** list of the **Mixture Properties** section.
 - The interface cannot have **Define variables for porous pellets** enabled in the **Pellet Chemistry** section.

Vapor Species in Equilibrium with Upstream Liquid

When a **Thermochemistry coupling** is used, select the vapor species which are in equilibrium with the liquid using the checkboxes next to the respective species. When a species is correctly coupled, the mass fraction variable as well as the chemical formula and compound name is indicated, for example as “wA, C3H6O (acetone)”. If the species is not correctly coupled, the corresponding label is “wA, Not coupled”. To inspect the coupling, use the **Go to source** button to navigate to the **Chemistry** interface. In the **Species Matching** section make sure that each species is coupled both to a mass fraction solved for as well as to a species **From thermodynamics**.

Other Species

For the species that are not in equilibrium with the liquid, specify the vapor pressure at the inlet. This could for example be species which are not soluble in the upstream liquid.

Liquid Phase

Use this section to specify the composition in the **Upstream liquid system** when using a **Thermochemistry coupling**.

Select a **Liquid phase concentration type** — **Mass fraction**, **Molar fraction**, or **Volume Fraction**. Use the **Composition** table to specify the corresponding concentration of each species in the liquid.

TRANSIENT INITIALIZATION

When prescribing boundary conditions in time-dependent problems, and in particular vapor pressure conditions, it is important not to introduce a discontinuity with respect to the conditions in the domain. When this happens the solver may struggle to converge and resort to small time steps. When using transient initialization a smooth step function, f_s , transitions the boundary condition, from the domain initial value, ω_0 , to the condition to be applied on the boundary, ω_{eq} , over a small time span Δt . The smoothed value applied on the boundary, ω_{bnd} , is defined as

$$\omega_{bnd} = \omega_{eq} f_s(t) + \omega_0 (1 - f_s(t))$$

where

$$f_s(t_0) = 0, f_s(t_0 + \Delta t) = 1 \quad (3-10)$$

When **Duration** is set to **Automatic** the initialization time is automatically defined as a fraction of the time between the initial and the last output

$$\Delta t = 1 \cdot 10^{-3} (t_{End} - t_0)$$

Select **User defined** to manually define the duration of the transient initialization Δt .

Vapor-Liquid Interface

Use this feature to account for evaporation or condensation at a vapor-liquid interface. The feature prescribes the vapor pressure as a function of the composition in the liquid and the temperature and pressure at the interface. Single species liquids as well as multi component liquid mixtures can be modeled. Local concentration gradients introduced

by the vapor pressure result in a species transfer to or from the interface. A vapor pressure higher than the partial pressure in the surrounding domain will result in evaporation of the species in question. Correspondingly, when the vapor pressure is lower than the surrounding partial pressure, condensation occurs.

The feature can be used on exterior boundaries to solve for mass transfer due to evaporation or condensation. The resulting mass flow and latent heat of evaporation are defined and announced and available for use in physics interfaces for fluid flow and heat transfer. The Vapor–Liquid Interface feature solves for mass transfer to or from a gas phase domain. To also model the resulting composition changes in an adjacent liquid phase, use the [Vapor–Liquid–Solution Interface](#).

To solve for the motion of the vapor–liquid interface, resulting from the phase transfer, use a fluid flow interface including a [Fluid–Fluid Interface](#) feature (described in the *CFD Module User’s Guide*). When solving for heat transfer apply the latent heat of evaporation using a [Boundary Heat Source](#) feature (described in the *Heat Transfer Module User’s Guide*).

The Vapor–Liquid Interface feature can also be used on stationary boundaries. This is for example efficient when the mass transfer is low and would only lead to insignificant variations of the interface position.

BOUNDARY SELECTION

Select the boundaries that represent an interface separating a vapor and liquid phase domain. The boundaries should be exterior to the domains where vapor transport modeled.

VAPOR EQUILIBRIUM

Use this section to select how the equilibrium vapor pressure is defined. The vapor pressure is applied on the vapor side of the interface and drives the condensation or evaporation.

Vapor pressure

In the **Vapor property** list, select **Vapor pressure** to prescribe the corresponding quantity at the vapor–liquid interface. It is also possible to instead prescribe the vapor phase [Mass Fraction](#).

The composition in the liquid phase is controlled from the **Liquid** list:

- Use **Water** to model condensation or evaporation to or from liquid water. In this case predefined expressions for the saturated vapor pressure and heat of evaporation are used.

- **Thermochemistry coupling** can be selected in models that contains a **Thermodynamic System** coupled to a **Chemistry** interface. In this case all thermodynamic properties needed at the interface (vapor pressure or fugacity, heat of evaporation, liquid phase density) are automatically added to the **Thermodynamic system**.
- Selecting **User defined** from the **Liquid** list, the equilibrium vapor pressure, $P_{\text{vap},i}$, and heat of vaporization $H_{\text{vap},i}$, for evaporating or condensing species can be manually defined.

Below the settings are detailed for the respective entries in the **Liquid** list.

- When **Water** is selected, use the **Water vapor species** list to identify the species solved for that corresponds to water vapor. The saturated water vapor pressure is prescribed to the species selected.
- For **Thermochemistry coupling** also select an interface from the **Chemistry** list. In order for a **Chemistry** interface to be available, it needs fulfill the following criteria:
 - The interface should include at least one species.
 - The interface should be set to consider a concentrated mixture. This is applied by selecting **Concentrated species** in the **Type** list at the top of the **Mixture Properties** section (in the settings window of the interface).
 - The interface should be coupled to a **Thermodynamics System**. In the **Mixture Properties** section, select **Thermodynamics** and chose an appropriate entry from the **Thermodynamic system** list.
 - The selected system needs to include both a vapor and a liquid phase.
 - Gas phase mixture properties should be produced by selecting **Gas** in the **Phase** list of the **Mixture Properties** section.
 - The interface cannot have **Define variables for porous pellets** enabled in the **Pellet Chemistry** section.

Evaporating/Condensing Species

When a **Thermochemistry coupling** is used, select the evaporating or condensing species using the checkboxes next to the respective species. When a species is correctly coupled, the mass fraction variable as well as the chemical formula and compound name is indicated, for example as “wA, C₃H₆O (acetone)”. If the species is not correctly coupled, the corresponding label is “wA, Not coupled”. To inspect the coupling, use the **Go to source** button to navigate to the **Chemistry** interface. In the **Species Matching** section make sure that each species is coupled both to a mass fraction solved for as well as to a species **From thermodynamics**.

Liquid Phase

Use this section to specify the liquid phase composition when using a **Thermochemistry coupling**.

Select a **Liquid phase concentration type** — **Mass fraction**, **Molar fraction**, or **Volume Fraction**. Use the **Composition** table to specify the corresponding concentration of each species in the liquid.

Mass Fraction

To define the vapor side condition directly, select **Mass fraction** from the **Vapor property** list. For each of the evaporating or condensing species, specify the equilibrium mass fraction, $\omega_{0,i}$, and heat of vaporization $H_{\text{vap},i}$.

TRANSIENT INITIALIZATION

When prescribing boundary conditions in time-dependent problems, and in particular vapor pressure conditions, it is important not to introduce a discontinuity with respect to the conditions in the domain. When this happens the solver may struggle to converge and resort to small time steps. When using transient initialization a smooth step function, f_s , transitions the boundary condition, from the domain initial value, ω_0 , to the condition to be applied in the boundary, ω_{eq} , over a small time span Δt . The smoothed value applied on the boundary, ω_{bnd} , is defined as

$$\omega_{\text{bnd}} = \omega_{\text{eq}} f_s(t) + \omega_0 (1 - f_s(t))$$

where

$$f_s(t_0) = 0, f_s(t_0 + \Delta t) = 1 \quad (3-11)$$

When **Duration** is set to **Automatic** the initialization time is automatically defined as a fraction of the time between the initial and the last output

$$\Delta t = 1 \cdot 10^{-3} (t_{\text{End}} - t_0)$$

Select **User defined** to manually define the duration of the transient initialization Δt .

Vapor–Liquid–Solution Interface

Use this feature to account for evaporation or condensation at a vapor–liquid interface when the liquid is composed of multiple species. The feature prescribes the vapor pressure as a function of the composition in the liquid and the temperature and pressure at the interface. Local concentration gradients introduced by the vapor

pressure results in a transport of species to or from the interface. A vapor pressure higher than the partial pressure in the surrounding domain will result in evaporation of the species in question. Correspondingly, in regions where the vapor pressure is lower than in the surrounding partial pressure, condensation occurs. The mass transfer to or from the vapor domain is accounted for by applying a corresponding fluxes for the species on the liquid side.

The feature can be used on interior boundaries to solve for mass phase transfer due to evaporation or condensation. The resulting mass flow and latent heat of evaporation are defined and announced and available for use in physics interfaces for fluid flow and heat transfer. The Vapor–Liquid–Solution Interface feature solves for the mass transfer to or from a gas phase as well as the composition changes in the adjacent liquid. When the liquid phase contains a single species it is more efficient to use the [Vapor–Liquid Interface](#) feature.

To solve for the motion of the vapor–liquid interface, resulting from the phase transfer, use a fluid flow physics interface including a [Fluid–Fluid Interface](#) feature (described in the *CFD Module User’s Guide*). When solving for heat transfer apply the latent heat of evaporation using a [Boundary Heat Source](#) feature (described in the *Heat Transfer Module User’s Guide*).

BOUNDARY SELECTION

Select the interior boundaries that represent an interface separating a vapor and liquid phase domain.

VAPOR EQUILIBRIUM

Use this section to select how the equilibrium vapor pressure is defined. The vapor pressure is applied on the vapor side of the interface and drives the condensation or evaporation.

Vapor pressure

In the **Vapor property** list, select **Vapor pressure** to prescribe the corresponding quantity at the vapor–liquid interface. It is also possible to instead prescribe the vapor phase [Mass Fraction](#).

The composition in the liquid phase is controlled from the **Liquid** list:

- Use **Water** to model condensation or evaporation to or from a liquid water domain. In this case predefined expressions for the saturated vapor pressure and heat of evaporation are used.

- **Thermochemistry coupling** can be selected in models that contains a **Thermodynamic System** coupled to a **Chemistry** interface. In this case all thermodynamic properties needed at the interface (vapor pressure or fugacity, heat of evaporation, liquid phase density) are automatically added to the **Thermodynamic system**.
- Selecting **User defined** from the **Liquid** list, the equilibrium vapor pressure, $P_{\text{vap},i}$, and heat of vaporization $H_{\text{vap},i}$, for evaporating or condensing species can be manually defined.

Below the settings are detailed for the respective entries in the **Liquid** list.

- When **Water** is selected, use the **Water vapor species** list to identify the species solved for that corresponds to water vapor. The saturated water vapor pressure is prescribed to the species selected.
- For **Thermochemistry coupling** also select an interface from the **Chemistry** list. In order for a **Chemistry** interface to be available, it needs fulfill the following criteria:
 - The interface should include at least one species.
 - The interface should be set to consider a concentrated mixture. This is applied by selecting **Concentrated species** in the **Type** list at the top of the **Mixture Properties** section (in the settings window of the interface).
 - The interface should be coupled to a **Thermodynamics System**. In the **Mixture Properties** section, select **Thermodynamics** and chose an appropriate entry from the **Thermodynamic system** list.
 - The selected system needs to include both a vapor and a liquid phase.
 - Gas phase mixture properties should be produced by selecting **Gas** in the **Phase** list of the **Mixture Properties** section.
 - The interface cannot have **Define variables for porous pellets** enabled in the **Pellet Chemistry** section.

Evaporating/Condensing Species

When a **Thermochemistry coupling** is used, select the evaporating or condensing species using the checkboxes next to the respective species. When a species is correctly coupled, the mass fraction variable as well as the chemical formula and compound name is indicated, for example as “wA, C3H6O (acetone)”. If the species is not correctly coupled, the corresponding label is “wA, Not coupled”. To inspect the coupling, use the **Go to source** button to navigate to the **Chemistry** interface. In the **Species Matching** section make sure that each species is coupled both to a mass fraction solved for as well as to a species **From thermodynamics**.

Liquid Side

Indicate the side in the interface that corresponds to a liquid using the red arrow in the Graphics interface. The arrow should point from the vapor into the liquid. Select **Reverse direction** when needed. Note, when modeling a moving interface, by using a [Fluid–Fluid Interface](#) feature (described in the *CFD Module User’s Guide*), the **Normal Direction** defined there needs to be aligned with the liquid side direction. When aligned the interface will move into the liquid during evaporation and into the vapor during condensation.

Mass Fraction

To define the vapor side condition directly, select **Mass fraction** from the **Vapor property** list. For each of the evaporating or condensing species, specify the equilibrium mass fraction, $\omega_{0,i}$, and heat of vaporization $H_{\text{vap},i}$.

TRANSIENT INITIALIZATION

When prescribing boundary conditions in time-dependent problems, and in particular vapor pressure conditions, it is important not to introduce a discontinuity with respect to the conditions in the domain. When this happens the solver may struggle to converge and resort to small time steps. When using transient initialization a smooth step function, f_s , transitions the boundary condition, from the domain initial value, ω_0 , to the condition to be applied in the boundary, ω_{eq} , over a small time span Δt . The smoothed value applied on the boundary, ω_{bnd} , is defined as

$$\omega_{\text{bnd}} = \omega_{\text{eq}} f_s(t) + \omega_0 (1 - f_s(t))$$

where

$$f_s(t_0) = 0, f_s(t_0 + \Delta t) = 1 \quad (3-12)$$

When **Duration** is set to **Automatic** the initialization time is automatically defined as a fraction of the time between the initial and the last output

$$\Delta t = 1 \cdot 10^{-3} (t_{\text{End}} - t_0)$$

Select **User defined** to manually define the duration of the transient initialization Δt .

Fluid (Porous Medium)

Use this node to specify the mass transport in a fluid phase filling the pores of a porous medium. It is used as a subnode to [Porous Medium](#), [Porous Catalyst](#), or [Packed Bed \(Moving Packed Bed\)](#).

MODEL INPUTS

Specify the temperature and pressure in the fluid. The temperature model input is used when calculating the density from the ideal gas law, but also when thermal diffusion is accounted for by supplying thermal diffusion coefficients. The pressure model input is used in the diffusional driving force in [Equation 3-6](#) (that is, when a [Maxwell–Stefan Diffusion Model](#) is used) and when calculating the density from the ideal gas law.

In a [Porous Catalyst](#) the temperature affecting is assumed to be the same in the fluid and the solid (thermal equilibrium). For temperature dependent reactions to use the same temperature as the fluid phase, the temperature is in this case given in the [Porous Catalyst](#) node.

Temperature

Select the source of the **Temperature** field T :

- Select **User defined** to enter a value or an expression for the temperature.
- When present, select a temperature defined by a Heat Transfer interface in the model. For example, select **Temperature (ht)** to use the temperature defined by the Heat Transfer in Fluids interface with the `ht` name.

Absolute Pressure

Select the source of the **Absolute pressure** p :

- Select **User defined** to enter a value or an expression for the absolute pressure.
- When present, select a pressure defined by a Fluid Flow interface present in the model. For example, select **Absolute pressure (spf)** to use the pressure defined in a Laminar Flow interface with `spf` as the name.

DENSITY

Use this section to define the density of the fluid phase, and to specify the molar masses of the participating species.

VISCOSITY

When the **Fluid** node is present in a **Packed Bed**, use this section to define the viscosity of the fluid phase. The fluid viscosity is used when solving for the flow inside the pellets. Select **From material** to use the viscosity defined in the material, or select **User defined** to enter a constant or expression in the corresponding input field.

Mixture Density

Select a way to define the density from the **Mixture density** list — **Ideal gas** or **User defined**:

- For **Ideal gas**, the density is computed from the ideal gas law in the manner of:

$$\rho = \frac{pM}{R_g T}$$

Here M is the mean molar mass of the mixture and R_g is the universal gas constant. The absolute pressure, p , and temperature, T , used corresponds to the ones defined in the **Model Inputs** section.

- For **User defined** enter a value or expression for the **Mixture density** ρ .

CONVECTION

Select the source of the **Velocity field** $\mathbf{u}$:

- Select **User defined** to enter values or expressions for the velocity components. This input is always available.
- Select a velocity field defined by a Fluid Flow interface that solves for the velocity of the fluid. For example, select **Velocity field (spf)** to use the velocity field defined by in a Single-Phase Flow, Laminar Flow interface with **spf** as the Name.

When the interface is used in a reacting flow multiphysics coupling, the velocity is automatically defined and the input is disabled.

DIFFUSION

Specify the species molecular and thermal diffusivities in fluid phase in the manner described for the **Fluid** node.

To account for the effect of porosity in the diffusivities, select an **Effective diffusivity model** — **Millington and Quirk model**, **Bruggeman model**, **Tortuosity model**, or **No correction**. Using one of the first four models, the effective transport factor, f_e , is defined from the porosity and the fluid tortuosity factor in the manner of:

$$f_e = \frac{\varepsilon_p}{\tau_F} \quad (3-13)$$

For **No correction**, the effective transport factor is set to one.

- For the **Millington and Quirk model**, the effective transport factor is $\tau_F = \varepsilon_p^{-1/3}$.

- For the **Bruggeman model**, the effective transport factor is $\tau_F = \varepsilon_p^{-1/2}$.
- For the **Tortuosity model**, specify the tortuosity factor τ_F . Select either **Isotropic** to define a scalar value, or **Diagonal** or **Symmetric** to define anisotropic tensor values.

The species diffusivities and mobilities are automatically adjusted for porous media transport using the effective transport factor.

MIGRATION IN ELECTRIC FIELD

This section is available when the **Migration in electric field** checkbox is selected in the [Transport Mechanisms](#) section of the interface. Select the source of the **Electric potential** V :

- Select **User defined** to enter a value or expression for the electric potential.
- When present, select an electric potential defined by an AC/DC interface that is present in the model. For example, select **Electric potential (ec)** to use the electric field defined an Electric Currents interface ec.

Settings for the mobilities are needed for the **Mixture-averaged** and **Fick's law** diffusion models. By default the mobility is set to be calculated based on the species diffusivities (adjusted by the **Effective diffusivity model** in the **Diffusion** section) using the **Nernst–Einstein relation**. To manually specify the mobilities, select **User defined** for the mobility $u_{m,w}$ and enter one value for each species.

The temperature (if you are using mobilities based on the Nernst–Einstein relation) is taken from the **Model Inputs** section.

PORE WALL INTERACTIONS

In porous media, it is possible to include **Pore Wall Interactions**. This accounts for species collisions with the surrounding media, that is, the pore walls.

For the **Fick's law** and **Mixture-averaged** diffusion models, the diffusion coefficient D_i^M is corrected with the **Wall diffusion** coefficient D_i^W in the following way

$$D_i^{MW} = \left(\frac{1}{D_i^M} + \frac{1}{D_i^W} \right)^{-1}$$

For the **Maxwell–Stefan** diffusion model, the following term is added to the species force balance

$$\dots + \frac{x_i}{D_{i,\text{eff}}^W} \left(\frac{\mathbf{j}_i}{\omega_i \rho} + \mathbf{u}_W \right) = \mathbf{d}_i$$

For weakly adsorbing gases, the **Knudsen** formula can be applied for computing the wall diffusivities, based on the **Pore diameter** d_{pore} (SI unit: m). For other cases, choose **User defined**.



For isobaric systems featuring concentration gradients in porous media, concentration gradients may give rise to nonzero fluid velocity as a result of the wall interactions. This effect can be modeled using the **Maxwell–Stefan** diffusion model in combination **Pore Wall Interactions**. For this case the wall velocity variable $\mathbf{u}_W$ (tcs.uw) can be used to contribute to the fluid momentum balance (either as a **Contributing Velocity** in **Darcy’s law**, or as a **Volume Force** in either the **Brinkman Equations** or **Free and Porous Media Flow**, **Brinkman** interfaces. The multiphysics **Reacting Flow** nodes incorporates this contribution from $\mathbf{u}_W$ automatically.

Adsorption

The adsorption process can be regarded as being under equilibrium if the adsorption rate is very fast and other mass transfer resistance is negligible. In such cases, the adsorption depends only on the amount of the adsorbent, concentration (or pressure) of adsorbate and temperature. This kind of isotherm adsorption process is implemented in Adsorption subfeature of the Porous Medium and Unsaturated Porous Medium features in the Transport of Diluted Species in Porous Media.

In general, the adsorption process deviates, to some extent, from its equilibrium state. This is dealt with in the adsorption kinetics. With the assumption of species adsorption process being independent of each other and the Linear Driving Force (LDF) model being used to represent the adsorbed species-solid (catalyst) mass transfer rate, the overall adsorption rate is

$$R_{\text{ads},i} = h_i(c_i - c_{\text{equ},i}) \quad (3-14)$$

where

$$c_{\text{equ},i} = \rho c_{\text{P},e,i}$$

here c_i is bulk mixture concentration of species i ; $c_{\text{equ},i}$ is the equilibrium concentration of the same species when adsorbed (amount per volume); h_i is the lumped mass transfer coefficient, which can be estimated from empirical equations based on the mass transfer mechanism; ρ is the dry bulk density of the material; and

$c_{P,e,i}$ is the adsorbed equilibrium concentration (amount per solid mass) defined from an isotherm adsorption model.

DOMAIN SELECTION

The domain selection inherits the all domain selections from its parent feature. Several Adsorption nodes can be added to the same domain.

ADSORPTION

Adsorption parameters can be set only for these species which are selected under Adsorbed Species section in the Porous Catalyst feature.

Select **Adsorption isotherm for individual species** checkbox to set different isotherm adsorption models for individual species. The equations for different models are shown under Equation section.

With **Adsorption isotherm for individual species** cleared, all species use the same isotherm model. In this case, the **Adsorption isotherm** direct down **Adsorption isotherm for individual species** is available, and specify an adsorption isotherm from one of models, **Langmuir**, **Freundlich**, **Toth**, **BET**, or **User defined**.

From the **Species** list, select these species which are adsorbate. With the species being selected as adsorbate, there are three kinds of parameters, Adsorption isotherm (available when checkbox **Adsorption isotherm for individual species** is selected), adsorption model parameters and mass transfer coefficient.

Select an adsorption isotherm for each adsorbate (available with **Adsorption isotherm for individual species** being selected).

Enter adsorption model parameters for species i , parameters are:

- For **Langmuir**:
Enter a **Langmuir constant** $k_{L,i}$ (SI unit: m^3/mol) and an **Adsorption maximum** $c_{p,\max,i}$ (SI unit: mol/kg).
- For **Freundlich**:
Enter a **Freundlich constant** $k_{F,i}$ (SI unit: mol/kg), a **Freundlich exponent** $N_{F,i}$ (dimensionless), and a **Reference concentration** $c_{\text{ref},i}$ (SI unit: mol/m^3).
- For **Toth**:
Enter an **Adsorption maximum** $c_{p,\max,i}$ (SI unit: mol/kg), a **Toth constant** $b_{T,i}$ (SI unit: m^3/mol), a **Toth exponent** $N_{T,i}$ (dimensionless).

- For **BET (Brunauer–Emmett–Teller)**:

Enter a **Monolayer adsorption capacity** $c_{0,i}$ (SI unit: mol/kg), a **Saturation concentration** $c_{S,i}$ (SI unit: mol/m³), and a **BET constant** $K_{B,i}$ (dimensionless).

- For **User defined** enter an **Adsorption isotherm** $c_{P,i}$ (SI unit: mol/kg):

The last parameter for adsorbate is the **Mass transfer coefficient** h_i (SI unit: m/s).

Porous Matrix

This node defines the porosity, when modeling transport in a [Porous Medium](#) or a [Porous Catalyst](#). The porosity corresponds to the volume fraction occupied by the species containing fluid.

- By default the porosity of the solid matrix, ϵ_p , is taken **From material**. The **Porous Matrix** node supports the use of a **Porous Material** added to the Materials branch. In this case the porosity is defined in the manner of

$$\epsilon_p = 1 - \sum_i \theta_{s,i} - \sum_i \theta_{\text{imf},i}$$

where $\theta_{s,i}$ and $\theta_{\text{imf},i}$ are the volume fractions given in **Solid** and **Immobile Fluids** subnodes under the **Porous Material** node.

- Select **From pellet bed densities** to compute the porosity using the (dry bulk) **Bed density** ρ_b and the (single phase) **Pellet density** ρ_{pe} . The porosity is then defined from

$$\epsilon_p = 1 - \frac{\rho_b}{\rho_{pe}}$$

- For **User defined**, enter a value or expression for the porosity.

DENSITY

In a [Porous Catalyst](#), the dry bulk density ρ of the porous medium is needed. This is used to compute mass phase transfer due to adsorption or desorption.

Select **Dry bulk density** when this quantity is known. Or select **Solid phase density** to instead compute the dry bulk density from the solid phase one. In both cases the density definition can be taken **From material** or **User defined**.

Surface Reaction

The Surface Reaction subnode under Porous Catalyst is used to define surface reaction source terms to the mass transport within the catalyst.

DOMAIN SELECTION

From the **Selection** list, choose the domains (from its parent feature) on which the surface reaction rate is defined.

Several Surface Reaction nodes can be used to account for different surface reactions in different parts for the modeling geometry.

SURFACE REACTION RATE FOR BULK SPECIES

The bulk species (defined as dependent variables in physics interface) can also take part in a surface reaction.

Select a surface reaction rate (SI unit: $\text{kg}/(\text{m}^2 \cdot \text{s})$) for species i from R_i^s list. Enter a reaction rate expression when **User defined** is selected from the list. Alternatively, if surface reaction rate expressions have been defined in a Chemistry interface, they can be chosen from the list.

SURFACE REACTION RATE FOR ADSORBED SPECIES

This section is available when at least one species is selected under the Adsorbed Species section in the Porous Catalyst feature.

Select a surface reaction rate (SI unit: $\text{kg}/(\text{m}^2 \cdot \text{s})$) for adsorbed species cads_i from $R_{\text{cads}_i}^s$ list. If **User defined** is selected from the list, enter a surface reaction rate expression. Alternatively, if surface reaction rate expressions have been defined in a Chemistry interface, they can be chosen from the list.

REACTION RATE FOR SURFACE SPECIES

This section is available when there is at least one species is added to the surface species table under the Surface Species section in the Porous Catalyst feature.

In the surface reaction rate table, there are three columns, **Surface species**, **Surface reaction rate**, and **Surface rate expression ($\text{kg}/(\text{m}^2 \cdot \text{s})$)**.

Select a surface reaction rate (SI unit: $\text{kg}/(\text{m}^2 \cdot \text{s})$) from the column **Surface reaction rate** for a surface species which is defined in the parent feature. With **User defined** being selected, the table cell at the same row under column **Surface rate expression ($\text{kg}/(\text{m}^2 \cdot \text{s})$)** is editable (otherwise it is read-only), and enter a surface reaction rate expression.

Alternatively, if surface reaction rate expressions have been defined in a Chemistry interface, they can be chosen from the list under column **Surface reaction rate**.

Electrode Surface Coupling

Use this node to define a flux boundary condition based on current densities of one or multiple [Electrode Reaction](#) (described in the Electrochemistry Module User's Guide) nodes in an Electrochemistry interface.

The flux is proportional to the current densities and the stoichiometric coefficients according to Faraday's law as defined by summation over the [Reaction Coefficients](#) subnodes. The molar fluxes are multiplied by the species molar masses to obtain the corresponding mass fluxes.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

MASS TRANSFER TO OTHER PHASES

When using the Reacting Flow coupling feature to model coupled momentum and mass transport, boundary conditions that results in a nonnegligible mass flux can be consistently handled by accounting for [The Stefan Velocity](#).

Select **Account for Stefan velocity** to update the Stefan velocity in accordance with the mass flux from the electrode reactions. One example that may benefit from this is when modeling gas diffusion electrodes.

Reaction

Use the **Reaction** node to specify the reaction kinetics for a single homogeneous chemical reaction. The resulting mass source terms for the species involved in the

reaction are automatically defined and added on the selected domains. For turbulent flow, the **Reaction** node includes the Eddy-dissipation turbulent-reactions model.



When licensed to use the Chemistry interface, multiple reactions can be defined at the same time, and the [Reaction Sources](#) node can be used to add the resulting mass sources.

See www.comsol.com/products/specifications/ for more details on availability.

REACTION RATE

Select a **Reaction rate** — **Automatic** (the default), or **User defined**. Selecting **Automatic** the laminar flow reaction rate is computed using the mass action law

For **User defined**, input a custom expression or constants for the **Reaction rate** r .

Specify the reaction stoichiometry by entering values for the stoichiometric coefficients (dimensionless) of each species. Enter negative values for reactants and positive values for products.

RATE CONSTANTS

When the **Use Arrhenius expressions** checkbox is not selected, input custom expressions or constants for the **Forward rate constant** k^f and **Reverse rate constant** k^r .

When the **Use Arrhenius expressions** checkbox is selected, enter values for the following parameters of the forward and reverse reactions:

- **Forward frequency factor** A^f and **Reverse frequency factor** A^r (dimensionless)
- **Forward activation energy** E^f and **Reverse activation energy** E^r
- **Forward temperature exponent** n^f and **Reverse temperature exponent** n^r (dimensionless)

TURBULENT FLOW

Note this section is only available when licensed to the CFD Module (see www.comsol.com/products/specifications/).

When the **Turbulent-reaction model** is set to **None**, laminar flow is assumed and the reaction source terms are defined from the reaction stoichiometry and reaction rates prescribed.

When the **Turbulent-reaction model** is set to **Eddy-dissipation**, turbulent flow will be accounted for in the reaction mass sources. In this case, enter values for the **Turbulent reaction model parameters** α_{ED} and β_{ED} (dimensionless).

The **Eddy-dissipation** model also requires an estimation of the turbulent mixing time of the fluid flow turbulence. When a Fluid Flow interface defining it is present in the model, it can be selected from the **Turbulence time scale** list. For example, select **Turbulence time scale (spf/fp1)** to use the time scale defined by the **Fluid Properties** node fp1 in a Turbulent Flow, k - ϵ interface with the **Name** set to spf.

REGULARIZATION

Select **Rate expressions** to regularize the individual rate expressions added to each species. If the mass fraction for a reactant species ω_i becomes smaller than its **Damping limits**, ω_i^{dl} , the rate expression, for species ω_i is reduced linearly. If $\omega_i \leq 0$ for a reactant species, the reaction rate contribution to that species is completely removed. Similarly, if the mass fraction for a product species ω_j becomes larger than $1 - \omega_j^{dl}$, the rate expression added is damped linearly. If $\omega_j \geq 1$ for a product species, the reaction rate contribution to that species is completely removed.

The default value for the **Damping limits**, ω_i^{dl} , is 10^{-6} , which is appropriate for most applications, but can require adjustment when working with for example catalytic trace species.

Regularization of the rate expressions adds considerable stability to the reaction expressions, but it should ideally only be used as a means to reach convergence. When reaction zones are not adequately resolved on the mesh, regularization can affect the mass balance.

ENTHALPY OF REACTION

Select a heat source for this reaction from the **H** list. The listed heat sources usually come from reactions of Chemistry interfaces. If there is no heat source for the reaction in the list, select **User defined**, then enter the heat source.

This heat source is accounted for in the Reacting Flow coupling.

Reaction Sources

Use the **Reaction Sources** node to define mass source terms from one or more chemical reactions. Both homogeneous reactions as well as heterogeneous reactions occurring in a porous media can be studied. The node also includes the possibility to pick up automatically defined source term definitions from a Chemistry interface.

REACTIONS

The combo box **Chemistry** lists all the Chemistry interfaces that are coupled to the current mass transport interface. The coupled chemistry interface should

- Be active
- Not have **Define variables for porous pellets** (under the **Pellet Chemistry** section) selected
- Have **Species solved for** (under the **Species Matching** section) set to the current mass transport interface.

Furthermore

- With a **Chemistry** interface selected, the species mass sources will be set to the corresponding reaction rates in **Chemistry** according to the **Species Matching** table. In the case of partial coupling, the mass sources for the uncoupled species will be set to 0.
- With **User defined** selected, specify mass source (SI unit: $\text{kg}/(\text{m}^3\cdot\text{s})$) for each species, or pick up a reaction rate variable from a **Chemistry** interface.

Select the **Mass transport to other phases** checkbox if mass is leaving or entering the fluid as a result of the reactions, for instance due to condensation or vaporization in a porous matrix. In this case the mass source for the species calculated from the mass constraint can also be specified. The net mass transfer corresponds to the sum of the mass sources for all species.

REACTING VOLUME

This section is available when the **Chemistry** list is set to **User defined**.

When specifying reaction sources for a species in porous media, the specified mass source may have the basis of the total volume, or the pore volume. For nonporous domains, the **Reacting Volume** setting has no impact.

- For **Total volume**, the reaction mass source expressions are specified per unit volume of the model domain.
- For **Pore volume**, the reaction mass source expressions are specified per unit volume pore space. In this case the reaction mass sources will be multiplied by the domain porosity ϵ_p (ϵ_p equals unity for nonporous domains).

When a coupled chemistry is selected from the **Chemistry** list, the porosity will be applied to the reaction source terms automatically.

HEAT SOURCE

The heat source will be defined as reaction heat from whole Chemistry if the **Chemistry** list is set to a coupled chemistry interface, otherwise it is defined as the specified value (entered by the user).

Initial Values

The **Initial Values** node adds initial values for the mass fractions that can serve as an initial condition for a transient simulation, or as an initial guess for a nonlinear solver. If required, add additional **Initial Values** nodes from the **Physics** toolbar.

INITIAL VALUES

The initial mass fractions can be specified using a number of quantities. Select the type of input from the **Mixture specification** list. Select:

- **Mass fractions** (the default) to enter mass fractions ($\omega_{0,\omega1}$ for example)
- **Mole fractions** to enter mole fractions ($x_{0,\omega1}$ for example)
- **Molar concentrations** (SI unit: mol/m^3) to enter molar concentrations ($c_{0,\omega1}$ for example)
- **Number densities** (SI unit: $1/\text{m}^3$) to enter number densities ($n_{0,\omega1}$ for example)
- **Densities** (SI unit: kg/m^3) to enter densities ($\rho_{0,\omega1}$ for example)

Enter a value or expression in the field for each species except for the one computed from the mass constraint.

INITIAL MIXTURE DENSITY

When the selecting **Molar concentrations**, **Number densities**, or **Densities** are selected from the **Mixture specification** list, the should also be specified. Select **Ideal gas** or **User defined** from the **Initial mixture density** list.

- For **Ideal gas**, also specify the **Initial pressure** p_0 and the **Initial Temperature** T_0 . Note that dependent variables solved for are evaluated to zero for initial values. When solving for pressure or temperature together with the mass fractions, apply the initial values from the corresponding interfaces here as well.
- For **User defined**, input a custom for the **Initial mixture density** ρ_{m0} .

Mass Fraction

The **Mass Fraction** node adds boundary conditions for the species mass fractions. For example, the following condition specifies the mass fraction of species i : $\omega_i = \omega_{i,0}$.

Set the mass fractions of all species except the one computed from the mass constraint. This ensures that the sum of the mass fractions is equal to one (see [Species](#)). This node is available for exterior and interior boundaries.

When defining conditions on boundaries between different phases, see the section [Prescribing Conditions on Fluid–Solid Interfaces](#).

MASS FRACTION

Specify the mass fraction for each species individually. Select the checkbox for the species to specify the mass fraction, and enter a value or expression in the corresponding field. To use another boundary condition for a specific species, click to clear the checkbox for the mass fraction of that species.

MASS TRANSFER TO OTHER PHASES

When using the Reacting Flow coupling feature to model coupled momentum and mass transport, boundary conditions that results in a nonnegligible mass flux can be consistently handled by accounting for [The Stefan Velocity](#).

Select **Account for Stefan velocity** to update the Stefan velocity in accordance with the prescribed mass fractions. Examples of cases that may benefit from this are, for example, when modeling surface reactions or phase change on an exterior boundary.

CONSTRAINT SETTINGS

To display this section, click the **Show** button () and select **Advanced Physics Options**.

Flux

The **Flux** node is available on exterior boundaries and can be used to specify the mass flux. The boundary mass flux for each species is defined in the manner of

$$-\mathbf{n} \cdot \mathbf{j}_i = j_{0,i} \quad (3-15)$$

where $\mathbf{n}$ denotes the outward pointing normal of the boundary. The prescribed flux $j_{0,i}$ can contain an arbitrary flux expression. It can, for example, be used to represent a *heterogeneous reaction* or a separation process occurring at the boundary. The flux

can, for example, be a function of ω_i , the temperature, the pressure or even the electric potential.

When defining conditions on boundaries between different phases, see the section [Prescribing Conditions on Fluid–Solid Interfaces](#).

CONVECTION

By default, the flux due to diffusion and migration in an electric field is prescribed. This is the appropriate flux condition when the relative velocity at the boundary is zero. When the fluid velocity is not equal to that of the boundary, for example due to a fluid injection, it is often convenient to prescribe the total flux including convection. To do so select **Include** in the **Convection** section. In this case the prescribed flux is defined as:

$$-\mathbf{n} \cdot (\mathbf{j}_i + \rho \mathbf{u} \omega_i) = j_0$$

MASS TRANSFER TO OTHER PHASES

When using the Reacting Flow coupling feature to model coupled momentum and mass transport, boundary conditions that results in a nonnegligible mass flux can be consistently handled by accounting for [The Stefan Velocity](#).

Select **Account for Stefan velocity** to update the Stefan velocity in accordance with the prescribed flux. Examples of cases that benefit from this are for example when modeling surface reactions or phase change on an exterior boundary. The Stefan velocity represents the mixture velocity resulting from mass fluxes at the wall. This section is not available when **Include** is selected in the **Convection** section.

INWARD FLUX

Specify the **Inward flux** for each species individually. Select the checkbox for the species to prescribe a flux for and enter a value or expression in the corresponding field. To use another boundary condition for a specific species, click to clear the checkbox for the flux of that species. Use a positive value for an inward flux.

External convection

Set **Flux type** to **External convection** to prescribe a mass flux to or from an exterior domain (not modeled) assumed to include convection. The exterior can for example include a forced convection to control the temperature or to increase the mass transport. In this case the prescribed mass flux corresponds to

$$j_{0,i} = k_{\omega,i}(\omega_{b,i} - \omega_i) \quad (3-16)$$

where $k_{o,i}$ is a mass transfer coefficient and $\omega_{b,i}$ is the bulk mass fraction, the typical mass fraction far into the surrounding exterior domain.

Inflow

The **Inflow** node adds a boundary condition for an inflow boundary, defining the composition of the mixture. The node is available for exterior boundaries. The mixture composition can be specified using the following quantities:

- Mass fractions: $\omega_i = \omega_{0,i}$
- The mole fraction: $x_i = x_{0,i}$
- The molar concentration: $c_i = c_{0,i}$
- The number density, which describes the number of particles per volume: $n_i = n_{0,i}$
- The density: $\rho_i = \rho_{0,i}$
- The mass flow rates.

The node requires input for all species. (The [Mass Fraction](#) node can be used to specify boundary mass fractions for a subset of the active species in the interface.)

INFLOW

Select a **Mixture specification**:

- **Mass fractions** (the default) to enter mass fractions ($\omega_{0,\omega 1}$, for example)
- **Mole fractions** to enter mole fractions ($x_{0,\omega 1}$ for example)
- **Molar concentrations** (SI unit: mol/m^3) to enter molar concentrations ($c_{0,\omega 1}$, for example)
- **Number densities** (SI unit: $1/\text{m}^3$) to enter number densities ($n_{0,\omega 1}$, for example)
- **Densities** (SI unit: kg/m^3) to enter densities ($\rho_{0,\omega 1}$, for example)
- **Mass flow rates** (SI unit: kg/s) to enter the total mass flow over the boundary ($J_{in,\omega 1}$, for example)

Enter a value or expression in the field for each species except for the one computed from the mass constraint.

CONSTRAINT SETTINGS

To display this section, click the **Show** button () and select **Advanced Physics Options**.

No Flux

The **No Flux** node is the default boundary condition available for exterior boundaries. It should be used on boundaries across which there is no mass flux, typically exterior solid walls where no surface reactions occur. The condition applied for each species corresponds to

$$-\mathbf{n} \cdot \mathbf{j}_i = 0$$

where $\mathbf{n}$ denotes the outward pointing normal of the boundary.

Out-of-Plane Flux

The out-of-plane flux node is used to prescribe a flux in an out-of-plane domain, and it can be used to reduce a model geometry to 2D, or even 1D, when the variation of mass fraction is small in one or more directions. This could be the case for example when the object to model is thin or slender. [Figure 3-1](#) shows examples of possible situations in which this type of geometry reduction can be applied.

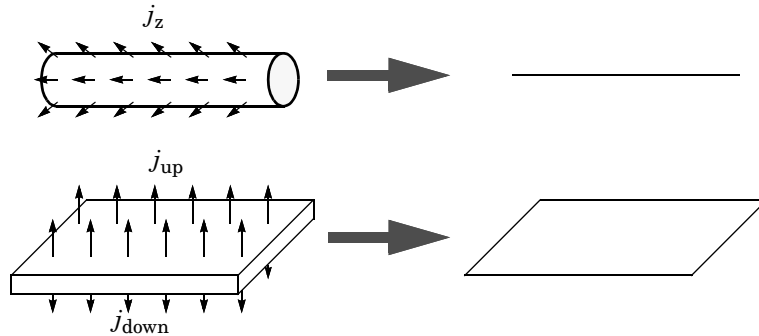


Figure 3-2: Geometry reduction from 3D to 1D (top) and from 3D to 2D (bottom).

For a 1D component this node adds a single out-of-plane mass flux $j_{0,z,i}$ for species i . For a 2D component two fluxes can be prescribed for each species; $j_{0,u,i}$ for the upside of the domain, and $j_{0,d,i}$ for the downside of the domain.

The fluxes are added as a source contribution to the right-hand side of the transport equation. For example, in 2D components with diffusion, convection, and out-of-plane flux present, the mass transfer of species i is defined by the mass conservation equation

$$\rho \frac{\partial}{\partial t}(\omega_i) + \rho(\mathbf{u} \cdot \nabla)\omega_i + \nabla \cdot \mathbf{j}_i = R_i + S_{\text{opf},i}, \quad (3-17)$$

where $S_{\text{opf},i}$ is the out-of-plane source for species i

$$S_{\text{opf},i} = \frac{j_{0,i}}{d_z} = \frac{j_{0,\text{u},i} + j_{0,\text{d},i}}{d_z}.$$

For external mass transfer on the upside and the downside of the domain, the out-of-plane flux is

$$j_{0,i} = k_{w,\text{u},i}(w_{\text{b},\text{u},i} - w_i) + k_{w,\text{d},i}(w_{\text{b},\text{d},i} - w_i)$$

where $k_{\omega,i}$ is a mass transfer coefficient and $\omega_{\text{b},i}$ is the bulk mass fraction, the typical mass fraction far into the surrounding exterior domain. The prescribed flux, j_0 , can include any arbitrary user-specified expressions. It can be a constant or a function of a dependent variable or independent variables.

UPSIDE INWARD FLUX

For 1D components, enter the **cross-sectional perimeter** P_c to get the out-of-plane flux

$$j_{0,i} = P_c j_{0,z,i}.$$

The default value of P_c is the circumference. Either keep the default value, for a circular cross-section shape, or edit the value to get a user-defined shape of the out-of-plane cross section.

The available options are **General inward flux** and **External convection**. Specify the **Inward flux** for each species individually. Select the **Species** checkbox for the species for which to specify the flux, and enter a value or expression in the corresponding field. To use another boundary condition for a specific species, click to clear the checkbox for the flux of that species. Use a positive value for an inward flux.

Set **Flux type** to **External convection** to prescribe a mass flux to or from an exterior domain (not modeled) assumed to include convection. The exterior can for example include a forced convection to control the temperature or to increase the mass transport. In this case the prescribed mass flux corresponds to

$$j_{0,i} = k_{\omega,i}(\omega_{\text{b},i} - \omega_i)$$

where $k_{\omega,i}$ is a mass transfer coefficient and $\omega_{\text{b},i}$ is the bulk mass fraction, the typical mass fraction far into the surrounding exterior domain.

DOWNSIDE INWARD FLUX

The available options are **General inward flux** and **External convection**. The settings are the same as for the [Upside Inward Flux](#) section.

FURTHER READING

For an example of using the Out-of-Plane Flux node, see this application example:



Cross-Flow Mass Transfer in a Thin Domain: Application Library path
Chemical_Reaction_Engineering_Module/Tutorials/thin_domain

Outflow

The **Outflow** node is the preferred boundary condition at outlets where the species are to be transported out of the model domain. It is useful, for example, in mass transport models where it is assumed that convection is the dominating effect driving the mass flow through the outflow boundary. This node is available for exterior boundaries. The boundary condition is applied to all species and corresponds to one of the following equations depending on the selected diffusion model:

- For the [Maxwell–Stefan Diffusion Model](#):

$$-\mathbf{n} \cdot \rho \omega_i \sum_k D_{ik} \mathbf{d}_k = 0$$

- For the [Mixture-Averaged Diffusion Model](#):

$$-\mathbf{n} \cdot \rho D_i^m \nabla \omega_i = 0$$

- For the [Fick’s Law Diffusion Model](#):

$$-\mathbf{n} \cdot \rho D_i^f \nabla \omega_i = 0$$

Symmetry

The **Symmetry** node can be used to represent boundaries where the species concentration is symmetric; that is, there is no mass flux in the normal direction across the boundary:

$$-\mathbf{n} \cdot \mathbf{N} = -\mathbf{n} \cdot (\rho \omega_i \mathbf{u} + \mathbf{j}_i) = 0$$

This boundary condition is identical to the [No Flux](#) node, but applies to all species and cannot be applied to individual species. The Symmetry node is available for exterior boundaries.

Flux Discontinuity

The **Flux Discontinuity** node represents a discontinuity in the mass flux across an interior boundary:

$$-\mathbf{n} \cdot (\mathbf{N}_d - \mathbf{N}_u) = N_0 \quad \mathbf{N} = (\rho \omega_i \mathbf{u} + \mathbf{j}_i)$$

where the value of N_0 specifies the size of the flux jump evaluated from the down to the upside of the boundary.

FLUX DISCONTINUITY

Specify the jump in species mass flux. Use a positive value for increasing flux when going from the downside to the upside of the boundary. The boundary normal points in the direction from the downside to the upside of an interior boundary and can be plotted for visualization.

Select the **Species** checkboxes to specify a flux discontinuity, and enter a value or expression for the **Flux discontinuity** N_0 (SI unit: $\text{kg}/(\text{m}^2 \cdot \text{s})$) in the corresponding field, $N_{0,w1}$, for example. To use a different boundary condition for a specific species, click to clear the checkbox for the flux discontinuity of that species.

Open Boundary

Use the **Open Boundary** node to set up mass transport across boundaries where both convective inflow and outflow can occur. Use the node to specify an exterior species composition on parts of the boundary where fluid flows into the domain. A condition equivalent to the [Outflow](#) node applies to the parts of the boundary where fluid flows out of the domain. The direction of the flow across the boundary is typically calculated by a Fluid Flow interface and is entered as [Model Inputs](#).

EXTERIOR COMPOSITION

Enter a value or expression for the species composition. Select:

- **Mass fractions** (the default) to enter mass fractions ($\omega_{0,w1}$, for example)
- **Mole fractions** to enter mole fractions ($x_{0,w1}$, for example)

- **Molar concentrations** (SI unit: mol/m^3) to enter molar concentrations ($c_{0,\omega 1}$, for example)
- **Number densities** (SI unit: $1/\text{m}^3$) to enter number densities ($n_{0,\omega 1}$, for example) and to describe the number of particles per volume $n = n_0$
- **Densities** (SI unit: kg/m^3) to enter densities ($\rho_{0,\omega 1}$, for example)



A concentration quantity other than the mass fractions can only be used when all species are defined.

Equilibrium Reaction

Use this node to model a reaction where the kinetics is assumed so fast that the equilibrium condition is fulfilled at all times. The node solves for an additional degree of freedom (the reaction rate R_{eq}) to fulfill the equilibrium condition at all times in all space coordinates.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

EQUILIBRIUM CONDITION

The **Chemistry** list includes all the chemistry interfaces which are coupled to the current mass transfer interface. The conditions for valid coupling are the same as that in the [Reaction Sources](#) feature. The **Chemistry** list can be set to a coupled chemistry or **None**.

- A coupled chemistry — The **Equilibrium reaction formula** is available when the **Chemistry** list is set to a coupled chemistry interface. All the equilibrium reaction formulas in the coupled chemistry will appear in the formula list. If you select one of them, all coefficients (stoichiometric coefficients, equilibrium constant) will automatically be taken from the coupled reaction. There are no source term contributions if the **Equilibrium reaction formula** list is set to **User defined**.
- **None** — If **None** is selected from the **Chemistry** list, an equilibrium reaction will be defined by settings from the fields **Equilibrium definition**, **Equilibrium constant**, and **Unit activity concentration**.

Selecting **Equilibrium constant** in the list, the following equilibrium condition based on the species activities and the law of mass action is used

$$K_{\text{eq}} = \frac{\prod_{i \in \text{products}} a_i^{v_i}}{\prod_{i \in \text{reactants}} a_i^{-v_i}}$$

where v_i are the stoichiometric coefficients and the species activities are defined from the concentration, c_i , and the unit activity concentration c_{a0} .

$$a_i = \frac{c_i}{c_{a0}}$$

Enter a value or expression for the dimensionless **Equilibrium constant** K_{eq} , and the **Unit activity concentration** C_{a0} .

Select **User defined** from the list to instead enter a manually defined **Equilibrium expression** E_{eq} .

STOICHIOMETRIC COEFFICIENTS

This section is available when the **Chemistry** list is set to **None**, and used to set stoichiometric coefficients for the specified equilibrium reaction.

Enter a value for the stoichiometric coefficient ν for all participating species. Use negative values for reactants and positive values for products in the modeled reaction. Species with a stoichiometric coefficient value of 0 are not affected by the **Equilibrium Reaction** node.

HEAT SOURCE

The heat source will be defined as follows,

- An equilibrium reaction selected from the **Equilibrium reaction formula** list. The heat source is defined as the equilibrium reaction (from the coupled chemistry) heat,
- **None** selected from the **Chemistry** list. Enter a value or expression for the heat source (SI unit: W/m^3).

Surface Equilibrium Reaction

Use this node to model an equilibrium reaction on a boundary (surface). The kinetics of the reaction is assumed so fast that the equilibrium condition is fulfilled at all times.

The node solves for an additional degree of freedom (the mass flux J_{eq}) to fulfill the equilibrium condition at all times in all space coordinates along the boundary.



This feature is only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

The settings for this node are the same as for [Equilibrium Reaction](#) except for the setting in the section below.

MASS TRANSFER TO OTHER PHASES

When using the Reacting Flow coupling feature to model coupled momentum and mass transport, boundary conditions that results in a nonnegligible mass flux can be consistently handled by accounting for [The Stefan Velocity](#). Select **Account for Stefan velocity** to update the Stefan velocity in accordance with the prescribed mass fractions. Examples of cases that may benefit from this are for example when modeling surface reactions or phase change on an exterior boundary.

The Nernst–Planck Equations Interface

Ionic solutions subjected to electric fields are affected by migration transport in addition to the existing mass transport by diffusion and convection. Migration implies that positive ions migrate from positive to negative potential, and vice versa for negatively charged ions. The flux of the ionic species i in the solution is given by the mass flux vector (SI unit: mol/(m²·s))

$$\mathbf{N}_i = -D_i \nabla c_i - z_i u_{m,i} F c_i \nabla V + c_i \mathbf{u} \quad (3-18)$$

where c_i denotes the concentration of species i (SI unit: mol/m³), D_i is the diffusion coefficient of species i (SI unit: m²/s), $\mathbf{u}$ is the solvent velocity (SI unit: m/s), F refers to Faraday's constant (SI unit: s·A/mol), V denotes the electric potential (SI unit: V), z_i is the charge number of the ionic species (dimensionless), and $u_{m,i}$ its ionic mobility (SI unit: s·mol/kg).

There are three physics interfaces for this type of transport. In the **Nernst–Planck Equations (npe)** interface () , found under the **Chemical Species Transport** branch () when adding a physics interface, the transport of every charged species is accounted for, and is solved for in combination with the electroneutrality condition.

The Nernst–Planck Equations interface supports modeling of transport by convection, diffusion, and migration of dissolved ionic species in 1D, 2D, and 3D as well as for axisymmetric models in 1D and 2D. The dependent variables are the species concentrations, c_i , and the electric potential, V . The physics interface is tailored for modeling of electrochemical cells and corrosion processes and offers a straightforward way to perform tertiary current distribution analysis.

The physics interface has the equations, boundary conditions, and rate expression terms for modeling mass transport of ionic species by convection, diffusion, and migration, solving for the species concentrations and the electric potential. The main feature is the Convection, Diffusion, and Migration node, which adds the equation for the species concentration and the potential and provides an interface for defining species properties and model inputs.

When this physics interface is added, these default physics nodes are also added to the **Model Builder — Convection, Diffusion and Migration, No Flux** (the default boundary condition for species concentrations), **Electric Insulation** (the default boundary

condition for the electric field), and **Initial Values**. Then, from the **Physics** toolbar, add other nodes that implement, for example, boundary conditions and rate expression terms. You can also right-click **Nernst–Planck Equations** to select physics features from the context menu.

SETTINGS

The **Label** is the default physics interface name.

The **Name** is used primarily as a scope prefix for variables defined by the physics interface. Refer to such physics interface variables in expressions using the pattern `<name>.<variable_name>`. In order to distinguish between variables belonging to different physics interfaces, the name string must be unique. Only letters, numbers, and underscores (`_`) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first physics interface in the model) is `npe`.

OUT-OF-PLANE THICKNESS

For 2D components, the **Thickness** field (default value: 1 m) defines a parameter for the thickness of the geometry perpendicular to the two-dimensional cross-section. The value of this parameter is used, among other things, to automatically calculate molar flow rates from the total molar flux.

CROSS-SECTIONAL AREA

For 1D components, enter a **Cross-sectional area** A_c (SI unit: m^2) to define a parameter for the area of the geometry perpendicular to the 1D component. The value of this parameter is used, among other things, to automatically calculate molar flow rates from the total molar flux. The default is 1 m^2 .

SPECIES

Select the species that the physics interface computes from the electroneutrality condition in [Equation 3-18](#). That is, its value comes from the fact that the net charge in every control volume is zero. Select the preferred species in the **From electroneutrality** list. To minimize the impact of any numerical errors, use the species with the highest concentration. By default, the software uses the first species.

DEPENDENT VARIABLES

Add or remove species in the model and also change the names of the dependent variables that represent the species concentrations.

Specify the **Number of species**. There must be at least two species. To add a single species, click the **Add species** button (). To remove a species, select it in the list and click the **Remove species** button (). Edit the names of the species directly in the table.



The species are dependent variables, and their names must be unique with respect to all other dependent variables in the model.

ADVANCED SETTINGS

To display this section, click the **Show More Options** button () and select **Advanced Physics Options**. Normally these settings do not need to be changed. Select the **Material balance form** — **Nonconservative** (the default) or **Conservative**.



Material Balance Formulation

DISCRETIZATION

To display all settings available in this section, click the **Show More Options** button () and select **Advanced Physics Options**.

For more information about these settings, see the *Discretization* section under [The Transport of Diluted Species Interface](#).

CONSISTENT AND INCONSISTENT STABILIZATION

To display this section, click the **Show More Options** button () and select **Stabilization**. Any settings unique to this physics interface are listed below.

The crosswind diffusion implementation used in the Nernst–Planck Equations interface is based on Codina ([Ref. 1](#)).

- When the **Crosswind diffusion** checkbox is selected, the **Lower gradient limit** g_{lim} (SI unit: mol/m^4) field defaults to $0.1 [\text{mol}/\text{m}^3] / \text{npe.helem}$, where npe.helem is the local element size.
- For both consistent stabilization methods, select an **Equation residual**. **Approximate residual** is the default setting and it means that derivatives of the diffusion tensor

components are neglected. This setting is usually accurate enough and is faster to compute. If required, select **Full residual** instead.

	• Domain and Boundary Nodes for the Nernst–Planck Equations Interface • Theory for the Nernst–Planck Equations Interface • Material Balance Formulation
	In the <i>COMSOL Multiphysics Reference Manual</i> see Table 2-4 for links to common sections and Table 2-5 to common feature nodes. You can also search for information: press F1 to open the Help window or Ctrl+F1 to open the Documentation window.

Domain and Boundary Nodes for the Nernst–Planck Equations Interface

The [Nernst–Planck Equations Interface](#) has these domain and boundary physics nodes available.

DOMAIN NODES

- [Species Properties](#)
- [Convection, Diffusion, and Migration](#)
- [Reactions](#)
- [Equilibrium Reaction](#) is described for the Transport of Diluted Species interface
- [Initial Values](#)

BOUNDARY CONDITIONS

	In the Nernst–Planck Equations interface, there are separate boundary conditions for the species concentrations and for the electric potential. This means that there are two active boundary conditions on each boundary. Many different combinations of boundary conditions can be set for the electric field and concentration fields on a boundary, but not all combinations are relevant.
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SPECIES CONCENTRATION BOUNDARY CONDITIONS

The following boundary conditions for the species concentrations are available on exterior boundaries:

- [Concentration](#)
- [Flux](#)
- [No Flux](#) (the default boundary condition for species concentrations)
- [Symmetry](#)
- [Inflow](#)
- [Outflow](#)

For interior boundaries, continuity in the concentration of all species is the default boundary condition. In addition, the [Flux Discontinuity](#) boundary condition is available on interior boundaries.

ELECTRIC POTENTIAL BOUNDARY CONDITIONS

The following boundary conditions for the electric potential are available on exterior boundaries:

- [Current Density](#)
- [Electric Insulation](#) (the default boundary condition for the electric potential)
- [Electric Potential](#) (described for the Electrostatics interface in the *COMSOL Multiphysics Reference Manual*)

For interior boundaries, continuity in the electric potential is the default boundary condition. In addition, the following boundary conditions are available on interior boundaries:

- [Current Discontinuity](#)
- [Electric Potential](#) (described for the Electrostatics interface in the *COMSOL Multiphysics Reference Manual*)

EDGES AND POINTS

The [Point Mass Source](#) and [Line Mass Source](#) features are also available and described for the Transport of Diluted Species interface.



The links to the physics nodes described in the *COMSOL Multiphysics Reference Manual* do not work in the PDF, only from the online help.



For axisymmetric models, COMSOL Multiphysics takes the axial symmetry boundaries (at $r = 0$) into account and automatically adds an **Axial Symmetry** node to the model that is valid on the axial symmetry boundaries only.

Species Properties

Use this node to define parameters specific to each of the species.

The node will change its label depending on what inputs are available in the node. If only **Charge** is available, it will be labeled “Species charges”.

CHARGE

Enter the charge number z_c (dimensionless, but requires a plus or minus sign) for each species.

ACTIVITY

This section is available when the **Species Activity** combo box is set to **Debye-Hückel** or **User defined**.

If **Debye-Hückel** is selected, enter the ion size number, a_0 , for each species. If **User defined** is selected, enter the activity coefficient, f , for each species.

Convection, Diffusion, and Migration

The **Convection, Diffusion, and Migration** is the main physics interface node which adds the concentration and electric field variables and the Nernst–Planck equations governing them.

MODEL INPUTS

The temperature model input is always available, and set to **Common model input** by default. Select the source of the **Temperature** field T :

- Select **User defined** to enter a value or an expression for the temperature (SI unit: K). This input is always available.
- If required, select a temperature defined by a Heat Transfer interface present in the component (if any). For example, select **Temperature (ht)** to use the temperature defined by the Heat Transfer in Fluids interface with the ht name.

CONVECTION

This section is available when the **Convection** checkbox is selected.

The velocity field for the convection of the solvent, which is used in the convective term, needs to be specified as feature input. Select the source of the **Velocity field $\mathbf{u}$** :

- Select **User defined** to enter values or expressions for the velocity components. This input is always available.
- Select a velocity field defined by a Fluid Flow interface present in the model (if any). For example, select **Velocity field (spf)** to use the velocity field defined by the **Fluid Properties** node fp1 in a Single-Phase Flow, Laminar Flow interface with spf as the Name.

DIFFUSION

Select a **Material** from the list (when applicable and available). The default is **None**.

Specify the **Diffusion coefficient D_i** for each species. This can be a scalar value for isotropic diffusion or a tensor describing anisotropic diffusion. Enter the values in the field (one for each species).

MIGRATION IN ELECTRIC FIELD

Specify the **Mobility** and the **Charge number** of the ionic species. These properties affect the migration of ionic species. By default the **Mobility** is set to be calculated based on the species diffusivities and the temperature using the **Nernst–Einstein relation**. To manually specify the mobility, select **User defined** for the mobility $u_{m,i}$ and enter one value for each species.

Enter the **Charge number z_i** (dimensionless, but requires a plus or minus sign) for each species.

Specify the temperature (if you are using mobilities based on the Nernst–Einstein relation) in the **Model Inputs** section.

Electric Insulation

The **Electric Insulation** node is the default boundary condition for the electric field. It provides a boundary condition for the electric field that means that the boundary is electrically insulated; that is, there is no current flowing across the boundary:

$$-\mathbf{n} \cdot \mathbf{i} = 0 \quad (3-19)$$

No Flux

The **No Flux** node is the default boundary condition on exterior boundaries. It should be used on boundaries across which there is no mass flux, typically exterior solid walls where no surface reactions occur. The condition applied for each species corresponds to

$$-\mathbf{n} \cdot (-D\nabla c - zu_{\text{m}}Fc\nabla V) = 0$$

where $\mathbf{n}$ denotes the outward pointing normal of the boundary.

CONVECTION

By default, the feature prescribes a vanishing flux due to diffusion and migration in an electric field. This is the appropriate no flux condition when the relative convective velocity at the boundary is zero. When the fluid velocity at the boundary is not equal to that of the boundary, it is often convenient to prescribe the total flux including the convection. To do this select **Include** in the **Convection** section. In this case the no flux condition is:

$$-\mathbf{n} \cdot (-D\nabla c - zu_{\text{m}}Fc\nabla V + \mathbf{u}c) = 0$$

Initial Values

The **Initial Values** node adds initial values for the concentrations c_i and the electric potential V that can serve as an initial condition for a transient simulation or as an initial guess for a nonlinear solver. Add more **Initial Values** nodes from the **Physics** toolbar.

INITIAL VALUES

Enter values or expressions for the initial value of the **Concentration** c_i (SI unit: mol/m³) (except the one that a mass constraint determines) and for the **Electric potential** V (SI unit: V). The default concentration values are 0 mol/m³ and the default electric potential is 0 V.

Reactions

In order to account for consumption or production of species, the **Reactions** node adds rate expressions R_i , which appear on the right-hand side of the species transport equations.

REACTIONS

Add a rate expression R_i (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$) for each of the species solved for. Enter a value or expression in the field of the corresponding species.

Concentration

The **Concentration** node adds a boundary condition for the species concentrations: for example the following condition specifies the concentration of species, c_i :

$$c_i = c_{i,0}$$



Set the concentration of all species except the one computed from the electroneutrality condition, which ensures that the net charge is zero. See [Species](#).

CONCENTRATION

Specify the concentration c_i (SI unit: mol/m^3) for each species individually. Select the checkbox for the species to specify the concentration, and enter a value or expression in the corresponding field. To use another boundary condition for a specific species, click to clear the checkbox for that species' mass fraction.

CONSTRAINT SETTINGS

To display this section, click the **Show More Options** button () and select **Advanced Physics Options**.

Flux

The **Flux** node can be used to specify the species flux across a boundary. The species flux, for example at the surface of an electrode, is defined as:

$$-\mathbf{n} \cdot (-D\nabla c - zu_{\text{m}} Fc\nabla V) = J_0$$

where $\mathbf{n}$ denotes the outward pointing normal of the boundary. The flux J_0 can be any function of concentration and electric potential (and indeed temperature if heat transfer is included in the model), often provided by a kinetic expression. In electrochemistry, this function usually depends on concentration and exponentially on the electric potential through either the Tafel equation or the *Butler-Volmer equation*.

CONVECTION

By default, the flux due to diffusion and migration in an electric field is prescribed. This is the appropriate flux condition when the relative velocity at the boundary is zero. When the fluid velocity is not equal to that of the boundary, it is often convenient to prescribe the total flux, including the convection. To do this select **Include** in the **Convection** section. When including the convection, the prescribed flux is defined as:

$$-\mathbf{n} \cdot (-D\nabla c - zu_m Fc\nabla V + \mathbf{u}c) = J_0$$

INWARD FLUX

Specify the inward species flux for each species individually. Select the checkbox for the species to prescribe a flux for and enter a value or expression for the **inward flux** J_0 . To use another boundary condition for a specific species, click to clear the checkbox for that species' flux. Use a positive value for an inward flux.

Symmetry

The **Symmetry** node can be used to represent boundaries where the concentration is symmetric; that is, there is no species flux in the normal direction across the boundary. This boundary condition is identical to that of the [No Flux](#) condition.

Inflow

The **Inflow** node adds a boundary condition for an inflow boundary, where the concentration for all species is specified in the manner of $c_i = c_{i,0}$.



This condition is identical to the [Concentration](#) node, but is applied to all species and cannot be applied to individual species.

CONCENTRATION

Specify the concentration c_i (SI unit: mol/m³) of each species individually by entering a value or expression in the corresponding field.

BOUNDARY CONDITION TYPE

The option **Concentration constraint** constrains the concentration values on the boundary by the use of pointwise constraints.

The other option, **Flux (Danckwerts)** can be used when the concentration at the boundary is not known, or when it varies in a nontrivial manner. This may, for example,

be useful when reactions with high reaction rates occur in the vicinity of the inlet. In this case the concentration far upstream of the boundary is instead prescribed. The **Flux (Danckwerts)** condition prescribes the total flux defined by the upstream concentration and the fluid velocity at the boundary.



Danckwerts Inflow Boundary Condition

CONSTRAINT SETTINGS

To display this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog.

Outflow

The **Outflow** condition assumes that convection and migration is the dominating transport mechanism across the outflow boundary, and therefore that diffusive transport can be ignored, that is:

$$\mathbf{n} \cdot (-D\nabla c) = 0$$

Flux Discontinuity

The **Flux Discontinuity** node represents a discontinuity in the species flux across an interior boundary:

$$-\mathbf{n} \cdot (\mathbf{N}_u - \mathbf{N}_d) = N_{0,i} \quad \mathbf{N} = (c_i \mathbf{u} + \mathbf{J}_i)$$

where the value of $N_{0,i}$ specifies the size of the flux jump, evaluated from the up to the down side of the boundary. This boundary condition is only available on interior boundaries.

FLUX DISCONTINUITY

Specify the jump in species flux, using a positive value for increasing flux when going from the down to the up side of the boundary. The normal direction (nx,ny,nz) ((nr,nz) in axisymmetric models) points in the direction from the downside toward the upside of an interior boundary and can be plotted for visualization.

Select the checkboxes for the species to specify a **Flux discontinuity** N_0 (SI unit: mol/(m²·s)) and enter a value or expression for the mass flux jump in the corresponding

field. To use another boundary condition for a specific species, click to clear the checkbox for that species' flux discontinuity.

Current Density

The **Current Density** node provides a boundary condition for the electric field which makes it possible to set the total current density at a boundary. The condition is defined as:

$$-\mathbf{n} \cdot \mathbf{i} = i_0 \quad (3-20)$$

where $\mathbf{n}$ denotes the normal vector to the boundary and i_0 is the inward current density.



It is common to combine this boundary condition with concentration conditions for the species. Such a combination is useful in cases where the flow in an electrochemical cell is vigorous and in cases where the whole cell is not treated in a model, most often when the opposite electrode is a far away. Half-cell investigations are very applicable for this type of condition.

INWARD CURRENT DENSITY

Specify the current density i_0 (SI unit: A/m²). Use a positive value for an inward current flow.

Current Discontinuity

The **Current Discontinuity** node represents a discontinuity in the current density across an interior boundary:

$$-\mathbf{n} \cdot (\mathbf{i}_u - \mathbf{i}_d) = i_0 \quad \mathbf{i} = F \sum_{k=1}^n z_i (-D_i \nabla c_i - z_i u_{m,i} F c_i \nabla V)$$

where the value of i_0 specifies the size of the current density jump evaluated from the up to the down side of the boundary. This boundary condition is only available on interior boundaries.

CURRENT DISCONTINUITY

Specify i_0 (SI unit: A/m^2), the jump in current density, using a positive value for increasing current density when going from the up to the down side of the boundary. The normal direction (n_x, n_y, n_z) ((n_r, n_z) in axisymmetric models) points in the direction from the downside toward the upside of an interior boundary and can be plotted for visualization.

Open Boundary

Use the **Open Boundary** node to set up mass transport across boundaries where both convective inflow and outflow can occur. Specify an exterior species concentration on parts of the boundary where fluid flows into the domain. A condition equivalent to the Outflow node applies to parts of the boundary where fluid flows out of the domain.



The direction of the flow across the boundary is typically calculated by a Fluid Flow interface, and is provided as a Model Input to the Nernst–Planck Equations interface.

The potential is assumed to be constant in the normal direction across the boundary.

EXTERIOR CONCENTRATION

Enter a value or expression for the exterior concentration $c_{0,e2}$ (SI unit: mol/m^3).

Reference for the Nernst–Planck Equations Interface

I. R. Codina, “A Discontinuity-capturing Crosswind-dissipation Stabilization for the Finite Element Solution of the Convection-diffusion Equation,” *Comput. Methods Appl. Mech. Eng.*, vol. 110, pp. 325–342, 1993.

The Nernst–Planck–Poisson Equations Interface

The **Nernst–Planck–Poisson Equations** interface () , found under the **Chemical Species Transport** branch () when adding a physics interface, is a predefined multiphysics interface for modeling transport of electrolyte species without the otherwise common assumption of local electroneutrality. This allows for simulating charge separation that typically arises close to an electrode surface, where ions in the electrolyte are attracted and repelled by unscreened excess charge on the electrode. The charge separation region, also called the diffuse double layer, normally extends a few nanometers away from the electrode surface into the electrolyte. The study of the charge separation is important to applications that consider very thin layers of electrolyte, such as electrochemical capacitors, atmospheric corrosion problems, ion-selective field effect transistors (ISFETs), and nanoelectrochemistry.

The Nernst–Planck–Poisson Equations predefined multiphysics interface adds () (with Migration enabled), which solves for the Nernst–Planck equations without charge neutrality, and [The Electrostatics Interface](#)() , that solves for the Poisson’s equation.

A () multiphysics node is also added. This node computes the local space charge, based on the local concentrations and species charges in the Transport of Diluted Species interface, and adds it to Poisson’s Equation in the Electrostatics interface.

Finally, a () multiphysics node is added that applies the potential dependent variable in the Electrostatics interface into the migration term of the Transport of Diluted Species interface.

	• The Multiphysics Branch in the <i>COMSOL Multiphysics Reference Manual</i>. • The Transport of Diluted Species Interface • The Electrostatics Interface
	<i>Diffuse Double Layer:</i> Application Library path Chemical_Reaction_Engineering_Module/Electrokinetic_Effects/ diffuse_double_layer

The Electrophoretic Transport Interface

The **Electrophoretic Transport (el)** interface (🧪), found under the **Chemical Species Transport** branch (🧪), is used to solve for the electrophoretic transport of an arbitrarily number of species in water-based system, subject to potential gradients. The species transported can be any combination of weak and strong acids and bases, ampholytes, and uncharged species. The transport of masses and charge is based on the Nernst–Planck equations for molecular transport, in combination with electroneutrality, dissociation equilibria for weak acids, bases, and ampholytes as well as the water autoionization reaction.

The physics interface can simulate most forms of electrophoresis modes, such as zone electrophoresis, isotachophoresis, isoelectric focusing, and moving boundary electrophoresis.

Gel electrophoresis can be simulated by the inclusion of immobile charged species.

The interface supports simulation in 1D, 2D, and 3D as well as for axisymmetric components in 1D and 2D.

The dependent variables are the electrolyte potential, and the molar concentrations of the included species, added individual by each species node in the model tree.



Zone Electrophoresis: Application Library path
**Chemical_Reaction_Engineering_Module/Electrokinetic_Effects/
zone_electrophoresis**

SETTINGS

The **Label** is the default physics interface name.

The **Name** is used primarily as a scope prefix for variables defined by the physics interface. Refer to such physics interface variables in expressions using the pattern <name>.<variable_name>. In order to distinguish between variables belonging to different physics interfaces, the name string must be unique. Only letters, numbers, and underscores (_) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first physics interface in the model) is **e1**.

DOMAIN SELECTION

If any part of the model geometry should not partake in the mass transfer model, remove that part from the selection list.

OUT-OF-PLANE THICKNESS



For 2D components, enter a value or expression for the out-of-plane **Thickness** d (SI unit: m). The value of d determines the size of the domain perpendicular to the modeled 2D cross section. This value yields, for example, the correct total current when the current density is obtained from a 2D simulation.

CROSS-SECTIONAL AREA



For 1D components, enter a **Cross-sectional area** A_c (SI unit: m^2) to define a parameter for the area of the geometry perpendicular to the 1D component. The value of this parameter is used, among other things, to automatically calculate the total current from the current density vector. The analogy is valid for other fluxes. The default is 1 m^2 .

TRANSPORT MECHANISMS

Mass transport due to diffusion and migration is always included. Use the checkboxes available under **Additional transport mechanisms** to control other transport mechanisms.

- By default, the **Convection** checkbox is selected. Clear the checkbox to disable convective transport.
- The **Mass transfer in porous media** checkbox activates functionality specific to species transport in porous media. When selected, the Porous Matrix Properties node can be added to a domain to specify the electrolyte volume fraction and tortuosity, and the Effective Transport Parameter Correction sections are enabled in the species nodes.

CONSISTENT STABILIZATION

To display this section, click the **Show More Options** button () and select **Stabilization** in the **Show More Options** dialog.

- When the **Crosswind diffusion** checkbox is selected, a weak term that reduces spurious oscillations is added to the transport equation. The resulting equation system is always nonlinear.
- For both **Streamline diffusion** and **Crosswind diffusion**, select an **Equation residual**. **Approximate residual** is the default and means that derivatives of the diffusivity are neglected. This setting is usually accurate enough and is computationally faster. If required, select **Full residual** instead.

INCONSISTENT STABILIZATION

To display this section, click the **Show More Options** button () and select **Stabilization** in the **Show More Options** dialog. By default, the **Isotropic diffusion** checkbox is not selected because this type of stabilization adds artificial diffusion and affects the accuracy of the original problem. However, this option can be used to get a good initial guess for under-resolved problems.

ADVANCED SETTINGS

To display this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog. Normally these settings do not need to be changed. Select a **Convective term** — **Nonconservative form** (the default) or **Conservative form**. Use the conservative formulation for compressible flow.

DISCRETIZATION

To display all settings available in this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog.

The concentration variables are set to use Linear elements by default.

The potential variable is set to use Quadratic elements by default.

The **Compute boundary fluxes** checkbox is activated by default so that COMSOL Multiphysics computes predefined accurate boundary flux variables. When this option is selected, the solver computes variables storing accurate boundary fluxes from each boundary into the adjacent domain.

If the checkbox is cleared, the COMSOL Multiphysics software instead computes the flux variables from the dependent variables using extrapolation, which is less accurate

in postprocessing results but does not create extra dependent variables on the boundaries for the fluxes.

The flux variables affected in the interface are:

- `<name>.nI1`, where `<name>` is the name of the interface (default is `e1`), set on the interface top node. This is the normal electrolyte current density.
- `<name>.ntflux_<species_name>` is the **Species name** (see [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#) below). This is the normal total flux for each species.

Also the **Apply smoothing to boundary fluxes** checkbox is available if the previous checkbox is selected. The smoothing can provide a more well-behaved flux value close to singularities.

For details about the boundary fluxes settings, see [Computing Accurate Fluxes](#) in the *COMSOL Multiphysics Reference Manual*.

Regarding the **Value type when using splitting of complex variables**, see [Splitting Complex-Valued Variables](#) in the *COMSOL Multiphysics Reference Manual*.

DEPENDENT VARIABLES

The dependent variable name for the electrolyte potential variable is `phi1` by default.

The name of the concentration dependent variables are named as `e1.xxx`, where `e1` is the name of the interface as set above, and the `xxx` string is controlled by the **Species name** setting on the individual species nodes.

FURTHER READING



- [Theory for the Electrophoretic Transport Interface](#)
- [Numerical Stabilization](#) in the *COMSOL Multiphysics Reference Manual*.
- In the *COMSOL Multiphysics Reference Manual*, see [Table 2-4](#) for links to common sections and [Table 2-5](#) for common feature nodes. You can also search for information: press F1 to open the **Help** window or Ctrl+F1 to open the **Documentation** window.

Common Settings for the Species Nodes in the Electrophoretic Transport Interface

The interface features the following species nodes, applicable on the domain level:

- [Ampholyte](#)
- [Fully Dissociated Species](#)
- [Uncharged Species](#)
- [Weak Acid](#)
- [Weak Base](#)

Each species node add a dependent variable for the concentration. The initial and boundary condition, as well as adding additional source reaction terms, for each species concentration is controlled by adding subnodes to the species nodes:

- [Concentration](#)
- [Flux](#)
- [Inflow](#)
- [Initial Concentration](#)
- [No Flux](#)
- [Outflow](#)
- [Species Source](#)

The [Ampholyte](#), [Weak Acid](#), and [Weak Base](#) nodes are dissociation species and may define an arbitrary number of dissociation steps. Each dissociation step is defined by its pK_a (the acid equilibrium constant) parameter. For the weak bases the pK_a refers to the acid constant of the conjugate acid. Each dissociation step adds one additional subspecies concentration variable so that the concentration dependent variable represents the sum of all subspecies, and initial and boundary conditions are defined with respect to this total concentration.

All species node have a setting for the **Species name**, which needs to be unique. The species name is used for naming of all related variables of the species. For species nodes not defining any subspecies, the concentration variables are named as `<name>.c_<species_name>` where `<name>` is the name of the interface (default is `e1`), set on the interface top node, and `<species_name>` is the **Species name**. For dissociation species nodes defining multiple subspecies, the concentration nodes are named as `<name>.c<X>_<species_name>` where `<X>` is the integer from 1 up to the

total number of subspecies. Note that the **Solvent** node automatically defines the concentration variables for protons (<name>.cH) and hydroxide ions (<name>.cOH).

All species except the **Uncharged Species** carry charge and contribute to the total electrolyte current which is used in the equation for solving the electrolyte potential.

The **Immobile Species** checkbox can be used to lock the concentration of a species, to, for instance, define the immobile charges in an ion-selective membrane or a gel. When the checkbox is enabled the concentration of the species is not added as a dependent variable to the model; instead the concentration will be set to the value provided in the **Concentration** field. The contribution to the electrolyte current for immobile species is zero.

Diffusion and Migration Settings

All species, except when the **Immobile Species** checkbox has been enabled, may be transported by diffusion, and charged species are also transported by migration in the electric field.

For dissociation species you can choose to set the transport parameters to be the **Same for all species** appearing in the different dissociation steps, or you can use **Individual** settings for each subspecies.

Typically the mobilities and diffusivities for small species are related by the **Nernst–Einstein relation**, and when this relation is enabled you can choose whether to specify either the **Diffusivity** (SI unit: m^2/s) or the **Mobility** (SI-unit: $\text{s}\cdot\text{mol}/\text{kg}$). The **Debye–Hückel–Henry** relation is commonly used for larger molecules, such as proteins.

Note: There are other definitions of the migration transport equations in the literature which use mobilities expressed in $\text{m}^2/(\text{V}\cdot\text{s})$, whereas COMSOL Multiphysics uses $\text{s}\cdot\text{mol}/\text{kg}$. To convert mobilities expressed in $\text{m}^2/(\text{V}\cdot\text{s})$ to the corresponding values in $\text{s}\cdot\text{mol}/\text{kg}$, you typically divide by the Faraday constant, F_{const} (about 96,485 C/mol).

EFFECTIVE TRANSPORT PARAMETER CORRECTION

If **Mass transfer in porous media** is enabled on the interface top node, you can use the settings of this section in the species nodes to account for the changed mass transport in a porous domain due to the lowered porosity and the increased tortuosity.

The default correction model is **Bruggeman**, which multiplies the diffusivity and mobility values by the porosity to the power of 1.5. The porosity of a domain is set by the [Porous Matrix Properties](#) node.

IONIC STRENGTH CONTRIBUTION

To display this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog.

The **Debye–Hückel–Henry** relation makes use of the ionic strength for calculating the species mobility from the diffusivity. All charged species contribute to the ionic strength, either assuming the species contributing to an **Ideal solution** or by using the **Linderstrøm-Lang** assumption. The latter is usually used for macromolecules.

Domain, Boundary, and Pair Nodes for the Electrophoretic Transport Interface

The interface has the following domain, boundary, and pair nodes, listed in alphabetical order, available from the **Physics** ribbon toolbar (Windows users), **Physics** context menu (Mac or Linux users), or by right-clicking to access the context menu (all users).



To add a node, go to the **Physics** toolbar, no matter what operating system you are using. Subnodes are available by clicking the parent node and selecting it from the **Attributes** menu.

- [Ampholyte](#)
- [Concentration](#)
- [Current](#)
- [Current Density](#)
- [Current Source](#)
- [Flux](#)
- [Fully Dissociated Species](#)
- [Inflow](#)
- [Initial Concentration](#)
- [Initial Potential](#)
- [Insulation](#)
- [No Flux](#)
- [Outflow](#)
- [Porous Matrix Properties](#)
- [Potential](#)
- [Protein](#)
- [Species Source](#)
- [Solvent](#)
- [Uncharged Species](#)
- [Weak Acid](#)
- [Weak Base](#)



For axisymmetric components, COMSOL Multiphysics takes the axial symmetry boundaries (at $r = 0$) into account and automatically adds an **Axial Symmetry** node that is valid on boundaries representing the symmetry axis.



In the *COMSOL Multiphysics Reference Manual*, see [Table 2-4](#) for links to common sections and [Table 2-5](#) for common feature nodes. You can also search for information: press F1 to open the **Help** window or Ctrl+F1 to open the **Documentation** window.

Solvent

The settings of this node are used to define the properties of the aqueous solvent.

If Convection is enabled on the interface top node, you can specify the **Velocity field** (m/s) as user defined input using analytical expressions or the velocity field variables solved for by a separate physics interface.

The Solvent node automatically defines the concentration variables and for protons (<name>.cH) and hydroxide ions (<name>.cOH), and the corresponding flux expressions. See [Diffusion and Migration Settings](#) for how to set up the transport parameters for the proton and hydroxide ions.

In the **Solvent Properties** section you can modify the **Dynamic viscosity** (Pa·s) and **Relative permittivity** (unitless) values. The **Built in** and default values are applicable to water. These parameters are used when calculating mobilities according to the **Debye–Hückel–Henry** relation in the species nodes.

In the **Water Self-Ionization** section you can change the default **Built in** expression for the **Water self-ionization constant**, pK_w (unitless), to any user-defined expression.

Porous Matrix Properties

Use this node, available if **Mass transfer in porous media** has been enabled on the interface top-node, to define the **Porosity** (electrolyte volume fraction) and **Tortuosity** of a domain.

The porosity should be a number between 0 and 1.

Fully Dissociated Species

Use this node to define a fully dissociated charged species, such as strong bases and acids.

The **Charge number**, Z_0 (unitless), defines the species charge.

See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#) and [Diffusion and Migration Settings](#).

Uncharged Species

Use this node to define a species that does not carry any charge, nor is impacted by the electric field.

See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#) and [Diffusion and Migration Settings](#).

Weak Acid

The **Weak acid** node supports multiple dissociation steps, where the acid of the first dissociation step is uncharged.

The species can be either **Monoprotic**, subject to one dissociation step only, or **Polyprotic**. For the latter case, any positive integer larger than one can be used to define the **Number of dissociation steps**.

See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#) and [Diffusion and Migration Settings](#).

Weak Base

The **Weak base** node supports multiple dissociation steps, where the base of the last dissociation step is uncharged.

The species can be either **Monoprotic**, subject to one dissociation step only, or **Polyprotic**. For the latter case, any positive integer larger than one can be used to define the **Number of dissociation steps**.

Note that the pK_a refers to the acid constant of the conjugate acid of the weak base.

See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#) and [Diffusion and Migration Settings](#).

Ampholyte

Use the **Ampholyte** node generically to define any species where the average charge depends on the pH of the solution. The **Dissociation model** may be based either on a set of **Equilibrium constants** or an **Average charge**.

When using the **Equilibrium constants** the **Base charge in last dissociation step**, Z_0 (unitless), needs to be set and be defined as an integer.

The **Average charge**, $\bar{Z}$ (unitless) parameter is typically a function of pH and may be added as an analytical function or an interpolation polynomial under **Definitions**. The average squared charge, used in the electrolyte potential equation, is calculated automatically based on $\bar{Z}$.

See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#) and [Diffusion and Migration Settings](#).

Protein

Use the **Protein** node to define macromolecules. The features of the Protein node are similar to the [Ampholyte](#) node, but with the default settings applicable for larger molecules.

See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#) and [Diffusion and Migration Settings](#).

Current Source

To make this node available, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog.

Use this node to add a current source in a domain. A current source may appear in a domain in homogenized porous electrode modeling, but should normally not be used.

Initial Potential

Use this node to specify the **Initial Value** of the electrolyte potential for the solver.

Current

The **Current** boundary condition sets the total current or average current density over a boundary. It will set a constant electrolyte potential along the given boundary, which satisfies the current value setting.

This node is typically used to model electrode surfaces or boundaries facing an electrolyte reservoir (containing an electrode).

When using the **Total current** option in 1D or 2D, the boundary area is based either on the **Cross-sectional area** (1D) or the **Out-of-Plane thickness** (2D) properties, set on the physics interface top node.

Current Density

Use the **Current Density** node to specify the current density distribution along a boundary.

This node is typically used to model electrode surfaces where the electrode kinetics depends on the electrolyte potential.

Note that using this node in 2D or 3D may result in an uneven potential distribution along the boundary. To mitigate such effects you can use the [Current](#) node instead.

Insulation

The **Insulation** boundary condition describes the walls of a cell or the boundaries of the cell that do not face an electrode (or a reservoir containing an electrode). The boundary condition imposes the following equation:

$$\mathbf{i}_l \cdot \mathbf{n} = 0$$

Potential

Add the **Potential** node on a boundary to set a fixed potential. This node is typically used to model electrode surfaces or boundaries facing an electrolyte reservoir.

The node sets the potential in the electrolyte, ϕ_l , to be equal to the **Boundary electrolyte potential**, $\phi_{l, \text{bnd}}$ (SI unit: V).

CONSTRAINT SETTINGS

To display this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog.

Species Source

To make this node available, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog.

This node may be added as a subnode to any species node. See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#).

In order to account for consumption or production of species, for example in porous electrodes or as a result of reversible reactions, the **Species Source** node adds source terms expressions S (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$) to the right-hand side of the transport equation of the parent species.

Initial Concentration

This node may be added as a subnode to any species node. See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#).

This node specifies the initial value for the **Concentration**, c (mol/m^3), of the parent species. This value serve as the initial condition for a transient simulation. The value also serves as a start guess for stationary problems.

You can use spatially dependent functions (such as smoothed step functions) available under **Definitions** when defining the **Concentration** expression to specify different concentrations in different parts of the geometry. You can also use additional **Initial Values** node and modify the **Selection** to set different values for different domains.

Concentration

This node may be added as a subnode to any species node. See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#)

This condition node adds a boundary condition for the parent species concentration. Use the node to, for instance, specify the inlet concentration at the boundary facing an electrolyte reservoir.

No Flux

This node may be added as a subnode to any species node. See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#).

This node is the default boundary condition on exterior boundaries. It represents boundaries where no mass flows in or out of the boundaries. Hence, the total flux is zero.

Flux

This node may be added as a subnode to any species node. See also [Common Settings for the Species Nodes in the Electrophoretic Transport Interface](#).

This node can be used to specify the species inward flux across a boundary. The flux can represent a flux from or into a much larger surrounding environment, a phase change, or a flux due to chemical reactions.

INWARD FLUX

Enter a value or expression for the species mass flux J_0 . Use a minus sign when specifying a flux directed out of the system.

External convection

Set **Flux type** to **External convection** to prescribe a flux to or from an exterior domain (not modeled) assumed to include convection. The exterior can for example include a forced convection to control the temperature or to increase the mass transport. In this case the prescribed mass flux corresponds to

$$J_0 = k_c(c_b - c)$$

where k_c is a mass transfer coefficient and c_b is the bulk concentration, the typical concentration far into the surrounding exterior domain.

Inflow

This node is available when you select the **Convection** checkbox on the physics interface **Settings** window.

Use this node to specify the species concentrations at an inlet boundary.

BOUNDARY CONDITION TYPE

The option **Concentration constraint** constrains the concentration values on the boundary by the use of pointwise constraints. When using this option the boundary condition is identical to the [Concentration](#) node.

The other option, **Flux (Danckwerts)** can be more stable and fast to solve when high reaction rates are anticipated in the vicinity of the inlet. Oscillations on the solutions can also be avoided in such cases. The latter condition uses a flux boundary condition based on the velocity across the boundary and the concentration values.

Outflow

Set this condition at outlets where species are transported out of the model domain by migration or fluid motion. It is assumed that migration and convection are the dominating transport mechanisms across outflow boundaries, and therefore that diffusive transport can be ignored; that is:

$$\mathbf{n} \cdot (-D\nabla c) = 0$$

The Surface Reactions Interface

The **Surface Reactions (sr)** interface () , found under the **Chemical Species Transport** branch () when adding a physics interface, is used to model the chemical reactions of surface and bulk species on a boundary. Surface species can be transported in the tangential direction of the surface by Fick's law whereas bulk species are assumed to be immobile on the surface.

The physics interface supports simulation of surface reactions on boundaries in 1D, 2D, and 3D as well as for axisymmetric components in 1D and 2D. The dependent variables are the surface concentrations, $c_{s,i}$ (SI unit: mol/m²) and the bulk concentrations, $c_{b,i}$ (SI unit: mol/m²).

When this physics interface is added, these default nodes are also added to the **Model Builder** — **Surface Properties**, **No Flux**, and **Initial Values**. Then, from the **Physics** toolbar, add other nodes that implement, for example, boundary conditions. You can also right-click **Surface Reactions** to select physics features from the context menu.

SETTINGS

The **Label** is the default physics interface name.

The **Name** is used primarily as a scope prefix for variables defined by the physics interface. Refer to such physics interface variables in expressions using the pattern <name>.<variable_name>. In order to distinguish between variables belonging to different physics interfaces, the name string must be unique. Only letters, numbers, and underscores (_) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first physics interface in the model) is sr.

DEPENDENT VARIABLES

Add or remove species and also change the names of the dependent variables that represent the species concentrations. Note that the names can be changed but the names of fields and dependent variables must be unique within a model.

Enter the **Number of surface species**. Use the **Add Surface Concentration** () and **Remove Surface Concentration** () buttons as needed. The resulting **Surface (adsorbed) species concentrations** cs1, cs2, cs3, ... are then listed in the table.

Enter the **Number of bulk species**. Use the **Add Bulk Concentration** () and **Remove Bulk Concentration** () buttons as needed. The resulting **Bulk concentrations** cb1, cb2, cb3, ... are then listed in the table.

DISCRETIZATION

To display all settings available in this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog.

CONSISTENT AND INCONSISTENT STABILIZATION

To display these sections, click the **Show More Options** button () and select **Stabilization** in the **Show More Options** dialog.

By default the **Compensate for boundary stretching** checkbox is selected for the [Surface Properties](#) node. This section is then used to stabilize the tangential mesh velocity term.

When the **Compensate for boundary stretching** checkbox is cleared (not selected), and for fixed geometries or moving geometries, the stabilization has no effect.

See [Surface Reaction Equations on Deforming Geometries](#) for more information.

	• Boundary, Edge, Point, and Pair Nodes for the Surface Reactions Interface • Theory for the Surface Reactions Interface
	<i>Protein Adsorption:</i> Application Library path Chemical_Reaction_Engineering_Module/Mixing_and_Separation/protein_adsorption

Boundary, Edge, Point, and Pair Nodes for the Surface Reactions Interface

The [Surface Reactions Interface](#) has these boundary, edge, point, and pair nodes, listed in alphabetical order, available from the **Physics** ribbon toolbar (Windows users),

Physics context menu (Mac or Linux users), or right-click to access the context menu (all users).



In general, to add a node, go to the **Physics** toolbar, no matter what operating system you are using. Subnodes are available by clicking the parent node and selecting it from the **Attributes** menu.

These nodes are described in this section:

- [Initial Values](#)
- [Reactions](#)
- [Surface Concentration](#)
- [Surface Properties](#)

All other available nodes are described for the Transport of Diluted Species interface. See [Domain, Boundary, and Pair Nodes for the Transport of Diluted Species Interface](#).



In the *COMSOL Multiphysics Reference Manual* see [Table 2-4](#) for links to common sections and [Table 2-5](#) to common feature nodes. You can also search for information: press F1 to open the **Help** window or Ctrl+F1 to open the **Documentation** window.

Surface Properties

Use the **Surface Properties** node to define the density of sites, the site occupancy number, and the surface diffusion.

SPECIES CONSERVATION ON DEFORMING GEOMETRY

The **Compensate for boundary stretching** checkbox is selected by default. Click to clear the checkbox if required.

SITES

Enter a value or expression for the **Density of sites** Γ_s (SI unit: mol/m²). The default is 2×10^{-5} mol/m².

Enter a **Site occupancy number** σ_i (dimensionless), indicating how many surface sites a surface species block upon adsorption.

This section is only present if the number of surface species is 1 or higher.

SURFACE DIFFUSION

Select a **Surface material** from the list. The default is **None**.

For each surface concentration species, the default **Diffusion coefficient** D_i (SI unit: m^2/s) is **User defined**. Select **Isotropic**, **Diagonal**, **Symmetric**, or **Full** depending on the characteristics of the surface, and then enter values or expressions in the field or matrix.

This section is only present if the number of surface species is 1 or higher.

BULK SPECIES

For each bulk species enter the **Molar mass** M_i (SI unit: kg/mol) and the **Density** ρ_i (SI unit: kg/m^3). The default molar mass is 0.144 kg/mol and the default density is 5320 kg/m^3 .

This section is only present if the number of bulk species is 1 or higher.



Surface Reaction Equations on Deforming Geometries

Initial Values

The **Initial Values** node allows the initial value or guess for the surface and bulk concentrations.

INITIAL VALUES

Based on the number of surface species and number of bulk species entered for the physics interface under Dependent Variables section, enter values for the same number of **Surface concentration** $c_s, c_{s2}, c_{s3}, \dots$ (SI unit: mol/m^2) and **Bulk concentration** $c_b, c_{b2}, c_{b3}, \dots$ (SI unit: mol/m^2) in each field.

Reactions

The **Reactions** node adds rate expression terms to the species transport equations in order to account for consumption or production of species due to reactions.

REACTION RATE FOR SURFACE SPECIES

Add a reaction rate expression, $R_{s,cs1,cs2,\dots}$ (SI unit: $\text{mol}/(\text{m}^2 \cdot \text{s})$), for each surface species taking part in a surface reaction.

This section is only present if the number of surface species is 1 or higher.

REACTION RATE FOR BULK SPECIES

Add a reaction rate expression, $R_{b,cb1,cb2\dots}$ (SI unit: $\text{mol}/(\text{m}^2\cdot\text{s})$), for each bulk species taking part in a surface reaction.

This section is only present if the number of bulk species is 1 or higher.

Surface Concentration

Use the **Surface Concentration** node to set the surface concentrations for one or more species on an edge (3D components) or a point (2D and 2D axisymmetric components).

SURFACE CONCENTRATION

Select each species checkbox as needed and enter a value or expression for each species concentration, $c_{s,0,cs1,cs2\dots}$ (SI unit: $\text{mol}/(\text{m}^2\cdot\text{s})$).

CONSTRAINT SETTINGS

To display this section, click the **Show More Options** button () and select **Advanced Physics Options** in the **Show More Options** dialog.

The Size-Based Population Balance Interface

The **Size-Based Population Balance** interface () , found in the **Precipitation and Crystallization** group () of the **Chemical Species Transport** branch () , is used to model a size distribution of particles, droplets or bubbles.

The interface supports simulation of the space and time-dependent size distribution and mass transport by convection and diffusion. The interface can be used in 0D, 1D, 2D, and 3D as well as for axisymmetric components in 1D and 2D. The dependent variables are the population number density of each size interval, accounting for interval length, n ($1/\text{m}^4$). The minimum number of dependent variables is five.

SETTINGS

The **Label** is the default physics interface name.

The **Name** is used primarily as a scope prefix for variables defined by the physics interface. Refer to such physics interface variables in expressions using the pattern `<name>.<variable_name>`. In order to distinguish between variables belonging to different physics interfaces, the name string must be unique. Only letters, numbers and underscores (`_`) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first physics interface in the model) is `pbsb`.

DOMAIN SELECTION

Select the domains of the model that the population balance model should be applied to.

OUT-OF-PLANE THICKNESS

For 2D components, the text field (default value: 1 m) defines a parameter for the thickness of the geometry perpendicular to the two-dimensional cross-section area. Both constant and varying thicknesses are supported.

CROSS-SECTIONAL AREA

For 1D components, the text field (default value: 1 m^2) defines a parameter for the cross-sectional area perpendicular to the 1D component. Both constant and varying areas are supported.

CONSISTENT STABILIZATION

To display this section, click the **Show** button () and select **Stabilization**.

Two consistent stabilization methods are available — **Streamline Diffusion** and **Crosswind Diffusion**. Both are active by default. When the **Streamline Diffusion** checkbox is selected a weak artificial diffusion in the streamline direction is added to the transport equation. When the **Crosswind Diffusion** checkbox is selected, a weak diffusive term orthogonal to the streamline direction that reduces spurious oscillations is added. The consistent stabilization methods do not perturb the original transport equation.

INCONSISTENT STABILIZATION

To display this section, click the **Show** button () and select **Stabilization**. One inconsistent stabilization method is available, isotropic diffusion. Isotropic diffusion can affect the accuracy of the original problem and the **Isotropic Diffusion** checkbox is not selected by default. For under resolved problems, this option can be used to get a good initial guess.

ADVANCED SETTINGS

To display this section, click the **Show** button () and select **Advanced Physics Options**. Select **Nonconservative** or **Conservative** from the **Material balance form** list. **Nonconservative** is selected by default and does usually not need to be changed.

PARTICLE PROPERTIES

The **Particle Properties** section is used to specify properties of the particles in the population. Enter a value or expression for the density and molar mass in the **Particle density** and **Particle molar mass** fields.

Select the particle shape from the **Particle shape** list. Select **Spherical** for spherical particles or **Cubic** for cubic particles. For **User defined**, specify an **Area shape factor** and a **Volume shape factor** in the corresponding fields.

These properties are used for calculating additional variables such as particle volume and when using certain predefined nucleation and growth rate models.

SIZE INTERVALS

Select the size discretization approach from the **Discretization** list. Select **Linear** to use a linear discretization approach. For a non-linear discretization you can select **Logarithmic** or **Geometric**. For all options, enter the number of size intervals to model in the **Number of intervals** field and the smallest and largest size to model in the

Minimum size and **Maximum size** fields. For the **Geometric** option, also enter a finite constant in the **Geometric constant** field that controls the size interval width.

NUCLEATION

Select an option from the **Nucleation rate** list to add a particle source to the smallest size interval. Three different options are available — **User defined**, **Arrhenius expression**, and **Homogeneous nucleation**.

GROWTH AND DISSOLUTION

Select an option from the list to apply a growth and dissolution rate to all intervals. The options that are available are — **User defined** and **Transport controlled**.

Select **User defined** to enter a value or expression. When this option is selected, the text field also accepts an expression written as a function of the particle size, `pop.L`. When solving, `pop.L` will be replaced by the discretized particle sizes. Separate text fields are available for growth and dissolution. A positive growth rate results in particles becoming larger while a positive dissolution rate results in particles becoming smaller.

Select **Transport controlled** to define a growth and dissolution rate that is limited by the transport of a dissolved species.

AGGREGATION

Select the **Include** checkbox to include aggregation. When selected, enter an aggregation kernel in the **Aggregation kernel** field. The field supports expressions that are functions of the two colliding particle sizes, `pop.Lj` and `pop.Lk`.

BREAKAGE

Select the **Include** checkbox to include breakage. When selected, enter a breakage rate constant and an average number of particles resulting from breakage in the **Breakage rate constant** and **Average number of resulting particles** fields, respectively.

DISSOLVED SPECIES PROPERTIES

This section is only visible when additional information is requested by the **Growth** or **Nucleation** section. This is when **Homogeneous nucleation** is selected in the **Nucleation** section or **Transport controlled** is selected in the **Growth** section. Enter a value or expression in the **Concentration**, **Equilibrium concentration**, and **Diffusion coefficient** fields.

DISCRETIZATION

To display all settings available in this section, click the **Show** button () and select **Advanced Physics Options**.

The **Compute boundary fluxes** checkbox is activated by default so that COMSOL Multiphysics computes the predefined accurate boundary flux variables storing accurate boundary fluxes from each boundary into the adjacent domain.

If the checkbox is cleared, the COMSOL Multiphysics software instead computes the flux variables from the dependent variables using extrapolation, which is less accurate in postprocessing results but does not create extra dependent variables on the boundaries for the fluxes.

The **Apply smoothing to boundary fluxes** checkbox is available if the previous checkbox is selected. The smoothing can provide a more well-behaved flux value close to singularities.

For details about the boundary fluxes settings, see [Computing Accurate Fluxes](#) in the *COMSOL Multiphysics Reference Manual*.

The **Value type when using splitting of complex variables** setting should in most pure mass transfer problems be set to **Real**, which is the default. It makes sure that the dependent variable does not get affected by small imaginary contributions, which can occur, for example when combining a Time Dependent or Stationary study with a frequency-domain study. For more information, see [Splitting Complex-Valued Variables](#) in the *COMSOL Multiphysics Reference Manual*.

Domain, Boundary, and Pair Nodes for the Size-Based Population Balance Interface

The interface has the following domain, boundary, and pair nodes, listed in alphabetical order, available from the **Physics** ribbon toolbar (Windows users), **Physics** context menu (Mac or Linux users), or by right-Clicking to access the context menu (all users).



To add a node, go to the **Physics** toolbar, no matter what operating system you are using. Subnodes are available by clicking the parent node and selecting it from the **Attributes** menu.

- [Additional Source](#)
- [Fluid](#)
- [Inflow](#)
- [Initial Values](#)
- [No Flux](#)
- [Number Density](#)
- [Outflow](#)



For axisymmetric components, COMSOL Multiphysics takes the axial symmetry boundaries (at $r = 0$) into account and automatically adds an **Axial Symmetry** node that is valid on boundaries representing the symmetry axis.



In the *COMSOL Multiphysics Reference Manual*, see [Table 2-4](#) for links to common sections and [Table 2-5](#) for common feature nodes. You can also search for information: press F1 to open the **Help** window or Ctrl+F1 to open the **Documentation** window.

Additional Source

Use the **Additional Source** node to add an additional source term on a domain.

ADDITIONAL SOURCE

To add an additional source term to a size interval, enter a value or expression in the corresponding text field.

Fluid

The **Fluid** node is added by default and is used to model the mass transfer in a fluid with the Size-Based Population Balance interface. The node adds all equations governing the diffusive and convective mass transport of the population. It also provides inputs for material properties. The **Fluid** node is not available in 0D.

MODEL INPUTS

The temperature model input is always available. Select the source of the **Temperature**. To enter a value or expression for the temperature, select the **User Defined** option. The **User Defined** option is always available.

CONVECTION

Under **Convection**, the velocity field for convective mass transfer can be specified. Select the source of the velocity field from the **Velocity Field** list. For **User Defined**, enter values or expressions for the velocity field components in the input fields. This option is always available.

A fluid flow interface can be selected as the source of the velocity field.

DIFFUSION

In the **Diffusion** section, select the source of the diffusion from the list. It can be set to **From material**, **User defined**, or **Define Individually**.

For **From material**, select a material from the **Material** list to access diffusion coefficients defined within that material. Select a diffusion coefficient from the **Diffusion coefficient** list. If you do not select a material with a diffusion coefficient, you can enter a value or expression in the **Diffusion coefficient** field. The value or expression can be a scalar value for isotropic diffusion or a tensor describing anisotropic diffusion. Select the appropriate tensor type — **Isotropic**, **Diagonal**, **Symmetric** or **Full**.

When the diffusion source is set to **User defined**, enter a value or expression in the **Diffusion coefficient** field. When this option is selected, the text field also accepts an expression written as a function of the particle size, pop.L .

Select **Define individually** to set the diffusion coefficient for each discretized size interval individually.

Inflow

Use the **Inflow** node to specify the number densities of different sizes at an inlet boundary.

If you want to specify the number densities of only certain size intervals, this can be done using the [Number Density](#) node instead.

The **Inflow** node is not available in 0D.

POPULATION NUMBER DENSITY

Select **User defined** from the list to enter a value or expression that is applied to all size intervals. When this option is selected, the text field also accepts an expression written as a function of the particle size, pop.L . Select **Define individually** instead to enter a separate value or expression for each size interval.

BOUNDARY CONDITION TYPE

The option **Number-density constraint** constrains the number density values on the boundary.

The other option, **Flux (Danckwerts)**, prescribes the total flux at the boundary defined by an upstream number density and fluid velocity at the boundary.

CONSTRAINT SETTINGS

To display this section, click the **Show More Option** button () and select **Advanced Physics Options**. You can find details about the different constraint settings in the section [Constraint Reaction Terms](#) in the *COMSOL Multiphysics Reference Manual*.

Initial Values

The **Initial Values** node specifies the initial values for the number densities of the different size intervals. These are used as an initial guess for a stationary solver or as initial conditions for a transient simulation.

POPULATION NUMBER DENSITY

Select either **User defined** or **Define individually** from the list. When **User defined** is selected, enter a value or expression for the initial value of the number density to be used for all size intervals. When **User defined** is selected, the text field also accepts an expression written as a function of the particle size, `pop.L`. Instead, when **Define individually** is selected, a number density can be entered for each dependent variable individually with the unit $1/\text{m}^4$. To instead specify number densities independent of interval length, select **Define number densities independent of interval length**. The number densities then have the unit $1/\text{m}^3$.

No Flux

The **No Flux** node is added by default and is the default boundary condition on exterior boundaries. It should be used on boundaries across which there is no mass flux. The condition is applied as:

$$-\mathbf{n} \cdot (-D\nabla n) = 0$$

CONVECTION

By default, the **No Flux** node prescribes a vanishing diffusive flux at the boundary. This is appropriate when the relative convective velocity at the boundary is zero. To include the convective flux, select **Include** in the **Convection** section.

When including convection, the total flux is prescribed instead and the no flux condition is:

$$-\mathbf{n} \cdot (-D\nabla n + \mathbf{u}n) = 0$$

The **No Flux** node is not available in 0D.

Number Density

The **Number Density** node adds a boundary condition for the number density at the boundary.

The **Number Density** node is not available in 0D.

POPULATION NUMBER DENSITY

Select **User defined** from the list to enter a value or expression that is applied to all size intervals. When this option is selected, the text field also accepts an expression written as a function of the particle size, `pop.L`. Select **Define individually** instead to define number densities individually for each size interval. For **Define individually**, select the checkbox for the number density of a specific size interval and enter a value or an expression in the corresponding field to specify the number density.

CONSTRAINT SETTINGS

To display this section, click the **Show More Option** button () and select **Advanced Physics Options**. You can find details about the different constraint settings in the section [Constraint Reaction Terms](#) in the *COMSOL Multiphysics Reference Manual*.

Outflow

Use the **Outflow** node to specify an outlet boundary where transport out of the model domain can occur. It is assumed that convection is the dominating transport mechanism across the boundary and the diffusive transport can be ignored, that is:

$$\mathbf{n} \cdot (D\nabla n) = 0$$

The **Outflow** node is not available in 0D.

The Vapor–Liquid Equilibrium Multiphysics Interfaces

The Vapor–Liquid Equilibrium interfaces are predefined multiphysics interfaces, available in the Chemical Reaction Engineering Module, that solve for mass transfer and fluid flow of gas mixtures in contact with liquids.

Selecting a Vapor–Liquid Equilibrium multiphysics interface under the **Chemical Species Transport** > **Vapor–Liquid Equilibrium** branch of the **Model Wizard**, a Fluid Flow interface and a **Transport of Concentrated Species in Vapor** interface are added to the Model Builder. In addition, the Multiphysics node is added, which includes the multiphysics coupling feature **Reacting Flow**. The **Reacting Flow** feature predefines and controls the couplings between the separate interfaces in order to facilitate easy set up of models.

The **Transport of Concentrated Species in Vapor Interface** includes three dedicated boundary features for vapor transfer. The **Vapor Inflow** feature can be used to prescribe inflow conditions resulting from an upstream system involving a liquid. The **Vapor–Liquid Interface** feature can be used to model condensation and evaporation to or from a gas mixture in contact with a liquid. In the case of a liquid with multiple components, the **Vapor–Liquid–Solution Interface** can be used to also model the composition in the liquid.

In this section:

- [Laminar Vapor Flow](#)
- [Laminar Two-Phase Flow](#)
- [Turbulent Flow, \$\kappa\$ – \$\epsilon\$](#)
- [Turbulent Flow, \$\kappa\$ – \$\omega\$](#)
- [Turbulent Flow, SST](#)
- [Turbulent Flow, Low Re \$\kappa\$ – \$\epsilon\$](#)

Laminar Vapor Flow

The **Laminar Vapor Flow** () version of the Vapor–Liquid Equilibrium multiphysics interface is used to simulate evaporation and condensation to or from a vapor–liquid

interface. The interface combines the Transport of Concentrated Species in Vapor and the Laminar Flow interfaces.

The Reacting Flow multiphysics coupling, which is added automatically, couples vapor mass transport and fluid flow.

Laminar Two-Phase Flow

The **Laminar Two Phase Flow** () version of the Vapor–Liquid Equilibrium multiphysics interface is used to simulate evaporation and condensation to or from a vapor–liquid interface, and to track the position of the interface. The interface combines the Transport of Concentrated Species in Vapor, the Laminar Flow, and the Moving Mesh interfaces.

The Reacting Flow multiphysics coupling, which is added automatically, couples vapor mass transport and fluid flow.

Turbulent Flow, κ – ϵ

The **Turbulent Flow, κ – ϵ** () version of the Vapor–Liquid Equilibrium multiphysics interface is used to simulate evaporation and condensation to or from a vapor–liquid interface. The interface combines the Transport of Concentrated Species in Vapor and the Turbulent Flow, κ – ϵ interfaces.

The Reacting Flow multiphysics coupling, which is added automatically, couples vapor mass transport and turbulent fluid flow.

This interface requires a license for the CFD Module or the Heat Transfer Module.

Turbulent Flow, κ – ω

The **Turbulent Flow, κ – ω , Concentrated Species** () version of the Vapor–Liquid Equilibrium multiphysics interface is used to simulate evaporation and condensation to or from a vapor–liquid interface. The interface combines the Transport of Concentrated Species in Vapor and the Turbulent Flow, κ – ω interfaces.

The Reacting Flow multiphysics coupling, which is added automatically, couples vapor mass transport and turbulent fluid flow.

This interface requires a license for the CFD Module.

Turbulent Flow, SST

The **Turbulent Flow, SST** () version of the Vapor–Liquid Equilibrium multiphysics interface is used to simulate evaporation and condensation to or from a vapor–liquid interface. The interface combines the Transport of Concentrated Species in Vapor and the Turbulent Flow, SST, interfaces.

The Reacting Flow multiphysics coupling, which is added automatically, couples vapor mass transport and turbulent fluid flow.

This interface requires a license for the CFD Module or the Heat Transfer Module.

Turbulent Flow, Low Re κ – ϵ

The **Turbulent Flow, Low Re κ – ϵ** () version of the Vapor–Liquid Equilibrium multiphysics interface is used to simulate evaporation and condensation to or from a vapor–liquid interface. The interface combines the Transport of Concentrated Species in Vapor and the Turbulent Flow, Low Re κ – ϵ interfaces.

The Reacting Flow multiphysics coupling, which is added automatically, couples vapor mass transport and turbulent fluid flow.

This interface requires a license for the CFD Module or the Heat Transfer Module.

The Reacting Flow, Concentrated Species Multiphysics Interface

Reacting Flow, Concentrated Species is a predefined multiphysics interface, available in the Chemical Reaction Engineering Module, that couples fluid flow with mass transport and reactions.

Selecting **Laminar Flow, Concentrated Species** under the **Chemical Species Transport > Reacting Flow** branch of the **Model Wizard** or **Add Physics** windows, a **Laminar Flow** interface and a **Transport of Concentrated Species** interface are added to the Model Builder.

In addition, the Multiphysics node is added, which includes the multiphysics coupling feature **Reacting Flow**. The **Reacting Flow** feature predefines and controls the couplings between the separate interfaces in order to facilitate easy set up of models.

In this section:

- [The Reacting Laminar Flow, Concentrated Species Interface](#)
- [The Reacting Flow Coupling Feature](#)
- [Physics Interface Features](#)

The Reacting Laminar Flow, Concentrated Species Interface

The **Reacting Laminar Flow** () multiphysics interface is used to simulate laminar flow coupled to species transport in a gas or liquid.

It combines the **Laminar Flow**, and **Transport of Concentrated Species** interfaces. The **Reacting Flow** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. The fluid flow can either be free flow or flow in a porous medium. The species transport supports both a mixture, where the concentrations are of comparable order of magnitude, and low-concentration solutes in a solvent.

The interface can be used for stationary and time-dependent analysis in 2D, 2D axial symmetry, and 3D.

ON THE CONSTITUENT PHYSICS INTERFACES

The equations solved by the **Laminar Flow** interface are the Navier–Stokes equations for conservation of momentum and the continuity equation for conservation of mass. A

Fluid Properties feature is active by default on the entire interface selection. A **Porous Medium** feature can be added in order to model flow in porous media by solving the Brinkman equations.

The **Transport of Concentrated Species** interface solves for an arbitrary number of mass fractions. The species equations include transport by convection, diffusion and, optionally, migration in an electric field.



The **Reacting Laminar Flow** interface triggers pseudo time stepping for the flow equations when **Use pseudo time stepping for stationary equation form** in the **Fluid Flow** interface is set to **Automatic from physics**.

The Reacting Flow Coupling Feature

Use the **Reacting Flow** () multiphysics coupling to simulate mass transport and reactions in a gas or liquid mixture where the fluid flow can be dependent on the mixture composition. When a **Heat Transfer** and a **Chemistry** interface are selected, use this coupling to simulate heat transfer additionally to mass transport and reactions.

The coupling adds the heat source of reaction when a heat transfer interface is selected. No additional heat source needs to be defined to account for it in the heat transfer interface. It also accounts for the multiphysics stabilization terms, for work due to pressure forces and viscous dissipation.

Select a **Fluid Flow** interface and a **Species transport** interface to couple fluid flow with mass transport. **Chemistry** and **Heat Transfer** are optional. They can be set to **None** when the coupling is used to simulate isothermal mixtures.

When a **Chemistry** interface is selected and **Heat Transfer** is set to **None**, fluid properties are taken from the Chemistry interface. Set the **Temperature** to evaluate the fluid properties synchronized with all the physics interfaces at the given temperature.

Select a **Chemistry** interface and **Heat Transfer** interface in order to account for the heat of reaction, enthalpy diffusion, viscous heating and mass fluxes contributing to the heat and energy balance.



When **Chemistry** is set to **None** and a **Heat Transfer** interface is selected, the coupling is solved in the same way as when no Heat Transfer interface is selected. Thermodynamic properties are required by the Heat Transfer interface.



The pressure, velocity, and temperature variables of the **Reacting Flow** coupling node are set to the **Common Model Input** values of the **Default Model Inputs** node on the complementary selection, that is, all domains except those from the **Selection** list. It allows to couple multiple fluid flow or transport of species interfaces with a single heat transfer interface. See [Default Model Inputs](#) in the *COMSOL Multiphysics Reference Manual* for details.

DOMAIN LEVEL SYNCHRONIZATION

The **Reacting Flow** coupling synchronizes the features from a Chemistry interface, Heat Transfer interface, Single-Phase Flow, or Brinkman Equations, interface and a Transport of Concentrated Species interface. When the Chemistry interface is not selected, the density in the Single-Phase Flow interface is automatically synchronized to the one defined by the Transport of Concentrated Species interface.

The velocity field used by the Transport of Concentrated Species interface and Heat Transfer interface is synchronized to the one computed in the Single-Phase Flow interface.

When a Chemistry interface is selected, the **Reacting Flow** coupling synchronizes the definition of the thermal conductivity, density, heat capacity, enthalpy, and dynamic viscosity with the other coupled physics interfaces. The reference temperature is taken from the Heat Transfer interface.

THE STEFAN VELOCITY

The **Reacting Flow** coupling feature automatically couples mass transfer on boundaries and applies a corresponding velocity contribution for the flow. Prescribing a net mass boundary flux in the **Transport of Concentrated Species** interface, either using a **Flux** or **Mass Fraction** feature, the **Reacting Flow** feature computes [The Stefan Velocity](#) and applies this in **Wall** features using the same selection.

MASS TRANSFER TO OTHER PHASES IN POROUS MEDIA

When coupled to the **Brinkman Equations** interface, the **Reacting Flow** node automatically computes the net mass source or sink in a **Reactions** (when **Mass transfer to other phases is enabled**) node in the **Transport of Concentrated Species** interface and adds the corresponding source/sink to the momentum equations of the **Fluid and Matrix Properties** domains.

TURBULENT MASS TRANSFER

When a turbulence model is used, the **Reacting Flow** coupling applies turbulence modeling for the mass transport in the following manners:

- Turbulent heat transfer and mass transport are added, defined from the turbulent viscosity and a turbulent Schmidt number (for more information, see [Turbulent Mass Transport Models](#)).
- Temperature and mass transport wall functions. When a turbulence model using wall functions is used for the fluid flow, the Reacting Flow coupling automatically adds wall functions for the temperature and mass transport on the same boundaries (for more information, see [Mass Transport Wall Functions](#)).

SETTINGS

The **Label** is the default multiphysics coupling feature name.

The **Name** is used primarily as a scope prefix for variables defined by the coupling node. Refer to such variables in expressions using the pattern `<name>.<variable_name>`. In order to distinguish between variables belonging to different coupling nodes or physics interfaces, the `name` string must be unique. Only letters, numbers, and underscores (`_`) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first multiphysics coupling feature in the model) is `nirf1`.

DOMAIN SELECTION

The **Reacting Flow** coupling is automatically defined on the intersection of the selections for the coupled interfaces. When modeling porous media transport a **Porous Medium** feature, applied on the same domains, is needed in all coupled transport interfaces. **Reacting Flow** coupling supports porous media transport in **Fluid flow** and **Species transport** interfaces.

The **Selection** list displays the domains where the coupling feature is active.

COUPLED INTERFACES

This section defines the physics involved in the multiphysics coupling. The **Fluid flow**, **Species transport**, **Chemistry**, **Heat Transfer** lists include all applicable physics interfaces.

The default values depend on how this coupling node is created.

- If it is added from the **Physics** ribbon (Windows users), **Physics** contextual toolbar (Mac and Linux users), or context menu (all users), then the first physics interface of each type in the component is selected as the default.
- If it is added automatically when a multiphysics interface is chosen in the **Model Wizard** or **Add Physics** window, then the two participating physics interfaces are selected.

You can also select **None** from a list to uncouple the node from a physics interface.

Click the **Go to Source** buttons () to move to the main physics interface node for the selected physics interface.

Click the **Show or Hide Physics Properties Settings** button () to toggle the display of physics properties settings affecting the coupling feature. When a turbulence model is used, turbulent heat and mass transfer is automatically accounted for (see the settings in the **Turbulence** section below). Using Reacting Flow, the heat and mass transfer treatment at walls follows that applied for the fluid flow. Therefore the **Wall treatment** setting is also displayed when using a turbulence model. For more information on turbulent mass transfer at walls, see the section [Mass Transport Wall Functions](#) in the *CFD Module User's Guide*.



If a physics interface is deleted and then added to the model again, then in order to reestablish the coupling, you need to choose the physics interface again from the **Fluid flow**, **Species transport**, **Chemistry** and **Heat Transfer** lists. This is applicable to all multiphysics coupling nodes that would normally default to the once present interface. See [Multiphysics Modeling Workflow](#) in the *COMSOL Multiphysics Reference Manual*.

MASS TRANSPORT TURBULENCE MODEL

When the fluid flow interface uses a turbulence model, select an option from the **Mass transport turbulence model** list — **Kays–Crawford**, **High Schmidt Number**, or **User-defined turbulent Schmidt number**.

For **User-defined turbulent Schmidt number**, enter a **Turbulent Schmidt number** Sc_T (dimensionless).

The turbulent mass transfer added to the mass fraction equations is defined as

$$N_{i,T} = -\frac{\mu_T}{Sc_T} \nabla \omega_i$$

where μ_T is the turbulent viscosity defined by the flow interface, and the turbulent Schmidt number, Sc_T , depends on the **Mass transport turbulence model** used.

HEAT TRANSFER TURBULENCE MODEL

This section is available when a Heat Transfer interface is selected and the fluid flow interface uses a turbulence model. When a RANS turbulence model is used, the conductive heat flux is defined as

$$q = -(k + k_T) \nabla T$$

with the turbulent thermal conductivity defined by the heat transport turbulence model. The turbulent Prandtl number models are available for all RANS turbulence models except for the Elliptic Blending R- ϵ , while the anisotropic turbulent thermal conductivity models can be used with RANS-RSM turbulence models only, which require the CFD Module. The **Turbulence model type** used by the fluid flow interface can be displayed by selecting the **Show or Hide Physics Property Settings** button at the right of the **Fluid flow** list.

When a RANS-RSM turbulence model (except Elliptic Blending R- ϵ) is selected in the fluid flow interface, the **Heat transport turbulence model** can be set to **Turbulent Prandtl number** (the default) or **Anisotropic turbulent thermal conductivity**. For RANS-EVM models, this option is not available since only turbulent Prandtl number models can be used. In this case, the subsequent options act as if **Turbulent Prandtl number** was selected. For Elliptic Blending R- ϵ , this option is not available either since only anisotropic turbulent thermal conductivity models can be used. In this case, the subsequent options act as if **Anisotropic turbulent thermal conductivity** was selected.

When **Turbulent Prandtl number** option is used, the turbulent thermal conductivity is

$$k_T = \frac{\mu_T C_p}{Pr_T}$$

where μ_T is defined by the flow interface, and Pr_T depends on the **Turbulent Prandtl number model**. Select an option from the **Turbulent Prandtl number model** list: **Kays–Crawford** (the default), **Extended Kays–Crawford**, or **User-defined turbulent Prandtl number**.

For **Extended Kays–Crawford**, enter a **Reynolds number at infinity** Re_∞ (dimensionless).

For **User-defined turbulent Prandtl number**, enter a **Turbulent Prandtl number** Pr_T (dimensionless).

When **Anisotropic turbulent thermal conductivity** option is used, the turbulent thermal conductivity is defined by the selected **Anisotropic turbulent thermal conductivity model** option: **Daly–Harlow (GGDH)** (the default), **Abe–Suga (High order GGDH)**, or **User-defined turbulent thermal conductivity**.

For **Daly–Harlow (GGDH)**, enter the **Daly–Harlow model coefficient** C_G (dimensionless).

For **Abe–Suga (High order GGDH)**, enter the **Abe–Suga model coefficient** C_H (dimensionless).

For **User-defined turbulent thermal conductivity**, select **Isotropic**, **Diagonal**, **Symmetric**, or **Full** based on the characteristics of the turbulent thermal conductivity, and enter a value or expression. For **Isotropic** enter a scalar which will be used to define a diagonal tensor. For the other options, enter values or expressions into the editable fields of the tensor.

The components of the turbulent thermal conductivity κ_T when given in tensor form are available as `ht.kappaTxx`, `ht.kappaTyy`, and so on (using the default name `ht`).



See [Turbulent Conductivity](#) for details.

With CFD Module, this section is also available when either of the LES interfaces RBVMWV or Smagorinsky is used. For these two LES models, additional diffusion terms, corresponding to the ones in the momentum equation, appear in the energy equation. The magnitude of these terms can be controlled by a turbulent Prandtl number.

When the **Wall treatment** option selected in the fluid flow interface is set either to **Wall functions** or **Automatic**, the **Thermal wall function** can be set to **Standard** (default) or **High viscous dissipation at wall**. The **Standard** option is suitable for most of the configurations. The **High viscous dissipation at wall** option accounts for viscous dissipation in the boundary layer which is not the case with the **Standard** option. This

is needed for accurate results in case of fast internal flow, especially if paths are narrow or the fluid very viscous.



See [Temperature Wall Functions](#) for details.

Physics Interface Features

Physics nodes are available from the **Physics** ribbon toolbar (Windows users), **Physics** context menu (Mac or Linux users), or right-click to access the context menu (all users).



In general, to add a node, go to the **Physics** toolbar, no matter what operating system you are using.

LAMINAR FLOW

The available physics features for The Laminar Flow interface are listed in the section [Domain, Boundary, Pair, and Point Nodes for Single-Phase Flow](#) in the *COMSOL Multiphysics Reference Manual*.

TRANSPORT OF CONCENTRATED SPECIES

The available physics features for The Transport of Concentrated Species interface are listed in the section [Domain, Boundary, and Pair Nodes for the Transport of Concentrated Species Interface](#).

The Reacting Flow, Diluted Species Multiphysics Interface

Reacting Flow, Diluted Species is a predefined multiphysics interface, available in the Chemical Reaction Engineering Module, that couples fluid flow with mass transport and reactions.

Selecting **Laminar Flow, Diluted Species** under the **Chemical Species Transport > Reacting Flow** branch of the **Model Wizard** or **Add Physics** windows, a **Laminar Flow** interface and a **Transport of Diluted Species** interface are added to the Model Builder.

In addition, the Multiphysics node is added, which includes the multiphysics coupling feature **Reacting Flow, Diluted Species**. The **Reacting Flow, Diluted Species** feature predefines and controls the couplings between the separate interfaces in order to facilitate easy set up of models.

In this section:

- [The Reacting Laminar Flow, Diluted Species Interface](#)
- [The Reacting Flow, Diluted Species Coupling Feature](#)
- [Physics Interface Features](#)

The Reacting Laminar Flow, Diluted Species Interface

The **Reacting Laminar Flow, Diluted Species** multiphysics interface () is used to simulate laminar flow coupled to transport of dilute solutes in a gas or liquid.

It combines the **Laminar Flow**, and **Transport of Diluted Species** interfaces. The **Reacting Flow, Diluted Species** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. The fluid flow can either be free flow or flow in a porous medium. The mass transfer solves for the development of one or several solutes dissolved in a gas or liquid solvent.

The interface can be used for stationary and time-dependent analysis in 2D, 2D axial symmetry, and 3D.

On the constituent physics interfaces:

The equations solved by the **Laminar Flow** interface are the Navier–Stokes equations for conservation of momentum and the continuity equation for conservation of mass. A

Fluid Properties feature is active by default on the entire interface selection. A **Porous Medium** feature can be added in order to model flow in porous media by solving the Brinkman equations.

The **Transport of Diluted Species** interface solves for an arbitrary number of concentrations. The species equations include transport by convection, diffusion and, optionally, migration in an electric field.

The Reacting Flow, Diluted Species Coupling Feature

The **Reacting Flow, Diluted Species** () multiphysics coupling is used to simulate mass transport and reactions in a gas or liquid.

DOMAIN LEVEL SYNCHRONIZATION

The **Reacting Flow, Diluted Species** coupling synchronizes the features from a Single-Phase Flow, or Brinkman Equations, interface and a Transport of Diluted Species interface. When added, the velocity field used by the Transport of Diluted Species interface is synchronized to the one computed in the fluid flow interface.

TURBULENT MASS TRANSFER

When a turbulence model is used, the **Reacting Flow, Diluted Species** coupling applies turbulence modeling for the mass transport in the following manners:

- Turbulent mass transport is added, defined from the turbulent viscosity and a turbulent Schmidt number (for more information, see [Turbulent Mass Transport Models](#)).
- Mass transport wall functions. When a turbulence model using wall functions is used for the fluid flow, the Reacting Flow coupling automatically adds wall functions for the mass transport on the same boundaries (for more information, see [Mass Transport Wall Functions](#)).

SETTINGS

The **Label** is the default multiphysics coupling feature name.

The **Name** is used primarily as a scope prefix for variables defined by the coupling node. Refer to such variables in expressions using the pattern <name>.<variable_name>. In order to distinguish between variables belonging to different coupling nodes or physics interfaces, the name string must be unique. Only letters, numbers, and underscores (_) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first multiphysics coupling feature in the model) is `rfd1`.

DOMAIN SELECTION

The **Reacting Flow, Diluted Species** coupling is automatically defined on the intersection of the selections for the coupled interfaces.

The **Selection** list displays the domains where the coupling feature is active.

COUPLED INTERFACES

This section defines the physics involved in the multiphysics coupling. The **Fluid flow** and **Species transport** lists include all applicable physics interfaces.

The default values depend on how this coupling node is created.

- If it is added from the **Physics** ribbon (Windows users), **Physics** contextual toolbar (macOS and Linux users), or context menu (all users), then the first physics interface of each type in the component is selected as the default.
- If it is added automatically when a multiphysics interface is chosen in the **Model Wizard** or **Add Physics** window, then the two participating physics interfaces are selected.

You can also select **None** from either list to uncouple the node from a physics interface. If the physics interface is removed from the **Model Builder**, for example **Laminar Flow** is deleted, then the **Fluid Flow** list defaults to **None** as there is nothing to couple to.

Click the **Go to Source** buttons () to move to the main physics interface node for the selected physics interface.



If a physics interface is deleted and then added to the model again, then in order to reestablish the coupling, you need to choose the physics interface again from the **Fluid flow** or **Species transport** lists. This is applicable to all multiphysics coupling nodes that would normally default to the once present interface. See [Multiphysics Modeling Workflow](#) in the *COMSOL Multiphysics Reference Manual*.

TURBULENCE

When the fluid flow interface uses a turbulence model, select an option from the **Mass transport turbulence model** list — **Kays–Crawford**, **High Schmidt Number**, or **User-defined turbulent Schmidt number**.

For **User-defined turbulent Schmidt number**, enter a **Turbulent Schmidt number** Sc_T (dimensionless).

The turbulent mass transfer added to the concentration equations is defined as

$$\mathbf{J}_{T,i} = -\frac{\mu_T}{Sc_T} \nabla c_i$$

where μ_T is the turbulent viscosity defined by the flow interface, and the turbulent Schmidt number, Sc_T , depends on the **Mass transport turbulence model** used.

Physics Interface Features

Physics nodes are available from the **Physics** ribbon toolbar (Windows users), **Physics** context menu (macOS or Linux users), or right-click to access the context menu (all users).



In general, to add a node, go to the **Physics** toolbar, no matter what operating system you are using.

LAMINAR FLOW

The available physics features for The Laminar Flow interface are listed in the section [Domain, Boundary, Pair, and Point Nodes for Single-Phase Flow](#) in the *CFD Module User's Guide*.

TRANSPORT OF DILUTED SPECIES

The available physics features for The Transport of Diluted Species interface are listed in the section [Domain, Boundary, and Pair Nodes for the Transport of Diluted Species Interface](#).

The Reacting Flow in Porous Media Multiphysics Interfaces

The Chemical Reaction Engineering Module includes a number of predefined multiphysics interfaces that couple fluid flow in porous media with mass transport and reactions in porous media.

Selecting a multiphysics interface under the **Chemical Species Transport > Reacting Flow in Porous Media** branch of the **Model Wizard** or **Add Physics** windows, a **Brinkman Equations** interface combined with either the **Transport of Diluted Species in Porous Media** interface or the **Transport of Concentrated Species** interfaces are added to the Model Builder.

In addition, the Multiphysics node is added, which includes the multiphysics coupling feature. The multiphysics coupling feature controls the coupling between the separate interfaces in order to facilitate easy setup of models.

In this section:

- The Reacting Flow in Porous Media, Transport of Diluted Species Interface
- The Reacting Flow in Porous Media, Transport of Concentrated Species Interface
- The Reacting Flow in Porous Catalysts, Transport of Diluted Species Interface
- The Reacting Flow in Porous Catalysts, Transport of Concentrated Species Interface
- The Reacting Flow in Packed Beds, Transport of Diluted Species Interface
- The Reacting Flow in Packed Beds, Transport of Concentrated Species Interface
- The Reacting Flow in Packed Beds, Transport of Diluted Species with Shrinking Core Model Interface
- The Reacting Flow in Packed Beds, Transport of Diluted Species Interface
- The Reacting Flow in Packed Beds, Transport of Concentrated Species Interface
- The Reacting Flow in Moving Packed Beds, Transport of Diluted Species with Shrinking Core Model Interface
- The Reacting Flow, Diluted Species Coupling Feature
- The Reacting Flow Coupling Feature
- Physics Interface Features

The Reacting Flow in Porous Media, Transport of Diluted Species Interface

The **Reacting Flow in Porous Media, Transport of Diluted Species** interface () is used to study the flow and chemical composition of a gas or liquid moving through the interstices of a porous medium.

It combines the **Brinkman Equations**, and **Transport of Diluted Species in Porous Media** interfaces. The **Reacting Flow, Diluted Species** multiphysics coupling feature, which is added automatically, couples the fluid flow and mass transport. A **Porous Material** node, including a **Fluid** and a **Solid** subnode, is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Diluted Species in Porous Media** interface computes the species concentration in free and porous media, assuming that the species are solutes, dissolved in a solvent of significantly higher concentration. A **Porous Medium** feature is active by default on the entire interface selection.

The Reacting Flow in Porous Media, Transport of Concentrated Species Interface

The **Reacting Flow in Porous Media, Transport of Concentrated Species** interface () is used to study the flow and chemical composition of a gas or liquid moving through the interstices of a porous medium. The fluid can consist of a mixture of species where the individual concentrations are of comparable order of magnitude.

It combines the **Brinkman Equations**, and **Transport of Concentrated Species** interfaces. The **Reacting Flow** multiphysics coupling feature, which is added automatically, couples the fluid flow and mass transport. A **Porous Material** node is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Concentrated Species in Porous Media** interface solves for an arbitrary number of mass fractions in free and porous media. In the current multiphysics interface a **Porous Medium** feature is active by default on the entire interface selection.

The Reacting Flow in Porous Catalysts, Transport of Diluted Species Interface

The **Reacting Flow in Porous Catalysts, Transport of Diluted Species** interface () is used to study the impact of adsorption and heterogeneous reactions involving chemical species, as a gas or liquid is moving through the interstices of a porous medium.

It combines the **Brinkman Equations** interface () and the **Transport of Diluted Species in Porous Catalysts** interface (). The **Reacting Flow, Diluted Species** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. A **Porous Material** node, including a **Fluid** and a **Solid** subnode, is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Diluted Species in Porous Catalysts** interface solves for the concentration of an arbitrary number of bulk species, and optionally, an arbitrary number of surface species and porous media. The bulk species are assumed to be solutes, dissolved in a solvent of significantly higher concentration. In the current multiphysics interface a **Porous Catalyst** feature is active by default on the entire interface selection.

The Reacting Flow in Porous Catalysts, Transport of Concentrated Species Interface

The **Reacting Flow in Porous Catalysts, Transport of Concentrated Species** interface () is used to study the impact of adsorption and heterogeneous reactions involving

chemical species, as a gas or liquid is moving through the interstices of a porous medium.

It combines the **Brinkman Equations** interface () and the **Transport of Concentrated Species in Porous Catalysts** interface (). The **Reacting Flow, Diluted Species** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. A **Porous Material** node is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Concentrated Species in Porous Catalysts** interface solves for the concentration of an arbitrary number of fluid phase mass fractions, and optionally, an arbitrary number of surface species in porous media. In the current multiphysics interface a **Porous Catalyst** feature is active by default on the entire interface selection.

The Reacting Flow in Packed Beds, Transport of Diluted Species Interface

The **Reacting Flow in Packed Beds, Transport of Diluted Species** interface () is used to model a reactor filled with porous catalytic pellets. In addition to interparticle mass transport, intraparticle transport of chemical species and reactions can be studied in detail.

It combines the **Brinkman Equations** interface () and a **Transport of Diluted Species in Packed Beds** interface (). The **Reacting Flow, Diluted Species** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. A **Porous Material** node, including a **Fluid** and a **Solid** subnode, is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Diluted Species in Packed Beds** interface solves for the concentration of an arbitrary number of bulk species, and optionally, an arbitrary number of surface species in porous pellets. The bulk species are assumed to be solutes, dissolved in a solvent of significantly higher concentration. In the current multiphysics interface a **Packed Bed** feature is active by default on the entire interface selection.

The Reacting Flow in Packed Beds, Transport of Concentrated Species Interface

The **Reacting Flow in Packed Beds, Transport of Concentrated Species** interface () is used to model a reactor filled with porous catalytic pellets. In addition to interparticle mass transport, intraparticle transport of chemical species and reactions can be studied in detail.

It combines the **Brinkman Equations** interface () and a **Transport of Concentrated Species in Packed Beds** interface (). The **Reacting Flow** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. A **Porous Material** node, including a **Fluid** and a **Solid** subnode, is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Concentrated Species in Packed Beds** interface solves for the concentration of an arbitrary number of fluid phase mass fractions, and optionally, an arbitrary number of surface species in porous pellets. In the current multiphysics interface a **Packed Bed** feature is active by default on the entire interface selection.

The Reacting Flow in Packed Beds, Transport of Diluted Species with Shrinking Core Model Interface

The **Reacting Flow in Packed Beds, Transport of Diluted Species with Shrinking Core Model** interface () is used to model a reactor filled with initially solid reactant particles. As the reactant particles are consumed by heterogeneous reactions, occurring on the core surface, the solid core shrinks leaving a porous layer around it. Apart from the core

consuming reactions, transport and reactions inside the porous reacted layer, as well as transport in the pore space around the particles can be studied in detail.

It combines the **Brinkman Equations** interface () and a **Transport of Diluted Species in Packed Beds** interface (). The **Reacting Flow, Diluted Species** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. A **Porous Material** node, including a **Fluid** and a **Solid** subnode, is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Diluted Species in Packed Beds** interface solves for the concentration of an arbitrary number of bulk species, and optionally, an arbitrary number of surface species in porous pellets. The bulk species are assumed to be solutes, dissolved in a solvent of significantly higher concentration. In the current multiphysics interface a **Packed Bed** feature, using the **Shrinking Core Model**, is active by default on the entire interface selection.

The Reacting Flow in Moving Packed Beds, Transport of Diluted Species Interface

The **Reacting Flow in Moving Packed Beds, Transport of Diluted Species** interface () is used to model reactors where the bed, comprised of packed porous catalytic pellets, move relative to the reactor. In addition to transport in the pore space between the pellets, intraparticle transport of chemical species and reactions can be studied in detail.

It combines the **Brinkman Equations** interface () and a **Transport of Diluted Species in Moving Packed Beds** interface (). The **Reacting Flow, Diluted Species** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. A **Porous Material** node, including a **Fluid** and a **Solid** subnode, is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Diluted Species in Moving Packed Beds** interface solves for the concentration of an arbitrary number of bulk species, and optionally, an arbitrary number of surface species in porous pellets. The bulk species are assumed to be solutes, dissolved in a solvent of significantly higher concentration. In the current multiphysics interface a **Moving Packed Bed** feature is active by default on the entire interface selection.

The Reacting Flow in Moving Packed Beds, Transport of Concentrated Species Interface

The **Reacting Flow in Moving Packed Beds, Transport of Concentrated Species** interface () is used to model reactors where the bed, comprised of packed porous catalytic pellets, move relative to the reactor. In addition to transport in the pore space between the pellets, intraparticle transport of chemical species and reactions can be studied in detail.

It combines the **Brinkman Equations** interface () and a **Transport of Concentrated Species in Moving Packed Beds** interface (). The **Reacting Flow** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. A **Porous Material** node, including a **Fluid** and a **Solid** subnode, is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Concentrated Species in Moving Packed Beds** interface solves for the concentration of an arbitrary number of fluid phase mass fractions, and optionally, an arbitrary number of surface species in porous pellets. In the current multiphysics interface a **Moving Packed Bed** feature is active by default on the entire interface selection.

The Reacting Flow in Moving Packed Beds, Transport of Diluted Species with Shrinking Core Model Interface

The **Reacting Flow in Packed Beds, Transport of Diluted Species with Shrinking Core Model** interface () is used to model a reactor filled with initially solid reactant particles. As the reactant particles are consumed by heterogeneous reactions, occurring on the core surface, the solid core shrinks leaving a porous layer around it. Apart from the core consuming reactions, transport and reactions inside the porous reacted layer, as well as transport in the pore space around the particles can be studied in detail.

It combines the **Brinkman Equations** interface () and a **Transport of Diluted Species in Moving Packed Beds** interface (). The **Reacting Flow, Diluted Species** multiphysics coupling, which is added automatically, couples fluid flow and mass transport. A **Porous Material** node, including a **Fluid** and a **Solid** subnode, is also added to the component.

The interface can be used for stationary and time-dependent analysis in 2D, 2Daxi, and 3D.

On the constituent physics interfaces:

The **Brinkman Equations** interface computes the fluid velocity and pressure fields of single-phase flow in porous media in the laminar flow regime. A **Porous Medium** feature is active by default on the entire interface selection.

The **Transport of Diluted Species in Moving Packed Beds** interface solves for the concentration of an arbitrary number of bulk species, and optionally, an arbitrary number of surface species in porous pellets. The bulk species are assumed to be solutes, dissolved in a solvent of significantly higher concentration. In the current multiphysics interface a **Moving Packed Bed** feature, using the **Shrinking Core Model**, is active by default on the entire interface selection.

The Reacting Flow, Diluted Species Coupling Feature

For the settings of this feature see [The Reacting Flow, Diluted Species Coupling Feature](#).

The Reacting Flow Coupling Feature

For the settings of this feature see [The Reacting Flow Coupling Feature](#).

Physics nodes are available from the **Physics** ribbon toolbar (Windows users), **Physics** context menu (Mac or Linux users), or right-click to access the context menu (all users).

BRINKMAN EQUATIONS

The available physics features for The Brinkman Equations interface are listed in the [Domain, Boundary, Point, and Pair Nodes for the Brinkman Equations Interface](#) section in the *CFD Module User's Guide*.

TRANSPORT OF DILUTED SPECIES

The available physics features for The Transport of Diluted Species interface are listed in the [Domain, Boundary, and Pair Nodes for the Transport of Diluted Species Interface](#) section.

TRANSPORT OF CONCENTRATED SPECIES

The available physics features for The Transport of Concentrated Species interface are listed in the [Domain, Boundary, and Pair Nodes for the Transport of Concentrated Species Interface](#) section.

The Nonisothermal Reacting Flow Multiphysics Interfaces

Nonisothermal Reacting Flow is a predefined multiphysics interface, available in the Chemical Reaction Engineering Module, that couples fluid flow and heat transfer with species transport and reactions.

Selecting an entry in the **Chemical Species Transport > Nonisothermal Reacting Flow** branch of the **Model Wizard**, or **Add Physics** windows, the following four physics interfaces are added to the Model Builder:

- **Chemistry**
- **Transport of Concentrated Species**
- **Laminar Flow**, or a turbulent flow equivalent like **Turbulent Flow, k- ϵ** .
- **Heat Transfer in Fluids**

In addition, the Multiphysics node is added, which includes the multiphysics coupling feature **Reacting Flow**. The **Reacting Flow** feature predefines and controls the couplings between the separate interfaces in order to facilitate setup of models.

In this section:

- [The Nonisothermal Reacting Laminar Flow Interface](#)
- [The Nonisothermal Reacting Turbulent Flow, k- \$\epsilon\$ Interface](#)
- [The Nonisothermal Reacting Turbulent Flow, k- \$\omega\$ Interface](#)
- [The Nonisothermal Reacting Turbulent Flow, SST Interface](#)
- [The Nonisothermal Reacting Turbulent Flow, Low Re k- \$\epsilon\$ Interface](#)
- [The Reacting Flow Coupling Feature](#)
- [Physics Interface Features](#)

The Nonisothermal Reacting Laminar Flow Interface

The **Nonisothermal Reacting Laminar Flow** () multiphysics interface is used to simulate laminar flow and heat transfer coupled to species transport in a gas or liquid moving either in a free flow or through the interstices of a porous medium.

It combines the **Chemistry**, **Transport of Concentrated Species**, **Laminar Flow**, and **Heat Transfer in Fluids** interfaces. The **Reacting Flow** multiphysics coupling, which is added

automatically, couples fluid flow, heat transfer, and mass transfer. The species transport supports both a mixture, where the concentrations are of comparable order of magnitude, and low-concentration solutes in a solvent.

The multiphysics coupling takes into account the heat of reaction in the fluid and at the surface of a porous catalyst, enthalpy diffusion, and contributing mass fluxes. In addition, the temperature dependency of the chemical properties and reactions are accounted for.

The interface can be used for stationary and time-dependent analysis in 2D, 2D axial symmetry, and 3D.

On the constituent physics interfaces:

The **Chemistry** interface defines thermodynamic properties and transport properties of the fluid. Provided that properties of each species have been defined, composition dependent mixture properties such as the heat capacity, the density, and the heat conduction are defined. The **Chemistry** interface also defines reaction rates for species involved in the chemical reactions added to the system.

The **Transport of Concentrated Species** interface solves for an arbitrary number of mass fractions. The species equations include transport by convection, diffusion and, optionally, migration in an electric field. A **Transport Properties** feature is active by default on the entire interface selection

The equations solved by the **Laminar Flow** interface are the Navier–Stokes equations for conservation of momentum and the continuity equation for conservation of mass. A **Fluid Properties** feature is active by default on the entire interface selection.

The **Heat Transfer** interface solves for conservation of energy. A **Fluid** feature is active by default on the entire interface selection.

The Nonisothermal Reacting Turbulent Flow, k - ϵ Interface

The **Nonisothermal Reacting Turbulent Flow, k - ϵ** () multiphysics interface is used to simulate flow in the turbulent regime, heat transfer, and species transport and in a gas or liquid moving either in a free flow or through the interstices of a porous medium. This interface requires a license for either the CFD Module or the Heat Transfer Module.

It combines the **Chemistry**, **Transport of Concentrated Species**, **Turbulent Flow, k - ϵ** , and **Heat Transfer in Fluids** interfaces. The **Reacting Flow** multiphysics coupling, which is added automatically, couples fluid flow, heat transfer and mass transfer. The species

transport supports both a mixture, where the concentrations are of comparable order of magnitude, and low-concentration solutes in a solvent.

The multiphysics coupling takes into account the heat of reaction in the fluid and at the surface of a porous catalyst, enthalpy diffusion, and contributing mass fluxes. In addition, the temperature dependency of the chemical properties and reactions are accounted for.

The interface can be used for stationary and time-dependent analysis in 2D, 2D axial symmetry, and 3D.

On the constituent physics interfaces:

The **Chemistry** interface defines thermodynamic properties and transport properties of the fluid. Provided that properties of each species have been defined, composition dependent mixture properties such as the heat capacity, the density, and the heat conduction are defined. The **Chemistry** interface also defines reaction rates for species involved in the chemical reactions added to the system.

The **Transport of Concentrated Species** interface solves for an arbitrary number of mass fractions. The species equations include transport by convection, diffusion and, optionally, migration in an electric field. Mass transfer close to walls is modeled using wall functions

The equations solved by the **Turbulent Flow, k - ϵ** interface are the Navier–Stokes equations for conservation of momentum and the continuity equation for conservation of mass. Turbulence effects are modeled using the standard two-equation k - ϵ model with realizability constraints. Flow close to walls is modeled using wall functions.

The **Heat Transfer** interface solves for conservation of energy. A **Fluid** feature is active by default on the entire interface selection. Heat transfer close to walls is modeled using wall functions.

The Nonisothermal Reacting Turbulent Flow, k - ω Interface

The **Nonisothermal Reacting Turbulent Flow, k - ω** () multiphysics interface is used to simulate flow in the turbulent regime, heat transfer, and species transport and in a gas or liquid moving either in a free flow or through the interstices of a porous medium. This interface requires a license for the CFD Module.

It combines the **Chemistry**, **Transport of Concentrated Species**, **Turbulent Flow, k - ω** , and **Heat Transfer in Fluids** interfaces. The **Reacting Flow** multiphysics coupling, which is

added automatically, couples fluid flow, heat transfer and mass transfer. The species transport supports both a mixture, where the concentrations are of comparable order of magnitude, and low-concentration solutes in a solvent.

The multiphysics coupling takes into account the heat of reaction in the fluid and at the surface of a porous catalyst, enthalpy diffusion, and contributing mass fluxes. In addition, the temperature dependency of the chemical properties and reactions are accounted for.

The interface can be used for stationary and time-dependent analysis in 2D, 2D axial symmetry, and 3D.

On the constituent physics interfaces:

The **Chemistry** interface defines thermodynamic properties and transport properties of the fluid. Provided that properties of each species have been defined, composition dependent mixture properties such as the heat capacity, the density, and the heat conduction are defined. The **Chemistry** interface also defines reaction rates for species involved in the chemical reactions added to the system.

The **Transport of Concentrated Species** interface solves for an arbitrary number of mass fractions. The species equations include transport by convection, diffusion and, optionally, migration in an electric field. Mass transfer close to walls is modeled using wall functions

The equations solved by the **Turbulent Flow, k- ω** interface are the Navier–Stokes equations for conservation of momentum and the continuity equation for conservation of mass. The fluid flow turbulence is modeled using the Wilcox revised k- ω model with realizability constraints. Flow close to walls is modeled using wall functions.

The **Heat Transfer** interface solves for conservation of energy. A **Fluid** feature is active by default on the entire interface selection. Heat transfer close to walls is modeled using wall functions.

The Nonisothermal Reacting Turbulent Flow, SST Interface

The **Nonisothermal Reacting Turbulent Flow, SST** () multiphysics interface is used to simulate flow in the turbulent regime, heat transfer, and species transport and in a gas or liquid moving either in a free flow or through the interstices of a porous medium. This interface requires a license for either the CFD Module or the Heat Transfer Module.

It combines the **Chemistry**, **Transport of Concentrated Species**, **Turbulent Flow**, **SST**, and **Heat Transfer in Fluids** interfaces. The **Reacting Flow** multiphysics coupling, which is added automatically, couples fluid flow, heat transfer and mass transfer. The species transport supports both a mixture, where the concentrations are of comparable order of magnitude, and low-concentration solutes in a solvent.

The multiphysics coupling takes into account the heat of reaction in the fluid and at the surface of a porous catalyst, enthalpy diffusion, and contributing mass fluxes. In addition, the temperature dependency of the chemical properties and reactions are accounted for.

The interface can be used for stationary and time-dependent analysis in 2D, 2D axial symmetry, and 3D.

On the constituent physics interfaces:

The **Chemistry** interface defines thermodynamic properties and transport properties of the fluid. Provided that properties of each species have been defined, composition dependent mixture properties such as the heat capacity, the density, and the heat conduction are defined. The **Chemistry** interface also defines reaction rates for species involved in the chemical reactions added to the system.

The **Transport of Concentrated Species** interface solves for an arbitrary number of mass fractions. The species equations include transport by convection, diffusion and, optionally, migration in an electric field. Mass transfer close to walls is modeled using wall functions

The equations solved by the **Turbulent Flow**, **SST** interface are the Navier–Stokes equations for conservation of momentum and the continuity equation for conservation of mass. The fluid flow turbulence is modeled using the SST model. The SST model is a low-Reynolds number model which means that it resolves the velocity, pressure, and concentrations all the way down to the wall. For that reason this physics interface is suited for studying mass transfer at high Schmidt numbers. The SST model depends on the distance to the closest wall, and the interface therefore includes a wall distance equation.

The **Heat Transfer** interface solves for conservation of energy. A **Fluid** feature is active by default on the entire interface selection. Heat transfer close to walls is modeled using wall functions.

The Nonisothermal Reacting Turbulent Flow, Low Re k- ϵ Interface

The **Nonisothermal Reacting Turbulent Flow, Low Re k- ϵ**  multiphysics interface is used to simulate flow in the turbulent regime, heat transfer, and species transport and in a gas or liquid moving either in a free flow or through the interstices of a porous medium. This interface requires a license for either the CFD Module or the Heat Transfer Module.

It combines the **Chemistry, Transport of Concentrated Species, Turbulent Flow, Low Re k- ϵ** , and **Heat Transfer in Fluids** interfaces. The **Reacting Flow** multiphysics coupling, which is added automatically, couples fluid flow, heat transfer and mass transfer. The species transport supports both a mixture, where the concentrations are of comparable order of magnitude, and low-concentration solutes in a solvent.

The multiphysics coupling takes into account the heat of reaction in the fluid and at the surface of a porous catalyst, enthalpy diffusion, and contributing mass fluxes. In addition, the temperature dependency of the chemical properties and reactions are accounted for.

The interface can be used for stationary and time-dependent analysis in 2D, 2D axial symmetry, and 3D.

On the constituent physics interfaces:

The **Chemistry** interface defines thermodynamic properties and transport properties of the fluid. Provided that properties of each species have been defined, composition dependent mixture properties such as the heat capacity, the density, and the heat conduction are defined. The **Chemistry** interface also defines reaction rates for species involved in the chemical reactions added to the system.

The **Transport of Concentrated Species** interface solves for an arbitrary number of mass fractions. The species equations include transport by convection, diffusion and, optionally, migration in an electric field. Mass transfer close to walls is modeled using wall functions

The equations solved by the **Turbulent Flow, Low Re k- ϵ** interface are the Navier–Stokes equations for conservation of momentum and the continuity equation for conservation of mass. The fluid flow turbulence is modeled using the AKN low-Reynolds number k- ϵ model. The low-Reynolds number model resolves the velocity, pressure, and concentrations all the way down to the wall. For that reason this physics interface is suited for studying mass transfer at high Schmidt numbers. The

AKN model depends on the distance to the closest wall, and the interface therefore includes a wall distance equation.

The **Heat Transfer** interface solves for conservation of energy. A **Fluid** feature is active by default on the entire interface selection. Heat transfer close to walls is modeled using wall functions.

The Nonisothermal Reacting Flow, LES RBVM Interface

The **Nonisothermal Reacting Flow, LES RBVM** () multiphysics interface is used to simulate flow in the turbulent regime, heat transfer, and species transport in a gas or a liquid moving in free flow. This interface requires licenses for the CFD Module and either the Chemical Reaction Engineering Module, or the Battery Design Module, or the Fuel Cell & Electrolyzer Module.

It combines the **Chemistry**, **Transport of Concentrated Species**, **LES RBVM**, and **Heat Transfer in Fluids** interfaces. The **Reacting Flow** multiphysics coupling, which is added automatically, couples fluid flow, heat transfer and mass transfer. The species transport supports a mixture, where the concentrations are of comparable order of magnitude.

The multiphysics coupling takes into account the heat of reaction in the fluid, enthalpy diffusion, and contributing mass fluxes. In addition, the temperature dependency of the chemical properties and reactions are accounted for.

The interface can be used for time-dependent analysis in 3D.

On the constituent physics interfaces:

The **Chemistry** interface defines thermodynamic properties and transport properties of the fluid. Provided that properties of each species have been defined, composition dependent mixture properties such as the heat capacity, the density, and the heat conduction are defined. The **Chemistry** interface also defines reaction rates for species involved in the chemical reactions added to the system.

The **Transport of Concentrated Species** interface solves for an arbitrary number of mass fractions. The species equations include transport by convection, diffusion and, optionally, migration in an electric field. Mass transfer close to walls can be modeled using wall functions. Crosswind diffusion is active by default for this interface, but can be deactivated if reaction fronts are adequately resolved in the simulation.

The equations solved by the **LES RBVM** interface are the continuity equations for conservation of mass and the Navier–Stokes equation, augmented by additional stress

terms, for conservation of momentum. Consistent modeling is used for the additional stresses. The interface is thus also applicable to laminar and transitional flows.

The **Heat Transfer** interface solves for conservation of energy. A **Fluid** feature is active by default on the entire interface selection. Heat transfer close to walls can be modeled using wall functions. Crosswind diffusion is active by default for this interface, but can be deactivated if reaction fronts are adequately resolved in the simulation.

The Nonisothermal Reacting Flow, LES RBVMWV Interface

The **Nonisothermal Reacting Flow, LES RBVMWV** () multiphysics interface is used to simulate flow in the turbulent regime, heat transfer, and species transport in a gas or a liquid moving in free flow. This interface requires licenses for the CFD Module and either the Chemical Reaction Engineering Module, or the Battery Design Module, or the Fuel Cell & Electrolyzer Module.

It combines the **Chemistry**, **Transport of Concentrated Species**, **LES RBVMWV**, and **Heat Transfer in Fluids** interfaces. The **Reacting Flow** multiphysics coupling, which is added automatically, couples fluid flow, heat transfer and mass transfer. The species transport supports a mixture, where the concentrations are of comparable order of magnitude.

The multiphysics coupling takes into account the heat of reaction in the fluid, enthalpy diffusion, and contributing mass fluxes. In addition, the temperature dependency of the chemical properties and reactions are accounted for.

The interface can be used for time-dependent analysis in 3D.

On the constituent physics interfaces:

The **Chemistry** interface defines thermodynamic properties and transport properties of the fluid. Provided that properties of each species have been defined, composition dependent mixture properties such as the heat capacity, the density, and the heat conduction are defined. The **Chemistry** interface also defines reaction rates for species involved in the chemical reactions added to the system.

The **Transport of Concentrated Species** interface solves for an arbitrary number of mass fractions. The species equations include transport by convection, diffusion and, optionally, migration in an electric field. Mass transfer close to walls can be modeled using wall functions. Crosswind diffusion is active by default for this interface, but can be deactivated if reaction fronts are adequately resolved in the simulation.

The equations solved by the **LES RBVMWV** interface are the continuity equations for conservation of mass and the Navier–Stokes equation, augmented by additional stress

terms, for conservation of momentum. Similar modeling as for the **RBVM** model is applied but the effect of the Reynolds stress is enhanced by an additional residual-based diffusion term. Since the modeling is consistent, the interface is also applicable to laminar and transitional flows.

The **Heat Transfer** interface solves for conservation of energy. A **Fluid** feature is active by default on the entire interface selection. Heat transfer close to walls can be modeled using wall functions. Crosswind diffusion is active by default for this interface, but can be deactivated if reaction fronts are adequately resolved in the simulation.

The Nonisothermal Reacting Flow, LES Smagorinsky Interface

The **Nonisothermal Reacting Flow, LES Smagorinsky** () multiphysics interface is used to simulate flow in the turbulent regime, heat transfer, and species transport in a gas or a liquid moving in free flow. This interface requires licenses for the CFD Module and either the Chemical Reaction Engineering Module, or the Battery Design Module, or the Fuel Cell & Electrolyzer Module.

It combines the **Chemistry**, **Transport of Concentrated Species**, **LES Smagorinsky**, and **Heat Transfer in Fluids** interfaces. The **Reacting Flow** multiphysics coupling, which is added automatically, couples fluid flow, heat transfer and mass transfer. The species transport supports a mixture, where the concentrations are of comparable order of magnitude.

The multiphysics coupling takes into account the heat of reaction in the fluid, enthalpy diffusion, and contributing mass fluxes. In addition, the temperature dependency of the chemical properties and reactions are accounted for.

The interface can be used for time-dependent analysis in 3D.

On the constituent physics interfaces:

The **Chemistry** interface defines thermodynamic properties and transport properties of the fluid. Provided that properties of each species have been defined, composition dependent mixture properties such as the heat capacity, the density, and the heat conduction are defined. The **Chemistry** interface also defines reaction rates for species involved in the chemical reactions added to the system.

The **Transport of Concentrated Species** interface solves for an arbitrary number of mass fractions. The species equations include transport by convection, diffusion and, optionally, migration in an electric field. Mass transfer close to walls can be modeled using wall functions. Crosswind diffusion is active by default for this interface, but can be deactivated if reaction fronts are adequately resolved in the simulation.

The equations solved by the **LES Smagorinsky** interface are the continuity equations for conservation of mass and the Navier–Stokes equation, augmented by a turbulent viscosity term, for conservation of momentum. The modeled term is inconsistent, which means that it does not vanish for a fully resolved flow.

The **Heat Transfer** interface solves for conservation of energy. A **Fluid** feature is active by default on the entire interface selection. Heat transfer close to walls can be modeled using wall functions. Crosswind diffusion is active by default for this interface, but can be deactivated if reaction fronts are adequately resolved in the simulation.

The Reacting Flow Coupling Feature

For the settings of this feature see [The Reacting Flow Coupling Feature](#).

Physics Interface Features

Physics nodes are available from the **Physics** ribbon toolbar (Windows users), **Physics** context menu (Mac or Linux users), or right-click to access the context menu (all users).



In general, to add a node, go to the **Physics** toolbar, no matter what operating system you are using.

CHEMISTRY

The available physics features for the Chemistry interface are listed in the section [Feature Nodes Available for the Chemistry Interface](#).

LAMINAR FLOW

The available physics features for The Laminar Flow interface are listed in the section [Domain, Boundary, Pair, and Point Nodes for Single-Phase Flow](#) in the *CFD Module User's Guide*.

TURBULENT FLOW

The available physics features for a single-phase flow interface using a turbulence model, for example the Turbulent Flow, k- ϵ interface, is listed in the section [Domain, Boundary, Pair, and Point Nodes for Single-Phase Flow](#) in the *CFD Module User's Guide*.

TRANSPORT OF CONCENTRATED SPECIES

The available physics features for the Transport of Concentrated Species interface are listed in the section [Domain, Boundary, and Pair Nodes for the Transport of Concentrated Species Interface](#).

HEAT TRANSFER

The available physics features for the Heat Transfer interface are listed in the section [The Heat Transfer Features](#) in the *COMSOL Multiphysics Reference Manual*.

The Dispersed Two-Phase Flow with Species Transport Multiphysics Interfaces

Dispersed Two-Phase Flow with Species Transport is a predefined multiphysics interface that couples dispersed two-phase flow with mass transport in both the continuous phase and in the dispersed phase.

Selecting **Laminar Flow** under the **Chemical Species Transport > Dispersed Two-Phase Flow with Species Transport** branch of the **Model Wizard** or **Add Physics** windows, a **Mixture Model**, **Laminar Flow** interface along with a **Continuous Phase Transport of Diluted Species** interface and a **Dispersed Phase Transport of Diluted Species** interface are added to the Model Builder.

In addition, the Multiphysics node is added, including the multiphysics coupling feature **Dispersed Two-Phase Flow, Diluted Species**. The **Dispersed Two-Phase Flow, Diluted Species** feature predefines and controls the couplings between the separate interfaces in order to facilitate easy setup of models.

In this section:

- [The Dispersed Two-Phase Laminar Flow with Species Transport Interface](#)
- [The Dispersed Two-Phase Flow, Diluted Species Coupling Feature](#)
- [Physics Interface Features](#)
- [The Continuous Phase Transport of Diluted Species Interface](#)
- [The Dispersed Phase Transport of Diluted Species Interface](#)

The Dispersed Two-Phase Laminar Flow with Species Transport Interface

This multiphysics interface () is used to simulate two-phase flow, in the laminar regime, coupled to transport of dilute solutes in each the two fluid phases. It combines the **Mixture Model**, **Laminar Flow**, the **Continuous Phase Transport of Diluted Species** interface and the **Dispersed Phase Transport of Diluted Species** interface.

The **Dispersed Two-Phase Flow, Diluted Species** multiphysics coupling, which is added automatically, couples the two-phase flow with mass transport. The mass transport interfaces solve for the development of one or several solutes present in the continuous and dispersed fluid phase. The interface can be used for stationary and time-dependent analysis in 2D, 2D axial symmetry, and 3D.

On the constituent physics interfaces:

- The **Mixture Model, Laminar Flow** interface solves one set of Navier–Stokes equations for the momentum of the mixture. The pressure distribution is calculated from a mixture-averaged continuity equation and the velocity of the dispersed phase is described by a slip model. The volume fraction of the dispersed phase is tracked by solving a transport equation for the volume fraction.
- The **Continuous Phase Transport of Diluted Species** interface solves for an arbitrary number of solute concentrations present in the continuous phase. The species equations include transport by convection, and diffusion within the phase, as well as reactions and solute extraction between the continuous and dispersed phase.
- The **Dispersed Phase Transport of Diluted Species** interface solves for an arbitrary number of solute concentrations present in the dispersed phase. The species equations include transport by convection, and diffusion within the phase, as well as reactions and solute extraction between the continuous and dispersed phase.

The Dispersed Two-Phase Flow, Diluted Species Coupling Feature

The **Dispersed Two-Phase Flow, Diluted Species** () multiphysics coupling is used to simulate the fluid flow of a liquid containing a dispersed phase, together with mass transport occurring in the continuous phase and the dispersed phase. The coupling feature also support mass transfer (solute extraction) between the phases.

DOMAIN LEVEL SYNCHRONIZATION

The **Dispersed Two-Phase Flow, Diluted Species** coupling synchronizes the features in a Mixture Model interface, with those in the transport interfaces for each of the phases (**Continuous Phase Transport of Diluted Species** and **Dispersed Phase Transport of Diluted Species**). When added, the velocity field and the volume fraction of each of the phases, computed by the Mixture Model interface, is synchronized to the species transport interface for each phase.

SOLUTE EXTRACTION

Extraction is the process where a species is transferred between two immiscible phases due to different relative solubilities in each phase. Physically this transfer process occurs at the phase separating interfaces, the bubble or droplet surfaces in the case of dispersed two phase flow.

Use the checkboxes to select which of the species that can be dissolved in both phases and is subjected to extraction. When multiple species are present in each phase, the species in the two mass transport interfaces are paired by position, from top to bottom, in the **Concentrations** list (in the **Dependent Variables** section when selecting an interface). During extraction, the species in the continuous phase physics interface is transferred into, or from, the corresponding species in the dispersed phase physics interface. The rate at which species is removed from the continuous phase is

$$R_{e,c} = -k_m(K_p c_c - c_d)a_s = -R_{e,d} \quad (3-21)$$

where c_c and c_d are the concentration in the continuous and the dispersed phase respectively. K_p is the partition coefficient determining the phase partitioning at equilibrium, a_s is the dispersed phase specific surface area (m^2/m^3), and k_m is a mass transfer coefficient governing the transport from the bubble or droplet surfaces to the bulk of each phase. Note same source term but with opposite sign is added for the corresponding species in the dispersed phase.

Assuming that the dispersed phase particles are spherical, the specific surface area is computed from the dispersed phase volume fraction ϕ_d and the particle radius r_p in the manner of

$$a_s = \frac{3\phi_d}{r_p} \quad (3-22)$$

When modeling phase transfer with a net mass change between the phases, and a corresponding change in the particle radius, **Solve for interfacial area** can be enabled in the settings of the coupled Mixture Model interface. In this case the resulting specific area is used in the extraction rate.

SETTINGS

The **Label** is the default multiphysics coupling feature name.

The **Name** is used primarily as a scope prefix for variables defined by the coupling node. Refer to such variables in expressions using the pattern `<name>.<variable_name>`. In order to distinguish between variables belonging to different coupling nodes or physics

interfaces, the **name** string must be unique. Only letters, numbers, and underscores (_) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first multiphysics coupling feature in the model) is **dds1**.

DOMAIN SELECTION

The **Dispersed Two-Phase Flow, Diluted Species** coupling is automatically defined on the intersection of the selections for the coupled interfaces.

The **Selection** list displays the domains where the coupling feature is active.

COUPLED INTERFACES

This section defines the physics involved in the multiphysics coupling. The lists **Dispersed Two-Phase Flow, Continuous phase species**, and **Dispersed phase species** include the applicable physics interfaces of each kind.

The default values depend on how this coupling node is created.

- If it is added from the **Physics** ribbon (Windows users), **Physics** contextual toolbar (macOS and Linux users), or context menu (all users), then the first physics interface of each type in the component is selected as the default.
- If it is added automatically when a multiphysics interface is chosen in the **Model Wizard** or **Add Physics** window, then the two participating physics interfaces are selected.

You can also select **None** from either list to uncouple the node from a physics interface. If the physics interface is removed from the **Model Builder**, for example **Mixture Model, Laminar Flow** is deleted, then the **Dispersed Two-Phase Flow** list defaults to **None** as there is nothing to couple to.

Click the **Go to Source** buttons () to move to the main physics interface node for the selected physics interface.



If a physics interface is deleted and then added to the model again, then in order to reestablish the coupling, you need to choose the physics interface again using the lists in the **Coupled Interfaces** section. This is applicable to all multiphysics coupling nodes that would normally default to the once present interface. See [Multiphysics Modeling Workflow](#) in the *COMSOL Multiphysics Reference Manual*.

TURBULENCE

When the fluid flow interface uses a turbulence model, select an option from the **Mass transport turbulence model** list — **Kays–Crawford**, **High Schmidt Number**, or **User-defined turbulent Schmidt number**.

For **User-defined turbulent Schmidt number**, enter a **Turbulent Schmidt number** Sc_T (dimensionless).

The turbulent mass transfer added to the concentration equations is defined as

$$\mathbf{J}_{T,i} = -\frac{\mu_T}{Sc_T} \nabla c_i$$

where μ_T is the turbulent viscosity defined by the flow interface, and the turbulent Schmidt number, Sc_T , depends on the **Mass transport turbulence model** used.

Physics Interface Features

Physics nodes are available from the **Physics** ribbon toolbar (Windows users), **Physics** context menu (macOS or Linux users), or right-click to access the context menu (all users).



In general, to add a node, go to the **Physics** toolbar, no matter what operating system you are using.

MIXTURE MODEL, LAMINAR FLOW

The available physics features for The Mixture Model Laminar Flow interface are listed in the section [Domain and Boundary Nodes for the Mixture Model Interfaces](#) in the *CFD Module User's Guide*.

CONTINUOUS PHASE TRANSPORT OF DILUTED SPECIES AND DISPERSED PHASE TRANSPORT OF DILUTED SPECIES

These interfaces are versions of the Transport of Diluted Species interface. The interfaces extends the functionality by supporting phase specific mass transport. Porous media transport, and migration in an electric field, are however not supported in the interfaces for phase specific mass transport. Apart from [Transport Properties](#), the available features are described in the section [Domain, Boundary, and Pair Nodes for the Transport of Diluted Species Interface](#).

The Continuous Phase Transport of Diluted Species Interface

The **Continuous Phase Transport of Diluted Species** interface is used to calculate the concentration of one or several a dilute solutes present in a continuous phase in a fluid with multiple immiscible phases. Transport and reactions of the species within the continuous phase can be handled with this interface. The driving forces for transport can be convection by the continuous phase velocity, diffusion by Fick's law, and turbulent diffusion, when the interface is used in [The Dispersed Two-Phase Flow, Diluted Species Coupling Feature](#).

The interface supports simulation of transport 1D, 2D, and 3D as well as for axisymmetric components in 1D and 2D. The dependent variable is the molar concentration, c . Modeling multiple species transport is possible, whereby the physics interface solves for the molar concentration, c_i , of each species i . It should be noted that the concentration solved for is the total concentration

$$c_i = \phi_c c_{c,i} \quad (3-23)$$

comprised of the phase volume fraction, ϕ_c , and the continuous phase concentration $c_{c,i}$. For each concentration solved for, the interface defines a variable `tds.phs_<concentration>` corresponding to the phase concentration ($c_{c,i}$). This can be used in boundary conditions and reaction rate expressions when the phase concentration is to be prescribed or used.

The Continuous Phase Transport of Diluted Species interface is a version of the Transport of Diluted Species interface. Except for the transport mechanism settings described below, the interface settings are identical to those available in [The Transport of Diluted Species Interface](#).

TRANSPORT MECHANISMS

Mass transport due to diffusion is always included. Use the checkboxes available under **Additional transport mechanisms** to control other transport mechanisms.

- By default, the **Convection** checkbox is selected. Clear the checkbox to disable convective transport.
- **Phase specific transport** is enabled by default. When enabled, the **Phase type** can be set to **Continuous** or **Dispersed**. Clear the **Phase specific transport** checkbox to disable the support for a volume fraction and phase specific transport.

The Dispersed Phase Transport of Diluted Species Interface

The **Dispersed Phase Transport of Diluted Species** interface is used to calculate the concentration of one or several dilute solutes present in a dispersed phase in a fluid with multiple immiscible phases. Transport and reactions of the species within the continuous phase can be handled with this interface. The driving forces for transport can be convection by the dispersed phase velocity, diffusion by Fick's law, and turbulent diffusion, when the interface is used in [The Dispersed Two-Phase Flow, Diluted Species Coupling Feature](#).

The interface supports simulation of species transport in 1D, 2D, and 3D as well as for axisymmetric components in 1D and 2D. The dependent variable is the molar concentration, c . Modeling multiple species transport is possible, whereby the physics interface solves for the molar concentration, c_i , of each species i . It should be noted that the concentration solved for is the total concentration

$$c_i = \phi_d c_{d,i} \quad (3-24)$$

comprised of the phase volume fraction, ϕ_d , and the dispersed phase concentration $c_{d,i}$. For each concentration solved for, the interface defines a variable `tds.phs_<concentration>` corresponding to the phase concentration ($c_{d,i}$). This can be used in boundary conditions and reaction rate expressions when the phase concentration is to be prescribed or used.

The Dispersed Phase Transport of Diluted Species interface is a version of the Transport of Diluted Species interface. Except for the transport mechanism settings described below, the interface settings are identical to those available in [The Transport of Diluted Species Interface](#).

TRANSPORT MECHANISMS

Mass transport due to diffusion is always included. Use the checkboxes available under **Additional transport mechanisms** to control other transport mechanisms.

- By default, the **Convection** checkbox is selected. Clear the checkbox to disable convective transport.
- **Phase specific transport** is enabled by default. When enabled, the **Phase type** can be set to **Continuous** or **Dispersed**. Clear the **Phase specific transport** checkbox to disable the support for a volume fraction and phase specific transport.

Use this node to define the volume fraction, the convective velocity, and the diffusion coefficient to use for the phase specific transport in a Continuous Phase Transport of Diluted Species interface or a Dispersed Phase Transport of Diluted Species interface.

Conservation of mass of a species i in an immiscible fluid phase is formulated as

$$\frac{\partial \phi \rho_i}{\partial t} + \nabla \cdot (\phi \rho_i \mathbf{u}) = \nabla \cdot [D_i \nabla (\phi \rho_i)] + R_{m,i} \quad (3-25)$$

where ϕ is the phase volume fraction, ρ_i the species density, and $\mathbf{u}$ the (mass averaged) velocity of the phase, D_i is the diffusion coefficient, and $R_{m,i}$ is the species mass source. Using the species density definition $\rho_i = c_{p,i} M_i$, where $c_{p,i}$ is the species phase concentration, and diving by the molar mass M_i , the conservation equation becomes

$$\frac{\partial \phi c_{p,i}}{\partial t} + \nabla \cdot (\phi c_{p,i} \mathbf{u}) = \nabla \cdot [D_i \nabla (\phi c_{p,i})] + R_i \quad (3-26)$$

Choosing to solve for the total concentration $c_i = \phi c_{p,i}$, the equation is

$$\frac{\partial c_i}{\partial t} + \nabla \cdot (c_i \mathbf{u}) = \nabla \cdot (D_i \nabla c_i) + R_i \quad (3-27)$$

The following equations are solved for the total species concentration in the continuous and the dispersed phase respectively

$$\frac{\partial c_{c,i}}{\partial t} + \nabla \cdot (c_{c,i} \mathbf{u}_c) = \nabla \cdot [(D_{c,i} + D_{T,i}) \nabla c_{c,i}] + R_{c,i} \quad (3-28)$$

$$\frac{\partial c_{d,i}}{\partial t} + \nabla \cdot (c_{d,i} \mathbf{u}_d) = \nabla \cdot [(D_{d,i} + D_{T,i}) \nabla c_{d,i}] + R_{d,i} \quad (3-29)$$

Here $D_{T,i}$ is a turbulent dispersion coefficient accounting for the enhanced transport in turbulent flow. It is added by the Dispersed Two-Phase, Diluted Species coupling feature when the Mixture Model interface is using a turbulence model.

MODEL INPUTS

The temperature model input is always available. Select the source of the **Temperature**. For **User defined**, enter a value or expression for the temperature in the input field. This input option is always available.

You can also select the temperature solved for by a Heat Transfer interface added to the model component. These physics interfaces are available for selection in the **Temperature** list.

VOLUME FRACTION

Specify the phase volume fraction. Select the source of the **Volume fraction, continuous phase** or the source of the **Volume fraction, dispersed phase**. When using a Dispersed Two-Phase Flow, Diluted Species coupling feature, the volume fraction is automatically coupled to the one computed by the Mixture Model interface in the same coupling.

CONVECTION

If transport by convection is active, the velocity field of the phase in question needs to be specified. Select the source of the **Velocity field, continuous phase** or the source of the **Velocity field, dispersed phase**. When using a Dispersed Two-Phase Flow, Diluted Species coupling feature, the velocity is automatically coupled to the one computed by the Mixture Model interface in the same coupling.

DIFFUSION

Use this section to specify diffusion coefficients describing the diffusion of each species in the respective phase.

Use the **Source** list to select to pick up diffusion coefficients defined in a material or a Chemistry interface.

Select **Material**, and a **Fluid material** to use a diffusion coefficient in a material available in the model. User-defined property groups including a diffusion coefficient output property are available for selection in the **Fluid diffusion coefficient** list.

Select **Chemistry** to use a diffusion coefficient from a Chemistry interface added to model component. Diffusion coefficients are automatically defined when **Calculate transport properties** is selected on the interface level of a Chemistry interface. All defined diffusion coefficients are available for selection in the **Fluid diffusion coefficient** list.

For **User defined**, enter a value or expression for the **Fluid diffusion coefficient** D_c of each species in the corresponding input field. This can be a scalar value for isotropic diffusion or a tensor describing anisotropic diffusion. Select the appropriate tensor type — **Isotropic**, **Diagonal**, **Symmetric**, or **Full** that describes the diffusion transport, and then enter the values in the corresponding element (one value for each species).

The Precipitation and Crystallization Multiphysics Interfaces

The **Precipitation and Crystallization** branch of **Chemical Species Transport** contains interfaces used to simulate precipitation of a chemical species in a fluid solution, and solve for the size distribution of the precipitate.

[The Precipitation and Crystallization Interface](#) and [The Precipitation and Crystallization in Fluid Flow Interface](#) are multiphysics interfaces that combines the equations for mass conservation of dissolved species with equations for the size distribution of a precipitate. For space-dependent models, equations describing the velocity and pressure of the solution are also included.

[The Size-Based Population Balance Interface](#) can be used to compute the population balance of particles in a fluid. The interface solves for the discretized size distribution of particles, with support for growth and nucleation terms. For space dependent models transport through convection and diffusion are supported. Examples of systems that can be studied include crystals in a liquid, gas bubbles in a liquid, and droplets in a gas.

The Precipitation and Crystallization Interface

The **Precipitation and Crystallization** () multiphysics interface is used to simulate precipitation of a chemical species in a solution and solve for the size distribution of the precipitate. The interface is only available in 0D.

The interface combines the **Reaction Engineering** and the **Size-Based Population Balance** interfaces.

The **Precipitation** multiphysics coupling is automatically added and couples solute species with the size distribution of particles to ensure mass conservation.

On the constituent physics interfaces:

- The **Reaction Engineering** interface solves for a variable number of species concentrations and reactions.
- The **Size-Based Population Balance** interface solves for a discretized size-distribution used to represent the precipitate. The interface can solve for the nucleation, growth, aggregation, and breakage of particles.

The Precipitation and Crystallization in Fluid Flow Interface

The **Precipitation and Crystallization in Fluid Flow** () multiphysics interface is used to simulate precipitation of a chemical species in a fluid solution and solve for the size distribution of the precipitate. The interface can be used for stationary and time-dependent analysis in 2D, 2D axial symmetry, and 3D.

The interface combines the **Laminar Flow**, the **Transport of Diluted Species**, and the **Size-Based Population Balance** interfaces.

The **Precipitation in Fluid Flow** multiphysics coupling is automatically added and couples the fluid flow to the species transport and a size distribution of particles. The species transport interface solves for one or multiple solute species in the reactants in a precipitation process. The population balance interface solves for the transport and size distribution of the precipitate.

On the constituent physics interfaces:

- The **Laminar Flow** interface solves the Navier–Stokes equations for conservation of momentum and the continuity equation for mass conservation.
- The **Transport of Diluted Species** interface solves for a variable number of species concentrations. The species equations include reactions, and transport by convection, and diffusion.
- The **Size-Based Population Balance** interface solves for a discretized size-distribution used to represent the precipitate. The interface solves for the nucleation, growth, aggregation, and breakage of particles, as well as transport by convection, and diffusion.

The Precipitation Coupling Feature

The **Precipitation** () multiphysics coupling can be used to simulate a space-independent system containing one or more solute species that are present in a precipitation process. The coupling feature adds an extra reaction to the precipitating species to ensure mass conservation.

SETTINGS

The **Label** is the default multiphysics coupling name.

The **Name** is used primarily as a scope prefix for variables defined by the coupling node. Refer to such variables in expressions using the pattern <name>.<variable_name>. In order to distinguish between variables belonging to different coupling nodes or physics

interfaces, the name string must be unique. Only letters, numbers, and underscore (_) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first multiphysics coupling in the model) is pffl.

PRECIPITANT

Each species defined in the **Reaction Engineering** interface, is available as a reactant in the precipitation and can be selected in the list. When a species is selected, an additional reaction rate is added to that species. The additional reaction rate is

$$R_{p,i} = \frac{\rho_p \sum (V_{av,j} \Delta L_j S_{T,j})}{M_p}$$

where ρ_p and M_p is the density and molar mass of the precipitate respectively, $V_{av,j}$ is the average volume, ΔL_j is the interval length and $S_{T,j}$ is the total source term ($1/\text{m}^4/\text{s}$) of size interval j of the population balance defined as

$$S_{T,j} = \left(B_0 \delta(L_j - L_0) - \frac{\partial(Gn_j)}{\partial L_j} + \frac{\partial(Dn_j)}{\partial L_j} + S_j \right)$$

The reaction rate is added under the assumption that the molar mass of the precipitant and precipitate are equal.

When the **Size-Based Population Balance** interface defines the nucleation, growth, or dissolution using the concentration and diffusion coefficient of a dissolved species, additional input fields may be available. Enter an equilibrium concentration in the **Equilibrium concentration** field.

PRECIPITATE

When coupled with the **Transport of Diluted Species** interface, enter the precipitate molar mass in the corresponding text field. When coupled with the **Reaction Engineering** interface, the **Molar mass** field is only available when the molar mass is not defined in the **Reaction Engineering** interface.

PRECIPITATE

Enter the precipitate molar mass in the corresponding text field. The **Molar mass** field is only available when the molar mass is not defined in the **Reaction Engineering** interface.

The Precipitation in Fluid Flow Coupling Feature

The **Precipitation in Fluid Flow** () multiphysics coupling is used to simulate a fluid flow containing one or multiple solute species that act as reactants in precipitation. The coupling feature accounts for the mass transport of species and precipitate in the fluid flow, as well as the consumption of selected species resulting from precipitation.

DOMAIN LEVEL SYNCHRONIZATION

The **Precipitation in Fluid Flow** coupling feature synchronizes the features of the fluid flow interface with the in the transport interfaces. When using the **Precipitation in Fluid Flow**, the velocity field used by the **Transport of Diluted Species** and **Size-Based Population Balance** interfaces are synchronized to the one computed by the **Laminar Flow** interface.

TURBULENT MASS TRANSFER

When a turbulence model is used, the **Precipitation in Fluid Flow** coupling applies turbulence modeling for the population balance and species transport interfaces in the following manners:

- Turbulent mass transport is added. The additional mass transport contribution is defined from the turbulent viscosity and a turbulent Schmidt number (for more information, see [Turbulent Mass Transport Models](#)).
- Mass transport wall functions are added if the turbulence model also uses wall functions. The coupling automatically adds wall functions for the mass transport on the same boundaries as those defined by the fluid flow interface (for more information, see [Mass Transport Wall Functions](#)).

Wall functions are not added for the **Size-Based Population Balance** interface.

SETTINGS

The **Label** is the default multiphysics coupling name.

The **Name** is used primarily as a scope prefix for variables defined by the coupling node. Refer to such variables in expressions using the pattern <name>.<variable_name>. In order to distinguish between variables belonging to different coupling nodes or physics interfaces, the name string must be unique. Only letters, numbers, and underscore (_) are permitted in the **Name** field. The first character must be a letter.

The default **Name** (for the first multiphysics coupling in the model) is pffl.

DOMAIN SELECTION

The selection of the **Precipitation in Fluid Flow** coupling feature is automatically defined on the intersection of the selections for the coupled interfaces.

PRECIPITANT

Each species defined in the **Transport of Diluted Species** interface, is available as a reactant in the precipitation and can be selected in the list. When a species is selected, an additional reaction rate is added to that species. For the added reaction rate, refer to the [Precipitant](#) section of the **Precipitation** interface.

PRECIPITATE

Enter the precipitate molar mass in the corresponding text field.

TURBULENCE

When the fluid flow interface uses a turbulence model, select an option from the **Mass transport turbulence model** list — **Kays-Crawford**, **High Schmidt Number**, or **User-defined turbulent Schmidt number**.

For **User-defined turbulent Schmidt number**, enter a **Turbulent Schmidt number** Sc_T (dimensionless).

The turbulent mass transfer added to the concentration equation is defined as

$$\mathbf{J}_{T,i} = -\frac{\mu_T}{Sc_T} \nabla c_i$$

where μ_T is the turbulent viscosity defined by the flow interface, and the turbulent Schmidt number, Sc_T depends on the **Mass transport turbulence model** used.

Theory for the Transport of Diluted Species Interface

The [Transport of Diluted Species Interface](#) provides a predefined modeling environment for studying the evolution of chemical species transported by diffusion and convection as well as migration due to an electric field. The physics interface assumes that all species present are dilute; that is, that their concentration is low compared to a solvent fluid or solid. As a rule of thumb, a mixture containing several species can be considered dilute when the concentration of the solvent is more than 90 mol%. Due to the dilution, mixture properties such as density and viscosity can be assumed to correspond to those of the solvent.

When studying mixtures that are not dilute, the mixture and transport properties depend on the composition, and a different physics interface is recommended. See [The Transport of Concentrated Species Interface](#) for more information.

Fick's law governs the diffusion of the solutes, dilute mixtures, or solutions, while the phenomenon of ionic migration is sometimes referred to as *electrokinetic flow*. The Transport of Diluted Species interface supports the simulations of chemical species transport by convection, migration, and diffusion in 1D, 2D, and 3D as well as for axisymmetric components in 1D and 2D.

In this section:

- [Adding Transport Through Migration](#)
- [Material Balance Formulation](#)
- [Crosswind Diffusion](#)
- [Danckwerts Inflow Boundary Condition](#)
- [Equilibrium Reaction Theory](#)
- [Mass Balance Equation](#)
- [Mass Sources for Species Transport](#)
- [Solving a Diffusion Equation Only](#)
- [Supporting Electrolytes](#)
- [The Packed Bed Feature](#)
- [References](#)

Note: Some features explained in this section require certain add-on modules. For details see www.comsol.com/products/specifications/

The section also includes the theory for [The Transport of Diluted Species in Porous Media Interface](#):

- [Adsorption](#)
- [Convection in Porous Media](#)
- [Diffusion in Porous Media](#)
- [Dispersion](#)
- [Mass Balance Equation for Transport of Diluted Species in Porous Media](#)
- [Mass Transport in Fractures](#)
- [Reactions](#)

Mass Balance Equation

The default node attributed to the Transport of Diluted Species interface models chemical species transport through diffusion and convection and solves the mass conservation equation for one or more chemical species i :

$$\frac{\partial c_i}{\partial t} + \nabla \cdot \mathbf{J}_i + \mathbf{u} \cdot \nabla c_i = R_i \quad (3-30)$$

[Equation 3-30](#) in its form above includes the transport mechanisms diffusion and convection. If *Migration in Electric Field* is activated (only available in some add-on products), the migration mechanism will be added to the equation as well. See more details in the section [Adding Transport Through Migration](#).

- c_i is the concentration of the species (SI unit: mol/m³)
- D_i denotes the diffusion coefficient (SI unit: m²/s)
- R_i is a reaction rate expression for the species (SI unit: mol/(m³·s))
- $\mathbf{u}$ is the mass averaged velocity vector (SI unit: m/s)
- $\mathbf{J}_i$ is the mass flux diffusive flux vector (SI unit: mol/(m²·s))

The mass flux relative to the mass averaged velocity, $\mathbf{J}_i$ (SI unit: mol/(m²·s)), is associated with the mass balance equation above and used in boundary conditions and flux computations. The Transport of Diluted Species interface always includes mass transport due to molecular diffusion. In this case the mass flux $\mathbf{J}_i$ defines the diffusive flux vector

$$\mathbf{J}_i = -D\nabla c_i \quad (3-31)$$

An input field for the diffusion coefficient is available.

When *Migration in Electric Fields* is activated, the migration term is also added to the diffusive flux vector as shown in the section [Adding Transport Through Migration](#).

The third term on the left side of [Equation 3-30](#) describes the convective transport due to a velocity field $\mathbf{u}$. This field can be expressed analytically or obtained from coupling the physics interface to one that solves for fluid flow, such as *Laminar Flow*. Note that all fluid flow interfaces solve for the mass averaged velocity.

On the right-hand side of the mass balance equation ([Equation 3-30](#)), R_i represents a source or sink term, typically due to a chemical reaction or desorption on a porous matrix. To specify R_i , another node must be added to the Transport of Diluted Species interface — the **Reaction** node for example, which includes an input field for specifying a reaction expression using the variable names of all participating species.

Equilibrium Reaction Theory

The feature Equilibrium Reaction is described in this section. A chemical equilibrium reaction system is defined by the stoichiometry of the reaction and the relation between the chemical activities of the chemical species participating in the reaction (the equilibrium condition).

The kinetics of the reaction is so fast that the equilibrium condition is fulfilled at all times in all space coordinates.

The equilibrium condition is commonly based on the stoichiometric coefficients, ν_i (dimensionless), of the reaction; the species activities of the reacting species a_i (dimensionless); and an equilibrium constant, K_{eq} (1) according to

$$K_{\text{eq}} = \frac{\prod_{i \in \text{products}} a_i^{\nu_i}}{\prod_{i \in \text{reactants}} a_i^{-\nu_i}}$$

where the species activities are defined as

$$a_i = \gamma_{c,i} \frac{c_i}{c_{a0}}$$

where c_{a0} (SI unit: mol/m³) is the standard molarity, and $\gamma_{c,i}$ (dimensionless) an activity coefficient.

Defining the stoichiometric coefficients positive for products and negative for reactants, the above equilibrium condition can also be written

$$K_{\text{eq}} = \prod_i a_i^{v_i}$$

The **Equilibrium Reaction** node solves for a reaction rate so that the equilibrium condition is always fulfilled in the domain. It is available for the Chemical Reaction Engineering Module; Battery Design Module; Corrosion Module; Electrochemistry Module; Electrodeposition Module; and Fuel Cell & Electrolyzer Module.



$\gamma_{e,i}$ is set to unity when the Equilibrium constant is selected in the Settings window. For nonunity activity coefficients, a user-defined equilibrium condition can be used.

EQUILIBRIUM REACTIONS AND INFLOW BOUNDARY CONDITIONS

Contradictory constraints arise if the boundary conditions for concentrations or activities are set so that the domain equilibrium condition is not fulfilled. Special treatment is therefore needed at Inflow boundaries, where the concentrations are set for all species in the mass transport interfaces.

One way of avoiding competing constraints on an inflow boundary is to add an additional reaction coordinate degree of freedom, solved for to create a set of modified inflow concentrations that fulfill the domain equilibrium condition. The reaction coordinate gives rise to a concentration shift, which is the offset to the inflow concentrations provided by the user. The shift for each species obeys the stoichiometry of the reaction and the equilibrium expression. The modified inflow concentrations are then used in the boundary conditions for the domain mass transport equations. The resulting modified inflow concentrations can be seen as the stationary solution for a batch reactor with the user inflow concentrations as initial concentrations. In addition, the domain reaction rate degree of freedom of the equilibrium reaction is constrained to zero on all Inflow boundaries.

EQUILIBRIUM REACTIONS AND CONCENTRATION BOUNDARY CONDITIONS

No special treatment is made with regards to input concentration values of the Concentration boundary node. Using this feature, you can explicitly set one or a set of concentrations, and the equilibrium condition acts on the rest of the concentrations.

However, there is no solution to the problem if more concentrations than the number of species minus the number of equilibrium reactions are set using this feature.

EQUILIBRIUM REACTIONS AND TIME-DEPENDENT SIMULATIONS

Spurious oscillations may occur in a time-dependent problem if the initial conditions do not fulfill the equilibrium condition. Since equilibrium reactions are assumed to be infinitely fast, the solution is to initialize the problem using an additional study step, solving for a stationary problem with all nonequilibrium reaction rates set to zero. Manual scaling of the reaction rate dependent variables is needed in this study step.

Material Balance Formulation

The default node attributed to [The Transport of Diluted Species Interface](#) assumes chemical species transport through diffusion and convection (depending on the modules licensed, a checkbox to activate migration is available) and implements the mass balance equation in [Equation 3-30](#).

Two formulations of the mass balance are available, these are referred to as the nonconservative and conservative form:

$$\text{nonconservative: } \frac{\partial c}{\partial t} + \nabla \cdot \mathbf{J}_i + \mathbf{u} \cdot \nabla c = R \quad (3-32)$$

$$\text{conservative: } \frac{\partial c}{\partial t} + \nabla \cdot \mathbf{J}_i + \nabla \cdot (c\mathbf{u}) = R \quad (3-33)$$

In these equations $\mathbf{J}_i$ is the diffusive flux vector, $\mathbf{u}$ is the solvent velocity field, and R is a production or consumption rate expression. The nonconservative is often more numerically robust, and therefore chosen by default for the interface. This formulation however disregards density changes in the fluid. This can be seen by expanding the conservative convection term using the chain rule. One of the resulting terms, $c\nabla \cdot \mathbf{u}$, equals zero for an incompressible fluid which results in the nonconservative form. To switch between the two formulations, click the **Show** button () and select **Advanced Physics Options**.

Solving a Diffusion Equation Only

Remove the convection term from [Equation 3-32](#) and [Equation 3-33](#) by clearing the Convection checkbox in the Transport Mechanisms section for [The Transport of Diluted Species Interface](#). The equation then becomes

$$\frac{\partial c}{\partial t} = \nabla \cdot \mathbf{J}_i + R$$

Mass Sources for Species Transport

Note: The features below are only available in a limited set of add-on products. For a detailed overview of which features are available in each product, visit www.comsol.com/products/specifications/

There are two types of mass sources in the Transport of Diluted Species interface: point sources and line sources.

POINT SOURCE

A point source is theoretically formed by assuming a mass injection/ejection, Q_c (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$), in a small volume δV and then letting the size of the volume tend to zero while keeping the total mass flux constant. Given a point source strength, $\dot{q}_{p,c}$ (SI unit: mol/s), this can be expressed as

$$\lim_{\delta V \rightarrow 0} \int_{\delta V} Q_c = \dot{q}_{p,c} \quad (3-34)$$

An alternative way to form a point source is to assume that mass is injected/extracted through the surface of a small object. Letting the object surface area tend to zero while keeping the mass flux constant results in the same point source. For this alternative approach, effects resulting from the physical object's volume need to be neglected.

The weak contribution

$$\dot{q}_{p,c} \text{test}(c)$$

is added at a point in the geometry. As can be seen from Equation 3-34, Q_c must tend to plus or minus infinity as δV tends to zero. This means that in theory the concentration also tends to plus or minus infinity.

Observe that “point” refers to the physical representation of the source. A point source can therefore only be added to points in 3D components and to points on the symmetry axis in 2D axisymmetry components. Other geometrical points in 2D components represent physical lines.

The finite element representation of [Equation 3-34](#) corresponds to a finite concentration at a point with the effect of the point source spread out over a region around the point. The size of the region depends on the mesh and on the strength of the source. A finer mesh gives a smaller affected region but also a more extreme concentration value. It is important not to mesh too finely around a point source since this can result in unphysical concentration values. It can also have a negative effect on the condition number for the equation system.

LINE SOURCE

A line source can theoretically be formed by assuming a source of strength $Q_{l,c}$ (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$), located within a tube with cross section δS and then letting δS tend to zero while keeping the total mass flux per unit length constant. Given a line source strength, $\dot{q}_{l,c}$ (SI unit: $\text{mol}/(\text{m} \cdot \text{s})$), this can be expressed as

$$\lim_{\delta S \rightarrow 0} \int_{\delta S} Q_{l,c} = \dot{q}_{l,c} \quad (3-35)$$

As in the point source case, an alternative approach is to assume that mass is injected/extracted through the surface of a small object. This results in the same mass source, but requires that effects resulting from the physical object's volume are neglected.

The weak contribution

$$q_{l,c} \text{test}(c)$$

is added on lines in 3D or at points in 2D (which represent cut-through views of lines). Line sources can also be added on the axisymmetry line in 2D axisymmetry components. It cannot, however, be added on geometrical lines in 2D since those represent physical planes.

As with a point source, it is important not to mesh too finely around the line source.



For feature node information, see [Line Mass Source](#) and [Point Mass Source](#).



For the Reacting Flow in Porous Media, Diluted Species interface, which is available with the CFD Module, Chemical Reaction Engineering Module, or Battery Design Module, these shared physics nodes are renamed as follows:

- The **Line Mass Source** node is available as two nodes, one for the fluid flow (**Fluid Line Source**) and one for the species (**Species Line Source**).
- The **Point Mass Source** node is available as two nodes, one for the fluid flow (**Fluid Point Source**) and one for the species (**Species Point Source**).

Adding Transport Through Migration

Note: Migration is only available in a limited set of add-on products. For a detailed overview of which features are available in each product, visit www.comsol.com/products/specifications/

In addition to transport due to convection and diffusion, the Transport of Diluted Species interface supports ionic species transport by *migration*. This is done by selecting the Migration in Electric Field checkbox under the Transport Mechanisms section for the physics interface. The mass balance then becomes:

$$\frac{\partial c_i}{\partial t} + \nabla \cdot (-D_i \nabla c_i - z_i u_{m,i} F c_i \nabla V + c_i \mathbf{u}) = R_i \quad (3-36)$$

where

- c_i (SI unit: mol/m³) denotes the concentration of species i
- D_i (SI unit: m²/s) is the diffusion coefficient of species i
- $\mathbf{u}$ (SI unit: m/s) is the fluid velocity
- F (SI unit: A·s/mol) refers to Faraday's constant
- V (SI unit: V) denotes the electric potential
- z_i (dimensionless) is the charge number of the ionic species, and
- $u_{m,i}$ (SI unit: mol·s/kg) is its ionic mobility

In this case the diffusive flux vector is

$$\mathbf{J}_i = -D_i \nabla c_i - z_i u_{m,i} F c_i \nabla V$$

The velocity, $\mathbf{u}$, can be a computed fluid velocity field from a Fluid Flow interface or a specified function of the spatial variables x , y , and z . The potential can be provided by an expression or by coupling the system of equations to a current balance, such as the Electrostatics interface. Sometimes it is assumed to be a supporting electrolyte present, which simplifies the transport equations. In that case, the modeled charged species concentration is very low compared to other ions dissolved in the solution. Thus, the species concentration does not influence the solution's conductivity and the net charge within the fluid.

The Nernst–Einstein relation can in many cases be used for relating the species mobility to the species diffusivity according to

$$u_{m,i} = \frac{D_i}{RT}$$

where R (SI unit: J/(mol·K)) is the molar gas constant and T (SI unit: K) is the temperature.

Note: In the *Nernst–Planck Equations* interface, the ionic species contribute to the charge transfer in the solution. It includes an *electroneutrality condition* and also computes the electric potential field in the electrolyte. For more information, see [Theory for the Nernst–Planck Equations Interface](#). This interface is included in the Chemical Reaction Engineering Module.

Supporting Electrolytes

In *electrolyte* solutions, a salt can be added to provide a high electrolyte conductivity and decrease the ohmic losses in a cell. These solutions are often called *supporting electrolytes*, buffer solutions, or carrier electrolytes. The added species, a negative and a positive ion pair, predominates over all other species. Therefore, the supporting electrolyte species can be assumed to dominate the current transport in the solution. In addition, the predominant supporting ions are usually selected so that they do not react at the electrode surfaces since the high conductivity should be kept through the process, that is, they should not be electroactive species. This also means that the concentration gradients of the predominant species in a supporting electrolyte are usually negligible.

Modeling and solving for a supporting electrolyte in the *Electrostatics* or *Secondary Current Distribution* interfaces will give a potential distribution that drives the migration in the Transport of Diluted Species Interface.

The current density vector is proportional to the sum of all species fluxes as expressed by Faraday's law:

$$\mathbf{i} = F \sum_i z_i \mathbf{N}_i$$

The electroneutrality condition ensures that there is always a zero net charge at any position in a dilute solution. Intuitively, this means that it is impossible to create a current by manually pumping positive ions in one direction and negative ions in the other. Therefore, the convective term is canceled out to yield the following expression for the electrolyte current density, where j denotes the supporting species:

$$\mathbf{i} = F \sum_j -z_j^2 u_{m,j} F c_j \nabla \phi \quad (3-37)$$

Equation 3-37 is simply Ohm's law for ionic current transport and can be simplified to

$$\mathbf{i} = -\kappa \nabla \phi \quad (3-38)$$

where κ is the conductivity of the supporting electrolyte. A current balance gives the current and potential density in the cell

$$\nabla \cdot \mathbf{i} = 0$$

which, in combination with Equation 3-38, yields:

$$\nabla \cdot (-\kappa \nabla \phi) = 0 \quad (3-39)$$

Equation 3-39 can be easily solved using the Electrostatics or Secondary Current Distribution interface and, when coupled to the Transport in Diluted Species interface, the potential distribution shows up in the migration term.

Crosswind Diffusion

Transport of diluted species applications can often result in models with a very high cell Péclet number — that is, systems where convection or migration dominates over diffusion. Streamline diffusion and crosswind diffusion are of paramount importance to obtain physically reasonable results. The Transport of Diluted Species interface

provides two crosswind diffusion options using different formulations. Observe that crosswind diffusion makes the equation system nonlinear even if the transport equation is linear.

DO CARMO AND GALEÃO

This is the formulation described in [Numerical Stabilization](#) in the *COMSOL Multiphysics Reference Manual*. The method reduces over- and undershoots to a minimum, even for anisotropic meshes.

In some cases, the resulting nonlinear equation system can be difficult to converge. This can happen when the cell Péclet number is very high and the model contains many thin layers, such as contact discontinuities. You then have three options:

- Refine the mesh, especially in regions with thin layers.
- Use a nonlinear solver with a constant damping factor less than one.
- Switch to the Codina crosswind formulation.

CODINA

The Codina formulation is described in [Ref. 1](#). It adds diffusion strictly in the direction orthogonal to the streamline direction. Compared to the do Carmo and Galeão formulation, the Codina formulation adds less diffusion but is not as efficient at reducing over- and undershoots. It also does not work as well for anisotropic meshes. The advantage is that the resulting nonlinear system is easier to converge and that underresolved gradients are less smeared out.

Danckwerts Inflow Boundary Condition

Constraining the composition to fixed values at an inlet to a reactor may sometimes result in issues with unreasonably high reaction rates or singularities at the inlet boundary. These problems may many times be mitigated by using a flux boundary condition instead, based on the upstream concentrations and the fluid velocity at the boundary. In chemical engineering, this type of flux boundary condition is also known as a Danckwerts condition.

Use the Danckwerts condition to specify inlet conditions for domains where high reaction rates are anticipated in the vicinity to the inlet ([Ref. 2](#)).

Given an upstream concentration $c_{i,0}$, the Danckwerts inflow boundary condition prescribed the total flux as

$$\mathbf{n} \cdot (\mathbf{J}_i + \mathbf{u}c_i) = \mathbf{n} \cdot (\mathbf{u}c_{i,0}) \quad (3-40)$$

Mass Balance Equation for Transport of Diluted Species in Porous Media

VARIABLY SATURATED POROUS MEDIA

The following equations for the molar concentrations, c_i , describe the transport of solutes in a variably saturated porous medium for the most general case, when the pore space is primarily filled with liquid but also contain pockets or immobile gas:

$$\begin{aligned} \frac{\partial}{\partial t}(\theta_l c_i) + \frac{\partial}{\partial t}(\rho c_{P,i}) + \frac{\partial}{\partial t}(\theta_g c_{G,i}) + \mathbf{u} \cdot \nabla c_i = \\ \nabla \cdot [(D_{D,i} + D_{e,i})\nabla c_i] + R_i + S_i \end{aligned} \quad (3-41)$$

On the left-hand side of Equation 3-41, the first three terms correspond to the accumulation of species within the liquid, solid, and gas phases, while the last term describes the convection due to the velocity field $\mathbf{u}$ (SI unit: m/s).

In Equation 3-41 c_i denotes the concentration of species i in the liquid (SI unit: mol/m³), $c_{P,i}$ the amount adsorbed to solid particles (moles per unit dry weight of the solid), and $c_{G,i}$ the concentration of species i in the gas phase.

The equation balances the mass transport throughout the porous medium using the porosity ϵ_p , the liquid volume fraction θ_l ; the dry bulk density, $\rho = (1 - \epsilon_p)\rho_s$, and the solid phase density ρ_s .

For saturated porous media, the liquid volume fraction θ_l is equal to the porosity ϵ_p , but for unsaturated porous media, they are related by the saturation s as $\theta_l = \epsilon_p s$. The resulting gas volume fraction in the case of an unsaturated porous medium is

$$\theta_g = \epsilon_p - \theta_l = (1 - s)\epsilon_p$$

On the right-hand side of Equation 3-41, the first term introduces the spreading of species due to mechanical mixing resulting from the porous media (dispersion), as well as from diffusion and volatilization to the gas phase. The dispersion tensor is denoted D_D (SI unit: m²/s) and the effective diffusion by D_e (SI unit: m²/s).

The last two terms on the right-hand side of Equation 3-41 describe production or consumption of the species; R_i is a reaction rate expression which can account for reactions in the liquid, solid, or gas phase, and S_i is an arbitrary source term, for example due to a fluid flow source or sink.

Adsorption

The time evolution of the adsorption, the solute transport to or from the solid phase, is defined by assuming that the amount of solute adsorbed to the solid, $c_{p,i}$, is a function of the concentration in the fluid c_i . This implies that the solute concentration in the liquid and solid phase are in instant equilibrium. The adsorption term can be expanded to give

$$\frac{\partial}{\partial t}(\rho c_{p,i}) = \rho \frac{\partial c_{p,i}}{\partial c_i} \frac{\partial c_i}{\partial t} - c_{p,i} \rho_s \frac{\partial \epsilon_p}{\partial t} = \rho K_{p,i} \frac{\partial c_i}{\partial t} - c_{p,i} \rho_s \frac{\partial \epsilon_p}{\partial t} \quad (3-42)$$

where $K_{p,i} = \partial c_{p,i} / \partial c_i$ is the adsorption isotherm.

Volatilization

Volatilization is the process where a solute species in the liquid is transported to the gas phase due to vaporization. Assuming that the amount of solute in the gas phase, $c_{G,i}$, is a linear function of the liquid phase concentration, the volatilization term is defined as

$$\frac{\partial}{\partial t} \theta_g c_{G,i} = \theta_g \frac{\partial c_{G,i}}{\partial c_i} \frac{\partial c_i}{\partial t} + k_{G,i} c_i \frac{\partial \theta_g}{\partial t} = \theta_g k_{G,i} \frac{\partial c_i}{\partial t} + k_{G,i} c_i \frac{\partial \theta_g}{\partial t} \quad (3-43)$$

where $k_{G,i} = \partial c_{G,i} / \partial c_i$ is the linear volatilization.

SATURATED POROUS MEDIA

In the case of transport in a saturated porous medium, $\theta_l = \epsilon_p$, and the governing equations are

$$\frac{\partial}{\partial t}(\epsilon_p c_i) + \frac{\partial}{\partial t}(\rho c_{p,i}) + \mathbf{u} \cdot \nabla c_i = \nabla \cdot [(D_{D,i} + D_{e,i}) \nabla c_i] + R_i + S_i \quad (3-44)$$

Convection in Porous Media

Convection (also called advection) describes the movement of a species, such as a pollutant, with the bulk fluid velocity. The velocity field $\mathbf{u}$ corresponds to a superficial volume average over a unit volume of the porous medium, including both pores and matrix. This velocity is sometimes called *Darcy velocity*, and defined as volume flow

rates per unit cross section of the medium. This definition makes the velocity field continuous across the boundaries between porous regions and regions with free flow.



The velocity field to be used in the Model Inputs section on the physics interface can, for example, be prescribed using the velocity field from a Darcy's Law or a Brinkman Equations interface.

The average linear fluid velocities $\mathbf{u}_a$, provides an estimate of the fluid velocity within the pores:

$$\mathbf{u}_a = \frac{\mathbf{u}}{\varepsilon_p} \quad \text{Saturated}$$

$$\mathbf{u}_a = \frac{\mathbf{u}}{\theta_l} \quad \text{Unsaturated}$$

where ε_p is the porosity and $\theta_l = s\varepsilon_p$ the liquid volume fraction, and s the saturation, a dimensionless number between 0 and 1.

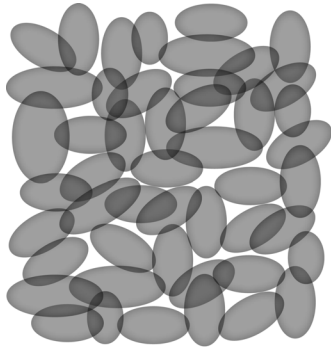


Figure 3-3: A block of a porous medium consisting of solids and the pore space between the solid grains. The average linear velocity describes how fast the fluid moves within the pores. The Darcy velocity attributes this flow over the entire fluid-solid face.

MATERIAL BALANCE FORMULATION

The Transport of Diluted Species interface in porous media includes two formulations of the material balance. The conservative formulation of the species equations in [Equation 3-41](#) is written as:

$$\frac{\partial}{\partial t}(\theta_l c_i) + \frac{\partial}{\partial t}(\rho c_{P,i}) + \frac{\partial}{\partial t}(\theta_g c_{G,i}) + \nabla \cdot \mathbf{u} c_i = \nabla \cdot [(D_{D,i} + D_{e,i}) \nabla c_i] + R_i + S_i \quad (3-45)$$

If the conservative formulation is expanded using the chain rule, then one of the terms from the convection part, $c_i \nabla \cdot \mathbf{u}$, would equal zero for an incompressible fluid and would result in the nonconservative formulation described in [Equation 3-41](#).

When using the nonconservative formulation, which is the default, the fluid is assumed incompressible and divergence free: $\nabla \cdot \mathbf{u} = 0$. The nonconservative formulation improves the stability of systems coupled to a momentum equation (fluid flow equation).



To switch between the two formulations, click the **Show** button () and select **Advanced Physics Options**. In the section **Advanced Settings** select either **Nonconservative** (the default) or **Conservative**. The conservative formulation should be used for compressible flow.

Diffusion in Porous Media

The effective diffusion in porous media, D_e , depends on the structure of the porous material and the phases involved. Dependent on whether the transport of diluted species occurs in free flow or in saturated or unsaturated porous media, the effective diffusivity is defined as:

$$\begin{aligned} D_e &= D_F && \text{Free Flow} \\ D_e &= \frac{\epsilon_p}{\tau_L} D_L && \text{Saturated Porous Media} \\ D_e &= \frac{\theta_l}{\tau_L} D_L && \text{Unsaturated Porous Media} \\ D_e &= \frac{\theta_l}{\tau_L} D_L + \frac{\theta_g}{\tau_G} k_G D_G && \text{Unsaturated with Volatilization} \end{aligned}$$

Here D_F , D_L , and D_G are the single-phase diffusion coefficients for the species diluted in a fluid, a pure liquid, and a gas phase, respectively (SI unit: m^2/s), and τ_F , τ_L , and τ_G are the corresponding tortuosity factors (dimensionless).

The tortuosity factor accounts for the reduced diffusivity due to the fact that the solid grains impede Brownian motion. The interface provides predefined expressions to

compute the tortuosity factors in partially saturated porous media according to the Millington and Quirk model ([Ref. 3](#)):

$$\tau_L = \theta_l^{-7/3} \epsilon_p^2, \tau_G = \theta_g^{-7/3} \epsilon_p^2$$

and Bruggeman model

$$\tau_L = \theta_l^{-5/2} \epsilon_p^2, \tau_G = \theta_g^{-5/2} \epsilon_p^2$$

For saturated porous media $\theta_l = \epsilon_p$. The fluid tortuosity for the Millington and Quirk model is

$$\tau_L = \epsilon_p^{-1/3}$$

and for the Bruggeman model the tortuosity is defined as

$$\tau_L = \epsilon_p^{-1/2}$$

User defined expressions for the tortuosity factor can also be applied.

Dispersion

The contribution of dispersion to the mixing of species typically overshadows the contribution from molecular diffusion, except when the fluid velocity is very low.

The spreading of mass, as a fluid travel through a porous medium is caused by several contributing effects. Local variations in fluid velocity lead to mechanical mixing referred to as dispersion occurs because the fluid in the pore space flows around solid particles, so the velocity field varies within pore channels. The spreading in the direction parallel to the flow, or *longitudinal dispersivity*, typically exceeds the *transverse dispersivity* from up to an order of magnitude. Being driven by the concentration gradient alone, molecular diffusion rate is low relative to the mechanical dispersion, except at very low fluid velocities.

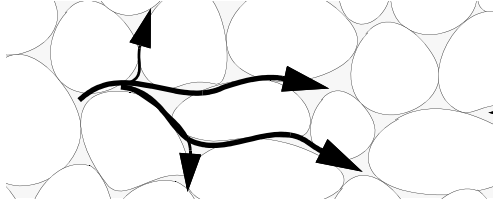


Figure 3-4: Spreading of fluid around solid particles in a porous medium.

The dispersion is controlled through the dispersion tensor D_D . The tensor components can either be given by user-defined values or expressions or derived from the directional dispersivities.

Using the longitudinal and transverse dispersivities in 2D, the dispersivity tensor components are (Ref. 5):

$$D_{Dii} = \alpha_L \frac{u_i^2}{|\mathbf{u}|} + \alpha_T \frac{u_j^2}{|\mathbf{u}|}$$

$$D_{Dij} = D_{Dji} = (\alpha_L - \alpha_T) \frac{u_i u_j}{|\mathbf{u}|}$$

In these equations, D_{Dii} (SI unit: m^2/s) are the principal components of the dispersivity tensor, and D_{Dji} and D_{Dij} are the cross terms. The parameters α_L and α_T (SI unit: m) specify the longitudinal and transverse dispersivities; and u_i (SI unit: m/s) stands for the velocity field components.

In order to facilitate modeling of stratified porous media in 3D, the tensor formulation by Burnett and Frind (Ref. 4) can be used. Consider a transverse isotropic media, where the strata are piled up in the z direction, the dispersivity tensor components are:

$$\begin{aligned}
D_{Lxx} &= \alpha_1 \frac{u^2}{|\mathbf{u}|} + \alpha_2 \frac{v^2}{|\mathbf{u}|} + \alpha_3 \frac{w^2}{|\mathbf{u}|} \\
D_{Lyy} &= \alpha_1 \frac{v^2}{|\mathbf{u}|} + \alpha_2 \frac{u^2}{|\mathbf{u}|} + \alpha_3 \frac{w^2}{|\mathbf{u}|} \\
D_{Lzz} &= \alpha_1 \frac{w^2}{|\mathbf{u}|} + \alpha_2 \frac{u^2}{|\mathbf{u}|} + \alpha_3 \frac{v^2}{|\mathbf{u}|} \\
D_{Lxy} &= D_{Lyx} = (\alpha_1 - \alpha_2) \frac{uv}{|\mathbf{u}|} \\
D_{Lxz} &= D_{Lzx} = (\alpha_1 - \alpha_3) \frac{uw}{|\mathbf{u}|} \\
D_{Lyz} &= D_{Lzy} = (\alpha_1 - \alpha_3) \frac{vw}{|\mathbf{u}|}
\end{aligned} \tag{3-46}$$

In Equation 3-46 the fluid velocities u , v , and w correspond to the components of the velocity field $\mathbf{u}$ in the x , y , and z directions, respectively, and α_1 (SI unit: m) is the longitudinal dispersivity. If z is the vertical axis, α_2 and α_3 are the dispersivities in the transverse horizontal and transverse vertical directions, respectively (SI unit: m). Setting $\alpha_2 = \alpha_3$ gives the expressions for isotropic media shown in Bear (Ref. 5 and Ref. 6).

Adsorption

As species travel through a porous medium they typically attach to (adsorb), and detach (desorb) from the solid phase, which slows chemical transport through the porous medium. Adsorption and desorption respectively reduces or increases species concentrations in the fluid. The adsorption properties vary between chemicals, so a plume containing multiple species can separate into components (Ref. 7). The Adsorption feature includes four predefined and one user-defined relationships to predict the solid concentrations, c_{pi} from the concentration in the liquid phase, c_i :

$$\begin{aligned}
c_p &= c_{p\max} \frac{K_L c}{1 + K_L c} && \text{Langmuir} \\
c_p &= K_F \left(\frac{c}{c_{\text{ref}}} \right)^N && \text{Freundlich} \\
c_p &= c_{p\max} \frac{b_T c}{(1 + (b_T c)^{N_T})^{1/N_T}} && \text{Toth} \\
c_p &= \frac{K_B c_0 c}{(c_S - c) \left(1 + (K_B - 1) \frac{c}{c_S} \right)} && \text{BET}
\end{aligned} \tag{3-47}$$

The above equations contains the following parameters:

- Freundlich ([Ref. 8](#))
 - Freundlich constant K_F (SI unit: $\cdot \text{mol/kg}$), Freundlich exponent N_F (dimensionless), and reference concentration c_{ref} (SI unit: mol/m^3).
- Langmuir ([Ref. 9](#))
 - Langmuir constant K_L (SI unit: m^3/mol), and adsorption maximum $c_{p\max}$ (SI unit: mol/kg).
- Toth ([Ref. 10](#))
 - Toth constant b_T (SI unit: m^3/mol), Toth exponent N_T (dimensionless), and adsorption maximum $c_{p\max}$ (SI unit: mol/kg).
- BET (Brunauer–Emmett–Teller, [Ref. 11](#))
 - BET constant K_B (dimensionless), the monolayer adsorption capacity c_0 (SI unit: mol/kg) and the saturation concentration, c_S (SI unit: mol/m^3).

These predefined expressions are adsorption isotherms that describe the amount of species sorbed to the solid. Applying the isotherms, as defined in the [Adsorption](#) section above, assumes that adsorption is at equilibrium locally. This implies that the species phase transfer, between the fluid and the solid phase (within the porous medium), is much faster than the transport within the fluid phase.

Using a [Species Source](#) feature, arbitrary expressions can be entered to define, for example, nonequilibrium and temperature-dependent adsorption laws, in the manner of Fetter ([Ref. 12](#)), or Bear and Verruijt ([Ref. 13](#)).

The retardation factor, $\mathbf{RF}$, describes how adsorption slows the solute velocity, $\mathbf{u}_c$, relative to the average linear velocity of the fluid, $\mathbf{u}_a$, as in

$$\text{RF} = 1 + \frac{\rho_b}{\theta} \frac{\partial c_p}{\partial c} = \frac{\mathbf{u}_a}{\mathbf{u}_c}$$

If the contaminant moves at the average linear velocity of the fluid for $\text{RF} = 1$. For $\text{RF} > 1$, the contaminant velocity is lower than the fluid velocity owing to residence time on solids.

Reactions

Chemical reactions of all types influence species transport in porous media. Examples include biodegradation, radioactive decay, transformation to tracked products, temperature- and pressure-dependent functions, exothermic reactions, and endothermic reactions. The reactions represent change in species concentration per unit volume porous medium per time. Reaction terms are used on the right-hand side of the governing equation to represent these processes. For reactions in a fluid phase, multiply the expression by the fluid volume fraction θ . Similarly, solid phase reaction expressions include the bulk density, ρ_b , and gas phase reactions include the gas volume fraction, α_v .

The following expressions define some common types of reactions:

$$\begin{aligned} R_{Li} &= -\theta \frac{\ln 2}{\lambda_{Li}} c_i && \text{Radioactive decay — liquid} \\ R_{Pi} &= -\rho_b \frac{\ln 2}{\lambda_{Pi}} \left(\frac{\partial c_{Pi}}{\partial c_i} \right) c_i && \text{Radioactive decay — solid} \\ R_{Gi} &= -\frac{\ln 2}{\lambda_{Gi}} \alpha_v \left(\frac{\partial c_{Gi}}{\partial c} \right) c_i && \text{Radioactive decay — gas} \\ R_{Lk} &= \theta \zeta_{Li} c_i && \text{Creation from parent } c_{Li} \text{ — liquid} \\ R_{Pk} &= \rho_b \zeta_{Pi} \left(\frac{\partial c_{Pi}}{\partial c_i} \right) c_i && \text{Creation from sorbed parent } c_{Pi} \text{ — solid} \\ R_{Gk} &= -\zeta_{Gv} \left(\frac{\partial c_{Gi}}{\partial c} \right) c_i && \text{Reaction — gas} \end{aligned}$$

where λ is the chemical half life, ζ is a reaction rate, and the subscripts L , P , and G denote liquid, solid, and gas phases, respectively. In the equations, the reactions either depend on liquid concentration c_i or solid phase concentrations c_{Pi} obtained using the sorption derivative with c_i or gas phase concentration c_{Gi} depending on the gas volume fraction, the volatilization, and the liquid concentration.

Reaction rates can vary with results from other equations in your model, such as temperature. For example, enter the Arrhenius rate law given in [Ref. 14](#):

$$\zeta_T = \zeta_R \exp \left[\frac{E_a(T - T_R)}{R_u T T_R} \right] \quad (3-48)$$

In [Equation 3-48](#), T denotes the current absolute temperature, T_R denotes the reference absolute temperature, E_a is the activation energy, and R_u is the universal gas constant.

Mass Transport in Fractures

When thin fractures occur in porous media, fluid flow tends to move faster along the fracture than in the surrounding media. The transport of chemical species therefore also occur also faster in the direction of the fractures.

The fluid flow in a fracture can be modeled using Darcy's law formulated in a thin sheet of porous medium (a fracture):

$$\mathbf{u} = \frac{\kappa}{\mu} \nabla_t p$$

Here $\mathbf{u}$ is the tangential Darcy velocity, κ is the fracture permeability, μ the fluid's dynamic viscosity, and $\nabla_t p$ is the tangential gradient of the fluid pressure.

The equation to solve for mass transport of species c_i in a thin fracture, embedded in a porous media, is derived from [Equation 3-41](#). The resulting equation is:

$$d_{fr} \left(\frac{\partial \rho_b c_{p,i}}{\partial t} + \frac{\partial \varepsilon_p c_i}{\partial t} + \nabla_t \cdot (D_{e,i} \nabla_t c_i) + \mathbf{u} \cdot \nabla_t c_i \right) = d_{fr} R_i + d_{fr} S_i + n_0 \quad (3-49)$$

Here d_{fr} is the fracture thickness, $c_{p,i}$ the amount of species adsorbed to (or desorbed from) the porous matrix (moles per unit dry weight of the solid), ε_p is the fracture porosity, and D_e is the effective diffusivity. The first two terms on the right hand side represent source terms from reactions, and n_0 corresponds to out-of plane flux from the adjacent porous domain.

In order to arrive at the tangential differential equation, the gradient is split into the contributions normal and tangential to the fracture:

$$\nabla c_i = \nabla_n c_i + \nabla_t c_i$$

The normal gradient is defined in the direction normal to the boundary representing the fracture and the tangential gradient is defined along the boundary. Assuming that the variations in the normal (thin) direction of the fracture are negligible compared to those in the tangential direction, the gradient is simplified as:

$$\nabla c_i = \nabla_t c_i$$

Using [The Transport of Diluted Species in Fractures Interface](#), the transport along fracture boundaries alone is solved for. In this case the transport in the surrounding porous media neglected and the out-of plane flux n_0 vanishes.



See [Fracture](#) for more information about the boundary feature solving [Equation 3-49](#). See [The Transport of Diluted Species in Fractures Interface](#) for more information about the physics interface solving the equation on boundaries only.

The Packed Bed Feature

The **Packed Bed** feature, available to users of the Chemical Reaction Engineering Module, adds a domain feature that can simulate regions filled with reactive porous pellets, as in packed bed reactors. The theory for the **Packed Bed** feature available in the **Transport of Diluted Species** interface is presented below.

The pellets in the bed are simulated with the Extra Dimension technology provided by the COMSOL software: A 1D, 2D, or 3D domain represents the bed volume. The extra dimension is the added radial microscale dimension inside each pellet.

There are two types of pellets, acting as porous catalyst and reactant (Shrinking Core) respectively. For the porous catalyst pellet, the whole radius is mapped to the extra dimension, while for the Shrinking Core pellet, the radial length of porous reacted outer-layer is mapped to the extra dimension.

PELLET: POROUS CATALYST

The species diffuse from packed bed into porous pellet where they go through catalytic reactions, then diffuse back into the packed bed.



For an example of how to use the Packed Bed, see the model example *A Multiscale 3D Packed Bed Reactor*, file path **Chemical_Reaction_Engineering_Module/Reactors_with_Porous_Catalysts/packed_bed_reactor_3d**

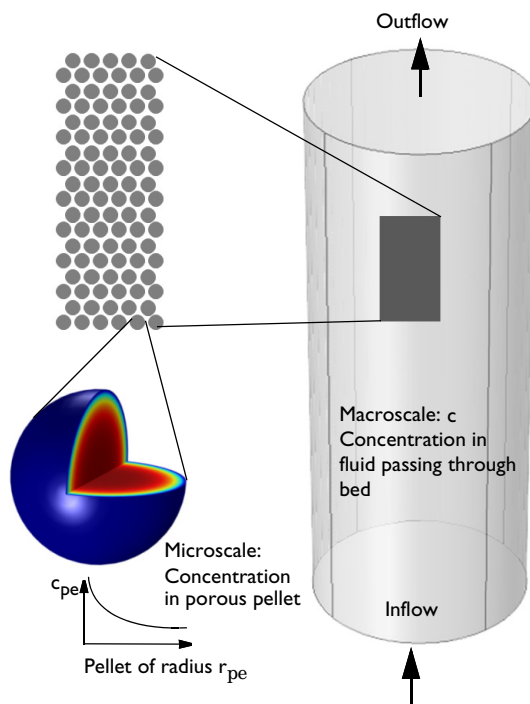


Figure 3-5: Schematic showing the macroscale (bed volume) and the microscale (pellet).

The transport and reaction equations inside the pellets are solved on an extra dimension attached to the 1D, 2D, or 3D physics interfaces, including axisymmetric cases.

The equations inside the spherical pellet are solved as spherical transport equations on nondimensional radial coordinates on the domain 0-1.

The model equations assume spherical particles (pellets) of a radius r_{pe} . Modeling assumptions for cylinders, flakes, and user-defined shapes can also be used. Consider the microscale concentration c_{pe} inside an individual porous pellet or pellets, and the macro-concentration c in the packed bed gas volume.

The pellet radius input can be:

The model equation for the bulk (macroscale) species is, for example:

$$\varepsilon_b \frac{\partial}{\partial t}(c_i) + \mathbf{u} \cdot \nabla c_i + \nabla \cdot (-D_{b,i} \nabla c_i) = R_i \quad (3-50)$$

- ε_b is the bed porosity.
- The dependent variable c for each chemical species i represents the interparticle concentration, that is, the concentration based on unit volume of fluid flowing between the pellets.
- R_i is the species interparticle reaction rate in the bed. This corresponds to reactions occurring in the macropores between the pellets in the bed. Note that R_i is the reaction rate per unit volume of bed.

Looking inside a pellet: Assuming no concentration variations in the space-angle (θ , φ) direction, but only in the radial (r) direction of the spherical pellet allows a spherically symmetric reaction-diffusion transport equation inside the pellet. If r_{dim} (SI unit: m) is the spatial radial coordinate in the pellet, and r_{pe} is the pellet radius, the nondimensional coordinate $r = r_{\text{dim}}/r_{\text{pe}}$ can be defined. The modeling domain on r goes from 0 to 1.

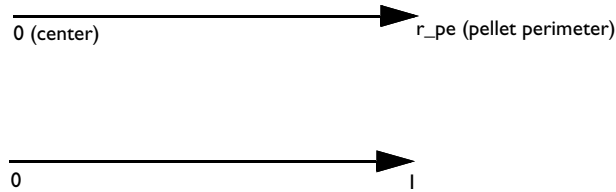


Figure 3-6: Modeling domain in a pellet for dimensional (top) and nondimensional (bottom) coordinates.

A shell mole balance across a spherical shell at radius r_{dim} (SI unit: m), and a subsequent variable substitution $r = r_{\text{dim}}/r_{\text{pe}}$ gives the following transport equation on the pellet domain $0 < r < 1$:

$$4\pi \left\{ r^2 r_{\text{pe}}^2 \epsilon_{\text{pe}} \frac{\partial}{\partial t} (c_{\text{pe},i}) + \nabla \cdot (-r^2 D_{\text{pe},i} \nabla c_{\text{pe},i}) = r^2 r_{\text{pe}}^2 R_{\text{pe},i} \right\} \quad (3-51)$$

- ϵ_{pe} is the pellet (microscale) porosity.
- $c_{\text{pe},i}$ is the intraparticle concentration of species i in moles/m³ of fluid volume inside the micropores.
- $R_{\text{pe},i}$ is the pellet reaction rate. This corresponds to reactions taking place inside the pellets. Note that the user input of $R_{\text{pe},i}$ is per unit volume of pellet.

Assuming that the pellet bed is homogenized, that is that pellets are present in an averaged sense in each point of the geometry, the equation to be solved in the packed bed is:

$$4\pi N \left\{ r^2 r_{\text{pe}}^2 \epsilon_{\text{pe}} \frac{\partial}{\partial t} (c_{\text{pe},i}) + \nabla \cdot (-r^2 D_{\text{pe},i} \nabla c_{\text{pe},i}) = r^2 r_{\text{pe}}^2 R_{\text{pe},i} \right\} \quad (3-52)$$

where N is the number of pellets per unit volume of bed:

$$N = \frac{(1 - \epsilon_b)}{V_{\text{pe}}}$$

Dividing Equation 3-51 by $4\pi r_{\text{pe}}^2$ (the pellet area times r^2), the governing equation inside each pellet is:

$$\epsilon_{\text{pe}} \frac{\partial}{\partial t} (c_{\text{pe},i}) + \frac{a}{r^2 r_{\text{pe}}^2} \nabla \cdot (-r^2 D_{\text{pe},i} \nabla c_{\text{pe},i}) = R_{\text{pe},i} \quad (3-53)$$

The effective diffusion coefficient in the equations above depends on the porosity ϵ_{pe} , tortuosity τ , and physical gas diffusivity D in the manner of

$$D_{\text{pe}} = \frac{\epsilon_{\text{pe}} D}{\tau}.$$

The available models for the porous media tortuosity are the one by Millington and Quirk (Ref. 3),

$$\tau = \epsilon_{\text{pe}}^{-1/3} \Rightarrow D_{\text{pe}} = \epsilon_{\text{pe}}^{4/3} D, \quad (3-54)$$

the Bruggeman model,

$$\tau = \varepsilon_{\text{pc}}^{-1/2} \Rightarrow D_{\text{pc}} = \varepsilon_{\text{pc}}^{3/2} D \quad (3-55)$$

and the Tortuosity model, where the tortuosity expression is entered as user-defined input.

These are readily used for both gaseous and liquid fluids along with various types of pellet shapes. For instance, the first model has been shown to fit mass transport in soil-vapor and soil-moisture well.

Equation 3-51 can be solved for two types of boundary conditions at the interface between the pellet surface and the fluid in this feature.

- **Continuous concentration:** assuming that all resistance to mass transfer to/from the pellet is within the pellet and no resistance to pellet-fluid mass transfer is on the bulk fluid side. The concentration in the fluid will thus be equal to that in the pellet pore just at the pellet surface: $c_{\text{pe},i} = c_i$. This constraint also automatically ensures flux continuity between the internal pellet domain and the free fluid domain through so-called reaction forces in the finite element formulation.
- **Film resistance (mass flux):** The flux of mass across the pellet-fluid interface into the pellet is possibly rate determined on the bulk fluid side by film resistance. The resistance is expressed in terms of a film mass transfer coefficient, h_{Di} , such that:

$$N_{i,\text{inward}} = h_{D,i}(c_i - c_{\text{pe},i}), \quad (3-56)$$

where $N_{i,\text{inward}}$ is the molar flux from the free fluid into a pellet and has the unit moles/(m²·s).

With the film resistance formulation above, the free fluid Equation 3-50 needs to be amended for flux continuity so that

$$\varepsilon_{\text{b}} \frac{\partial}{\partial t}(c_i) + \mathbf{u} \cdot \nabla c_i + \nabla \cdot (-D_{b,i} \nabla c_i) = R_i - N_{i,\text{inward}} S_{\text{b}} \quad (3-57)$$

where S_{b} (SI unit: m²/m³) is the specific surface area exposed to the free fluid of the packed bed (not including the inside of the pores).

For the case of randomly packed spherical pellets, the specific surface area exposed to the free fluid is (Ref. 15):

$$S_{\text{b}} = \frac{3}{r_{\text{pc}}}(1 - \varepsilon_{\text{b}}) \quad (3-58)$$

The mass transfer coefficient in Equation 3-56 can be computed from the fluid properties and flow characteristics within the porous media. For this, the Sherwood, Sh , number defined as the ratio between the convective mass transfer coefficient and the diffusive mass transfer coefficient is often used:

$$Sh = \frac{hL}{D}$$

where L is a characteristic length (for spheres, typically the radius), and D is the diffusion coefficient in the fluid. From the Sherwood number definition, the mass transfer coefficient can be computed.

Three commonly used empirical expressions for the calculation of the Sherwood number are the Frössling relation (Ref. 16):

$$Sh = 2 + 0.552Re^{1/2}Sc^{1/3}, \quad (3-59)$$

which was measured on pellets in the size region 1 mm, the Rosner relation (Ref. 17)

$$Sh = Sc^{0.4}(0.4Re^{1/2} + 0.2Re^{2/3}), \quad (3-60)$$

and the Garner and Keey relation (Ref. 16)

$$Sh = 0.94Re^{1/2}Sc^{1/3}, \quad (3-61)$$

which was measured for Re numbers greater than 250.

All three depend on the Reynolds, Re , and Schmidt, Sc , numbers. The first describing the fluid flow regime (laminar versus turbulent) and the second, the ratio between the viscous diffusion rate and the molecular (mass) diffusion rate. In the expressions, properties such as velocity, u , dynamic viscosity, μ , and density, ρ , of the fluid are included.

$$Re = \frac{\rho u L}{\mu}$$

$$Sc = \frac{\mu}{\rho D}$$

Surface Species

Surface species correspond to species bound to the solid interface within the porous pellet, which is also in contact with the pore fluid. The surface species hence only exists within the pellet. The surface species are assumed to be immobile, and the

concentration is only dependent on the reaction rate of surface reactions involving the species:

$$\frac{\partial c_{\text{pes}}}{\partial t} = R_{\text{pes}}^s \quad (3-62)$$

It can be noted that there is no mass flux of surface species within the pellet or across the pellet outer surface and the bulk fluid. The unit of surface species concentration c_{pes} is amount per area, and the corresponding surface reaction rate unit is amount per area and time.

When surface species are present, the equation governing transport of bulk species inside the each pellet is:

$$\epsilon_{\text{pe}} \frac{\partial}{\partial t} (c_{\text{pe}, i}) + \frac{a}{r_{\text{pe}}^2} \nabla \cdot (-r^2 D_{\text{pe}, i} \nabla c_{\text{pe}, i}) = R_{\text{pe}, i} + S_{\text{b, reac}} R_{\text{pe}}^s \quad (3-63)$$

The second term on the right hand side represents species production or removal due to surface reactions in the pellet. It is composed of the reactive specific area $S_{\text{b, reac}}$, the area per volume available for surface reactions, and the surface reaction rate R_{pe}^s .

It should be noted that the surface reaction rate for a bulk species R_{pe}^s and the surface reaction rate for a surface species R_{pes}^s are related but not identical since the species may represent different molecules.

Nonspherical Particles

For nonspherical pellets (of any shape), the relations above can be applied approximately by reinterpreting the pellet radius r_{pe} as

$$r_{\text{pe}} = 3 \frac{V_{\text{pe}}}{A_{\text{pe}}} \quad (3-64)$$

(Ref. 18), where V_{pe} and A_{pe} are the volume and external surface, respectively, of a single pellet of any shape. Since the specific surface S_{pe} (SI unit: m^2/m^3) of one pellet is defined as

$$S_{\text{pe}} = \frac{A_{\text{pe}}}{V_{\text{pe}}}$$

it follows by insertion of Equation 3-64 that

$$S_{\text{pe}} = \frac{3}{r_{\text{pe}}}.$$

For a packed bed of which the packing has a porosity ε_b , the specific surface of the bed will be

$$S_b = S_{pe}(1 - \varepsilon_b) = \frac{A_{pe}}{V_{pe}}(1 - \varepsilon_b)$$

or

$$S_b = \frac{3}{r_{pe}}(1 - \varepsilon_b) \quad (3-65)$$

for any pellet shape. Now r_{pe} and S_b can be calculated for any shape and inserted in equations [Equation 3-51](#) and [Equation 3-57](#). Some common specific shapes have automatic support:

Cylinders

For cylindrical shapes, applying [Equation 3-64](#) gives

$$r_{pe} = 3 \frac{V_{pe}}{A_{pe}} = 3 \frac{\pi r_{cyl}^2 L_{cyl}}{2\pi r_{cyl} L_{cyl} + 2\pi r_{cyl}^2} = \frac{1}{\frac{2}{3r_{cyl}} + \frac{2}{3L_{cyl}}} \quad (3-66)$$

It is common practice to assume that the top and bottom surface of cylindrical pellets have negligible effect on the mass transfer to and from the internals of the pellet, or, $r_{cyl} \ll L_{cyl}$. [Equation 3-66](#) then simplifies to

$$r_{pe} = \frac{3}{2} r_{cyl}$$

and [Equation 3-65](#) to

$$S_b = \frac{2}{r_{cyl}}(1 - \varepsilon_b)$$

Flakes

The derivation for a disc-shaped catalyst pellet is exactly the same as for cylindrical pellets, except that the assumption is reversed about the end surfaces and the envelope surface: $r_{flake} \gg w_{flake}$, where w_{flake} is the thickness of the disc. This gives

$$r_{pe} = \frac{3}{2} w_{flake}$$

and

$$S_b = \frac{2}{w_{\text{flake}}} (1 - \epsilon_b) .$$

Surface Reaction

Surface reactions can also be simulated inside the pellet. Surface species are introduced in pellet by adding them in the Surface Species section of the Pellets node available under Packed Bed.

A bulk species can take part in both volumetric and surface reactions. The total reaction rate for a bulk species within a pellet is defined as:

$$R = R_{\text{pe}} + R_{\text{pe},s} \cdot S_{\text{b, reac}}$$

Here R_{pe} is the reaction rate for bulk reactions occurring inside the pellet. $R_{\text{pe},s}$ and S_b are the reaction rate and the reactive specific surface area for a surface reaction occurring inside the pellet, on the interface between the solid matrix and the fluid.

Heat Source

The heat source of endothermic or exothermic reactions inside the pellet needs to be accounted for in the heat transfer on the bulk level if the heat balance is not solved within the pellet. Thermal equilibrium is assumed in each pellet, and the source is averaged across the pellet:

$$Q_{\text{bed}} = \frac{\int Q_{\text{pe}} dV}{V_{\text{pe}}} (1 - \epsilon_b) \text{ (W/m}^3\text{)}$$

If there are multiple pellet sizes i in the bed the heat source computed by summing over all sizes:

$$Q_{\text{bed}} = \sum_i (Q_{\text{bed}, i} \cdot \theta_{v,i})$$

Here $\theta_{v,i}$ is the volume fraction of pellet i in the pellet mix.

PELLET: SHRINKING CORE

Species diffuse from packed bed through porous reacted shell layer onto the pellet core surface where they take part in heterogeneous reactions. With the solid species inside pellet core consumed in the reactions, the unreacted core is shrinking, and at the same time the reaction products diffuse through the reacted layer back into the packed bed.

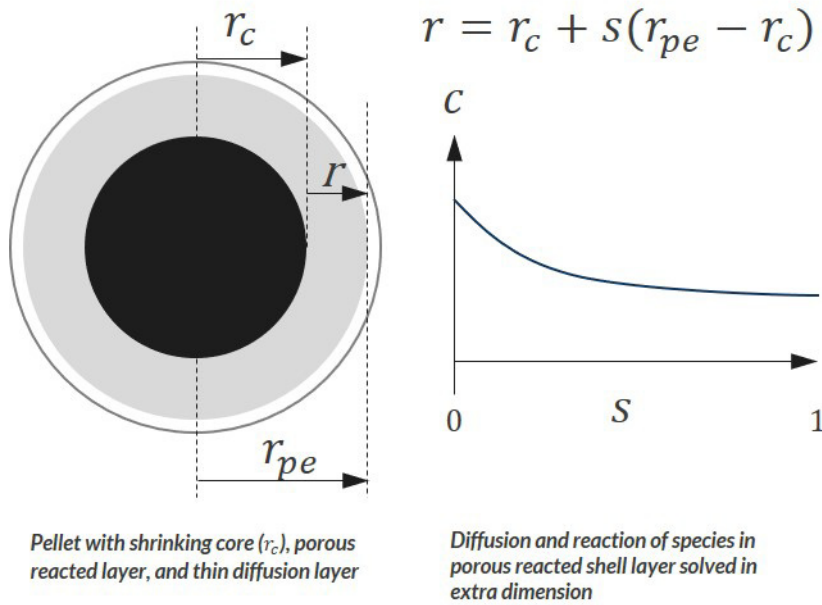


Figure 3-7: Schematic showing Shrinking Core Model.

The core shrinking rate depends on the heterogeneous reaction rate and the properties of pellet,

$$\frac{dr_c}{dt} = R_{\text{solid}} \frac{M_{\text{solid}}}{\rho_{\text{pe}}} \quad (3-67)$$

- r_c is the pellet core radius (m)
- R_{solid} is the surface reaction rate ($\text{mol}/(\text{m}^2 \cdot \text{s})$) for solid species
- M_{solid} is the solid species molar mass
- ρ_{pe} is the pellet core density

If there are two or more species, the term ($R_{\text{solid}} M_{\text{solid}}$) is mass weighted, assuming same consuming pace is held along the r direction for all solid species.

The transport and reaction equations inside the porous reacted shell (layer) are solved on a normalized dimensionless extra dimension S (0-1) which is based on the thickness (r) of the reacted layer.

The general form of the mass transport equation inside reacted layer is

$$\frac{\partial c}{\partial t} = \nabla \bullet D \nabla c + R$$

The shell mole balance across a spherical shell at radius r ($r_c \leq r \leq r_{pe}$),

$$\int_{r_c}^{r_{pe}} 4\pi r^2 \frac{\partial c}{\partial t} dr = \int_{r_c}^{r_{pe}} \frac{\partial}{\partial r} 4\pi r^2 D \frac{\partial c}{\partial r} dr + \int_{r_c}^{r_{pe}} 4\pi r^2 R dr$$

and weak form,

$$\int_{r_c}^{r_{pe}} 4\pi r^2 \left(-\frac{\partial c}{\partial t} \hat{c} - D \frac{\partial c}{\partial r} \frac{\partial \hat{c}}{\partial r} + R \hat{c} \right) = 0 \quad (3-68)$$

Defining extra dimension coordinate s as

$$s = \frac{r - r_c}{r_{pe} - r_c}$$

The integration can be written as

$$\int_{r_c}^{r_{pe}} (...) dr = \int_0^1 (...) \frac{dr}{ds} ds \quad (3-69)$$

For time derivative in moving coordinate system,

$$c_{\text{TIME}} = c_t + \frac{dc}{dx} x_{\text{TIME}} \quad (3-70)$$

On the extra dimension(s), the mass transport equation to be solved inside the reacted layer is (from Equation 3-68, 3-69, and 3-70).

$$4\pi N_{pe} r^2 \left(-\epsilon_{pr} \left(r_{\text{TIME}} - \frac{dc}{dr} r_{\text{TIME}} \right) \hat{c} - D \frac{\partial c}{\partial r} \frac{\partial \hat{c}}{\partial c} + R \hat{c} \right) \frac{dr}{ds} = 0$$

where ϵ_{pr} is the porosity inside the reacted pellet.

For the pellet shapes, diffusion process and heat sources, they are similar to that for porous catalyst pellet. The shape can be both spherical and nonspherical particles, and the characteristics and properties of the pellet are defined in the Pellet node under Porous Material. The heat sources are contributed from the bulk reactions and Shrinking Core reactions (surface reactions).

Shrinking Core Reactions

The pellet core consuming rate depends on the rate of Shrinking Core Reactions, Equation 3-67. The solid species in the pellet core take part in the heterogeneous reactions as reactants. The pellet core is shrinking with the proceedings of the surface reactions. If there are two or more solid species, the shrinking rate is the mass weighted consuming rate, and here it is supposed that all solid species shrinks at the same pace along the radial direction.

Boundary Conditions

Outer Surface

There are two types of boundary conditions on interface between the pellet outer surface and the fluid in the bed: Continuous concentration and Film resistance (mass flux). They are the same as that for catalyst pellet.

Core Surface

The core consuming reactions always exist on the core surface. Besides the solid species in the pellet core, there could be also bulk transport species that participate in the heterogeneous reactions. The surface reaction rate for the bulk species is a kind of the mass flux ($J_{pe,i}$) between the core surface and the porous reacted layer. The mass source for species $C_{pe,i}$ is

$$R_{pe,i} = S_{s, \text{area}} \left(J_{pe,i} + c_{pe,i} \varepsilon_{pr} \frac{dr}{dT} \right)$$

where

$$S_{s, \text{area}} = 4\pi r_c^2$$

References

1. R. Codina, "A discontinuity-capturing crosswind-dissipation for the finite element solution of the convection-diffusion equation," *Comput. Methods Appl. Mech. Eng.*, vol. 110, pp. 325–342, 1993.
2. P.V. Danckwerts, "Continuous flow systems: Distribution of residence times," *Chem. Eng. Sci.*, vol. 2, no. 1, 1953.
3. R.J. Millington and J.M. Quirk, "Permeability of Porous Solids," *Trans. Faraday Soc.*, vol. 57, pp. 1200–1207, 1961.

4. R.D. Burnett and E.O. Frind, “An Alternating Direction Galerkin Technique for Simulation of Groundwater Contaminant Transport in Three Dimensions: 2. Dimensionality Effects,” *Water Resour. Res.*, vol. 23, no. 4, pp. 695–705, 1987.
5. J. Bear, *Hydraulics of Groundwater*, McGraw Hill, 1979.
6. J. Bear, *Dynamics of Fluids in Porous Media*, Elsevier Scientific Publishing, 1972.
7. D.M. Mackay, D.L. Freyberg, P.V. Roberts, and J.A. Cherry, “A Natural Gradient Experiment on Solute Transport in a Sand Aquifer: 1. Approach and Overview of Plume Movement,” *Water Resour. Res.*, vol. 22, no. 13, pp. 2017–2030, 1986.
8. H.M.F. Freundlich, “Über die Adsorption in Lösungen” (in German), *Z. Phys. Chem.*, vol. 57, pp. 385–470, 1906
9. I. Langmuir, “The Constitution and Fundamental Properties of Solids and Liquids, Part I Solids,” *J. Am. Chem. Soc.*, vol. 38, pp. 2221–2295, 1916.
10. J. Toth, “State equation of the solid gas interface layer,” *Acta Chem. Acad. Hung.* vol. 69, pp. 311–317, 1971.
11. S. Brunauer, P.H. Emmett, E. Teller, “Adsorption of gases in multimolecular layers,” *J. Am. Chem. Soc.*, vol. 60, no. 2, pp. 309–319, 1938.
12. C.W. Fetter, *Contaminant Hydrogeology*, Prentice Hall, 1999.
13. J. Bear and A. Verruijt, *Modeling Groundwater Flow and Pollution*, D. Reidel Publishing, 1994.
14. I. Langmuir, “Chemical Reactions at Low Temperatures,” *J. Amer. Chem. Soc.*, vol. 37, 1915.
15. J.M. Coulson and J.F. Richardson, *Chemical Engineering*, vol. 2, 4th ed., Pergamon Press, Oxford, U.K., 1991.
16. J.M. Coulson and J.F. Richardson, *Chemical Engineering*, vol. 1, 4th ed., Pergamon Press, Oxford, U.K., 1991.
17. D.E Rosner, *Transport Processes in Chemically Reacting Flow Systems*, ISBN-13: 978-1483130262, Butterworth–Heinemann, 1986.
18. R.B. Bird, W.E. Stewart, and E.N. Lightfoot, *Transport Phenomena*, 2nd ed., John Wiley & Sons, Inc., 2007.

Theory for the Transport of Concentrated Species Interface

The Transport of Concentrated Species Interface theory is described in this section:

- [Multicomponent Mass Transport](#)
- [Multicomponent Gas Diffusion: Maxwell–Stefan Description](#)
- [Multicomponent Diffusivities](#)
- [Multicomponent Diffusion: Mixture-Averaged Approximation](#)
- [Multispecies Diffusion: Fick’s Law Approximation](#)
- [Multicomponent Thermal Diffusion](#)
- [The Packed Bed Feature](#)
- [Vapor Transport](#)
- [Regularization of Reaction Rate Expression](#)
- [References for the Transport of Concentrated Species Interface](#)

Multicomponent Mass Transport

Suppose a reacting flow consists of a mixture with $i = 1, \dots, Q$ species and $j = 1, \dots, N$ reactions. Equation 3-4 then describes the mass transport for an individual species:

$$\frac{\partial}{\partial t}(\rho\omega_i) + \nabla \cdot (\rho\omega_i\mathbf{u}) = -\nabla \cdot \mathbf{j}_i + R_i \quad (3-71)$$

where, ρ (SI unit: kg/m^3) denotes the mixture density and $\mathbf{u}$ (SI unit: m/s) the mass averaged velocity of the mixture. The remaining variables are specific for each of the species, i , being described by the mass transfer equation:

- ω_i is the mass fraction (1)
- $\mathbf{j}_i$ (SI unit: $\text{kg}/(\text{m}^2 \cdot \text{s})$) is the mass flux relative to the mass averaged velocity, and
- R_i (SI unit: $\text{kg}/(\text{m}^3 \cdot \text{s})$) is the rate expression describing its production or consumption.

The relative mass flux vector $\mathbf{j}_i$ can include contributions due to molecular diffusion, mass flux due to migration in an electric field, and thermal diffusion.

Summation of the transport equations over all present species gives Equation 3-72 for the conservation of mass

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (3-72)$$

assuming that

$$\sum_{i=1}^Q \omega_i = 1, \quad \sum_{i=1}^Q \mathbf{j}_i = 0, \quad \sum_{i=1}^Q R_i = 0$$

Using the mass conservation equation, the species transport for an individual species, i , is given by:

$$\rho \frac{\partial}{\partial t}(\omega_i) + \rho(\mathbf{u} \cdot \nabla)\omega_i = -\nabla \cdot \mathbf{j}_i + R_i \quad (3-73)$$

$Q - 1$ of the species equations are independent and possible to solve for using Equation 3-73. To compute the mass fraction of the remaining species, COMSOL Multiphysics uses the fact that the sum of the mass fractions is equal to 1:

$$\omega_1 = 1 - \sum_{i=2}^Q \omega_i \quad (3-74)$$

Multicomponent Gas Diffusion: Maxwell–Stefan Description

The Maxwell–Stefan formulation for a multicomponent mixture defines $N - 1$ (with N being the number of active species in the model) force balances according to

$$-\sum_{k \neq i} \frac{x_i x_k}{D_{ik}} \left(\frac{\mathbf{j}_i}{\omega_i \rho} - \frac{\mathbf{j}_k}{\omega_k \rho} \right) = \mathbf{d}_k - \sum_{k \neq i} \frac{x_i x_k}{D_{ik}} \left(\frac{D_k^T}{\omega_k \rho} - \frac{D_i^T}{\omega_i \rho} \right) \nabla \ln T \quad (3-75)$$

where

- D_{ik} (SI unit: m^2/s) are the binary Maxwell–Stefan diffusivities
- T (SI unit: K) is the temperature
- D_i^T (SI unit: $\text{kg}/\text{m}\cdot\text{s}$) are the thermal diffusion coefficients, and
- $\mathbf{d}_k$ (SI unit: $1/\text{m}$) is the diffusional driving force acting on species k .

By inverting the above set of equations, the mass flux relative to the mass average velocity, $\mathbf{j}_i$, can be defined by the generalized Fick equations (Ref. 1):

$$\mathbf{j}_i = -\rho\omega_i \sum_{k=1}^Q \tilde{D}_{ik} \mathbf{d}_k - D_i^T \nabla \ln T \quad (3-76)$$

where $\tilde{D}_{ik}$ (SI unit: m^2/s) are the multicomponent Fick diffusivities.

For ideal gas mixtures the diffusional driving force is (Ref. 1)

$$\mathbf{d}_k = \frac{1}{cR_gT} \left[\nabla p_k - \omega_k \nabla p - \rho_k \mathbf{g}_k + \omega_k \sum_{l=1}^Q \rho_l \mathbf{g}_l \right] \quad (3-77)$$

where

- c (SI unit: mol/m^3) is the total molar concentration
- R_g is the universal gas constant $8.314 \text{ J}/(\text{mol}\cdot\text{K})$
- p (SI unit: Pa) is the total pressure
- p_k (SI unit: Pa) is the partial pressure, and
- ρ_k (SI unit: kg/m^3) is the density of species k , and
- $\mathbf{g}_k$ (SI unit: m/s^2) is an external force (per unit mass) acting on species k . In the case of an ionic species the external force arises due to the electric field.



When an external force applies equally to all species (such as for gravity), the last two terms disappear.

As can be seen in Equation 3-76 and Equation 3-77, the total diffusive flux for the species depends on the gradients of all species concentrations, temperature, and pressure as well as any external force on the individual species.

Using the ideal gas law, $p = cR_gT$, and the definition of the partial pressures, $p_k = x_k p$, the equation can be written as

$$\mathbf{d}_k = \nabla x_k + \frac{1}{p} \left[(x_k - \omega_k) \nabla p - \rho \omega_k \mathbf{g}_k + \omega_k \sum_{l=1}^Q \rho \omega_l \mathbf{g}_l \right] \quad (3-78)$$

The mole fraction x_k is given by

$$x_k = \frac{\omega_k}{M} M \quad (3-79)$$

and the mean molar mass M (SI unit: kg/mol) by

$$\frac{1}{M} = \sum_{i=1}^Q \frac{\omega_i}{M_i}$$

When using the Maxwell–Stefan diffusion model, the transport equations for the species' mass are

$$\begin{aligned} \rho \frac{\partial}{\partial t}(\omega_i) + \rho(\mathbf{u} \cdot \nabla)\omega_i &= \nabla \cdot \left(\rho \omega_i \sum_{k=1}^Q \tilde{D}_{ik} \mathbf{d}_k + D_i^T \frac{\nabla T}{T} \right) + R_i \\ \mathbf{d}_k &= \nabla x_k + \frac{1}{P} \left[(x_k - \omega_k) \nabla P - \rho \omega_k \mathbf{g}_k + \omega_k \sum_{l=1}^Q \rho \omega_l \mathbf{g}_l \right] \end{aligned} \quad (3-80)$$

Multicomponent Diffusivities

The multicomponent Fick diffusivities, $\tilde{D}_{ik}$, are needed to solve [Equation 3-80](#). The diffusivities are symmetric

$$\tilde{D}_{ik} = \tilde{D}_{ki}$$

and are related to the multicomponent Maxwell–Stefan diffusivities, D_{ik} , through the following relation ([Ref. 2](#))

$$\frac{x_i x_k}{D_{ik}} = -\omega_i \omega_k \frac{\sum_{j \neq i} (\text{adj} B_i)_{jk}}{\sum_{j \neq i} \tilde{D}_{ij} (\text{adj} B_i)_{jk}}, \quad (B_i)_{kj} = \tilde{D}_{kj} - \tilde{D}_{ij}, \quad i \neq j \quad (3-81)$$

where $(\text{adj} B_i)_{jk}$ is the jk^{th} component of the adjoint of the matrix B_i .

For low-density gas mixtures, the multicomponent Maxwell–Stefan diffusivities, D_{ij} , can be replaced with the binary diffusivities for the species pairs that are present.

Solving for Equation 3-81 leads to a number of algebraic expressions for each of the components in the multicomponent Fick diffusivity matrix. For two- and three-component systems, these are implemented and solved directly by COMSOL Multiphysics. For instance, the component $\tilde{D}_{12}$ in a ternary system is given by:

$$\tilde{D}_{12} = \frac{-\frac{\omega_1(\omega_2 + \omega_3)}{x_1 D_{23}} - \frac{\omega_2(\omega_1 + \omega_3)}{x_2 D_{13}} + \frac{\omega_3^2}{x_3 D_{12}}}{\frac{x_1}{D_{12} D_{13}} + \frac{x_2}{D_{12} D_{23}} + \frac{x_3}{D_{23} D_{13}}}$$

For four components or more, COMSOL Multiphysics obtains the multicomponent Fick diffusivities numerically through matrix inversion derived from the matrix properties defined in Ref. 2. The program starts with the multicomponent Maxwell–Stefan diffusivity matrix D_{ik} to compute the multicomponent Fick diffusivity matrix, $\tilde{D}_{ik}$, using the following equation:

$$\tilde{D}_{ij} = N_{ij} - q \quad (3-82)$$

where ij , the indices in the matrices $\tilde{D}$ and N , range from 1 to the number of species, Q .

The elements of the matrix N in Equation 3-82 are defined as

$$N_{ij} = (P^{-1})_{ij} \quad (3-83)$$

where P^{-1} is the inverse of a matrix P defined as

$$P_{ij} = \frac{\omega_i \omega_j}{q} - \tilde{C}_{ij}$$

The matrix $\tilde{C}$ in turn is defined as

$$\tilde{C}_{ij} = \begin{cases} \frac{x_i x_j}{D_{ij}} & i \neq j \\ -\sum_{k \neq j} \tilde{C}_{ik} & i = j \end{cases}$$

The term q in Equation 3-82 is a scalar value that provides numerical stability and should be of the same order of magnitude as the multicomponent Maxwell–Stefan diffusion coefficients. The physics interface therefore defines q as the sum of the multicomponent Maxwell–Stefan diffusion coefficients:

$$q = \sum_{i=1}^{n-1} \left(\sum_{j=i+1}^n D_{ij} \right)$$



This definition for q works well in most cases. In rare cases, it might be necessary to change the value to obtain convergence.

Because the multicomponent Fick diffusivity matrix is symmetric, it is sufficient to apply Equation 3-83 to the upper triangle elements of $\bar{D}$. The remaining elements are obtained by swapping the indices in the matrix.

Multicomponent Diffusion: Mixture-Averaged Approximation

The mixture-averaged diffusion model assumes that the relative mass flux due to molecular diffusion is governed by a Fick's law type approximation.

Assuming that the diffusive flux, relative to the mass averaged velocity, is proportional to the mole fraction gradient, the mass flux is defined as

$$\mathbf{j}_{\text{md}, i} = -\rho_i D_i^m \frac{\nabla x_i}{x_i} \quad (3-84)$$

Here ρ_i is the density, and x_i the mole fraction of species i . Using the definition of the species density and mole fraction

$$\rho_i = \rho \omega_i, \quad x_i = \frac{\omega_i}{M_i} M$$

Equation 3-84 can be expressed in terms of the mass fractions (ω_i) in the manner of

$$\mathbf{j}_{\text{md}, i} = -\left(\rho D_i^m \nabla \omega_i + \rho \omega_i D_i^m \frac{\nabla M}{M} \right)$$

Using Equation 3-84 together with the Maxwell–Stefan equations, where isobaric and isothermal conditions have been assumed, the following expression for the mixture-averaged diffusion coefficients can be derived (Ref. 3):

$$D_i^m = \frac{1 - \omega_i}{\sum_{k \neq i}^N \frac{x_k}{D_{ik}}} \quad (3-85)$$

If instead the diffusive flux (relative to the mass averaged velocity) is assumed proportional to the mass fraction gradient, the mass flux is defined as

$$\mathbf{j}_{\text{md},i} = -\rho_i D_i^{m*} \frac{\nabla \omega_i}{\omega_i} = -\rho D_i^{m*} \nabla \omega_i \quad (3-86)$$

For this assumption, using Equation 3-86 together with the Maxwell–Stefan equations, also assuming isobaric and isothermal conditions, the following expression for the mixture-averaged diffusion coefficients can be derived (Ref. 3):

$$\frac{1}{D_i^{m*}} = \sum_{k \neq i}^N \frac{x_k}{D_{ik}} + \frac{x_i}{1 - \omega_i} \sum_{k \neq i}^N \frac{\omega_k}{D_{ik}} \quad (3-87)$$

THE MIXTURE-AVERAGED DIFFUSION COEFFICIENT

When using the mixture-averaged model, the diffusion is proportional to a single diffusion coefficient. The coefficient describes the diffusion of species i relative to the remaining mixture and is referred to as the mixture-averaged diffusion coefficient. The coefficient is explicitly given in terms of the multicomponent Maxwell–Stefan diffusivities D_{ik} . As a consequence, no matrix inversion operation is required as for the Maxwell–Stefan diffusion model (when using four or more species). For low-density gas mixtures, the D_{ik} components can be replaced by the binary diffusivities for the species pairs present.

MIXTURE DIFFUSION CORRECTION

It should be noted that models for the multicomponent diffusivities based on Fick’s law are approximations. One drawback of this is that the net diffusive mass flux, the sum of all mass fluxes relative to the mass averaged velocity, is not constrained to zero. To account for this we add a flux, defined by a correction velocity $\mathbf{u}_c$, to enforce a zero net diffusive flux in the manner of:

$$\sum_{i=1}^N \mathbf{j}_{\text{md},i} = \sum_{i=1}^N \rho \omega_i (\mathbf{u}_{\text{d},i} + \mathbf{u}_c) = 0 \quad (3-88)$$

Hence the correction velocity is:

$$\mathbf{u}_c = -\sum_{i=1}^N \omega_i \mathbf{u}_{\text{d},i} \quad (3-89)$$

Here $\mathbf{u}_{\text{d},i}$ is the diffusion velocity resulting from the flux assumption in Equation 3-84 or Equation 3-86. Note that the correction velocity is a constant correction (same for all species), but varies in space.

Using the correction velocity together with Equation 3-84 (flux proportional to the mole fraction gradient), the resulting diffusive flux is

$$\mathbf{j}_{\text{md},i} = -\rho D_i^m \nabla \omega_i - \rho \omega_i D_i^m \frac{\nabla M}{M} + \rho \omega_i \sum_{i=1}^N \frac{M_i}{M} D_i^m \nabla x_i \quad (3-90)$$

If instead Equation 3-86 is used (flux proportional to the mass fraction gradient), the resulting diffusive flux is

$$\mathbf{j}_{\text{md},i} = -\rho D_i^{m*} \nabla \omega_i + \rho \omega_i \sum_{i=1}^N D_i^{m*} \nabla \omega_i \quad (3-91)$$

ADDITIONAL TRANSPORT MECHANISMS

Apart from molecular diffusion, transport due to thermal diffusion and migration of charged species in an electric field can be accounted for by adding the following terms to the diffusive flux:

$$\mathbf{j}_i = \nabla \cdot \left(D_i^T \frac{\nabla T}{T} + \rho \omega_i z_i u_{m,i} F \nabla \phi \right)$$

where

- D_i^T (SI unit: kg/(m·s)) is the thermal diffusion coefficient
- z_i (dimensionless) is the charge number
- $u_{m,i}$ the mobility of the i^{th} species, and
- ϕ (SI unit: V) is the electric potential.

Multispecies Diffusion: Fick's Law Approximation

Using a *Fick's law approximation*, the relative mass flux due to molecular diffusion is governed by

$$\mathbf{j}_{\text{md},i} = -\rho_i D_{i,kl}^F \frac{\nabla x_i}{x_i} \quad (3-92)$$

when assuming that the diffusive flux is proportional to the mole fraction gradient. If instead assuming that it is proportional to the mass fraction it becomes

$$\mathbf{j}_{\text{md},i} = -\rho D_{i,kl}^F \nabla \omega_i \quad (3-93)$$

In the equations above $D_{i,kl}^F$ represents a general diffusion matrix (SI unit: m^2/s) describing the diffusion of species i into the mixture. This form makes it possible to use any diffusion coefficient, matrix, or empirical model based on Fick's law. For example, in situations when the mass transport is not dominated by diffusion, an alternative is to use the diffusion coefficients at infinite dilution,

$$D_{i,kl}^F = D_i^0$$

These coefficients are typically more readily available compared to the binary diffusion diffusivities, especially for liquid mixtures.

The mixture diffusion correction described above for the mixture-averaged diffusion can also be applied in this case. Correspondingly, the resulting diffusive flux is

$$\mathbf{j}_{\text{md},i} = -\rho D_{i,kl}^F \nabla \omega_i - \rho \omega_i D_{i,kl}^F \frac{\nabla M}{M} + \rho \omega_i \sum_{i=1}^N \frac{M_i}{M} D_{i,kl}^F \nabla x_i \quad (3-94)$$

when using [Equation 3-92](#) and

$$\mathbf{j}_{\text{md},i} = -\rho D_{i,kl}^F \nabla \omega_i + \rho \omega_i \sum_{i=1}^N D_{i,kl}^F \nabla \omega_i \quad (3-95)$$

when instead using [Equation 3-93](#).

When using the Fick's Law approximation, [Additional Transport Mechanisms](#) can be accounted for in the same manner as described above for the mixture-averaged approximation.

Multicomponent Thermal Diffusion

Mass diffusion in multicomponent mixtures due to temperature gradients is referred to as the *Soret effect*. This occurs in mixtures with high temperature gradients and large variations in molecular weight (or size) of the species. Typically species with high molecular weight accumulate in lower temperature regions while the diffusion due to the Soret effect transports species with low molecular weight to higher temperature regions. In COMSOL Multiphysics, thermal diffusion is included by prescribing the thermal diffusion coefficients D_i^T . In a multicomponent mixture, the sum of the thermal diffusion coefficients is zero:

$$\sum_{i=1}^Q D_i^T = 0$$

In porous media, the diffusion models may be extended to account for species collisions with the pore walls.

For the **Fick's law** or the **Mixture-averaged** diffusion models, the diffusion coefficient D_i^M is corrected with the **Wall diffusion** coefficient D_i^W in the following way

$$D_i^{MW} = \left(\frac{1}{D_i^M} + \frac{1}{D_i^W} \right)^{-1}$$

For the **Maxwell–Stefan** diffusion model (see Ref. 9), the following term is added to the species force balance (Equation 3-75)

$$\dots + \frac{x_i}{D_{i, \text{eff}}^W} \left(\frac{\mathbf{j}_i}{\omega_i \rho} + \mathbf{u}_W \right) = \mathbf{d}_i + \dots$$

The wall velocity $\mathbf{u}_W$, representing the velocity of the fluid in a representative region adjacent to the pore wall, is here defined as

$$\mathbf{u}_W = - \frac{\frac{\nabla p_A}{p_A} + \sum_k \frac{x_k}{D_{k, \text{eff}}^W} \frac{\mathbf{j}_k'}{\omega_k \rho}}{\sum_k \frac{x_k}{D_{k, \text{eff}}^W}} \quad (3-96)$$

where the modified fluxes $\mathbf{j}_k'$ have been calculated using the modified driving forces

$$\mathbf{d}_k' = \nabla x_k + \frac{1}{p} x_k \nabla p \quad (3-97)$$

It should be noted that $\mathbf{u}_W$ can become nonzero also in the absence of pressure gradients. For a proper description of the fluid momentum balance, $\mathbf{u}_W$ needs to be added either as a contributing velocity (Darcy's law) or as a corresponding volume force $\mathbf{F} = \mu \kappa^{-1} \mathbf{u}_W$ (Brinkman equations).

The Maxwell–Stefan model formulated in this way correctly captures the Klinkenberg effect, and has large similarities with other models in literature incorporating Knudsen diffusion such as the Dusty Gas Model and the (Modified) Binary Friction Model. In addition, for the limit of vanishing permeabilities ($\kappa \rightarrow 0$), the formulation reduces to the Lightfoot model for membrane transport.

The Packed Bed Feature

The Packed Bed feature, available to users of the Chemical Reaction Engineering Module, adds a domain feature that can simulate regions filled with reactive porous pellets, as in packed bed reactors. Below the theory for the Packed Bed feature in the Transport of Concentrated Species interfaces is presented.

The pellets in the bed are simulated with the Extra Dimension technology provided by the COMSOL software: A 1D, 2D, or 3D domain represents the bed volume. The extra dimension is the added radial microscale dimension inside each pellet.

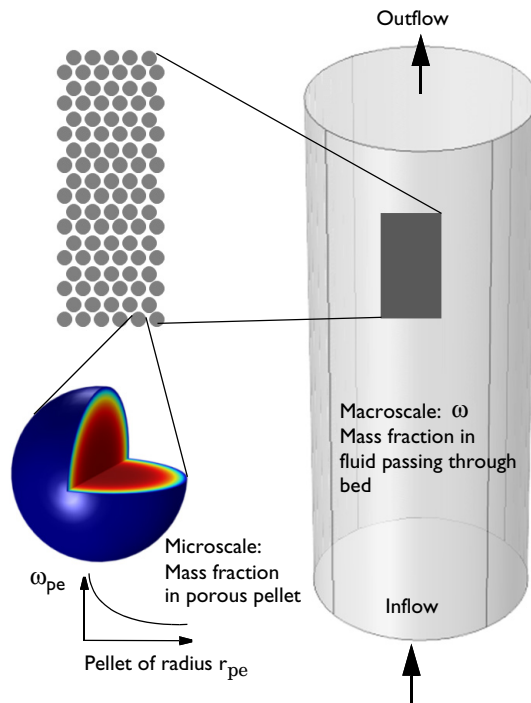


Figure 3-8: Schematic showing the macroscale (bed volume) and the microscale (pellet).

The equations of mass and momentum transfers inside the pellets are solved on an extra dimension attached to the 1D, 2D, or 3D physics interfaces, including axisymmetric cases.

The equations inside the spherical pellet are solved as a spherical transport equations on a nondimensional radial coordinate on the domain 0-1.

The model equations assume spherical particles (pellets) of a radius r_{pe} . Modeling assumptions for cylinders, flakes, and user-defined shapes can also be used. Consider the microscale mass fraction $\omega_{pe,i}$ inside an individual porous pellet or pellets, and the macro mass fraction ω_i in the packed bed volume.

The pellet radius input can be:

- One uniform pellet radius, which can be space dependent $f(x, y, z)$.
- Binary, ternary, and so on, mix up to 5 radii. The user inputs a table with the mix of sizes (1 mm, 2 mm, for example), and volume fraction of each. Different chemical reactions per pellet size can be specified.

The model equations in the macroscale (base geometry) for the species mass fraction i and velocity (Darcy's Law) are, for example:

$$\epsilon_b \rho \frac{\partial \omega_i}{\partial t} + \nabla \cdot \mathbf{J}_i + \rho \mathbf{u} \cdot \nabla \omega_i = R_i \quad (3-98)$$

$$\frac{\partial}{\partial t}(\epsilon_b \rho) + \nabla \cdot (\rho \mathbf{u}) = Q_m \quad (3-99)$$

- ϵ_b is the bed porosity.
- The dependent variable, ω_i , represents the interparticle mass fraction (for each chemical species i) in the fluid flowing between the pellets.
- $\mathbf{J}_i$ is the diffusive mass flux of species i .
- R_i is the species reaction rate ($\text{kg}/(\text{m}^3 \cdot \text{s})$) in the bed. This corresponds to reactions occurring in the pore space outside of the pellets. Note, that R_i is the reaction rate per unit volume of bed.
- Q_m is the mass source ($\text{kg}/(\text{m}^3 \cdot \text{s})$).

Looking inside a pellet: Assuming no concentration variations in the space-angle (θ, ϕ) direction, but only in the radial (r) direction of the spherical pellet allows a spherically symmetric reaction-diffusion transport equation inside the pellet. If r_{dim} (SI unit: m) is the spatial radial coordinate in the pellet, and r_{pe} is the pellet radius, the nondimensional coordinate $r = r_{dim}/r_{pe}$ can be defined. The modeling domain on r goes from 0 to 1.

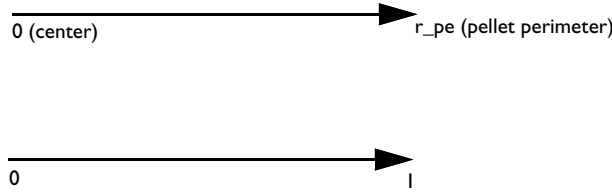


Figure 3-9: Modeling domain in a pellet for dimensional (top) and nondimensional (bottom) coordinates.

A shell mass balance across a spherical shell at radius r_{dim} (SI unit: m), and a subsequent variable substitution $r = r_{\text{dim}}/r_{\text{pe}}$ gives the following mass transport equation on the pellet domain $0 < r < 1$, Equation 3-50 becomes

$$4\pi N \left\{ r^2 r_{\text{pe}}^2 \epsilon_{\text{pe}} \frac{\partial}{\partial t} (\omega_{\text{pe},i}) + \nabla \cdot (r^2 \mathbf{J}_{\text{pe},i}) + r^2 r_{\text{pe}} (\rho \mathbf{u})_{\text{pe}} \cdot (\nabla \omega_{\text{pe},i}) \right\} = \quad (3-100)$$

$$4\pi N \left\{ r^2 r_{\text{pe}}^2 R_{\text{pe},i} \right\}$$

- N is the number of pellets per unit volume of bed, $N = \frac{(1 - \epsilon_b)}{V_{\text{pe}}}$, V_{pe} is the pellet volume.
- ϵ_{pe} is the pellet (microscale) porosity.
- $\omega_{\text{pe},i}$ is the intraparticle species mass fraction of fluid volume element inside the pore channel in the pellet.
- $\mathbf{J}_{\text{pe},i}$ is the mass flux for species i inside pellet. It is similar to the mass flux in Transport of Concentrated Species.
- $\mathbf{u}_{\text{pe}}$ is the velocity of mass bulk in the pellet.
- $R_{\text{pe},i}$ is the pellet reaction rate for species i . This corresponds to reactions taking place inside the pellets. Note that the user input of $R_{\text{pe},i}$ is per unit volume of pellet.

The velocity $\mathbf{u}$ is calculated from the pressure gradient:

$$\mathbf{u}_{\text{pe}} = -\frac{\kappa_{\text{pe}}}{\mu_{\text{pe}}} \nabla p_{\text{pe}}$$

- κ_{pe} is the permeability inside pellet.
- μ_{pe} is the dynamic viscosity inside pellet.

The equation for pressure p_{pe} (based on Equation 3-99) in the pellet domain, $0 < r < 1$, is

$$4\pi N \left\{ r_{\text{pe}}^2 \epsilon_{\text{pe}}^2 \frac{\partial p_{\text{pe}}}{\partial t} + r^2 \nabla \cdot (r^2 (\rho \mathbf{u})_{\text{pe}}) = r_{\text{pe}}^2 Q_{\text{m}} \right\} \quad (3-101)$$

The mass flux $\mathbf{J}_i$ in Equation 3-51, with the diffusion model being Mixture-averaged, is

$$J_{\text{pe},i} = - \left(\rho_{\text{pe}} D_i^m \nabla \omega_{\text{pe},i} + \rho_{\text{pe}} \omega_{\text{pe},i} D_i^m \frac{\nabla M_{\text{pe},n}}{M_{\text{pe},n}} + D_{\text{e},i}^T \frac{\nabla T_{\text{pe}}}{T_{\text{pe}}} \right) \quad (3-102)$$

where $D_{\text{e},i}^T$ is the effective thermal diffusion coefficient, and D_i^m is

$$D_i^m = \frac{1 - \omega_{\text{pe},i}}{\sum_{k \neq i} \frac{x_{\text{pe},k}}{D_{\text{e},ik}}}$$

If the diffusion model is Maxwell–Stefan, the equation for mass flux $\mathbf{J}_i$ is

$$J_{\text{pe},i} = - \left(\rho_{\text{pe}} \omega_{\text{pe},i} \sum_k \bar{D}_{\text{e},ik} \mathbf{d}_k + D_{\text{e},i}^T \frac{\nabla T_{\text{pe}}}{T_{\text{pe}}} \right) \quad (3-103)$$

where $\mathbf{d}_k$ is

$$\mathbf{d}_k = \nabla x_{\text{pe},k} + \frac{1}{p_{\text{pe}}} [(x_{\text{pe},k} - \omega_{\text{pe},k}) \nabla p_{\text{pe}}]$$

where $D_{\text{e},ik}$ is the effective binary diffusion coefficient. It depends on the porosity ϵ_{pe} , tortuosity τ , and binary diffusion coefficient D_{ik} :

$$D_{\text{e},ik} = \frac{\epsilon_{\text{pe}} D_{ik}}{\tau}$$

The available models for the porous media tortuosity are the one by Millington and Quirk (Ref. 4):

$$\tau = \varepsilon_{\text{pe}}^{-1/3} \Rightarrow D_{e,ik} = \varepsilon_{\text{pe}}^{4/3} D_{ik}; \quad (3-104)$$

the Bruggeman model:

$$\tau = \varepsilon_{\text{pe}}^{-1/2} \Rightarrow D_{e,ik} = \varepsilon_{\text{pe}}^{3/2} D_{ik}; \quad (3-105)$$

and the Tortuosity model, where the tortuosity expression is entered as user-defined input.

These models are readily used for both gaseous and liquid fluids along with various types of pellet shapes.

Equation 3-51 and Equation 3-101 can be solved for two types of boundary conditions at the interface between the pellet surface and the fluid in this feature.

- **Mass fraction constraint:** assuming that all resistance to mass transfer to/from the pellet is within the pellet and no resistance to pellet-fluid mass transfer is on the bulk fluid side. The mass fraction and pressure in the fluid will thus be equal to that in the pellet pore just at the pellet surface: $\omega_{\text{pe},i} = \omega_i$ and $p_{\text{pe}} = p_A$. These constraints also automatically ensure flux continuity between the internal pellet domain and the free fluid domain through so-called reaction forces in the finite element formulation.
- **Film resistance (mass flux):** The flux of mass across the pellet-fluid interface into the pellet is possibly rate determined on the bulk fluid side by film resistance. The resistance is expressed in terms of a film mass transfer coefficient, h_{Di} , such that:

$$N_{i,\text{inward}} = h_{\text{Di}}(c_i - c_{\text{pe},i})M_i, \quad (3-106)$$

where $N_{i,\text{inward}}$ is the mass flux from the free fluid into a pellet and has the unit $\text{kg}/(\text{m}^2 \cdot \text{s})$, M_i is the molar mass.

With the film resistance formulation above, the equation for mass transfer (Equation 3-51) needs to be amended for flux continuity so that

$$4\pi N \left\{ r_{\text{pe}}^2 \varepsilon_{\text{pe}} \frac{\partial}{\partial t} (\omega_{\text{pe},i}) + \nabla \cdot (r_{\text{pe},i}^2 J_{\text{pe},i}) + r_{\text{pe}}^2 (\rho \mathbf{u})_{\text{pe}} \cdot (\nabla \omega_{\text{pe},i}) \right\} = \quad (3-107)$$

$$4\pi N \left\{ r_{\text{pe}}^2 (R_{\text{pe},i} - N_{i,\text{inward}} S_b) \right\}$$

where S_b (SI unit: m^2/m^3) is the specific surface area exposed to the free fluid of the packed bed (not including the inside of the pores).

For the case of randomly packed spherical pellets, the specific surface area exposed to the free fluid is (Ref. 5):

$$S_b = \frac{3}{r_{\text{pc}}} (1 - \epsilon_b) \quad (3-108)$$

The total mass source Q_{mass} is

$$Q_{\text{mass}} = \sum_i N_{i, \text{inward}} S_b$$

The equation for pressure (Equation 3-101) becomes

$$4\pi N \left\{ r_{\text{pc}}^2 \epsilon_{\text{pc}}^2 \frac{\partial p_{\text{pc}}}{\partial t} + r_{\text{pc}}^2 \nabla \cdot (r_{\text{pc}}^2 (\rho \mathbf{u})_{\text{pc}}) = r_{\text{pc}}^2 \epsilon_{\text{pc}}^2 (Q_m - Q_{\text{mass}}) \right\} \quad (3-109)$$

The mass transfer coefficient in Equation 3-56 can be computed from the fluid properties and flow characteristics within the porous media. For this, the Sherwood, Sh , number defined as the ratio between the convective mass transfer coefficient and the diffusive mass transfer coefficient is often used:

$$Sh = \frac{hL}{D}$$

where L is a characteristic length (for spheres, typically the radius), and D is the diffusion coefficient in the fluid. From the Sherwood number definition, the mass transfer coefficient can be computed.

Three commonly used empirical expressions for the calculation of the Sherwood number are the Frössling relation (Ref. 6):

$$Sh = 2 + 0.552 Re^{1/2} Sc^{1/3}, \quad (3-110)$$

which was measured on pellets in the size region 1 mm, the Rosner relation (Ref. 7)

$$Sh = Sc^{0.4} (0.4 Re^{1/2} + 0.2 Re^{2/3}), \quad (3-111)$$

and the Garner and Keey relation (Ref. 6)

$$Sh = 0.94 Re^{1/2} Sc^{1/3}, \quad (3-112)$$

which was measured for Re numbers greater than 250.

All three depend on the Reynolds, Re, and Schmidt, Sc, numbers. The first describing the fluid flow regime (laminar versus turbulent) and the second, the ratio between the viscous diffusion rate and the molecular (mass) diffusion rate. In the expressions, properties such as velocity, u , dynamic viscosity, μ , and density, ρ , of the fluid are included.

$$\text{Re} = \frac{\rho u L}{\mu}$$

$$\text{Sc} = \frac{\mu}{\rho D}$$

NONSPHERICAL PARTICLES

For nonspherical pellets (of any shape), the relations above can be applied approximately by reinterpreting the pellet radius r_{pe} as

$$r_{\text{pe}} = 3 \frac{V_{\text{pe}}}{A_{\text{pe}}} \quad (3-113)$$

(Ref. 8), where A_{pe} is the external surface of a single pellet of any shape. Since the specific surface S_{pe} (SI unit: m^2/m^3) of one pellet is defined as

$$S_{\text{pe}} = \frac{A_{\text{pe}}}{V_{\text{pe}}}$$

it follows by insertion of Equation 3-64 that

$$S_{\text{pe}} = \frac{3}{r_{\text{pe}}}.$$

For a packed bed of which the packing has a porosity ϵ_b , the specific surface of the bed will be

$$S_b = S_{\text{pe}}(1 - \epsilon_b) = \frac{A_{\text{pe}}}{V_{\text{pe}}}(1 - \epsilon_b)$$

or

$$S_b = \frac{3}{r_{\text{pe}}}(1 - \epsilon_b) \quad (3-114)$$

for any pellet shape. Now r_{pe} and S_b can be calculated for any shape and inserted in equations [Equation 3-51](#), [Equation 3-101](#), [Equation 3-57](#) and [Equation 3-109](#). Some common specific shapes have automatic support:

Cylinders

For cylindrical shapes, applying [Equation 3-64](#) gives

$$r_{pe} = 3 \frac{V_{pe}}{A_{pe}} = 3 \frac{\pi r_{cyl}^2 L_{cyl}}{2\pi r_{cyl} L_{cyl} + 2\pi r_{cyl}^2} = \frac{1}{\frac{2}{3r_{cyl}} + \frac{2}{3L_{cyl}}} . \quad (3-115)$$

It is common practice to assume that the top and bottom surface of cylindrical pellets have negligible effect on the mass transfer to and from the internals of the pellet, or, $r_{cyl} \ll L_{cyl}$. [Equation 3-66](#) then simplifies to

$$r_{pe} = \frac{3}{2} r_{cyl}$$

and [Equation 3-65](#) to

$$S_b = \frac{2}{r_{cyl}} (1 - \epsilon_b)$$

Flakes

The derivation for a disc-shaped catalyst pellet is exactly the same as for cylindrical pellets, except that the assumption is reversed about the end surfaces and the envelope surface: $r_{flake} \gg w_{flake}$, where w_{flake} is the thickness of the disc. This gives

$$r_{pe} = \frac{3}{2} w_{flake}$$

and

$$S_b = \frac{2}{w_{flake}} (1 - \epsilon_b) .$$

HEAT SOURCE

The heat source of endothermic or exothermic reactions inside the pellet needs to be accounted for in the heat transfer on the bulk level if the heat balance is not solved within the pellet. In this case, thermal equilibrium is assumed in each pellet. To account for the heat evolution in the bulk, the source is averaged across the pellet:

$$Q_{\text{bed}} = \frac{\int Q_{\text{pc}} dV}{V_{\text{pc}}} (1 - \varepsilon_b) \text{ (W/m}^3\text{)}$$

If there are multiple pellet sizes i in the bed the heat source computed by summing over all sizes:

$$Q_{\text{bed}} = \sum_i (Q_{\text{bed},i} \cdot \theta_{v,i})$$

Here $\theta_{v,i}$ is the volume fraction of pellet i in the pellet mix.

Vapor Transport

Three boundary features are available for vapor transport; the [Vapor Inflow](#), the [Vapor-Liquid Interface](#), and the [Vapor-Liquid-Solution Interface](#) feature. All three features prescribe the equilibrium vapor pressure for one or several vapor species, either at an inlet connected to an upstream system involving liquids, or at a vapor-liquid interface. Evaporation or condensation occurs when the partial pressure in the adjacent gas domain differs from the vapor pressure at the interface. Below the conditions prescribed at the interface and the resulting heat of evaporation is defined.

VAPOR PRESSURE

Water

The saturated water pressure $p_{\text{sat,w}}$ (Pa) at a liquid water surface is defined as a function of temperature T using the correlation:

$$p_{\text{sat,w}}(T) = 610.7 \cdot 10^{(7.5(T - 273.15))/(T - 35.85)}$$

The resulting vapor molar concentration (mol/m³) at the interface is

$$c_{\text{eq,w}} = \frac{p_{\text{eq,w}}(T)}{RT}$$

and the corresponding mass fraction

$$\omega_{\text{eq,w}} = \frac{c_{\text{eq,w}} M_w}{\rho}$$

where R is the universal gas constant, M_w is the molar mass of water, and p the vapor density.

Thermochemistry Coupling

This option can be used when coupled to a Thermodynamic System through a Chemistry interface. In this case the Thermodynamics node can provide not only the thermodynamic functions for physicochemical properties in vapor phase, but also the fugacity and density for the liquid phase. The thermodynamic functions are affected by the thermodynamics models selected in the system. Most prominently, the fugacity is dependent on the liquid phase model in the Thermodynamics System.

For a multicomponent vapor–liquid interface at equilibrium, the fugacity (Pa) of all components should be the same in both the vapor and the liquid phase

$$f_i^L(T, p, \omega_{l,i}) = f_i^V(T, p, \omega_{v,i})$$

Here f_i^L is the fugacity in the liquid phase, and f_i^V the fugacity in the vapor phase for component i . The fugacity in each phase depends on the temperature and pressure, as well as on its mass fraction in the liquid ($\omega_{l,i}$) and vapor phase ($\omega_{v,i}$), respectively. Using the Thermochemistry coupling, functions for the fugacity $f_i = f_i^L = f_i^V$ at the interface are created automatically.

Using the fugacity instead of the pressure, the vapor molar concentration and the vapor mass fraction at the vapor–liquid interface are defined as

$$c_{\text{eq},i} = \frac{f_i}{RT}$$

and

$$\omega_{0,i} = \frac{f_i M_i}{p}$$

HEAT OF VAPORIZATION

Water

The heat flux due to vaporization or condensation of water is

$$Q_{\text{vap},w} = -(\Delta H_{\text{vap},w})(\mathbf{j}_w + \rho \mathbf{u}_s \omega_w) \cdot \mathbf{n}$$

where $\Delta H_{\text{vap},w}$ denotes the heat of vaporization of water, $\mathbf{j}_w$ is the water mass flux at the interface, $\mathbf{u}_s$ is the Stefan velocity, and $\mathbf{n}$ a normal unit vector.

Other Liquids

The heat flux due to vaporization or condensation of a multicomponent solution is

$$Q_{\text{vap}} = \sum (\Delta H_{\text{vap},i}) j_{m,i} = -\sum (\Delta H_{\text{vap},i}) ((\mathbf{j}_i + \rho \mathbf{u}_s \omega_i) \cdot \mathbf{n})$$

where $\Delta H_{\text{vap},i}$ is heat of vaporization of species i . The evaporative heat flux is defined and announced by the vapor–liquid interface feature for use when modeling heat transfer. It can for example be applied using a Boundary Heat Source feature in Heat Transfer in Fluids Interface.

Regularization of Reaction Rate Expression

The flexibility of COMSOL means that a reaction can result in nearly any reaction rate contribution. But a reactant, with mass fraction ω_i , can physically only be consumed if $\omega_i > 0$. Correspondingly a product can only be formed when $\omega_i < 1$. The Reaction feature in Transport of Concentrated Species Interface contains a way to regularize the rate expression and comply with these restrictions.

When selected, the original reaction rate contribution, R_i^c , is replaced by

$$R_i = \frac{1}{2} \frac{R_i^c - |R_i^c|}{\max(\omega_i, \omega_i^{\text{dl}})} \max(\omega_i, 0) + \frac{1}{2} \frac{R_i^c + |R_i^c|}{\max(1 - \omega_i, \omega_i^{\text{dl}})} \max(1 - \omega_i, 0) \quad (3-116)$$

If the mass fraction for a reactant species ω_i becomes smaller than its **Damping limit**, ω_i^{dl} , the rate expression is reduced linearly. If $\omega_i \leq 0$ for a reactant species, the reaction rate contribution to that species is completely removed. Similarly, if the mass fraction for a product species ω_j becomes larger than $1 - \omega_j^{\text{dl}}$, the rate expression added is damped linearly. If $\omega_j \geq 1$ for a product species, the reaction rate contribution to that species is completely removed.

The default value for the **Damping limit**, ω_i^{dl} , is 10^{-6} , which is appropriate for most applications, but can require adjustment when working with for example catalytic trace species.

Regularization of the rate expressions adds considerable stability to the reaction expressions, but it should ideally only be used as a means to reach convergence. When reaction zones are not adequately resolved on the mesh, regularization can affect the mass balance.

References for the Transport of Concentrated Species Interface

1. C.F. Curtiss and R.B. Bird, “Multicomponent Diffusion,” *Ind. Chem. Res.*, vol. 38 pp. 2515–2522, 1999.
2. R.B. Bird, W.E. Stewart, and E.N. Lightfoot, *Transport Phenomena*, 2nd ed., John Wiley & Sons, 2005.
3. R.J. Kee, M.E. Coltrin, and P. Glarborg, *Chemically Reacting Flow*, John Wiley & Sons, 2003.
4. R.J. Millington and J.M. Quirk, “Permeability of Porous Solids,” *Trans. Faraday Soc.*, vol. 57, pp. 1200–1207, 1961.
5. J.M. Coulson and J.F. Richardson, *Chemical Engineering*, vol. 2, 4th ed., Pergamon Press, Oxford, U.K., 1991.
6. J.M. Coulson and J.F. Richardson, *Chemical Engineering*, vol. 1, 4th ed., Pergamon Press, Oxford, U.K., 1991.
7. D.E Rosner, *Transport Processes in Chemically Reacting Flow Systems*, ISBN-13: 978-1483130262, Butterworth-Heinemann, 1986.
8. R.B. Bird, W.E. Stewart, and E.N. Lightfoot, *Transport Phenomena*, 2nd ed., John Wiley & Sons, Inc., 2007.
9. H. Ekström, “Combining wall interactions, fluid momentum balances and the Maxwell-Stefan equations for gas transport in porous media: An alternative approach,” *Int. J. Thermofluids*, vol. 21, 100534, 2024.

Theory for the Electrophoretic Transport Interface

The [Electrophoretic Transport Interface](#) models transport of charged and uncharged species, and in addition sets up a charge balance equation for the electrolyte potential.

The species concentrations are denoted, c_i (SI unit: mol/m^3), and the potential, ϕ_l (SI unit: V).

The species are transported by diffusion, migration, and (optionally) convection according to the Nernst–Planck set of equations. The total flux of species i is denoted $\mathbf{N}_i$ (SI unit: $\text{mol}/(\text{m}^2 \cdot \text{s})$) according to

$$\mathbf{N}_i = -D_i \nabla c_i - z_i u_{m,i} F c_i \nabla \phi_l + \mathbf{u} c_i = \mathbf{J}_i + \mathbf{u} c_i \quad (3-117)$$

where D_i (SI unit: m^2/s) is the diffusion coefficient, z_i (1) the corresponding charge, $u_{m,i}$ (SI unit: $\text{s} \cdot \text{mol}/\text{kg}$) the mobility and $\mathbf{u}$ (SI unit: m/s) the velocity vector. $\mathbf{J}_i$ denotes the molar flux relative to the convective transport (SI unit: $\text{mol}/(\text{m}^2 \cdot \text{s})$). For a detailed description of the theory of these equations and the different boundary conditions, see [The Transport of Diluted Species Interface](#).

CHARGE BALANCE EQUATION

The current vector, $\mathbf{i}_l$ (A/m^2), is defined as

$$\mathbf{i}_l = \sum_i z_i \mathbf{N}_i \quad (3-118)$$

where the summation is made over all species in the electrolyte.

The governing equation for the electrolyte potential is

$$\nabla \cdot \mathbf{i}_l = Q_l \quad (3-119)$$

where Q_l (SI unit: A/m^3) is the electrolyte current source stemming from, for example, porous electrode reactions. For nonporous electrode domains this source term is usually zero.

PROTON AND HYDROXIDE CONCENTRATIONS AND ELECTRONEUTRALITY

Assuming the total number of species to be $N + 2$, the assumption of electroneutrality is

$$\sum_{i=1}^{N+2} z_i c_i = 0 \quad (3-120)$$

In water-based systems the species H^+ and OH^- are always present. The auto ionization reaction for water is



This reaction is fast, following the equilibrium relation

$$c_{H^+} c_{OH^-} = K_w \times 1 \text{ mol}^2 \text{dm}^{-6} \quad (3-122)$$

where $K_w \approx 10^{-14}$.

Now, the electroneutrality condition, including the two additional species H^+ and OH^- , reads

$$c_{H^+} - c_{OH^-} + \sum_{i=1}^N z_i c_i = 0 \quad (3-123)$$

Combining these two equations results in the following algebraic expressions for the concentrations of H^+ and OH^- .

$$c_{H^+} = -\frac{\Sigma}{2} + \sqrt{\frac{\Sigma^2}{4} + K_w \times 1 \text{ mol}^2 \text{dm}^{-6}} \quad (3-124)$$

and

$$c_{OH^-} = \frac{\Sigma}{2} + \sqrt{\frac{\Sigma^2}{4} + K_w \times 1 \text{ mol}^2 \text{dm}^{-6}} \quad (3-125)$$

where

$$\Sigma = \sum_{i=1}^N z_i c_i \quad (3-126)$$

MASS BALANCE EQUATIONS FOR DISSOCIATION SPECIES

Assume a set of species S_i describing k dissociation steps from

$$S_k^{z_0+k} \leftrightarrow S_{k-1}^{z_0+k-1} + H^+ \quad K_{a,1} = \frac{[H^+][S_{k-1}^{z_0+k-1}]}{[S_k^{z_0+k}]} \quad (3-127)$$

to

$$S_k^{z_0+1} \leftrightarrow S_0^{z_0} + H^+ \quad K_{a,k} = \frac{[H^+][S_0^{z_0}]}{[S_k^{z_0+1}]} \quad (3-128)$$

where z_0 is the charge (valence) of species S_0 (which has no dissociable protons) and $K_{a,j}$ is the acid (equilibrium) constant of the j th dissociation reaction. The brackets “[]” here represent the species activity. The charge of each species is always deductible from the index i according to z_{0+i} and will be dropped from now on.

If the proton activity is known, any species S_m may be expressed using any other species S_l according to

$$[S_m] = \frac{[H]^{m-l}[S_l]}{\prod_{j=k-m+1}^{k-l} K_{a,j}} \quad (3-129)$$

if $m > l$ and

$$[S_m] = [H]^{m-l}[S_l] \prod_{j=k-l+1}^{k-m} K_{a,j} \quad (3-130)$$

if $l > m$.

Setting $m = i$ and denoting the flux of species i by $\mathbf{N}_i$ using equation [Equation 3-117](#), the mass balance equation for the concentration c_i of each subspecies i in the dissociation chain is

$$\frac{\delta c_i}{\delta t} + \nabla \cdot \mathbf{N}_i = R_{\text{eq},i,k-i} - R_{\text{eq},i,k-i+1} + R_i \quad (3-131)$$

where $R_{\text{eq},i,j}$ is the reaction source stemming from the j th dissociation step (with $R_{\text{eq},i,k+1} = 0$), and R_i any additional reaction sources.

The reaction source contributions from the dissociation steps are generally not known, but may be canceled by taking the sum of all mass balance equations, resulting in

$$\sum_{i=0}^k \left(\frac{\delta c_i}{\delta t} + \nabla \cdot \mathbf{N}_i \right) = \sum_{i=0}^k R_i \quad (3-132)$$

AVERAGE IONIZATION FORMULATION FOR LARGE MOLECULES

Large protein molecules are typically subject to a large number of dissociation steps.

As an alternative way to a long chain of dissociation steps for describing protein transport, you can instead formulate the protein transport based on the average number of protons $\bar{v}$ (1) removed from the molecule. $\bar{v}$ is typically provided as a function of pH, based on experimental data. The average charge of the species then is $\bar{z} = z_0 - \bar{v}$, so that the flux of the species is written as:

$$\mathbf{N}_i = -D \nabla c_i - \bar{z} u_m F c_i \nabla \phi_l + \mathbf{u} c_i \quad (3-133)$$

and similarly the addition to the charge neutrality condition is $\bar{z} c_i$.

When considering the contribution to the current and the charge balance equation you need to take into account that the squared average charge, $\bar{z}^2 = (z_0 - \bar{v})^2$, is not equal to the “average squared charge”, $\overline{z^2} = z_0^2 + 2z_0 \bar{v} + \bar{v}^2 = \bar{z}^2 - \bar{v}^2 + \bar{v}^2$ (Ref. 1).

The addition to the current density vector is

$$\mathbf{i}_l = \dots - F(\bar{z} D_i \nabla c_i + \overline{z^2} u_m c_i \nabla \phi_l) \quad (3-134)$$

The average number of protons removed from the proton typically depends on the pH. If the average number of removed protons depend only on the pH, the averaged squared number of protons removed can be written as

$$\overline{v^2} = -c_{H^+} \frac{d}{dc_{H^+}} \bar{v} + \bar{v}^2 \quad (3-135)$$

And from this you can derive the average squared charge according to

$$\overline{z^2} = -c_{H^+} \frac{d}{dc_{H^+}} \bar{v} + \bar{z}^2 = -c_{H^+} \frac{d}{dc_{H^+}} \bar{z} + \bar{z}^2 \quad (3-136)$$

DIFFUSIVITY–MOBILITY RELATIONS

The Stokes radius r of a molecule is related to the diffusivity according to

$$r = \frac{kT}{6\pi\mu D} \quad (3-137)$$

where μ (SI unit: Pa·s) is the dynamic viscosity and k the Boltzmann constant.

For small molecules, there is the frequently used Nernst–Einstein relation between the diffusivity and the mobility

$$u_m = \frac{D}{RT} \quad (3-138)$$

For larger molecules, such as proteins, the mobility may instead be calculated based on the Debye–Hückel–Henry expression (Ref. 2) according to

$$u_m = \frac{ef(\kappa r)}{6\pi\mu F(1 + \kappa r)} = \frac{Df(\kappa r)}{RT(1 + \kappa r)} \quad (3-139)$$

where κ (1/m) is the Debye parameter, which depends on the ionic strength of the solution, is defined for ideal solutions as

$$\kappa^2 = \frac{2e^2 N_A}{\epsilon \epsilon_0 kT} \sum_{i=1}^N z_i^2 c_i \quad (3-140)$$

where ϵ is the dielectric constant of the fluid and ϵ_0 the permittivity of free space. (z_i^2 should be used if available in the formula above when calculating the ionic strength).

The function f above is based on a sigmoidal function so that it ranges from 1 for $\kappa r = 0$ to 1.5 for $\kappa r = \infty$. Note that the Debye–Hückel–Henry expression approaches the Nernst–Einstein mobility as $r \rightarrow 0$.

For larger molecules (macro ions), where the distance between the charges is large compared to $1/\kappa$, the Linderström–Lang approximation postulates a smaller contribution of to the ionic strength so that the z -valent ion behaves as a monovalent ion with a z -fold concentration. For an assemble of $N - M$ smaller molecules and M macro ions, the Debye parameter then is defined as

$$\kappa^2 = \frac{2e^2 N_A}{\epsilon \epsilon_0 kT} \left(\sum_{i=1}^{N-M} z_i^2 c_i + \sum_{i=N-M+1}^N \text{abs}(z_i) c_i \right) \quad (3-141)$$

REFERENCES

1. R.A. Moshier, D.A. Saville, and W. Thormann, *The Dynamics of Electrophoresis*, VCH Verlagsgesellschaft mbH, Weinheim, Germany, 1992.
2. R.A. Moshier, P. Gebauer, J. Caslavská, and W. Thormann, “Computer Simulation and Experimental Validation of the Electrophoretic Behavior of Proteins. 2. Model Improvement and Application to Isotachophoresis,” *Anal. Chem.*, vol. 64, pp. 2991–2997, 1992.

Theory for the Surface Reactions Interface

The [Surface Reactions Interface](#) solves for an arbitrary number of surface concentrations, $c_{s,i}$ (SI unit: mol/m²), referring to species adsorbed on a reactive surface, and an arbitrary number of bulk concentrations $c_{b,k}$ (SI unit: mol/m³), referring to the species in the solid material that constitutes the bulk of the reactive surface.

In this section:

- [Governing Equations for the Surface Concentrations](#)
- [Governing Equations for the Bulk Concentrations](#)
- [ODE Formulations for Surface Concentrations](#)
- [Surface Reaction Equations on Deforming Geometries](#)
- [Reference for the Surface Reactions Interface](#)

Governing Equations for the Surface Concentrations

Transport of adsorbed species occurs in the tangential direction along the surface. The Surface Reactions interface models the tangential flux in the surface dimension, the *surface molar flux*, $\mathbf{N}_{t,i}$ (SI unit: mol/(m·s)) (the subscript t refers to the tangential direction), as governed by diffusion according to Fick's law:

$$\mathbf{N}_{t,i} = -D_{s,i} \nabla_t c_{s,i}$$

where $D_{s,i}$ (SI unit: m²/s) is the surface diffusion coefficient for species i .

The governing equation for the surface concentrations is written as

$$\frac{\partial c_{s,i}}{\partial t} = -\nabla_t \cdot \mathbf{N}_{t,i} + R_{s,i} \quad (3-142)$$

where $R_{s,i}$ (SI unit: mol/(m²·s)) is the sum of all sources due to surface reactions and adsorption/desorption phenomena.

Of frequent interest for surface reaction kinetics are the fractional surface coverages, θ_i (dimensionless), of the species (with index i).

Writing the density of sites of the surface as Γ_s (SI unit: mol/m²) and the site occupancy number for each species as σ_i (dimensionless), the surface coverages can be calculated from

$$\theta_i = \frac{\sigma_i c_{s,i}}{\Gamma_s}$$

(The site occupancy number accounts for the situation when a large species covers more than one site on the surface.)

For the case of monolayer adsorption, the sum of all fractional coverages of free and adsorbed sites is unity, and hence the fraction of free sites on the surface, θ_* , can be calculated from

$$\theta_* = 1 - \sum_i \theta_i$$

Governing Equations for the Bulk Concentrations

BULK SPECIES RATES AND GROWTH VELOCITY

The reaction rate for a bulk species of index k , being added to the bulk due to surface reactions is defined as $R_{b,k}$ (SI unit: mol/(s·m²)).

The reaction rate in mass basis, $r_{b,k}$ (SI unit: kg/(s·m²)) for species k , is given by

$$r_k = M_k R_{b,k}$$

with M_k (SI unit: kg/mol) being the molar mass of the species.

Based on this, the species contribution to the bulk growth velocity, v_k (SI unit: m/s), is given by

$$v_{b,k} = \frac{r_{b,k}}{\rho_k}$$

where ρ_k is the species density (SI unit: kg/m³).

TOTAL RATES AND VELOCITY

To get the corresponding total values, a summation for all bulk species is made according to

$$R_{b, \text{tot}} = \sum_k R_{b, k}$$

$$r_{b, \text{tot}} = \sum_k r_{b, k}$$

$$v_{b, \text{tot}} = \sum_k v_{b, k}$$

FRACTIONAL RATES AND VELOCITY

The fractional deposition rates are calculated by

$$R_{b, \text{frac}, k} = \frac{R_{b, k}}{R_{b, \text{tot}}}$$

$$r_{b, \text{frac}, k} = \frac{r_{b, k}}{r_{b, \text{tot}}}$$

$$v_{b, \text{frac}, k} = \frac{v_{b, k}}{v_{b, \text{tot}}}$$



The fractional deposition rates can be used as measures of the local fractional composition in a deposited layer at a certain thickness or time.

ACCUMULATED BULK QUANTITIES

By solving for the bulk species concentrations for each time step, the accumulated mass and thickness changes of the bulk material can be calculated.

The bulk concentration, $c_{b, k}$ (SI unit: mol/m^2), for species k is governed by the equation

$$\frac{\partial c_{b, k}}{\partial t} = R_{b, k} \quad (3-143)$$

The bulk concentration in mass basis, $m_{b, k}$ (SI unit: kg/m^2) for a species k , can be derived from

$$m_{b, k} = M_k c_{b, k}$$

leading to the bulk thickness $s_{b,k}$ (SI unit: m) of species k according to

$$s_{b,k} = \frac{m_{b,k}}{\rho_k}$$

TOTAL BULK QUANTITIES

The total bulk concentration is calculated according to

$$c_{b,\text{tot}} = \sum_k c_{b,k}$$

The deposited total mass, $m_{b,\text{tot}}$ (SI unit: kg/m²) is

$$m_{b,\text{tot}} = \sum_k m_{b,k}$$

and the total bulk thickness, s_{tot} (SI unit: m) is

$$s_{b,\text{tot}} = \sum_k s_k$$

FRACTIONAL QUANTITIES

The fractional bulk concentration, bulk mass, and thickness (all dimensionless) are calculated according to

$$c_{b,\text{frac},k} = \frac{c_{b,k}}{c_{b,\text{tot}}}$$

$$m_{b,\text{frac},k} = \frac{m_{b,k}}{m_{b,\text{tot}}}$$

$$s_{b,\text{frac},k} = \frac{s_{b,k}}{s_{b,\text{tot}}}$$

ODE Formulations for Surface Concentrations

The surface of a 1D geometry has a 0 dimension, hence no surface transport can be modeled and the governing equation is reduced to an ODE:

$$\frac{\partial c_{s,i}}{\partial t} = R_{s,i}$$

Surface Reaction Equations on Deforming Geometries

If a Surface Reaction interface is used in conjunction with a Deformed Geometry or a Moving Mesh interface, the boundary concentration can either be assumed to be transported with the deforming geometry (moving mesh), with no compensation for the stretching (the Compensate for boundary stretching checkbox is not selected), or the species can be assumed to “float” on the mesh (the Compensate for boundary stretching checkbox is selected, which is the default). In the latter case the following is assumed in regard to the coupling between the surface species and bulk species and the mesh movement:

- Expansion or contraction of the boundary dilutes or increases concentration of the species, respectively, so that the surface integral (in spatial coordinates) of the species is kept constant.
- Tangential mesh movement has no impact on the local concentration in spatial coordinates, that is, the tangential transport of surface and bulk species does not move with the mesh in the tangential direction.

In order to comply with the additional contributions to the mass balance, equations are added. First, the following terms are added to the right-hand side of [Equation 3-142](#) and [Equation 3-143](#), respectively:

$$-c_{s,i} \frac{\partial}{\partial t} \ln \partial A$$

$$-c_{b,k} \frac{\partial}{\partial t} \ln \partial A$$

where ∂A is the infinitesimal mesh area segment (area scale factor). The above terms account for the concentration change due to a fractional area change.

Second, the resulting unwanted tangential convection, imposed by the mesh movement, is compensated for by the adding following terms to the right-hand side of [Equation 3-142](#) and [Equation 3-143](#), respectively:

$$\nabla_t \cdot (c_{s,i} \mathbf{v}_{t, \text{mesh}})$$

$$\nabla_t \cdot (c_{b,k} \mathbf{v}_{t, \text{mesh}})$$

where $\mathbf{v}_{t,\text{mesh}}$ is the mesh velocity in the tangential direction.

This convectional term needs often to be stabilized using methods such as streamline diffusion or isotropic diffusion.

Reference for the Surface Reactions Interface

1. R.J. Kee, M.E. Coltrin, and P. Glarborg, *Chemically Reacting Flow*, Chapter 11, John Wiley & Sons, 2003.

Theory for the Nernst–Planck Equations Interface

The Nernst–Planck Equations Interface theory is described in this section:

- [Governing Equations for the Nernst–Planck Formulation](#)
- [Material Balance Formulation](#)

Governing Equations for the Nernst–Planck Formulation

The following version of the Nernst–Planck equation treats transport by convection, diffusion, and migration of each dissolved species:

$$\frac{\partial c_i}{\partial t} + \nabla \cdot (c_i \mathbf{u}) = -\nabla \cdot (-D_i \nabla c_i - u_{m,i} F c_i \nabla V) + R_i = -\nabla \cdot \mathbf{J}_i + R_i \quad (3-144)$$

where $\mathbf{J}_i$ is the molar flux relative to the convective transport, and R_i (SI unit: mol/(m³·s)) is the reaction term. The velocity, $\mathbf{u}$, is equal to the velocity of the solvent. This implies that the solute’s contribution to the solvent’s velocity, through shear or any other forces, is negligible in comparison to the solvent’s contribution to the solute. [Equation 3-144](#) introduces one variable for the concentration of each of the dissolved species and the electric potential, V .

Apart from the transport equations, the physics interface assumes that the *electroneutrality condition* holds:

$$\sum z_i c_i = 0 \quad (3-145)$$

that is that the net charge in every control volume is zero. This means that $Q - 1$, where Q is the number of species present, can be solved for using [Equation 3-144](#). The remaining species concentration is computed from the electroneutrality condition. This means that boundary conditions for this species cannot be specified, although it takes part in the boundary condition descriptions for the current density. Often, the species chosen to be computed from electroneutrality is the oppositely charged ion, to the electroactive species, from a supporting electrolyte.

To compute the electric potential, taking the transport of every charged species into account, create the following linear combination of the n mass balance equations, where each mass balance equation is weighted with the factor Fz_i :

$$F \sum_{i=1}^n z_i \frac{\partial c_i}{\partial t} + \nabla \cdot \left(F \sum_{i=1}^n z_i \mathbf{N}_i \right) = F \sum_{i=1}^n z_i R_i \quad (3-146)$$

The first term in Equation 3-146 is zero, which can be shown by taking the time derivative of the electroneutrality condition. The expression under the divergence operator is the total current density vector, defined by:

$$\mathbf{i} = F \sum_{i=1}^n z_i (-D_i \nabla c_i - z_i u_{m,i} F c_i \nabla V) \quad (3-147)$$

Notice that no convective term is included in the expression for the current density, which is also a result of the electroneutrality condition.

It is now possible to rewrite Equation 3-146 as:

$$\nabla \cdot \mathbf{i} = F \sum_{i=1}^n z_i R_i \quad (3-148)$$

This equation states the conservation of electric charge and is the one solved for to compute the electric potential.

Equation 3-144, Equation 3-145, and Equation 3-148 are sufficient for describing the potential and concentration distribution in an electrochemical cell or in an electrolyte subjected to an electric field.

A useful observation from Equation 3-147 is that the ionic conductivity, defined in absence of concentration gradients, is implicitly given by:

$$\kappa = F^2 \sum_{i=1}^n z_i^2 c_i u_{m,i}$$

Furthermore, the potential gradient caused by the presence of a concentration gradient under situations with zero current becomes:

$$\nabla V = \frac{\sum_{i=1}^n (-D_i \nabla c_i)}{\sum_{i=1}^n z_i u_{m,i} F c_i}$$

In the field of electrochemistry, this is known as the junction potential.

Material Balance Formulation

When modeling mass transport in the Nernst–Planck Equations interface, two formulations are available for the material balance: a conservative and a nonconservative formulation. [Equation 3-144](#) shows the conservative formulation.

In the nonconservative formulation the velocity is moved out of the divergence operator, which improves the stability of systems coupled to a momentum balance (fluid-flow equation). This is only applicable to systems that contain incompressible fluids. In these fluids, the following continuity equation is satisfied:

$$\nabla \cdot \mathbf{u} = 0 \quad (3-149)$$

and [Equation 3-144](#) can be rewritten as:

$$\frac{\partial c_i}{\partial t} + \nabla \cdot (-D_i \nabla c_i - z_i u_{m,i} F c_i \nabla V) + \mathbf{u} \cdot \nabla c_i = R_i \quad (3-150)$$

which can be called the nonconservative formulation and is the default formulation in the Nernst–Planck Equations interface.

Specify the equation formulation in the Advanced Settings section of the Nernst–Planck Equations interface.

Theory for the Size-Based Population Balance Interface

The [Size-Based Population Balance Interface](#) interface provides a framework for modeling the evolution of a size-based population balance equation in time and space. The interface can be used to describe a population of solid particles, droplets, gas bubbles, and more. In this section, the term “particle” will be used to encompass all of these.

In this section:

- [Population Balance Equation](#)
- [Discretization of the Size Coordinate](#)
- [Particle Nucleation](#)
- [Particle Growth](#)
- [Particle Dissolution](#)
- [Particle Aggregation](#)
- [Particle Breakage](#)
- [Mass Transport](#)
- [References for the Size-Based Population Balance Interface](#)

Population Balance Equation

The population balance equation describes the balance of particles in a population. A discretized formulation is implemented across a number of different size intervals i according to

$$\begin{aligned} \frac{\partial n_i}{\partial t} + \nabla \cdot \mathbf{J}_i + \mathbf{u} \cdot \nabla n_i = & - \frac{\partial(G_i n_i)}{\partial L_i} + \frac{1}{\Delta L_i} B \delta(L_i - L_0) \\ & + B_{\text{agg}, i} - D_{\text{agg}, i} + B_{\text{break}, i} - D_{\text{break}, i} + S_i \end{aligned} \quad (3-151)$$

- n_i is the number density in a specific size interval (SI unit: $1/\text{m}^4$)
- G_i is the growth rate (SI unit: m/s)
- L_i is the end-point size of interval i (SI unit: m)

- L_0 is the smallest size solved for (SI unit: m)
- B is the nucleation rate (SI unit: $1/\text{m}^3/\text{s}$)
- $B_{\text{agg},i}$ is the aggregation birth term (SI unit: $1/\text{m}^4/\text{s}$)
- $D_{\text{agg},i}$ is the aggregation death term (SI unit: $1/\text{m}^4/\text{s}$)
- $B_{\text{break},i}$ is the breakage birth term (SI unit: $1/\text{m}^4/\text{s}$)
- $D_{\text{break},i}$ is the breakage death term (SI unit: $1/\text{m}^4/\text{s}$)
- S_i is an additional source term (SI unit: $1/\text{m}^4/\text{s}$)
- $\mathbf{J}_i$ is the diffusive mass flux vector (SI unit: $1/\text{m}^3/\text{s}$)
- $\mathbf{u}$ is the velocity vector (SI unit: m/s)

In the above equation, δ is the Dirac delta function. The diffusive flux, $\mathbf{J}_i$, is written as

$$\mathbf{J}_i = -D_i \nabla n_i \quad (3-152)$$

where D_i (SI unit: m^2/s) is the diffusion coefficient for particles in size interval i .

Discretization of the Size Coordinate

To solve the population balance equation, the size parameter is discretized into smaller size intervals. The interface provides three different options for discretization of the internal coordinate, linear discretization, geometric discretization, and logarithmic discretization. Linear discretization divides the size coordinate into evenly sized intervals. The size at the end of interval i is calculated according to

$$L_i = L_0 + i\Delta L \quad (3-153)$$

and

$$\Delta L_i = L_i - L_{i-1} \quad (3-154)$$

where I is the total number of intervals and L_I and L_0 are the largest and smallest size solved for respectively.

Geometric discretization instead results in intervals with an increasing interval length as the size coordinate increases. This takes the form

$$L_i = L_{\min} + 2^{(i-I)/q} (L_I - L_{\min}) \quad (3-155)$$

and where an additional point $L_{\min}$, only used for discretization is defined as (Ref. 1)

$$L_{\min} = \frac{L_0 - 2^{(-I)/q} L_I}{1 - 2^{(-I)/q}} \quad (3-156)$$

The last option, logarithmic discretization gives a discretization on the form

$$L_i = \exp\left(\ln(L_0) + \frac{i}{I} \ln\left(\frac{L_I}{L_0}\right)\right) = L_0 \left(\frac{L_I}{L_0}\right)^{i/I} \quad (3-157)$$

Further on, we will not only see the size at the right-hand boundary of the interval, L_i , being used for calculating growth and dissolution rates but also the size at the center of the interval, denoted $L_{\text{mid},i}$, being used for calculating aggregation and breakage contributions. $L_{\text{mid},i}$ is taken as the size at the exact mid point of the interval as

$$L_{\text{mid},i} = \frac{L_i + L_{i-1}}{2} \quad (3-158)$$

Particle Nucleation

Particle nucleation describes the birth of new particles. The rate at which new particles are introduced into the system is written as

$$\frac{\partial n_i}{\partial t} = \frac{1}{\Delta L_i} B \delta(L_i - L_0) \quad (3-159)$$

In this equation, B is the nucleation rate (SI unit: $1/\text{m}^3/\text{s}$). The birth of new particles through nucleation is assumed to only occur at the smallest particle size, L_0 . This is shown with the Dirac delta function, δ . Predefined options to express the nucleation rate B are according to an Arrhenius expression

$$B = A \exp\left(-\frac{E_a}{k_B T}\right) \quad (3-160)$$

or a homogeneous nucleation expression ([Ref. 2](#))

$$B = \frac{3}{2} D_S (c N_A)^{7/3} \sqrt{\frac{\gamma_S}{k_B T}} V_m \exp\left(\frac{-16\pi}{3} \frac{\gamma_S^3}{T^3 k_B^3 \ln\left(\frac{c}{c^*}\right)^2} \frac{V_m^2}{c^*}\right) \quad (3-161)$$

where

- D_S is the solute species diffusion coefficient (SI unit: m^2/s)

- N_A is the Avogadro number (SI unit: 1/mol)
- γ_S is the surface energy (SI unit: J/m²)
- k_B is the Boltzmann constant (SI unit: J/K)
- V_m is the molecular volume (SI unit: m³)
- T is the temperature (SI unit: K)

Particle Growth

The particle growth source that is added to the number density of each bin is used to describe the flux of particles across the boundaries to adjacent size intervals resulting from a continuous change in size. With finite differences the growth source term can be approximated as

$$\frac{\partial(G_i n_i)}{\partial L} = \frac{f_i - f_{i-1}}{\Delta L_i} \quad (3-162)$$

where f_i (SI unit: 1/(m³·s)) is the growth flux at the right-side boundary of interval i and G_i is the growth rate (SI unit: m/s). The most basic way of expressing the boundary fluxes is with a first-order upwind approach ([Ref. 3](#) and [Ref. 4](#))

$$f_i = G_i n_i \quad (3-163)$$

As we will see later, this way of expressing the fluxes is needed for the first and last size intervals. For the inner intervals we turn to higher order approaches, namely Weighted Essentially Non-Oscillatory (WENO) schemes. If we assume that the mapping between cell average and boundary values to be linear, we can construct a number of polynomials, q , to approximate the number density at the boundaries. The WENO scheme relies on combining a number of these polynomials to get a convex combination of lower order approximations on the form

$$W_1 q_1 + W_2 q_2 + \dots + W_N q_N \quad (3-164)$$

where $W_1, W_2, \dots, W_N$ denotes weights that fulfill

$$W_1 + W_2 + \dots + W_N = 1 \quad (3-165)$$

Using this, we can write the boundary fluxes as ([Ref. 5](#))

$$f_i = G_i \sum_{m=0}^{r-1} W_{m,i}^r q_{m,i}^r \quad (3-166)$$

where the order of the polynomial is given by $2r-1$. A polynomial, q_i , describing the number density at the right-side boundary of interval i can be written as

$$q_{m,i}^r = \sum_{j=0}^{r-1} C_{m,j,i}^r n_{i-m+j} \quad (3-167)$$

The weights in Equation 3-166 are written as (Ref. 5)

$$W_{m,i}^r = \frac{\frac{c_{m,i}^r}{(S_{m,i}^r + \Delta L_i)^2}}{\sum_{k=0}^{r-1} \frac{c_{m,i}^r}{(S_{m,i}^r + \Delta L_i)^2}} \quad (3-168)$$

where S is a measure of smoothness

$$S_{m,i}^r = \sum_{k=1}^{r-1} \left(\Delta L_i \int_{L_{i-1}}^{L_i} \left(\frac{\partial^k n}{\partial L^k} \right)^2 dL \right) \quad (3-169)$$

The constants $C_{m,j,i}^r$ can be calculated using (Ref. 6)

$$C_{m,j,i}^r = \Delta L_{i-m+j} \sum_{n=j+1}^r \frac{\sum_{l=0}^r \prod_{q=0}^r (L_i - L_{i-m+q-1})}{r \prod_{l=0}^{n-1} (L_{i-m+n-1} - L_{i-m+l-1})} \quad (3-170)$$

while the constants $c_{m,i}^r$ can be derived by ensuring that they fulfill Equation 3-171 and Equation 3-172 (Ref. 5).

$$q_{m,i}^{2r-1} = \sum_{m=0}^{r-1} c_{m,i}^r q_{m,i}^r \quad (3-171)$$

$$\sum_{m=0}^{r-1} c_{m,i}^r = 1 \quad (3-172)$$

Equation 3-173 shows the summarized interval boundary fluxes.

$$\text{if } i = 0 \quad : \quad f_0 = 0 \quad (3-173)$$

$$\text{if } i = 1 \quad : \quad f_1 = G_1 n_1$$

$$\text{if } i = 2 \quad : \quad f_2 = G_2 \sum_{m=0}^1 W_{m,2}^2 q_{m,2}^2$$

$$\text{if } i = N-1 \quad : \quad f_{N-1} = G_{N-1} \sum_{m=0}^1 W_{m,N-1}^2 q_{m,N-1}^2$$

$$\text{if } i = N \quad : \quad f_N = G_N n_N$$

$$\text{else} \quad : \quad f_i = G_i \sum_{m=0}^2 W_{m,i}^3 q_{m,i}^3$$

For the growth rate, G_i , it is possible to use a custom expression or a predefined transport controlled expression available as

$$G = \frac{k_a}{3k_v} \frac{\text{Sh} D_s M_p}{\rho_p L} (c - c^*) \quad (3-174)$$

where k_a and k_v are the area and volume shape factors, Sh is the Sherwood number, D_s (SI unit: m^2/s) is the solute species diffusion coefficient, M_p (SI unit: kg/mol) is the particle molar mass, ρ_p (SI unit: kg/m^3) is the particle density, c (SI unit: mol/m^3) is the species concentration in the solution and c^* (SI unit: mol/m^3) is the equilibrium concentration of the species. Equation 3-174 is based on the mass transport equation where the driving force is the concentration difference between the bulk solution, c and the concentration at the particle surface, assumed to be c^* .

Particle Dissolution

The continuous shrinking of particles through for example dissolution is modeled using the same equations as growth but with a reversed flux direction (see [Particle Growth](#)). This means that [Equation 3-167](#) becomes

$$q_{m,i}^r = \sum_{j=0}^{r-1} C_{m,j,i}^r n_{i+m-j} \quad (3-175)$$

where [Equation 3-175](#) instead describes the number density at the left-side boundary of interval i and D_i is the dissolution rate at the same boundary.

Particle Aggregation

When the concentration of particles is high there is the possibility for them to collide and combine into larger particles. This is what we call aggregation, agglomeration or coagulation. The process can be described by the Smoluchowski coagulation equation

$$\frac{\partial n_v(v)}{\partial t} = \frac{1}{2} \int_0^v K(v-\Lambda, \Lambda) n_v(v-\Lambda) n_v(\Lambda) d\Lambda - \int_0^\infty K(v, \Lambda) n_v(v) n_v(\Lambda) d\Lambda \quad (3-176)$$

where v and Λ are particle volumes and K is the aggregation kernel. The equation only considers binary aggregation. The first right-hand side term describes the source (or birth) of new particles with a volume v that result from the aggregation of smaller particles with volumes of Λ and $v-\Lambda$. The second term describes the sink (or death) of particles with a volume v as they together with any other particle of a volume between 0 and infinity form a particle with a volume larger than v .

To get an equation that is compatible with the size-based formulation, we rewrite [Equation 3-176](#) as ([Ref. 1](#) and [Ref. 7](#))

$$\begin{aligned} \frac{\partial n(L)}{\partial t} = & \frac{L^2}{2} \int_0^L \frac{1}{(L^3 - \lambda^3)^{2/3}} K(\sqrt[3]{L^3 - \lambda^3}, \lambda) n(\sqrt[3]{L^3 - \lambda^3}) n(\lambda) d\lambda \\ & - n(L) \int_0^\infty K(L, \lambda) n(\lambda) d\lambda \end{aligned} \quad (3-177)$$

The volume of the two aggregating particles, Λ and $v-\Lambda$, have been replaced by the particle sizes λ and $(L^3 - \lambda^3)^{1/3}$. The discretized form of [Equation 3-177](#) becomes

$$\begin{aligned} \frac{\partial n_i}{\partial t} = & \frac{L_i^{2^{i-1}}}{2} \sum_j \frac{1}{(L_{\text{mid},i}^3 - L_{\text{mid},j}^3)^{2/3}} K(\sqrt[3]{L_{\text{mid},i}^3 - L_{\text{mid},j}^3}, L_j) \\ & \times n(\sqrt[3]{L_{\text{mid},i}^3 - L_{\text{mid},j}^3}) n_j \Delta L_j - n_i \sum_{j=1}^I K(L_{\text{mid},i}, L_{\text{mid},j}) n_j \Delta L_j \end{aligned} \quad (3-178)$$

To ensure volume conservation on a discretized grid, the equation is rewritten to include weights as (Ref. 7 and Ref. 8)

$$\begin{aligned} \frac{\partial n_i}{\partial t} = & \sum_{(j,k) \in S_i} \left(1 - \frac{1}{2} \delta_{j,k} \right) K_{j,k} n_j n_k \frac{\Delta L_j \Delta L_k}{\Delta L_i} \frac{L_{\text{mid},j}^3 + L_{\text{mid},k}^3}{2L_{\text{mid},i}^3 - L_{\text{mid},j}^3 - L_{\text{mid},k}^3} \\ & - \sum_{j=1}^{\infty} K_{i,j} n_i n_j \Delta L_j \frac{L_{\lambda_{i,j}}^3}{2L_{\lambda_{i,j}}^3 - L_{\text{mid},i}^3 - L_{\text{mid},j}^3} \end{aligned} \quad (3-179)$$

where $K_{j,k}$ is the aggregation kernel for two colliding particles belonging to intervals j and k and δ_{ij} is the Kronecker delta defined as

$$\delta_{ij} = \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j \end{cases} \quad (3-180)$$

Furthermore, the set S_i contains all index combinations of j and k such that two particles of sizes $L_{\text{mid},j}$ and $L_{\text{mid},k}$ can form a larger particle with a size that lies within size interval i .

$$S_i = \left\{ (j, k) \in N^2 : L_{i-1} \leq \sqrt[3]{L_{\text{mid},j}^3 + L_{\text{mid},k}^3} < L_i \right\} \quad (3-181)$$

The index $\lambda_{i,j}$, in Equation 3-179, represents the interval that ensures that the two colliding particles from intervals i and j are included in the index pair set $S_{\lambda_{i,j}}$.

Note that with this size-based formulation of the coagulation equation, it is implicitly assumed that all three particles, the two colliding particles and the resulting particle, have the same shape. Or at least that the three sizes are selected such that when we calculate the volume of the particles using the same volume shape factor, the volume is conserved.

Breakage of particles is described using the fragmentation equation. The continuous fragmentation equation on a volume basis is written as (Ref. 7)

$$\frac{\partial n_v(v)}{\partial t} = \int_v^\infty S(\Lambda) b(v|\Lambda) n_v(\Lambda) d\Lambda - S(v) n_v(v) \quad (3-182)$$

The first right-hand side term describes the source (or birth) of new particles as larger particles with a volume from v to infinity break into two or more smaller particles including one with a volume of v . The second term, the sink (or death), describes the loss of particles with the volume v due to breakage. Rewritten on a size basis Equation 3-182 becomes

$$\frac{\partial n(L)}{\partial t} = \int_L^\infty S(\lambda) b(L|\lambda) n(\lambda) d\lambda - S(L) n(L) \quad (3-183)$$

Here, S is a selection function and b is a probability density function. Note that these functions are a function of volume in Equation 3-182 but a function of size in Equation 3-183. The discretized formulation is written as (Ref. 7)

$$\frac{\partial n_i}{\partial t} = \sum_{j=1}^{\infty} S(L_{\text{mid},j}) n_j \Upsilon_j^B \frac{\Delta L_j}{\Delta L_i} \int_{L_{i-1}}^{p_i} b(x|L_{\text{mid},j}) dx - S(L_{\text{mid},i}) n_i \Upsilon_i^D \quad (3-184)$$

where the birth weight term is written as

$$\Upsilon_i^B = \begin{cases} \frac{L_i^3 \left(\int_0^{L_{\text{mid},i}} b(x|L_{\text{mid},i}) dx - 1 \right)}{\sum_{j=1}^i \left[(L_{\text{mid},i}^3 - L_j^3) \int_{L_{j-1}}^{L_j} b(x|L_{\text{mid},i}) dx \right]}, & \text{if } i \neq 1 \\ L_i^3 \left(\int_0^{L_{\text{mid},i}} b(x|L_{\text{mid},i}) dx - 1 \right), & \text{if } i = 1 \end{cases} \quad (3-185)$$

death weight term is written as

$$\Upsilon_i^D = \frac{\Upsilon_i^B}{L_{\text{mid},i}^3} \sum_{j=1}^i \left[L_{\text{mid},j}^3 \int_{L_{j-1}}^{p_j} b(x|L_{\text{mid},i}) dx \right] \quad (3-186)$$

and where

$$p_j^i = \begin{cases} L_i, & \text{if } i \neq j \\ L_{\text{mid}, i}, & \text{if } i = j \end{cases} \quad (3-187)$$

The selection and probability density function are implemented as (Ref. 7)

$$S(L) = S_0 L^3 \quad (3-188)$$

and

$$b(L|\lambda) = 3L^2 \frac{\gamma}{\lambda^3} \left(\frac{L^3}{\lambda^3} \right)^{\gamma-2} \quad (3-189)$$

where S_0 is a constant controlling the breakage rate and γ is the average number of resulting particles from the breakage of a larger particle.

Mass Transport

The mass transport terms of the population balance equation describe the convective and diffusive mass transport. The transport equation can be written as

$$\frac{\partial n_i}{\partial t} + \nabla \cdot \mathbf{J}_i + \mathbf{u} \cdot \nabla n_i = S_{T,i} \quad (3-190)$$

In this equation $\mathbf{J}_i$ is the diffusive flux, $\mathbf{u}$ is the solvent velocity field, and $S_{T,i}$ is the source term containing all sources of particles in interval i . The above equation is the transport equation on nonconservative form. This is often more numerically robust and chosen by default. This formulation disregards density changes in the fluid. To switch between the two formulations, click the **Show** button () and select **Advanced Physics Options**.

References for the Size-Based Population Balance Interface

1. M. Ölander, *Numerical Simulations for Battery Recycling*, master's thesis, KTH, Royal Institute of Technology, 2023.
2. H. Y. Tang, S. Rigopoulos, and G. Papadakis, "On the interaction of turbulence with nucleation and growth in reaction crystallisation," *J. Fluid Mech*, vol. 944, p. A48, 2022.

3. S. Qamar, M.P. Elsner, I.A. Angelov, G. Warnecke, and A. Seidel-Morgenstern, “A comparative study of high resolution schemes for solving population balances in crystallization,” *Comput. Chem. Eng.*, vol. 30, no. 6–7, pp. 1119–1131, 2006.
4. B. Koren, “A robust upwind discretization method for advection, diffusion and source terms,” *Notes Numer. Fluid Mech.*, vol. 45, pp. 117–138, 1993.
5. W. Zhang, T. Przybycien, J. Schmölder, S. Leweke, and E. von Lieres, “Solving crystallization/precipitation population balance models in CADET, part 1: Nucleation growth and growth rate dispersion in batch and continuous modes on nonuniform grids,” *Comput. Chem. Eng.*, vol. 183, 2024.
6. H. Dong, C. Lu, and H. Yang, “The finite volume WENO with Lax-Wendroff scheme for nonlinear system of Euler equations,” *Mathematics*, vol. 10, pp. 211, 2018.
7. W. Zhang, T. Przybycien, J. M. Breuer, and E. von Lieres, “Solving crystallization/precipitation population balance models in CADET, Part II: Size-based Smoluchowski coagulation and fragmentation equations in batch and continuous modes,” *Comput. Chem. Eng.*, vol. 192, 2025.
8. G. Kaur, J. Kumar, S. Heinrich, “A weighted finite volume scheme for multivariate aggregation population balance equations,” *Comput. Chem. Eng.*, vol. 101, pp. 1–10, 2017.

Theory for the Vapor–Liquid Equilibrium Interfaces

The following sections describe theory applicable for the Vapor–Liquid Equilibrium Interfaces:

- [Vapor–Liquid Equilibrium](#)
- [Evaporation and Condensation](#)
- [Reference for the Vapor–Liquid Equilibrium Interfaces](#)

Vapor–Liquid Equilibrium

For a vapor and liquid system, thermodynamic equilibrium is reached when the fugacities are the equal in both phases (for each species i)

$$f_i^V(T, P, y_i) = f_i^L(T, P, x_i). \quad (3-191)$$

Here, f_i^V denotes the vapor phase fugacity, and f_i^L is the liquid phase fugacity. The fugacity is a measure of the chemical potential and relates to the tendency of a substance to prefer one of the phases over the other. In general, the fugacity in each phase depends on the temperature, T , the absolute pressure, P , as well as composition. The composition is usually expressed by the molar fractions in each phase, y_i and x_i , respectively. When accounting for nonideal behavior, the fugacity equality in [Equation 3-191](#) can be formulated as (see [Ref. 1](#))

$$\hat{\phi}_i y_i P = \gamma_i x_i \phi_i^{\text{sat}} P_i^{\text{sat}} \exp \left[\int_{P_i^{\text{sat}}}^P \frac{V_i^L}{RT} dP \right]. \quad (3-192)$$

Here the fugacity coefficient $\hat{\phi}_i$ accounts for nonideality in the gas, while the activity coefficient γ_i , accounts for nonideality in the liquid phase. Both these coefficients are functions of the temperature, pressure, and fraction of species i in the respective solutions. For the liquid, ϕ_i^{sat} and P_i^{sat} denote the fugacity coefficient and the vapor pressure, both for a pure species i . The superscript ‘sat’ indicates saturation conditions. The exponential term on the right hand side is the Poynting correction factor, including the molar liquid volume $V_i^L = M_i / \rho^L$. Here, M_i is the molar mass of species i , and ρ^L is the liquid density. The correction factor describes the effect of pressure on

the liquid fugacity. For low pressures, the Poynting correction factor is close to one, and two simplified relations can be formulated.

- In the case of an ideal gas mixture and an ideal liquid solution, the fugacity coefficient and the activity coefficient are both one, and Equation 3-192 reduces to Raoult's law:

$$y_i P = x_i P_i^{\text{sat}} \quad (3-193)$$

- For an ideal gas mixture, but a nonideal liquid solution, Equation 3-192 reduces to the modified Raoult's law:

$$y_i P = \gamma_i x_i P_i^{\text{sat}}. \quad (3-194)$$

Assuming that the thermodynamic equilibrium prevails at the vapor–liquid interface, the molar fraction to prescribe on the vapor side of the interface is in general (from Equation 3-192)

$$y_i = \frac{f_i^{\text{L}}(T, P, x_i)}{\phi_i(T, P, y_i)P}. \quad (3-195)$$

The [Vapor–Liquid-Solution Interface](#) feature uses separate composition variables for the two phases along the droplet surface. To enforce a thermodynamic equilibrium, the composition on the vapor side is prescribed from Equation 3-195 using the liquid composition, the temperature, and the pressure at the surface. When the feature is coupled to a thermodynamics system, functions for both the liquid phase fugacity, and the vapor phase fugacity are available. It is also possible to prescribe a vapor pressure for each species. In this case, the system is assumed ideal and the vapor molar fraction is

$$y_i = \frac{x_i P_i^{\text{sat}}}{P}. \quad (3-196)$$

The [Vapor–Liquid Interface](#) uses the same formulation for the vapor–liquid equilibrium, but the composition in the liquid phase is user-specified. It can be used to model vapor transport to or from a liquid where the composition changes are assumed small.

The [The Transport of Concentrated Species in Vapor Interface](#) interface solves for the mass fractions denote ω . For a species i , the boundary condition applied on the vapor side of the droplet surface is

$$\omega_i^V = \frac{y_i M_i}{M_n^V} = \frac{x_i P_i^{\text{sat}}}{P} \frac{M_i}{M_n^V}, \quad (3-197)$$

where M_n^V is the mean molar mass of the vapor.

Using the ideal gas law of the form

$$P = \frac{\rho^V RT}{M_n^V}, \quad (3-198)$$

the vapor side concentration corresponds to

$$c_i^V = \frac{\omega_i^V \rho^V}{M_i} = \frac{x_i P_i^{\text{sat}}}{P} \frac{\rho^V}{M_n^V} = \frac{x_i P_i^{\text{sat}}}{RT}. \quad (3-199)$$

Evaporation and Condensation

Vapor–liquid equilibrium is enforced by prescribing the mass fractions in accordance with [Equation 3-195](#). The resulting vapor pressure drives evaporation or condensation across the vapor–liquid interface. When the vapor pressure at the interface is higher than the ambient vapor pressure, mass evaporates from the liquid into the vapor. If instead the vapor pressure drops below the ambient vapor pressure, for example due to a decrease in temperature, vapor will instead condense into liquid. These processes will proceed until the vapor pressure at the interface and the ambient vapor pressure are in equilibrium.

The total mass flux transported across the interface ($\text{kg}/(\text{m}^2 \cdot \text{s})$) can be computed by summing all fluxes of evaporated species in the manner of

$$j^V = \sum_i j_i^V = \sum_i \mathbf{n} \cdot (\mathbf{j}_i^V + \rho^V \mathbf{u}^V \omega_i^V). \quad (3-200)$$

Here $\mathbf{j}_i^V$ denotes the diffusive mass flux resulting from enforcing the equilibrium conditions in [Equation 3-195](#), $\mathbf{n}$ is the surface normal, and $\mathbf{u}^V$ the mass averaged vapor velocity at the interface. Fluxes from species that does not evaporate or condensate should not be included in the sum. Assuming that the interface is stationary the total mass flux is used to define a Stefan velocity to be applied at the interface

$$j^V = \rho^V u_s \mathbf{n}. \quad (3-201)$$

For two-phase flow, the relationship governing the velocities of the vapor and the liquid at the surface is

$$\mathbf{u}^L = \mathbf{u}^V + j^V \left(\frac{1}{\rho^L} - \frac{1}{\rho^V} \right) \mathbf{n}, \quad (3-202)$$

where the mass transfer across the interface corresponds to the total vapor mass flux j^V . It can be noted that a nonzero mass flux is necessary for the phase velocities to differ. The difference is prescribed in the normal direction of the surface.

For two-phase flow with evaporation, the position of the vapor–liquid interface needs to also be updated. When using a Moving Mesh to track the interface position, the relationship between the mesh velocity and the liquid velocity is

$$\mathbf{u}_{\text{mesh}} \cdot \mathbf{n} = \left(\mathbf{u}^L - \frac{j^V}{\rho^L} \right) \mathbf{n}. \quad (3-203)$$

The mesh velocity is the velocity of the vapor–liquid interface. In a situation without evaporation or condensation, the normal mesh velocity equals the normal liquid velocity. Which in turn is equal to the vapor velocity according to [Equation 3-202](#). For a non-zero mass flux on the other hand, the interface normal mesh velocity differs from that of the liquid phase.

To conserve mass during phase transfer, the normal mass flux on the liquid side should equal that of the vapor side. The normal diffusive mass fluxes on the liquid side are therefore defined as

$$\mathbf{n} \cdot \mathbf{J}_i^L = j^V \left(\omega_i^L - \frac{j_i^V}{j^V} \right). \quad (3-204)$$

The heat of evaporation at the vapor–liquid interface (W/m^2) is defined as

$$Q_{\text{vap}} = \sum_i j_i^V \Delta H_{\text{vap},i} \quad (3-205)$$

where $\Delta H_{\text{vap},i}$ is the species heat of evaporation (J/kg). When coupled to a thermodynamic system, functions for the heat of evaporation are added automatically.

Reference for the Vapor–Liquid Equilibrium Interfaces

1. B.E. Poling, J.M. Prausnitz, and J.P. O’Connell, *The Properties of Gases and Liquids*, 5th ed., McGraw Hill, 2000.

Theory for the Reacting Laminar Flow Interface

The following sections describe theory applicable for the Reacting Flow interface:

- [Pseudo Time Stepping for Mass Transport](#)
- [The Stefan Velocity](#)
- [The Chemical Reaction Rate](#)
- [Nonisothermal Reacting Flow](#)
- [References for the Reacting Flow Interfaces](#)

Pseudo Time Stepping for Mass Transport

In order to improve the solution robustness, pseudo time stepping can be used for the Transport of Concentrated Species interface when solving a stationary model. Using pseudo time stepping, a fictitious time derivative term:

$$\rho \frac{\omega_i - \text{nojac}(\omega_i)}{\Delta \tilde{t}}$$

is added to the left-hand side of the mass fraction equations. Here ρ is the fluid mixture density, ω_i is the mass fraction (dimensionless) of species i , and $\Delta \tilde{t}$ is the *pseudo time step*. Since $\omega_i - \text{nojac}(\omega_i)$ is always zero, this term does not affect the final solution. It does, however, affect the discrete equation system and effectively transforms a nonlinear iteration into a time step of size $\Delta \tilde{t}$.



For a description of the pseudo time step term for the Navier–Stokes equations and the pseudo time step see [Pseudo Time Stepping for Laminar Flow Models](#) and [Pseudo Time Stepping](#) in the *COMSOL Multiphysics Reference Manual*.

The Stefan Velocity

Heterogeneous reactions on fluid-solid surfaces can affect the mass, momentum, and energy balances at the interface separating the fluid and the solid. On the reacting surface, the production or destruction rate, $r_{s,i}$ (SI unit: $\text{mol}/(\text{m}^2 \cdot \text{s})$), of a fluid phase

species is balanced by the total mass flux of the species. The mass balance for species i on a boundary representing a fluid-solid interface is given by

$$\mathbf{n} \cdot (\mathbf{j}_i + \rho \omega_i \mathbf{u}) = r_{s,i} M_i \quad (3-206)$$

Here, $\mathbf{n}$ is the unit normal pointing out of the fluid domain, $\mathbf{u}$ is the mass averaged velocity of the fluid mixture (SI unit: m/s), $\mathbf{j}_i$ denotes the mass flux of species i relative to the mixture (typically due to diffusion), and M_i is the species molar mass (SI unit: kg/mol). Summing the mass balances at the surface, over all species, results in an effective mixture velocity:

$$\mathbf{n} \cdot \rho \mathbf{u}_s = \sum_{i=1}^N r_{s,i} M_i \quad (3-207)$$

referred to as the Stefan velocity, here denoted $\mathbf{u}_s$. To reach Equation 3-207 the fact that the sum of all mass fractions is one, and that the sum of all relative diffusive fluxes is zero, was used.

Equation 3-207 implies that surface reactions result in a net flux between the surface and the domain. A net flux in turn corresponds to an effective convective velocity at the domain boundary; the Stefan velocity. It should be noted here that when solving for mass transport inside a fluid domain, an outer boundary of the domain corresponds to a position just outside of the actual physical wall (on the fluid side). The domain boundary does not coincide with the physical wall.

In most reacting flow models, the species mass fractions in the fluid domain are solved for without including the surface concentrations (mol per area) on exterior walls. One reason for this is that the surface reaction rates are often not known. In this case, surface reactions can be modeled either by applying a mass flux or prescribing the mass fraction, or a combination of both, on fluid boundaries adjacent to the reacting surface. The Stefan velocity on a fluid domain boundary is then defined as the net mass flux resulting from the boundary conditions applied:

$$\mathbf{n} \cdot \rho \mathbf{u}_s = \sum_{i=1}^N \mathbf{n} \cdot (\mathbf{j}_{0,i} + \rho \omega_{0,i} \mathbf{u}) \quad (3-208)$$

Here, the first term contains contributions from boundary conditions prescribing the mass flux, while the second contains contributions from boundary conditions prescribing the mass fractions. Contributions to the Stefan velocity can be added by

selecting **Account for Stefan velocity** in the Flux or Mass Fraction features in The Transport of Concentrated Species interface.

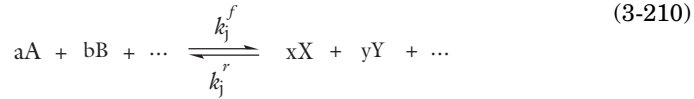
The resulting Stefan velocity based on mass transport boundary conditions is computed as

$$\mathbf{n} \cdot \mathbf{u}_s = \frac{\sum_{i=1}^N \mathbf{n} \cdot \mathbf{j}_{0,i}}{\rho \left(1 - \sum_i \omega_{i,0} \right)} \quad (3-209)$$

Using a Reacting Flow interface, the Stefan velocity, defined in the manner of Equation 3-209, is automatically computed and applied on boundaries corresponding to walls in the coupled fluid flow interface. The Stefan velocity is prescribed in the wall normal direction on the wall selection.

The Chemical Reaction Rate

For laminar flow, or when the **Turbulent-reaction model** is set to **None** in a **Reaction** feature (in Transport of Concentrated Species), the default (Automatic) reaction rate used by the **Reaction** feature is based on the mass action law. Consider a general reaction belonging to a set of j reactions and involving i species:



For such a reaction set, the reaction rates r_j (SI unit: $\text{mol}/(\text{m}^3 \cdot \text{s})$), can be described by the mass action law:

$$r_j = k_j^f \prod_{i \in \text{react}} c_i^{-v_{ij}} - k_j^r \prod_{i \in \text{prod}} c_i^{v_{ij}} \quad (3-211)$$

Here, k_j^f and k_j^r denote the forward and reverse rate constants, respectively. The concentration of species i is denoted c_i (SI unit: mol/m^3). The stoichiometric coefficients are denoted v_{ij} , and are defined to be negative for reactants and positive for products. In practice, a reaction seldom involves more than two species colliding in a reacting step, which means that a kinetic expression is usually of order 2 or less (with respect to the involved concentrations).

In addition to the concentration dependence, the temperature dependence can be included by using the predefined Arrhenius expressions for the rate constants:

$$k = AT^n \exp\left(-\frac{E}{R_g T}\right)$$

Here, A denotes the frequency factor, n the temperature exponent, E the activation energy (SI unit: J/mol) and R_g the gas constant, 8.314 J/(mol·K). The pre-exponential factor, including the frequency factor A and the temperature factor T^n , is given the units $(\text{m}^3/\text{mol})^{\alpha-1}/\text{s}$, where α is the order of the reaction (with respect to the concentrations).

Nonisothermal Reacting Flow

When a Chemistry interface and a Heat Transfer interface are used in the Nonisothermal Reacting Flow multiphysics coupling, the heat equation is modified to account for reactions and diffusion.

In presence of chemical reactions, the enthalpy (SI unit: J/kg) is a function of temperature, pressure and the mixture composition. For a mixture of N components, the mass fractions verify the condition $\omega_1 + \dots + \omega_N = 1$. Therefore the enthalpy is a function of the temperature, pressure, and only $N - 1$ mass fractions:

$$\hat{H} = \hat{H}(T, p, \omega_1, \dots, \omega_{N-1})$$

Assuming the same velocity for the convection term of all components, the mass fractions follow the equation:

$$\rho \left(\frac{\partial}{\partial t} (\omega_i) + (\mathbf{u} \cdot \nabla) \omega_i \right) = -\nabla \cdot \mathbf{j}_i + R_i$$

On the other hand, using the chain rule, the enthalpy variation is written:

$$\rho \frac{D\hat{H}}{Dt} = \rho \frac{\partial \hat{H}}{\partial T} \bigg|_{p, \omega_j} \frac{DT}{Dt} + \rho \frac{\partial \hat{H}}{\partial p} \bigg|_{T, \omega_j} \frac{Dp}{Dt} + \rho \sum_{i=1}^{N-1} \frac{\partial \hat{H}}{\partial \omega_i} \bigg|_{T, p, \omega_{j \neq i}} \frac{D\omega_i}{Dt}$$

which can be rewritten using the partial molar enthalpies:

$$\bar{H}_i = \frac{\partial \hat{H}}{\partial n_i} \bigg|_{T, p, n_{j \neq i}}$$

$$\rho \frac{D\hat{H}}{Dt} = \rho \left. \frac{\partial \hat{H}}{\partial T} \right|_{p, \omega_j} \frac{DT}{Dt} + \rho \left. \frac{\partial \hat{H}}{\partial p} \right|_{T, \omega_j} \frac{Dp}{Dt} + \sum_{i=1}^N \frac{\bar{H}_i}{M_i} (-\nabla \mathbf{j}_i + R_i)$$

Additionally, the heat balance equation in terms of internal energy is written:

$$\rho \frac{DE}{Dt} = -\nabla \cdot \mathbf{q} + Q_{\text{vd}} + Q$$

where Q_{vd} is the heat source from viscous dissipation, and Q represents additional sources. The heat flux $\mathbf{q}$ consists in a term of conduction and a term of diffusion:

$$\mathbf{q} = \mathbf{q}_{\text{cond}} + \mathbf{q}_{\text{diff}} = -k \nabla T + \sum_{i=1}^N \frac{\bar{H}_i}{M_i} \mathbf{j}_i$$

Enthalpy and internal energy are linked through the formula $E = \hat{H} - \frac{p}{\rho}$, we can thus rewrite the heat balance equation in terms of enthalpy:

$$\rho \left. \frac{\partial \hat{H}}{\partial T} \right|_{p, \omega_j} \frac{DT}{Dt} + \left(\rho \left. \frac{\partial \hat{H}}{\partial p} \right|_{T, \omega_j} - 1 \right) \frac{Dp}{Dt} + \sum_{i=1}^N \frac{\bar{H}_i}{M_i} (-\nabla \mathbf{j}_i + R_i) = -\nabla \cdot \mathbf{q} + Q_{\text{vd}} + Q$$

Introducing the heat capacity at constant pressure C_p (SI unit: J/(kg·m³), and the coefficient of thermal expansion α_p (SI unit: 1/K) gives:

$$\rho C_p \frac{DT}{Dt} - \alpha_p T \frac{Dp}{Dt} + \sum_{i=1}^N \frac{\bar{H}_i}{M_i} (-\nabla \mathbf{j}_i + R_i) = -\nabla \cdot \mathbf{q} + Q_{\text{vd}} + Q$$

The term with the pressure derivative is the work of pressure forces Q_p . The enthalpy variation due to the gradient of species mass flux can be partly simplified with the diffusion part of the heat flux. The heat source of reaction can be written in terms of enthalpies of reaction H_i instead of partial enthalpies (see [Ref. 1](#)). The temperature derivative term can be expanded to make the convective term appear. We finally obtain the temperature equation for a free flow with N species and M reactions:

$$\rho C_p \left[\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T \right] + \nabla \cdot (-k \nabla T) = Q_{\text{diff}} + Q_{\text{react}} + Q_{\text{vd}} + Q_p + Q$$

$$Q_{\text{diff}} = - \sum_{i=1}^N \mathbf{j}_i \cdot \nabla \left(\frac{\bar{H}_i}{\bar{M}_i} \right)$$

$$Q_{\text{react}} = - \sum_{i=1}^M H_i r_i$$

$$Q_{\text{vd}} = (-p\mathbf{I} + \boldsymbol{\tau}) : \nabla \mathbf{u}$$

$$Q_p = \alpha_p T \left[\frac{\partial p}{\partial t} + (\mathbf{u} \cdot \nabla) p \right]$$

For reacting flows moving through the interstices of a porous medium, the heat sources of reaction and diffusion are only accounted for the temperature of the fluid phase. While the porosity is already accounted for in the diffusion term through the mass flux, the heat source of reaction has to be multiplied by the porosity. Also, in presence of a porous catalyst, both reactions in the bulk and reactions at the surface are accounted for in the reaction source term:

$$Q_{\text{react}} = -\varepsilon_p \sum_{i=1}^{M_{\text{bulk}}} H_i r_i - S_{\text{area}} \sum_{i=1}^{M_{\text{surf}}} H_i^s r_i^s$$

where ε_p is the porosity of the medium (SI unit: 1), S_{area} is the specific surface area (SI unit: 1/m), and r_i^s is the surface reaction rate (SI unit: mol/(m²·s)).

An additional source term is added to model convective heat transfer between the fluid and the porous matrix to get the temperature equation for nonisothermal reacting flows in porous media:

$$\rho_f C_{p,f} \left[\frac{\partial T_f}{\partial t} + (\mathbf{u} \cdot \nabla) T_f \right] + \nabla \cdot (-k_f \nabla T_f) = Q_{\text{diff}} + Q_{\text{react}} + Q_{\text{vd}} + Q_p + Q_{s,f} + Q$$

This equation is the one that is solved when the **Porous medium type** of the [Porous Medium](#) feature is set to **Local thermal nonequilibrium**, since both fluid and porous matrix temperatures are solved separately. With the **Local thermal equilibrium** approach, the equation is written for the common temperature, and in terms of effective material properties for the time derivative term and conduction term:

$$(\rho C_p)_{\text{eff}} \frac{\partial T}{\partial t} + \rho_f C_{p,f} (\mathbf{u} \cdot \nabla) T + \nabla \cdot (-k_{\text{eff}} \nabla T) = Q_{\text{diff}} + Q_{\text{react}} + Q_{\text{vd}} + Q_p + Q$$

References for the Reacting Flow Interfaces

1. H.A. Jakobsen, *Chemical Reactor Modeling: Multiphase Reactive Flows*, Springer, 2008.

Fluid Flow Interfaces

The functionality for simulating fluid flow in free, porous media is found under the **Fluid Flow** branch () when adding a physics interface.

The documentation of the available interfaces and features can be found in the *CFD Module User's Guide* in the chapters [Single-Phase Flow Interfaces](#) and [Porous Media and Subsurface Flow Interfaces](#).

Modeling Fluid Flow

In this section:

- [Available Physics Interfaces](#)
- [Coupling to Other Physics Interfaces](#)

Available Physics Interfaces

This module primarily models the transport and conversion of mass in chemical reactors and other reacting systems. Many of these applications include transport through convection, and the module includes a number of physics interfaces for the simulation of fluid flow, particularly within porous media. As with all other physical descriptions simulated by COMSOL Multiphysics, any description of fluid flow can be directly coupled to any other physical process. This is particularly relevant for systems based on turbulent or multiphase flow, which are supported by the CFD Module. For detailed descriptions of the fluid-flow interfaces listed below, see the *CFD Module User's Guide*.

The **Fluid Flow** branch () has a number of subbranches to describe momentum transport. The **Laminar Flow Interface** (), found under the **Single-Phase Flow** branch () when adding a physics interface, is used to model laminar flow described by the Navier–Stokes equations. It is also possible to extend this physics interface to study nonisothermal or non-Newtonian flow.

The **Creeping Flow Interface** (), also under the **Single-Phase Flow** branch (), is used to model flow fluid flows at very low Reynolds numbers, also referred to as Stokes flow. This typically occurs in fluid systems with high viscosity or small geometrical length scales (for example in microfluidics and MEMS devices).

The rest of the available physics interfaces for fluid flow are found under the **Porous Media and Subsurface Flow** branch ()

The **Darcy's Law Interface** () is used to model fluid movement through interstices in a porous medium where a homogenization of the porous and fluid media into a single medium is done. This physics interface combines the continuity equation and an equation of state for the pore fluid (or gas), can be used to model low velocity flows, for which the pressure gradient is the major driving force.

Darcy's law can be used in porous media where the fluid is mostly influenced by the frictional resistance within the pores. When the size of the interstices are larger and the fluid is influenced by itself, the kinetic potential from fluid velocity, pressure, and gravity must be also be considered, and [The Brinkman Equations Interface](#) () is the appropriate physics interface to use.

The Brinkman equations extend Darcy's law to describe the dissipation of the kinetic energy by viscous shear, similar to the Navier–Stokes equation. Consequently, they are well suited to transitions between slow flow in porous media, governed by Darcy's law, and fast flow in channels described by the Navier–Stokes equations. The equations and boundary conditions that describe these types of phenomena are also found in [The Free and Porous Media Flow, Brinkman Interface](#) ().

[The Free and Porous Media Flow, Darcy Interface](#) () models porous media flow connected to free flow domains. This multiphysics interface couples the Laminar Flow interface with the Darcy's Law interface over their common boundaries.

The next section briefly discusses coupling to other physics interfaces.

Coupling to Other Physics Interfaces

Often, you are simulating applications that couple fluid flow in free porous media to another type of phenomenon described in another physics interface. This can include chemical reactions and mass transport, as covered in the Chemical Species Transport branch, or energy transport in porous media found in the Heat Transfer branch.

More advanced descriptions of fluid flow, such as turbulent and multiphase flow, can be found in the CFD Module. More extensive descriptions of heat transfer, such as that in turbulent flow or systems including radiation, can be found in the Heat Transfer Module. Furthermore, some applications that involve electrochemical reactions and porous electrodes, particularly in electrochemical power source applications, are better handled by the Battery Design Module.

Heat Transfer Interfaces

The functionality for simulating heat transfer in free and porous media is found under the **Heat Transfer** branch () when adding a physics interface.

The documentation of all features in the Heat Transfer in Porous Media interface can be found in the *Heat Transfer Module User's Guide*, in the chapter [Heat Transfer Interfaces](#).

Modeling Heat Transfer

Available Physics Interfaces

Heat transfer is an important part of chemical reaction engineering. Most chemical reactions either require or produce heat, which in turn affects both the reactions themselves and other physical processes connected to the system.

The heat transfer branch includes a number of physics interfaces for the simulation of heat transfer, particularly in porous media. As with all other physical descriptions simulated by COMSOL Multiphysics, any description of heat transfer can be directly coupled to any other physical process. This is particularly important for systems based on chemical reactions and mass transfer, along with fluid flow.

You can manipulate these physics interfaces with respect to including source and sink terms or, in other words, account for exothermic or endothermic chemical reactions.



If you also have the Heat Transfer Module there are additional Heat Transfer interface features, such as radiation, available.

Coupling Heat Transfer with Other Physics Interfaces

[The Reaction Engineering Interface](#) includes support for setting up and simulating chemical reactions, including reaction kinetics through the temperature-dependent Arrhenius expression and mass action law. This physics interface also has support for including and calculating thermodynamic data as temperature-dependent expressions. The Reaction Engineering interface can generate a space-dependent model using any heat transfer interface to benefit from the bidirectional dependence between heat transfer and chemical reactions.

More advanced descriptions of fluid flow, such as turbulent and multiphase flow, require the CFD Module. In addition, the Heat Transfer Module also includes more detailed descriptions and tools for simulating energy transport, such as radiation.

Thermodynamics

This chapter describes how you can use the thermodynamics functionality to define thermo-physical and transport properties. The properties in turn can be used when simulating chemical reaction systems, or any type of transport model involving mass transfer, fluid flow, or heat transfer.

In this chapter:

- [Using Thermodynamic Properties](#)
- [Thermodynamic Models and Theory](#)

Using Thermodynamic Properties

In this section:

- [Workflow for Thermodynamics Property Calculations](#)
- [Thermodynamics](#)
- [Thermodynamic System](#)
- [External Thermodynamic Packages](#)
- [External Thermodynamic System](#)
- [Predefined System](#)
- [Exporting and Importing Thermodynamic Systems](#)
- [Species Property](#)
- [Mixture Property](#)
- [Equilibrium Calculation](#)
- [Generate Chemistry](#)
- [Generate Material](#)
- [Coupling with the Reaction Engineering and the Chemistry Interfaces](#)
- [Evaluating a Property Function in a Physics Interface](#)
- [User-Defined Species](#)
- [References](#)

Workflow for Thermodynamics Property Calculations

The thermodynamic properties database, together with the thermodynamic calculator, make it possible to evaluate physical properties of fluids such as heat capacity, thermal conductivity, density, and diffusivity, as well as equilibrium compositions of two phases at equilibrium. The properties can be calculated for pure fluids, mixtures, and for two-phase fluid systems consisting of pure species as well as mixtures.

Using the thermodynamics functionality involves the following main steps:

- 1 Add the **Thermodynamics** node and create a thermodynamic system to add properties for a selected set of chemical compounds, referred to as chemical species in this manual. A thermodynamic system can be created in the following ways:
 - A built-in **Thermodynamic System** uses the built-in, or user-defined, property database with native functionality for thermodynamic property calculations.
 - An **External Thermodynamic System** uses properties from an installed external package provider. For more information on how to enable the use of external packages see the [Installing External Thermodynamic Packages — An Example](#).
 - A **Predefined System** uses the built-in database with native functionality to set up one of the following common systems: Dry air, Moist air, or Water-steam.
- 2 Once a thermodynamic system is set up, it can be used to define a multitude of property functions. The [Generate Material](#) option provides a fast route to defining the material properties most commonly required for fluid flow, heat transfer and mass transfer modeling. By specifying the composition and phase of the mixture, a **Material** node can automatically be set up. The material node can include properties such as density, heat capacity, heat capacity ratio, thermal conductivity and viscosity.
- 3 Users of the Reaction Engineering, or Chemistry interface can couple chemical species in these interfaces with the chemical species in the thermodynamic system. When [Coupling with the Reaction Engineering and the Chemistry Interfaces](#) the required property parameters and functions are automatically added and visualized as nodes under the corresponding thermodynamic system node.
- 4 As an alternative to the functions automatically set up when using Generate Material or coupling to, for example, Chemistry, functions or constants can be set up manually for any thermodynamic system. This is accomplished using the thermodynamic features [Species Property](#), [Mixture Property](#) or [Equilibrium Calculation](#). You can, for example, create a **Species Property** function describing the density of a fluid, and use this function in a fluid-flow interface. With **Mixture Property** you can define mixture functions, that is, functions that depend on the composition of a mixture, describing for example density, enthalpy, or heat capacity.

The functionalities described in Step 2– 4 generate functions and constants, defined by a thermodynamic system, that can be used in any physics interface in COMSOL Multiphysics. The functions are used to evaluate fluid properties that depend on variables such as temperature, pressure, and mixture composition. For more information, see [Evaluating a Property Function in a Physics Interface](#).

To access the functionality for thermodynamic calculations, right-click the **Global Definitions** node in the **Model Builder** tree and select **Thermodynamics** (). When the model includes a Component, **Thermodynamics** is also available in the **Physics** toolbar.

Using the **Reaction Engineering** or **Chemistry** interfaces, **Thermodynamics** is also available on the **Reaction Engineering** or **Chemistry** toolbar.

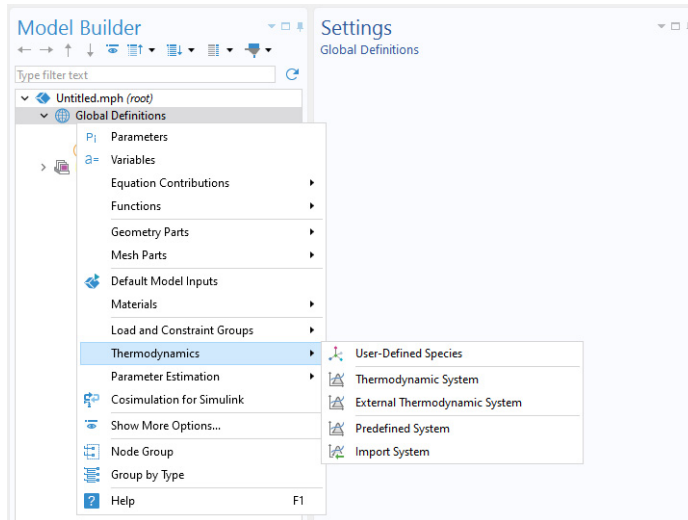


Figure 6-1: The Thermodynamics is always available under Global Definitions.

Thermodynamic properties given as parameters or functions are collected in a so called thermodynamic system. Each thermodynamic system includes a set of available species and phases, as well as different models used for the thermodynamic property calculations, for example an equation of state such as Peng–Robinson.

Select **Thermodynamic System** in the context menu to add a thermodynamic system that uses the built-in database included with COMSOL Multiphysics.

As an alternative, selecting **External Thermodynamic System** can make use of an installed thermodynamics software to make the corresponding calculations.

The option **Predefined System**, allows you to set up one of the following systems: Dry air, Moist air, or Water-steam.

It is also possible to import previously created and stored thermodynamic systems by selecting **Import Thermodynamic System** (see [Exporting and Importing Thermodynamic Systems](#)).

In addition to creating thermodynamic systems, you can also define [User-Defined Species](#) in Thermodynamics. Use **User-Defined Species** to add new species that are not available in the COMSOL database. You can also edit available species in the database.

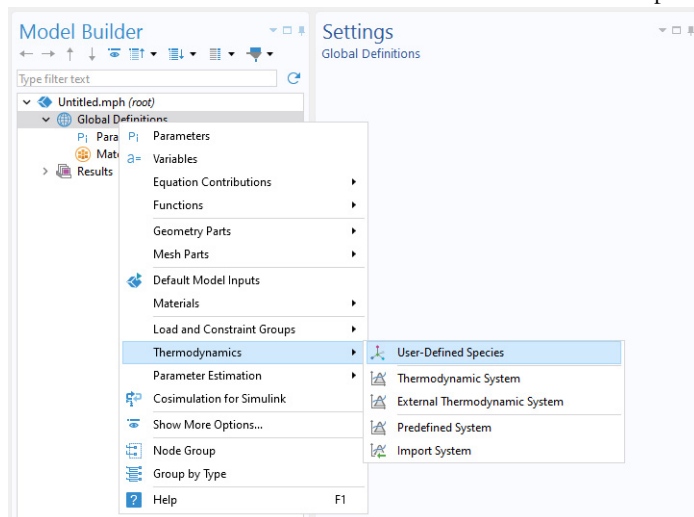


Figure 6-2: Adding Thermodynamic Systems to the Thermodynamics node. Note that external package providers have been installed.

When a thermodynamic system has been added, thermodynamic and transport properties are defined by functions. These can be used to describe properties for pure species, mixtures, and for gas–liquid (or liquid–liquid) systems at equilibrium.

Thermodynamic System

A thermodynamic system is used to describe properties of pure species and mixtures of chemical compounds for liquids, gases, liquid–vapor equilibria, and liquid–liquid equilibria. It specifies the available species and the phases (states of aggregation) that are present in the modeled system. It also defines and evaluates the functions for thermodynamic and transport properties of the chemical system, that is, the species and mixture properties for liquids, gases, and phase equilibria.

To create a **Thermodynamic System**, start the **Thermodynamic System Wizard** in one of the following ways:

- Right-click the **Global Definitions** node, select **Thermodynamics** and choose **Thermodynamic System**.
- Using the Reaction Engineering interface, choose **Thermodynamic System** from the **Thermodynamics** section of the **Reaction Engineering** toolbar.
- Correspondingly, using the Chemistry interface, choose **Thermodynamic System** from the **Thermodynamics** section of the **Chemistry** toolbar.

The **Thermodynamic System Wizard** contains the following steps:

- 1 [Select System](#)
- 2 [Select Species](#)
- 3 [Select Thermodynamic Model](#)

SELECT SYSTEM

Use the **Select System** step in the wizard to define the phases in the modeled system. You can select **Gas**, **Liquid**, **Vapor-liquid**, **Vapor-liquid-liquid**, or **Liquid-liquid**. The names of the phases in the Selected system table can be changed by editing the element in the Name column. Click the **Next** button (➞) to proceed to the next step in the wizard.

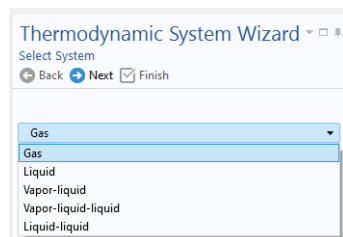


Figure 6-3: Thermodynamic System Wizard: Select System.

SELECT SPECIES

Use the **Species Type filter text** field to search among the available species in the **COMSOL** database or the [User-Defined Species](#). Species can be searched for by typing the name, CAS number, or the chemical formula. Using a formula, all species whose chemical formula includes a combination of the given atoms are listed. For example, when searching for ammonia NH_3 or H_3N can be used.

Select one or more species in the list and click the **Add Selected** button (+) to add species to the **Selected species** table. You can also press the **Enter** key to add either the

first species in the list, or the species that have been selected. To remove species from the Selected species table, use either the Remove Selected button () or the Remove All button (), both located below the table. Click the **Next** button () to proceed to the next step in the wizard.

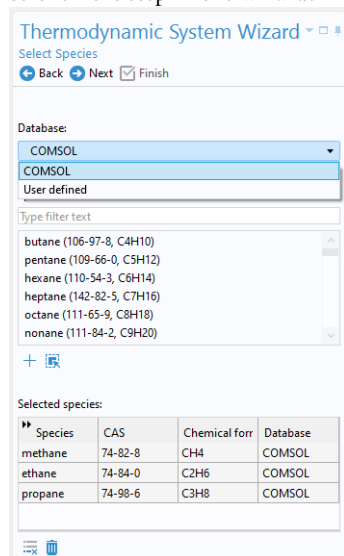


Figure 6-4: Thermodynamic System Wizard: Select Species.

SELECT THERMODYNAMIC MODEL

The Thermodynamic model controls the relations between the intensive thermodynamic properties of the system such as pressure, specific volume, and temperature. It also controls how extensive properties like enthalpy, entropy, or the heat capacity of the system are computed. Furthermore, the Thermodynamic model specifies the default models used to compute transport properties.

The following [Thermodynamic Models](#) are available for all phases:

- [Peng–Robinson](#)
- [Peng–Robinson \(Twu\)](#)
- [Soave–Redlich–Kwong](#)
- [Soave–Redlich–Kwong \(Graboski–Daubert\)](#)

For a system including only gas phase, or for liquid phase systems with thermodynamic models requiring an equation of state, the [Ideal Gas Law](#) is added to the available models in the **Gas phase model** list or **Equation of state** list.

When the system includes one or more liquid phases, the following additional models are also available (in the **Liquid phase model** list):

- [Chao–Scader \(Grayson–Streed\)](#)
- [Wilson](#)
- [NRTL](#)
- [UNIFAC VLE](#)
- [UNIQUAC](#)
- [Regular Solution](#)
- [Extended Regular Solution](#)
- [Ideal Solution](#)

These models correspond to activity models.

To calculate properties of water and steam, use the [Water \(IAPWS\)](#) model. This is available for systems that include a single water species.

For recommendations regarding the appropriate thermodynamic model, based on the available phases and the system pressure level, see the section [Selecting the Right Thermodynamic Model](#).

Select the **Advanced options** checkbox in order to manually control the models used for thermodynamic properties, transport properties, and surface tension. When this checkbox is selected, all available property models are shown in the property model table. The available property models are dependent on the phases available in the thermodynamic system.

When only a gas phase is present, models are available for the following properties:

- Gas diffusivity
- Gas thermal conductivity
- Gas viscosity

For a single-phase liquid, or for a two-phase liquid–liquid system, there are models for the following properties:

- Liquid diffusivity
- Liquid diffusivity at infinite dilution
- Liquid thermal conductivity

- Liquid viscosity
- Liquid volume

For systems with both vapor and liquid phases, all of the above properties are available, as well as gas–liquid surface tension.

The screenshot shows the 'Thermodynamic System Wizard' dialog box. At the top, it says 'Select Thermodynamic Model' with 'Back', 'Next', and 'Finish' buttons. Below, 'Liquid phase model' and 'Gas phase model' are both set to 'Soave-Redlich-Kwong'. The 'Advanced options' section is checked and contains a table of property models.

Property	Model
Gas diffusivity	Automatic
Gas-liquid surface tension	Ideal
Gas thermal conductivity	Kinetic theory
Gas viscosity	Brokaw
Liquid diffusivity	Wesselingh-Krishna
Liquid diffusivity at infinite...	Automatic
Liquid-liquid surface tension	None
Liquid thermal conductivity	Ideal
Liquid viscosity	Mass logarithmic mixin
Liquid volume	Soave-Redlich-Kwong

Figure 6-5: Thermodynamic System Wizard: Advanced thermodynamic model options available for a Vapor–liquid system.

Click the **Finish** button () to exit the **Thermodynamic System Wizard** and add the corresponding system under the **Thermodynamics** node. Note that the default node label reflects the available phase in the system. For example, when creating a vapor–liquid system, a node labeled **Vapor–Liquid System** is added. The name of the system can be changed using the Label text field in the Settings window for the system.

SETTINGS

Selecting a [Thermodynamic System](#) or [Predefined System](#) node shows its **Settings** window with a description of the package content and the available properties. Editing

the text in the Label text field changes the name of the system. Below follows a description of the thermodynamic system settings.

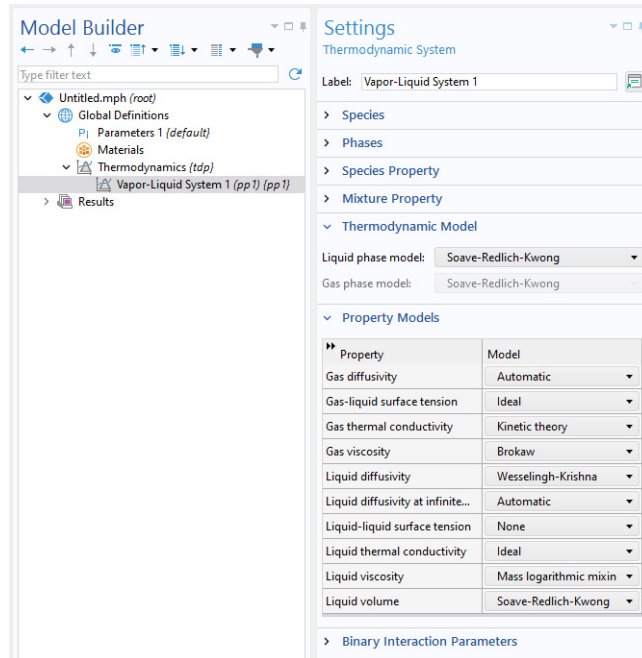


Figure 6-6: Thermodynamic System Settings.

Species

Lists the species included in the thermodynamic system. You can change the list of species by right-clicking the **Thermodynamic System** node and selecting [Add or Remove Species](#).

Phases

Lists the phases included in the thermodynamic system. You can add or remove phases, such as gas, liquid, or liquid–vapor, by right-clicking the **Thermodynamic System** node and selecting [Define System](#).

Species Property

Lists the parameters and functions that describe thermodynamics and transport properties for pure species. Such functions may describe density, heat capacity, thermal conductivity, or other thermodynamic and transport properties. For more information, see [Species Property](#).

Mixture Property

Lists the available mixture property functions. Note that you have to have a mixture model defined in order to use these functions. You can define a mixture property by right-clicking the **Thermodynamic System** node and selecting [Mixture Property](#).

Thermodynamic Model

When a liquid is present, use the **Liquid phase model** list to select the thermodynamic model for this phase.

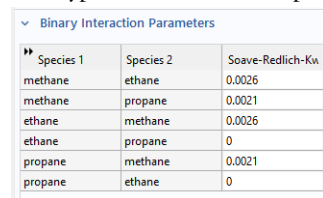
When a gas is present, use the **Gas phase model** list to select the thermodynamic model for this phase. If both gas and liquid phases are available, and the **Liquid phase model** is set to an equation of state, then the **Gas phase model** is set to the same model automatically.

Property Models

Displays the settings for the available individual property models. The property models can be changed by selecting different values from the combo box in the Model column. For example, for **Gas thermal conductivity** you can select **Kinetic theory** or **Ideal** from the corresponding combo box. You can also select the property model in the last step of the thermodynamic system wizard by selecting the **Advanced option** checkbox, see [Select Thermodynamic Model](#).

Binary Interaction Parameters

This section can be used to inspect and edit binary interaction parameters. Select a pair of species (first and second column, Species 1 and Species 2) as shown in [Figure 6-7](#) to edit the corresponding binary interaction parameter currently in use. In cases that the thermodynamic model has more than one type of binary interaction parameter, each type of the interaction parameter will be shown in a column in the table.



Binary Interaction Parameters		
Species 1	Species 2	Soave-Redlich-Kw
methane	ethane	0.0026
methane	propane	0.0021
ethane	methane	0.0026
ethane	propane	0
propane	methane	0.0021
propane	ethane	0

Figure 6-7: Entering the Binary Interaction Parameters for the Soave–Redlich–Kwong model.

When a binary interaction parameter is not available in the database for a pair of species, the parameter value is set to zero (default value).

MODIFYING AN EXISTING SYSTEM

It is possible to modify an existing thermodynamic system, for example by changing the included species or phases. To do so, right-click the **Thermodynamic System** node and select [Add or Remove Species](#), or [Define System](#) from the context menu. Both of these options are available for systems created using Thermodynamic System. For a **Predefined System**, the option **Add or Remove Species** is available for Dry Air and Moist Air. For a Water-steam system, neither of the two options are available.

When the properties of a [User-Defined Species](#) have been changed, select [Update From Species](#) for the changes to take effect in the thermodynamic system.

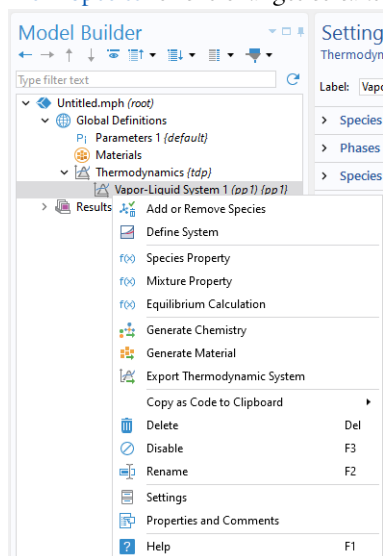


Figure 6-8: Modifying the species and phases of a thermodynamic system.

Add or Remove Species

Selecting **Add or Remove Species** takes you to the [Select Species](#) step in the **Thermodynamic System Wizard**. Use the **Add Species** button () to add a species to the **Selected Species** list. To remove a species, select it in the list and use the **Remove Species** button ().

Make sure to click the **Finish** button () in order for the changes to take effect.

Note that adding or removing species in a thermodynamic system that is currently coupled to a Reaction Engineering interface or a Chemistry interface breaks this coupling. See the [Coupling with the Reaction Engineering and the Chemistry Interfaces](#) for how to update this coupling.

Define System

Selecting **Define System** takes you to the [Select System](#) step in the **Thermodynamic System Wizard**. You can select the desired system from the list.

Make sure to click the **Finish** button (☒) button in order for the changes to take effect.

Note that changing the phase(s) in a thermodynamic system that is currently coupled to a Reaction Engineering interface or a Chemistry interface breaks this coupling. See the [Coupling with the Reaction Engineering and the Chemistry Interfaces](#) section for how this coupling can be updated.

Update From Species

This option is available in a **Thermodynamic System** that includes a [User-Defined Species](#). Use it to update the system following a change in the species properties.

WARNING INFORMATION

A sanity check is always performed when a **Thermodynamic System** is created. If any problems are found, a Warning Information node listing the problems is added under the package. One example when this occurs is when a parameter required for the thermodynamic model, typically a binary interaction parameter, is not available in the database.

External Thermodynamic Packages

The thermodynamics functionality includes support for creating properties from external thermodynamic packages. In order to use external packages, the external software has to be CAPE-OPEN compliant. Thermodynamics in COMSOL Multiphysics supports the CAPE-OPEN Thermo 1.1 and the CAPE-OPEN Thermo 1.0 standards.

CAPE-OPEN is an open interface standard for chemical process simulation software ([Ref. 1](#)). This standard defines rules and interfaces that allows linking between compliant software. Some examples of CAPE-OPEN compliant software are:

- COCO/TEA (Free of charge package provider; see [Ref. 2](#)).
- Multiflash™ (KBC Advanced Technologies plc).
- VMGThermo™ (Virtual Materials Group, Inc.).
- Simulis® Thermodynamics (ProsSim SA).

- Aspen Properties, aspenONE[®] (Aspen Technology, Inc.).
- COMThermo[®] packages (comes with Aspen HYSYS[®]).

Accessing external thermodynamic systems allows species properties, mixture properties, as well as phases equilibrium calculations (gas–vapor, liquid–liquid) to be obtained from external software, such as the ones listed above.

INSTALLING EXTERNAL THERMODYNAMIC PACKAGES — AN EXAMPLE

TEA is a thermodynamics thermodynamic system that handles the physical and thermodynamic property calculations for the simulation environment COCO. The property data bank contains more than 190 commonly used chemicals, and the package exhibits more than 100 property calculation routines with their analytical or numerical derivatives. COCO is maintained by AmsterCHEM and is free to download from the Internet (Ref. 2).

To use the external physical and thermodynamic property calculations from COCO/TEA in COMSOL Multiphysics models, you need to go through the following steps:

- 1 Download and install COCO, which includes the TEA thermodynamic system manager. The software is available from www.cocosimulator.org/index_download.html.
- 2 Create and configure a thermodynamic system template that handles physical and thermodynamics calculations needed for your model. If you have already created a package template earlier, or if an adequate thermodynamic system already exists in the installation, this step is not needed.
- 3 Create an [External Thermodynamic System](#) node as detailed in the next section. The installed packages are available in the [Select Property Package](#) step of the Thermodynamic System Wizard.

External Thermodynamic System

The following assumes that you have installed external thermodynamic packages from a CAPE-OPEN compliant software in the manner outlined in [Installing External Thermodynamic Packages — An Example](#).

Locate the **Thermodynamics** node and right-click to add an **External Thermodynamic System** (Figure 6-2). Using a **Reaction Engineering** or **Chemistry** interface, you can also add an external package by choosing **External Thermodynamic System** from the **Thermodynamics** section of the **Reaction Engineering** or **Chemistry** toolbar.

SELECT PROPERTY PACKAGE

All installed packages, from providers such as COCO/TEA, Aspen Properties, and so on, are available in the **Thermodynamic System Wizard** (Figure 6-9). Select one of the packages and click the **Finish** button () in the upper right corner of the wizard in order to add the package as an **External Thermodynamic System** node under the **Thermodynamics** node.

When adding the package its default label reflects the included phases. For example, a node labeled **Vapor–Liquid System (External)** is created when adding an external package containing a vapor and a liquid phase. The name of the system can be changed using the Label text field in the Settings window for the system.

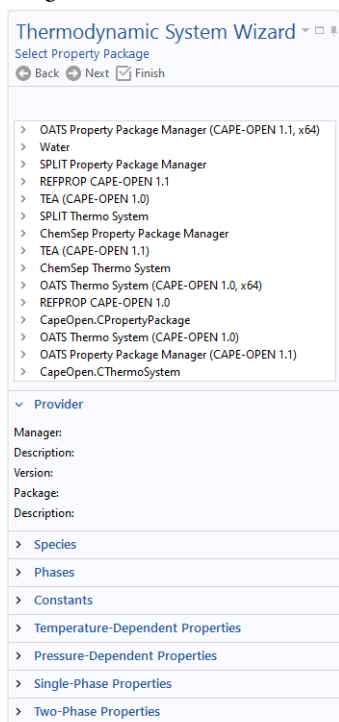


Figure 6-9: Available external thermodynamic systems. Use the Thermodynamic System Wizard to browse the contents of the installed external thermodynamic systems on your system. The example shows the packages shipped with the COCO provider.

SETTINGS

Selecting the **External Thermodynamic System** node shows the thermodynamic system settings with sections for available species, phases, species properties, and mixture

properties. Editing the text in the Label text field changes the name of the system. Below follows a description of the thermodynamic system settings.

Provider

Contains information about the external thermodynamic system provider such as version, CAPE-OPEN compliance, and a short description of the external software that provides the thermodynamic system.

Species

Lists the species included in the selected thermodynamic system.

Phases

Lists the phases included in the selected thermodynamic system for example gas, liquid, gas–vapor, or liquid–liquid.

Species Properties

Lists the parameter values or the functions that describe species properties, for example molar mass, and properties available for pure compounds, such as density as a function of temperature.

Mixture Properties

Lists the available functions that describe mixture properties. One example is the density of a nonideal mixture as a function of composition.

Settings
External Thermodynamic System

Label: Vapor-Liquid System (External) 2

Provider

Manager: COCO_TEA.ThermoPack
Version: Thermo System version 1.0
Property package: HDA
Description: <no description>

Species

Name	CAS	Chemical formula
Hydrogen	1333-74-0	H2
Methane	74-82-8	CH4
Benzene	71-43-2	C6H6
Toluene	108-88-3	C7H8
Biphenyl	92-52-4	C12H10

Phases

Name	State
Vapor	Vapor
Liquid	Liquid

Species Property

Property: Acentric factor, Charge, Critical compressibility factor

Mixture Property

Property: Density, Enthalpy, Enthalpy of formation, Entropy

Figure 6-10: Settings window for an External Thermodynamic System.



The thermodynamic functionality supports package *persistence*. This means that an MPH file for a model that uses a thermodynamic system stores that package definition in the file. You can use the MPH file to access a thermodynamic system defined in the corresponding model without having to store the thermodynamic system as a separate file.

Predefined System

A **Predefined System** uses the built-in database with native functionality to set up one of the following common systems; dry air, moist air, or water and steam.

To create a **Predefined System**, start the **Thermodynamic System Wizard** in one of the following ways:

- Right-click the **Global Definitions** node, select **Thermodynamics** and choose **Predefined System**.
- Using the Reaction Engineering interface, choose **Predefined System** from the **Thermodynamics** section of the **Reaction Engineering** toolbar.
- Correspondingly, using the Chemistry interface, choose **Predefined System** from the **Thermodynamics** section of the **Chemistry** toolbar.

The **Thermodynamic System Wizard** that opens is identical to the wizard used to create a [Thermodynamic System](#), but with the important difference that all settings have been entered for the chosen predefined system. The wizard contains the following steps:

- 1 [Select System](#)
- 2 [Select Species](#)
- 3 [Select Thermodynamic Model](#)

SELECT SYSTEM

In the Select System window, choose one of the predefined systems in the list. Choose among **Dry air**, **Moist air**, and **Water-steam**. Click the Next button (➞) to proceed to the next step in the wizard.

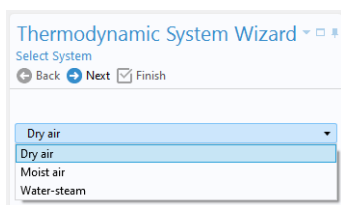


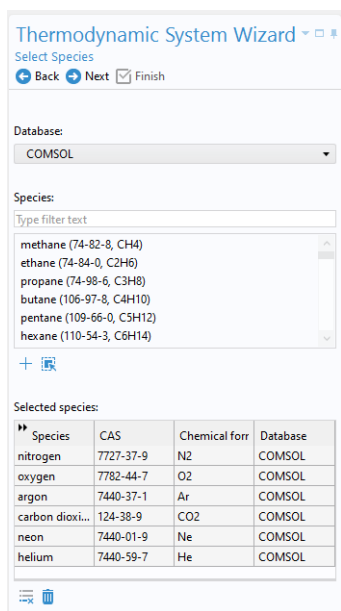
Figure 6-11: Thermodynamic System Wizard: Select Predefined System.

SELECT SPECIES

If Dry or Moist air was chosen in the Select System step, the **Selected species** table is populated with the species relevant for the chosen system. For Dry air the species are: nitrogen, oxygen, argon, carbon dioxide, neon, and helium. For Moist air the list also

contains water. If instead Water-steam was chosen in the Select System step, the Next button in that step guides you directly to the Select Thermodynamic Model pane. No additions of species can be done in the wizard.

To add more species to the Dry air and Moist air systems, use the **Species Type filter text** field to search among the available species in the **COMSOL** database or the [User-Defined Species](#). Species can be searched for by typing the name, CAS number, or the chemical formula. Select one or more species in the list and click the **Add Selected** button () to add species to the **Selected species** table. To remove species from the Selected species table, use either the Remove Selected button () or the Remove All button (), both located below the table. Click the **Next** button () to proceed to the next step in the wizard.



Thermodynamic System Wizard ▾ □ ⓘ

Select Species

⏮ Back ⏭ Next ☒ Finish

Database:

COMSOL ▾

Species:

Type filter text

methane (74-82-8, CH4)
ethane (74-84-0, C2H6)
propane (74-98-6, C3H8)
butane (106-97-8, C4H10)
pentane (109-66-0, C5H12)
hexane (110-54-3, C6H14)

+ 

Selected species:

Species	CAS	Chemical forr	Database
nitrogen	7727-37-9	N2	COMSOL
oxygen	7782-44-7	O2	COMSOL
argon	7440-37-1	Ar	COMSOL
carbon diox...	124-38-9	CO2	COMSOL
neon	7440-01-9	Ne	COMSOL
helium	7440-59-7	He	COMSOL

Figure 6-12: Thermodynamic System Wizard: Select Species.

SELECT THERMODYNAMIC MODEL

This last step of the Thermodynamic System Wizard allows you to choose what thermodynamic model to use. For the system Dry air and Moist air, the Ideal Gas law is chosen by default. For Water-steam, the model Water (IAPWS) is used. For more information about this step in the Thermodynamic System Wizard, see [Select Thermodynamic Model](#) for Thermodynamic System.

Click the **Finish** button () to exit the **Thermodynamic System Wizard** and add the corresponding **Predefined System** under the **Thermodynamics** node. Note that the default node label reflects the chosen predefined system.

SETTINGS

Selecting a **Predefined System** node shows its **Settings** window with a description of the package content and the available properties. For information about the content in the Settings window, see [Settings](#) for **Thermodynamic System** and **Predefined System**.

MODIFYING A PREDEFINED SYSTEM

Please see [Modifying an Existing System](#) for information about how to modify an existing thermodynamic system.

Exporting and Importing Thermodynamic Systems

EXPORT THERMODYNAMIC SYSTEM

Right-click the thermodynamic system node for the thermodynamic system that you want to export ([Figure 6-8](#)) and select **Export Thermodynamic System**. Use the **Select File** window to store the thermodynamic system.

Note that only the thermodynamic system definition is exported, for example the underlying species data (from the database), the included phases, and the applied thermodynamic models. Property values and functions created using the thermodynamic system are not stored.

IMPORT THERMODYNAMIC SYSTEM

Right-click the **Thermodynamics** node and select **Import Thermodynamic System** ([Figure 6-2](#)). Use the **Select File** window to navigate to a thermodynamic system file and click **Open**.

It is possible to manually modify the species properties, such as parameters and temperature-dependent functions, in the created Thermodynamic System by using **Export Thermodynamic System** and **Import Thermodynamic System**. In the saved xml file from **Export thermodynamic system**, there are all parameters and temperature-dependent properties for species, which are saved inside a `<Compound CompID="name"></Compound>` block. For example, to edit the acentric factor of methane from 0.01141 to 0.2 perform the following steps

- Open the saved xml file for the thermodynamic system

- Find the compound methane which starts with the line:

```
<Compound CompID="methane">
```

- Find the line with the acentric factor and change it to

```
<AcentricFactor >0.2</AcentricFactor>.
```

To use the system with the updated species, right-click the Thermodynamics node and select Import System.

For temperature-dependent functions, for example for the (saturated) liquid density, the database input is of the form:

```
<Density>
  <Phase>Liquid</Phase>
  <Coefficients>Tlb;a0;a1;a2;a3;Tub</Coefficients>
  <Data>Tlb;f(Tlb);Tub;f(Tub)</Data>
<Comment></Comment>
</Density>
```

Here T_{lb} and T_{ub} defines the lower and upper bound for a temperature range, and a_1 to a_4 are the corresponding coefficients for a cubic polynomial as $f(T) = a_0 + a_1T + a_2T^2 + a_3T^3$ fitted for that range. Multiple sequential ranges can be added by appending temperature ranges and coefficient sets. The `<Data></Data>` block can be used to input data points directly.

The unit for temperature is K. The unit used for the temperature-dependent properties (fitted polynomials functions) are defined in the table below:

TABLE 6-1: UNIT FOR TEMPERATURE-DEPENDENT PROPERTIES.

PROPERTY	UNIT
Liquid density	mol/m ³
Ideal heat capacity	J/mol/K
Ln vapor pressure	I
Heat of vaporization	J/mol
Vapor viscosity	Pa·s
Ln liquid viscosity	I
Vapor thermal conductivity	W/m/K
Liquid thermal conductivity	W/m/K
Surface tension	N/m

Species Property

A **Species Property** is used to define and compute a pure species property. The available properties consist of both parameters and functions. Some examples of available parameters are molar mass, Lennard Jones diameter, and dipole moment. Some examples of available functions are density, enthalpy, heat capacity, and viscosity. The property functions created are either dependent on temperature alone, or both on temperature and pressure. For all property functions, the first order derivative with respect to temperature and, when applicable, with respect to pressure are automatically defined. The second-order derivatives with respect to temperature and pressure are available for property functions of density and molar volume.

Right-click the relevant **Thermodynamic System** node (see [Figure 6-8](#)), **Predefined System** node, or the relevant **External Thermodynamic System** node, and select **Species Property** () to start the **Species Property Wizard**.

The **Species Property Wizard** contains the following steps:

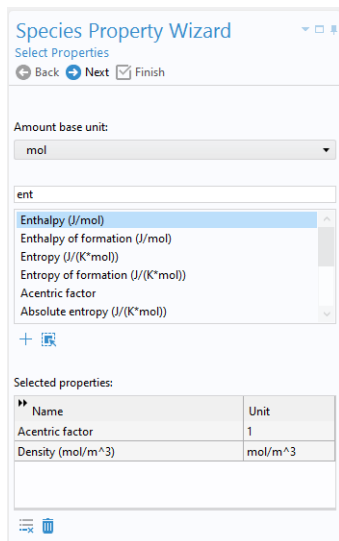
- 1 [Select Properties](#)
- 2 [Select Phase](#)
- 3 [Select Species](#)
- 4 [Species Property Overview](#)

SELECT PROPERTIES

First use the **Amount base unit** list to define the base unit. Select **mol** or **kg**.

Use the filter to search among the available properties. Select one or more properties in the list and click the **Add Selected** button () to add them to the **Selected properties** list.

Click the **Next** button (→) to proceed to the next step, selecting the phase.



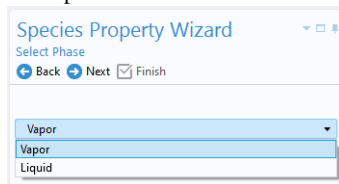
The screenshot shows the 'Species Property Wizard' window, specifically the 'Select Properties' pane. At the top, there are navigation buttons: 'Back' (disabled), 'Next' (active), and 'Finish' (disabled). Below this, the 'Amount base unit' is set to 'mol'. A text input field contains 'ent'. A list of properties is shown, with 'Enthalpy (J/mol)' selected. Other properties include 'Enthalpy of formation (J/mol)', 'Entropy (J/(K*mol))', 'Entropy of formation (J/(K*mol))', 'Acentric factor', and 'Absolute entropy (J/(K*mol))'. Below the list is a '+ Add' button. The 'Selected properties' section contains a table:

Name	Unit
Acentric factor	1
Density (mol/m^3)	mol/m^3

Figure 6-13: Species Property Wizard; Select Properties pane.

SELECT PHASE

Use the list to specify the phase for the species property. The available phases correspond to the ones defined for the thermodynamic system.



The screenshot shows the 'Species Property Wizard' window, specifically the 'Select Phase' pane. At the top, there are navigation buttons: 'Back' (disabled), 'Next' (active), and 'Finish' (disabled). Below this, a dropdown menu is open, showing 'Vapor' as the selected phase. Other options in the list are 'Vapor' and 'Liquid'.

Figure 6-14: Species Property Wizard; Select Phase pane.

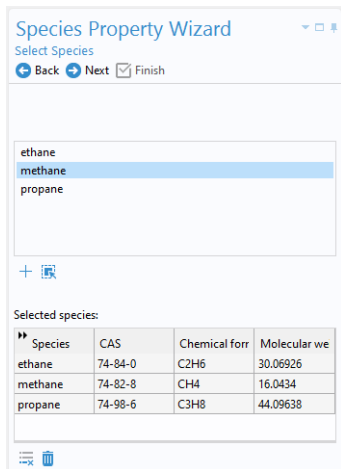
Note that for parameters (constants) this step is skipped.

Click the **Next** button (→) to proceed to the next step, selecting the species.

SELECT SPECIES

Select one or more of the species available in the thermodynamic system and use the **Add Selected** button (+) to add them to the **Selected species** table. One property function is created for each of the selected species.

Click the **Next** button (➡) to proceed to the Species Property Overview in the wizard.



Species Property Wizard

Select Species

Back Next Finish

ethane
methane
propane

+ [icon]

Selected species:

Species	CAS	Chemical formula	Molecular weight
ethane	74-84-0	C ₂ H ₆	30.06926
methane	74-82-8	CH ₄	16.0434
propane	74-98-6	C ₃ H ₈	44.09638

Figure 6-15: Species Property Wizard; the Select Species step.

SPECIES PROPERTY OVERVIEW

The final step of the wizard presents an overview of the parameters and functions defined for the corresponding property. Click the **Finish** button (☑) to exit the wizard and add the species property to the current thermodynamic system.

THE SPECIES NODE UNDER A SYSTEM

When creating a species properties, the resulting parameters and functions are collected under a node with the species name. You can create new functions from an

existing species node under the system. Right-click the species and select **Species Property** to start the **Species Property Wizard**.

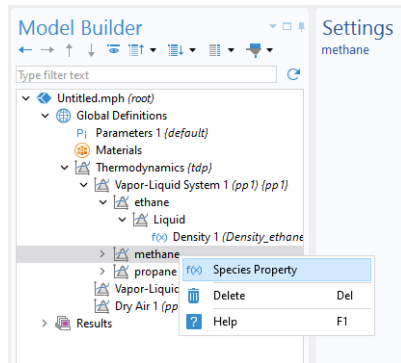


Figure 6-16: Create a Species Property from an existing species node.

SETTINGS

Selecting a **Species Property** node shows the settings for such property.

Settings
Species Property
Plot Create Plot

Label: Density 1
Function name: Density_ethane_Liquid11

Definition

Property name: Density
Unit: mol/m^3
Phase: Liquid

Species

Name
ethane

Arguments

Name	Unit	Description
temperature	K	Temperature
pressure	Pa	Pressure

Derivatives

Name
Density_ethane_Liquid11_Dtemperature
Density_ethane_Liquid11_Dpressure

Second derivatives

Name
Density_ethane_Liquid11_Dtemperature_Dtemperature
Density_ethane_Liquid11_Dtemperature_Dpressure
Density_ethane_Liquid11_Dpressure_Dpressure

> Plot Parameters

Figure 6-17: Species Property settings.

Definition:

Shows the definition of a property that is defined as a parameter or a function, for example the name of the parameter or function.

You can use the **Parameter name** or **Function name** fields to specify or change the name of a parameter or a function. The section also provides information about the property type and the species it is defined for.

For a parameter, this section also displays the numerical value.

For a function, this section displays the arguments for the functions and the first-order derivatives with respect to these arguments.

Plot Parameters

Available for property functions in order to plot a selected function for a given set of argument values.

Apply a **Lower limit** and an **Upper limit** for each argument, and click the **Plot** button () to plot the function using the given argument range. You can also click the **Create Plot** button () in order to create a plot group, under the **Results** node.

Properties Window

To see the reference for constant or temperature-dependent functions, right-click on the function and select **Properties**. This opens the Properties window.

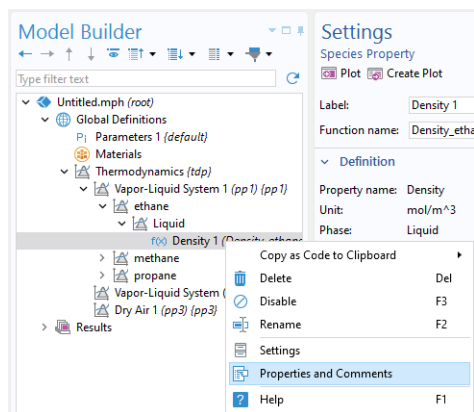


Figure 6-18: Show constant and temperature-dependent properties references for species functions.

The reference for the constant or temperature-dependent property is available in the **Comments** field in the **Node Properties** section.

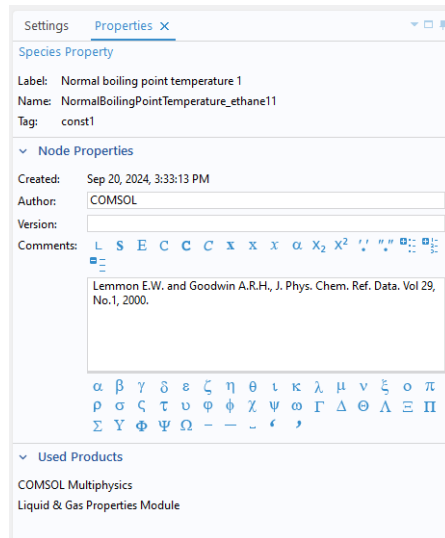


Figure 6-19: Properties window for species function.

Mixture Property

A **Mixture Property** is used to compute a property function that depends on the concentration of the species in a thermodynamic system. Some examples of the available property functions are density, enthalpy, heat capacity, and surface tension.

Apart from the composition, the mixture property functions are also dependent on temperature and pressure. The first-order derivatives with respect to temperature and pressure are automatically defined. The second-order derivatives with respect to temperature and pressure are available for property functions of density and molar volume.

Right-click the relevant **Thermodynamic System** node (see Figure 6-8), **Predefined System** node, or the relevant **External Thermodynamic System** node, and select **Mixture Property** to start the **Mixture Property Wizard**.

The **Mixture Property Wizard** contains the following steps:

- 1 Select Properties
- 2 Select Phase

3 Select Species

4 Mixture Property Overview

SELECT PROPERTIES

The settings in this step corresponds to those for [Select Properties](#) in the **Species Property Wizard**. Click the **Next** button () to proceed to the next step.

SELECT PHASE

Use this list to specify the phase, among the ones available in the system, for the selected mixture property.

Some properties require that the system consists of two phases, for example surface tension. A two-phase system may consist of a combination of liquid–vapor or liquid–liquid phases.

SELECT SPECIES

First select the **Species composition** base unit to be used for function arguments. Select **Mole fraction** or **Mass fraction**.

Select the species to be included in the list. Use the **Add All** button () to add all species in the thermodynamic system. It is also possible to select a subset of the available species. In that case use the **Add Selected** button () to add species. The **Selected species** table is updated as you add species.

Click the **Next** button () to proceed to the next step in the wizard.

MIXTURE PROPERTY OVERVIEW

The final step of the wizard displays an overview of the functions that the wizard has added. Click the **Finish** button () to exit the wizard and add the properties to the current thermodynamic system.

SETTINGS

Selecting a desired **Mixture Property** node to display its settings window.

THE MIXTURE NODE UNDER A SYSTEM

When creating mixture properties, the resulting functions are collected under the **Mixture** node. You can create new functions from an existing **Mixture** node. Right-click

the node and select **Mixture Property** from the context menu to start the **Mixture Property Wizard**.

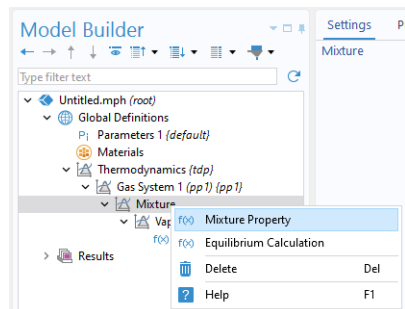


Figure 6-20: Mixture properties can be created by right-clicking a Mixture node under a thermodynamic system.

Equilibrium Calculation

The **Equilibrium Calculation** functionality is used to compute the resulting equilibrium conditions for a mixture of a set of species and phases.

Equilibrium calculations are often used for processes with vapor–liquid equilibrium (VLE), so-called flash calculations. A typical process that requires flash calculations is a distillation process, where a multiphase feed stream is separated into a vapor and a liquid product, and where the concentrations of the species in each phase are required. Equilibrium calculations involve combining the VLE-equations with material balances for the chemical species, and in some cases with an energy balance. Flash calculations are typically used to obtain estimates of:

- Bubble point at given temperature.
- Bubble point at given pressure.
- Dew point at given temperature.
- Dew point at given pressure.
- Equilibrium conditions at given pressure and temperature.
- Equilibrium conditions at given pressure and enthalpy.
- Equilibrium conditions at given pressure and entropy.
- Equilibrium conditions at given internal energy and specific volume.

The last three examples are often considered more difficult, since they require energy balances and relations for computing enthalpy and entropy. Thermodynamics in

COMSOL Multiphysics can handle all of the above cases using the equilibrium calculation functionality. Phase envelopes, bubble point, and dew point can be calculated for any number of species.

Right-click the relevant **Thermodynamic System** node (see [Figure 6-8](#)), **Predefined System** node, or the relevant **External Thermodynamic System** node, and select **Equilibrium Calculation** to start the **Equilibrium Calculation Wizard**.

The **Equilibrium Calculation Wizard** consists of the following steps:

- 1 [Select species](#)
- 2 [Equilibrium Specifications](#)
- 3 [Equilibrium Function Overview](#)

The **Equilibrium Calculation** settings include three types of equilibrium functions.

- The first function type is used to detect whether a phase is present in the system and includes “exist” in its name.
- The amount function computes the total amount of material in each phase.
- The phase composition functions compute the mass or mole fraction of each species in each phase, depending on the selected base unit for the equilibrium calculation.

SELECT SPECIES

Select one or more of the species available in the thermodynamic system and use the **Add Selected** button () to add them to the **Selected species** table. Click the **Next** button () to proceed to the next step in the wizard.

EQUILIBRIUM SPECIFICATIONS

Use the **Amount base unit** list to use either **kg** or **mol** as base unit.

Select two **Equilibrium conditions** that define the current equilibrium, for example a given pressure and a temperature. These equilibrium conditions are used as arguments in the equilibrium functions, in addition to the composition (overall fractions of species).

The available equilibrium conditions are: **Temperature**, **Pressure**, **Phase fraction**, **Energy** (or **Internal energy of formation**), **Enthalpy** (or **Enthalpy of formation**), **Specific volume**, **Density**, and **Entropy** (or **Entropy of formation**). For chemical reactions, it is

recommended to use **Enthalpy of formation**, **Entropy of formation**, or **Internal energy of formation**, since they account for heat of reactions.

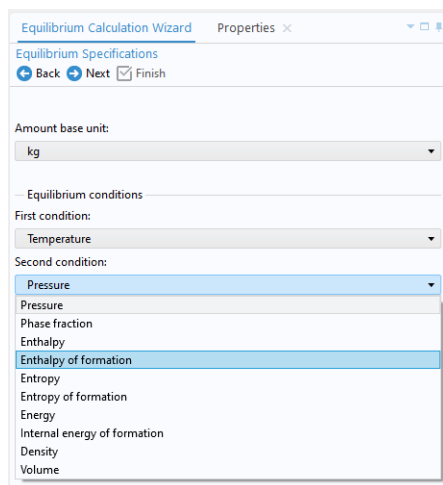


Figure 6-21: Equilibrium Calculation Wizard; Equilibrium Specifications pane.

Selecting **Phase fraction** as one of the equilibrium conditions activates the **Solution type** input field. This can be used to indicate the direction of the desired solution, which is of great use especially near critical points. The options **Undefined**, **Normal**, or **Retrograde** define different directions for the search of the solution to the equilibrium equations. Using **Normal** means that the derivative of the vapor phase fraction with respect to temperature (at constant pressure and composition) is kept positive and the derivative of the vapor phase fraction with respect to pressure (at constant temperature and composition) is kept negative. Using **Retrograde** means that the opposite sign of the previous mentioned derivatives are enforced.

For a single species, the critical point is the highest pressure and temperature at which two phases (liquid and vapor) are distinguishable. However, for some multispecies systems, the critical point is a point between the dew point and the bubble point. In this case, the critical point does not represent the maximum pressure or the maximum temperature of vapor–liquid coexistence. This phenomenon is known as retrograde condensation. This means that under isothermal conditions, when the pressure decreases, some of the vapor condenses into liquid instead of expanding or vaporizing. An example of such system is formation of liquid hydrocarbons in a gas reservoir as the pressure decreases below the dew point pressure. In this case, setting **Solution type** to **Normal** or **Retrograde** may not be sufficient to distinguish between the two solutions.

Note that the **Solution type** setting is only available for a built-in thermodynamic systems. For external thermodynamic systems, the corresponding functionality needs to be supplied by the thermodynamic software provider. For instance, the COCO/TEA provider does not support the **Normal** or **Retrograde** options. In those cases, the **Solution type** should be **Undefined**.

Click the **Next** button () to proceed to the next step.

EQUILIBRIUM FUNCTION OVERVIEW

In this step, you can review all the functions including units and arguments. Click the **Finish** button () to exit the wizard and add functions at equilibrium state to the current thermodynamic system.

THE EQUILIBRIUM CALCULATION NODE

When creating an **Equilibrium Calculation**, the resulting functions are collected under the **Mixture** node. You can create new functions from an existing **Mixture** node. Right-click it and select **Equilibrium Calculation** to start the **Equilibrium Calculation Wizard** (see [Figure 6-20](#)).

SETTINGS

Selecting an **Equilibrium Calculation** node displays the settings including the property functions, see [Figure 6-22](#).

Plot Parameters:

Can be used to evaluate and plot the equilibrium function for a range of argument values. First specify which of the included functions to plot. Then apply a **Lower limit** and **Upper limit** for each argument and click the **Plot** button (📊).

Settings Properties ×

Equilibrium Calculation

Plot Create Plot

Label: Equilibrium Calculation 2

Definition

Property name: Equilibrium calculation

Functions

Name	Unit	Description
Flash1_2_PhaseExist_Gas	1	Presence of Gas phase
Flash1_2_PhaseAmount_Gas	kg	Amount in Gas phase
Flash1_2_PhaseCompositi...	kg/kg	Fraction of ethane in Gas phase
Flash1_2_PhaseCompositi...	kg/kg	Fraction of methane in Gas phase
Flash1_2_PhaseCompositi...	kg/kg	Fraction of propane in Gas phase

Name:

Arguments

Name	Unit	Description
temperature	K	Temperature
pressure	Pa	Pressure
ethane	kg	Amount ethane
methane	kg	Amount methane

Phases

Gas

Species

ethane

methane

propane

Plot Parameters

Select a function: Flash1_2_PhaseExist_Gas

Argument	Lower limit	Upper limit
temperature	298.15	373.15
pressure	101325	101325
ethane	1	1
methane	1	1
propane	1	1

Figure 6-22: Settings window; Equilibrium Calculation.

Generate Chemistry

The Chemistry interface includes functionality to automatically define and use mixture properties by interfacing to a Thermodynamic System node. How to manually set this

coupling up is described in [Coupling with the Reaction Engineering and the Chemistry Interfaces](#).

The **Generate Chemistry Wizard** can be used to directly create a Chemistry interface from an existing Thermodynamics System node. The resulting Chemistry interface is automatically populated with the selected species, and is set up to be fully coupled to the thermodynamics. This means for example that functions for mixture properties, like the density enthalpy, are automatically defined and used in the generated Chemistry interface.

Right-click the relevant **Thermodynamic System** node, **Predefined System** node, or **External Thermodynamic System** node, and select **Generate Chemistry** to start the **Generate Chemistry Wizard**.

The **Generate Chemistry Wizard** contains the following steps:

- 1 [Select Species](#)
- 2 [Chemistry Settings](#)

SELECT SPECIES

Select the species to be included in the list. Use the **Add All** button () to add all species in the thermodynamic system. It is also possible to select a subset of the available species. In that case use the **Add Selected** button () to add species. The **Selected species** table is updated as you add species.

Click the **Next** button () to proceed to the next step in the wizard.

CHEMISTRY SETTINGS

The table at the top shows the selected species and their respective chemical formula. For each species, a corresponding Species feature will be added in the Chemistry interface. As an option, use the **Name in Chemistry** columns to specify another name of the species in Chemistry. This means that the species name could be different in Thermodynamics and Chemistry for a species. By default the chemical formula is used.

Component

Use the **Component** list to select the model component to which the Chemistry interface will be added. The already existing components are shown in the top of the list. A new component can be created by selecting the desired space dimension.

Mass Transfer

The **Mass transfer** list controls the intended mixture type for the Chemistry interface. Select **Diluted species** to model a mixture where the species are assumed diluted in a

high concentration solvent. For this choice, also select **Solvent**. Optionally, select **Concentrated species** to model a mixture where all species are assumed to contribute to the mixture properties. In the resulting Chemistry interface the **Type** setting in the Mixture Properties section is defined accordingly.

Phase

Select the Phase of the mixture from the **Phase** list. Included in the list are the phases available in the Thermodynamic System from which the Chemistry is generated.

Click the **Finish** button () to create the desired Chemistry interface and exit the wizard.

Generate Material

Once a thermodynamic system is defined, it can be used to define a multitude of property functions. The Generate Material option provides a fast route to defining the material properties most commonly required for fluid flow, heat transfer and mass transfer. By specifying the composition and the phase of the mixture, a Material node can automatically be set up. The material node can include properties such as the density, heat capacity, heat capacity ratio, thermal conductivity, viscosity, and the diffusivity. Two types of property functions can be used, either functions from the underlying thermodynamics, or interpolation functions created from these. The interpolation functions are computationally more efficient since it is cheap in terms of CPU-time to evaluate polynomials, whereas the thermodynamics functions are more complex and potentially solve implicit systems iteratively. Interpolation functions are appropriate when the temperature and pressure is not expected to vary wildly, and a sensible number of nodes is needed to reproduce the behavior, then the cost is paid upfront during the creation of the material. The computational savings are typically most pronounced for larger space-dependent models of higher dimensions. To validate that the interpolation error is acceptable, create two materials, one with, and one without interpolation functions, and compare the material properties in the relevant region of (T, p)-space.

Right-click the relevant **Thermodynamic System** node (or an **External Thermodynamic System** node or **Predefined System** node), and select **Generate Material** to start the **Generate Material Wizard**.

The **Generate Material Wizard** contains the following steps:

- 1 [Select Phase](#)
- 2 [Select Species](#)

3 Select Properties

4 Define Material

SELECT PHASE

Use this list to specify the phase of the material. Select among the ones available in the system. This step of the Generate Material Wizard is not available for the **Predefined Systems Dry Air** and **Moist Air**.

Click the **Next** button () to proceed to the next step in the wizard.

SELECT SPECIES

Select the species to be included in the list. Use the **Add All** button () to add all species in the thermodynamic system. It is also possible to select a subset of the available species. In that case use the **Add Selected** button () to add species. The **Selected species** table is updated as you add species.

Notice that when using Generate Material for the **Predefined System Water-steam** this step is not included in the wizard.

Material Composition

Specify the mixture composition in terms of the **Mole fractions** or **Mass fractions** of all species. The fractions should sum to one. This setting is only available for systems with more than one component, and for materials with more than one selected species.

Click the **Next** button () to proceed to the next step in the wizard.

SELECT PROPERTIES

Use this step to select the material properties to be included.

For a liquid phase material the following properties are available:

- Density, (ρ)
- Diffusion coefficient at infinite dilution
- Heat capacity (C_p)
- Thermal conductivity (k)
- Viscosity (μ)

For a gas phase material, the Heat capacity ratio ($\gamma = C_p/C_v$) is also available.

To add a property to the **Selected properties** list click the **Add Selected** button ().

To remove properties, use the **Remove Selected** button () or the **Remove All** button ()

When adding the **Diffusion coefficient at infinite dilution**, also select which of the species is the solvent. The material will include one diffusion coefficient for each of the remaining (solute) species.

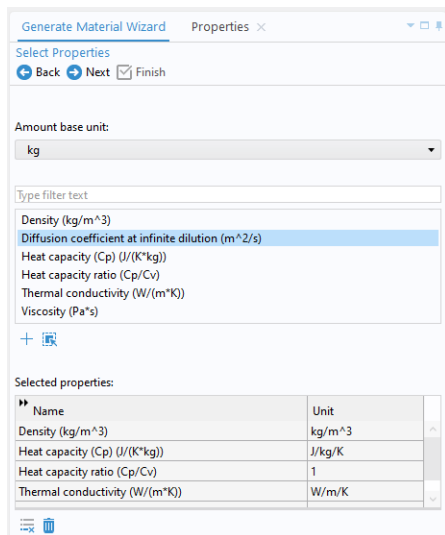


Figure 6-23: Generate Material Wizard; Select Properties pane with Diffusion coefficient at infinite dilution added.

Additional Properties

The additional properties below are defined from the selected properties and added to the output properties of the material.

For a liquid phase material the isobaric coefficient of thermal expansion is defined as

$$\alpha_{\text{iso}} = -\frac{1}{\rho} \frac{\partial \rho}{\partial T} \quad (6-1)$$

For a gas phase material, the speed of sound c , and the isentropic compressibility χ_f are also defined as

$$c = \sqrt{\gamma \frac{\partial p}{\partial \rho}}, \quad \chi_f = \frac{1}{\gamma \rho} \frac{\partial \rho}{\partial p} \quad (6-2)$$

Click the **Next** button (➡) to proceed to the next step.

DEFINE MATERIAL

Component

Use the **Component** list to select where the Material will be added. The already existing components are shown in the top of the list. A new component can be created by selecting the desired space dimension. Select **Global** at the end of the list to add the material under **Global Definitions**. A global material can be utilized in any component, through the use of a Material Link. It can also be used to define phase specific properties when linked to a subnode of a **Porous Material**.

Generate Material Wizard Properties ×

Define Material

Back Next Finish

Component:

2D: New

Function type:

Interpolation

Interpolation Points:

☒ Same for all functions ☐ Individual

Maximum number of interpolation points:

Normal (400)

Ranges and interpolation points:

	Low:	High:	Number of points:
Temperature, K:	273	373	20
Pressure, Pa:	101325	201325	15

Pressure form: Pa

Figure 6-24: Generate Material Wizard; Define Material pane.

Function Type

The material can be set up to use functions defined under the thermodynamics system, or interpolation functions defined from these. The interpolation functions are significantly faster to evaluate, especially for space dependent models, but requires that the temperature range and the pressure range are specified.

Select **Thermodynamics** to base the material properties on functions in the current thermodynamic system. The required functions are created, if not already present, and

added to the system. When evaluating material properties, the underlying thermodynamics functions are called.

Select **Interpolation** to generate interpolation functions for the material properties. In this case the required functions are first created and added to the system. Next, the functions are evaluated in a number of sampling points over a given range, for both temperature and pressure, leading to a grid of (T, p) pairs. The result of the evaluation is stored and used to define interpolation functions for all material properties.

For the **Interpolation points**, select **Same for all functions** to use the same number of sampling points for all functions. By selecting **Individual**, the number of sampling points can be given per function. The latter can for example be used to increase the number of points for a function that is known to vary faster than the rest of the functions.

Select one of the preset levels for the **Maximum number of interpolation points**. This is used to provide a maximum size of the underlying data for the interpolation functions. The **Number of points** for temperature multiplied by the **Number of points** for pressure should not exceed the **Maximum number of interpolation points**. If the total number of sampling points for any function, varying temperature and pressure, is exceeded, a warning is shown.

There are two types (or forms) of input values for pressure, **Pa** and **log10(Pa)**. The later is often used in cases where the pressure changes in a very large range, for example, for a system from atmosphere to a vacuum system. The **log10(Pa)** is expressed as $\log_{10}(p/1[\text{Pa}])$, here p is pressure ([Pa]).

When the selection from **Pressure form** changes, the corresponding input fields for pressure will also follow the change, for both **Same for all functions** and **Individual**, respectively.

Define the temperature range and pressure range for the temperature and pressure by providing the **Low** and **High** end values. To neglect the variation in either temperature or pressure, give the same value for the low and high end value. This can for example be used when the pressure variation is known to be negligible. For each range, also define the number of evaluation points in the **Number of points** field. The underlying functions are evaluated at uniformly distributed positions across the temperature and pressure (uniform logarithmic values for **log10(Pa)**) range.

When **Individual** has been selected for the **Interpolation points**, the **Ranges and interpolation points** can be specified per function to be created.

Click the **Finish** button () to create the specified Material and exit the wizard.

Coupling with the Reaction Engineering and the Chemistry Interfaces

Using the **Reaction Engineering** or **Chemistry** interfaces, all species property parameters and property functions required by these interfaces can be created automatically by coupling to an existing **Thermodynamic System**, **Predefined System**, or **External Thermodynamic System** under **Thermodynamics**. Examples of species properties that can be created automatically are molar mass, heat capacity, enthalpy, and entropy for each species. Parameters and functions for these properties are created by the thermodynamic system. The **Reaction Engineering** and **Chemistry** interfaces can also be used to define transport properties for the resulting mixture (all species in the interface). When coupled, the following mixture properties can be automatically created: heat capacity, density, thermal conductivity, and dynamic viscosity.



The Reaction Engineering interface and the Chemistry interface are only available in a limited set of add-on products. See www.comsol.com/products/specifications/ for more details on availability.

Note that using a thermodynamic system significantly increases the modeling capabilities in the **Reaction Engineering** and **Chemistry** interfaces. All ideal and nonideal thermodynamic models, for gases and liquids, are directly available and also automatically updated by editing the settings for the thermodynamic system in use. The **Chemistry** interface can furthermore be used to make the mixture properties readily available in space-dependent models for modeling of mass transport, heat transfer, or fluid flow.

REACTION ENGINEERING

Mixture Properties

You can couple a Reaction Engineering interface with an existing thermodynamic system in the **Reaction Engineering** interface's settings window.

You need to have at least one species defined in the Reaction Engineering interface in order to couple it to a thermodynamic system. You can make this coupling in the Settings window for the **Reaction Engineering** interface by selecting the **Thermodynamics** checkbox in the **Mixture Properties** section.

Select a thermodynamic system from the **Thermodynamic system** list. Use the **Phase** list to select the phase to be used in the **Reaction Engineering** interface.

Mixture Properties
☒ Thermodynamics
 Thermodynamic system:
 Gas System 1 (pp1) (pp1)
 Phase:
 Gas
 Reactor pressure:
 Ideal gas law

Figure 6-25: Coupling Reaction Engineering Interface with Thermodynamics.

Species Matching

The **Species Matching** section is activated when the **Thermodynamics** checkbox is selected in the **Mixture Properties** section; see above. The species in the Reaction Engineering interface can be matched to a species in the thermodynamic system. This step ensures that the arguments in the thermodynamic system functions are correctly defined.

Use the lists in the **From Thermodynamics** column to match each species in the interface to a species in the coupled thermodynamic system.

For each species matched, the required property parameters and functions are added under to the corresponding thermodynamic system.

When all species are matched, the interface is considered fully coupled and functions representing mixture properties, such as density, are also added automatically under the corresponding thermodynamic system.

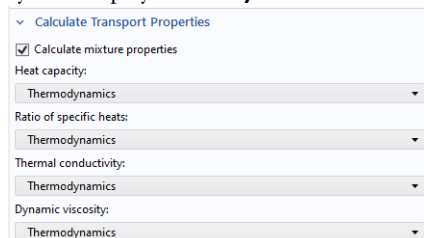
Species	From Thermodynamics
Ar	Ar

Figure 6-26: Matching the species in Reaction Engineering to those in the corresponding thermodynamic system.

Calculate Transport Properties

When the interface is fully coupled to a thermodynamic system, property functions for the mixture transport properties are created automatically when the **Calculate mixture**

properties checkbox is selected. The properties calculated by the thermodynamic system display **Thermodynamics** in the corresponding list; see below.



Calculate Transport Properties

☒ Calculate mixture properties

Heat capacity:

Thermodynamics

Ratio of specific heats:

Thermodynamics

Thermal conductivity:

Thermodynamics

Dynamic viscosity:

Thermodynamics

Figure 6-27: Select the Calculate mixture properties checkbox when coupled to a thermodynamic system.

CHEMISTRY

Mixture Properties

You can couple a Chemistry interface with an existing thermodynamic system in the **Chemistry** interface's settings window. You can make this coupling in the Settings window for the **Chemistry** interface by selecting the **Thermodynamics** checkbox in the **Mixture Properties** section. You need to have at least one species defined in the Chemistry interface in order to couple it to a thermodynamic system.

Select a thermodynamic system from the **Thermodynamic system** list. Use the **Phase** list to select the phase to be used in the **Chemistry** interface.

Species Matching

The **Species Matching** section is activated when the **Thermodynamics** checkbox is selected in the **Mixture Properties** section. Here you can match the variables for the concentrations, and by this calculate mixture properties (transport and thermodynamic properties). For information on how to specify the dependent variables to be used, see [Species Matching](#) in [The Chemistry Interface](#) documentation.

You can match the species in the Chemistry interface with those in the corresponding thermodynamic system in the **Species matching** section's table. Use the lists in the column with the title **From Thermodynamics** to match each species in the Chemistry interface to a species in the corresponding thermodynamic system. This ensures that the composition function arguments in the thermodynamic system are correctly defined.

For each species matched, the species property parameters and functions required by the Chemistry interface are automatically created and added under the corresponding thermodynamic system.

When all species are matched, the interface is fully coupled to Thermodynamics and functions representing mixture properties, such as the density, are added automatically under the thermodynamic system that the interface is coupled to.

▼ Mixture Properties

Type:
 Diluted species ▼

☒ Thermodynamics

Thermodynamic system:
 Dry Air 1 (pp2) (pp2) ▼

Phase:
 Gas ▼

Density:
 Thermodynamics ▼

▼ Species Matching

— Bulk species

Species solved for:
 User defined ▼

Spec	Type	Molar concentra	Value (mol/m ³)	From Thermodynarr
------	------	-----------------	-----------------------------	-------------------

Figure 6-28: Matching concentration variables in a Chemistry interface, as well as species in a coupled Thermodynamic System.

Calculate Transport Properties

When the interface is fully coupled to a thermodynamic system, property functions for the mixture transport properties are created automatically when the **Calculate mixture properties** checkbox is selected. The properties calculated by the thermodynamic system display **Thermodynamics** in the corresponding list; see Figure 6-27.

MIXTURE PROPERTY DEFINITIONS WHEN COUPLING TO THERMODYNAMICS

When all species in a physics interface (**Reaction Engineering** or **Chemistry**) are matched to the corresponding species in a thermodynamic system, the mixture properties are calculated based on the composition of the mixture. For example, consider $Z_{\text{mix}}(T, P, n_1, \dots, n_m)$, which denotes a generic extensive mixture property for a mixture of m number of species. The property function's arguments are the temperature T , the pressure P , and the number of moles, n , for each species.

When the system is fully matched, the mixture property Z_{mix} is defined as

$$Z_{\text{mix}} = \sum_{i=1}^m n_i \bar{Z}_i \quad (6-3)$$

where $\bar{Z}_i$ is the partial molar of species i :

$$Z_i = \left(\frac{\partial Z_{\text{mix}}(T, P, n_1, n_2, \dots, n_m)}{\partial n_i} \right)_{T, P, n_{j \neq i}} \quad (6-4)$$

The definition of partial molar properties can be rewritten using mole fraction derivatives as:

$$Z_i(T, P, n_1, \dots, n_m) = \bar{Z}_i(T, P, x_1, \dots, x_m) = \quad (6-5)$$

$$\left(Z_{\text{mix}}(T, P, x_1, \dots, x_m) + \left(\frac{\partial Z_{\text{mix}}(T, P, x_1, \dots, x_m)}{\partial x_i} \right)_{T, P, x_{i \neq j}} - \sum_{i=1}^m \left(x_i \left(\frac{\partial Z_{\text{mix}}(T, P, x_1, \dots, x_m)}{\partial x_i} \right)_{T, P, x_{i \neq j}} \right) \right)$$

When the system is partially coupled, which means that some but not all species have been coupled, the mixture property is instead calculated assuming ideal mixing:

$$Z_{\text{mix}} = \sum_{i=1}^m n_i Z_i(T, P) \quad (6-6)$$

using the pure species properties $Z_i = Z_{\text{mix}}(T, P, x_i = 1, 0, \dots, 0)$.

Evaluating a Property Function in a Physics Interface

The functions set up using a thermodynamic system in Thermodynamics can be used directly in any physics interfaces in COMSOL Multiphysics, in the same way as for standard functions. Simply type in the function name with the proper arguments in the appropriate field of the model to use the function. [Figure 6-29](#) shows an example of this from the Reaction Engineering interface. In this example, the **Species Thermodynamic Expressions** section gets input for the molar entropy, the enthalpy and the heat capacity of toluene. These functions, generated using **Species Property** for a

thermodynamic system containing toluene, have been entered in the corresponding input fields using the appropriate arguments.

Thermodynamic Expressions

User defined

Heat capacity at constant pressure:
 C_p Cp_benzene(T, p) J/(mol·K)

Molar enthalpy:
 h h_benzene(T, p) J/mol

Molar entropy:
 s 0 J/(mol·K)

Figure 6-29: The enthalpy and heat capacity of toluene are evaluated using property calculations through a thermodynamics system. In this case, the functions are temperature and pressure dependent properties for a pure component.

User-Defined Species

Use this feature to specify the properties of a new species not available in the included COMSOL database. This feature can also be used to edit a database species, by using an existing species as a template, to create a new one. A User-Defined Species can be included in a Thermodynamic System feature, or Predefined System feature, together with predefined database species and other user-defined species.

The data needed for a user-defined species includes both material properties, such as the molar mass and the vapor pressure, and properties for specific thermodynamic models or transport models (see [Thermodynamic Models and Theory](#)). For instance, in order to use the [UNIFAC](#) thermodynamic model, you need to define UNIFAC groups for the new species.

Note that the thermodynamics calculator includes measures to handle missing species properties, sometimes by applying approximations. For instance, if the Wilson volume is not available, the liquid volume at normal boiling point is used instead. If data for this is also missing, it is estimated from the saturated liquid density correlation.

When creating a user-defined species it is recommended to add the following common material properties:

- Molecular mass
- Critical properties
- Acentric factor
- Absolute entropy
- Standard enthalpy of formation
- Ideal gas heat capacity

- Vapor pressure
- Heat of vaporization
- Saturated liquid density
- Transport properties

EXAMPLE MODEL

For an example of using a User-Defined Species see this application example:



Dissociation in a Tubular Reactor: Application Library path
Chemical_Reaction_Engineering_Module/Thermodynamics/dissociation

THE USER-DEFINED SPECIES WIZARD

To create a **User-Defined Species**, start the **User-Defined Species Wizard** in one of the following ways:

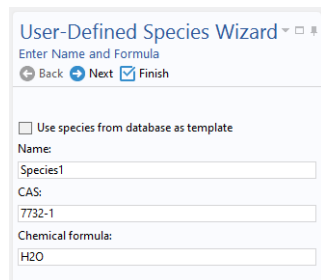
- Right-click the **Global Definitions** node, select **Thermodynamics** and choose **User-Defined Species**.
- If the model already includes the **Thermodynamics** node, right-click it and choose **User-Defined Species**.
- Using the **Reaction Engineering** or **Chemistry** interface, choose **User-Defined Species** from the **Thermodynamics** section of the **Reaction Engineering** or **Chemistry** toolbar.

The **User-Defined Species Wizard** contains the following steps:

- 1 [Enter Name and Formula](#)
- 2 [Enter Parameters](#)
- 3 [Define Properties](#)

ENTER NAME AND FORMULA

Enter the **Name**, **CAS** number, and the **Chemical formula** for the new species.



User-Defined Species Wizard ▾ □ #

Enter Name and Formula

⏮ Back ⏭ Next ⏹ Finish

☐ Use species from database as template

Name:

Species1

CAS:

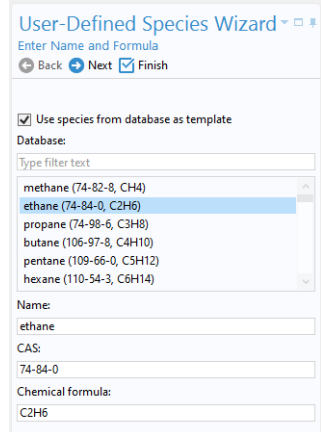
7732-1

Chemical formula:

H2O

Figure 6-30: User-Defined Species Wizard; Enter Name and Formula.

If you want to edit a species in the COMSOL database, select the **Use species from database as template** checkbox. Then locate the species to edit in the list. The filter text field can be used to search among the available species.



User-Defined Species Wizard ▾ □ #

Enter Name and Formula

⏮ Back ⏭ Next ⏹ Finish

☒ Use species from database as template

Database:

Type filter text

- methane (74-82-8, CH4)
- ethane (74-84-0, C2H6)
- propane (74-98-6, C3H8)
- butane (106-97-8, C4H10)
- pentane (109-66-0, C5H12)
- hexane (110-54-3, C6H14)

Name:

ethane

CAS:

74-84-0

Chemical formula:

C2H6

Figure 6-31: User-Defined Species Wizard; edit an available species by selecting *Use species from database as template*.

Click the **Next** button (⏭) to proceed to the next step.

ENTER PARAMETERS

Define the necessary parameter for the new species.

Constants

Use this table to define material constants such as molar mass, critical temperature, and standard enthalpy of formation.

User-Defined Species Wizard ▾ □

Enter Parameters

⬅ Back ➡ Next ☑ Finish

— Constants

Name	Values	Unit
Absolute entropy	229.6500...	J/mol/K
Critical compressibility fa...	0.279	1
Critical pressure	4880000	Pa
Critical temperature	305.42	K
Critical volume	0.0001455	m ³ /mol
Dipole moment	0	C-m
Heat of combustion (hig...	-1560880...	J/mol
Lennard-Jones diameter (...)	4.443e-10	m
Lennard-Jones energy (p...	215.7	K
Liquid volume at normal...	4.395902...	m ³ /mol
Molecular mass	30.069	g/mol
Normal boiling point tem...	184.56	K
Standard enthalpy of for...	-84741	J/mol
Standard molar enthalpy...	0	J/mol
Standard molar enthalpy...	0	J/mol
Standard molar enthalpy...	0	J/mol
Standard molar enthalpy...	0	J/mol
Standard molar entropy (i...	0	J/mol/K

Figure 6-32: User-Defined Species Wizard; enter material constants.

Structure Information

If you want to use the **UNIFAC** model, specify the structure formula, the UNIFAC groups and their number occurrences.

Structure information

Structure formula:

CH₃CH₃

UNIFAC group:

Group	Occurrences
CH ₃	2

+ 

Figure 6-33: User-Defined Species Wizard; specify the structure information in the Enter Parameters step.

Model Parameters

Specify the parameters for the thermodynamic models and transport models in use.

Model parameters		
Name	Values	Unit
Acentric factor	0.098	1
Chao-Seader acentric fac...	0.1064	1
Chao-Seader liquid volu...	6.8e-05	m^3/mol
Chao-Seader solubility p...	12379.3085...	J^0.5/m^1.5
COSTALD acentric factor	0.0983	1
COSTALD volume param...	0.0001458	m^3/mol
Fuller diffusion volume	45.66	cm^3
Parachor	8.8676e-06	kg^0.25*m...
Peng-Robinson (Tiwu) pa...	0.311041	1
Peng-Robinson (Tiwu) pa...	0.866279	1
Peng-Robinson (Tiwu) pa...	1.29869	1
Rackett parameter	0.2808	1
Solubility parameter	13500	J^0.5/m^1.5
Stockmayer delta param...	0	1
UNIQVAC Q parameter	1.696	1
UNIQVAC R parameter	1.802	1
Wilke-Chang association...	1	1
Wilson volume parameter	0	m^3/mol

Figure 6-34: User-Defined Species Wizard; specify model parameters.

Click the **Next** button (➡) to proceed to the next step.

DEFINE PROPERTIES

The last step in the wizard is to add temperature-dependent properties for the new species. All temperature-dependent properties are defined using cubic polynomials on the form

$$f(T) = a_0 + a_1T + a_2T^2 + a_3T^3 \quad (6-7)$$

Each property can consist of an arbitrary number of temperature intervals, each using the above form. Click the **Add** button (+) under the table for a specific property to add an interval.

The following temperature-dependent properties are necessary in order to compute the basic set of mixture properties when including the **User-Defined Species** in a thermodynamic system:

- Ideal gas heat capacity
- Liquid density
- Vapor pressure
- Heat of vaporization

Note that some thermodynamic properties, such as the enthalpy and entropy, of a species or mixture, are estimated from the ideal gas heat capacity and depends on the thermodynamic model applied for the system (see [Thermodynamic Properties Definitions](#)).

Click the **Finish** button (☒) in the top of the Settings window to finalize the species and add it to the **User-Defined Species** node under **Thermodynamics**.

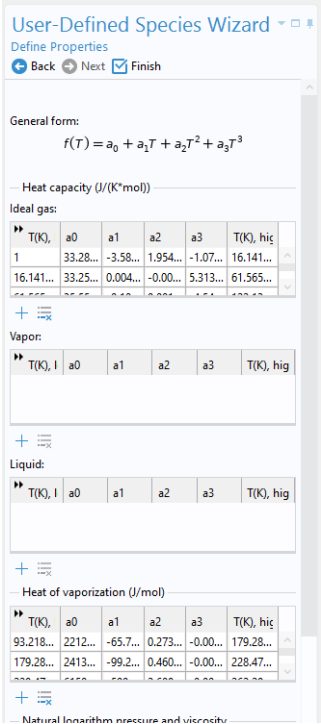


Figure 6-35: User-Defined Species Wizard; specify temperature-dependent properties.

SETTINGS

Selecting a species node under **User-Defined Species** shows the Settings window including the definitions of all species properties. Properties are categorized into

sections as seen in [Figure 6-36](#) below. All properties can be edited by expanding the corresponding section.



When a species properties has been changed, any **Thermodynamic System** that includes the species needs to be updated for the changes to take effect (see [Update From Species](#)).



Figure 6-36: The Settings window for a User-Defined Species.

References

1. www.colan.org
2. Cape Open to Cape Open Simulation Environment, www.cocosimulator.org/index_download.html

Thermodynamic Models and Theory

In this section:

- [Introduction](#)
- [Thermodynamic Models](#)
- [Selecting the Right Thermodynamic Model](#)
- [Species Property References](#)
- [Thermodynamic Properties Definitions](#)
- [Standard Enthalpy of Formation and Absolute Entropy Terms](#)
- [Reference State](#)
- [Transport Properties](#)
- [Surface Tension](#)
- [References](#)

Introduction

In this chapter, we review the theory behind the thermodynamic properties database and its functions. The thermodynamic models in the database are available for single phase, gas or liquid, and phase equilibrium systems for two or more phases such as vapor–liquid equilibrium (VLE), vapor-liquid–liquid equilibrium (VLLE) and liquid–liquid equilibrium (LLE).

Thermodynamic Models

In the following sections, the available thermodynamic models are described:

- [Equation of State Models](#)
- [Liquid Phase Models](#)

EQUATION OF STATE MODELS

An equation of state (EOS) model is an algebraic relation between the absolute pressure (P), the molar volume (V), and the absolute temperature (T). An equation of state that is at least cubic in volume and in the form of

$$Z \equiv \frac{PV}{RT} = f_v(V, T) \quad (6-8)$$

where Z is the compressibility factor and R the universal gas constant, can be used to describe both gases and liquids. The cubic equations of state are widely used to describe the gaseous and liquid volumetric behavior of pure fluids and also all other properties with extensions to mixtures. A comprehensive comparison of different cubic EOS models can be found in [Ref. 2](#) and [Ref. 3](#).

The equations of state models available in the thermodynamic properties database are:

- [Ideal Gas Law](#)
- [Peng–Robinson](#)
- [Peng–Robinson \(Twu\)](#)
- [Soave–Redlich–Kwong](#)
- [Soave–Redlich–Kwong \(Graboski–Daubert\)](#)
- [Water \(IAPWS\)](#)

Ideal Gas Law

The ideal gas law equation of state is

$$P = \frac{RT}{V} \quad (6-9)$$

As the name suggests, the ideal gas law is only applicable to gases. In fact, its use is limited to gases at low to moderate pressures.

Peng–Robinson

The classical Peng–Robinson (PR) equation of state [Ref. 5](#) is given by

$$P = \frac{RT}{V-b} - \frac{a\alpha}{V^2 + 2bV - b^2} \quad (6-10)$$

where for species i ,

$$a_i = \Omega_A \frac{R^2 T_c^2}{P_c} \quad (6-11)$$

$$b_i = \Omega_B \frac{RT_c}{P_c} \quad (6-12)$$

$$\Omega_A = \frac{1}{3} + \frac{1}{3}\Omega_B(4 + 10\Omega_B) \quad (6-13)$$

$$\Omega_B = \left(\beta - \frac{63}{1024\beta} - 0.03125 \right) \quad (6-14)$$

$$\beta = \left(\frac{351}{32768} + \frac{27}{2048} \sqrt{2} \right)^{\frac{1}{3}} \quad (6-15)$$

The alpha function, α_i , is given by

$$\alpha_i = [1 + (0.37464 + 1.54226\omega_i - 0.26992\omega_i^2)(1 - \sqrt{T_{r,i}})]^2 \quad (6-16)$$

For mixtures

$$a\alpha = \sum_i \sum_j x_i x_j (1 - k_{PR,i,j}) \sqrt{(a\alpha)_i (a\alpha)_j} \quad (6-17)$$

$$b = \sum_i x_i b_i \quad (6-18)$$

The binary interaction parameters (BIPs), k_{PR} , are symmetric with zeros in the diagonal:

$$k_{PR} = k_{PR,i,j} \quad (6-19)$$

$$k_{PR,i,j} = 0 \quad (6-20)$$

When binary interaction parameters are missing in the database for a set of species, the value is set to zero (a warning node is created). The values for critical temperature, T_c , critical pressure, P_c , and acentric factor, ω_i must be specified for all species.

Peng–Robinson (Twu)

For the Twu modification (Ref. 6) of the Peng–Robinson model, the alpha function, α_i , is replaced by

$$\alpha_i = T + R^{(N_i(M_i - 1))} \exp(L_i(1 - T_{r,i}^{N_i M_i})) \quad (6-21)$$

The binary interaction parameters k_{PR} , are used for the Twu modification. Acentric factor, ω_i , is not used in this model but critical temperature and critical pressure must be specified for all species. The species specific fit parameters L_i , M_i , N_i can be determined by fitting the pure species phase equilibrium to the vapor pressure curve.

Soave–Redlich–Kwong

The classical Soave–Redlich–Kwong equation of state [Ref. 7](#) is given by

$$P = \frac{RT}{V-b} - \frac{a\alpha}{V(V+b)} \quad (6-22)$$

with

$$\alpha = \frac{1}{\sqrt{T}} \quad (6-23)$$

and for the pure species i ,

$$a_i = \Omega_A \frac{R^2 T_c^{2.5}}{P_c} \quad (6-24)$$

$$b_i = \Omega_B \frac{RT_c}{P_c} \quad (6-25)$$

$$\Omega_A = \frac{1}{3} + \frac{1}{3}\Omega_B(3 + 3\Omega_B) \quad (6-26)$$

$$\Omega_B = \frac{1}{27}2^{\frac{1}{3}}27^{\frac{2}{3}} - \frac{1}{3} \quad (6-27)$$

For mixtures

$$a = \sum_i \sum_j x_i x_j (1 - k_{\text{SRK},i,j}) \sqrt{a_i a_j} \quad (6-28)$$

$$b = \sum_i x_i b_i \quad (6-29)$$

The binary interaction parameters, k_{SRK} , are symmetric with zero in the diagonal:

$$k_{\text{SRK},i,j} = k_{\text{SRK},j,i} \quad (6-30)$$

$$k_{\text{SRK},i,i} = 0 \quad (6-31)$$

If a value is missing for $k_{\text{SRK},i,j}$ in the database, it is set to zero (a warning node is created). The values for critical temperature, T_c , critical pressure, P_c , and acentric factor, ω_i , must be specified for all species. The Soave–Redlich–Kwong equation of

state is a version of [Equation 6-22](#) modified by Soave [Ref. 8](#), where for pure species i , the alpha function is modified to

$$\alpha_i = [1 + (0.480 + 1.574\omega_i - 0.176\omega_i^2)(1 - \sqrt{T_{r,i}})]^2 \quad (6-32)$$

Soave–Redlich–Kwong (Graboski–Daubert)

The Graboski and Daubert [Ref. 9](#) modification of the Soave–Redlich–Kwong equation of state replaces alpha function by

$$\alpha_i = [1 + (0.48508 + 1.55174\omega_i - 0.1561\omega_i^2)(1 - \sqrt{T_{r,i}})]^2 \quad (6-33)$$

except for H_2 , where it is replaced by

$$\alpha_{H_2} = 1.202 \exp(-0.30228T_{r,H_2}) \quad (6-34)$$

Water (IAPWS)

The International Association for the Properties of Water and Steam (IAPWS) provides a set of correlations to compute properties of water in different states. The correlations available in COMSOL correspond to the version named Industrial Formulation 1997 (IF-97) [Ref. 10](#) - [Ref. 12](#). The correlations are valid in the following ranges:

$$273.15 \text{ K} \leq T \leq 1073.15 \text{ K} \quad p \leq 100 \text{ MPa} \quad (6-35)$$

$$1073.15 \text{ K} \leq T \leq 2273.15 \text{ K} \quad p \leq 50 \text{ MPa} \quad (6-36)$$

The following water and steam properties, available by creating a [Species Property](#), are computed using the IAPWS correlations: density, volume, enthalpy, internal energy, Helmholtz energy, heat capacity at constant pressure, and heat capacity at constant volume. The amounts present in the vapor and liquid phase can be computed using an [Equilibrium Calculation](#).

Other parameters and thermodynamic properties available for the system are provided from the COMSOL database.

LIQUID PHASE MODELS

Activity Coefficient Models

Activity coefficient models are used to describe liquid mixtures at low to moderate pressures and temperatures where the equations of state are inadequate. When using activity coefficient models, the partial fugacity for species i in the liquid phase is defined as

$$\hat{f}_{i,l} = x_i \gamma_i \phi_{i,sat} P_{i,sat} F_i \quad (6-37)$$

which yields

$$\hat{\phi}_{i,l} = \gamma_i \phi_{i,sat} \frac{P_{i,sat}}{P} F_i \quad (6-38)$$

where the activity coefficient, γ_i , describes the nonideal liquid phase and $\hat{\phi}_{i,sat} P_{i,sat}$ is the fugacity at the vapor–liquid phase boundary at equilibrium for the pure species i . The Poynting correction, F_i , describes the pure species fugacity deviation from the boiling curve and can be expressed as

$$F_i = \exp \left(\int_{P_{i,sat}}^P \frac{V_{i,l}}{RT} dp \right) \quad (6-39)$$

For an incompressible liquid, an approximation of the Poynting correction can be done according to

$$F_i \approx \exp \left(\frac{V_{i,l}}{RT} (P - P_{i,sat}) \right) \quad (6-40)$$

The Poynting correction can often be ignored for moderate pressure. Hence, Equation 6-38 can be expressed as

$$\hat{\phi}_{i,l} = \gamma_i \phi_{i,sat} \frac{P_{i,sat}}{P} \quad (6-41)$$

If the vapor phase is considered ideal, then $\phi_{i,sat} = 1$ and the above equation reduces to

$$\hat{\phi}_{i,l} = \gamma_i \frac{P_{i,sat}}{P} \quad (6-42)$$

This reduction can be selected explicitly in case the vapor phase is not ideal.

The logarithmic transformation of Equation 6-38 is:

$$\ln \hat{\phi}_{i,l} = \ln \gamma_i + \ln \phi_{i,sat} + \ln P_{i,sat} - \ln P + \ln F_i \quad (6-43)$$

Ideal Solution

For an ideal solution the activity coefficient is equal to one, which gives:

$$\ln \gamma_i = 0 \quad (6-44)$$

Regular Solution

The Scatchard–Hildebrand equation [Ref. 14](#) for a nonpolar mixture is

$$\ln \gamma_i = \frac{V_i(\delta_i - \delta_{av})^2}{RT} \quad (6-45)$$

where V_i is species molar volume and δ_i is species solubility parameter, and δ_{av} is

$$\delta_{av} = \frac{\sum_i (x_i V_i \delta_i)}{\sum_i (x_i V_i)} \quad (6-46)$$

The volume parameter, V_i , is set equal the liquid volume, $V_{i,l,b}$ at normal boiling point which must be specified for all species. The solubility parameter, δ_i must be specified for all species and can be estimated from the normal heat of vaporization, $\Delta H_{vap,i}$ and the liquid volume at normal boiling point as

$$\delta_i \equiv \sqrt{\frac{\Delta H_{vap,i}}{V_{i,l,b}}} \quad (6-47)$$

Extended Regular Solution

The extended Scatchard–Hildebrand equation adds the Flory and Huggins correction to the regular solution model [Ref. 16](#):

$$\ln \gamma_i = \frac{V_i(\delta_i - \delta_{av})^2}{RT} + \ln(\theta_i) + 1 - \theta_i \quad (6-48)$$

where

$$\theta_i = \frac{V_i}{\sum_i (x_i V_i)} \quad (6-49)$$

Wilson

Wilson [Ref. 17](#) derived his activity coefficient model from a consideration of probabilities of neighboring molecules in a liquid

$$\ln \gamma_i = 1 - \sum_j \left(\frac{x_j \Lambda_{j,i}}{\sum_k x_k \Lambda_{j,k}} \right) - \ln \left(\sum_j x_j \Lambda_{i,j} \right) \quad (6-50)$$

$$\Lambda_{i,j} = \frac{V_{w,j}}{V_{w,i}} \exp \left(-\frac{\lambda_{i,j}}{T} \right) \quad (6-51)$$

where the Wilson volume parameter, $V_{w,i}$, is a species-specific parameter describing volume. If the volume parameter is not available, the liquid volume at normal boiling point is used. The Wilson binary interaction parameters $\lambda_{i,j}$ are specified in terms of absolute temperature. The binary interaction parameter matrix is nonsymmetric and with zeros in its diagonal. All off-diagonal values must be specified

$$\lambda_{i,i} = 0 \quad (6-52)$$

$$\Lambda_{i,i} = 1 \quad (6-53)$$

NRTL

Renon and Prausnitz (Ref. 18) formulated a three parameter activity coefficient model that is able to describe liquid–liquid equilibrium; the nonrandom two-liquid (NRTL) model:

$$\ln \gamma_i = \frac{\sum_j x_j \tau_{j,i} G_{j,i}}{\sum_j x_j G_{j,i}} + \sum_j \left(\frac{x_j G_{j,i}}{\sum_k x_k G_{k,j}} \left(\tau_{i,j} - \frac{\sum_k x_k \tau_{k,j} G_{k,j}}{\sum_k x_k G_{k,j}} \right) \right) \quad (6-54)$$

$$\tau_{i,j} = \frac{A_{i,j}}{T} \quad (6-55)$$

$$G_{i,j} = \exp(-\alpha_{i,j} \tau_{i,j}) \quad (6-56)$$

The three parameters are $A_{i,j}$, $A_{j,i}$, and $\alpha_{i,j}$. A more general form is implemented here:

$$\tau_{i,j} = \frac{A_{i,j}}{T} + B_{i,j} \quad (6-57)$$

$$G_{i,j} = \exp(\beta_{i,j} - \alpha_{i,j} \tau_{i,j}) \quad (6-58)$$

The binary interaction parameters, $A_{i,j}$, are specified in terms of absolute temperature. The diagonal values are zero and the matrix is nonsymmetric. All off-diagonal values must be specified.

The binary interaction parameters, $B_{i,j}$, have values of zero on the diagonal and the matrix is nonsymmetric. For each pair of species, at least $A_{i,j}$ or $B_{i,j}$ should be specified.

The randomness parameters, $\alpha_{i,j}$, have values of zero on the diagonal and the matrix is symmetric. All off-diagonal values must be specified. Alternatively one can set the more generic form directly specifying parameter $\beta_{i,j}$ for which diagonal values are zero and the matrix is nonsymmetric. For each pair of species at least $\alpha_{i,j}$ or $\beta_{i,j}$ should be specified.

If any value for these parameters is missing in the database, it is set to zero (warning node is created).

$$A_{i,i} = 0 \quad (6-59)$$

$$B_{i,i} = 0 \quad (6-60)$$

$$\alpha_{i,i} = 0 \quad (6-61)$$

$$\alpha_{j,i} = \alpha_{i,j} \quad (6-62)$$

$$\tau_{i,i} = 0 \quad (6-63)$$

$$G_{i,i} = 1 \quad (6-64)$$

$$\beta_{i,i} = 0 \quad (6-65)$$

UNIQUAC

Abrams and Prausnitz followed up with another two-liquid model known as Universal Quasi Chemical equation (UNIQUAC) (see [Ref. 19](#)), which is formulated in terms of two activity coefficients:

$$\ln \gamma_i = \ln \gamma_{i, \text{comb}} + \ln \gamma_{i, \text{res}} \quad (6-66)$$

The first term is the combinatorial part contributes to the Gibbs free energy originating from size and shape effects as

$$\ln \gamma_{i, \text{comb}} = 1 - \phi_i + \ln \phi_i - \frac{z}{2} q_i \left(1 + \frac{\phi_i}{\theta_i} + \ln \left(\frac{\phi_i}{\theta_i} \right) \right) \quad (6-67)$$

and the second term is the residual part from chemical interactions between the molecules,

$$\ln \gamma_{i,\text{res}} = q_i \left(1 - \ln \left(\frac{\sum_j x_j q_j \tau_{j,i}}{\sum_j x_j q_j} \right) - \sum_j \frac{x_j q_j \tau_{j,i}}{\sum_k x_j q_j \tau_{k,j}} \right) \quad (6-68)$$

where

$$\phi_i = \frac{r_i}{\sum_j x_j r_j} \quad (6-69)$$

$$\theta_i = \frac{q_i}{\sum_j x_j q_j} \quad (6-70)$$

$$\tau_{i,j} = \exp\left(\frac{-\Delta E_{i,j}}{T}\right) \quad (6-71)$$

The coordination number is taken equal to $z = 10$. The binary interaction energy parameters, $\Delta E_{i,j}$, are specified in terms of absolute temperature (K). The diagonal values are zero and the matrix is nonsymmetric. All off-diagonal values must be specified.

$$\Delta E_{i,i} = 0 \quad (6-72)$$

$$\tau_{i,i} = 1 \quad (6-73)$$

The volume parameters, r_i , and surface area parameters, q_i , are model-specific parameters for each species. If the parameters are not specified, they can be derived from the van der Waals volume, $V_{\text{VDW},i}$, and area, $A_{\text{VDW},i}$, respectively

$$r_i = \frac{V_{\text{VDW},i}}{0.01517 \cdot 10^{-3}} \quad (6-74)$$

$$q_i = \frac{A_{\text{VDW},i}}{2.5 \cdot 10^{-5}} \quad (6-75)$$

For all species r_i or $V_{VDW,i}$ and q_i or $A_{VDW,i}$ must be specified.

UNIFAC

The UNIQUAC Functional-group Activity Coefficients (UNIFAC; see Ref. 20) uses the same equations as UNIQUAC but the parameters are constructed from group contributions. The model can be used if UNIQUAC parameters are not available for all species. The activity coefficients are calculated from Equation 6-66. The combinatorial part follows from equation Equation 6-67, where

$$r_i = \sum_k v_{k,i} r_k \quad (6-76)$$

$$q_i = \sum_k v_{k,i} q_k \quad (6-77)$$

where r_k and q_k are the values for group k in species i , and $v_{k,i}$ is the number of occurrences of group k in molecule. The residual term in Equation 6-66 is calculated from a summation over functional groups:

$$\ln \gamma_{i, \text{res}} = \sum_k v_{k,i} (\ln \gamma_{k, \text{res}} - \ln \gamma_{i, k, \text{res}}) \quad (6-78)$$

The values for $\ln(\gamma_{k, \text{res}})$ are calculated from the mixture containing all species at a specified composition x . The values for $\ln(\gamma_{i, k, \text{res}})$ are calculated for a mixture of group k considering only pure species i . Both are defined, for functional group k , by

$$\ln \gamma_{k, \text{res}} = q_k \left(1 - \ln \left(\frac{\sum_l x_l q_l \tau_{l,k}}{\sum_l x_l q_l} \right) - \sum_j \frac{x_j q_j \tau_{j,k}}{\sum_m x_m q_m \tau_{m,l}} \right) \quad (6-79)$$

where x_l and x_m are the compositions of functional group l and m in the mixture

$$x_m = \frac{\sum_i x_i v_{m,i}}{\sum_i \left(x_i \sum_k v_{k,i} \right)} \quad (6-80)$$

For the calculation of a pure species' i residual activity we get:

$$x_m = \frac{v_{m,i}}{\sum_k v_{k,i}} \quad (6-81)$$

The volume parameters r_k and surface area parameters q_k are model-specific parameters for each group. The binary interaction between groups k and m is

$$\tau_{k,m} = \exp\left(-\frac{A_{k,m}}{T}\right) \quad (6-82)$$

The binary interaction parameters, $A_{k,m}$, are specified in terms of absolute temperature. The diagonal values are zero, the matrix is nonsymmetric. All off-diagonal values must be specified.

The default group and interaction parameters are those published by the UNIFAC consortium (Ref. 21 through Ref. 26), with added groups from Balslev and Abildskov (Ref. 27) but can be modified per package or database. The groups must be specified for all species. Note that the interaction parameter matrix is sparse, and a package can only be used if all interaction parameters for all used groups are specified.

Chao–Seader (Grayson–Streed)

The Chao–Seader model Ref. 13 correlates liquid phase partial coefficients for pure species, for use of hydrogen and hydrocarbon mixtures at elevated pressure and temperatures. It is expressed by

$$\ln \phi_{i,l} = \ln \gamma_i + \ln \phi_{i,l,0} \quad (6-83)$$

The activity is based on the Scatchard–Hildebrand equation Ref. 14 and presented in Equation 6-45 to Equation 6-47. Chao–Seader specific values for liquid volume, V_i , and solubility parameter, δ_i , are used.

If V_i is unspecified, it can be estimated by

- a** molar volume at normal boiling point
- b** van der Waals volume
- c** from the Rackett model, Equation 6-80, at normal boiling point temperature.

If the Chao–Seader specific solubility parameter, δ_i , is not specified, the generic solubility parameter is used.

The fugacity coefficient for pure species i is correlated as

$$\log \phi_{i,l,0} = \log \phi_{i,l,1} + \omega_i \log \phi_{i,l,2} \quad (6-84)$$

$$\log \phi_{i,l,1} = A_{i,0} + \frac{A_{i,1}}{T_{r,i}} + A_{i,2}T_{r,i} + A_{i,3}T_{r,i}^2 + A_{i,4}T_{r,i}^3 \quad (6-85)$$

$$+ (A_{i,7}T_{r,i}^2 + A_{i,6}T_{r,i} + A_{i,5})P_{r,i} + (A_{i,9}T_{r,i} + A_{i,8})P_{r,i}^2 - \log P_{r,i}$$

$$\log \phi_{i,l,2} = A_{i,10} + \frac{A_{i,12}}{T_{r,i}} + A_{i,11}T_{r,i} + A_{i,13}T_{r,i}^3 + A_{i,14}(P_{r,i} - 0.6) \quad (6-86)$$

A Chao–Seader specific acentric factor ω_i is used. If it is unknown, it can be set equal to the generic acentric factor. The parameter values are taken from the later publication of Grayson and Streed [Ref. 15](#) and given in [Table 6-2](#).

TABLE 6-2: CHAO–SEADER GRAYSON–STREED FUGACITY COEFFICIENT PARAMETERS.

	H ₂	CH ₄	OTHER		ALL
A _{i,0}	1.50709	1.36822	2.05135	A _{i,10}	-4.23893
A _{i,1}	2.74283	-1.54831	-2.10899	A _{i,11}	8.65808
A _{i,2}	-0.02110			A _{i,12}	-1.22060
A _{i,3}	0.00011	0.02889	-0.19396	A _{i,13}	-3.15224
A _{i,4}		-0.01076	0.02282	A _{i,14}	-0.02500
A _{i,5}	0.008585	0.10486	0.08852		
A _{i,6}		-0.02529			
A _{i,7}			-0.00872		
A _{i,8}			-0.00353		
A _{i,9}			0.00203		

The corresponding vapor phase model is Redlich–Kwong equation of state. The Chao–Seader (Grayson–Streed) model is valid when

- 255 K < T < 533 K
- P < 6.89 MPa
- P_r < 0.8
- $0.5 < T_{r,i} < 1.3$, for all hydrocarbons except CH₄
- $x_{\text{CH}_4} < 0.3$
- For systems that include CH₄ or H₂S:

$$\sum_i x_i T_{r,i} < 0.93$$

The enthalpy, entropy, and Gibbs free energy can be calculated from [Equation 6-109](#) to [Equation 6-118](#).

LIQUID VOLUME MODELS

When an activity coefficient model or the Chao–Seader (Grayson–Streed) model is used, a liquid volume model must be explicitly selected. It is also possible to assign a liquid volume model when an equation of state is used.

For liquids the density is defined as the reciprocal of the liquid volume:

$$\rho_l = \frac{1}{\bar{V}_l} \quad (6-87)$$

Using a cubic equation of state, the solution has 1, 2, or 3 different roots for the volume. The liquid density is defined as the root producing the highest density, and the vapor density as the root producing the lowest density.

Equation of State

When an equation of state is selected as the liquid phase model, the liquid volume is by default set to be calculated using the same equation of state model. The other liquid volume models; [Ideal Mixture](#), [COSTALD](#), and [Rackett](#) are also available.

Note, the vapor phase model and the liquid volume cannot use different equation of state models.

Ideal Mixture

For an ideal mixture the liquid volume is computed from the pure species densities (corresponding to the saturated liquid density):

$$V_{l,m} = \sum_i \frac{x_i}{\rho_{i,l}} \quad (6-88)$$

In this case the pure species densities corresponds to the saturated liquid density, which is available as a temperature-dependent correlation for all species in the built-in database.

COSTALD

Hankinson and Thomson [Ref. 28](#) presented the Corresponding States Liquid Density (COSTALD) equation as

$$\frac{V}{V_{mix}} = V_{r,ref}(1 - \omega V_{r,\delta}) \quad (6-89)$$

$$V_{r,ref} = 1 + a(1 - T_r)^{1/3} + b(1 - T_r)^{2/3} + c(1 - T_r) + d(1 - T_r)^{4/3} \quad (6-90)$$

$$V_{r,\delta} = \frac{e + fT_r + gT_r^2 + hT_r^3}{T_r - 1.00001} \quad (6-91)$$

$$V_{mix} = \frac{1}{4} \left(\sum_i x_i V_i + 3 \left(\sum_i x_i V_i^{2/3} \right) \left(\sum_i x_i V_i^{1/3} \right) \right) \quad (6-92)$$

$$T_c = \frac{\left(\sum_i x_i \sqrt{T_{c,i} V_i} \right)^2}{V_{mix}} \quad (6-93)$$

$$T_r = \begin{cases} \frac{T}{T_c} & T < T_c \\ 1 & T \geq T_c \end{cases} \quad (6-94)$$

$$\omega = \sum_i x_i \omega_i \quad (6-95)$$

where the volume, V_i , and acentric factor, ω_i for species i are model specific parameters. If the COSTALD volume parameter is unspecified, it is estimated from the van der Waals volume when $V_{VDW,i} > 0.3 \times 10^{-3} \text{ m}^3/\text{mol}$

$$V_i = 5.385 V_{VDW,i} - 5.1022 V_{VDW,i}^2 + 79.524 V_{VDW,i}^3 - 99.316 V_{VDW,i}^4 + 100.88 V_{VDW,i}^5 - 1152.7 V_{VDW,i}^6 \quad (6-96)$$

otherwise, it can be set equal to the critical volume as

$$V_i = V_{c,i} \quad (6-97)$$

If the COSTALD acentric factor, ω_i is not specified, it can be set equal to the generic acentric factor for species i . A critical temperature, $T_{c,i}$ must be specified for all species.

The correlation parameter values are:

TABLE 6-3: REDUCED VOLUME CORRELATION PARAMETERS IN COSTALD.

a	-1.52816	e	-0.296123
b	1.43907	f	0.386914

TABLE 6-3: REDUCED VOLUME CORRELATION PARAMETERS IN COSTALD.

c	-0.81446	g	-0.0427258
d	0.190454	h	-0.0480645

Rackett

The Rackett equation [Ref. 29](#) computes the liquid density at the saturation point, and can be used to describe liquid density at any pressure using the assumption that the liquid is incompressible. The equation and its condition can be expressed as:

$$V_l = \left(R \sum_i \frac{x_i T_{c,i}}{(MW)_i P_{c,i}} \right) \left(\sum_i Z_{r,i} \right)^{(1 + (1 - T_r^2)^{2/7})} \left(\sum_i x_i (MW)_i \right) \quad (6-98)$$

$$T_c = \sum_i x_i T_{c,i} \quad (6-99)$$

$$T_r = \begin{cases} \frac{T}{T_c} & T < T_c \\ 1 & T \geq T_c \end{cases} \quad (6-100)$$

Critical temperatures, $T_{c,i}$, critical pressures, $P_{c,i}$, and molecular weights, M_i , must be specified for all species. The model parameter, $Z_{r,i}$, must be specified for all species. If the value is not available it can be set to the critical compressibility factor:

$$Z_{r,i} = Z_{c,i} \quad (6-101)$$

Selecting the Right Thermodynamic Model

Which thermodynamic model is the most appropriate depends on a number of things, for example the nature of the properties to be used, the operational pressure and temperature, and not least the availability of the required model parameters. One general introduction to how to select thermodynamic models, as well as estimating missing parameters and validating properties can be found in [Ref. 77](#).

Below is a decision tree that can be used as a guide for choosing the thermodynamic model:

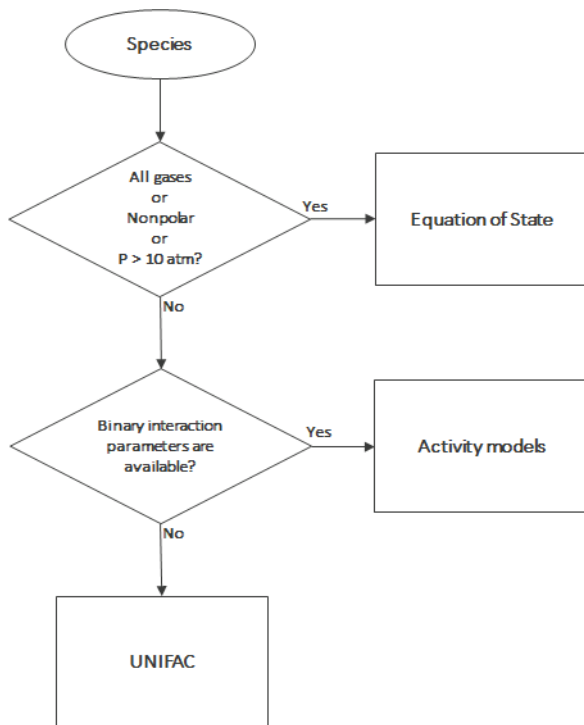


Figure 6-37: Decision tree to select thermodynamics model.

Species Property References

All references used for the included constants and temperature-dependent properties are available in the database file. The reference comments can be inspected in the settings window by right-clicking a species property node (under a **Thermodynamic System**) and selecting **Properties**.

Thermodynamic Properties Definitions

The thermodynamic properties provided are listed as species and mixture property in a Property package.

Ideal Gas

The ideal gas law is independent of composition and determines V at given T and P . Density can be calculated from

$$\rho = \frac{1}{V} \quad (6-102)$$

The partial fugacity coefficients

$$\ln \hat{\phi}_i = 0 \quad (6-103)$$

The ideal gas enthalpy for mixture

$$H_{ig} = \sum_i x_i \left(H_{i,ig,T_{ref}} + \int_{T_{ref}}^T C_{P,i,ig} dT \right) \quad (6-104)$$

where $H_{i,ig,T_{ref}}$ relates the enthalpy of an ideal gas to the enthalpy at the selected reference state for species i .

The ideal gas entropy for species i

$$S_{ig} = \sum_i x_i \left(S_{i,ig,T_{ref}} - R \ln x_i + \int_{T_{ref}}^T \frac{C_{P,i,ig}}{T} dT \right) - R \ln \left(\frac{P}{P_{ref}} \right) \quad (6-105)$$

where $S_{i,ig,T_{ref}}$ is the entropy of an ideal gas to the entropy of species at the selected reference state.

The Gibbs free energy follows from

$$G_{ig} = H_{ig} - TS_{ig} \quad (6-106)$$

Equation of State

The equation of state determines V at given x , T , and P . Density can be expressed as [Equation 6-102](#). The partial fugacity coefficients are derived from

$$\ln \hat{\phi}_i = \frac{1}{RT} \int_0^P \left(\hat{V}_i - \frac{RT}{P} \right) dP \quad (6-107)$$

where $\hat{V}_i$ is the partial molar volume. The enthalpy, entropy, and Gibbs free energy follow from the partial fugacity coefficients and the ideal gas contributions as:

$$H = H_{ig} - RT^2 \sum_i x_i \left(\frac{\partial \ln \hat{\phi}_i}{\partial T} \right) \quad (6-108)$$

$$S = S_{ig} - R \sum_i x_i \left(\ln \hat{\phi}_i + T \frac{\partial \ln \hat{\phi}_i}{\partial T} \right) \quad (6-109)$$

$$G = G_{ig} - RT \sum_i x_i \ln \hat{\phi}_i \quad (6-110)$$

Heat Capacity

The heat capacity at constant pressure is calculated from

$$C_P = \left(\frac{\partial H}{\partial T} \right)_{P,x} \quad (6-111)$$

It is available when the enthalpy is available. The heat capacity at constant volume is defined by

$$C_v = \left(\frac{\partial H}{\partial T} \right)_{v,x} \quad (6-112)$$

The relationship between the heat capacity at constant pressure and constant volume can be expressed as

$$C_P - C_v = T \left(\frac{\partial V}{\partial T} \right)_{P,x} \left(\frac{\partial P}{\partial T} \right)_{v,x} \quad (6-113)$$

where v is a function of T and P . A Taylor expansion of v at constant composition gives

$$\Delta v = \left(\frac{\partial V}{\partial T} \right)_{P,x} \Delta T + \left(\frac{\partial V}{\partial P} \right)_{T,x} \Delta P = 0 \quad (6-114)$$

Rearranging equations above gives

$$C_v = C_P + \frac{T \left(\frac{\partial V}{\partial T} \right)_{P,x}^2}{\left(\frac{\partial V}{\partial P} \right)_{T,x}} \quad (6-115)$$

C_v is available if C_P and volume are available and fluid is compressible by means of volume is pressure dependent. For ideal gas [Equation 6-113](#) express as

$$C_p - C_v = R \quad (6-116)$$

Specific heat capacity ratio is defined as

$$\gamma = \frac{C_p}{C_v} \quad (6-117)$$

Activity Coefficient Models

The enthalpy, entropy, and Gibbs free energy follow from the activity coefficients and the ideal gas contributions in [Equation 6-104](#) – [Equation 6-106](#). The activity coefficients describe the deviation of chemical potentials from the ideal liquid phase, so heats of vaporization need to be accounted for

$$H = \sum_i x_i \left(H_{i,ig, T_{ref}} + \int_{T_{ref}}^T C_{P,i,ig} dT - \Delta H_{i,vap} - RT^2 \frac{\partial \ln \gamma_i}{\partial T} - RT^2 \frac{\partial \ln \hat{\phi}_{i,sat}}{\partial T} \right) \quad (6-118)$$

$$S = \sum_i x_i \left(S_{i,ig, T_{ref}} - R \ln x_i + \int_{T_{ref}}^T \frac{C_{P,i,ig}}{T} dT - R \ln \left(\frac{P_{i,sat}}{P_{ref}} \right) + \ln \gamma_i + T \frac{\partial \ln \gamma_i}{\partial T} + \ln \hat{\phi}_{i,sat} + T \frac{\partial \ln \hat{\phi}_{i,sat}}{\partial T} + R \frac{\Delta H_{i,vap}}{T} \right) \quad (6-119)$$

$$G = \sum_i x_i \left(H_{i,ig, T_{ref}} + \int_{T_{ref}}^T C_{P,i,ig} dT - T \left(S_{i,ig, T_{ref}} + \int_{T_{ref}}^T \frac{C_{P,i,ig}}{T} dT \right) + RT \left(\ln x_i + \ln \gamma_i + \ln \hat{\phi}_{i,sat} + \ln \frac{P_{i,sat}}{P_{ref}} \right) \right) \quad (6-120)$$

Note that if the vapor phase is ideal, then the saturated fugacity, $\hat{\phi}_{i,sat}$, contribution can be ignored.

Other Properties

Partial fugacity is calculated from

$$\hat{f}_i = x_i \hat{\phi}_i P \quad (6-121)$$

Internal energy is calculated from

$$U = H - PV + \sum_i x_i U_{i,ig,\text{ref}} \quad (6-122)$$

where $U_{i,ig,\text{ref}}$ is the enthalpy of an ideal gas to the species enthalpy at the selected reference state.

Helmholtz energy is calculated from

$$A = U - TS = G - PV \quad (6-123)$$

The K -values for phases p and q are taken from

$$K_{i,p,q} = \frac{\hat{\phi}_{i,q}}{\hat{\phi}_{i,p}} \quad (6-124)$$

If only liquid phases are defined, the K -value calculation is reduced to

$$K_{i,p,q} = \frac{\gamma_{i,q}}{\gamma_{i,p}} \quad (6-125)$$

$$M_{\text{mix}} = \sum_i x_i M_i \quad (6-126)$$

Standard Enthalpy of Formation and Absolute Entropy Terms

Formation terms are not included in the enthalpy, entropy, or internal energy. If the enthalpy of formation is specified for all species, EnthalpyF (H_F) represent the enthalpy including the formation terms. If the absolute entropy of all species are specified then the EntropyF (S_F) includes the absolute entropy.

At reference conditions, the heat of any reaction relates to the heat of formation as

$$\Delta H_{\text{reac,ref}} = \sum_i \nu_i (\Delta H_{f,\text{ref}}) \quad (6-127)$$

Note that for reacting flow or a heat balance in a reactor when the heat of reaction is explicitly taken into account, the enthalpy should not include the heat of formation.

The following enthalpy expression should be used:

$$H_F = H + \sum_i x_i \Delta H_{f,i,\text{ref}} \quad (6-128)$$

$H_{f,i,\text{ref}}$ is calculated such that it matches the reference state of the package, even if the species have a different reference state. Similarly, the absolute entropy of each species is taken into account according to

$$S_F = S + \sum_i x_i S_{\text{abs},i} \quad (6-129)$$

The entropy balance over a process that includes reactions should include either the entropy of reaction and use S , or use S_F without entropy of reaction.

In database, a set of parameters introduced to define Absolute entropy and Standard enthalpy of formation for ideal gas and different phases (Vapor, liquid, Solid, and Aqueous infinite dilution). One can use these parameters to calculate Gibbs free energy of reaction when different phases are available (See Electrode reaction).

SPECIES GIBBS FREE ENERGY OF FORMATION

As the COMSOL ThermoDB does not include entropy of formation of species, entropy Equation 6-129 and in consequence Gibbs free energy Equation 6-110 are estimated from absolute entropy's value. Therefore, the absolute value of entropy of formation and Gibbs free energy of formation of the species are not correct. However, for our applications these absolute values are not required while the value of entropy and Gibbs free energy change of reaction (the difference value) are correct.

If user needs the absolute value of entropy of formation, it is possible to estimate it by:

$$S_i = \Delta S_{f,i} = \sum_j \nu_j S_{\text{abs},j} \quad (6-130)$$

where the entropy of formation of species i is calculated from its elemental constituent j . For example, entropy of formation of ammonia at 298 K is



$$\Delta S_{f,NH_3} = 2S_{\text{abs},NH_3} - (S_{\text{abs},N_2} + 3S_{\text{abs},H_2}) \quad (6-132)$$

The values from RHS of Equation 6-132 can be extracted from COMSOLThermoDB. This value should be used in Equation 6-128 to estimate the absolute value of Gibbs free energy of species.

Reference State

The reference temperature, pressure, and phase can be specified for each package and for species within a package. The reference phase can be a real phase or an ideal gas. If a reference temperature, pressure, or phase is unspecified for a particular species, the package reference temperature, pressure, or phase is used for that species.

The reference values for enthalpy, $H_{i,\text{ref}}$, entropy, $S_{i,\text{ref}}$, and internal energy, $U_{i,\text{ref}}$, are calculated so that the pure species enthalpy, entropy, and internal energy are equal to zero at reference conditions.

The reference values for enthalpy, including formation terms, and entropy, including absolute terms, are calculated such that the corresponding property for pure species i has a value equal to the specified formation term at the species reference conditions.

Transport Properties

This section includes definitions of the models available in for thermal conductivity, viscosity, and diffusivity:

- [Thermal Conductivity](#)
- [Viscosity](#)
- [Diffusivity](#)

THERMAL CONDUCTIVITY

Vapor

Ideal

The thermal conductivity correlations is according to:

$$\lambda_v = \sum_i x_i \lambda_{i,v} + \Delta\lambda_{v,P} \quad (6-133)$$

The pressure correction $\Delta\lambda_{v,P}$ is calculated from the method of Stiel and Thodos, see [Ref. 30](#), which is applicable for $p_r < 3$, but is less accurate for H_2 , strongly polar gases, and gases with a high degree of hydrogen bonding, such as H_2O and NH_3

$$\Delta\lambda_{v,P} = \frac{P_c^{2/3}}{\sqrt{M} T_c^{1/6} Z_c^5} A(\exp((B p_r) + C)) \quad (6-134)$$

$$\rho_r = \frac{\rho}{\rho_c} \quad (6-135)$$

$$\rho_c = \frac{1}{V_c} \quad (6-136)$$

The mixing rules are as suggested by Yoricane, see [Ref. 31](#):

$$V_c = \sum_i x_i \sum_j x_j V_{c,i,j} \quad (6-137)$$

$$T_c = \frac{\sum_i x_i \sum_j x_j V_{c,i,j} T_{c,i,j}}{V_c} \quad (6-138)$$

$$\omega = \sum_i x_i \omega_i \quad (6-139)$$

$$Z_c = 0.291 - 0.08\omega \quad (6-140)$$

$$P_c = \frac{RT_c Z_c}{V_c} \quad (6-141)$$

$$M = \sum_i x_i M_i \quad (6-142)$$

where the binary constants are

$$V_{c,i,j} = \frac{1}{8} (V_{c,i}^{1/3} + V_{c,j}^{1/3})^3 \quad (6-143)$$

$$T_{c,i,j} = \sqrt{T_{c,i} T_{c,j}} \quad (6-144)$$

The Stiel and Thodos coefficients are

TABLE 6-4: PRESSURE CORRECTION PARAMETERS.

	A	B	C
$\rho_r < 0.5$	$A_1 = 2.702E8$	$B_1 = 0.535$	$C_1 = -1$
$0.5 \leq \rho_r < 2.0$	$A_2 = 2.528E8$	$B_2 = 0.670$	$C_2 = -1.069$
$\rho_r \geq 2.0$	$A_3 = 0.574E8$	$B_3 = 1.155$	$C_3 = 2.016$

However, in order to ensure 0th-order continuity at $\rho_r = 0.5$ and $\rho_r = 2.0$, the following coefficients for $0.5 \leq \rho_r < 2.0$ are recalculated from

$$A_2 = -\frac{A_1 C_1 + A_1 \exp(0.5 B_1) - A_3 C_3 - A_3 \exp(2.0 B_3)}{\exp(2.0 B_2) - \exp(0.5 B_1)} \quad (6-145)$$

$$C_2 = -\frac{\frac{A_1 C_1 \exp(2.0 B_2) + A_1 \exp(0.5 B_1) \exp(2.0 B_2)}{A_1 C_1 + A_1 \exp(0.5 B_1) - A_3 C_3 - A_3 \exp(2.0 B_3)} - \frac{A_3 C_3 \exp(2.0 B_2) + A_3 \exp(0.5 B_2) \exp(2.0 B_3)}{A_1 C_1 + A_1 \exp(0.5 B_1) - A_3 C_3 - A_3 \exp(2.0 B_3)}}{(6-146)$$

The vapor thermal conductivity correlation must be available for all species. Also critical volumes, $V_{c,i}$, critical temperatures, $T_{c,i}$, molecular weights M_i , and acentric factors ω_i must be specified for all species.

Kinetic Theory

Lindsay and Bromley (see Ref. 32) provided an equation for the interaction parameters of the method of Wassiljewa (see Ref. 33) based on the kinetic theory, to provide mixture thermal conductivity from pure species values

$$\lambda_v = \sum_i \left(\frac{x_i \lambda_{i,v}}{\sum_j x_j \phi_{i,j}} \right) + \Delta \lambda_{v,P} \quad (6-147)$$

$$\phi_{i,j} = \frac{1}{4} \left(1 + \sqrt{\frac{\eta_{i,v} \left(\frac{M_j}{M_i} \right)^{3/4} \left(T + \left(\frac{3}{2} \right) T_{b,i} \right)}{\eta_{j,v} \left(\frac{M_i}{M_j} \right)^{3/4} \left(T + \left(\frac{3}{2} \right) T_{b,j} \right)}} \right)^2 \left(\frac{T + \sqrt{\left(\frac{9}{4} \right) T_{b,i} T_{b,j}}}{T + \left(\frac{3}{2} \right) T_{b,i}} \right) \quad (6-148)$$

where the pressure correction $\Delta \lambda_{v,P}$ is calculated from Equation 6-134. Both vapor thermal conductivity correlation $\lambda_{i,v}$ and the vapor viscosity correlation $\eta_{i,v}$ must be available for all species. In addition, all normal boiling points $T_{b,i}$, molecular weights M_i , critical volumes $V_{c,i}$, critical temperatures $T_{c,i}$, and acentric factors ω_i must be specified.

Water (IAPWS)

The International Association of the Properties of Water and Steam recommend an equation Ref. 34 which is valid in the following range:

$$273.15 \text{ K} \leq T \leq 773.15 \text{ K} \quad p \leq 100 \text{ MPa} \quad (6-149)$$

$$773.15 \text{ K} \leq T \leq 923.15 \text{ K} \quad p \leq 70 \text{ MPa} \quad (6-150)$$

$$923.15 \text{ K} \leq T \leq 1073.15 \text{ K} \quad p \leq 40 \text{ MPa} \quad (6-151)$$

Liquid

The following mixture models are available for liquid thermal conductivity

Ideal

To calculate the mixture liquid thermal conductivity, $\lambda_{l,m}$, the values of pure liquid thermal conductivity correlations are mixed ideally

$$\lambda_{l,m} = \sum_i x_i \lambda_{i,l} \quad (6-152)$$

The pressure dependence is based on the work of Missenard ([Ref. 51](#)) where

$$\frac{\lambda_{l,m,P}}{\lambda_{l,m}} = 1 + Q P_r^{0.7} \quad (6-153)$$

where Q is correlated as

$$Q = -0.013693477 + 0.057427673 T_r + 0.001617831 (T_r)(\ln P_1) \quad (6-154)$$

and the following mixing rules are used

$$T_r = \frac{T}{T_c} \quad (6-155)$$

$$P_r = \frac{P}{P_c} \quad (6-156)$$

$$P_c = \frac{RT_c Z_c}{V_c} \quad (6-157)$$

$$V_c = \sum_i x_i V_{c,i} \quad (6-158)$$

$$T_c = \sum_i x_i T_{c,i} \quad (6-159)$$

$$Z_c = \sum_i x_i Z_{c,i} \quad (6-160)$$

All liquid thermal conductivity correlations must be specified. All values for critical temperatures, $T_{c,i}$, critical volumes, $P_{c,i}$ and critical compressibility factors, $Z_{c,i}$ must be specified for all species i .

Power Law

The values of pure liquid vapor thermal conductivity correlations are mixed according to the following power law

$$\frac{1}{\lambda_{l,m}^2} = \sum_i \frac{x_i}{\lambda_{l,i}} \quad (6-161)$$

All liquid thermal conductivity correlations must be specified. The model is valid for pure compound thermal conductivity values that are no more apart than a factor of 2 (Ref. 52 and Ref. 53). The pressure dependence is introduced using Equation 6-153 through Equation 6-160. All values for critical temperatures, $T_{c,i}$, critical volumes, $P_{c,i}$ and critical compressibility factors, $Z_{c,i}$ must be specified for all species i .

Local Composition

The local composition model by Rowley (Ref. 53) uses an ideal and excess contribution

$$\lambda_{l,m} = \lambda_{l,\text{ideal}} + \lambda_{l,\text{excess}} \quad (6-162)$$

The ideal part is based on mass fractions

$$\lambda_{l,\text{ideal}} = \sum_i \omega_i \lambda_{i,l} \quad (6-163)$$

$$\omega_i = \frac{x_i M_i}{\sum_j x_j M_j} \quad (6-164)$$

The excess term is based on NRTL local concentrations

$$\lambda_{l, \text{excess}} = \frac{\sum_i \sum_j \omega_i \omega_j G_{j,i} (\lambda_{j,i} - \lambda_{i,l})}{\sum_j \omega_j G_{j,i}} \quad (6-165)$$

where $G_{j,i}$ follows from Equation 6-58. The binary interaction terms follow from

$$\lambda_{i,j} = \lambda_{j,i} = \frac{\bar{\omega}_i \bar{\omega}_{i,i} \lambda_{i,l} + \bar{\omega}_j \bar{\omega}_{j,j} \lambda_{j,l}}{\bar{\omega}_i \bar{\omega}_{i,i} + \bar{\omega}_j \bar{\omega}_{j,j}} \quad (6-166)$$

which is symmetric, and on the diagonal,

$$\lambda_{i,i} = \lambda_{i,l} \quad (6-167)$$

and

$$\bar{\omega}_{i,i} = \bar{\omega}_i (\bar{\omega}_i + \bar{\omega}_j G_{j,i}) \quad (6-168)$$

with $\bar{\omega}_i$ is the composition in the binary mixture of species i and j and the local composition is equi-molar

$$\bar{\omega}_i = \frac{M_i \sqrt{G_{j,i}}}{M_i \sqrt{G_{j,i}} + M_j \sqrt{G_{i,j}}} \quad (6-169)$$

All liquid thermal conductivity correlations must be specified. The pressure dependence is introduced using Equation 6-153 – Equation 6-160. All values for critical temperatures $T_{c,i}$, critical volumes $P_{c,i}$, critical compressibility factors $Z_{c,i}$, and molecular weights M_i must be specified for all compounds i . In addition, all NRTL binary interaction parameters $A_{i,j}$ must be specified. Unspecified values for NRTL interaction parameters $B_{i,j}$ are set to zero. The randomness parameters $\alpha_{i,j}$ have values of zero on the diagonal and the matrix is symmetric. All off-diagonal values must be specified. NRTL model is presented in Equation 6-54 to Equation 6-65.

Local Composition (Modified)

Rowley (Ref. 52) adapted the local composition model by replacing the mixing rule in Equation 6-166 by the following

$$\lambda_{i,j} = \lambda_{j,i} = \frac{M_i (\bar{\omega}_i \bar{\omega}_{i,i}) \lambda_{i,l} + M_j (\bar{\omega}_j \bar{\omega}_{j,j}) \lambda_{j,l}}{M_i (\bar{\omega}_i \bar{\omega}_{i,i}) + M_j (\bar{\omega}_j \bar{\omega}_{j,j})} \quad (6-170)$$

which he found to produce better model predictions in most cases where both the Local Composition model and Power Law model have trouble. However, the model is not as generally applicable; for instance, systems containing H₂O are not well described by this model due to the low molecular weight of H₂O.

Water (IAPWS)

The International Association of the Properties of Water and Steam recommend an equation [Ref. 34](#) which is valid in the following range:

$$273.15 \text{ K} \leq T \leq 773.15 \text{ K} \quad p \leq 100 \text{ MPa} \quad (6-171)$$

$$773.15 \text{ K} \leq T \leq 923.15 \text{ K} \quad p \leq 70 \text{ MPa} \quad (6-172)$$

$$923.15 \text{ K} \leq T \leq 1073.15 \text{ K} \quad p \leq 40 \text{ MPa} \quad (6-173)$$

VISCOSITY

Vapor

Wilke

Wilke, see [Ref. 35](#), based his method for mixture viscosity of the vapor phase on kinetic theory:

$$\eta_v = \sum_i \left(\frac{x_i \eta_{i,v}}{\sum_j x_j \Psi_{i,j}} \right) \quad (6-174)$$

$$\Psi_{i,j} = \frac{\left(1 + \sqrt{\frac{\eta_{i,v}}{\eta_{j,v}}} \left(\frac{M_j}{M_i} \right)^{1/4} \right)^2}{\sqrt{8 \left(1 + \frac{M_i}{M_j} \right)}} \quad (6-175)$$

The vapor viscosity correlation $\eta_{i,v}$ must be available for all species. In addition, all molecular weights M_i must be specified.

Brokaw

Brokaw (see [Ref. 36](#)) uses the same basic equation as Wilke ([Equation 6-174](#)). However, [Equation 6-175](#) is replaced by

$$\psi_{i,j} = S_{i,j} A_{i,j} \sqrt{\frac{\eta_{i,v}}{\eta_{j,v}}} \quad (6-176)$$

and the interaction parameter is defined as

$$A_{i,j} = \frac{\beta_{i,j}}{\sqrt{RM_{i,j}}} \left(1 + \frac{RM_{i,j} - RM_{i,j}^{0.45}}{2(1 + RM_{i,j}) + \frac{1 + RM_{i,j}^{0.45}}{(\sqrt{\beta_{i,j}})(1 + RM_{i,j})}} \right) \quad (6-177)$$

where

$$\beta_{i,j} = \left(\frac{4M_i M_j}{(M_i + M_j)^2} \right)^{1/4}, \quad RM_{i,j} = \frac{M_i}{M_j} \quad (6-178)$$

The vapor viscosity correlation, $\eta_{i,v}$ must be available for all species i . In addition, all molecular weights M_i must be specified. If Lennard–Jones energy ϵ_i (see [Ref. 37](#)) Stockmayer’s polar parameter $\delta_{s,i}$ ([Ref. 38](#) and [Ref. 39](#)) are specified for both species i and j then

$$S_{i,j} = \frac{1 + \sqrt{\frac{T^2}{\epsilon_i \epsilon_j}} + \frac{1}{4} \delta_{s,i} \delta_{s,j}}{\sqrt{1 + \frac{T}{\epsilon_i} + \frac{1}{4} \delta_{s,i}} \sqrt{1 + \frac{T}{\epsilon_j} + \frac{1}{4} \delta_{s,j}}} \quad (6-179)$$

Otherwise,

$$S_{i,j} = 1 \quad (6-180)$$

Davidson

The Davidson method, see [Ref. 40](#), requires fewer compound specific parameters than Brokaw, while reported accuracy is almost as good, and in the case of H_2 , even surpasses it. The Davidson model only requires molar masses and the viscosities of the pure gases. The model is based on fluidity, which is defined to be the reciprocal viscosity.

$$f = \frac{1}{\eta} \quad (6-181)$$

The fluidity of the mixture is then calculated as

$$f = \sum_{i,j} \left(\frac{y_i y_j}{\sqrt{\mu_i} \sqrt{\mu_j}} \cdot E_{i,j}^A \right) \quad (6-182)$$

where y_i is the momentum fraction of species i ; $E_{i,j}$ is the momentum transfer coefficient of the species pair i, j ; and A is an empirical species-independent parameter set to 1/3. The momentum fraction is given by

$$y_i = \frac{x_i \sqrt{M_i}}{\sum_j (x_j \sqrt{M_j})} \quad (6-183)$$

and the momentum transfer coefficient is taken as

$$E_{i,j} = \frac{2 \sqrt{M_i} \sqrt{M_j}}{M_i + M_j} \quad (6-184)$$

High Pressure Modification

To account for the effect of pressure on vapor viscosity, a pressure correction can be applied. The pressure dependence is based on kinetic gas theory, which adds the following term to the vapor viscosity:

$$\Delta \eta_{v,P} = \left(\frac{\sqrt{M_t} P_c^{2/3}}{T_c^{1/6}} \right) \left(\frac{10^{-7}}{101325^{2/3}} \zeta \right) \quad (6-185)$$

where ξ is calculated from the correlation of Jossi ([Ref. 41](#)), which is applicable for $\rho_r < 3.0$. It is less accurate for H_2 , strongly polar gases and gases with a high degree of hydrogen bonding such as H_2O and NH_3 .

The correction factor is due to using pressure, atm, and viscosity, cP, units in Jossi's correlation. It is expressed as:

$$\begin{aligned} (\xi + 1)^{\frac{1}{4}} = & 1.0230 + 0.23364 \rho_r + 0.58533 \rho_r^2 \\ & - 0.40758 \rho_r^3 + 0.093324 \rho_r^4 \end{aligned} \quad (6-186)$$

$$\rho_r = \frac{\rho}{\rho_c} \quad (6-187)$$

$$\rho_c = \frac{1}{V_c} \quad (6-188)$$

The following mixture rules are used

$$P_c = \frac{RT_c Z_c}{V_c}, \quad V_c = \sum_i x_i V_{c,i}, \quad T_c = \sum_i x_i T_{c,i} \quad (6-189)$$

$$Z_c = \sum_i x_i Z_{c,i}, \quad M_t = \sum_i x_i M_i \quad (6-190)$$

The values for critical volumes, $V_{c,i}$, critical temperatures, $T_{c,i}$, critical compressibility factors, $Z_{c,i}$ and molecular weights, M_i must be specified for all species i .

The high pressure correction is available for the Wilke, Brokaw, and Davidson mixture models. The vapor viscosity follows from

$$\eta_v = \eta_{v, \text{Wilke}} + \Delta\eta_{v, P} \quad (6-191)$$

where $\eta_{v, \text{Wilke}}$ is calculated from [Equation 6-174](#).

Pedersen Corresponding States Model

The corresponding states viscosity model of Pedersen ([Ref. 42](#) and [Ref. 43](#)) applies to both vapor and liquid phases of hydrocarbon mixtures. The selected reference species is CH_4 .

The CH_4 viscosity is calculated from [Ref. 44](#), modified by Pedersen and Fredenslund ([Ref. 45](#)) to avoid issues below 91 K where CH_4 becomes solid

$$\eta_{\text{CH}_4} = \eta_{\text{CH}_4, 0} + \rho_{\text{CH}_4} \eta_{\text{CH}_4, 1} + F_1 \eta_{\text{CH}_4, 2} + F_2 \eta_{\text{CH}_4, 3} \quad (6-192)$$

where

$$F_1 = \frac{H+1}{2} \quad (6-193)$$

$$F_2 = \frac{1-H}{2} \quad (6-194)$$

$$H = \frac{\exp(\Delta T) - \exp(-\Delta T)}{\exp(\Delta T) + \exp(-\Delta T)} \quad (6-195)$$

$$\Delta T = T - 91 \text{ K} \quad (T \in K) \quad (6-196)$$

Here, ρ_{CH_4} is used in g/cm^3 ; for the mass-mole conversion of ρ_{CH_4} , a molecular weight of $M_{\text{CH}_4} = 16.042568$ g/mol is used.

The dilute gas part is given by

$$\eta_{\text{CH}_4, 0} = \frac{-L_1}{T} + \frac{L_2}{T^{2/3}} - \frac{L_3}{T^{1/3}} + L_4 - L_5 T^{1/3} + L_6 T^{2/3} - L_7 T + L_8 T^{4/3} - L_9 T^{5/3} \quad (6-197)$$

The first density correction for the moderately dense gas is given by

$$\eta_{\text{CH}_4, 1} = L_{10} - L_{11} \left(1.4 - \ln \frac{T}{168 \text{ K}} \right)^2 \quad (6-198)$$

The remainder is given by the empirical correlation

$$\eta_{\text{CH}_4, 2} = \exp \left(\frac{L_{12}}{T} - L_{13} \right) \left(\exp \left((10 \sqrt{\rho_{\text{CH}_4}}) \left(L_{14} - \frac{L_{15}}{T^{3/2}} \right) + \left(\frac{\rho_{\text{CH}_4} - \rho_{c, \text{CH}_4}}{\rho_{c, \text{CH}_4}} \right) (\sqrt{\rho_{\text{CH}_4}}) \left(L_{16} + \frac{L_{17}}{T} + \frac{L_{18}}{T^2} \right) \right) - 1 \right) \quad (6-199)$$

The correction term for solid CH_4

$$\eta_{\text{CH}_4, 3} = \exp \left(\frac{L_{19}}{T} - L_{20} \right) \left(\exp \left((10 \sqrt{\rho_{\text{CH}_4}}) \left(L_{21} - \frac{L_{22}}{T^{3/2}} \right) + \left(\frac{\rho_{\text{CH}_4} - \rho_{c, \text{CH}_4}}{\rho_{c, \text{CH}_4}} \right) (\sqrt{\rho_{\text{CH}_4}}) \left(L_{23} + \frac{L_{24}}{T} + \frac{L_{25}}{T^2} \right) \right) - 1 \right) \quad (6-200)$$

with the values of the parameters L_1 through L_{25} are listed in [Table 6-5](#) below:

TABLE 6-5: METHANE VISCOSITY NUMERICAL COEFFICIENTS.

L_1	$2.090975 \cdot 10^5$	L_{10}	1.696985927	L_{19}	9.74602
L_2	$2.647269 \cdot 10^5$	L_{11}	0.133372346	L_{20}	44.6055
L_3	$1.472818 \cdot 10^5$	L_{12}	188.73011594	L_{21}	18.0834
L_4	47167.40	L_{13}	10.35060586	L_{22}	4126.66
L_5	9491.827	L_{14}	17.571599671	L_{23}	0.976544
L_6	1219.979	L_{15}	3019.3918656	L_{24}	81.8134
L_7	96.27993	L_{16}	0.042903609488	L_{25}	15649.9

TABLE 6-5: METHANE VISCOSITY NUMERICAL COEFFICIENTS.

L_8	4.274152	L_{17}	145.29023444		
L_9	0.08141531	L_{18}	6127.6818706		

Here, ρ_{CH_4} is used in g/cm^3 ; the critical density is given by $\rho_{c,\text{CH}_4} = 0.16284 \text{ g}/\text{cm}^3$. The following equation by McCarty (Ref. 46) is solved for the density of CH_4

$$\begin{aligned}
 P = & N_1 \rho_{\text{CH}_4} T + \rho_{\text{CH}_4}^2 \left(-N_2 T + N_3 \sqrt{T} - N_4 + \frac{N_5}{T} - \frac{N_6}{T^2} \right) \\
 & + \rho_{\text{CH}_4}^3 \left(N_7 T - N_8 + \frac{N_9}{T} - \frac{N_{10}}{T^2} \right) + \rho_{\text{CH}_4}^4 \left(-N_{11} T + N_{12} + \frac{-N_{13}}{T} \right) + N_{14} \rho_{\text{CH}_4}^5 \\
 & + \rho_{\text{CH}_4}^6 \left(-\frac{N_{15}}{T} + \frac{N_{16}}{T^2} \right) + N_{17} \frac{\rho_{\text{CH}_4}^7}{T} + \rho_{\text{CH}_4}^8 \left(\frac{N_{18}}{T} + \frac{N_{19}}{T^2} \right) + N_{20} \frac{\rho_{\text{CH}_4}^8}{T} \\
 & + \exp(-N_{21} \rho_{\text{CH}_4}^2) \left(\rho_{\text{CH}_4}^3 \left(\frac{N_{22}}{T^2} - \frac{N_{23}}{T^3} \right) + \rho_{\text{CH}_4}^5 \left(\frac{N_{24}}{T^2} + \frac{N_{25}}{T^4} \right) + \rho_{\text{CH}_4}^7 \left(\frac{N_{26}}{T^2} + \frac{N_{27}}{T^3} \right) \right. \\
 & \left. + \rho_{\text{CH}_4}^9 \left(-\frac{N_{28}}{T^2} - \frac{N_{29}}{T^4} \right) + \rho_{\text{CH}_4}^{11} \left(-\frac{N_{30}}{T^2} + \frac{N_{31}}{T^3} \right) + \rho_{\text{CH}_4}^{13} \left(\frac{N_{32}}{T^2} - \frac{N_{33}}{T^3} + N_{34} T^4 \right) \right)
 \end{aligned} \quad (6-201)$$

where ρ_{CH_4} is used in mol/l .

TABLE 6-6: COEFFICIENTS IN THE FUNCTIONAL FORM OF THE MCCARTY EOS.

N_1	0.08205616	N_{13}	2.8685285973	N_{25}	$1.6428375992 \cdot 10^6$
N_2	0.018439486666	N_{14}	$0.11906973942 \cdot 10^{-3}$	N_{26}	0.21325387196
N_3	1.0510162064	N_{15}	0.0085315715699	N_{27}	37.791273422
N_4	16.057820303	N_{16}	3.8365063841	N_{28}	$0.1185701681 \cdot 10^{-4}$
N_5	848.44027562	N_{17}	$0.24986828379 \cdot 10^{-4}$	N_{29}	31.630780767
N_6	42738.409106	N_{18}	$0.57974531455 \cdot 10^{-5}$	N_{30}	$0.4100678294 \cdot 10^{-5}$
N_7	$0.76565285254 \cdot 10^{-3}$	N_{19}	0.0071648329297	N_{31}	0.0014870043284
N_8	0.48360724197	N_{20}	$0.12577853784 \cdot 10^{-3}$	N_{32}	$3.151226153 \cdot 10^{-9}$
N_9	85.195473835	N_{21}	0.0096	N_{33}	$0.2167077474 \cdot 10^{-5}$
N_{10}	16607.434721	N_{22}	22240.102466	N_{34}	$0.2400055107 \cdot 10^{-4}$
N_{11}	$0.37521074532 \cdot 10^{-4}$	N_{23}	$1.4800512328 \cdot 10^6$		
N_{12}	0.028616309259	N_{24}	50.498054887		

With the viscosity and density of CH_4 defined, the viscosity of any mixture, η_m , can be calculated from the corresponding states principle

$$\eta_m = \left(\frac{T_c}{T_{c, \text{CH}_4}} \right)^{-\frac{1}{6}} \left(\frac{P_c}{P_{c, \text{CH}_4}} \right)^{2/3} \left(\frac{M_m}{M_{\text{CH}_4}} \right)^{1/2} \left(\frac{\alpha}{\alpha_{\text{CH}_4}} \right) \eta_{\text{CH}_4, P_0, T_0} \quad (6-202)$$

where the CH_4 viscosity $\rho_{\text{CH}_4, P_0, T_0}$ is calculated at temperature T_0 and pressure P_0 :

$$T_0 = \left(\frac{T_{c, \text{CH}_4}}{T_c} \right) T \left(\frac{\alpha_{\text{CH}_4}}{\alpha} \right) \quad (6-203)$$

$$P_0 = \left(\frac{P_{c, \text{CH}_4}}{P_c} \right) P \left(\frac{\alpha_{\text{CH}_4}}{\alpha} \right) \quad (6-204)$$

The following mixing rules are used for the critical properties, see [Ref. 47](#):

$$T_c = \frac{\sum_i \sum_j x_i x_j \beta_{i,j} \sqrt{T_{c,i} T_{c,j}}}{\sum_i \sum_j x_i x_j \beta_{i,j}} \quad (6-205)$$

$$P_c = \frac{8 \sum_i \sum_j x_i x_j \beta_{i,j} \sqrt{T_{c,i} T_{c,j}}}{\left(\sum_i \sum_j x_i x_j \beta_{i,j} \right)^2} \quad (6-206)$$

$$\beta_{i,j} = \left(\left(\frac{T_{c,i}}{P_{c,i}} \right)^{1/3} + \left(\frac{T_{c,j}}{P_{c,j}} \right)^{1/3} \right)^3 \quad (6-207)$$

The parameter α is

$$\alpha = 1 + 7.378 \times 10^{-3} \rho_r^{1.847} M_m^{0.5173} \quad (6-208)$$

$$\rho_r = \frac{\rho_{\text{CH}_4, P_0, T_0}}{\rho_{c, \text{CH}_4}} \quad (6-209)$$

where

$$T_0 = \left(\frac{T_{c, \text{CH}_4}}{T_c} \right) T \quad (6-210)$$

$$P_0 = \left(\frac{P_{c, \text{CH}_4}}{P_c} \right) P \quad (6-211)$$

with $\rho_{c, \text{CH}_4} = 0.16284 \text{ g/cm}^3$. For CH_4

$$\alpha_{\text{CH}_4} = 1 + 7.378 \times 10^{-3} \rho_r^{1.847} M_{\text{CH}_4}^{0.5173} \quad (6-212)$$

where Equation 6-209 is used. The mixture molecular weight is a function of the weight-averaged molecular weight and the number-averaged molecular weight

$$M_m = 1.304 \times 10^{-4} (M_W^{2.303} - M_N^{2.303}) + M_N \quad (6-213)$$

$$M_W = \frac{\sum_i x_i M_i^2}{M_N} \quad (6-214)$$

$$M_N = \sum_i x_i M_i \quad (6-215)$$

where the power in Equation 6-213 is determined by fitting to experimental viscosity data.

Note that pure species vapor viscosity correlations $\eta_{i,v}$ are not required. However, for each species i , molecular weight M_i , critical temperature, $T_{c,i}$, and critical pressure, $P_{c,i}$ must be specified.

Water (IAPWS)

The International Association of the Properties of Water and Steam recommend an equation Ref. 48–Ref. 49 for industrial application which is valid in the following range:

$$273.16 \text{ K} \leq T \leq 1173.15 \text{ K} \quad 0 \leq p \leq p_t \quad (6-216)$$

$$T_m(p) \leq T \leq 1173.15 \text{ K} \quad p_t \leq p \leq 300 \text{ MPa} \quad (6-217)$$

$$T_m(p) \leq T \leq 873.15 \text{ K} \quad 300 \text{ MPa} \leq p \leq 350 \text{ MPa} \quad (6-218)$$

$$T_m(p) \leq T \leq 433.15 \text{ K} \quad 350 \text{ MPa} \leq p \leq 500 \text{ MPa} \quad (6-219)$$

$$T_m(p) \leq T \leq 373.15 \text{ K} \quad 500 \text{ MPa} \leq p \leq 1000 \text{ MPa} \quad (6-220)$$

where T_m is the pressure dependent melting temperature and p_t is the triple-point pressure. In accordance with industrial application recommendations, critical region correction is not applied.

Liquid

The following mixture models are available for liquid viscosity.

Molar Logarithmic Mixing

The values of pure species log liquid viscosity, $\ln \eta_{i,l}$ are mixed ideally using mole fractions x_i

$$\ln \eta_{l,m} = \sum_i x_i \ln \eta_{i,l} \quad (6-221)$$

where $\eta_{m,l}$ is the mixture viscosity of liquids.

Mass Logarithmic Mixing

The values of pure species log liquid viscosity correlation are mixed ideally using the weight fractions ω_i

$$\ln \eta_{l,m} = \sum_i \omega_i \ln \eta_{i,l} \quad (6-222)$$

Pedersen Corresponding States Model

The [Pedersen Corresponding States Model](#) described above for the gas phase viscosity also applies to the liquid phase. Pure species liquid viscosity correlations are not required. However, for each species i , molecular weight M_i , critical temperature $T_{c,i}$, and critical pressure $P_{c,i}$, must be specified.

Cubic mixing

The mixture viscosity is defined using the cubic root average in terms of the mole fractions x_i

$$\eta_{l,m} = \left(\sum_i x_i \eta_{i,l}^{1/3} \right)^3 \quad (6-223)$$

The model is noted in [Ref. 50](#) to provide reasonable results for hydrocarbon mixtures of similar components.

The model requires that the log liquid viscosity correlation is available for all species i .

Cubic mass mixing

The mass fraction equivalent of the previous model is

$$\eta_{l,m} = \left(\sum_i \omega_i \eta_{i,l}^{1/3} \right)^3 \quad (6-224)$$

The model requires that the log liquid viscosity correlation is available for all species i .

Water (IAPWS)

The International Association of the Properties of Water and Steam recommend an equation [Ref. 48–Ref. 49](#) for industrial application which is valid in the following range:

$$273.16\text{K} \leq T \leq 1173.15\text{K} \quad 0 \leq p \leq p_t \quad (6-225)$$

$$T_m(p) \leq T \leq 1173.15\text{ K} \quad p_t \leq p \leq 300\text{ MPa} \quad (6-226)$$

$$T_m(p) \leq T \leq 873.15\text{ K} \quad 300\text{ MPa} \leq p \leq 350\text{ MPa} \quad (6-227)$$

$$T_m(p) \leq T \leq 433.15\text{ K} \quad 350\text{ MPa} \leq p \leq 500\text{ MPa} \quad (6-228)$$

$$T_m(p) \leq T \leq 373.15\text{ K} \quad 500\text{ MPa} \leq p \leq 1000\text{ MPa} \quad (6-229)$$

where T_m is the pressure dependent melting temperature and p_t is the triple-point pressure. In accordance with industrial application recommendations, critical region correction is not applied.

DIFFUSIVITY

Two types of diffusion coefficients are supported. Diffusion coefficients in infinitely diluted systems, and Maxwell–Stefan diffusion coefficients.

For dilute systems, the binary diffusion coefficient $D_{i,j}^0$ represent the diffusivity of species i in a medium consisting of pure species j . This corresponds to the Fickian diffusion coefficient.

For any mixture, the binary Maxwell–Stefan diffusion coefficient $D_{i,j}$, represents the inverse drag coefficient of species i moving past species j ([Ref. 54–Ref. 57](#)). This

property is referred to as the Maxwell–Stefan diffusivity. The Maxwell–Stefan diffusivity is symmetric, $D_{i,j} = D_{j,i}$, and the diagonal elements $D_{i,i}$ are not used.

Gas Phase Diffusion Coefficient at Infinite Dilution

The following models are available for the diffusion coefficients at infinite dilution in the vapor phase:

- [Fuller–Schettler–Giddings](#)
- [Wilke–Lee](#)

Automatic When the Gas diffusivity property model is set to Automatic, the [Fuller–Schettler–Giddings](#) model is used, provided that the Fuller diffusion volume is known for both species (i and j), otherwise the [Wilke–Lee](#) model is used.

Fuller–Schettler–Giddings

Fuller and others ([Ref. 58](#)) modified the Chapman–Enskog relation to correlate binary diffusion coefficient for species i and j in the vapor phase according to the Fuller–Schettler–Giddings (FGS) model:

$$D_{i,j}^0 = \frac{1.01325 \times 10^{-2} T^{1.75} \sqrt{\frac{1}{M_i} + \frac{1}{M_j}}}{P \left(\left(\sum_i v \right)^{\frac{1}{3}} + \left(\sum_j v \right)^{\frac{1}{3}} \right)^2} \quad (6-230)$$

where T denotes the temperature (K), M_i the molecular weight of species i (g/mol) and P is the pressure (Pa). v_i are the atomic diffusion volumes (Fuller diffusion volume, cm³), which are estimated using group contribution for each species ([Ref. 59](#)):

TABLE 6-7: ATOMIC AND STRUCTURAL DIFFUSION VOLUME INCREMENTS.

GROUP	CONTRIBUTION
C	15.9
H	2.31
O	6.11
N	4.54
F	14.7
Cl	21
Br	21.9

TABLE 6-7: ATOMIC AND STRUCTURAL DIFFUSION VOLUME INCREMENTS.

GROUP	CONTRIBUTION
I	29.8
S	22.9
Aromatic Ring	-18.3
Heterocyclic Ring	-18.3

For some simple molecules the values below, determined from regression, are used:

TABLE 6-8: DIFFUSION VOLUMES OF ATOM AND SIMPLE MOLECULES.

SPECIES	FULLER DIFFUSION VOLUME
He	2.67
Ne	5.98
Ar	16.2
Kr	24.5
Xe	32.7
H ₂	6.12
D ₂	6.84
N ₂	18.5
O ₂	16.3
CO	18
CO ₂	26.9
N ₂ O	35.9
NH ₃	20.7
H ₂ O	13.1
SF ₆	71.3
Cl ₂	38.4
Br ₂	69
SO ₂	41.8
Air	19.7

Wilke–Lee

Wilke and Lee (Ref. 60) also modified the Chapman–Enskog relation to correlate binary diffusion coefficient in vapor phase according to:

$$D_{i,j}^0 = 10^{-2} \times \frac{3.03 - \left(\frac{0.98}{\sqrt{MW_{i,j}}} \right) T^{1.5}}{P \sigma_{i,j}^2 \Omega_D \sqrt{MW_{i,j}}} \quad (6-231)$$

where

$$MW_{i,j} = \frac{2}{\frac{1}{M_i} + \frac{1}{M_j}} \quad (6-232)$$

The length scale $\sigma_{i,j}$ for the interaction is taken from the Lennard–Jones diameter parameters of species i and j :

$$\sigma_{i,j} = \frac{(\sigma_i^{LJ} + \sigma_j^{LJ})}{2} \quad (6-233)$$

If σ_i^{LJ} is not specified in the database, it is instead estimated from:

$$\sigma_i^{LJ} = 1.18 \times 10^{-8} V_{i,l,b}^{\frac{1}{3}} \quad (6-234)$$

where $V_{i,l,b}$ is the molar volume of species at normal boiling point.

The collision integral Ω_D is evaluated from (Ref. 61):

$$\begin{aligned} \Omega_D = & 1.06036 T^{*-0.15610} + 0.19300 \exp (-0.47635 T^*) \\ & + 1.03587 \exp (-1.52996 T^*) + 1.76474 \exp (-3.89411 T^*) \end{aligned} \quad (6-235)$$

with

$$T^* = \frac{T}{\epsilon_{i,j}} \quad (6-236)$$

The energy scale $\epsilon_{i,j}$ for the interaction is taken from the Lennard–Jones energy parameters of species i and j :

$$\epsilon_{i,j} = \sqrt{\epsilon_i^{LJ} \epsilon_j^{LJ}} \quad (6-237)$$

If ϵ_i^{LJ} is not specified in the database, it is instead estimated from:

$$\frac{\epsilon_i^{LJ}}{k} = 1.15 T_{i,l,b} \quad (6-238)$$

where k is the Boltzmann constant and $T_{i,l,b}$ is the normal boiling point temperature.

Gas Phase Maxwell–Stefan Diffusivity

For gas phase diffusion the Maxwell–Stefan diffusivities are defined from the models for gas phase diffusivity at infinite dilution

$$D_{i,j} = D_{i,j}^0 \quad (i \neq j) \quad (6-239)$$

All models for gas phase diffusion at infinite dilution are symmetric, the diffusivity of species i in species j equals that of species j in species i . In addition, under the ideal gas assumption, the Maxwell–Stefan diffusion coefficient matches the Fick diffusion coefficient. This implies that the Maxwell–Stefan gas diffusivities provided are independent of composition.

Liquid Phase Diffusion Coefficients at Infinite Dilution

The following models are available for the diffusion coefficients at infinite dilution in liquid phase:

- Wilke–Chang
- Tyn–Calus
- Hayduk–Minhas
- Siddiqi–Lucas
- Erkey–Rodden–Akgerman

Automatic When the Liquid diffusivity at infinite dilution property model is set to Automatic, a selection for each solute i in solvent j will be made from the models according to the following rules. Except in the case of a temperature correlation and the Erkey–Rodden–Akgerman model, all of these rules require that the log liquid viscosity correlation, $\ln \eta_j$ is available in the database for species j and that the liquid volume at normal boiling point, $V_{i,l,b}$ is available for the species i .

- If a temperature correlation is available for species i and j , it is used.
- If the solvent is water, the Siddiqi–Lucas correlation (for aqueous systems) is used.
- If the solute and solvent are both normal paraffins, the Hayduk–Minhas correlation (for normal paraffins) is used.
- If the solvent is a normal paraffin, and the liquid density correlation for the solvent is available, and the solute is hydrogen, carbon-monoxide or carbon-dioxide, the Erkey–Rodden–Akgerman correlation (for normal paraffins) is used.

- If parachors P_i and P_j are both available, and the liquid volume at normal boiling point, $V_{i,l,b}$ is available for the solvent, the [Tyn–Calus](#) correlation is used.
- If parachors P_i and P_j are both available, the [Hayduk–Minhas](#) correlation is used.
- If the solute and solvent are both organic molecules and the liquid volume at normal boiling point, $V_{i,l,b}$ is available for the solvent, the [Siddiqi–Lucas](#) method is used.
- If the molecular weight M_j is available for the solvent, the [Wilke–Chang](#) correlation is used.
- If the solvent is a normal paraffin, and the liquid density correlation for the solvent is available, and the solute is a normal paraffin, the [Erkey–Rodden–Akgerman](#) correlation (for normal paraffins) is used.
- If the solvent is a normal paraffin, and the liquid density correlation for the solvent is available, and the Lennard–Jones diameter for the solute is available, the [Erkey–Rodden–Akgerman](#) correlation (for normal paraffins) is used.

For the [Siddiqi–Lucas](#) method, a molecule is considered organic if it has at least one C atom bound to anything other than O or C atoms. This is determined by the SMILES formula, if available. If, for any pair of species, the required input data for none of the above models is available, the entire property liquid diffusion coefficient at infinite dilution is not available.

Wilke–Chang

The correlation by Wilke and Chang ([Ref. 62](#)) for liquid phase diffusion coefficients at infinite dilution is:

$$D_{i,j}^0 = 1.858795959 \times 10^{-18} \frac{T \sqrt{\phi_j M_j}}{\eta_j V_{i,l,b}^{0.6}} \quad (6-240)$$

Molecular weight, M_j , and log liquid viscosity correlation, $\ln \eta_j$, for species j and liquid molar volume at normal boiling point for species i , $V_{i,l,b}$, is required. The Wilke–Chang association parameter ϕ_j , if unavailable, is set to:

TABLE 6-9: ASSOCIATION PARAMETERS FOR SOLVENT.

SOLVENT	ASSOCIATION PARAMETER
Water	2.26 (Ref. 63)
Methanol	1.9
Ethanol	1.5
Others	1

Species are identified by their CAS number or SMILES formula. The Wilke–Chang correlation is not suitable for diffusion of water. If water is the solute, the correction suggested by Kooijman (Ref. 64) is applied where liquid molar volume of water at normal boiling point, $V_{\text{water},l,b}$ is multiplied by 4.5.

Tyn–Calus

The Correlation by Tyn and Calus (Ref. 65) for liquid diffusion coefficients at infinite dilution reads:

$$D_{i,j}^0 = 8.93 \times 10^{-16} \frac{T \left(V_{i,l,b}^{\frac{1}{6}} \right) P_j^{0.6}}{\eta_j \left(V_{j,l,b}^{\frac{1}{3}} \right) P_i^{0.6}} \quad (6-241)$$

The log liquid viscosity correlation $\ln \eta_j$ should be available for species j , The liquid volume at normal boiling point, $V_{i,l,b}$ and parachor, P_i should be available for both species i and j .

If the solvent is nonpolar (dipole moment is zero), and the solvent is methanol, ethanol or 1-butanol or if the solvent is a mono-hydroxy alcohol, both the liquid volume at normal boiling point, $V_{j,l,b}$ and the parachor, P_j are corrected by a factor of $8 \times 10^3 \eta_j$.

If the solute is water, both the liquid volume at normal boiling point, $V_{i,l,b}$ and parachor, P_i for the solute are corrected by a factor 2. This factor also applies if it is detected from the SMILE formula that the species is an organic acid (a carboxyl group is found), except in the cases where the solvent is water, methanol, or n-butanol.

Hayduk–Minhas

Hayduk and Minhas (Ref. 66) suggested three different correlations for liquid diffusion coefficients at infinite dilution.

Aqueous Solutions: in case the solvent is water (derived from CAS number or SMILES formula), the correlation reads:

$$D_{i,j}^0 = 1.25 \times 10^{-12} T^{1.52} \left(\frac{0.07244359601}{V_{i,l,b}^{0.19}} - 0.292 \right) \times (\eta_j \times 10^3)^{\left(\frac{9.58 \times 10^{-6}}{V_{i,l,b}} - 1.12 \right)} \quad (6-242)$$

Normal paraffin solutions: In case both the solute and solvent are normal paraffins (derived from SMILES formula), the correlation reads:

$$D_{i,j}^0 = 7.30889362 \times 10^{-16} T^{1.47} \left(\frac{(\eta_j \times 10^3)^{\left(\frac{10.2 \times 10^{-6}}{V_{i,l,b}} - 0.791 \right)}}{V_{i,l,b}^{0.71}} \right) \quad (6-243)$$

For all other systems, the correlation is:

$$D_{i,j}^0 = 3.8934239 \times 10^{-16} \frac{T^{1.29} \sqrt{P_j}}{\eta_j^{0.92} (V_{j,l,b}^{0.23} P_i^{0.42})} \quad (6-244)$$

If the solvent is nonpolar (dipole moment is zero), and the solvent is methanol, ethanol or 1-butanol or if the solvent is a mono-hydroxy alcohol, both the liquid volume at normal boiling point, $V_{j,l,b}$ and the parachor, P_j are corrected by a factor of $8 \times 10^3 \eta_j$.

If the solute is water, both the liquid volume at normal boiling point, $V_{i,l,b}$ and parachor, P_i for solute are corrected by a factor 2. This factor also applies if it is detected from the SMILES formula that the species is an organic acid (a carboxyl group is found), except in the cases where the solvent is water, methanol, or n-butanol.

Siddiqi–Lucas

Siddiqi and Lucas (Ref. 67) suggested correlations for liquid diffusion coefficients at infinite dilution for aqueous system (including gases) and for organic solutions. For normal paraffins systems, they recommended the Hayduk–Minhas model.

For aqueous solutions:

$$D_{i,j}^0 = 1.295436901 \times 10^{-17} \frac{T}{\eta_j^{1.026} (V_{i,l,b}^{0.5473})} \quad (6-245)$$

For all other systems, the equation for organic solution is used as:

$$D_{i,j}^0 = 1.459473761 \times 10^{-15} \frac{T(V_{i,l,b}^{0.265})}{\eta_j^{0.907} (V_{i,l,b}^{0.45})} \quad (6-246)$$

Where T is the temperature, $V_{i,l,b}$ is the liquid volume at normal boiling point and η_j is the viscosity of the solvent.

Erkey–Rodden–Akgerman

The correlation by Erkey and others (Ref. 68) for liquid diffusion coefficients at infinite dilution in normal paraffins is:

$$D_{i,j}^0 = 94.5 \times 10^{-9} \frac{\sqrt{T}(V - V_j^{\text{ref}}) \times 10^6}{M_i^{0.239} M_j^{0.781} (\sigma_i \sigma_j \times 10^{-20})^{1.134}} \quad (6-247)$$

where the reference volume is

$$V_j^{\text{ref}} = \frac{1}{\sqrt{2}} b_j N_{\text{Av}} \sigma_j^3 \quad (6-248)$$

N_{Av} is Avogadro's number, and the deviation from closest packing volume is given by

$$b_j = 1.206 + 0.0632 \left(\frac{\sigma_i}{\sigma_j} \right) \quad (6-249)$$

The correlation is fitted to normal paraffins, hydrogen, carbon monoxide and carbon dioxide diffusing in normal paraffins. The molecular weight M_i should be available for both solute and solvent. The Lennard–Jones diameter, σ_i is estimated from Bondi group contribution method (Ref. 69), and for some species are taken from Ref. 68 and Ref. 70 as:

TABLE 6-10: MOLECULAR DIAMETERS FOR SPECIES.

SPECIES	$\sigma \times 10^{10}$ (m)
H ₂	2.92
CO	3.72
CO ₂	3.97
n-C _j H _{2j+2}	$(21.82 + 32.44 * j)^{1/3}$

To prevent the diffusion coefficients from becoming negative, the minimum difference of $(V - V_j^{\text{ref}})$ is considered to be 10^{-12} mol/m³.

Liquid Phase Maxwell–Stefan Diffusivity

The Maxwell–Stefan liquid diffusion coefficients are calculated from the liquid diffusion coefficients at infinite dilution. The diagonal values should be ignored and are set to zero. The Vignes (Ref. 71) model for diffusion in binary solutions can be extended to multicomponent systems (Ref. 72):

$$D_{i,j} = \prod_{k=1}^n \mathcal{D}_{i,j,x_k \rightarrow 1}^{x_k} \quad (6-250)$$

where $\mathcal{D}_{i,j,x_k \rightarrow 1}$ denotes the Maxwell–Stefan liquid diffusion coefficient for species i and j in the limited of pure species k . If $k = j$, it represents a binary system of species i and j where i is infinitely diluted. In addition, at infinite dilution the thermodynamics factor (activity) (Ref. 57) becomes unity and the Maxwell–Stefan diffusivity equals to Fick diffusivity:

$$D_{i,j,x_j \rightarrow 1} = D_{i,j}^0 = D_{i,j}^0 \quad (6-251)$$

Similarly,

$$D_{i,j,x_i \rightarrow 1} = D_{j,i}^0 = D_{j,i}^0 \quad (6-252)$$

Models for the $\mathcal{D}_{i,j,x_k \rightarrow 1}$ for $k \neq i$ and $k \neq j$ remain to be defined in such a way that symmetry is ensured:

$$D_{i,j} = D_{j,i} \quad (6-253)$$

and to ensure continuity if both species i and j vanish:

$$\lim_{x_i \rightarrow 0} \mathcal{D}_{i,j,x_k \rightarrow 1} = \lim_{x_j \rightarrow 0} \mathcal{D}_{i,j,x_k \rightarrow 1} \quad (6-254)$$

For binary systems, Equation 6-250 reduces to the Vignes interpolation formula.

Wesselingh–Krishna

Wesselingh and Krishna (Ref. 72) proposed:

$$D_{i,j,x_k \rightarrow 1} = \sqrt{D_{i,j}^0 D_{j,i}^0} \quad (6-255)$$

leading to:

$$D_{i,j} = (D_{i,j}^0)^{(1+x_j-x_i)/2} (D_{j,i}^0)^{(1+x_i-x_j)/2} \quad (6-256)$$

Kooijman–Taylor

Kooijman and Taylor (Ref. 73) found on a limited number of systems that this provides better results:

$$D_{i,j,x_k \rightarrow 1} = \sqrt{D_{i,k}^0 D_{j,k}^0} \quad (k \neq i, j) \quad (6-257)$$

leading to:

$$D_{i,j} = (D_{i,j}^0)^{x_j} (D_{j,i}^0)^{x_i} \prod_{\substack{k=1 \\ k \neq i,j}}^n (\sqrt{D_{i,k}^0 D_{j,k}^0})^{x_k} \quad (6-258)$$

Wesselingh and Bollen (Ref. 74) asserted that this is a reasonable estimate.

Krishna–van Baten

Krishna and van Baten (Ref. 75), on the basis of data obtained from molecular dynamic simulations, proposed the following Vignes-based (Ref. 71) interpolation:

$$D_{i,j,x_k \rightarrow 1} = (D_{i,k}^0)^{\left(\frac{x_i}{x_i+x_j}\right)} (D_{j,k}^0)^{\left(\frac{x_j}{x_i+x_j}\right)} \quad (k \neq i, j) \quad (6-259)$$

The model reduces to the Kooijman–Taylor model for $x_i = x_j$. The value of $D_{i,j,x_k \rightarrow 1}$ is undefined in the limit of both $x_i \rightarrow 0$ and $x_j \rightarrow 0$. From a physical point of view this is inconsequential as the value cancels out in the expressions obtained for the fluxes using the Maxwell–Stefan equations. Nevertheless, in order to obtain well-defined values and composition derivatives of $D_{i,j,x_k \rightarrow 1}$ itself, the equation is modified to:

$$D_{i,j,x_k \rightarrow 1} = (D_{i,k}^0)^{\left(\frac{x_i + \epsilon_x}{x_i + x_j + 2\epsilon_x}\right)} (D_{j,k}^0)^{\left(\frac{x_j + \epsilon_x}{x_i + x_j + 2\epsilon_x}\right)} \quad (k \neq i, j) \quad (6-260)$$

where $\epsilon_x = 10^{-10}$ is taken as a small composition. The limiting case for both $x_i \rightarrow 0$ and $x_j \rightarrow 0$ also reduces to the Kooijman–Taylor model. This equation leads to:

$$D_{i,j} = (D_{i,j}^0)^{x_j} (D_{j,i}^0)^{x_i} \prod_{\substack{k=1 \\ k \neq i,j}}^n \left((D_{i,k}^0)^{\left(\frac{x_i + \epsilon_x}{x_i + x_j + 2\epsilon_x} \right)} (D_{j,k}^0)^{\left(\frac{x_j + \epsilon_x}{x_i + x_j + 2\epsilon_x} \right)} \right)^{x_k} \quad (6-261)$$

Surface Tension

The available surface tension models are presented below.

GAS–LIQUID SURFACE TENSION

Ideal

The gas–liquid surface tension is predicted by ideally mixing the pure species correlations. It is independent of pressure, vapor temperature or composition.

$$\sigma_{vl} = \sum_i x_{i,l} \sigma_{i,vl} \quad (6-262)$$

where the vapor–liquid surface tension correlations, $\sigma_{i,vl}$, must be specified for all species i , and are evaluated at the temperature of the liquid phase.

Winterfeld

Following Winterfeld ([Ref. 76](#)), the vapor–liquid surface tension is predicted by mixing the pure species correlations according to

$$\sigma_{vl} = \frac{\sum_i \sum_j \frac{x_{i,l} x_{j,l} \sqrt{\sigma_{i,vl} \sigma_{j,vl}}}{\rho_{i,l} \rho_{j,l}}}{\left(\sum_i \frac{x_{i,l}}{\rho_{i,l}} \right)^2} \quad (6-263)$$

It is independent of pressure, vapor temperature and composition. The vapor–liquid surface tension correlations and liquid density correlations must be specified for all species i , and are evaluated at the temperature of the liquid phase.

References

1. A. Akerberg, *CFD analyses of the gas flow inside the vessel of a hot isostatic press*, Master of Science Thesis, KTH School of Industrial Engineering and Management, Stockholm, Sweden, 2012.

2. M.A. Trebble and P.R. Bishnoi, “Accuracy and consistency comparisons of ten cubic equations of state for polar and non-polar compounds,” *Fluid Ph. Equilib.*, vol. 29, pp. 465–474, 1986.
3. B.E. Poling, J.M. Prausnitz, and J.P. O’Connell, *The Properties of Gases and Liquids*, McGraw Hill, international edition, 2007.
4. M.C. Clapeyron, “Mémoire sur la puissance motrice de la chaleur,” *Journal de l’École polytechnique*, vol. 23, pp. 153–190, 1834 (in French).
5. D.Y. Peng and D. Robinson, “A new two-constant equation of state,” *Industrial and Engineering Chemistry: Fundamentals*, vol. 15, pp. 59–64, 1976.
6. C.H. Twu, J.E. Coon, and J.R. Cunningham, “A new generalized alpha function for a cubic equation of state, Part 1, Peng–Robinson equation,” *Fluid Phase Equilibria*, vol. 105, pp. 49–59, 1995.
7. O. Redlich and J.N.S. Kwong, “On the thermodynamics of solutions an equation of state, fugacities of gaseous solutions,” *Chem. Rev.*, vol. 44, no. 1, pp. 233–244, 1949.
8. G. Soave, “Equilibrium constants from a modified Redlich–Kwong equation of state,” *Chem. Eng. Sci.*, vol. 27, no. 6, pp. 1197–1203, 1972.
9. M.S. Graboski and T.E. Daubert, “A modified Soave equation of state for phase equilibrium calculations, 3, systems containing hydrogen,” *Ind. Eng. Chem. Process. Des. Dev.*, vol. 18, no. 2, pp. 300–306, 1979.
10. J.R. Cooper, “Revised Release on the IAPWS Industrial Formulation 1997 for the Thermodynamic Properties of Water and Steam,” *The International Association for the Properties of Water and Steam*, IAPWS R7-97, 2012.
11. K. Watanabe, “Revised Supplementary Release on Backward Equations for the Functions $T(p,h)$, $v(p,h)$ and $T(p,s)$, $v(p,s)$ for Region 3 of the IAPWS Industrial Formulation 1997 for the Thermodynamic Properties of Water and Steam,” *The International Association for the Properties of Water and Steam*, Kyoto, Japan, Sep. 2004.
12. F. Marsik, “Supplementary Release on Backward Equations for Specific Volume as a Function of Pressure and Temperature $v(p,T)$ for Region 3 of the IAPWS Industrial Formulation 1997 for the Thermodynamics Properties of Water and Steam,” *The International Association for the Properties of Water and Steam*, Santorini, Greece, July 2005.

13. K.C. Chao and J.D. Seader, "A general correlation of vapor-liquid equilibria in hydrocarbon mixtures," *AIChE J.*, vol. 7, no. 4, pp. 598–605, 1961.
14. J.H. Hildebrand and R.L. Scott, "The solubility of non-electrolytes," *J. Phys. Chem.*, vol. 55, no. 4, pp. 619–620, 1951.
15. H.G. Grayson and C.W. Streed, "Vapor-liquid equilibria for high temperature, high pressure hydrogen-hydrocarbon systems," *6th World Petroleum Congress*, 19–26 June, Frankfurt am Main, Germany, IV, pp. 169–180, 1963.
16. J.H. Hildebrand, J.M. Prausnitz, and R.L. Scott. *Regular and Related Solutions*, Van Nostrand Reinhold Co., 1970.
17. G.M. Wilson, "Vapor-liquid equilibrium. xi. a new expression for the excess free energy of mixing," *J. Am. Chem. Soc.*, vol. 86, pp. 127–130, 1964.
18. H. Renon and J.M. Prausnitz, "Local compositions in thermodynamic excess functions for liquid mixtures," *AIChE J.*, vol. 14, no. 1, pp. 135–144, 1968.
19. D.S. Abrams and J.M. Prausnitz, "Statistical thermodynamics of liquid mixtures: A new expression for the excess Gibbs energy of partly or completely miscible systems," *AIChE J.*, vol. 21, no. 1, pp. 116–128, 1975.
20. A. Fredenslund, R.L. Jones, and J.M. Prausnitz, "Group-contribution estimation of activity coefficients in nonideal liquid mixtures," *AIChE J.*, vol. 21, no. 6, pp. 1086–1099, 1975.
21. S. Skjold-Joergensen, B. Kolbe, J. Gmehling, and P. Rasmussen, "Vapor-liquid equilibria by unifac group contribution. revision and extension," *Ind. Eng. Chem. Process. Des. Dev.*, vol. 18, no. 4, pp. 714–722, 1979.
22. J. Gmehling, P. Rasmussen, and A. Fredenslund, "Vapor-liquid equilibria by unifac group contribution. revision and extension. 2," *Ind. Eng. Chem. Process. Des. Dev.*, vol. 21, no. 1, pp. 118–127, 1982.
23. E.A. Macedo, U. Weidlich, J. Gmehling, and P. Rasmussen, "Vapor-liquid equilibria by unifac group contribution. 3. Revision and extension," *Ind. Eng. Chem. Process. Des. Dev.*, vol. 22, no. 4, pp. 676–678, 1983.
24. D. Tiegs, J. Gmehling, P. Rasmussen P, and A. Fredenslund, "Vapor-liquid equilibria by unifac group contribution. 4. Revision and extension," *Ind. Eng. Chem. Process. Des. Dev.*, vol. 26, no. 1, pp. 159–161, 1987.

25. H.K. Hansen, P. Rasmussen, A. Fredenslund, M. Schiller, and J. Gmehling, "Vapor-liquid equilibria by unifac group contribution. 5. Revision and extension," *Ind. Eng. Chem. Process. Des. Dev.*, vol. 30, no. 10, pp. 2352–2355, 1991.
26. R. Wittig, J. Lohmann, and J. Gmehling, "Vapor-liquid equilibria by unifac group contribution. 6. Revision and extension," *Ind. Eng. Chem. Process. Des. Dev.*, vol. 42, no. 1, pp. 183–188, 2003.
27. K. Balslev and J. Abildskov, "Unifac parameters for four new groups," *Ind. Eng. Chem. Res.*, vol. 41, pp. 2047–2057, 2002.
28. R.W. Hankinson and G.H. Thomson, "A new correlation for saturated densities of liquid and their mixtures," *AIChE J.*, vol. 25, no. 4, pp. 653–663, 1979.
29. H.G. Rackett, "Equation of state for saturated liquids," *J. Chem. Eng. Data*, vol. 15, no. 4, pp. 514–517, 1970.
30. L.I. Stiel and G. Thodos, "The thermal conductivity of nonpolar substances in the dense gaseous and liquid regions," *AIChE J.*, vol. 10, no. 1, pp. 26–30, 1964.
31. M. Yorzane, S. Yoshimura, H. Masuoka, and H. Yoshida, "Thermal conductivities of binary gas mixtures at high pressures: nitrogen-oxygen, nitrogen-argon, carbon dioxide-argon, and carbon dioxide-methane," *Industrial & Engineering Chemistry Fundamentals*, vol. 22, no. 4, pp. 458–463, 1983.
32. A.L. Lindsay and L.A. Bromley, "Thermal conductivity of gas mixtures," *Industrial and Engineering Chemistry*, vol. 42, no. 8, pp. 1508–1511, 1950.
33. A. Wassiljewa, "Heat conduction in gas mixtures," *Physikalische Zeitschrift*, vol. 5, no. 22, pp. 737–742, 1904.
34. J.R. Cooper, "Release on the IAPS Formulation 1985 for the Thermal Conductivity of Ordinary Water Substance," *The International Association for the Properties of Water and Steam*, Berlin, Germany, Sep. 2008.
35. C.R. Wilke, "A viscosity equation for gas mixtures," *J. Chem. Phys.*, vol. 18, no. 4, pp. 517–520, 1950.
36. R.S. Brokaw, "Approximate formulas for the viscosity and thermal conductivity of gas mixtures. ii," *J. Chem. Phys.*, vol. 42, no. 4, pp. 1140–1147, 1965.
37. J.E. Lennard-Jones, "On the determination of molecular fields 1. from the variation of the viscosity of a gas with temperature," *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, vol. 106, pp. 441–462, 1924.

38. W.H. Stockmayer, "Second virial coefficients of polar gases," *J. Chem. Phys.*, vol. 9, pp. 398–402, 1941.
39. F.M. Mourits and F.H.A. Rummens, "A critical evaluation of Lennard–Jones and Stockmayer potential parameters and of some correlation methods," *Can. J. Chem. Eng.*, vol. 55, pp. 3007–3020, 1977.
40. T.A. Davidson, "A Simple and Accurate Method for Calculating Viscosity of Gaseous Mixtures," *Report of Investigations 9456*, United States Department of the Interior, Bureau of Mines, 1993.
41. J.A. Jossi, L.I. Stiel, and G. Thodos, "The viscosity of pure substances in the dense gaseous and liquid phases," *AIChE J.*, vol. 8, no. 1, pp. 59–63, 1962.
42. K.S. Pedersen and P.L. Christensen, "Phase Behavior of Petroleum Reservoir Fluids," *CRC Press/Taylor & Francis Group*, 2007.
43. K.S. Pedersen, A. Fredenslund, P.L. Christensen, and P. Thomassen, "Viscosity of crude oils," *Chem. Eng. Sci.*, vol. 39, no. 6, pp. 1011–1016, 1984.
44. H.J.M. Hanley, W.M. Haynes, and R.D. McCarty, "The viscosity and thermal conductivity coefficients for dense gaseous and liquid methane," *J. Chem. Phys.*, vol. 6, no. 2, pp. 597–609, 1977.
45. K.S. Pedersen and A. Fredenslund, "An improved corresponding states model for the prediction of oil and gas viscosities and thermal conductivities," *Chem. Eng. Sci.*, vol. 42, no. 1, pp. 182–186, 1987.
46. R.D. McCarty, "A modified Benedict-Webb-Rubin equation of state for methane using recent experimental data," *Cryogenics*, pp. 276–280, May 1974.
47. S. Murad and K.E. Gubbins, "Corresponding states correlation for thermal conductivity of dense fluids," *Chem. Eng. Sci.*, vol. 32, no. 5, pp. 499–505, 1977.
48. J.R. Cooper, "Revised Release on the IAPWS Formulation 2008 for the Viscosity of Ordinary Water Substance," *The International Association for the Properties of Water and Steam*, Berlin, Germany, Sep. 2008.
49. K. Dacic, "Revised Release on the Pressure along the Melting and Sublimation Curves of Ordinary Water Substance," *The International Association for the Properties of Water and Steam*, Plzen, Czech Republic, Sep. 2011.
50. J. Gabitto and M. Barrufet, "Experimental and theoretical determination of heavy oil viscosity under reservoir conditions," *Technical reports*, Office of Fossil Energy, USDOE Office of Fossil Energy, United States, 2002.

51. A. Missenard, "Conductivité thermique des liquides organiques d'une série ou d'un 'groupe de liquides'," *Revue Générale de Thermique*, vol. 101, no. 5, pp. 649–660, 1970 (in French).
52. R.L. Rowley, G.L. White, and M. Chiu, "Ternary liquid mixture thermal conductivities," *Chem. Eng. Sci.*, vol. 43, no. 2, pp. 361–371, 1988.
53. R.L. Rowley, "A local composition model for multicomponent liquid mixture thermal conductivities," *Chem. Eng. Sci.*, vol. 37, no. 6, pp. 897–904, 1982.
54. W.D. Niven, "The Scientific Papers of James Clerk Maxwell," *Cambridge University Press*, 1890.
55. J.C. Maxwell, "On the dynamical theory of gases," *Philos. Trans. R. Soc.*, vol. 157, pp. 49–88, 1867.
56. J. Stefan, "Über das Gleichgewicht und die Bewegung, insbesondere die Diffusion von Gasgemengen," *Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften Wien, 2te Abteilung a*, vol. 63, pp. 63–124, 1871.
57. R. Taylor and R. Krishna, "Multicomponent Mass Transfer," *Wiley-Interscience*, 1993.
58. E.N. Fuller, P.D. Schettler and J.C. Giddings, "A new method for prediction of binary gas-phase diffusion coefficients," *Industrial and Engineering Chemistry*, vol. 5, pp. 19–27, 1966.
59. E.N. Fuller, K. Ensley and J.C. Giddings, "Diffusion of halogenated hydrocarbons in helium. The effect of structure on collision cross sections," *J. Phys. Chem.*, vol. 73, no. 11, pp. 3679–3685, 1969.
60. C.R. Wilke and C.Y. Lee, "Estimation of diffusion coefficients for gases and vapors," *Industrial and Engineering Chemistry*, vol. 47, no. 6, pp. 1253–1257, 1955.
61. P.D. Neufeld, A.R. Janzen, and R.A. Aziz, "Empirical equations to calculate 16 of the transport collision integrals $\Omega^{(l,s)*}$ for the Lennard–Jones (12-6) potential," *J. Chem. Phys.*, vol. 57, pp. 1100–1102, 1972.
62. C.R. Wilke and P. Chang, "Correlation of diffusion coefficients in dilute solutions," *AIChE J.*, vol. 1, no. 2, pp. 264–270, 1972.
63. W. Hayduk and H. Laudie, "Prediction of diffusion coefficients for nonelectrolytes in dilute aqueous solutions," *AIChE J.*, vol. 20, no. 3, pp. 611–615, 1974.

64. H.A. Kooijman, "A modification of the Stokes-Einstein equation for diffusivities in dilute binary mixture," *Ind. Eng. Chem.*, vol. 41, pp. 3326–3328, 2002.
65. M.T. Tyn and W.F. Calus, "Diffusion coefficients in dilute binary mixtures," *J. Chem. Eng. Data*, vol. 20, no. 1, pp. 106–109, 1975.
66. W. Hayduk and B.S. Minhas, "Correlation for prediction of molecular diffusivities in liquids," *Can. J. Chem. Eng.*, vol. 60, no. 2, pp. 295–299, 1983.
67. M.A. Siddiqi and K. Lucas, "Correlations for prediction of diffusion in liquids," *Can. J. Chem. Eng.*, vol. 64, pp. 839–843, 1986.
68. C. Erkey, J.B. Rodden and A. Akgerman, "A correlation for predicting diffusion coefficients in alkanes," *Can. J. Chem. Eng.*, vol. 68, pp. 661–665, 1990.
69. A. Bondi, "van der Waals Volumes and Radii," *J. Phys. Chem.*, vol. 68, no. 3, pp. 441–451, 1964.
70. J.O. Hirschfelder, C.F. Curtiss and R.B. Bird, "The Molecular Theory of Gases and Liquids," *Wiley-Interscience*, 1964.
71. A. Vignes, "Diffusion in Binary Solutions," *Ind. Eng. Chem. Fundam.*, vol. 5, pp. 189–199, 1966.
72. J.A. Wesselingh and R. Krishna, "Mass Transfer," *Ellis Horwood Ltd*, 1990, ISBN: 0-13-553165-9.
73. H.A. Kooijman and R. Taylor, "Estimation of diffusion coefficients in multicomponent liquid systems," *Ind. Eng. Chem.*, vol. 30, pp. 1217–1222, 1991.
74. J.A. Wesselingh and A.M. Bollen, "Multicomponent diffusivities from the free volume theory," *Chem. Eng. Res. Des.*, vol. 75, no. 6, pp. 590–602, 1997.
75. R. Krishna and J.M. van Baten, "The Darken relation for multicomponent diffusion in liquid mixtures of linear alkanes: An investigation using molecular dynamics (MD) simulations," *Ind. Eng. Chem.*, vol. 44, pp. 6939–6947, 2005.
76. P.H. Winterfeld, L.E. Scriven, and H.T. Davis, "An approximate theory of interfacial tension of multicomponent systems: Applications binary liquid-vapor tensions," *AIChE J.*, vol. 24, no. 6, pp. 1010–1014, 1978.
77. E.C. Carlson, "Don't Gamble With Physical Properties For Simulation," *Chem. Eng. Prog.*, vol. 92, no. 10, pp. 35–46, 1996.

Glossary

This [Glossary of Terms](#) contains application-specific terms used in the Chemical Reaction Engineering Module software and documentation. For finite element modeling terms, mathematical terms, and geometry and CAD terms, see the glossary in the *COMSOL Multiphysics Reference Manual*. For references to more information about a term, see the index.

Glossary of Terms

absorption Uptake of a gas into the bulk of a liquid. Gas absorption takes place for example in the liquid of a scrubber tower where an up-streaming gas is washed by a down-going flow of a scrubber solution.

adsorption Attachment of a molecule or atom to a solid surface. Adsorption involves a chemical bond between the adsorbed species and the surface.

anisotropy Variation of a transport property in different directions in a material. Is often obtained from homogenization of regular structures, for example, monolithic structures in tubular reactors.

Arrhenius rate equation Expression that relates the rate constant of a chemical reaction to the exponential of the temperature.

batch reactor Reactor characterized by its operation, which means that the reactor does reaches steady state.

bipolar plate Electrically conducting plate connected to the anode on one side and to the cathode on the other side in an electrochemical cell.

Brinkman equations Extension of Darcy's law in order to include the transport of momentum through shear in porous media flow.

Butler-Volmer equation Expression that relates the reaction rate of an electron transfer reaction on an electrode surface to the exponential of the overpotential. The equation can be derived from the Arrhenius rate equation by accounting for the contribution of the electric potential to the activation energy.

boundary layer Region in a fluid close to a solid surface. This region is characterized by large gradients in velocity and is often treated with approximate methods because it is difficult to geometrically resolve the large gradients found there.

chemisorption See *adsorption*.

continuous reactor Reactor that operates without interruption. This type of reactor is characterized by its steady-state operation.

diffusion layer Fictitious layer in a fluid close to a solid surface where a chemical reaction takes place. The flux of species perpendicular to the surface in this layer is dominated by diffusion.

Darcy's law Equation that gives the velocity vector as proportional to the pressure gradient. Often used to describe flow in porous media.

electrokinetic flow Transport of fluid or charged particles within a fluid by means of electric fields.

electroneutrality condition Condition that states that the sum of charges in a control volume in an electrolyte should be zero.

electrophoresis Migration of charged electrolyte ions in an electric field.

electroosmosis Onset of a flow due to the application of an external electric field or due to the formation of an electric field created by ion transport in membranes, for example.

Euler flow Flow at high velocities, where incompressibility of the fluid is of importance whereas the influence of viscous momentum transport is negligible.

Fick's laws The first law relates the concentration gradients to the diffusive flux of a solute infinitely diluted in a solvent. The second law introduces the first law into a differential material balance for the solute.

fully developed laminar flow Laminar flow along a channel or pipe that only has velocity components in the main direction of the flow. The velocity profile perpendicular to the flow does not change downstream in the flow.

Hagen–Poiseuille equation See *Poiseuille's law*.

heterogeneous reaction Reaction that takes place at the interface between two phases.

homogeneous reaction Reaction that takes place in the bulk of a solution.

law of the wall See *wall function*.

Maxwell-Stefan equations Set of equations that describe the diffusion of solutes and solvent in a concentrated solution. In such a solution, the solutes interact with each other and with the solvent.

migration The transport of charged species in an electrolyte due to the electric force imposed by the electric field.

monolithic reactor Catalytic reactor made of one single piece of solid material. Incorporates a catalytic structure in its often porous structure.

Navier–Stokes equations Equations for the momentum balances coupled to the equation of continuity for a Newtonian incompressible fluid.

Nernst–Planck equation Equation that describes the flux of an ion through diffusion, convection, and migration in an electric field. The equation is valid for diluted electrolytes.

Newtonian flow Flow characterized by a constant viscosity or a viscosity that is independent of the shear rate in the fluid.

physisorption See *adsorption*.

Poiseuille’s law Equation that relates the mass rate of flow in a tube as proportional to the pressure difference per unit length and to the fourth power of the tube radius. The law is valid for fully developed laminar flow.

Soret effect Mass diffusion in multicomponent mixtures due to temperature gradients.

specific surface area Internal surface area of a porous structure given in area per unit volume, which yields the unit one over unit length. Often used to characterize the structure of porous catalysts.

streamline-diffusion stabilization A numerical technique for stabilization of the numeric solution to a convection-dominated PDE by artificially adding diffusion in the direction of the streamlines.

surface molar flux the tangential flux in the surface dimension as governed by diffusion according to Fick’s law.

wall function Semi-empirical expression for the anisotropic flow close to a solid surface used in turbulence models. Often based on negligible variations in pressure gradient in the direction tangential to the surface.

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